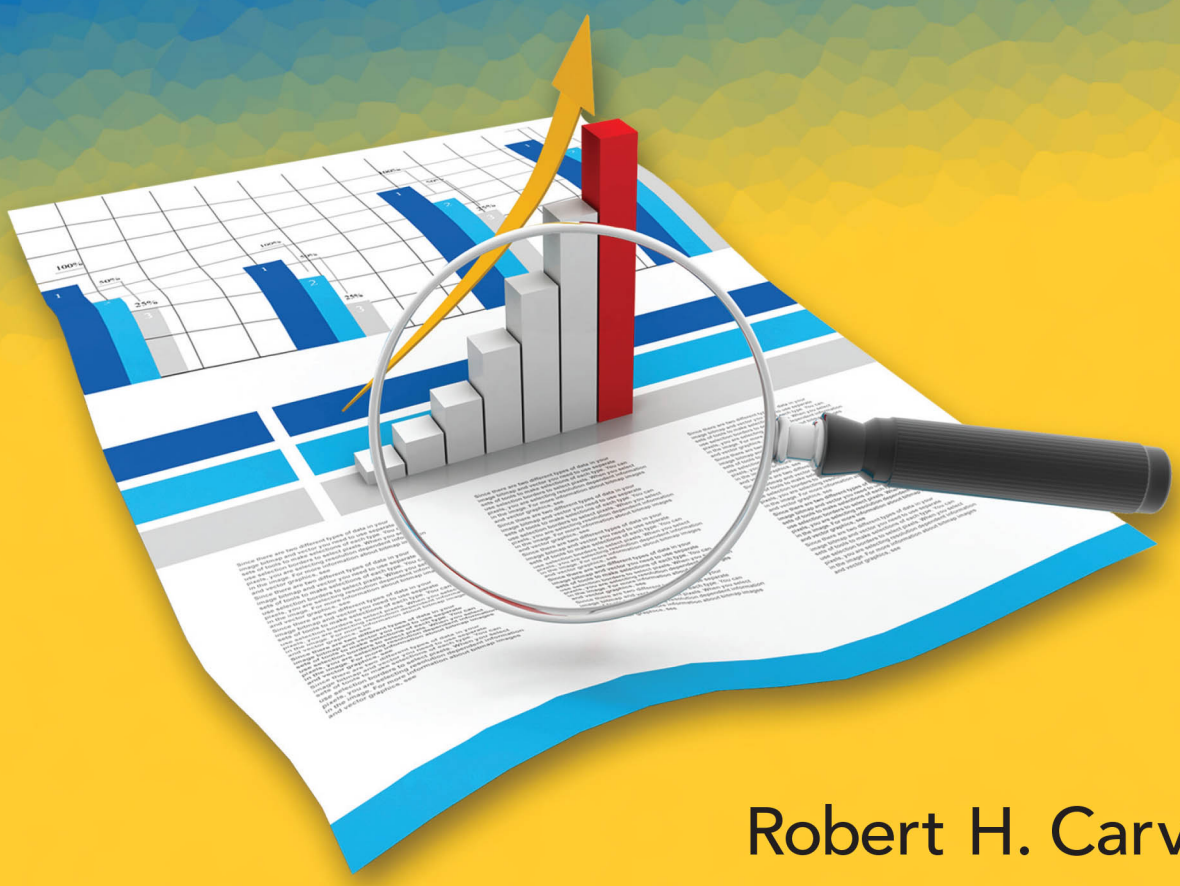
# Practical Data Analysis with JMP®

*Third Edition*



Robert H. Carver

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# About This Book

## What Does This Book Cover?

### Purpose: Learning to Reason Statistically

We live in a world of uncertainty. Today more than ever before, we have vast resources of data available to shed light on crucial questions. But at the same time, the sheer volume and complexity of the "data deluge" can distract and overwhelm us. The goal of applied statistical analysis is to work with data to calibrate, cope with, and sometimes reduce uncertainty. Business decisions, public policies, scientific research, and news reporting are all shaped by statistical analysis and reasoning. Statistical thinking is an essential part of the boom in "big data analytics" in numerous professions. This book will help you use and discriminate among some fundamental techniques of analysis, and it will also help you engage in statistical thinking by analyzing real problems. You will come to see statistical investigations as an iterative process and will gain experience in the major phases of that process.

To be an effective analyst or consumer of other people's analyses, you must know how to use these techniques, when to use them, and how to communicate their implications. Knowing how to use these techniques involves mastery of computer software like JMP. Knowing when to use these techniques requires an understanding of the theory underlying the techniques and practice with applications of the theory. Knowing how to effectively communicate with consumers of an analysis or with other analysts requires a clear understanding of the theory and techniques, as well as clarity of expression, directed toward one's audience.

There was a time when a first course in statistics emphasized abstract theory, laborious computation, and small sets of artificial data—but not practical data analysis or interpretation. Those days are thankfully past, and now we can address all three of the skill sets just cited.

### Scope and Structure of This Book

As a discipline, statistics is large and growing; the same is true of JMP. One paperback book must limit its scope, and the content boundaries of this book are set intentionally along several dimensions.

First, this book provides considerable training in the basic functions of JMP 15. JMP is a full-featured, highly interactive, visual, and comprehensive package. The book assumes that you have the software at your school or office. The software's capabilities extend far beyond an introductory course, and this book makes no attempt to "cover" the entire program. The book introduces students to its major platforms and essential features and should leave students with

sufficient background and confidence to continue exploring on their own. Fortunately, the Help system and accompanying manuals are quite extensive, as are the learning resources available online at http://www.jmp.com.

Second, the chapters largely follow a traditional sequence, making the book compatible with many current texts. As such, instructors and students will find it easy to use the book as a companion volume in an introductory course. Chapters are organized around core statistical concepts rather than software commands, menus, or features. Several chapters include topics that some instructors might view as "advanced"—typically when the output from JMP makes it a natural extension of a more elementary topic. This is one way in which software can redefine the boundaries of introductory statistics.

Third, nearly all the data sets in the book are real and are drawn from those disciplines whose practitioners are the primary users of JMP software. Inasmuch as most undergraduate programs now require coursework in statistics, the examples span major areas in which statistical analysis is an important path to knowledge. Those areas include engineering, life sciences, business, and economics.

Fourth, each chapter invites students to practice the habits of thought that are essential to statistical reasoning. Long after readers forget the details of a particular procedure or the options available in a specific JMP analysis platform, this book may continue to resonate with valuable lessons about variability, uncertainty, and the logic of inference.

Each chapter concludes with a set of "Application Scenarios," which lay out a problem-solving or investigative context that is in turn supported by a data table. Each scenario includes a set of questions that implicitly require the application of the techniques and concepts presented in the chapter.

## New in the Third Edition

This edition preserves much of the content and approach of the earlier editions, while updating examples and introducing new JMP features. As in the second edition, there are three review chapters (Chapters 5, 9, and 17) that pause to recap concepts and techniques. One of the perennial challenges in learning statistics is that it is easy to lose sight of major themes as a course progresses through a series of seemingly disconnected techniques and topics. Some readers should find the review chapters to be helpful in this respect. The review chapters share a single large data set of World Development Indicators, published by the World Bank.

The scope and sequence of chapters is basically the same as the prior edition. There is some additional new material about the importance of documenting one's work with an eye toward reproducibility of analyses, as well as production of presentation-ready reporting. The second edition was based on JMP 11, and since that time, platforms have been added or modified, and some functionality has relocated in the menu system. This edition captures those changes.

Some of the updated data tables are considerably larger than their counterparts in earlier editions. This creates the opportunity to demonstrate methods for meaningful graphs when data density and overplotting become issues. I also use some of the larger data tables to introduce machine learning practices like partitioning a data set into training and validation sets.

JMP Projects are introduced in Chapter 2 and used throughout the book. Projects are a way to organize, preserve, and document multiple analyses using multiple data tables. They naturally support a logical and reproducible workflow. Using projects is a way for newcomers to establish good habits and for JMP veterans to be more efficient.

Other additions and amendments include:

- Early introduction of more data types, Header Graphs, and JMP Public.
- Expanded use of Subset, Global and Local Data Filters and Animate. In the prior editions, for example, the set of data tables included some subsets of larger tables. Because data preparation is such an important part of the analytical cycle, readers learn to perform filtering and subsetting functions on their own.
- The Recode command has evolved since JMP 11, as have the lessons using Recode. Readers will learn why and how to recode a column.
- In the Regression chapters, coverage of the Profiler has expanded, and I have added the Partition Platform to the discussion of variable selection. The Fit Curve platform also makes its first appearance, as do temporary variable transformations.
- For JMP Pro users, there is a brief treatment of the Formula Depot to facilitate comparison of models.
- In Chapter 21 on Design of Experiments, we meet Definitive Screening Designs.
- In Chapter 22, Variability Charts have been added.
- Simulators and calculators previously supplied as JSL scripts in earlier editions have been bundled among JMP's teaching demonstrations in the Help system. The text now reflects this very useful change.

# Is This Book for You?

## Intended Audience

This book is intended to supplement an introductory college-level statistics course with real investigations of some important and engaging problems. Each chapter presents a set of self-paced exercises to help students learn the skills of quantitative reasoning by performing the types of analyses that typically form the core of a first course in applied statistics. Students can learn and practice the software skills outside of class. Instructors can devote class time to statistics and statistical reasoning, rather than to rudimentary software instruction. Both students and teachers can direct their energies to the practice of data analysis in ways that inform students' understanding of the world through investigations of problems that matter in various fields of study.

Though written with undergraduate and beginning graduate students in mind, some practitioners might find the book helpful on the job and are well-advised to read the book selectively to address current tasks or projects. Chapters 1 and 2 form a good starting point

before reading later sections. Appendix B (online for this edition) covers several data management topics that might be helpful for readers who undertake projects involving disparate data sources.

## Prerequisites

No prior statistical knowledge is presumed. A basic grounding in algebra and some familiarity with the Mac OS or Windows environment are all you need in advance. An open, curious mind is also helpful.

## A Message for Instructors

I assume that most teachers view class time as a scarce resource. One of my goals in writing this book was to strive for clarity throughout so that students can be expected to work through the book on their own and learn through their encounters with the examples and exercises. This book may be especially welcome for instructors using an inverted, or flipped, classroom approach.

Instructors might selectively use exercises as in-class demonstrations or group activities, interspersing instruction or discussion with computer work. More often, the chapters and scenarios can serve as homework exercises or assignments, either to prepare for other work, to acquire skills and understanding, or to demonstrate progress and mastery. Finally, some instructors might want to assign a chapter in connection with an independent analysis project. Several of the data tables contain additional variables that are not used within chapters. These variables might form the basis for original analyses or explorations.

The bibliography may also aid instructors seeking additional data sources or background material for exercises and assignments. Tips for classroom use of JMP are also available at the book's website, accessible through the author's page at support.sas.com/carver.

## A Message for Students

Remember that the primary goal of this book is to help you understand the concepts and techniques of statistical analysis. JMP provides an ideal software environment to do just that. Naturally, each chapter is "about" the software and at times you will find yourself focusing on the details of a JMP analysis platform and its options. If you become entangled in the specifics of a problem, step back and try to refocus on the main statistical ideas rather than software issues.

This book should augment, but not replace, your primary textbook or your classroom time. To get the maximum benefit from the book, work mindfully and carefully. Read through a chapter before you sit down at the computer. Each chapter will require approximately 30 minutes of computer time; work at your own pace and take your time. Remember that variability is omnipresent, so expect that the time you need to complete a chapter may be more or less than 30 minutes.

The Application Scenarios at the end of each chapter are designed to reinforce and extend what you have learned in the chapter. The questions in this section are designed to challenge you.

Sometimes, it is obvious how to proceed with your analysis; sometimes, you will need to think a bit before you issue your first command. The idea is to engage in statistical thinking, integrating what you have learned throughout your course. There is much more to data analysis than finding a numerical answer, and these questions provide an opportunity to do realistic analysis. Because the examples use real data, don't expect to find neat "pat" results; computations won't typically come out to nice round numbers.

JMP is a large program designed for diverse user needs. Many of the features of the software are beyond the scope of an introductory course, and therefore this book does not discuss them. However, if you are curious or adventurous, you should explore the menus and Help system as well as the JMP website. You might find a quicker, more intuitive, or more interesting way to approach a problem. For most of the topics addressed in the book, you will see an introduction. There is almost always more to know.

## What Should You Know about the Examples?

Real statistical investigations begin with pressing, important, or interesting questions, rather than with a set of techniques. Researchers do not begin a study by saying "Today is a good day to compute some standard deviations." Instead, they pose questions that can be pursued by analyzing data and follow a relatively straightforward protocol to refine the question, generate or gather suitable data, apply appropriate methods, and interpret their findings. The chapters in this book present questions that I hope you will find interesting, and then rely on the data tables provided to search for answers. The questions and analyses become progressively more challenging through the book.

### Software Used to Develop the Book's Content

The book was developed using pre-production versions of JMP15 Pro. The essential examples work with JMP. Whenever a section illustrates JMP Pro functionality, that fact is clearly announced.

### Example Data

As previously noted, each of the data tables referenced within the book contains real data, much of it downloaded from public websites. There are 45 different data tables, most of which have been updated for this edition. Readers should download all of the JMP data tables via the author page at support.sas.com/carver. Appendix A describes each file and its source. Many of the tables include columns (variables) in addition to those featured in exercises and examples. These variables might be useful for projects or other assignments.

## Where Are the Exercise Solutions?

Solutions to the scenario questions are available via the author page at support.sas.com/carver. Instructors who adopt the book will be able to access all solutions. Students and other readers can find solutions to the even-numbered problems at the same site.

# Thanks and Acknowledgments

This first edition of this book began at the urging of Curt Hinrichs, the Academic Program Manager for JMP. This led to conversations with Julie Palmieri, Editor-in-Chief at SAS Press at the time, after which the project started to take shape. I have had the great good fortune to work with a different editor for each edition: Stephenie Joyner, Sian Roberts, and, most recently, Catherine Connolly have kept this little trolley on the tracks.

At SAS Press, so many people have contributed to the planning and execution of the book in its development. For this edition, Sian Roberts is now Publisher. Suzanne Morgen handled the copyediting, Denise Jones the production, Robert Harris the cover design, and Missy Hannah the marketing effort.

Earlier editions were shaped and tended by Shelley Sessoms, Stacey Hamilton, Shelly Goodin, Mary Beth Steinbach, Cindy Puryear, Brenna Leath, Brad Kellam, Candy Farrell, Patrice Cherry, and Jennifer Dilley. My enduring thanks go to them all.

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I am especially thankful for the care and attention of those people who have reviewed this and the prior editions. Technical reviews of the current edition were provided by Mark Bailey, Duane Hayes, and Kristen Bradford. Mark has been the constant among reviewers, having made invaluable recommendations to all three editions. Performing double duty on the first two editions were Tonya Mauldin and Sue Walsh. Fang Chen, Paul Marovich, and Volker Kraft rounded out the many superb reviewers. Collectively, their critiques tightened and improved this book, and whatever deficiencies that may remain are entirely mine.

Naturally, the completion of a book requires time, space, and an amenable environment. I want to express public thanks to three institutions that provided facilities, time, and atmospherics suitable for steady work on this project. My home institution, Stonehill College, was exceptionally supportive, particularly through the efforts of Provost Joe Favazza and my chairperson, Debra Salvucci, and Department Administrative Assistant Carolyn McGuinness. Colleagues Dick Gariepy and Michael Salé generously tested several chapters and problems in their classrooms, and Jan Harrison and Susan Wall of our IT Department eased several technical aspects of this project as well.

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Special acknowledgment also goes to former Stonehill students from BUS207 (Intermediate Statistics) who "road tested" several chapters, and very considerable thanks to three students who assisted greatly in shaping prose and examples, as well as developing solutions to scenario problems: Frank Groccia, Dan Bouchard, and Matt Arey. Later students in BUS206 (Quantitative Analysis for Business) at Stonehill also class-tested several chapters and exercises.

Several of the data tables came through the gracious permission of their original authors and compilers. I gratefully acknowledge the permission granted by my good friend George Aronson for the Maine SW table; by Prof. Max A. Little for the Parkinson's disease vocal data; by Prof. Jesper Rydén for the Sonatas data table (from which the Haydn and Mozart tables were extracted); by Prof. John Holcomb for the North Carolina birth weight data; and by Prof. I-Cheng Yeh for the Concrete table and the two subsets from that data.

In recent years, my thoughts about what is important in statistics education have been radically reshaped by colleagues in the ISOSTAT listserv and the Consortium for the Advancement of Undergraduate Statistics Education (CAUSE) and the U.S. Conference on Teaching Statistics that CAUSE organizes every two years. The May 2013 CAUSE-sponsored workshop "Teaching the Statistical Investigation Process with Randomization-Based Inference" given by Beth Chance, Allan Rossman, and Nathan Tintle influenced some of the changes in my presentation of inference. Over an even longer period, our local group of New England Isolated Statisticians and the great work of the ASA's Section on Statistics Education influence me daily in the classroom and at the keyboard.

Finally, it is a pleasure to thank my family. My sons, Sam and Ben, keep me modest and regularly provide inspiration and insight. My wife Donna—partner, friend, wordsmith extraordinaire—has my love and thanks for unflagging encouragement, support, and warmth. This book is dedicated to them.

## We Want to Hear from You

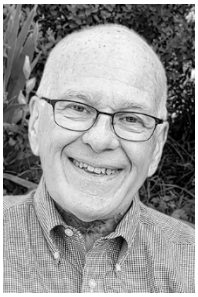
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# About The Author



Robert Carver is Professor Emeritus of Business Administration at Stonehill College in Easton, Massachusetts, and he recently retired as a senior lecturer at the Brandeis University International Business School in Waltham, Massachusetts. At both institutions, he was instrumental in establishing programs in Data Science and Business Analytics, taught courses on quantitative methods in addition to general management courses, and won teaching awards at both schools. His primary research interest is statistics education. A JMP user since 2006, Carver holds an AB in political science from Amherst College in Amherst, Massachusetts, and an MPP and PhD in public policy from the University of Michigan at Ann Arbor.

Learn more about this author by visiting his author page at support.sas.com/carver. There you can download free book excerpts, access example code and data, read the latest reviews, get updates, and more.

# Chapter 1: Getting Started: Data Analysis with JMP

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## Overview

Statistical analysis and visualization of data have become an important foundation of decision making and critical thinking. Professionals in numerous walks of life—from medicine to government, from science to sports, from commerce to public health—all rely on the analysis of data to inform their work. In this first chapter, we take our first steps into the important and rapidly growing practice of data analysis.

## Goals of Data Analysis: Description and Inference

The central goal of this book is to help you build your capacity as a statistical thinker through progressive experience with the techniques and approaches of data analysis, specifically by using the features of JMP. As such, before using JMP, we will begin with some remarks about activities that require data analysis.

People gather and analyze data for many different reasons. Engineers test materials or new designs to determine their utility or safety. Coaches and owners of professional sports teams track their players' performance in different situations to structure rosters and negotiate salary offers. Chemists and medical researchers conduct clinical trials to investigate the safety and efficacy of new treatments. Demographers describe the characteristics of populations and market segments. Investment analysts study recent market data to fine-tune investment portfolios. Increasingly, "smart" devices continuously generate high volumes of data touching on varying topics. All of the individuals who are engaged in these activities have consequential, pressing needs for information, and they turn to the techniques of statistics to meet those needs.

There are two basic types of statistical analysis: *description* and *inference*. We perform descriptive analysis in order to summarize or describe an ongoing process or the current state of a population—a group of individuals or items that is of interest to us. Sometimes we can collect data from every individual in a *population* (every professional athlete in a sport, or every firm in which we currently own stock), but more often we are dealing with a subset of a population—with a *sample* from the population. A sample is simply a subset. When we study ongoing processes, we nearly always deal with samples.

If a company reviews the records of all its client firms to summarize last month's sales to all customers, the summary will describe the population of customers. If the same company wants to use that summary information to make a forecast of sales for next month, the company needs to engage in inference. When we use available data to make a conclusion about something that we cannot observe, or about something that has not happened yet, we are drawing an inference. As we will come to understand, inferential thinking requires risk-taking. Learning to measure and minimize the risks involved in inference is a central part of the study of statistics.

## Types of Data

The practice of statistical analysis requires *data*—when we "do" analysis, we are analyzing data. It's important to understand that analysis is just one phase in a statistical study. Later in this chapter, we will look at some data collected and reported by the World Population Division of the United Nations. Specifically, we will analyze the estimated life expectancy at birth for nations around the world in 2017. This set of data is a portion of a considerably larger collection spanning many years and assembled by numerous national and international agencies.

In this example, we have five *variables* that are represented as five columns within a data table. A variable is an attribute that we can count, measure, or record. The variables in this example are a 3-letter code, country name, region, year, and life expectancy. Typically, we will capture multiple *observations* of each variable—whether we are taking repeated measurements of stock prices or recording facts from numerous respondents in a survey or individual countries around the globe. Each observation (often called a case or subject in survey data) occupies a row in a data table. In this example, the *observational units* are countries.

> Whenever we analyze a data set in JMP, we will work with a *data table*. The *columns* of the table contain different variables, and the *rows* of the table contain observations of each variable. In your statistics course, you will probably use the terms data set, variable, and observation (or case). In JMP, we more commonly speak of data tables, columns, and rows.
>
> Throughout this book, we will work with data organized into tables. The columns of the tables contain variables (for example, year, name, price) and the rows of the tables represent the individual items in the sample.

One of the organizing principles that you will notice in JMP is the differentiation among *data types* and *modeling types.* Most columns that you will work with in this book are all either

*numeric* or *character* data types, much like data in a spreadsheet are numeric or labels. JMP has two other major data types—Row States and Expressions—to be discussed later.

In your statistics course, you might be learning about the distinctions among different types of quantitative and qualitative (or categorical) data. Before we analyze any data, we will want to understand clearly whether a column is *quantitative* or *categorical*. JMP helps us keep these distinctions straight by using different *modeling types*. In the first several chapters, we will work with three modeling types:

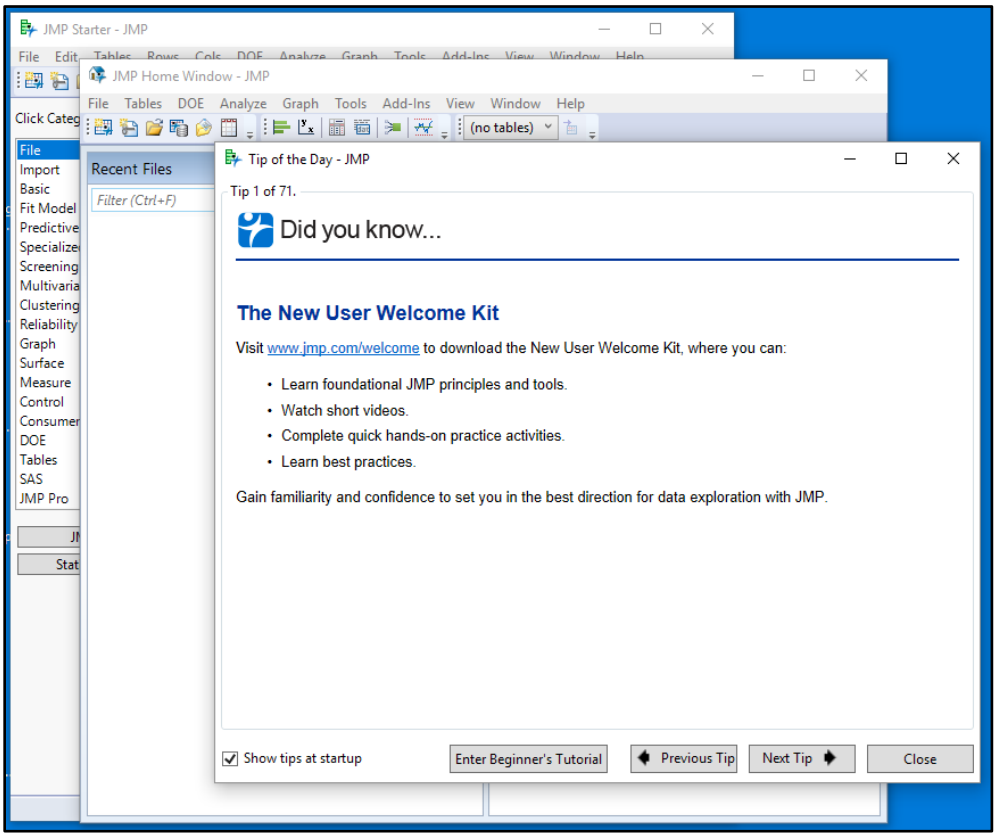
- *Continuous* columns are inherently quantitative. They are numeric so that you can meaningfully compute sums, averages, and so on. Continuous variables can assume an infinite number of values. Most measurements and financial figures are continuous data. Estimated average life expectancies (in years) are continuous.
- *Ordinal* columns reflect attributes that are sequential in nature or have some implicit ordering (for example, small, medium, large). Ordinal columns can be either numeric or character data.
- *Nominal* columns simply identify individuals or groups within the data. For example, if we are analyzing health data from different countries, we might want to label the nations and/or compare figures by continent. With our Life Expectancy 2017 data, both the names of countries and their continental regions are nominal columns. Nominal variables can also be numeric or character data. Names are nominal, as are postal codes or telephone numbers.

As we will soon see, understanding the differences among these modeling types clarifies how JMP treats our data and presents us with choices.

## Starting JMP

Whether you are using a Windows-based computer or a Macintosh, JMP works in very similar ways. All of the illustrations in this book were generated in a Windows environment. Find JMP[1] among your programs and launch it. You will see the opening screen shown in Figure 1.1. The software opens a **Tip of the Day** window each time you start the software (assuming no initial default settings have been changed). These are informative and helpful. You can choose to turn off the automatic messages by clearing the **Show tips at startup** check box in the lower left part of the window. You will be well advised to click the **Enter Beginner's Tutorial** button sooner rather than later to get a helpful introduction to the program (perhaps you should do so now or after reading this chapter). After you have read the tip of the day, click **Close**.

**Figure 1.1: The JMP Opening Screen**




JMP provides an extensive set of tutorials for users that illustrate many of the features of the software. Readers are encouraged to investigate the tutorials on their own. Find the full list of tutorials in the Help menu.

**You will also see** the **JMP Starter** window, which is an annotated menu of major functions. It is worth your time to explore the **JMP Starter** window by navigating through its various choices to get a feel for the wide scope of capabilities that the software offers. As a new user, though, you might find the range of choices to be overwhelming.

In this book, we will tend to close the JMP Starter window and use the menu bar at the top of the screen to make selections. Finally, look at the **JMP Home Window**. The home window is divided into two panes that can help you keep track of recently used files and currently open windows. You can customize this view, but this book shows the standard two-pane layout.

# A Simple Data Table

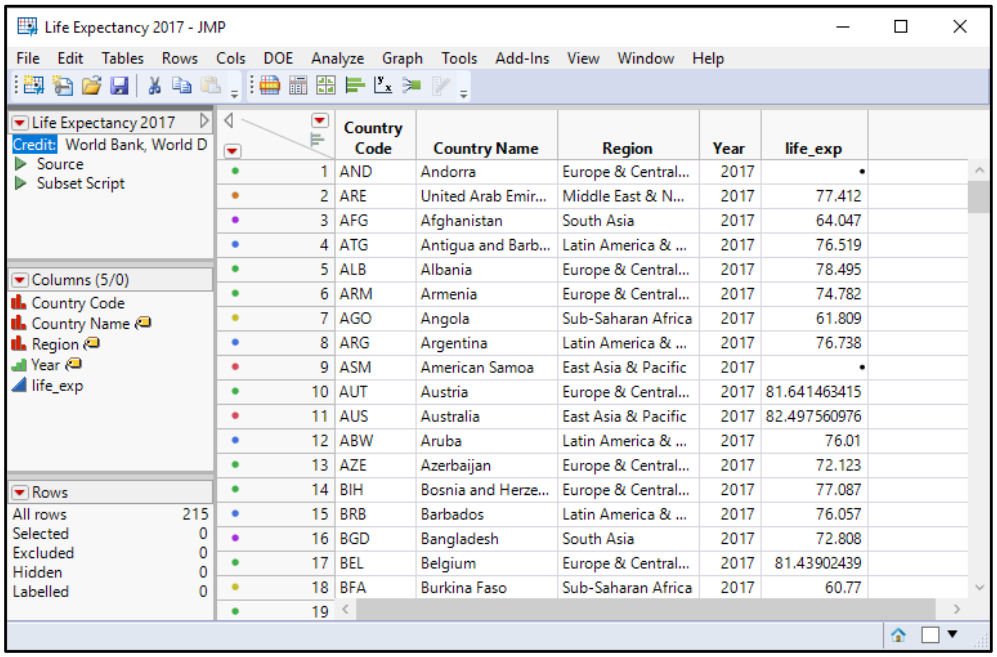
In this book, we will most often work with data that has already been entered and stored in a file, much like you would type and store a paper in a word-processing file or data in a spreadsheet file. In Chapter 2, you will see how to create a data table on your own.

We will start with the U.N. life expectancy data mentioned earlier. Within the Home Window, do this:

1. Click **File ► Open**.
2. Navigate your way to the folder of data tables that accompany this book.[2]
3. Select the file called **Life Expectancy 2017** and click **Open**.

The data table appears in Figure 1.2. Notice that there are four regions in this window including three vertically arranged *panels* on the left, and the *data grid* on the right.

**Figure 1.2: The Life Expectancy 2017 Data Table**



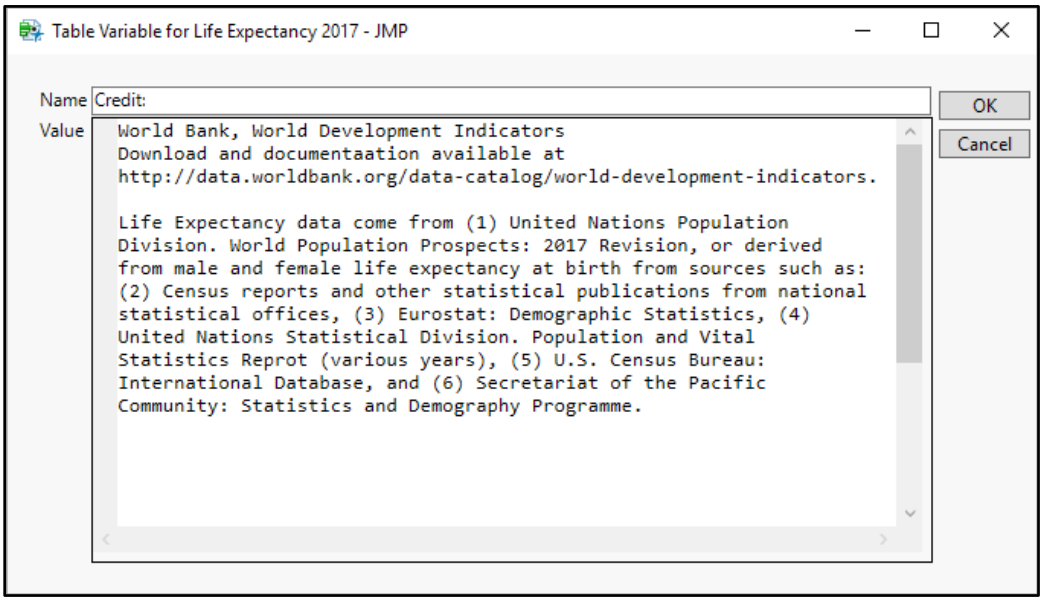

The three panels provide *metadata* (descriptive information about the data in the table), which is created at the time the data table was saved and can be altered. At this early stage, it is helpful to understand the purpose of each panel.

Beginning at the top left, we find the *Table* panel, which displays the name of the data table file as well as optional information provided by the creator of the table. You will see a small red triangle pointing downward next to the table name.

> *Red triangles* indicate a context-sensitive menu, and they are an important element in JMP. We will discuss them more in later chapters, but you should expect to make frequent use of these little red triangles.

Just below the red triangle, there is a note describing the data and identifying its source. You can open that note (called a *Table variable*) just by double-clicking on the word "Credit," the first line within the Table panel. Figure 1.3 shows the note for this table. A table variable contains metadata about the entire table.

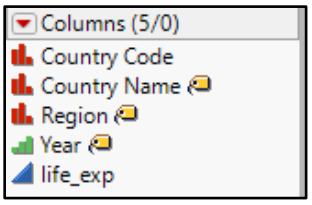
**Figure 1.3: Table Variable Dialog Box**




The second and third lines of the **Table** panel include a green arrow. Green arrows indicate that there is a script embedded in the data table. In this case, the script lists the steps to extract this set of data from a much larger data table called **WDI**, and one can reproduce the subsetting process by running the script. We will use the full WDI data table in future chapters.

Below the **Table** panel is the **Columns** panel, shown in Figure 1.4, which lists the column names, JMP *modeling types*, and other information about the columns.

**Figure 1.4: The Columns Panel**



There are several important things to notice in the Columns panel. The notation **(5/0)** at the top of the panel tells us that there are five columns in this data table, and that none of them are *selected* at the moment. In a JMP data table, we can select one or more columns or rows for special treatment, such as using the *label* property in the second, third, and fourth columns so that country names, regions, and the year will be displayed within graphs. There is much more to learn about the idea of selection and *column properties*, and we will return to it later in this chapter.

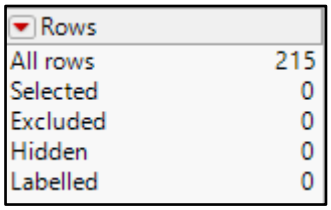
The panel lists the columns by name. To the left of the names are icons indicating the modeling type. In this example, the first three red icons (these look like bar charts) identify **Country Code**, **Country Name**, and **Region** as nominal data. The "price tag" icons indicate that these variables can act as labels to specifically identify observations that are displayed in a graph.

The green ascending bar icon next to **Year** indicates that year is to be analyzed as an ordinal variable. In this data table, all observations are from the same year, 2017, but in the original data set, we have annual observations from 1990 through 2018. Hence, this is an ordinal variable.

Finally, the blue triangle next to **life_exp** identifies the column as continuous data. Remember, it makes sense to perform calculations with continuous data.

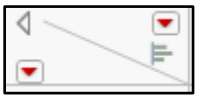
At the bottom left, we find the **Rows** panel (Figure 1.5), which provides basic information about the number of rows (in this case 215, for 215 countries). Like the other two panels, this one provides quick reference information about the number of rows and their *states*.

**Figure 1.5: Rows Panel**




The top entry indicates that there are 215 observations in this data table. The next four entries refer to four basic row states in a JMP data table. Initially, all rows share the same state, in that none have been selected, excluded, hidden, or labeled. Row states enable us to control whether individual observations appear in graphs, are incorporated into calculations, or whether they are highlighted in various ways.

The *Data Grid* area of the data table is where the data reside. It looks like a familiar spreadsheet format, and it just contains the raw data for any analysis. Generally speaking, each column of a table contains either a raw data value (for example, a number, date, or text) or the entire column contains a formula or the result of a computation. Unlike a spreadsheet, each cell in a JMP data table column must be consistent in this sense. You will not find some rows of a column representing one type of data and other rows representing a different type.


In the upper left corner of the data grid, you will see the region shown here. There is a triangular *disclosure button* (pointing to the left side here in Windows; on a Macintosh, it is an arrowhead ▶). Disclosure buttons enable you to expand or contract the amount of information displayed on the screen. The disclosure button shown here lets you temporarily hide the three panels discussed above.

4. Try it out! Click the disclosure button to hide and then reveal the panels. Also click the red triangles and the Header Graphs icon and notice what happens.

The red triangles offer you menu alternatives that will not mean much at this point, but which we will discuss in the next section. The red triangle in the upper right corner (above the diagonal line) relates to the columns of the grid, and the one in the lower left corner to the rows.

Below the right-hand red triangle is a small icon that looks like a bar chart. This opens thumbnail descriptive graphs for each column.

The top row of the grid contains the column names, and the left-most column contains row numbers. The cells contain the data.

## Graph Builder: An Interactive Tool to Explore Data

Our main interest within this data table is how life expectancy varies around the world. Variation is so common as to be unremarkable, but the very fact that they vary is what leads us to analyze them. We can imagine many reasons that life expectancy varies around the world; there are differences in nutrition, wealth, access to health care and clean water, education, political stability, and so on. Are there systematic differences in different parts of the world?

We have a table displaying all 215 countries, but it is difficult to detect patterns by scanning up and down a long list. As a first step in analysis, we will make some simple graphs to summarize the table information visually. Software affords us many options to visualize a set of data and can help us discover errors in the recording of the raw data, locate important patterns of variability, or identify possible connections between and among variables. JMP's **Graph Builder** is an intuitive, interactive platform for visualization.

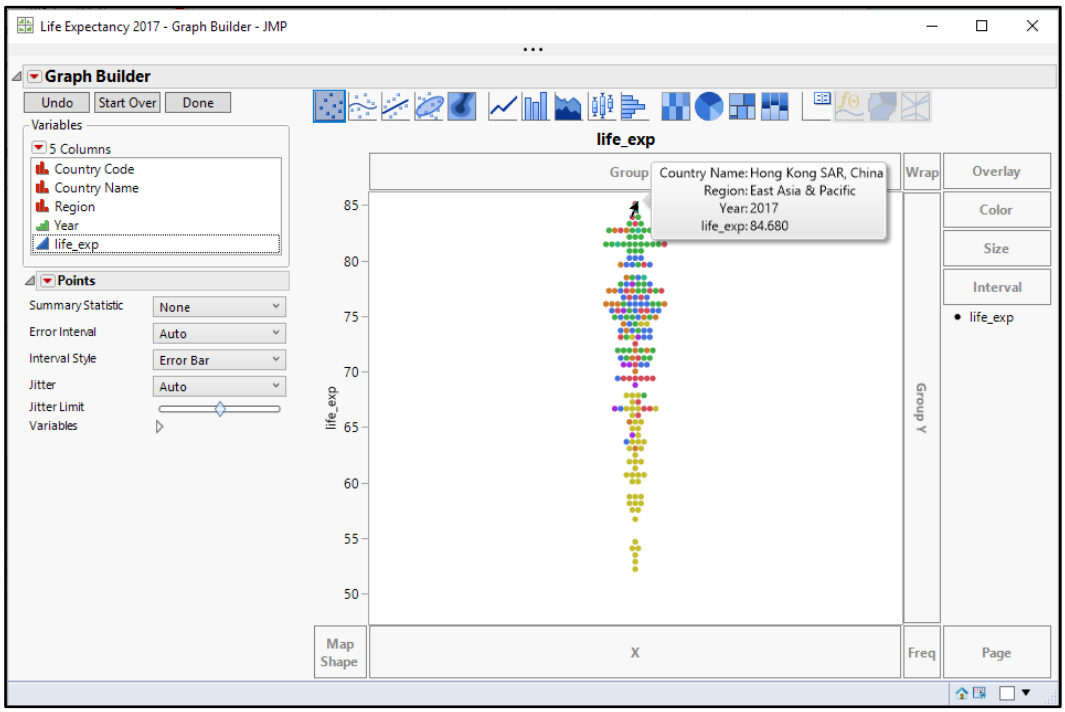
1. From the **Life Expectancy 2017** data table window, click **Graph ► Graph Builder**.

The graph builder gives us a Cartesian plane on which we can create a JMP visualization representing multiple columns in a single visual display. There are numerous options available, but in this first example, we will look at just a few.

In analyzing this set of data, our primary interest lies in the variation of life expectancy. Following one of JMP's conventions, we will think of this column as our Y variable.

2. To display **life_exp** on the Y axis, click the l**ife_exp** column in the panel of **Variables**, and drag it to the vertical **Y** drop zone in the Graph Builder window. When you do this, your screen should look like Figure 1.6.

**Figure 1.6: Using the Graph Builder**




In this graph, each dot represents the value for one country. If you move your cursor to any dot and hover, the name of the country and other data appear. So, for instance, we find that Hong Kong enjoyed the longest life expectancy. Notice that the reported life expectancies lie between approximately 53 years and 85 years, with a large number of countries enjoying life expectancies above 65 years.

By default, JMP *jitters* the points in this graph (see the drop-down menu next to **Jitter** just below the list of variables). This spreads the points apart to the left and right, so that identical or similar values do not overlap in the graph.

3. Click the Jitter menu and select None. You will see why jittering has its advantages. Explore the other options as well.

Now let's see how the values compare across different regions in the world. In the data table, we have already assigned a different color to each region, but have not provided a legend to explain the color coding.

4. One way to produce a legend is to drag **Region** to the **Color** drop zone.

Each global region is colored so that all the countries in East Asia and Pacific, for example, are red. This immediately reveals that nearly all the countries with short life expectancy are in Sub-

Saharan Africa. This fact was not at all obvious from the initial data table; that is what visualization can do for us.

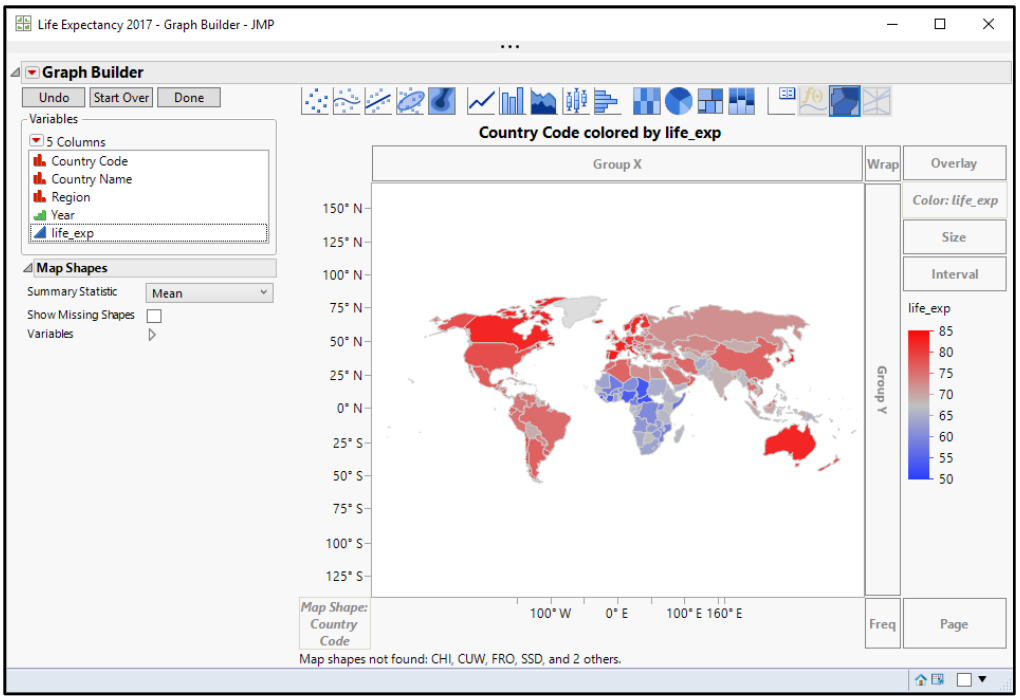
5. Now move the cursor back to the list of columns and once again choose **Region**, and this time drag it to the **Group X** drop zone at the top of the tableau.

When you do this, you will now have seven adjacent small graphs showing the values from each region. As you examine these graphs, you might notice that the values vary vertically within each region and that the patterns of variation are similar in some regions but dramatically different in others. The study of descriptive statistics largely revolves around common patterns of variation, comparisons of those patterns, and deviations from those patterns. Here again, it is very evident that the nations of Sub-Saharan Africa largely have the shortest life expectancies in the world. What other general patterns emerge?

Because the data are reported geographically, another useful way to examine the patterns is to overlay them on a map. Doing so magnifies a few key points.

6. In **Graph Builder**, click the **Start Over** button in the upper left.
7. Drag the **Country Code** column to the lower left of **Graph Builder** into the drop zone labeled **Map Shape**.
8. Now drag **life_exp** over the map and release the mouse button. Alternatively, you might drag **life_exp** into the **Color** drop zone. Your map should now look like Figure 1.7. At this point, click the **Done** button.

**Figure 1.7: Map of the World Colored by Life Expectancy**



As the legend to the right indicates, the colors shaded dark red enjoy the longest life expectancies and dark blue countries have the shortest life expectancies. This map is an alternative method to see how life expectancy varies around the world.

Please note two limitations of this graph. You might have spotted a white "hole" in the center of Africa. These are countries for which JMP found no data in our data table. Additionally, there is a notation at the bottom of the graph indicating that JMP did not recognize some of the country abbreviations, and hence did not display them on the map.

## Using an Analysis Platform

Of course, data analysis is not limited to graphing and mapping—there are numbers to be crunched, and JMP will do the heavy computational work. We have many pages ahead of us to learn how to request and to interpret many useful computations. With this set of data, we will summarize life expectancy in different parts of the world. Don't worry about the details of these steps. The goal right now is just for you to see a typical JMP platform and its output.

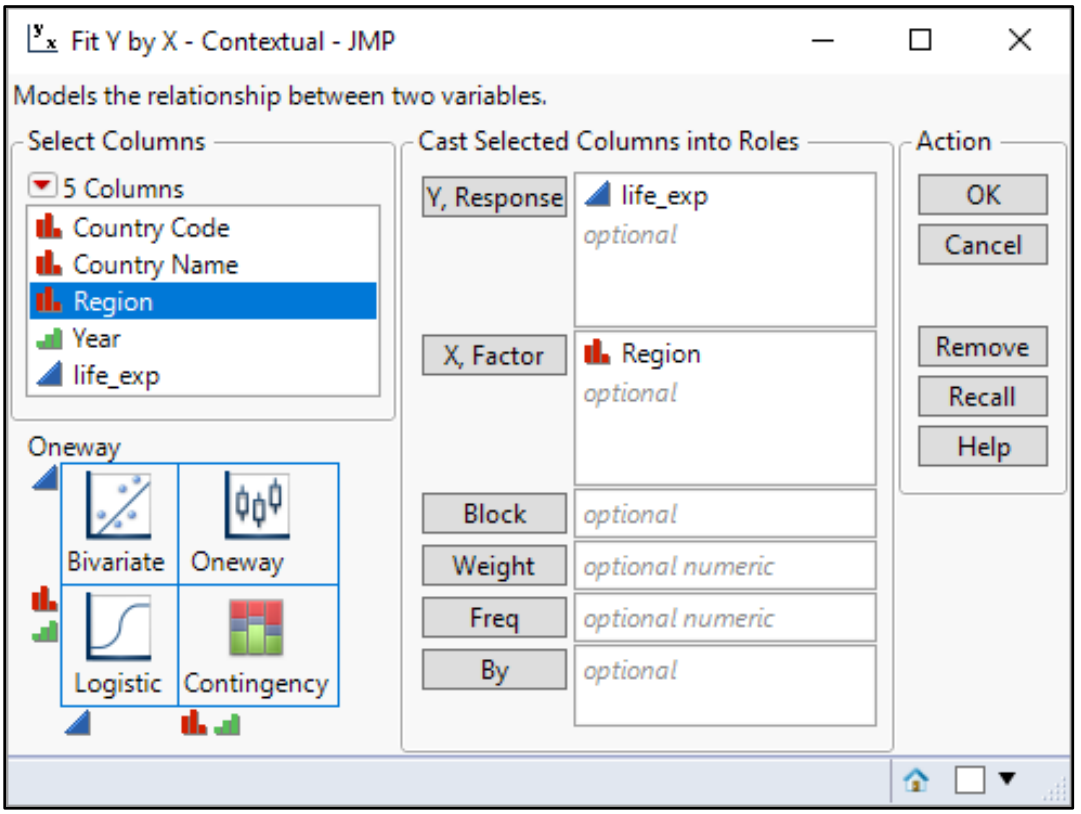
> Windows users: the next instruction asks you to select an option from the Analyze menu, but there is no visible menu bar in the Graph Builder window. At the top of the window, just above **Graph Builder**, find the gray horizontal bar with three dots. (See Figure 1.7.) Hover over the bar and the menus will appear.

1. Select **Analyze ► Fit Y by X**. This *analysis platform* lets us plot one variable (life expectancy) versus another (region).

Why "fit" Y by X? Analysts often speak of *fitting* an abstract or theoretical *model* to a set of data. We can think of models as common or standard patterns of variation, and the process of model fitting begins with exploring how a Y column varies across categories or values of an X column.

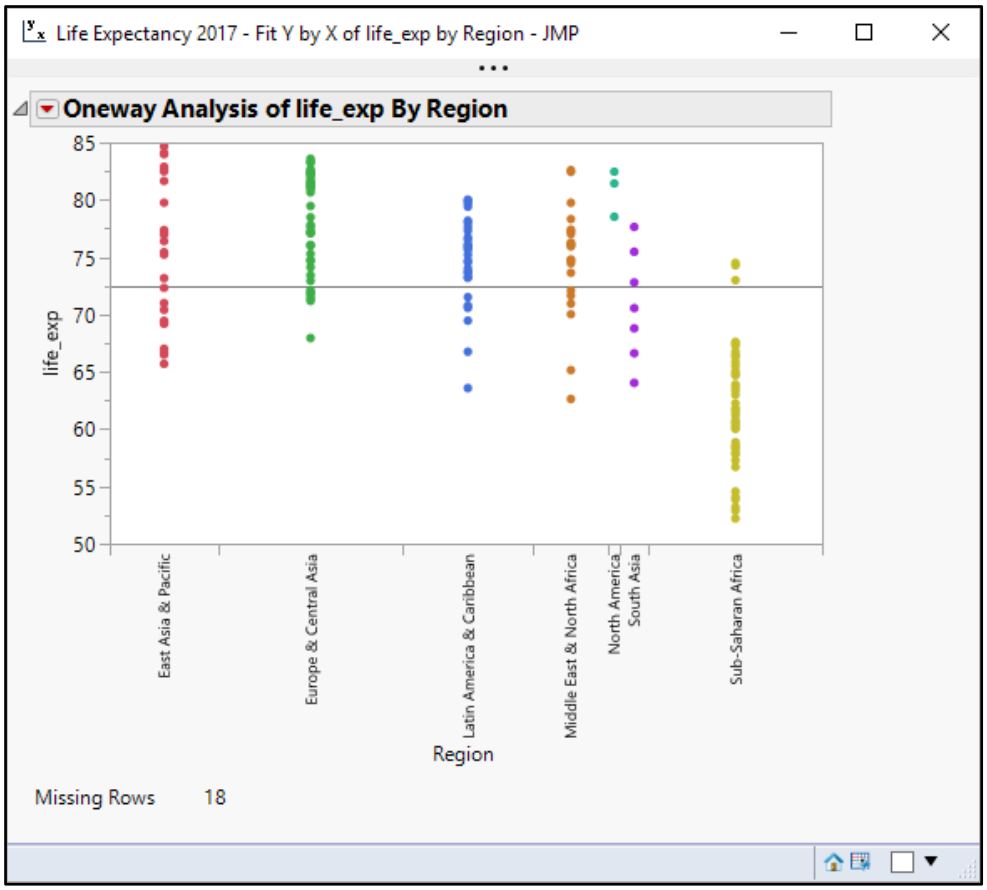
2. In this dialog box (Figure 1.8), we will cast **life_exp** as the **Y** or **Response** variable[3] and **Region** as the **X** variable, or **Factor** variable. Click **OK**.

**Figure 1.8: Fit Y by X Dialog Box**




By design, the initial output of a JMP analysis platform includes one or more graphs. In this case, the initial report includes only a graph, as shown in Figure 1.9.

**Figure 1.9: Initial Report of Life Expectancy by Region**




We saw a very similar graph earlier in Graph Builder; in this graph, the points are not initially jittered. There are two additional features in this graph: the horizontal line between 70 and 75 years is the *mean* (average) of the values. Also, in the lower left we learn that there are 18 Missing Rows, which just says that 18 countries did not report life expectancy data.

The overall grand mean of all countries does not really describe any of the continental regions. We might want to dig deeper and display the regional averages.

3. Click the red triangular hotspot in the upper left next to **Oneway Analysis**, and choose **Display Options**, and check **Connect Means**. If you wish, invoke the **Points Jittered** option as well.

Look again at the modified graph. The new blue line on your graph represents the mean life expectancy of the countries in each region. As a group, the nations of North America appear to have the longest life expectancies, whereas countries in South Asia and Sub-Saharan Africa have far shorter life expectancies. The visual comparison of means is revealing, but suppose we want to know the numerical values of the seven averages.

4. Click the red triangle once more, and this time choose **Means and Std Dev** (standard deviations).

This will generate a table of values beneath the graph, as shown in Figure 1.10. For the current discussion, we will focus our attention only on the first three columns. Later in the book, we will learn the meaning of the other columns. This table (below) reports the mean and number of countries for each region.

**Figure 1.10: Table of Means and Standard Deviations**

**Means and Std Deviations**

| Level | Number | Mean | Std Dev | Std Err Mean | Lower 95% | Upper 95% |
|---|---|---|---|---|---|---|
| East Asia & Pacific | 30 | 74.720036 | 6.0147549 | 1.098139 | 72.474089 | 76.965982 |
| Europe & Central Asia | 51 | 77.864288 | 4.2721959 | 0.5982271 | 76.662713 | 79.065862 |
| Latin America & Caribbean | 36 | 75.099116 | 3.6820815 | 0.6136802 | 73.853279 | 76.344953 |
| Middle East & North Africa | 21 | 74.739338 | 4.888584 | 1.0667765 | 72.514081 | 76.964595 |
| North America | 3 | 80.817154 | 2.0388427 | 1.1771264 | 75.752389 | 85.88192 |
| South Asia | 8 | 70.82375 | 4.4716165 | 1.5809552 | 67.085385 | 74.562115 |
| Sub-Saharan Africa | 48 | 62.031701 | 5.2576047 | 0.7588699 | 60.505051 | 63.55835 |

Missing Rows 18

## Row States

Our data table contains 1,075 cells: five variables with 215 observations each, arrayed in five columns and 215 rows. One guiding principle in statistical analysis is that we generally want to use all of our data. We do not casually discard or omit any portion of the data that we have collected (often at substantial effort or expense). There are times, however, that we might want to focus attention on a portion of the data table or examine the impact of a small number of extraordinary observations.

By default, when we analyze one or more variables using JMP, every observation is included in the resulting graphs and computations. You can use **row states** to confine the analysis to particular observations or to highlight certain observations in graphs.

There are four fundamental row states in JMP. Rows can be:

- *Selected*: selected rows appear bolded or otherwise highlighted in a graph.
- *Excluded*: when you exclude rows, those observations are temporarily omitted from calculated statistics such as the mean. The rows remain in the data table, but as long as they are excluded, they play no role in any computations.
- *Hidden*: when you hide rows, those observations do not appear in graphs, but are included in any calculations such as the mean.
- *Labeled*: The row numbers[4] of any labeled rows display next to data points in some graphs for easily identifying specific points. The user (you) can designate specific columns that contain useful labels.

Let's see how the row states change the output that we have already run by altering the row states of rows 3 and 4.

1. First, arrange the open windows so that you can clearly see both the **Fit Y by X report** window and the data table and click anywhere in the data table window to make it the active window.
2. Move your cursor into the column of row numbers in the data table. Within this column your cursor will become a "fat cross"  . Select rows 3 and 4 by clicking and dragging on the row numbers 3 and 4. You will see the two rows highlighted within the data table.

Look at your graph. Almost all the points are dim, except for two bright dots—one above the mean value of Latin America & Caribbean and the other well below the mean of South Asia. That is the effect of selecting these rows. Notice also that the **Rows** panel in the Life Expectancy data window now shows that two rows have been selected.

3. Click on another row, and then drag your mouse slowly down the column of row numbers. Do you notice the rows highlighted in the table and the corresponding data points "lighting up" in the graph?
4. Press Esc or click in the triangular area above the row numbers in the data table to deselect all rows.

Next, we will *exclude* two observations and show that the calculated statistics change when they are omitted from the computations. To see the effect, we first need to instruct JMP to automatically recalculate statistics when the data table changes.

5. Click the red triangle next to **Oneway Analysis** in the report window and choose **Redo ► Automatic Recalc**.
6. Now let's exclude rows 3 and 4 from the calculations. To do this, first select them as you did before.
7. Select **Rows ► Exclude/Unexclude** (you can also find this option by clicking the red triangle above the row numbers; in Windows, you could also right-click). This will exclude the rows.

Now look at the analysis output. The number of observations in the Latin America & Caribbean region drops from 36 to 35, and the mean value for that group has changed very slightly. Likewise, in South Asia we have 7 rather than 8 observations, and mean life expectancy increased from 70.82375 years to 71.791857 years. Toggle between the exclude and unexclude states of these two rows until you understand clearly what happens when you exclude observations.

8. Finally, let's *hide* the rows. First, be sure to unexclude rows 3 and 4 so that all points appear in the graph and in the calculations. If you are not sure if you have reversed earlier actions, choose **Rows ► Clear Row States** and then confirm in the **Rows** panel that the four row state categories show 0 rows.
9. Once again select rows 3 and 4 and choose **Rows ► Hide/Unhide.** This will hide the rows (check out the very cool dark glasses icon).

Look closely at the graph and at the table of means. It is a subtle change, but the two bright dots are gone, leaving only dim points. The numbers in the table of means are unaffected by hiding points. If you toggle the **Hide/Unhide** state, you will notice the dark points come and go, but the number of observations in each region is stable.

10. Before continuing to the next section, clear all row states again (as in Step 4 above). This will also remove the color coding of regions. To recolor, choose **Rows ► Color of Mark by Column…,** select **Region** and click OK.

## Exporting and Sharing JMP Reports

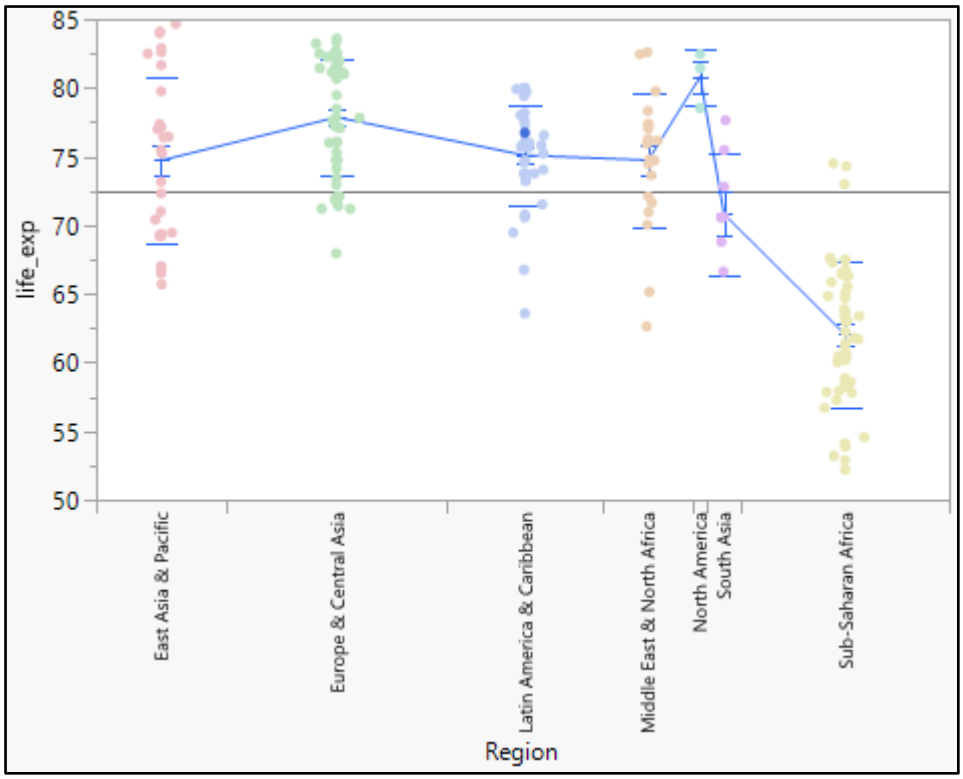
As a statistics student, you may often want to include some of your results within a paper or presentation that you are preparing for class. As we finish this first lesson, we will start with a quick way to capture output and transfer it to your paper or slides. Then we will introduce several other options to reach a wider audience.

To follow along, first open your word processing or slide creation software, then write a sentence introducing the graph that you have been working with. Next, return to the JMP Oneway Analysis report window.

Our analysis includes a graph and a table. To copy the graph only for your document, do this:

1. Select **Tools ► Selection**. Your cursor will now become an open cross. You could also click the open (“fat”) cross button from the menu icon bar.
2. Move the open cross to the lower left of the graph away from the X axis and click. This should highlight the entire graph. If it doesn’t, click and drag across the graph until it is entirely selected.
3. Select **Edit ► Copy**. Alternatively, you can right-click and select Copy.
4. Now move to your document or slide and paste your copied graph.

**Figure 1.11: A Graph Pasted from JMP**




The graph should look like the one shown above in Figure 1.11. Note that the graph will look slightly different from its appearance within JMP, but this demonstration should illustrate how very easy it is to incorporate JMP results into a document.

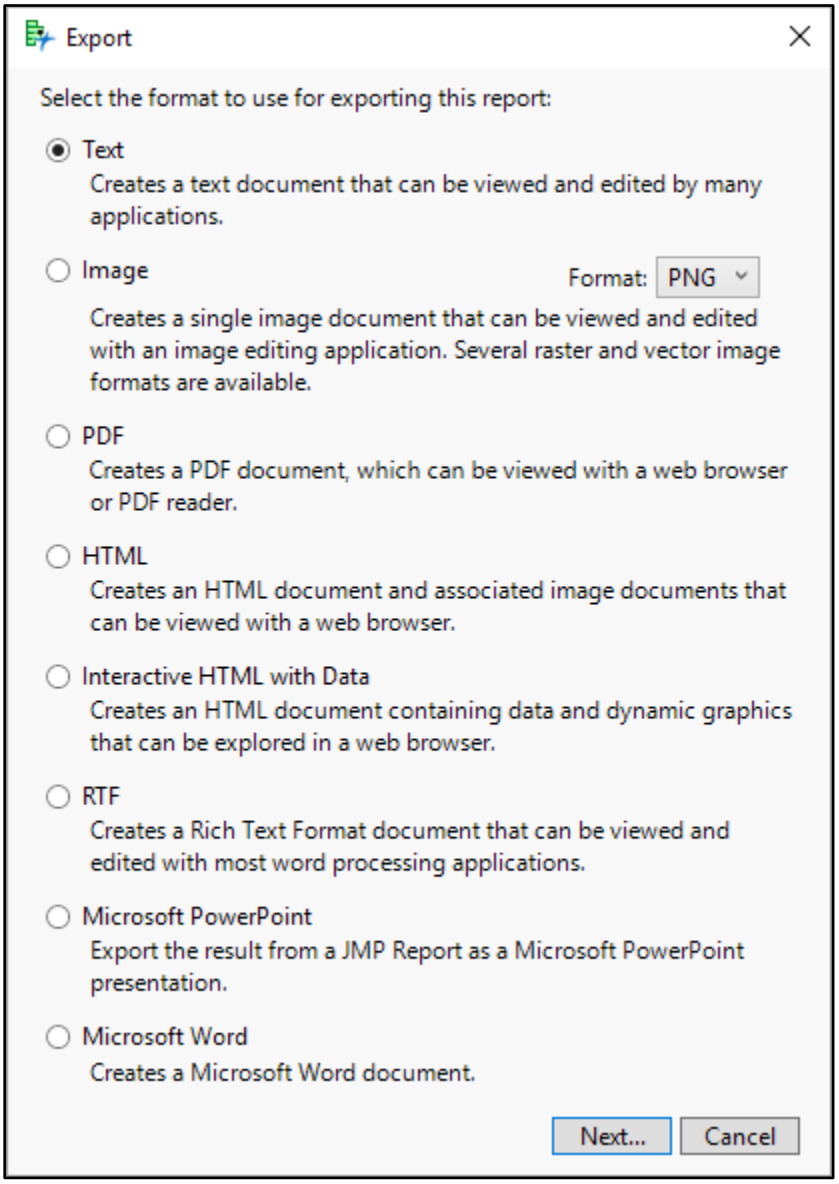
This approach preserves a static image of the graph but loses the interactivity of the labels that pop up when you hover over a point. Moreover, if you have many reports to copy, this method does not scale efficiently.

Let's look at other options. This section provides an overview, and we will apply some of these methods in later exercises.

5. Go to the Tools menu and choose the arrow tool in place of the selection tool.
6. Make sure the **Fit Y by X** window is active and choose **File ► Export**. As shown in Figure 1.12, JMP provides eight formatting options. You may explore these on your own, but now let's make an Interactive HTML document. HTML documents can be viewed by any web browser.
7. Click the **Interactive HTML with Data** button, and **Next**. This opens a typical file save dialog box. You should select a location to store the report file.

Soon, a browser window will open showing a fully interactive version of your report. The disclosure buttons work, as do hover labels.

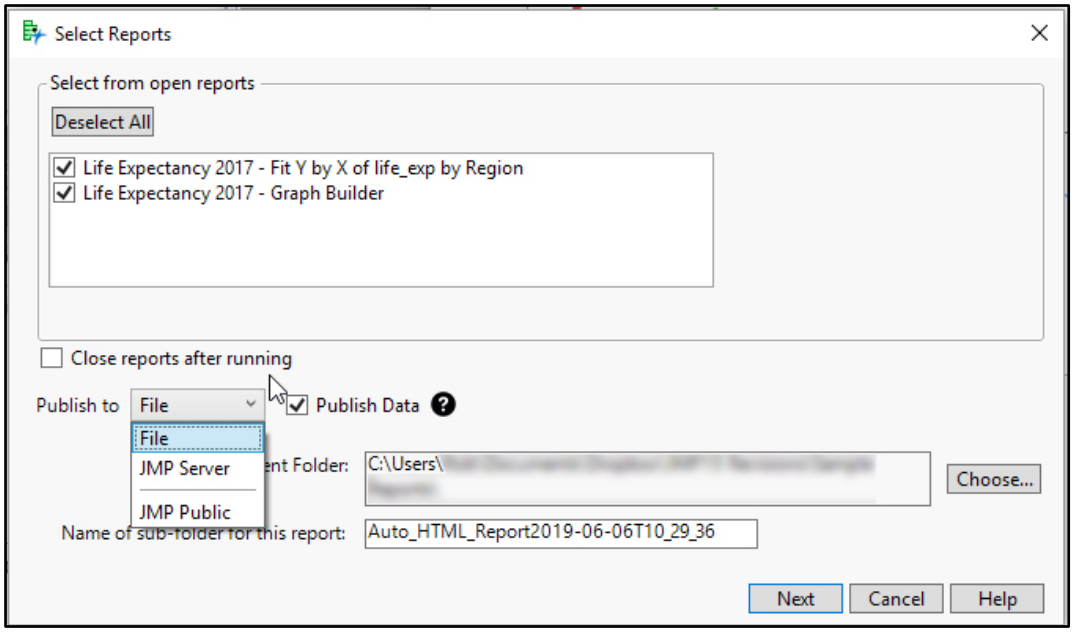
**Figure 1.12: Export Format Options**




To share multiple reports efficiently, there is another option.

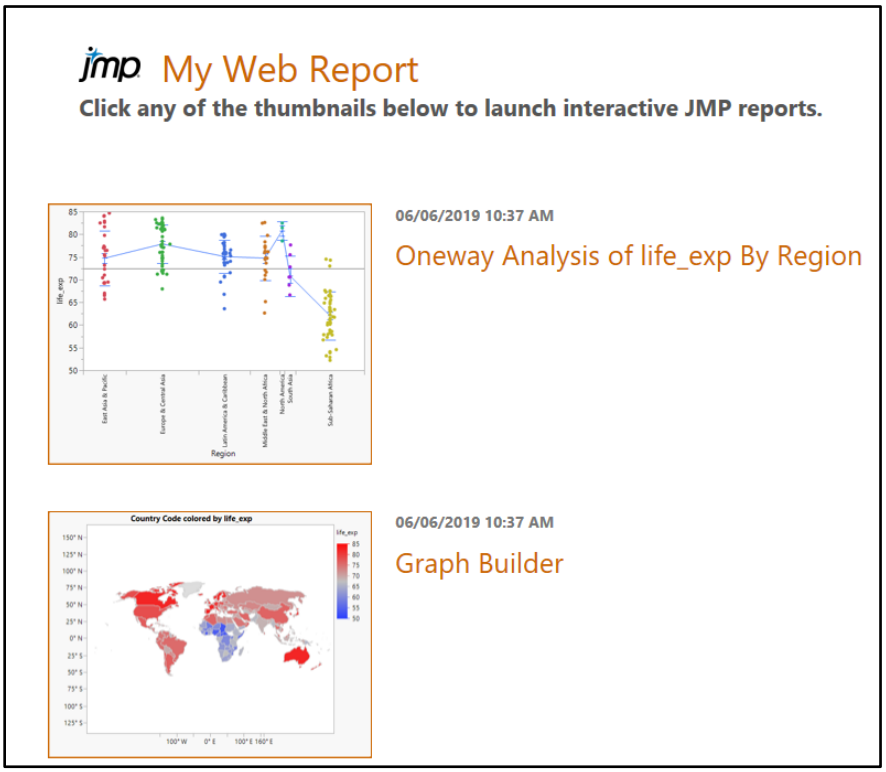
8. Choose **File ► Publish**. This opens a dialog box as shown in Figure 1.13 in which you can specify which of your open reports to publish, choose to save them locally in a file, publish them to a JMP server in your enterprise, or make them available globally at JMP Public.[5] After making your choice, specify a destination for the file.

**Figure 1.13: Select Reports for Publication**




After choosing a file location, another browser window will open with an interactive web report that looks like Figure 1.14.

**Figure 1.14: A Web Report Containing Multiple JMP Reports**




## Saving and Reproducing Your Work

As you work with any software, you should get in the habit of saving your work as you go. JMP supports several types of files and enables you to save separate portions of a session along the way or bundle all related files and reports into a JMP Project (to be illustrated in Chapter 2). You have already seen that data tables are files; we have modified the **Life Expectancy 2017** data table and might want to save it.

For now, let's preserve the commands that created our Fit Y by X report. Although we have seen some ways to save the final report image, we can save the series of commands within the data table. In that way, we can reproduce the report in the future without saving a space-consuming graphic. Here's how:

1. Go to the **Fit Y by X** report window and click the red triangle at the upper left. Select **Save Script ► To Data Table….**
2. In the small dialog box that opens, assign a name to this report or accept the suggested name, and click **OK**.
3. Now switch to the **Life Expectancy 2017** data table and look in the **Table** panel in the upper left. You should find another green arrow next to the report name you just assigned. Click that arrow, and a fresh copy of the report will open.

Alternatively, you can save the *session script*, which essentially is a transcript of the session coded in the *JMP Scripting Language* (JSL)—all of the commands that you issued, as well as their results. Later, when you restart JMP, you can open the script file, run it, and your screen will be exactly as you left it.

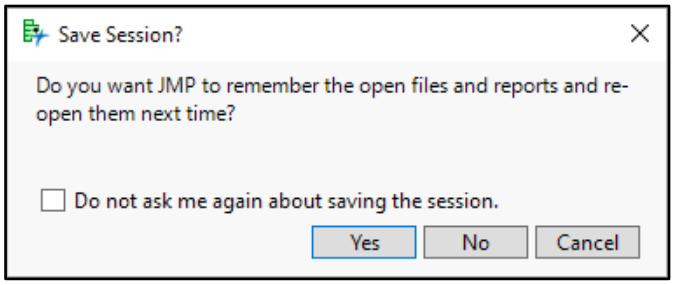
4. Select **File ► Save Session Script.** In the dialog box, choose a directory in which to save this JSL file, give the file a name, and click OK.

## Leaving JMP

We have covered a lot of ground in this first session, and it's time to quit.

1. Select **File ► Exit JMP.**

Answer **No** to the question about saving changes to the data. Then you will see this dialog box:




In this case, you can click **No**. In future work, if you want to take a break and resume later where you left off, you might want to click **Yes**. The next time you start the program, everything will look as it did when you quit.

Remember to run the Beginner's Tutorial from the **Tip of the Day** (or under the **Help** menu) before moving on to Chapter 2.

---

[1] This book's illustrations and examples are all based on JMP Pro 15.0. In most instances, we show the default settings that come with JMP when it is newly installed.

[2] All of the data tables used in this book are available from http://support.sas.com/publishing/authors/carver.html. If you are enrolled in a college or university course, your instructor may have posted the files in a special directory. Check with your instructor.

[3] In Chapter 4, we will study *response variables* and *factors*. In this chapter, we are getting a first look at how analysis platforms operate.

[4] Columns can contain labels (for example, the name of respondent or country name), which are also displayed when a row is labeled.

[5] Consult JMP Help for details about JMP Public. In the JMP Documentation Library, open "Using JMP", navigate to section 10 "Save and Share Data" and find "Create a Web Report".

# Chapter 2: Data Sources and Structures

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Representativeness and Sampling ..... 25
Cross-Sectional and Time Series Sampling ..... 28
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## Overview

This chapter is about data. More specifically, it is about how we make choices when we gather or generate data for analysis in a statistical investigation, and how we store that data in JMP files for further analysis. This chapter introduces some data storage concepts that are further developed in Appendix B. How and where we gather data are the foundation for what types of conclusions and decisions we can draw. After reading this chapter, you should have a solid foundation to build upon.

## Populations, Processes, and Samples

We analyze data because we want to understand something about variation within a *population*—a collection of people or objects or phenomena that we are interested in. Sometimes we are more interested in the variability of a *process*—an ongoing natural or artificial activity (like the occurrences of earthquakes or fluctuating stock prices). We identify one or more *variables* of interest and either count or measure them.

In most cases, it is impossible or impractical to gather data from every single individual within a population or every instance from an ongoing process. Marine biologists who study communication among dolphins cannot possibly measure every living dolphin. Manufacturers wanting to know how quickly a building material degrades in the outdoors cannot destroy 100% of their products through testing or they will have nothing left to sell. Thanks to huge databases and powerful software, financial analysts interested in future performance of a stock can analyze every single past trade for that stock, but they cannot analyze trades that have not yet occurred.

Instead, we typically analyze a *sample* of individuals selected from a population or process. Ideally, we choose the individuals in such a way that we can have some confidence that they are a “second-best” substitute or stand-in for the entire population or ongoing process. In later

chapters, we will devote considerable attention to the adjustments that we can make before generalizing sample-based conclusions to a larger population.

As we begin to learn about data analysis, it is important to be clear about the roles and relationship of a population and a sample. In many statistical studies, the situation is as follows (this example refers to a population, but the same applies to a process):

- We are interested in the variation of one or more attributes of a population. Depending on the scenario, we may wish to anticipate the variation, influence it, or just understand it better.
- Within the context of the study, the population consists of individual *observational units*, or *cases*. We cannot gather data from every single individual within the population.
- Hence, we choose some individuals from the population and observe them *to generalize about the entire population.*

We gather and analyze data from the sample of individuals instead of doing so for the whole population. The individuals within the sample are *not* the group that we are ultimately interested in knowing about. We really want to learn about the variability within the population. When we use a sample instead of a population, we run some risk that the sample will misrepresent the population—and that risk is at the center of statistical reasoning.

Depending on just what we want to learn about a process or population, we also concern ourselves with the method by which we generate and gather data. Do we want to characterize or describe the extent of temperature variation that occurs in one part of the world? Or do we want to understand how patients with a disease respond to a specific dosage of a medication? Or do we want to predict which incentives are most likely to induce consumers to buy a product?

There are two types of generalizations that we might eventually want to be able to make, and in practice, statisticians rely on *randomization* in data generation as the logical path to generalization. If we want to use sample data to characterize patterns of variation in an entire population or process, it is valuable to select sample cases using a probability-based or random process. If we ultimately want to draw definitive conclusions about how variation in one variable is caused or influenced by another, then it is essential to randomly control or assign values of the suspected causal variable to individuals within the sample.

Therefore, the design strategy in sampling is terrifically important in the practical analysis of data. We also want to think about the time frame for sampling. If we are interested in the current state of a population, we should select a *cross-section* of individuals at one time. On the other hand, if we want to see how a process unfolds over time, we should select a *time series*

sample by observing the same individual repeatedly at specific intervals. Some samples (often referred to as *panel data*) consist of a list of individuals observed repeatedly at a regular interval.

## Representativeness and Sampling

If we plan to draw general conclusions about a population or process from one sample, it's important that we can reasonably expect the sample to *represent* the population. Whenever we rely on sample information, we run the risk that the sample could misrepresent the population. (In general, we call this *sampling error.*) Statisticians have several standard methods for choosing a sample. No one method can guarantee that a single sample accurately represents the population, but some methods carry smaller risks of sampling error than others. What's more, some methods have *predictable* risks of sampling error, while others do not. As you will see later in the book, if we can predict the extent and nature of the risk, then we can generalize from a sample; if we cannot, we sacrifice our ability to generalize. JMP can accommodate different methods of representative sampling, both by helping us to select such samples and by taking the sampling method into account when analyzing data. At this point, we focus on understanding different approaches to sampling by examining data tables that originated from different designs. We will also take a first look at using JMP to select representative samples. In Chapters 8 and 21, we will revisit the subject more deeply.

### Simple Random Sampling

The logical starting point for a discussion of representative sampling is the *simple random sample* (SRS). Imagine a population consisting of N elements (for example, a lake with N = 1,437,652 fish), from which we want to take an SRS of $n$ = 200 fish. With a little thought, we recognize that there are many different 200-fish samples that we might draw from the lake. If we use a sampling method that ensures that all 200-fish samples have the same chance of being chosen, then any sample we take with that method is an SRS. Essentially, we depend on the probabilities involved in random sampling to produce a representative sample.

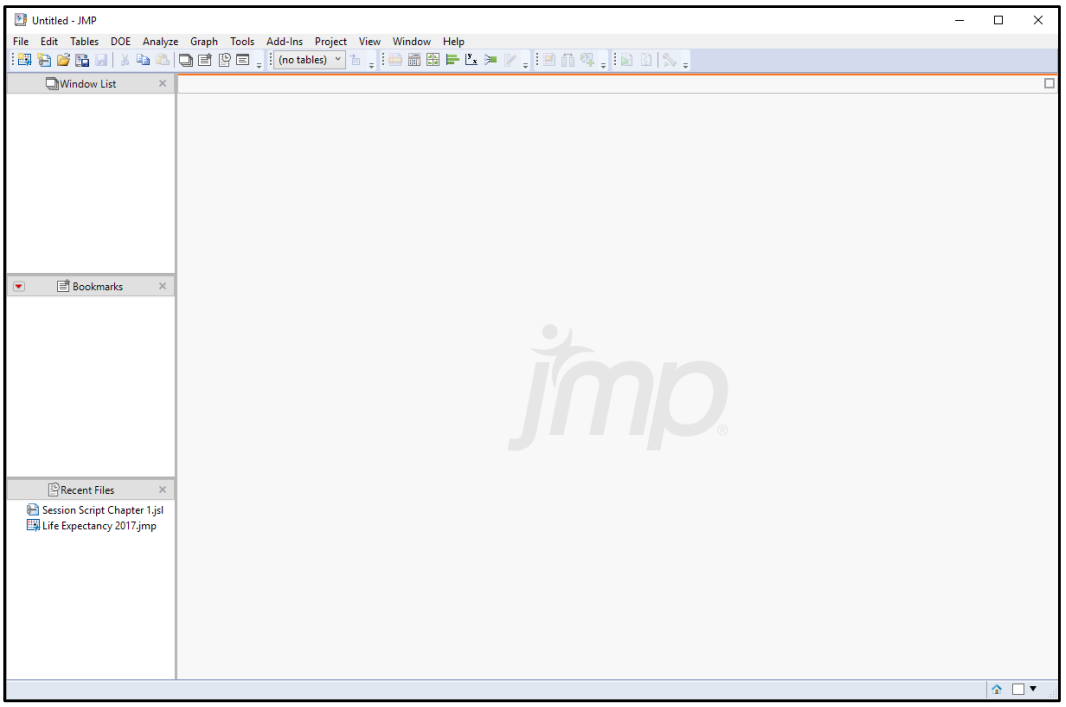
Simple random sampling requires that we have a *sampling frame*, or a list of all members of a population. The sampling frame could be a list of students in a university, firms in an industry, or members of an organization. To illustrate, we will start with a list of the countries in the world and see one way to select an SRS. For the sake of this example, suppose we want to draw a simple random sample of 20 countries for in-depth research.

There are several ways to select and isolate a simple random sample drawn from a JMP data table. In this illustration, we will first randomly select 20 rows, and then proceed to move them into a new data table. This is not the most efficient method, but it emphasizes the idea of random selection, and it introduces two useful commands.

> In this chapter, we will work with several data tables. As such, this is an opportunity to introduce **JMP Projects**. A project keeps track of data and report files together.

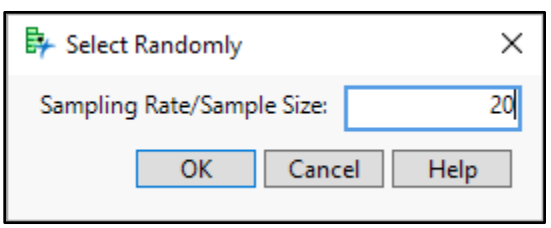
1. First, we will create a Project to store the work that we are about to do. **File ▶ New ▶ Project**. A blank project window opens. (See Figure 2.1.)

**Figure 2.1: A Project Window**




2. **File ▶ Open.** Select the data table called **World Nations**. This table lists the countries in the world as of 2017, as identified by the United Nations and the World Bank. Notice that the data table opens within the project window, and the Windows List in the upper left contains **World Nations**.
3. Select **Rows ▶ Row Selection ▶ Select Randomly…**. A small dialog box opens (Figure 2.2) asking either for a sampling rate or a sample size. If you enter a value between 0 and 1, JMP understands it as a rate. A number large than 1 is interpreted as a sample size, *n*. Enter **20** into the dialog box and click **OK**.

**Figure 2.2: Specifying a Simple Random Sample Size of 20**

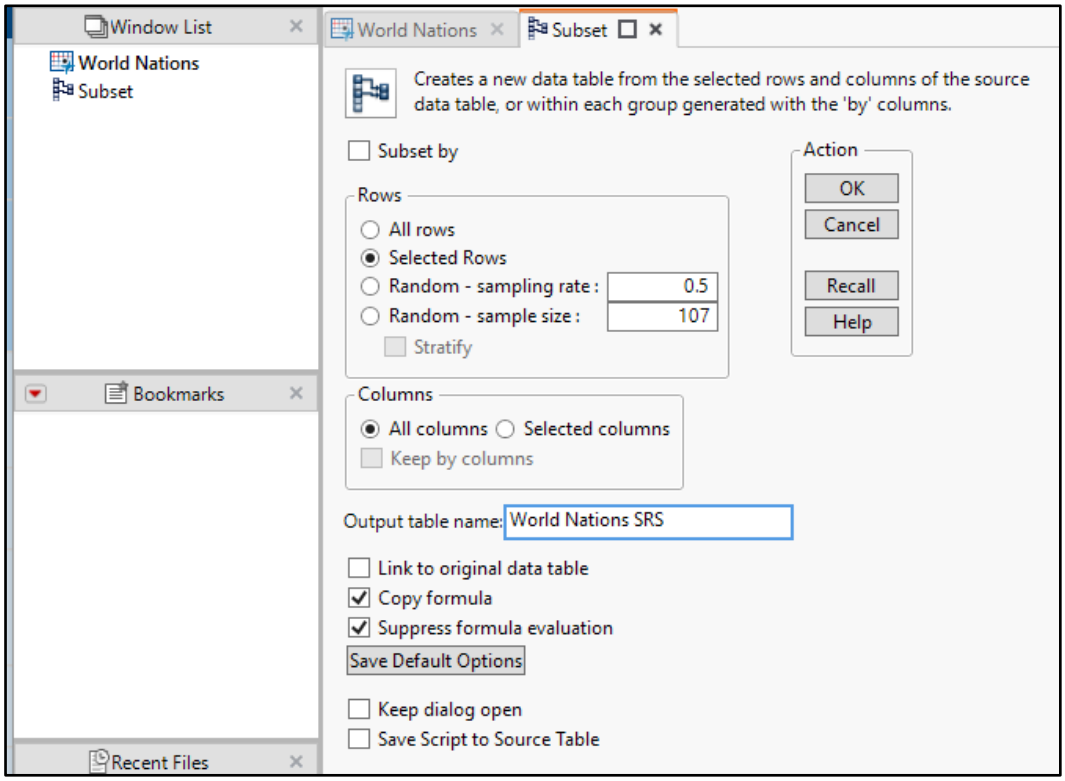



JMP randomly selects 20 of the 215 rows in this table. When you look at the Rows panel in the Data Table window, you will see that 20 rows have been selected. As you scroll down the list of countries, you will see that 20 selected rows are highlighted. If you repeat Step 2, a different list

of 20 rows will be highlighted because the selection process is random. If you compare notes with others in your class, you should discover that their samples comprise different countries. With the list of 20 countries now chosen, let's create a new data table containing just the SRS of 20 countries.

4. Select **Tables ▶ Subset**. This versatile dialog box (see Figure 2.3) enables us to build a table using the just-selected rows, or to randomly sample directly. Note that the dialog box opens in a fresh tab within the Project, and as an item in the Window List.
5. As shown in the figure, choose **Selected Rows** and then change the **Output table name** to **World Nations SRS**. Then click **OK**.

**Figure 2.3: Creating a Subset of Selected Rows**




In the upper left corner of the data table window, there is a green arrow with the label Source. JMP inserts the JSL script that created the subset. Readers wishing to learn more about writing JMP scripts should right-click the green arrow, choose Edit, and see what a JSL script looks like.

Before moving to the next example, let's save the Project. Before saving a project, all documents within the project must be saved individually. We just created a new data table; save it now.

6. **File ▶ Save**. The default name is **World Nations SRS**, which is fine. Place it in a folder of your choice.

7. **File ▶ Save Project**. Choose a location and name this project **Chap_02.** Then, click **OK**. Among your Recent Files list, you will now find Chap_02.jmpprj.

### Other Types of Random Sampling

As noted previously, simple random sampling requires that we can identify and access all *N* elements within a population. Sometimes this is not practical, and there are several alternative strategies available. It is well beyond the scope of this chapter to discuss these strategies at length, but Chapter 8 provides basic coverage of some of these approaches.

### Non-Random Sampling

This book is about practical data analysis, and in practice, many data tables contain data that were *not* generated by anything like a random sampling process. Most data collected within businesses and nonprofit organizations come from the normal operations of the organization rather than from a carefully constructed process of sampling. The data generated by "Internet of Things" devices, for example, are decidedly non-random. We can summarize and describe the data within a non-random sample but should be very cautious about the temptation to generalize from such samples. Whether we are conducting the analysis or reading about it, we always want to ask whether a given sample is likely to misrepresent the population or process from which it came. Voluntary response surveys, for example, are very likely to mislead us if only highly motivated individuals respond. On the other hand, if we watch the variation in stock prices during an uneventful period in the stock markets, we might reasonably expect that the sample could represent the process of stock market transactions.

### Big Data

You might have heard or read about "Big Data"—high volume raw data generated by numerous electronic technologies like cell phones, supermarket scanners, radio-frequency identification (RFID) chips, or other automated devices. The world-spanning, continuous stream of data carries huge potential for the future of data analytics and presents many ethical, technical, and economic challenges. In general, data generated in this way are not random in the conventional sense and don't neatly fit into the traditional classifications of an introductory statistics course. Big data can include photographic images, video, or sound recordings that don't easily occupy columns and rows in a data table. Furthermore, streaming data is neither cross-sectional nor time series in the usual sense.

## Cross-Sectional and Time Series Sampling

When the research concerns a population, the sampling approach is often *cross-sectional*, which is to say the researchers select individuals from the population at one period of time. Again, the individuals can be people, animals, firms, cells, plants, manufactured goods, or anything of interest to the researchers.

When the research concerns a process, the sampling approach is more likely to be *time series* or *longitudinal*, whereby a single individual is repeatedly measured at regular time intervals. A

great deal of business and economic data is longitudinal. For example, companies and governmental agencies track and report monthly sales, quarterly earnings, or annual employment. The major distinction between time series data and streaming data is whether observations occur according to a pre-determined schedule or whether they are event-driven (for example, when a customer places a cell phone call).

*Panel studies* combine cross-sectional and time series approaches. In a panel study, researchers repeatedly gather data about the same group of individuals. Some long-term public health studies follow panels of individuals for many years; some marketing researchers use consumer panels to monitor changes in taste and consumer preferences.

## Study Design: Experimentation, Observation, and Surveying

If the goal of a study is to demonstrate a cause-and-effect relationship, then the ideal approach is a *designed experiment*. The hallmark features of an experiment are that the investigator controls and manipulates the values of one or more variables, randomly assigns *treatments* to observational units, and then observes changes in the *response* variable. For example, engineers in the concrete industry might want to know how varying the amount of different additives affects the strength of the concrete. A research team would plan an experiment in which they would systematically vary specific additives and conditions, then measure the strength of the resulting batch of concrete.

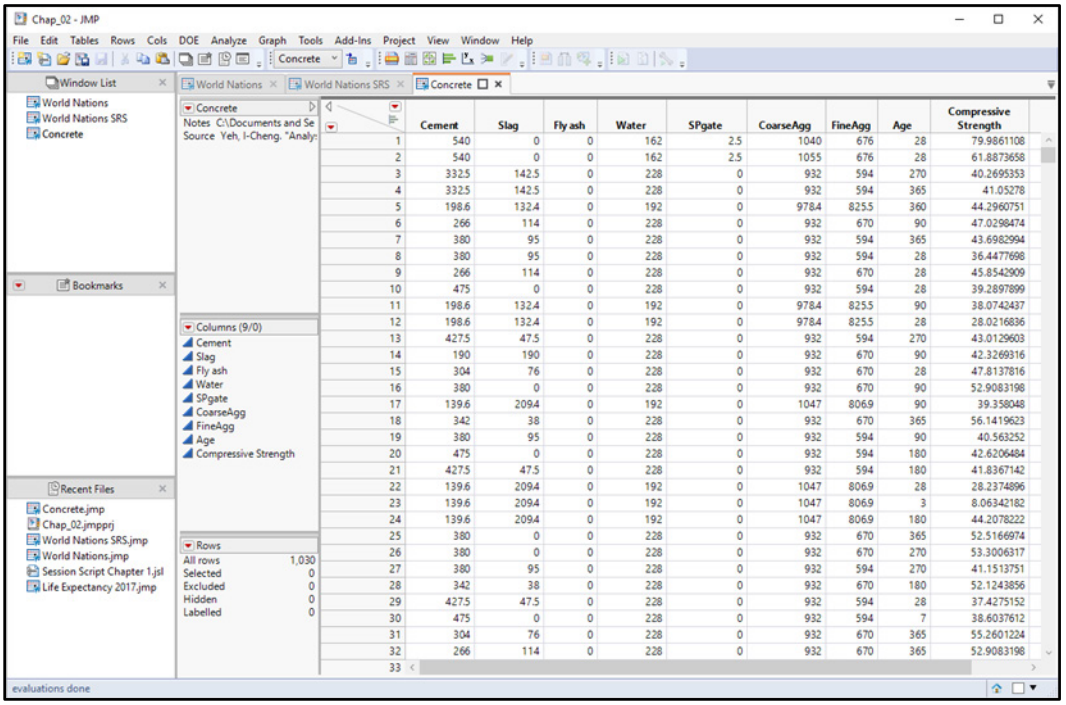
Similarly, consider a large retail company that has a "customer loyalty" program, offering discounts to its regular customers who present their bar-coded key tags at the check-out counter. Suppose the firm wants to nudge customers to return to their stores more frequently and generates discount coupons that can be redeemed if the customer visits the store again within so many days. The marketing analysts in the company could design an experiment in which they vary the size of the discount and the expiration date of the offer, issue the different coupons to randomly chosen customers, and then see when customers return.

### Experimental Data—An Example[1]

In an experimental design, the causal variables are called *factors,* and the outcome variable is called the *response* variable. As an illustration of a data table containing experimental data, open the data table called Concrete. Professor I-Cheng Yeh of Chung-Hua University in Taiwan measured the *compressive strength* of concrete prepared with varying formulations of seven different component materials. Compressive strength is the amount of force per unit of area, measured here in megapascals that the concrete can withstand before failing. Think of the concrete foundation walls of a building; they need to be able to support the mass of the building without collapsing. The purpose of the experiment was to develop an optimal mixture to maximize compressive strength.

1. Select **File ▶ Open**. Choose **Concrete** and click **OK**.

**Figure 2.4: The Concrete Data Table (Within Project)**

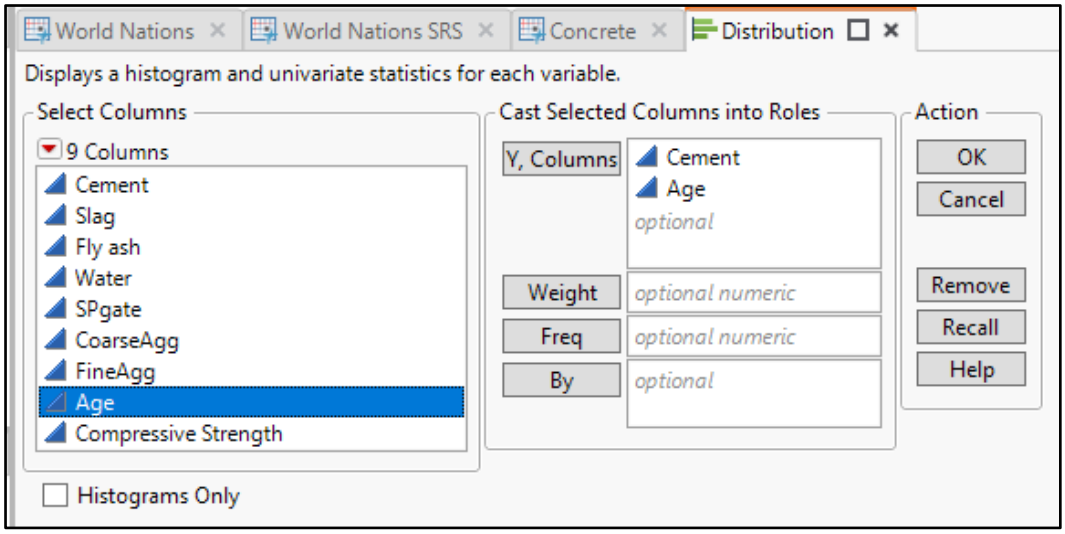



| | Cement | Slag | Fly ash | Water | SPgate | CoarseAgg | FineAgg | Age | Compressive Strength |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 540 | 0 | 0 | 162 | 2.5 | 1040 | 676 | 28 | 79.9861108 |
| 2 | 540 | 0 | 0 | 162 | 2.5 | 1055 | 676 | 28 | 61.8873658 |
| 3 | 332.5 | 142.5 | 0 | 228 | 0 | 932 | 594 | 270 | 40.2695353 |
| 4 | 332.5 | 142.5 | 0 | 228 | 0 | 932 | 594 | 365 | 41.05278 |
| 5 | 198.6 | 132.4 | 0 | 192 | 0 | 978.4 | 825.5 | 360 | 44.2960751 |
| 6 | 266 | 114 | 0 | 228 | 0 | 932 | 670 | 90 | 47.0298474 |
| 7 | 380 | 95 | 0 | 228 | 0 | 932 | 594 | 365 | 43.6982994 |
| 8 | 380 | 95 | 0 | 228 | 0 | 932 | 594 | 28 | 36.4477698 |
| 9 | 266 | 114 | 0 | 228 | 0 | 932 | 670 | 28 | 45.8542909 |
| 10 | 475 | 0 | 0 | 228 | 0 | 932 | 594 | 28 | 39.2897899 |
| 11 | 198.6 | 132.4 | 0 | 192 | 0 | 978.4 | 825.5 | 90 | 38.0742437 |
| 12 | 198.6 | 132.4 | 0 | 192 | 0 | 978.4 | 825.5 | 28 | 28.0216836 |
| 13 | 427.5 | 47.5 | 0 | 228 | 0 | 932 | 594 | 270 | 43.0129603 |
| 14 | 190 | 190 | 0 | 228 | 0 | 932 | 670 | 90 | 42.3269316 |
| 15 | 304 | 76 | 0 | 228 | 0 | 932 | 670 | 28 | 47.8137816 |
| 16 | 380 | 0 | 0 | 228 | 0 | 932 | 670 | 90 | 52.9083198 |
| 17 | 139.6 | 209.4 | 0 | 192 | 0 | 1047 | 806.9 | 90 | 39.358048 |
| 18 | 342 | 38 | 0 | 228 | 0 | 932 | 670 | 365 | 56.1419623 |
| 19 | 380 | 95 | 0 | 228 | 0 | 932 | 594 | 90 | 40.563252 |
| 20 | 475 | 0 | 0 | 228 | 0 | 932 | 594 | 180 | 42.6206484 |
| 21 | 427.5 | 47.5 | 0 | 228 | 0 | 932 | 594 | 180 | 41.8367142 |
| 22 | 139.6 | 209.4 | 0 | 192 | 0 | 1047 | 806.9 | 28 | 28.2374896 |
| 23 | 139.6 | 209.4 | 0 | 192 | 0 | 1047 | 806.9 | 3 | 8.06342182 |
| 24 | 139.6 | 209.4 | 0 | 192 | 0 | 1047 | 806.9 | 180 | 44.2078222 |
| 25 | 380 | 0 | 0 | 228 | 0 | 932 | 670 | 365 | 52.5166974 |
| 26 | 380 | 0 | 0 | 228 | 0 | 932 | 670 | 270 | 53.3006317 |
| 27 | 380 | 95 | 0 | 228 | 0 | 932 | 594 | 270 | 41.1513751 |
| 28 | 342 | 38 | 0 | 228 | 0 | 932 | 670 | 180 | 52.1243856 |
| 29 | 427.5 | 47.5 | 0 | 228 | 0 | 932 | 594 | 28 | 37.4275152 |
| 30 | 475 | 0 | 0 | 228 | 0 | 932 | 594 | 7 | 38.6037612 |
| 31 | 304 | 76 | 0 | 228 | 0 | 932 | 670 | 365 | 55.2601224 |
| 32 | 266 | 114 | 0 | 228 | 0 | 932 | 670 | 365 | 52.9083198 |
| 33 | | | | | | | | | |

The first seven columns in the data table in Figure 2.4 represent factor variables. Professor Yeh selected specific quantities of the seven component materials and then tested the compressive strength as the concrete aged. The eighth column, **Age**, shows the number of days elapsed since the concrete was formulated, and the ninth column is the response variable, **Compressive Strength**. In the course of his experiments, Professor Yeh repeatedly tested different formulations, measuring compressive strength after varying numbers of days. To see the structure of the data, let's look more closely at the two columns.

2. Choose **Analyze ▶ Distribution**.
3. As shown in Figure 2.5, select the **Cement** and **Age** columns and click **OK**.

**Figure 2.5: The Distribution Dialog Box**



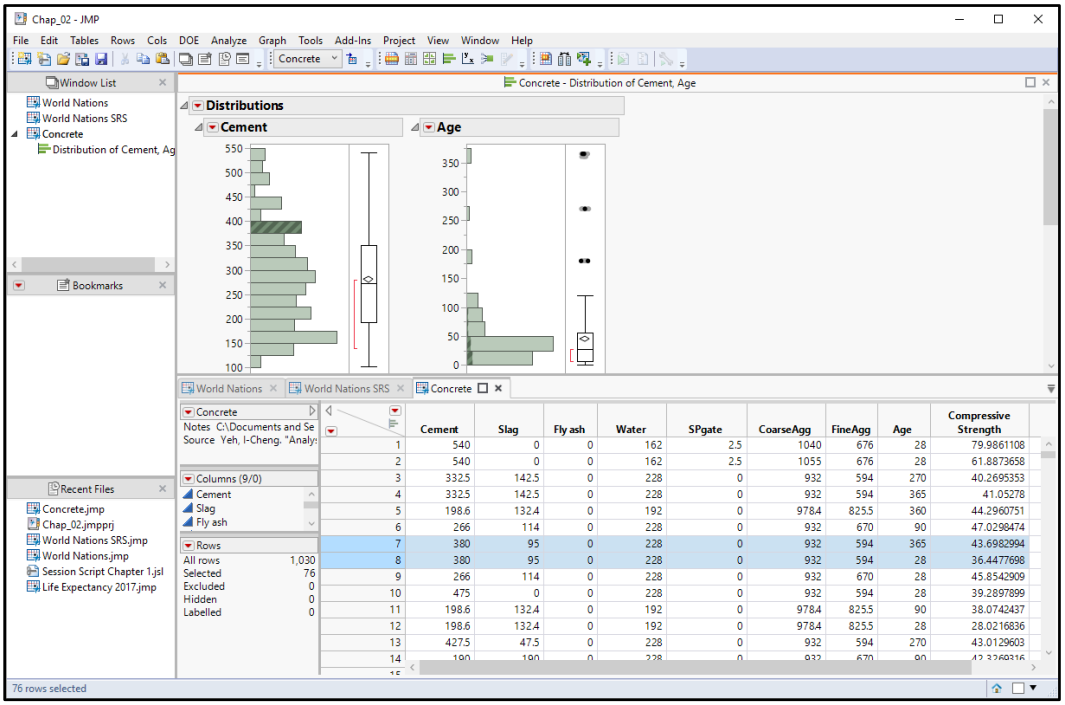

The Distribution platform generates two graphs and several statistics. We will study these in detail in Chapter 3. For now, you just need to know that the graphs, called *histograms*, display the variation within the two data columns. The lengths of the bars indicate the number of rows corresponding to each value. For example, there are many rows with concrete mixtures containing about 150 kg of cement, and very few with 100 kg.

4. Move your cursor over the **Cement** histogram and click the bar corresponding to values just below 400 kg of cement.

When you click that bar, the entire bar darkens, indicating that the rows corresponding to mixtures with that quantity (380 kg of cement, it turns out) are selected. Additionally, small portions of several bars in the **Age** histogram are also darkened, representing the same rows.

5. To see the relationships between a histogram bar, the data table, and the other histogram, we want to make the **Distribution Report** and **Data Table** visible at the same time. Move your cursor over the tab labeled **Concrete – Distribution of Cement, Age**. Click and drag downward; release the button over the **Dock Above** zone. The windows should look like Figure 2.6. Save your project once more.

**Figure 2.6: Select Rows by Selecting One Bar in a Graph**




Finally, look at the data table. Within the data grid, several visible rows are highlighted. All of them share the same value of **Cement**. Within the **Rows** panel, we see that we have now altered the row state of 76 rows by selecting them.

Look more closely at rows 7 and 8, which are the first two of the selected rows. Both represent seven-part mixtures following the same recipe of the seven materials. Observation #7 was taken at 365 days and observation #8 at 28 days. These observations are *not* listed chronologically, but rather are a randomized sequence typical of an experimental data table.

Row 16 is the next selected row. This formulation shares several ingredients in the same proportion as rows 7 and 8, but the amounts of other ingredients differ. This is also typical of an experiment. The researcher selected and tested different formulations. Because the investigator, Professor Yeh, manipulated the factor values, we find this type of repetition within the data table. And because Professor Yeh was able to select the factor values deliberately and in a controlled way, he was able to draw conclusions about which mixtures will yield the best compressive strength.

6. Before proceeding to the next section, restore the Distribution report to its original position. Move your cursor over the title bar **Concrete – Distribution of Cement, Age**. Click, drag, and release near the center of the screen over the **Dock Tab** drop zone.

## Observational Data—An Example

Of course, experimentation is not always practical, ethical, or legal. Medical and pharmaceutical researchers must follow extensive regulations, can experiment with dosages and formulations of medications, and may randomly assign patients to treatment groups, but cannot randomly expose patients to diseases. Investors do not have the ability to manipulate stock prices (if they do and get caught, they go to prison).

Now open the data table called **Stock Index Weekly**. This data table contains time series data for six different major international stock markets during 2008, the year in which a worldwide financial crisis occurred. A stock *index* is a weighted average of the prices of a sample of publicly traded companies. In this table, we find the weekly index values for the following stock market indexes, as well as the average number of shares traded per week:

- Nikkei 225 Index – Tokyo
- FTSE100 Index – London
- S&P 500 Index – New York
- Hang Seng Index – Hong Kong
- IGBM Index – Madrid
- TA100 Index – Tel Aviv

The first column is the observation date. Note that the observations are basically every seven days after the second week. The other columns simply record the index and volume values as they occurred. It was not possible to set or otherwise control any of these values.

## Survey Data—An Example

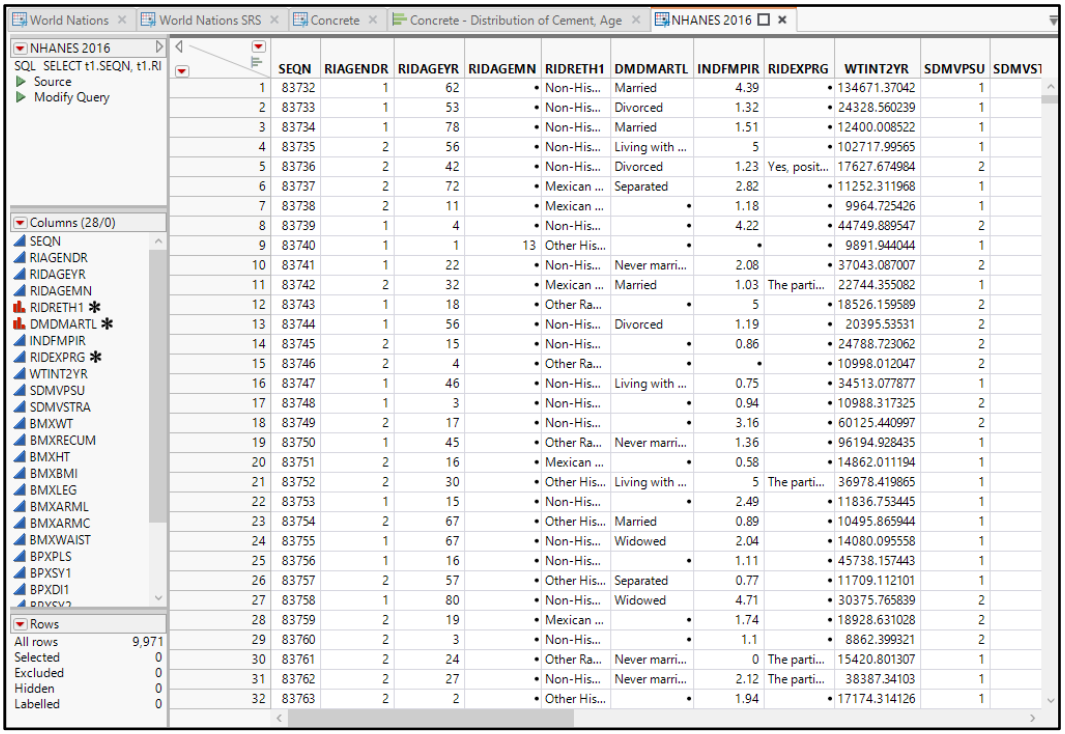
Survey research is conducted in the social sciences, in public health, and in business. In a survey, researchers pose carefully constructed questions to respondents, and then the researchers or the respondents themselves record the answers. In survey data, we often find *coding schemes* where categorical values are assigned numeric codes or other abbreviations. Sometimes continuous variables, like income or age, are converted to ordinal columns.

As an example of a well-designed survey data table, we have a small portion of the responses provided in the 2006 administration of the National Health and Nutrition Examination Survey (NHANES). NHANES is an annual survey of people in the U.S. that looks at their diet, health, and lifestyle choices. The Centers for Disease Control (CDC) posts the raw survey data for public access.[2]

1. Open the **NHANES 2016** data table.

As shown in Figure 2.7, this data table contains some features that we have not seen before. As usual, each column holds observations for a single variable, but these column names are not very informative. Typical of many large-scale surveys, the NHANES website provides both the data and a *data dictionary* that defines each variable, including information about coding and units of measurement. For our use in this book, we have added some notations to each column.

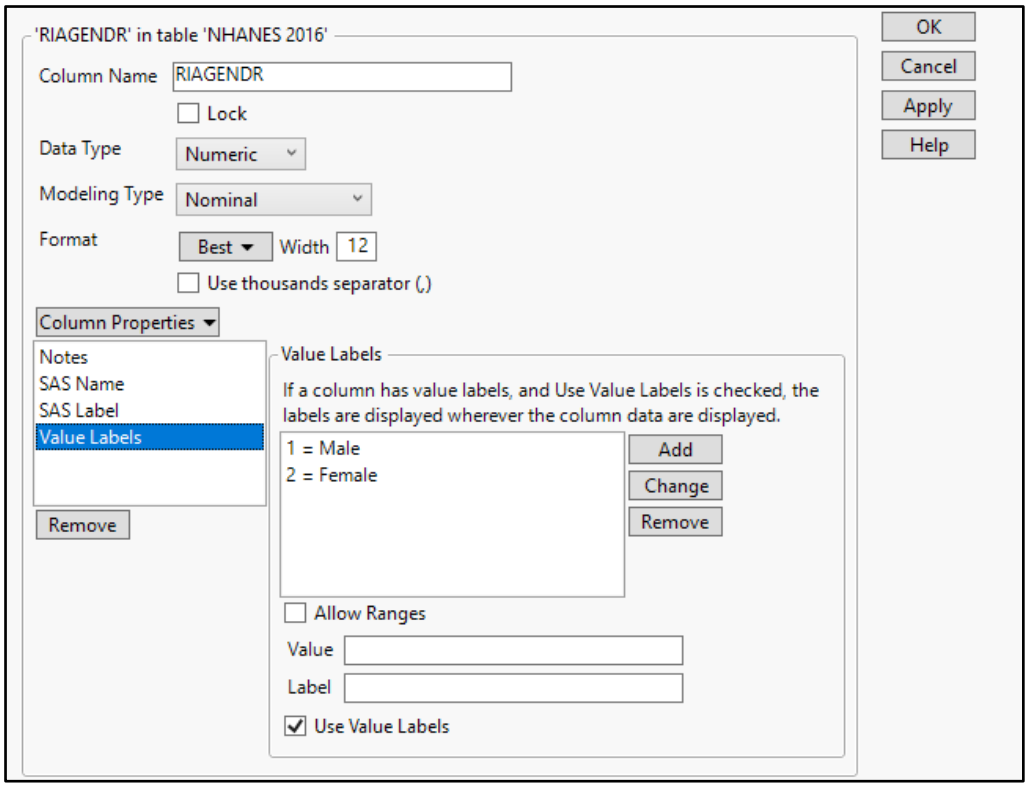
**Figure 2.7: NHANES Data Table**




| | SEQN | RIAGENDR | RIDAGEYR | RIDAGEMN | RIDRETH1 | DMDMARTL | INDFMPIR | RIDEXPRG | WTINT2YR | SDMVPSU | SDMVST |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 83732 | 1 | 62 | • | Non-His... | Married | 4.39 | • | 134671.37042 | 1 | |
| 2 | 83733 | 1 | 53 | • | Non-His... | Divorced | 1.32 | • | 24328.560239 | 1 | |
| 3 | 83734 | 1 | 78 | • | Non-His... | Married | 1.51 | • | 12400.008522 | 1 | |
| 4 | 83735 | 2 | 56 | • | Non-His... | Living with ... | 5 | • | 102717.99565 | 1 | |
| 5 | 83736 | 2 | 42 | • | Non-His... | Divorced | 1.23 | Yes, posit... | 17627.674984 | 2 | |
| 6 | 83737 | 2 | 72 | • | Mexican ... | Separated | 2.82 | • | 11252.311968 | 1 | |
| 7 | 83738 | 2 | 11 | • | Mexican ... | • | 1.18 | • | 9964.725426 | 1 | |
| 8 | 83739 | 1 | 4 | • | Non-His... | • | 4.22 | • | 44749.889547 | 2 | |
| 9 | 83740 | 1 | 1 | 13 | Other His... | • | • | • | 9891.944044 | 1 | |
| 10 | 83741 | 1 | 22 | • | Non-His... | Never marri... | 2.08 | • | 37043.087007 | 2 | |
| 11 | 83742 | 2 | 32 | • | Mexican ... | Married | 1.03 | The parti... | 22744.355082 | 1 | |
| 12 | 83743 | 1 | 18 | • | Other Ra... | • | 5 | • | 18526.159589 | 2 | |
| 13 | 83744 | 1 | 56 | • | Non-His... | Divorced | 1.19 | • | 20395.53531 | 2 | |
| 14 | 83745 | 2 | 15 | • | Non-His... | • | 0.86 | • | 24788.723062 | 2 | |
| 15 | 83746 | 2 | 4 | • | Other Ra... | • | • | • | 10998.012047 | 2 | |
| 16 | 83747 | 1 | 46 | • | Non-His... | Living with ... | 0.75 | • | 34513.077877 | 1 | |
| 17 | 83748 | 1 | 3 | • | Non-His... | • | 0.94 | • | 10988.317325 | 2 | |
| 18 | 83749 | 2 | 17 | • | Non-His... | • | 3.16 | • | 60125.440997 | 2 | |
| 19 | 83750 | 1 | 45 | • | Other Ra... | Never marri... | 1.36 | • | 96194.928435 | 1 | |
| 20 | 83751 | 2 | 16 | • | Mexican ... | • | 0.58 | • | 14862.011194 | 1 | |
| 21 | 83752 | 2 | 30 | • | Other His... | Living with ... | 5 | The parti... | 36978.419865 | 1 | |
| 22 | 83753 | 1 | 15 | • | Non-His... | • | 2.49 | • | 11836.753445 | 1 | |
| 23 | 83754 | 2 | 67 | • | Other His... | Married | 0.89 | • | 10495.865944 | 1 | |
| 24 | 83755 | 1 | 67 | • | Non-His... | Widowed | 2.04 | • | 14080.095558 | 1 | |
| 25 | 83756 | 1 | 16 | • | Non-His... | • | 1.11 | • | 45738.157443 | 1 | |
| 26 | 83757 | 2 | 57 | • | Other His... | Separated | 0.77 | • | 11709.112101 | 1 | |
| 27 | 83758 | 1 | 80 | • | Non-His... | Widowed | 4.71 | • | 30375.765839 | 2 | |
| 28 | 83759 | 2 | 19 | • | Mexican ... | • | 1.74 | • | 18928.631028 | 2 | |
| 29 | 83760 | 2 | 3 | • | Non-His... | • | 1.1 | • | 8862.399321 | 2 | |
| 30 | 83761 | 2 | 24 | • | Other Ra... | Never marri... | 0 | The parti... | 15420.801307 | 1 | |
| 31 | 83762 | 2 | 27 | • | Non-His... | Never marri... | 2.12 | The parti... | 38387.34103 | 1 | |
| 32 | 83763 | 2 | 2 | • | Other His... | • | 1.94 | • | 17174.314126 | 1 | |

2. In the **Columns** panel, move your cursor to highlight the column named **RIAGENDR** and right-click. Select **Column info…** to open the dialog box shown in Figure 2.8.

**Figure 2.8: Customizing Column Metadata**




When this dialog box opens, we find that this column holds **numeric** data, and its modeling type is **nominal**—which might seem surprising since the rows in the data table all say **Male** or **Female**. In fact, the lower portion of the column info dialog box shows us what is going on here. This is an example of *coded data:* the actual values within the column are 1s and 2s, but when displayed in the data table, a 1 appears as the word Male and a 2 as the word Female. This recoding is a *column property* within JMP.

The asterisk next to **RIAGENDR** in the **Columns** panel indicates that this column has one or more special column properties defined.

3. At the bottom of the dialog box, clear the check box marked **Use Value Labels** and click **Apply**. Now, look in the Data Table (**NHANES 2016** tab) and see that the column displays 1s and 2s rather than the value labels.
4. Click the word **Notes** under **Column Properties** (middle left of the dialog box). You will see a brief description of the contents of this column. As you work with data tables and create your own columns, you should get into the habit of annotating columns with informative notes.

In this table, we also encounter *missing observations* once again. Missing data is a common issue in survey data and it crops up whenever a respondent does not complete a question or an interviewer does not record a response. In a JMP data table, a black dot (·) indicates missing

numeric data; missing character data is just a blank. As a general rule, we want to think carefully about missing observations and *why* they occur in a particular set of data.

For instance, look at row 1 of this data table. We have missing observations for several variables, including the columns representing respondent's age in months (RIDAGEMN) and pregnancy status (RIDEXPRG). Further inspection shows that respondent #83732 was a 62-year-old male.

## Creating a Data Table

In this book, we will almost always analyze the data tables that are available at https://support.sas.com/en/books/authors/robert-carver.html. Once you start to practice data analysis on your own, you will often need to create your own data tables. Refer to Appendix B ("Data Management") for details about entering data from a keyboard, transferring data from the web, reading in an Excel worksheet, or assembling a data table from several other data tables.

## Raw Case Data and Summary Data

This book is accompanied by more than 50 data tables, most of which contain "casewise" data, meaning that each row of the data table refers to one observational unit. For example, each row of the NHANES table represents a person. Each row of the Concrete table is one batch of concrete at a moment in time. Like most statistical software, JMP is intended to analyze raw data. However, sometimes data come to us in a partially processed or summarized state. It is still possible to construct a data table and do some limited analysis of this type of data.

For example, consider public opinion surveys that have been reported in the news. Yale University and George Mason University, collaborating in the Yale Program on Climate Change Communication, published "Politics & Global Warming, April 2019."[3] The research team used "a nationally representative survey (*N* = 1,291 registered voters)" (Leiserowitz *et al.*, p. 4). Among other issues, they asked respondents if they think that global warming is happening. Among all voters, 70% expressed the view that global warming is real. The researchers broke down the total sample, as shown in Table 2.1 (created from the report's Executive Summary).

**Table 2.1: Sample of U.S. Voters' Belief that Global Warming Is Happening**

| Voter Group | Percentage agreeing |
|---|---|
| Liberal Democrats | 95 % |
| Moderate/Conservative Democrats | 87 |
| Liberal/Moderate Republicans | 63 |
| Conservative Republicans | 38 |

Tables like this are common in news reports. We could easily transfer this into a data table with one major caveat that often confuses introductory students. It is crucial to understand how the layout of this table relates to its content. In JMP, we usually expect each row to represent one observational unit, each column to represent one variable, and each cell to contain one data value. This table does not satisfy these assumptions.

To see why this is not a raw data table, we should go back and think about how the raw data were generated. We know that respondents to this question—the observational units—were the 1,291 registered voters in the survey sample. The interviewers asked them many questions, and Table 2.1 tallies the responses to two of the questions. First, people reported their voting habits. The second variable is the responses those people gave when asked the question, "Do you think that global warming is happening?" Respondents could agree, disagree, or offer no opinion. Because their responses were categorical rather than numeric, this is a nominal variable.

So, how does this clarify the contents of Table 2.1? The first column lists four voter groups, ordered from most liberal to most conservative; the cell entries are unambiguously categorical. The second column is a little tricky. It appears to be continuous data, looking very much like numbers. However, these numbers are not measurements of observational units (people). They *summarize* some of the answers provided by the respondents, indicating the fraction of each voter group expressing agreement to the global warming question. More precisely, the values are *relative frequencies* of each "level" of the ordinal voter group variable. In short, this table represents one ordinal and one nominal variable, summarizing the responses of nearly 1,300 individuals who are "invisible" when the information is presented this way. In later chapters, we will learn how to use columns of frequencies. For now, our introduction to data types and sources is complete.

## Application

Now that you have completed all the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. Use your primary textbook or the daily newspaper to locate a table that summarizes some data. Read the article accompanying the table and identify the variable, data type, and observational units.
2. Return to the **Concrete** data table. Browse through the column notes, and explain what variables these columns represent: **Cement**, **SPgate,** and **FineAgg**.
3. In the **NHANES 2016** data table, one nominal variable appears as continuous data within the **Columns** panel. Find the misclassified variable and correct it by changing its modeling type to nominal.
4. The **NHANES 2016** data table was assembled by scientific researchers. Why don't we consider this data table to be experimental data?
5. Find the data table called **Military**. We will use this table in later chapters. This table contains rank, gender, and race information about more than a million U.S. military personnel. Use a technique presented in this chapter to create a data table containing a random sample of 500 individuals from this data table.

6. In Chapter 3, we will work with a data table called **NIKKEI225**. Open this table and examine the columns and metadata contained within the table. Write a short paragraph describing the contents of the table, explaining how the data were collected (experiment, observation, or survey), and define each variable and its modeling type.
7. In later chapters, we will work with a data table called **Earthquakes**. Open this table and examine the columns and metadata contained within the table. Write a short paragraph describing the contents of the table, explain how the data were collected (experiment, observation, or survey), and define each variable and its modeling type.
8. In later chapters, we will work with a data table called **Tobacco Use**. Open this table and examine the columns and metadata contained within the table. Write a short paragraph describing the contents of the table, explain how the data were collected (experiment, observation, or survey), and define each variable and its modeling type.
9. Open the **Dolphins** data table, which we will work with in a later chapter. What are the variable(s), observational units, and data types represented in this table?
10. Open the data table **TimeUse**, which we will analyze more fully in later chapters. Write a few sentences to explain the contents of the columns named **marst**, **empstat**, **sleeping**, and **telff**.
11. Open the **States** data table, which contains statistics about the 50 U.S. states and the District of Columbia. Write a short paragraph describing the contents of the table and, in particular, define the columns called **poverty**, **fed_spend2010**, and **homicide**.

---

[1] In Chapter 21, we will learn how to design experiments. In this chapter, we will concentrate on the nature of experimental data.

[2] Visit http://www.cdc.gov/nchs/surveys.htm to find the NHANES and other public-use survey data. Though the topic is beyond the scope of this book, readers engaged in survey research will want to learn how to conduct a database query and import the results into JMP. Interested readers should consult the section on "Importing Data" in Chapter 2 of the *JMP User Guide.*

[3] Visit https://climatecommunication.yale.edu/publications/politics-global-warming-april-2019/ to read the full report.

# Chapter 3: Describing a Single Variable

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## Overview

Once we have framed some research questions and gathered relevant data, the next phase of an investigation is to examine the variability in the data. The goal of descriptive analysis is to summarize where things stand with each variable. In fact, the term *statistics* comes from the practice of characterizing the state of political affairs through the reporting of facts and figures. This chapter presents several standard tools that we can use to examine how a variable varies, to describe the pattern of variation that it exhibits, and to look for departures from the overall pattern as well.

## The Concept of a Distribution

Data analysis generally focuses on one or more variables—attributes of the individual observations. When we speak of a variable's distribution, we are referring to a pattern of values. The distribution describes the different values the variable can assume, and how often it assumes each value.

In our first example, we will continue to consider the variability of life expectancy around the world. The data that we will use come to us from the World Bank. In Chapter 1, we used a small portion of this data set for 2017. Now we will look at more years.

## Variable Types and Their Distributions

> In Chapter 2, we did our work in a JMP Project. Get in the habit of using a project for each chapter.

1. Select **File ▶ New ▶ Project**.
2. Select **File ▶ Open**, select the **Life Expectancy** data table, and click **Open**.

Before doing any analysis, make sure that you can answer these questions:

- What population does this data table represent?
- What is the source of the data?
- How many variables are in the table?
- What data type is each variable?
- What does each variable represent?
- How many observations are there?

Take special note of the way this data table has been organized. We have 12 annual observations for each country, spaced at 5-year intervals, and they are *stacked* one upon the other. Not surprisingly, JMP refers to this arrangement as stacked data.

As in Chapter 1, we will raise some questions about how life expectancy at birth varies in different parts of the world. There are far too many observations for us get a general sense of the variation simply by scanning the table visually. We need some sensible ways to find the patterns among the large number of rows. We will begin our analysis by looking at the *nominal* variable called **Region**.

Statisticians generally distinguish among four types of data:

| Categorical Types | Quantitative Types |
|---|---|
| Nominal | Interval |
| Ordinal | Ratio |

One reason that it is important to understand the differences among data types is that we analyze them in different ways. In JMP, we differentiate between nominal, ordinal, and continuous data. Nominal and ordinal variables are categorical, distinguishing one observation from another in some qualitative, non-measurable way. Interval and ratio data are both numeric. Interval variables are artificially constructed, like a temperature scale or stock index, with arbitrarily chosen zero points. Most measurement data are considered ratio data because

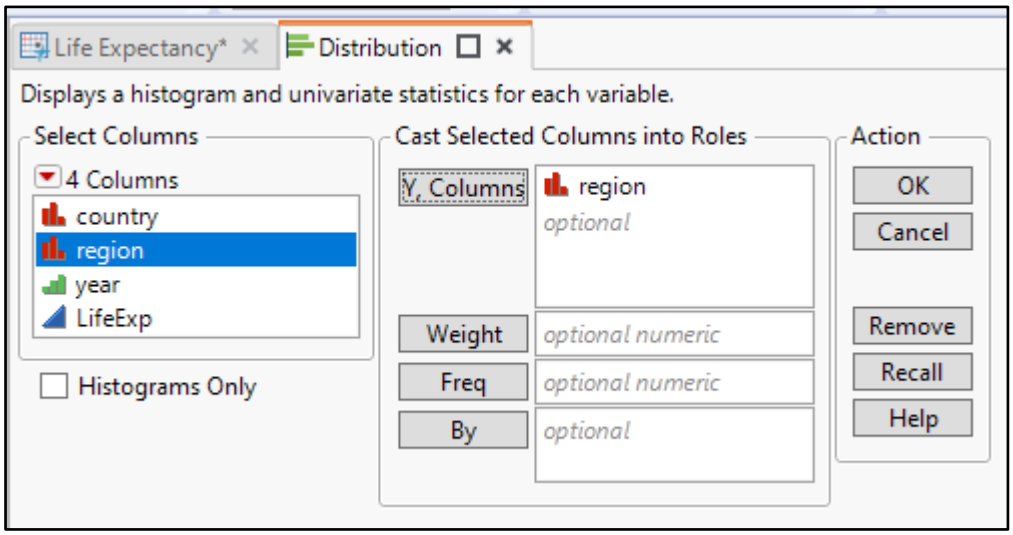
ratios of values are meaningful. For example, a film that lasts 120 minutes is twice as long as one lasting 60 minutes. In contrast, 120 degrees Celsius is *not* twice as hot as 60 degrees Celsius.

## Distribution of a Categorical Variable

In its reporting, the World Bank identifies each country of the world with a continental region. There are seven regions, each with a different number of countries. The variable **Region** is *nominal*—it literally names a country's general location on earth. Let's get familiar with the different regions and see how many countries are in each. In other words, let's look at the distribution of **Region**.

1. Select **Analyze** ▶ **Distribution**. In the **Distribution** dialog box (Figure 3.1), select the variable **region** as the **Y, Columns** variable. Click **OK**.

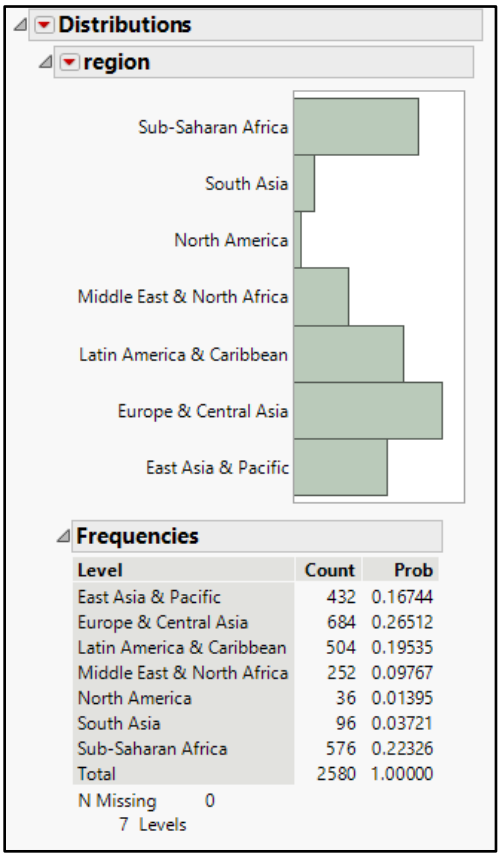
**Figure 3.1: Distribution Dialog Box**




> Anytime you want to assign a column to a role in a JMP dialog box, you have three options: you can highlight the column name in the **Select Columns** list and click the corresponding role button, you can double-click the column name, or you can click-drag the column name into the role box.

The result appears in Figure 3.2. JMP constructs a simple *bar chart* listing the six continental regions and showing a rectangular bar corresponding to the number of times the name of the region occurs in the data table. Though we cannot immediately tell from the graph alone exactly how many countries are in each, North America clearly has the fewest countries and Europe and Central Asia has the most.

**Figure 3.2: Distribution of Region**




Below the graph is a *frequency distribution* (titled **Frequencies**), which provides a more specific summary. Here we find the name of each region, and the number of times each regional name appears in our table. For example, "East Asia & Pacific" occurs 432 times. As a proportion of the whole table, 16.7% of the rows (Prob. = 0.16744) represent countries in that region.

At this point, you might wisely pause and say, "Wait a second. Can there possibly be 432 countries in the East Asia and the Pacific region?" And you would be right. Remember that we have stacked data, with 13 rows representing 12 years of data devoted to each country. Therefore, there are 432/12 = 36 countries in the region.

> Even though JMP handles the heavy computational or graphical tasks, always think about the data and its context and ask yourself if the results make sense to you.

## Using the Data Filter to Temporarily Narrow the Focus

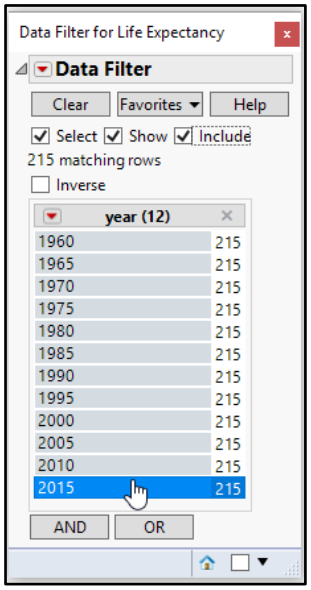
Because we know each country appears repeatedly in this data table, let's choose just one year's data to obtain a clearer picture of regional variation. We can specify rows to display in a graph by

using the **Data Filter.** This is a tool that allows us to select rows that satisfy specific conditions such as only displaying data rows from the year 2010.

> This chapter illustrates the use of the **Data Filter** to temporarily select rows in a data table for all active analyses. This is known as the global Data Filter. Alternatively, when you click the red triangles in most analysis reports, you will find a **Script** option with a local Data Filter that applies only to the current report. The local Data Filter is illustrated in later chapters, but curious readers should explore it at any time.

1. To see the effects of the Data Filter, we will instruct JMP to automatically update the graph and recalculate the frequencies. Click the red arrow next to **Distributions** and choose **Redo ▶ Automatic Recalc**.
2. Select **Rows ▶ Data Filter**. In the list of **Columns**, select **year** and click the **Add** button.
3. The dialog box takes on a new appearance (Figure 3.3). It now displays a list of years contained in the table. Near the top of the dialog box, check **Show** and **Include** so that only the rows that we select for **2010** will appear in all graphs and be included in any computations. Other rows will be hidden and excluded.

**Figure 3.3: Choosing 2010 in the Data Filter**




4. Scroll down the list of **Year** levels and highlight **2015**. As noted in the dialog box, this selects 215 rows and temporarily suppresses the others.
5. Minimize the **Data Filter**. If you look in the data table of Life Expectancy, you will see that most rows now have two icons (  ) indicating that they are excluded and hidden. The rows from 2015 are highlighted and will remain so until we clear the **Data Filter** or take another action that selects other rows.

## Using Graph Builder to Explore Categorical Data Visually

In Chapter 1, we met the Graph Builder, and we will use it throughout this book. It is most useful when working with multiple variables, but even with a single nominal variable, it provides a quick way to generate multiple views of the same data. Because interactivity is such an important feature of the tool, this section of the chapter provides few step-by-step directions. You should interact with the tool and think about the extent to which different graphing formats and options communicate the information content of the variable called **region**.

1. Select **Graph ▶ Graph Builder.** The **region** column identifies groups of countries. Drag it to the **X** drop zone.

> Within Graph Builder, you can freely reposition a column from one drop zone to another. Hover the cursor over the column name until the cursor changes to the hand shape, then click-drag it to any other drop zone. What's more, there is also an Undo button.
>
> You can also use the same variable in more than one drop zone. For example, you might also color the bars by region.

With Region on the X axis, you will see seven clumps of black points above the seven region names. This is not very informative.

At the top of Graph Builder is a selector bar of icons (see Figure 3.4) representing different graph types. The graphing options available depend on the type of data we have placed on the graph. Hence, some icons are dimmed, but with Region on the X axis, we can opt for any of the highlighted option.

**Figure 3.4: Graphing Options for a Nominal Column**



2. Spend some time using different graphing formats. Which ones do you think do the best job of clearly and fully summarizing the number of countries within each region?
3. For this example, let's use a bar chart (seventh option from the left). There is considerable research demonstrating that most people find this simple graph type easy to interpret accurately. Then click **Done**.

It is always good practice to help a reader by giving a graph an informative descriptive title. The default title "region," though accurate, is not very helpful. In JMP, it is easy to alter the titles of graphs and other results.

4. Move your cursor to the title **region** just above the graph and double-click. You can now customize the title of this chart to make it more informative. Type **Observations per Region**, replacing "Chart" as the title.
5. We have done a bit of work on our project. Let's save it now as **Chap_03**.

With most categorical data, JMP automatically reports values in alphabetical sequence[1] (East Asia & Pacific, Europe & Central Asia, and so on). We can revise the order of values to suit our purposes as well. Suppose that we want to list the regions approximately from West to East, North to South. In that case, we might prefer a sequence as follows:

**North America**
**Latin America & Caribbean**
**Europe & Central Asia**
**Middle East & North Africa**
**Sub-Saharan Africa**
**South Asia**
**East Asia & Pacific**

To change the default sequence of categorical values (whether nominal or ordinal), we return to the **Life Expectancy** data table.

6. Select **region** from the data grid or the columns panel, right-click, and select **Column Info**.
7. Click **Column Properties** and select **Value Order**.
8. Select a value name and use the **Move Up** and **Move Down** buttons to revise the value order to match what we have chosen. Then, click **OK**.

Now return to **Graph Builder** and look at the bar chart. You will see that customizing the value order within the data table re-orders the X axis. The effect should speak for itself.

9. Experiment with the other charting options by clicking the red arrow and choosing **Show Control Panel** and then selecting various graph types.

With categorical data, your choices are limited. Still, it's worth a few minutes to become familiar with them. When you are through exploring, restore the graphic to a bar chart and leave it open. We will return to this graph in a few pages.

## Distribution of a Quantitative Variable

The standard graphing choices expand considerably when we have quantitative data—particularly for continuous variables or discrete variables with many possible values. We will want to summarize a large collection of values in a way that shows where observations tend to cluster.

As a way of visualizing the distribution of a continuous variable, the most commonly used graph is a *histogram*. A histogram is basically a bar chart with values of the variable on one axis and frequency on the other. Let's illustrate.

In our data set, we have estimated life expectancy at birth for each country for 13 different years. We just used the Data Filter to isolate the data for 2015, so let's continue to explore the state of the world in 2015.

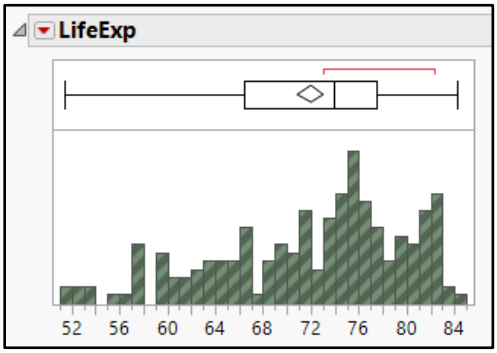
## Using the Distribution Platform for Continuous Data

As before, we will first use the **Distribution** platform to do most of the work here.

1. Select **Analyze ▶ Distribution.** Cast **LifeExp** into the role of **Y, Columns** and click OK.
2. When the distribution window opens, click the red triangle next to **Distributions**, and select **Stack**. This will re-orient the output horizontally making it a bit easier to interpret.

The histogram (Figure 3.5) is one representation of the distribution of life expectancy around the world in 2015, and it gives us one view of how much life expectancy varies. Above the histogram is a *box plot* (also known as a box-and-whiskers plot), which will be explained later in this chapter.

**Figure 3.5: A Typical Histogram**




As in the bar charts that we have studied earlier, there are two dimensions in the graph. Here, the horizontal axis displays values of the variable and the vertical axis displays the *frequency* of each small interval of values. For example, we can see that only a few countries have projected life expectancies of 51 to 54 years, but many have life expectancies between 74 and 78 years.

When we look at a histogram, we want to develop the habit of looking for four things: the *shape*, the *center* (or central tendency), the *dispersion* of the distribution, and *unusual observations*. The histogram can very often clearly represent these three aspects of the distribution.

**Shape:** Shape refers to the symmetry of the histogram and to the presence of peaks in the graph. A graph is symmetric if you could find a vertical line in the center defining two sides that are mirror images of one another. In Figure 3.5, we see an *asymmetrical* graph. There are few observations in the *tails* on the left, and most observations clump together on the right side. We say this is a *left-skewed* (or negatively skewed) distribution.

Many distributions have one or more peaks—data values that occur more often than the other values. Here we have a distinct peak around 75 to 76 years, and others closer to 72 and 83. Some

distributions have multiple peaks, and some have no distinctive peaks at all. In short, we might describe the shape of this distribution as "multi-peaked and left-skewed."

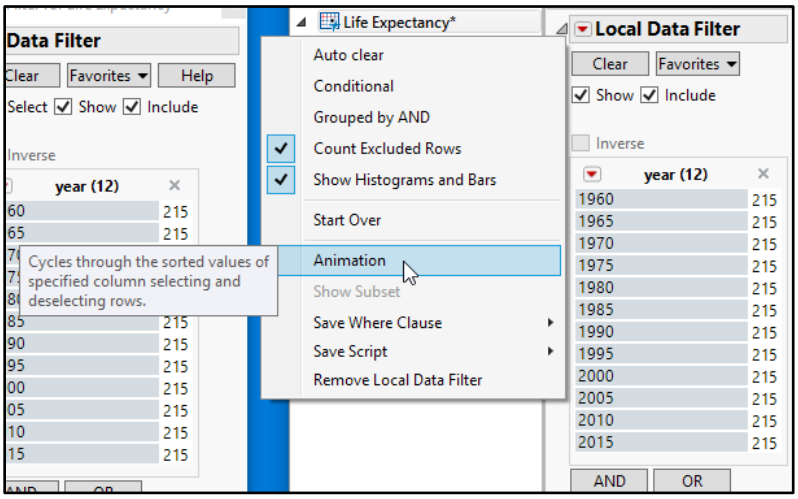
**Center** (*or central tendency*): Where do the values congregate on the number line? In other words, what values does this variable typically assume? As you might already know, there are several definitions of center as reflected in the mean, median, and mode statistics. Visually, we might think of the center of a histogram as the halfway point of the horizontal axis (the median, which is approximately 74 years in this case), as the highest-frequency region (the highest peak near 75), perhaps as a type of visual balancing point (the mean, which is approximately 72), or in some other way. Any of these interpretations have legitimacy, and all respond to the question in slightly different ways.

**Dispersion** (*or spread*): While the concept of center focuses on the typical, the concept of spread focuses on departures from the typical. The question here is, "how much does this variable vary?" and again there are several reasonable ways to respond. We might think in terms of the lowest and highest observed values (from about 40 to 85), in terms of a vicinity of the center (for example, "life expectancy tends to vary in most countries between about 65 and 85"), or in some other relative sense.

**Unusual Observations**: We can summarize the variability of a distribution by citing its shape, center, and dispersion, but in some distributions, there may be a small number of observations that deviate substantially from the pattern. In 2015, there was no such grouping, but let's explore the shifts in the distribution over time and also find some unusual observations.

3. Re-open the global data filter (**Rows ▶ Data Filter**). Click **Clear**.
4. Click the red triangle next to **Distributions** and choose **Redo ▶ Automatic Recalc**. You will see the histogram change and might notice that it now represents more observations—we are looking at all twelve years of data.
5. Again, click the red triangle next to **Distributions** and choose **Local Data Filter**; choose **year** and click **Add.**
6. Rather than choosing one year, click the red triangle next to **Local Data Filter** and choose **Animation**, as shown in Figure 3.6. This will step through the twelve years, briefly selecting each one and changing the histogram for each year.

**Figure 3.6: Animating a Local Data Filter**



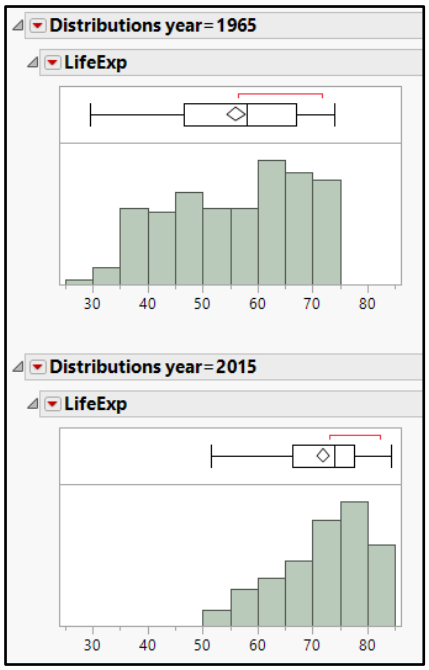

7. In the **Animation Controls**, click the blue "play" arrow and watch what happens. Take special notice of how life expectancy has tended to improve from 1960 through 2015.
8. After a few cycles, pause the animation in the year 1995.

Look at the box plot above the histogram. There are two dots at the far left end; these represent two nations with extraordinarily brief life expectancies. We refer to such values as *outliers*.

9. Hover the cursor over the left-most point in the box plot. You will see a pop-up note that this is Rwanda, with a life expectancy of only 31.977 years in 1995, reflecting the genocide that took place in 1994.

Often, it's easier to think about shape, center, dispersion, and outliers by comparing two distributions. For example, Figure 3.7 shows two histograms using the life expectancy data from 1965 and 2015. We might wonder how human life expectancy changed during a 50-year period, and in these two histograms, we can note differences in shape, center, dispersion and unusual observations.

**Figure 3.7: Comparing Two Distributions**




To create the results shown in Figure 3.7, do the following:

10. Return to the original **Life Expectancy** data table.
11. Re-open the Data Filter dialog box (either choose **Windows** and find the filter or **Rows ▶ Data Filter**). Clear the **Select** check box but leave **Show** and **Include** checked.
12. Hold down the **Ctrl** key and highlight **1965** and **2015**.
13. From the menu bar, choose **Analyze ▶ Distribution**.
14. Select **LifExp** as **Y**, just as you did earlier.
15. Cast **year** into the role of **By** and click **OK**.

This creates the two distributions with vertically oriented histograms. When you look at them, notice that the axis of the first one runs from 25 to 75 years, and the axis on the second graph runs from 50 to 85 years.

To facilitate the comparison, it is helpful to orient the histograms horizontally in a stacked arrangement and to set the axes to a uniform scale, an option that is available in the red triangle menu next to **Distributions.** This makes it easy to compare their shapes, centers, and spreads at a glance.

16. In the **Distribution** report, while pressing the Ctrl key, click the uppermost red triangle and select **Uniform Scaling**.

If you click the red triangle without pressing the Ctrl key, the uniform scaling option would apply only to the upper histogram. Pressing the Ctrl key has the effect of applying the choice to all graphs in the window.

17. Hold down the Ctrl key, click the red triangle once again, and choose **Stack**.

The histograms on your screen should now look like Figure 3.7. How does the shape of the 1965 distribution compare to that of the 2015 distribution? What might have caused these changes in the shape of the distribution?

We see that people tend to live longer now than they did in 1965. The location (or central tendency) of the 2015 distribution is to the right side of the 1965 distribution. Additionally, these two distributions also have quite different spreads (degrees of dispersion). We can see that the values were far more spread out in 1965 than they are in 2015 and that there were no outliers in either year. What does that reveal about life expectancy around the world during the past 50 years?

## Taking Advantage of Linked Graphs and Tables to Explore Data

When we construct graphs, JMP automatically links all open tables and graphs. If we select rows either in the data table or in a graph, JMP selects and highlights those rows in all open windows.

1. Within the 2015 life expectancy histogram, place the cursor over the right-most bar and click. While pressing the Shift key, also click the adjacent bar. Now you should have selected the two bars representing life expectancies of over 75 years. How many rows are now selected? Look in the **Rows** panel of the **Data Table** window.
2. Now find the first window with the Distribution of **Region** (second tab in your project). Notice that some bars are partially highlighted. When you selected the two bars in the histogram, you were indirectly selecting a group of countries. These countries are grouped within the bar chart as shown, revealing the parts of the world where people tend to live longest.

## Customizing Bars and Axes in a Histogram

When we use the **Distribution** platform to analyze a continuous variable, JMP determines how to divide the variable axis and how to create “bins” for grouping observations. These automatic choices can affect the appearance of the distribution and there are several ways to customize the appearance of a histogram.

We can alter the number of bars in the histogram, creating new boundaries between groups of observations and shifting observations from one bar to the next.

1. Move back to the **Distribution** report tab. Click anywhere in a blank area of the 2015 histogram to de-select the two right bars
2. Choose **Tools ▶ Grabber**.

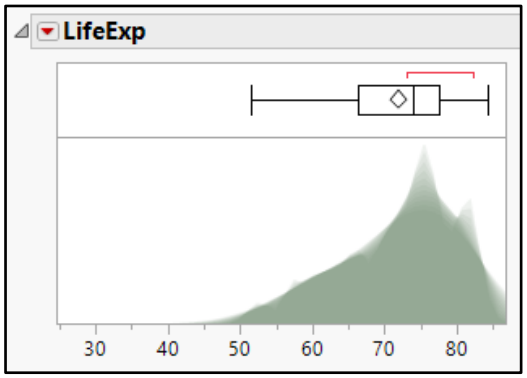
3. Position the hand  anywhere over the bars in the 2015 histogram beneath the box plot, and click-drag the tool straight up and down. In doing so, you will change the number and width of the bars, sometimes dramatically changing the shape of the graph.

Think about this: the apparent shape of the distribution depends on the number of bars we create. By default, the software chooses an initial number of bars, or *bins*, to categorize the continuous variable. However, that initial choice should not be the final word. As we adjust the number of bins, we should watch closely to see how the shape changes, looking for a rendering that accurately and honestly displays the overall pattern of variation.

One way to resolve the issue is by using a *shadowgram*. A shadowgram visually averages a large number of bin widths into a diffuse image with no distinct bars at all. Here is how:

4. Click the red triangle next to **LifeExp** in the 2015 histogram.
5. Choose **Histogram Options ▶ Shadowgram**. Figure 3.8 shows the result.

**Figure 3.8: A Shadowgram for a Continuous Variable**




You should notice that there are several Histogram Options. While you are here, explore them—see what there is to see.

We can also change the scale of the horizontal axis interactively. Initially, JMP set the left and right endpoints, and the limits changed when we chose uniform scaling. Suppose we want the axis to begin at 30 and end at 85.

6. Move the cursor to the left end of the horizontal axis, and notice that the hand now points to the left (this is true whether you have previously chosen the hand tool or not). Click and drag the cursor slowly left and right, and see that you are scrunching or stretching the axis values. Stop when the minimum value is 30.
7. Move the cursor to the right end of the axis, and similarly set the maximum at 100 years just by dragging the cursor.

Finally, we can "pan" along the axis. Think of the border around the graph as a camera's viewfinder through which we see just a portion of the entire infinite axis.

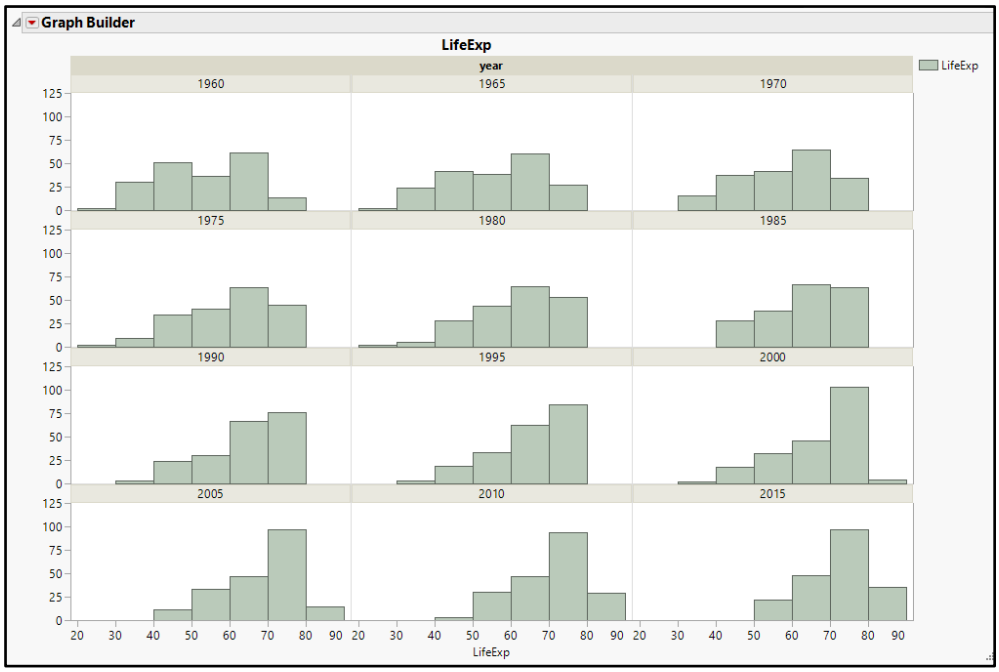
8. Without holding the mouse button, move the cursor toward the middle of the axis until the hand points upward. Now click and drag to the left or right, and you will pan along the axis.
9. Alternatively, rather than clicking and dragging to change axis attributes, you can directly edit all "Axis Settings" by double-clicking on the axis itself. This opens a dialog box where you can specify a variety of settings.

## Exploring Further with the Graph Builder

Our original data table contains values for 12 years, and we have now compared the variation in life expectancy for two years. The Graph Builder can allow us to make a quick visual comparison over 12 years.

1. First, we want to clear our earlier filtering so that we can now access all years. Choose **Rows ▶ Clear Row States** to deselect, show, and include all rows.
2. Select **Graph ▶ Graph Builder**.
3. Drag **LifeExp** to the **X** drop zone.
4. Find the menu bar at the top of the **Graph Builder** window and locate the **Histogram** button  near the center. Click it.
5. Drag **Year** to the **Wrap** drop zone and click the **Done** button. Your graph should look like Figure 3.9.

**Figure 3.9: Longer Lives in Most of the World, 1960 to 2015**



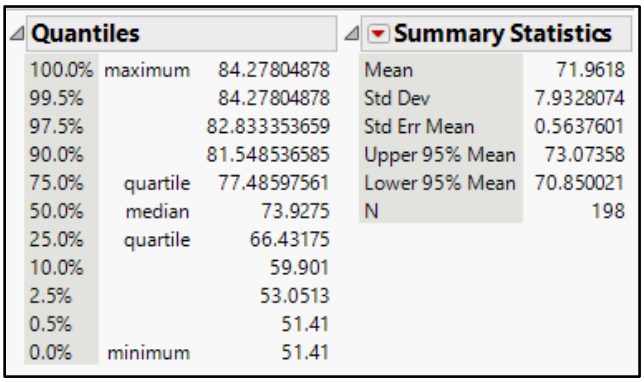

What do you see as you inspect these small multiple histograms? Can you see life expectancies gradually getting longer in most countries? There were two peaks in 1960: many countries with short lives, and many with longer lives. The lower peak slowly flattened out as the entire distribution has crept rightward.

## Summary Statistics for a Single Variable

Graphs are an ideal way to summarize a large data set and to communicate a great deal of information about a distribution. We can also describe variation in a quantitative variable with *summary statistics* (also called summary measures or descriptive statistics). Just as a distribution has shape, center, and dispersion, we have summary statistics that capture information about the shape, center, or dispersion of a variable.

Let's look back at the distribution report for our sample of 2015 life expectancies in 198 countries of the world. Just to the right of the histogram, we find a table of *Quantiles* followed by a list of *Summary Statistics*.

**Figure 3.10: Quantiles and Summary Statistics**



**Quantiles**

| | | |
|---|---|---|
| 100.0% | maximum | 84.27804878 |
| 99.5% | | 84.27804878 |
| 97.5% | | 82.833353659 |
| 90.0% | | 81.548536585 |
| 75.0% | quartile | 77.48597561 |
| 50.0% | median | 73.9275 |
| 25.0% | quartile | 66.43175 |
| 10.0% | | 59.901 |
| 2.5% | | 53.0513 |
| 0.5% | | 51.41 |
| 0.0% | minimum | 51.41 |

**Summary Statistics**

| | |
|---|---|
| Mean | 71.9618 |
| Std Dev | 7.9328074 |
| Std Err Mean | 0.5637601 |
| Upper 95% Mean | 73.07358 |
| Lower 95% Mean | 70.850021 |
| N | 198 |

**Quantile** is a generic term; you might be more familiar with *percentiles*. When we sort observations in a data set, divide them into groups of observations, and locate the boundaries between the groups, we establish quantiles. When there are 100 such groups, the boundaries are called percentiles. If there are four such groups, we refer to quartiles.

For example, we find that the 90th percentile is 81.54 years. This means that 90% of the observations have life expectancies shorter than 81.54 years. JMP also labels five quantiles known as the *five-number summary*. They identify the minimum, maximum, 25th percentile (1st quartile or Q1), 50th percentile (median), and 75th percentile (3rd quartile or Q3). Of the 198 countries listed in the data table, one-fourth have life expectancies shorter than 66.43 years, and one-fourth have life expectancies longer than 77.49 years.

**Summary Statistics** refer to the common descriptive statistics shown in Figure 3.10. At this stage in your study of statistics, three of these statistics are useful, and the other three should wait until Chapter 8.

- The *mean* is the simple arithmetic average of the observations, usually denoted by the symbol $\bar{X}$ and computed as follows:

  $$\bar{X} = \frac{\sum_i X_i}{n}$$

  Along with the *median*, it is commonly used as a measure of central tendency; in a symmetric distribution, the mean and median are quite close in value. When a distribution is strongly left-skewed like this one, the mean will tend to be smaller than the median. In a right-skewed distribution, the opposite will be true.

- The *standard deviation* (**Std Dev**) is a measure of dispersion, and you might think of it as a typical distance of a value from the mean of the distribution. It is usually represented by the symbol *s*, and is computed as follows:

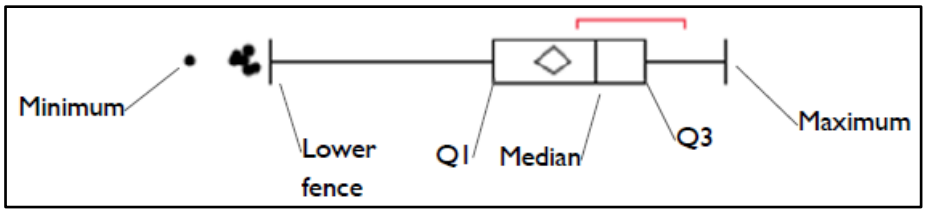
  $$s = \sqrt{\frac{\sum_i (X_i - \bar{X})^2}{(n-1)}}$$

We will have more to say about the standard deviation in later chapters, but for now, please note that it must be greater than or equal to zero, and that highly dispersed variables have larger standard deviations than consistent variables.

- *n* refers to the number of observations in the sample.

## Outlier Box Plots

Now that we have discussed the five-number summary, we can interpret a box plot. The key to interpreting an outlier box plot is to recognize that it is a diagram of the five-number summary. Here is a typical example:




In a box plot, there is a rectangle with an intersecting line. Two edges of the rectangle are located at the first (Q1) and third (Q3) quartile values, and the line is located at the median. In other words, the rectangular box spans the *interquartile range* (IQR). Extending from the ends of the box are two lines called *whiskers*. In a distribution that is free of outliers, the whiskers reach to the minimum and maximum values. Otherwise, the plot limits the reach of the whiskers by the *upper* and *lower fences, which* are located 1.5 IQRs from each quartile. In this illustration, we have a cluster of seven low-value outliers.

JMP also adds two other features to the box plot. One is a diamond that represents the location of the mean. If you imagine a vertical line through the vertices of the diamond, you have located the mean. The other two vertices are positioned at the upper and lower *confidence limits* of the mean. We will discuss those in Chapter 11.

The second additional feature is a red bracket above the box. This is the *shortest half bracket*, representing the smallest part of the number line comprising 50% of the cases. We can divide the observations in half in different ways. The median gives the upper and lower halves; the IQR box gives the middle half. This bracket gives the shortest half.

A box plot very efficiently conveys information about the center, symmetry, dispersion, and outliers for a single distribution. When we compare box plots across several groups or samples, the results can be quite revealing. In the next chapter, we will look at such box plots and other ways of summarizing two variables at a time.

# Application

Now that you have completed all of the activities in this chapter, use the techniques that you have learned to respond to these questions.

1. *Scenario:* We will continue our analysis of the variation in life expectancy at birth in 2015. Reset the Data Filter to show and include 2015.
    a. When we first constructed the **Life Exp** histogram, we described it as multi-peaked and left-skewed. Use the hand tool to increase and reduce the number of bars. Adjust the number of bars so that there are two prominent peaks. Describe what you did, and where the peaks are located.
    b. Rescale the axes of the same histogram and see if you can emphasize the two peaks even more (in other words, have them separated distinctly). Describe what you did to make these peaks more distinct and noticeable.
    c. Based on what you have seen in these exercises, why is it a good idea to think critically about an analyst's choice of scale in a reported graph?
    d. Highlight a few of the left-most bars in the histogram for **LifeExp** and look at the **Distribution** report for **region**. Which continent or continents are home to the countries with the shortest life expectancies in the world? What might account for this?
2. *Scenario:* Now let's look at the distribution of life expectancy 25 years before 2015. Use the **Data Filter** to choose the observations from 1990.
    a. Use the Distribution platform to summarize **Region** and **LifeExp** for this subset. In a few sentences, describe the distribution of **LifeExp** in 1990.
    b. Compare the five-number summaries for life expectancy in 1990 and in 2015. Comment on what you find.
    c. Compare the standard deviations for life expectancy in 1990 and 2015. Comment on what you find.
    d. You will recall that in 2015, the mean life expectancy was shorter than the median, consistent with the left-skewed shape. How do the mean and median compare in the 1990 data?
3. *Scenario:* The data file called **Sleeping Animals** contains data about the size, sleep habits, lifespan, and other attributes of different mammalian species.
    a. Construct box plots for **Lifespan** and **TotalSleep**. For each plot, explain what the landmarks on each plot tell you about the distribution of each variable. Comment on noteworthy features of the plot.
    b. Which distribution is more symmetric? Explain specifically how the graphs and descriptive statistics helped you come to a conclusion.
    c. According to the data table, "Man" has a maximum life span of 100 years. Approximately what percent of mammals in the data set live less than 100 years?
    d. Sleep hours are divided into "dreaming" and "non-dreaming" sleep. How do the distributions of these types of sleep compare?

e. Select the species that tend to get the most total sleep. Comment on how those species compare to the other species in terms of their predation, exposure, and overall danger indexes.

f. Now use the **Distribution** platform to analyze the body weights of these mammals. What's different about this distribution in comparison to the other continuous variables that you have analyzed thus far?

g. Select those mammals that sleep in the most exposed locations. How do their body weights tend to compare to the other mammals? What might explain this comparison?

4. *Scenario:* When financial analysts want a benchmark for the performance of individual equities (stocks), they often rely on a "broad market index" such as the S&P 500 in the U.S. There are many such indexes in stock markets around the world. One major index on the Tokyo Stock Exchange is the Nikkei 225, and this set of questions refers to data about the monthly values of the Nikkei 225 from December 31, 2013 through December 31, 2018. In other words, our data table called **NIKKEI225** reflects monthly market activity for a five-year period.

   a. The variable called **Volume** is the total number of shares traded per month (in millions of shares). Describe the distribution of this variable.

   b. The variable called **Change%** is the monthly change, expressed as a percentage, in the closing value of the index. When **Change%** is positive, the index increased that month. When the variable is negative, the index decreased that month. Describe the distribution of this variable.

   c. Use the **Quantiles** to determine approximately how often the Nikkei declines. (Hint: What percentile is 0?)

   d. Use **Graph Builder** to make a **Line Graph** ($6^{th}$ icon in the icon bar) that shows adjusted closing prices over time. Then, use the Distribution platform to create a histogram of adjusted closing prices. Each graph summarizes the **Adj Close** variable, but each graph presents a different view of the data. Comment on the comparison of the two graphs.

   e. Now make a line graph of the monthly percentage changes over time. How would you describe the pattern in this graph?

5. *Scenario:* Anyone traveling by air understands that there is always some chance of a flight delay. In the United States, the Department of Transportation monitors the arrival and departure time of every flight. The data table **Airline Delays** contains a sample of 51,603 flights for four airlines destined for three busy airports.
    a. The variable called **DEST** is the airport for the flight destination. Describe the distribution of this variable.
    b. The variable called **Arr Delay** is the actual arrival delay, measured in minutes. A positive value indicates that the flight was late, and a negative value indicates that the flight arrived early. Describe the distribution of this variable.
    c. Notice that the distribution of **Arr Delay** is skewed. Based on your experience as a traveler, why should we have anticipated that this variable would have a skewed distribution?
    d. Use the **Quantiles** to determine approximately how often flights in this sample were delayed. (Hint: Approximately what percentile is 0?)
6. *Scenario:* For many years, it has been understood that tobacco use leads to health problems related to the heart and lungs. The **Tobacco Use** data table contains data about the prevalence of tobacco use and of certain diseases around the world.
    a. Use an appropriate technique from this chapter to summarize and describe the variation in tobacco usage (**TobaccoUse**) around the world.
    b. Use an appropriate technique from this chapter to summarize and describe the variation in cancer mortality (**CancerMort**) around the world.
    c. Use an appropriate technique from this chapter to summarize and describe the variation in cardiovascular mortality (**CVMort**) around the world.
    d. You have now examined three distributions. Comment on the similarities and differences in the shapes of these three distributions.
    e. Summarize the distribution of the region variable and comment on what you find.
    f. We have two columns containing the percentage of males and females around the world who use tobacco. Create a summary for each of these variables and explain how tobacco use among men compares to that among women.

7. *Scenario:* The **States** data table contains measures and attributes for the 50 U.S. states and the District of Columbia.
   a. The population of the state as estimated by the United States Census Bureau is in **pop2018est.** Summarize the data in this column, commenting on the center, shape, and spread of the distribution. Note any outliers.
   b. Construct box plots for **owner-occ** and **poverty.** For each plot, explain what the landmarks tell you about the distribution of each variable and comment on noteworthy features of the plot.
   c. The column **mean_Income** is the mean household income, and **med_income_17** is the median household income in the state. Use an appropriate technique from this chapter to summarize the data in these two columns and comment on what you see. Why do you think mean incomes are consistently greater than median incomes?
   d. The column **homicide** is the rate of homicide deaths per 100,000 persons in the state. Summarize the responses and comment.
   e. The column **soc_sec** is the number of people receiving Social Security benefits within the state. Use an appropriate technique to summarize the distribution of this variable. Identify the outlying states and suggest a reason for the fact that these states are outliers.
   f. Compare the distributions of **unemp2010** and **unemp2017** and comment on what you find.

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[1] There are exceptions to this general principle, as with months of the year or days of the week, for example.

# Chapter 4: Describing Two Variables at a Time

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## Overview

Some of the most interesting questions in statistical inquiries involve *covariation*: how does one variable change when another variable changes? After working through the examples in this chapter, you will know some basic approaches to *bivariate* analysis, that is, the analysis of two variables at a time.

## Two-by-Two: Bivariate Data

Chapter 3 covered techniques for summarizing the variation of a single variable: Univariate distributions. In many statistical investigations, we are interested in how two variables vary together and, in particular, how one variable varies in response to the other. For example, nutritionists might ask how consumption of carbohydrates affects weight loss or marketers might ask whether a demographic group responds positively to an advertising strategy. In these cases, it's not sufficient to look at one univariate distribution or even to look at the variation in each of two key variables separately. We need methods to describe the *covariation of bivariate* data, which is to say we need methods to summarize the ways in which two variables vary together.

The organization of this chapter is simple. We have been classifying data as categorical or continuous. If we focus on two variables in a study and conceive of one variable as a *response* to the other *factor*, there are four possible combinations to consider, shown in Table 4.1. The next three sections discuss three of the four possibilities: We might have two categorical variables, two continuous variables, a continuous response with a categorical factor, or a categorical response to a continuous factor.

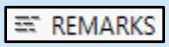
**Table 4.1: Chapter Organization—Bivariate Factor-Response Combinations**

| | Continuous Factor | Categorical Factor |
|---|---|---|
| Continuous Response | Third section to follow | Second section to follow |
| Categorical Response | See Chapter 19 | Next Section |

In this chapter, we will introduce several common methods for three ways to pair bivariate data. The first examples relate to a serious issue in civil (non-military) air travel: the periodic collisions between wildlife and commercial airplanes. According to the U.S. Federal Aviation Administration (FAA) so-called wildlife-aircraft strikes have cost hundreds of lives in the past century and account for significant financial losses as well in damage to aircraft. These collisions present environmental, public safety, and business issues for many interested parties. The FAA maintains a database to monitor the incidence of wildlife-aircraft strikes. From 2010 through April 2019, the database contains nearly 117,000 reports of strikes in North America. The state reporting the largest number of events was California.

For this chapter, we will use a subset of the database, looking only at bird strikes associated with three California airports: Los Angeles International, Sacramento (the state capital), and San Francisco International. All of the available data is in the data table called **FAA Bird Strikes CA**. This data table contains 36 columns providing attributes for each of 3,411 bird strikes at or near the three airports.

1. Open the **FAA Bird Strikes CA** data table now and scroll through the columns. Our analysis will use several columns, each of which we will explain as we work through the examples.

The REMARKS column in this data table has an icon that we have not previously seen: 

This column contains *Unstructured Text*, which is free-form character data. Specifically, remarks are the written description of the wildlife strike as provided by the person who filed the original report.

Although it is well beyond the scope of this book, interested readers should check out the **Text Explorer** platform in the **Analyze** menu or visit www.jmp.com for demonstrations and examples of text analysis.

This data table also has a large amount of *missing data.* Although the FAA attempts to record a full set of variables for each strike, sometimes the data is not known to the individual reporting the incident. In a JMP data table, missing categorical data appears as a blank cell, and missing continuous data is a dot.

In a bivariate analysis, we think in terms of a pair of observations for each bird strike. JMP will only analyze those incidents that have a complete pair of observations for whichever two columns that we select.

# Describing Covariation: Two Categorical Variables

At what point in a flight do bird strikes most often occur? Was it the same at all three airports? In our data table, we have two variables that identify the airport (the **Airport code** and **City**) and another ordinal variable identifying the **Phase of the Flight** at which the strike happened. We can use JMP to investigate the covariation of these categorical variables using a few different approaches.

1. Select **Analyze ▶ Distribution.** You should be quite familiar with this dialog box by now. Select the **AIRPORT** and **PHASE_OF_FLT** (phase of flight) columns, cast them into the **Y** role, and click **OK**.

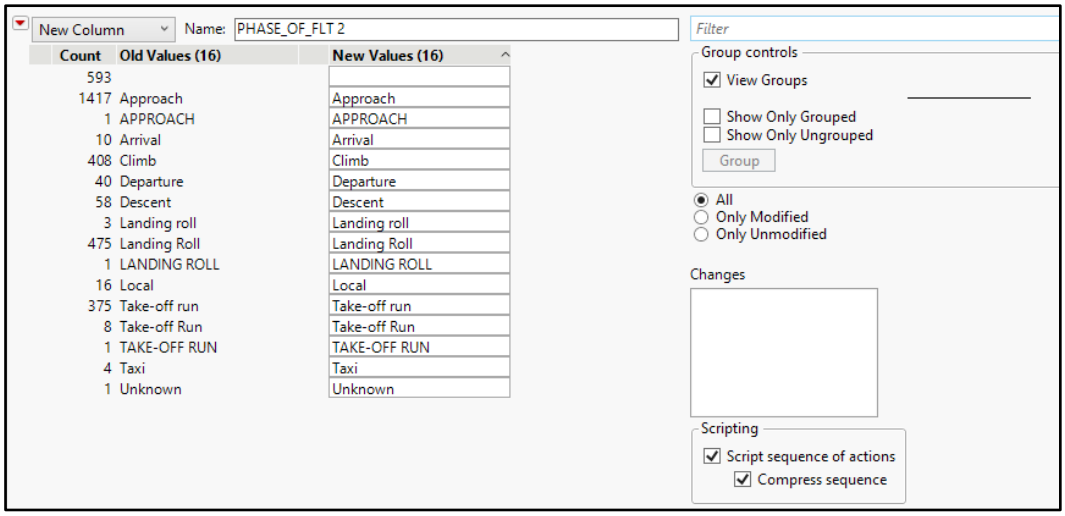
JMP produces two univariate graphs with accompanying frequency tables for each of the two columns. In the default state, they reveal no information about patterns of *co*variation, but there are a few important things to notice. Look at the phase of flight data and see that several of the levels of this column are repeated, but with different capitalization. Our analysis will be simplified if we can treat APPROACH and Approach as the same thing.

## A Digression: Recoding a Variable and Changing Value Order

When data are recorded by many individuals over a long time period, sometimes variations in spelling, punctuation, or capitalization creep into a set of data. Prior to analysis, we typically want to clean up and standardize such variation when we find it.

1. Make the data table the active tab of your project and select the column **PHASE_OF_FLT**.
2. Select **Cols ▶ Recode.** The Recode dialog box (Figure 4.1) lets us replace current values with new ones.

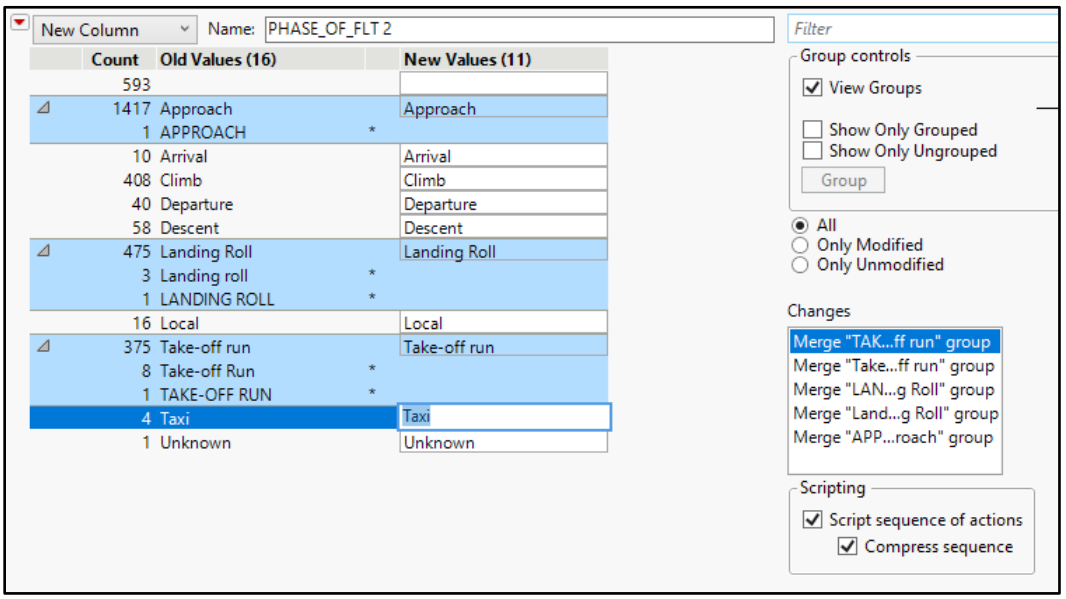
**Figure 4.1: Recode Dialog Box**



We have the option to preserve the original data and create a new column, and then individually type in text to replace values that we want to standardize. In this case, we see that 1,417 incidents happened on Approach and 1 happened on APPROACH.

3. In the **New Values (16)** column, replace **APPROACH** with **Approach**. Notice the changes in the dialog box that indicate that two old values that were previously treated as different are now grouped together.
4. Continue by recoding all the Landing Roll and Take-off run levels. When finished, your dialog box should look like Figure 4.2. Then click **Recode**.

**Figure 4.2: Recoded Values**



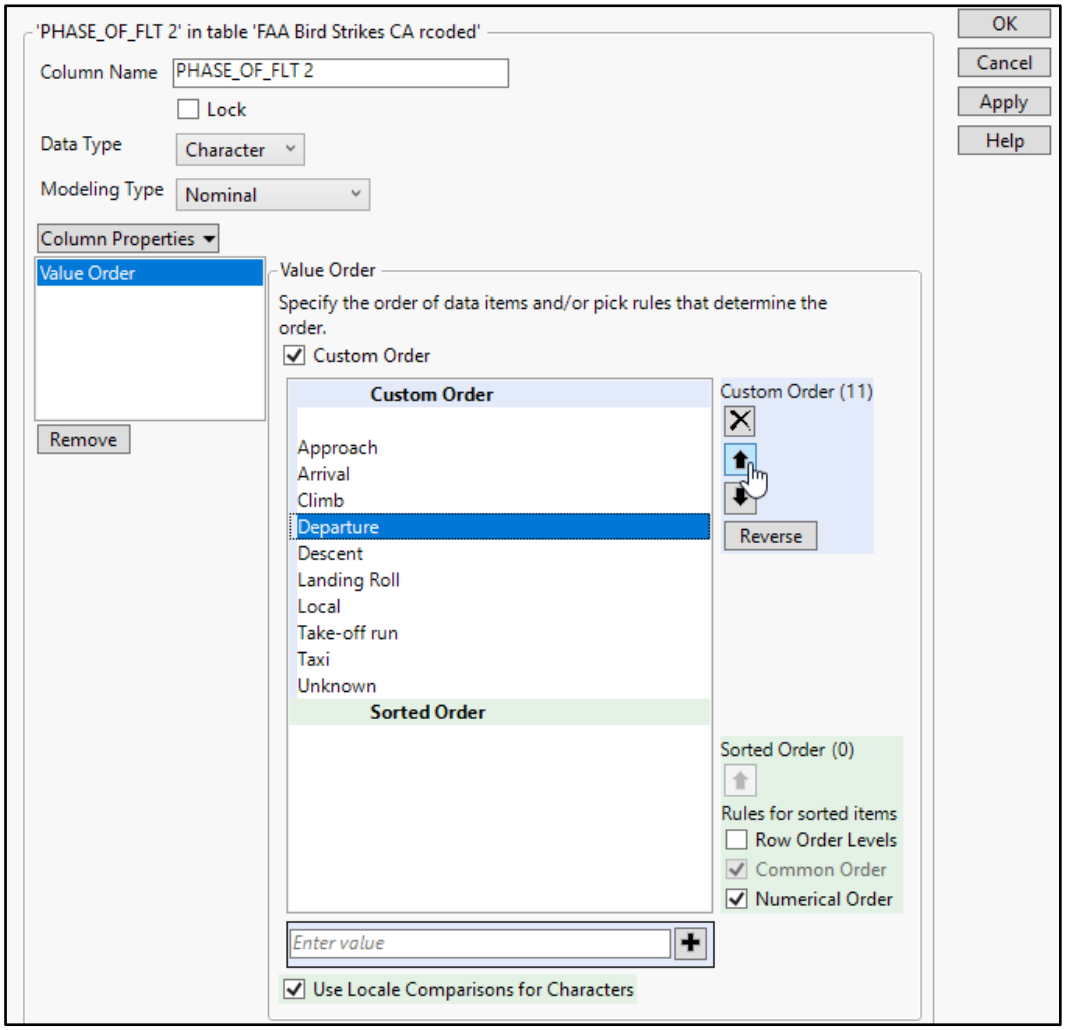

Also notice that the flight phase values are in alphabetical order. It might be more useful to change the order to correspond to the chronology of a flight, and fortunately this is easy to do. The basic chronology of the phases in this data table is as follows:

- Departure
- Taxi (the FAA distinguishes between Taxi-out and Taxi-in, but here we just have “Taxi”, which occurs 4 times)
- Take-off run
- Climb
- Descent
- Approach
- Landing Roll
- Arrival

- Local (rare; apparently refers to strikes noted at an airport)
- Unknown

5. In the **Columns** pane of the data window, right-click on the column **PHASE_OF_FLT 2** and select **Column info…**
6. Click **Column Properties** and select **Value Order**.
7. As shown in Figure 4.3, highlight **Departure** in the **Custom Order** list of values, and use the black arrows to move it to precede **Approach**. Note the blank line above **Approach**, representing missing values. We can leave that as the first item in the list.

**Figure 4.3: Defining a Custom Value Order**




8. Continue using the up and down arrows to place the value labels in the desired order. When you finish, click **OK**.

## Back to the bivariate analysis

1. Again, select **Analyze** ▶ **Distribution** and cast **AIRPORT** and **PHASE_OF_FLT 2** columns into the **Y** role, and click **OK**.

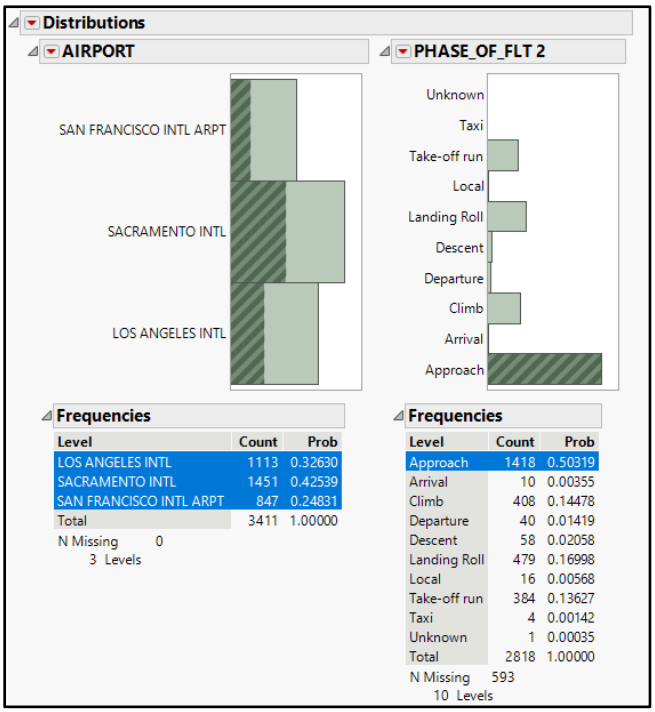
Wildlife strikes are rare during the taxi phase (in either direction between the terminal and runway) and Arrival phase. We also should note that, although the FAA database does contain strike reports while flights are en route, these three airports did not report any such strikes. Consequently, the en route phase of flight does not even appear in the right-hand panel of this graph.

In contrast, more than 50% of the strikes occurred during the approach phase of the flight. Clicking the **Approach** bar highlights the bar in the right-hand graph and also highlights all approach-related observations in the left-hand graph. If the relevant dynamics are similar at the three airports, then we would expect approximately half of each of the three bars in the left chart to be darkened. However, this is not the case. We will explore this curious pattern shortly.

2. Click the bar representing **Approach** (to the airport just prior to landing), we see something interesting, as shown in Figure 4.4.

Before going further, look at the frequency tables under each graph. Each table tallies the number of observations in each column category as well as the number of *missing observations*, and the number of distinct values (*levels*) within the column. The data table identifies the airport city for every bird strike, but only has data about the flight phase for 2,818 incidents. The other 593 are missing—they are unknown to history.

**Figure 4.4: Two Linked Univariate Distributions**



The issue of missing data is quite common in observational and survey data, though introductory courses often bypass it as a topic. In a bivariate analysis, we will need two values for each observation. If one is present, but the other is missing, then that observation will be omitted from the analysis.

It behooves the analyst to think about *why* a value is missing, and whether missing observations share a common cause. The very fact that observations are missing could be informative in its own right. For the current discussion, it is only important to realize that univariate analyses might include observations that are excluded in a related bivariate analysis.

3. We can generate a different view of the same data by using the **By** feature of the **Distribution** platform. Click the red triangle next to the word **Distributions** in the current window and select **Redo ▶ Relaunch Analysis**. This reopens the dialog box that we used earlier. The two columns still appear in the **Y, Column** box.
4. Drag the column name **Airport** from its current position in the **Y, columns** box into the **By** box and click **OK**.

The ability to re-launch and modify an analysis is a very handy feature of JMP, encouraging exploration and discovery—and allowing for easy correction of minor errors. Experiment with this feature as you continue to learn about JMP.

5. Before discussing the results, let's make a further modification to enhance readability. Click the uppermost red triangle icon in the new window, and select **Stack**. This orients the graphs horizontally and stacks all of the results vertically. Your results now look like Figure 4.4. On your screen, you need to scroll to see the results for San Francisco; in the figure below, we show only L.A. and Sacramento.

**Figure 4.5: Distribution of Phase of Flight BY City**

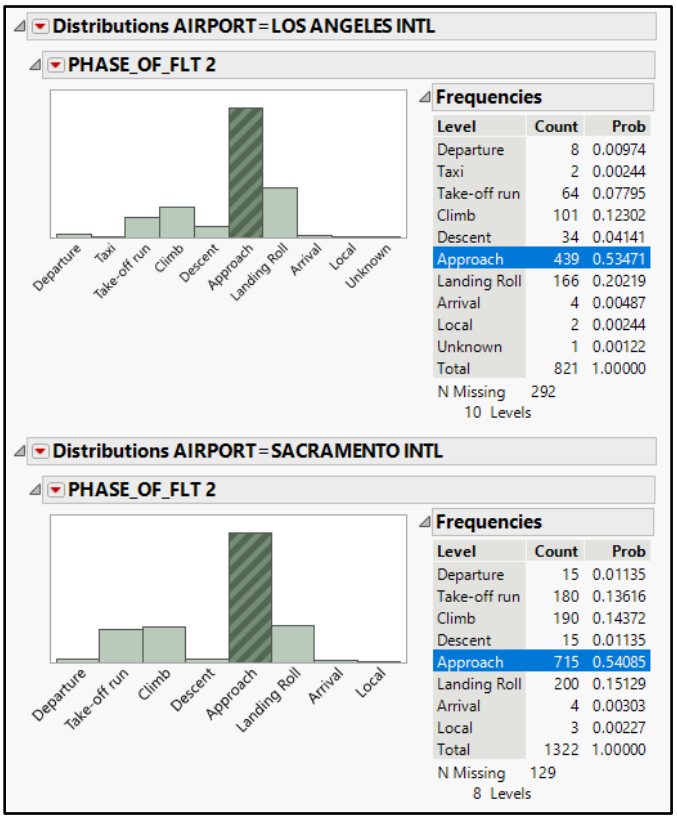



Figure 4.5 shows the results for two of the three airports. In general, the relative frequency of strikes is similar at both airports, though strikes during the Take-off run were more common in Sacramento than in Los Angeles, and just the opposite is true for strikes during the Landing Roll.

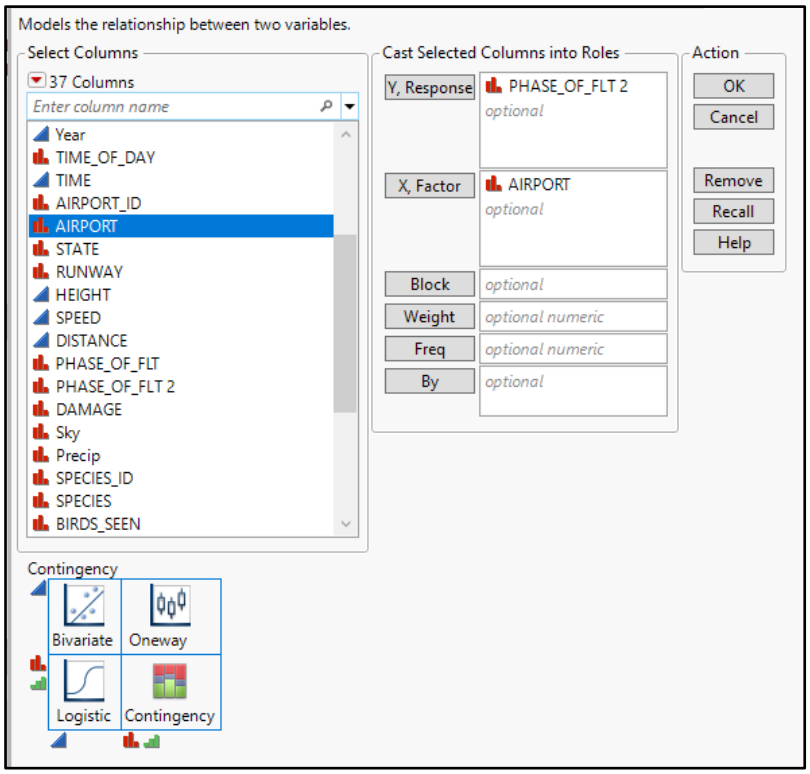
Based on Figure 4.4, we might have had the impression that none of the airports experienced half of the strikes during approach, despite the fact that just over 50% of all strikes occur during approach. We now see that for both L.A. and Sacramento, more than half of the strikes do occur during approach.

Why the difference between the two graphs? Missing observations. In L.A., we have Phase of Flight information about 821 strikes, but not for another 292 strikes. In other words, we have complete pairs of data for slightly less than three-quarters of all strikes (there were 821 + 292 = 1,113 strikes in L.A.). In contrast, the Phase of Flight was recorded for 91% of the strikes in Sacramento and 80% in San Francisco.

Another common way to display covariation in categorical variables is a *crosstabulation* (also known as a two-way table, a crosstab, a joint-frequency table, or a contingency table). JMP provides two different platforms that create crosstabs, and we will look at one of them here.

6. Select **Analyze ▶ Fit Y by X**. Select **PHASE_OF_FLT 2** as **Y, Response** and **Airport** as **X, Factor** as shown in Figure 4.6.

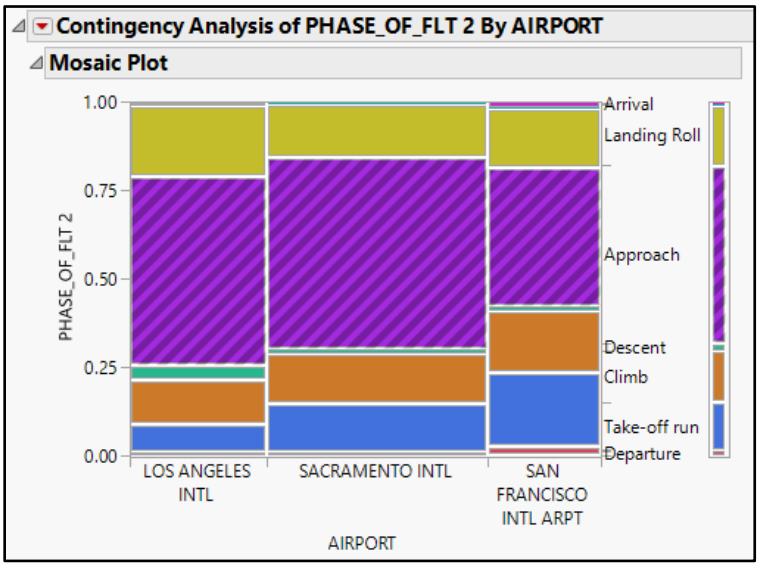
**Figure 4.6: Fit Y by X Contextual Platform**




The **Fit Y by X** platform is contextual in the sense that JMP selects an appropriate bivariate analysis depending on the modeling types of the columns that you cast as Y and X. The four possibilities are shown in the lower left of the dialog box.

Figures 4.7 and 4.8 show the results of this analysis. We find a *mosaic plot*—essentially a visual crosstab—and a contingency table. In the mosaic plot, the vertical axis represents phase of flight, and the horizontal represents airports. The width and height of the rectangles in the mosaic are determined by the number of events in each category, and colors indicate the different phases of flight.

**Figure 4.7: A Mosaic Plot of Wildlife Strikes at Three Airports**




In this graph, the width of the vertical bars makes it clear that Sacramento had the greatest number of reported strikes that included phase of flight information, and San Francisco had the fewest. Though the data table includes 10 distinct phases of flight, we see labels for only the seven most commonly appearing in the data. Finally, the height of boxes is comparatively consistent across cities, indicating that strikes tend to occur during the same phases of flight at the three airports, with some minor variations.

The plot provides a clear visual impression, but if we want to dig into the specific numerical differences across regions, we turn to the **Contingency Table** (Figure 4.8). You might have heard tables like these called crosstabs or crosstabulations, or perhaps joint-frequency tables or two-way tables. All of these terms are synonymous.

**Figure 4.8: A Crosstabulation of the Bird Strike Data**

**Contingency Table**

| AIRPORT | Count / Total % / Col % / Row % | Departure | Taxi | Take-off run | Climb | Descent | Approach | Landing Roll | Arrival | Local | Unknown | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | PHASE_OF_FLT 2 | | | | | | | | | | |
| LOS ANGELES INTL | | 8 | 2 | 64 | 101 | 34 | 439 | 166 | 4 | 2 | 1 | 821 |
| | | 0.28 | 0.07 | 2.27 | 3.58 | 1.21 | 15.58 | 5.89 | 0.14 | 0.07 | 0.04 | 29.13 |
| | | 20.00 | 50.00 | 16.67 | 24.75 | 58.62 | 30.96 | 34.66 | 40.00 | 12.50 | 100.00 | |
| | | 0.97 | 0.24 | 7.80 | 12.30 | 4.14 | 53.47 | 20.22 | 0.49 | 0.24 | 0.12 | |
| SACRAMENTO INTL | | 15 | 0 | 180 | 190 | 15 | 715 | 200 | 4 | 3 | 0 | 1322 |
| | | 0.53 | 0.00 | 6.39 | 6.74 | 0.53 | 25.37 | 7.10 | 0.14 | 0.11 | 0.00 | 46.91 |
| | | 37.50 | 0.00 | 46.88 | 46.57 | 25.86 | 50.42 | 41.75 | 40.00 | 18.75 | 0.00 | |
| | | 1.13 | 0.00 | 13.62 | 14.37 | 1.13 | 54.08 | 15.13 | 0.30 | 0.23 | 0.00 | |
| SAN FRANCISCO INTL ARPT | | 17 | 2 | 140 | 117 | 9 | 26 | 113 | 2 | 11 | 0 | 675 |
| | | 0.60 | 0.07 | 4.97 | 4.15 | 0.32 | 9.3 | 4.01 | 0.07 | 0.39 | 0.00 | 23.95 |
| | | 42.50 | 50.00 | 36.46 | 28.68 | 15.52 | 18.6 | 23.59 | 20.00 | 68.75 | 0.00 | |
| | | 2.52 | 0.30 | 20.74 | 17.33 | 1.33 | 39.1 | 16.74 | 0.30 | 1.63 | 0.00 | |
| Total | | 40 | 4 | 384 | 408 | 58 | 1416 | 479 | 10 | 16 | 1 | 2818 |
| | | 1.42 | 0.14 | 13.63 | 14.48 | 2.06 | 50.32 | 17.00 | 0.35 | 0.57 | 0.04 | |

Across the top of this contingency table, we find the values of the **PHASE_OF_FLT 2** column, and the rows correspond to the three airports. Each cell of the table contains four numbers representing the number of countries classified within the cell, as well as percentages computed for the full table, for the column, and for the row.

For example, look at the highlighted cell of the table in Figure 4.8. The numbers and their meanings are as follows:

| | |
|---|---|
| 113 | 113 reported incidents were strikes during the landing roll at San Francisco (SFO). |
| 4.01 | 4.01% of all 2,818 reported incidents fall into this cell (landing roll at SFO). |
| 23.59 | 23.59% of the 479 landing roll events occurred at SFO. |
| 16.74 | 16.74% of the 675 events at SFO occurred during landing roll. |

## Describing Covariation: One Continuous, One Categorical Variable

We have just been treating phase of flight as a response variable, inquiring whether the prevalence of strikes during a particular phase varies depending on which airport was the site of the incident. With only minor differences, we found that strikes occur most often during the approach phase of the flight. We might wonder what it is about the approach phase that might account for the relatively high proportion of bird strikes—perhaps the altitude? The speed? With the data available to us here, we can begin to answer such questions.

To illustrate, let's examine the distribution of flight speeds during the different flight phases. Consider that the observational units in our data table are incidents involving wildlife strikes. We are not observing all flights, and we aren't observing incident-free flights (which constitute the vast majority of air travel). Hence, we can only describe aspects of those flights that did strike birds.

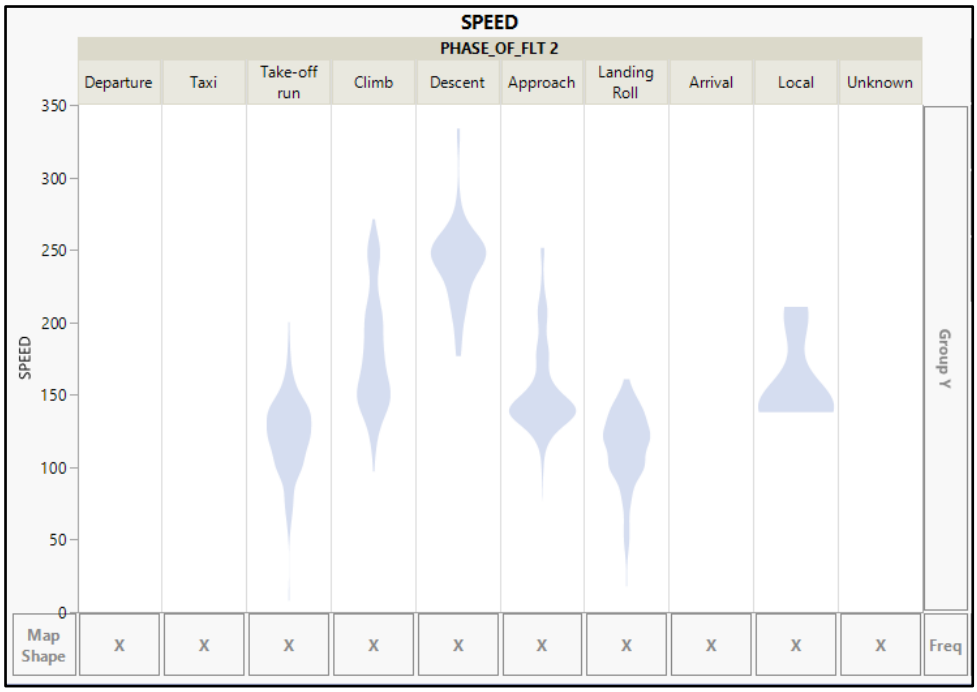
> We have mentioned missing data before. This is another form of missing data, and it highlights another good habit of statistical thinking. For any set of data, it is wise to ask "what is *not* here?" The FAA Wildlife Strike database, by its nature, only contains data from flights that did have a bird strike. These flights are extraordinary in the context of all flights in the U.S.

1. Clear all row states (**Rows** menu).
2. Open the **Graph Builder** (**Graph** menu). Drag **SPEED** to the **Y** drop zone, and **PHASE_OF_FLT 2** to the **Group X** drop zone at the top.

What do you see in your graph? As we might expect, the columns of jittered points tend to rise and fall with the familiar phases of flight. You might also notice that each of the speed distributions is asymmetric; there tend to be concentrations of many points at a relatively high or low speed. To visualize this more distinctly, we could make box plots or histograms of Speed by Phase of Flight. Instead, let's learn a new type of graph:

3. In the menu bar at the top of Graph Builder, click the **Contour** icon (  ). This will create a *violin plot*, so called because some of the resulting shapes (see Figure 4.9) look a bit like violins.

**Figure 4.9: Violin Plot of Aircraft Speeds by Phase of Flight**




A violin plot shows the range of values for a variable; for example, those strikes that occurred during take-off run had recorded speeds between 50 and 200 mph. The narrow portion of the violin indicates very few takeoff strikes below 75 mph. The bulges in the violin indicate higher frequencies. It appears that strikes during takeoff are relatively common in the vicinity of 120 to 140 mph. Looking across all phases, one might say that except during descent, strikes are relatively frequent at speeds between approximately 120 and 160 mph regardless of flight phase.

For a more refined set of numerical summaries, do this:

4. Select **Analyze ▶ Fit Y by X**. Select **SPEED** as the **Y** column and **PHASE_OF_FLT 2** as **X** and click **OK**.
5. For a numerical comparison, let's look at some quantiles. In the **Oneway Analysis** report, click the red triangle and select **Quantiles.**
6. Below the graph should look like Figure 4.10.

**Figure 4.10: Quantiles for Speed by Phase of Flight**

PHASE_OF_FLT 2

**Quantiles**

| Level | Minimum | 10% | 25% | Median | 75% | 90% | Maximum |
|---|---|---|---|---|---|---|---|
| Taxi | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| Take-off run | 10 | 91 | 100 | 120 | 140 | 150 | 200 |
| Climb | 100 | 140 | 150 | 170 | 200 | 250 | 270 |
| Descent | 180 | 210 | 230 | 250 | 250 | 264 | 332 |
| Approach | 80 | 130 | 135 | 140 | 160 | 200 | 250 |
| Landing Roll | 20 | 80 | 100 | 120 | 130 | 140 | 160 |
| Local | 140 | 140 | 140 | 142.5 | 186.25 | 210 | 210 |

Missing Rows 1999

The table reports seven quantiles for each flight phase. We readily see strong similarities between speeds at the time of strikes for take-off run and landing roll, phases when the aircraft is on the ground. Quantiles during climb and approach are also comparable, but speeds during descent are uniformly higher than other phases.

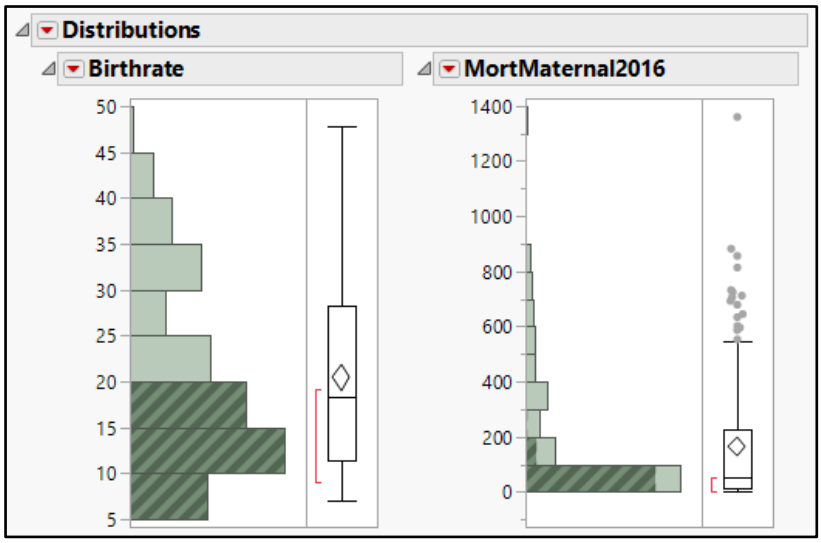
## Describing Covariation: Two Continuous Variables

To illustrate the standard bivariate methods for continuous data, we will now shift to a different set of data. In earlier chapters, we looked at variation in life expectancy around the world. We'll now look at data related to variation in birth rates and the risks of childbirth in different nations as of 2017. We will rely on a data table with five continuous columns. Two of the columns measure the relative frequency of births in each country, and the other three measure risks to mothers and babies around birth. Initially, we will look at two of the variables: the columns labeled **BirthRate** and **MortMaternal2016**. A country's annual birth rate is defined as the number of live births per 1,000 people in the country. The maternal mortality figure is the average number of mothers who die as a result of childbirth, per 100,000 births. At the time of this writing, the most current birth rate data was for 2017, but for maternal mortality, it was 2016.

As we did in the previous sections, let's start by simply looking at the univariate summaries of these two variables.

1. Open the **Birthrate 2017** data table.
2. Select **Analyze ▶ Distribution**. Cast **BirthRate** and **MortMaternal** in the **Y** role and click **OK**.
3. Within your **BirthRate** histogram, select different bars or groups of bars, and notice which bars are selected in the maternal mortality histogram. The results should look much like Figure 4.11.

**Figure 4.11: Linked Histograms of Two Continuous Distributions**

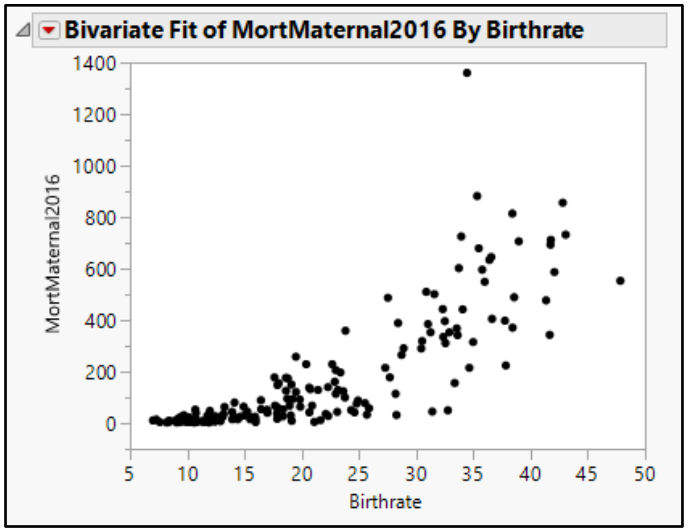



The general tendency is that countries with low birth rates also have low maternal mortality rates, but as one rate increases so does the other. We can see this tendency more directly in a *scatterplot*, an X-Y graph showing ordered pairs of values.

4. In the data table, press the Esc key to clear the de-select rows that you selected by clicking on histogram bars. Then choose **Analyze ▶ Fit Y by X**. Cast **MortMaternal16** as **Y** and **BirthRate** as **X** and click **OK**.

Your results will look like those shown in Figure 4.12.

**Figure 4.12: A Scatterplot**



By now you also have had enough experience with **Graph Builder** to know that you can easily create a similar graph with that tool. You should feel free to explore data with **Graph Builder**.

This graph provides more information about the ways in which these two variables tend to vary together. First, it is very clear that these two variables do indeed vary together in a general pattern that curves upward from left to right; maternal mortality increases at an accelerating rate as the birth rate increases, though there are some countries that depart from the pattern. Many countries are concentrated in the lower left, with low birth rates and relatively low maternal mortality.

As we will learn in Chapters 15 through 19, there are powerful techniques available for building models of patterns like the one visible in Figure 4.12. At this early point in your studies, the curvature of the pattern presents some unwelcome complications. Figure 4.13 shows another pair of variables whose relationship is more linear. We will investigate this relationship and meet three common numerical summaries of such bivariate covariation.

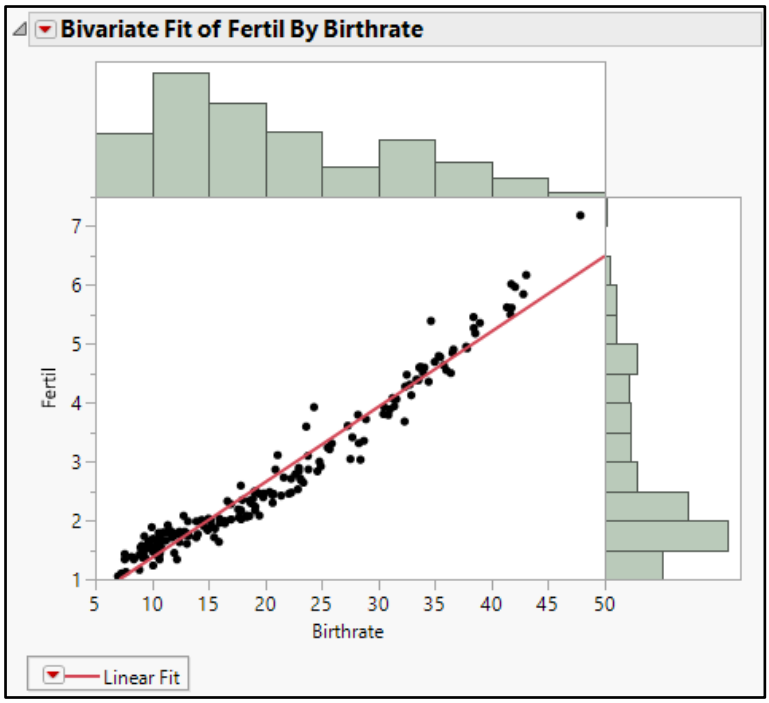
5. Click the red triangle next to the word **Bivariate** in the current window and select **Redo ▶ Relaunch Analysis**.
6. Remove **MortMaternal2016** from the **Y, Response** role and replace it with **Fertil**. This will produce a scatterplot (seen in modified fashion in Figure 4.13).

Earlier, we noted that birth rate counts the number of live births per 1,000 people in a country. Another measure of the frequency of births is the *fertility rate*, which is the mean number of children that would be born to a woman during her lifetime in each country.

When we look at this relationship in a scatterplot, we see that the points fall in a distinctive upward sloping pattern that is generally straight. We can also calculate three different numerical summaries to characterize the relationship. Each of the statistical measures compares the pattern of points to a perfectly straight line. The first summary is the equation of a straight line that follows the general trend of the points. (See Chapter 15 for a full explanation.) The second summary is a measure of the extent to which variation in X is associated with variation in Y, and the third summary measures the strength of the linear association between X and Y.

7. In the scatterplot of **Fertility** versus **Birthrate**, click the red triangle next to **Bivariate Fit** and select **Fit Line**.
8. Then click the red triangle again and select **Histogram Borders**.

**Figure 4.13: Scatterplot with Histogram Borders and Line of Best Fit**




Now your results look like Figure 4.13. The consequence of these two customizations is that along the outside borders of the bivariate scatterplot, we see the univariate distributions of each of our two columns. Additionally, we see a red *fitted line* that approximates the upward pattern of the points.

Below the graph, we find the equation of that line:

**Fertil = 0.0845109 + 0.1282129*BirthRate**

The slope of this line describes how these two variables co-vary. If we imagine two groups of countries whose birth rates differ by one birth per 1,000 people, the group with the higher birth rate would average 0.128 more births per woman.

9. Below the linear fit equation, you will find the **Summary of Fit** table (not shown here). Locate the first statistic called **Rsquare**.

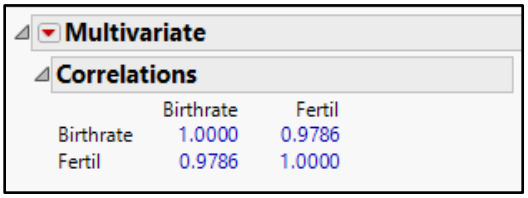
*R square* ($r^2$) is a goodness-of-fit measure; for now, think of it as the proportion of variation in Y that is associated with X. If fertility were perfectly and completely determined as a linear function of birth rate, then R square would equal 1.00. In this instance, R square equals 0.958; approximately 96% of the cross-country variability in fertility rates is accounted for by differences in birth rates.

A third commonly used summary for bivariate continuous data is called *correlation*, which measures the strength of the linear association between the two variables. The coefficient of correlation, symbolized by the letter *r*, is the square root of $r^2$ (if the slope of the best-fit line is

negative, then $r$ is $-\sqrt{r^2}$). As such, $r$ always lies within the interval [–1, +1]. Values near the ends of the interval indicate strong correlations, and values near zero are weak correlations.

10. Select **Analyze** ▶ **Multivariate Methods** ▶ **Multivariate**. Cast both columns (**BirthRate** and **Fertil**) as **Y** and click **OK**. The upper part of your results will look like Figure 4.14.

**Figure 4.14: A Correlation Matrix**




We find the correlation between birth rate and fertility rate to equal 0.9786, which is a very strong correlation. JMP uses a color scheme when reporting correlations. Strong positive values are in bright blue, and strong negative correlations are in bright red. Weaker values take on paler hues of blue and red, passing through gray near zero.

## Scatter Plots for Very Large Data Tables

As the number of rows in a data table grows, we might encounter a common graphing problem called *overplotting*. When we attempt to plot many points in a fixed area, the points can become so densely packed that they overlap. At times, we may have multiple instances of the same X-Y pair, so that the graph gives the misimpression that there is a single point when there really might be several.

It is beyond the scope of this chapter to provide a full treatment of graphing large data sets, but this is an apt moment to look at a few strategies.

We have already encountered jittering in the univariate context. Jittering spreads points slightly to reveal when multiple points are at a single location. When the number of rows grows very large, jittering might not be enough.

To illustrate, we will consider the data in **NHANES 2016**, which contains detailed health information about nearly 10,000 people. We will use this data table extensively in our study of regression analysis. We first saw this data in Chapter 2. As part of the National Health and Nutrition Examination Survey, researchers measured the blood pressure of subjects multiple times. We will examine the covariation of systolic blood pressure (the “top” number) between the first and second readings for each person.

1. Open the data table **NHANES 2016**.
2. In Graph Builder, drag **BPXSY1** to the **X** drop zone and **BPXSY2** to the **Y** drop zone. This will treat the second reading as a function of the first reading.

Note that Graph Builder initially displays a point cloud and a smoother. It is obvious that we have many points that are densely packed in an elongated elliptical pattern. Try suppressing the display of points by clicking the left-most button in the menu bar. When there are so many points, one option for clarity is to simply show their trend without plotting every point. As we did

earlier to create Figure 4.9, choosing the Contour option in the menu bar allows us to see the linear trend as well as regions that are most dense. The darkest contours here show the most common systolic pressure readings.

What if we want to see all 10,000 points more distinctly? Here are three options, among several others.

1. Click the left-most menu button to display all the points.
2. On the right side of the display, right-click over the legend ● **BPXSY2**
3. From the menu, select **Marker** to open a menu of marker shapes. Select an open circle or an X rather than a solid dot. After experimenting with a few shapes, return to the solid dot.

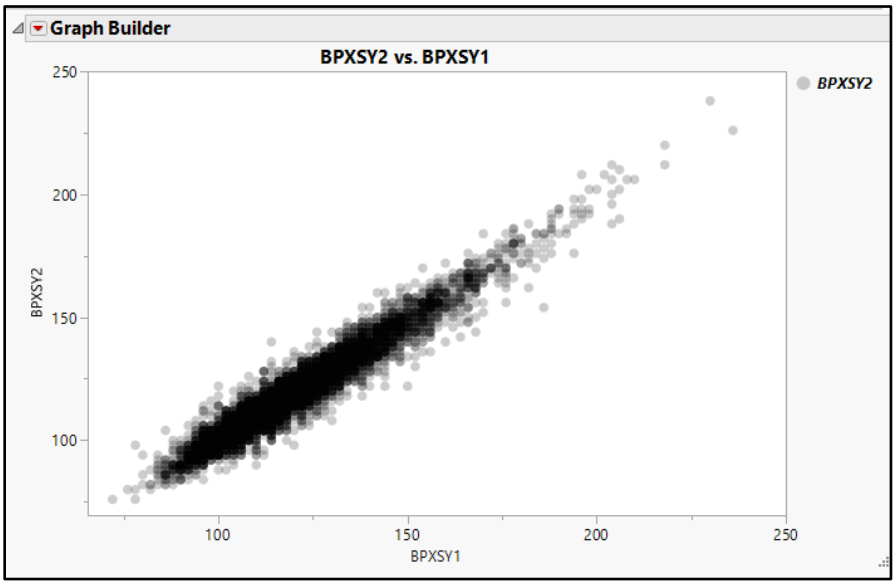
Depending on the density of the points, sometimes an open marker is preferable since it can show small spaces between nearby points, rather than having one occlude the other. In some graphs, reducing the size of each marker can achieve the same goal.

4. Again, right-click over the legend ● **BPXSY2** and choose **Marker Size** to see the choices. The Preferred Size in this graph happens to be **3, Large**. Try making them larger and smaller, noting how larger points tend to obscure one another in some parts of the graph. For the sake of the next step, leave the markers at size **5, XXL**.

Similarly, rendering points as translucent helps with the overplotting problem. If we make the points partially transparent, then locations with multiple points will just appear darker in the graph. This effectively communicates which graph locations are most prevalent.

5. Right-click over the legend again and choose **Transparency.** By default, points are opaque with a transparency parameter of 1. Try setting transparency to 0.2, 0.5, and 0.8. Figure 4.15 shows this scatterplot with a transparency parameter of 0.2.

**Figure 4.15: Using Transparency to Reveal Overplotting**




## More Informative Scatter Plots

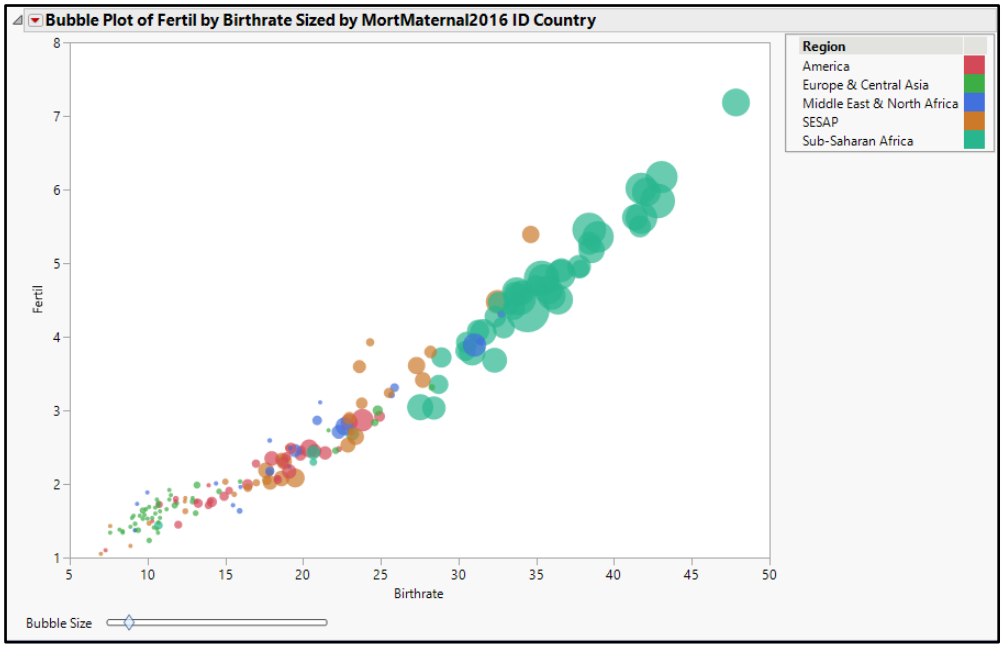
This chapter has introduced ways to depict and summarize bivariate data, but sometimes we want or need to incorporate more than two variables into an analysis. We already know that with **Graph Builder,** we can color points to represent another variable, and we can wrap or overlay additional variables in a graph. We can use a **By** column in a **Fit Y by X** analysis to add a third factor into an investigation.

Another very useful visualization tool called a bubble plot provides a way to visualize as many as seven columns in one graph. This section provides a brief first look at bubble plots in JMP. We'll use the tool further in Chapter 5.

Think of a bubble plot as a "scatterplot plus… ." In addition to the X-Y axes, the size and color of points can represent variables. Other variables can be used to interactively label points, and in cases where we have repeated measurements over time, the entire graph can be animated through time. Try this first exercise with the 2017 birth rate data:

1. **Graph ▶ Bubble Plot**. Cast **Fertil** as **Y** and **Birthrate** as **X**, just as before.
2. Next, cast **Country** as **ID**, **Region** as **Coloring**, and **MortMaternal2016** as **Sizes**. Click **OK**.
3. In the lower left of the **Bubble Plot** window, find the slider control next to **Bubble Size** and slide it to the left until your graph looks like Figure 4.16.

**Figure 4.16: Enhancing a Scatterplot by Using a Bubble Plot**




Compare this graph to Figure 4.13. The Y and X variables are the same, but this picture also conveys further descriptive generalizations; the highest birth and fertility rates are in sub-Saharan Africa, where maternal mortality rates are also the highest in the world. As you move your cursor around the graph, you will see hover labels identifying the points. You can also notice that countries with similar birth and fertility rates might have quite different maternal mortality rates (bubble sizes). Bubble plots pack a great deal of information into a compact, data-rich display.

# Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* We will continue to examine the World Development Indicators data in **BirthRate 2017**. We will broaden our analysis to work with other variables:
   - **Provider:** Source of maternity leave benefits (public, private, or a combination of both)
   - **Fertil:** Average number of births per woman during child-bearing years
   - **MortUnder5:** Deaths of children under 5 years per 1,000 live births
   - **MortInfant:** Deaths of infants per 1,000 live births
   a. Create a mosaic plot and contingency table for the **Provider** and **Region** columns. Report on what you find.
   b. Use appropriate methods to investigate how fertility rates vary across regions of the world. Report on what you find.
   c. Create a scatterplot for **MortUnder5** and **MortInfant**. Report the equation of the fitted line and the R-square value, and explain what you have found.
   d. Is there any noteworthy pattern in the covariation of **Provider** and **MatLeave90+**? Explain what techniques you used, and what you found.
2. *Scenario:* How do prices of used cars differ, if at all, in different areas of the United States? How do the prices of used cars vary according to the mileage of the cars? Our data table **Used Cars** contains observational data about the listed prices of three popular compact car models in three different metropolitan areas in the U.S. The cities are Phoenix, AZ; Portland, OR; and Raleigh-Durham-Chapel Hill, NC. The car models are the Chrysler PT Cruiser Touring Edition, the Honda Civic EX, and the Toyota Corolla LE. The cars were all two years old at the time.
   a. Create a scatterplot of price versus mileage. Report the equation of the fitted line, the R-square value, and the correlation coefficient, and explain what you have found.
   b. Use the **Graph Builder** to see whether the relationship between price and mileage differs across different car models.
   c. Describe the distribution of prices across the three cities in this sample.
   d. Within this sample, are the different car models equally favored in the three different metropolitan areas? Discuss your analysis and explain what you have found.

3. *Scenario:* High blood pressure continues to be a leading health problem in the U.S. We have a data table (**NHANES 2016**) containing survey data from nearly 10,000 people in the U.S. in 2017. For this analysis, we will focus on only the following variables:
   - **RIAGENDR:** Respondent's gender
   - **RIDAGEYR:** Respondent's age in years
   - **RIDRETH1:** Respondent's racial or ethnic background
   - **BMXWT:** Respondent's weight in kilograms
   - **BPXPLS:** Respondent's resting pulse rate
   - **BPXSY1:** Respondent's systolic blood pressure ("top" number in BP)
   - **BPXD1:** Respondent's diastolic blood pressure ("bottom" number in BP)

   a. Create a scatterplot of systolic blood pressure versus age. Within this sample, what tends to happen to blood pressure as people age?
   b. Compute and report the correlation between systolic and diastolic blood pressure. What does this correlation tell you?
   c. Use either a bubble plot to incorporate gender (Color) and pulse rate (Bubble Size) into the graph. Comment on what you see.
   d. Compare the distribution of systolic blood pressure in males and females. Report on what you find.
   e. Compare the distribution of systolic blood pressure by racial/ethnic background. Comment on any noteworthy differences that you find.
   f. Create a scatterplot of systolic blood pressure and pulse rate. One might suspect that higher pulse rate is associated with higher blood pressure. Does the analysis bear out this suspicion?

4. *Scenario:* Despite well-documented health risks, tobacco is used widely throughout the world. The **Tobacco** data table provides information about the several variables for 133 different nations in 2005, including these:
   - **TobaccoUse:** Prevalence of tobacco use (%) among adults 18 and older (both sexes)
   - **Female:** Prevalence of tobacco use among females, 18 and older
   - **Male:** Prevalence of tobacco use among males, 18 and older
   - **CVMort:** Age-standardized mortality rate for cardiovascular diseases (per 100,000 population in 2002)
   - **CancerMort:** Age-standardized mortality rate for cancers (per 100,000 population in 2002)

a. Compare the prevalence of tobacco use across the regions of the world, and comment on what you see.
b. Create a scatterplot of cardiovascular mortality versus prevalence of tobacco use (both sexes). Within this sample, describe the relationship, if any, between these two variables.
c. Create a scatterplot of cancer mortality versus prevalence of tobacco use (both sexes). Within this sample, describe the relationship, if any, between these two variables.
d. Compute and report the correlation between male and female tobacco use. What does this correlation tell you?
e. Create a bubble plot to modify your scatterplot from item c above to augment the display to incorporate region (color) and cardiovascular mortality (bubble size). Comment on what you find in the graph.

5. *Scenario:* Since 2003, the U.S. Bureau of Labor Statistics has been conducting the biennial American Time Use Survey. Census workers use telephone interviews and a complex sampling design to measure the amount of time people devote to various ordinary activities. We have some of the survey data in the data table called **TimeUse**. Our data table contains observations from more than 43,191 different respondents in 2003, 2007, and 2017. For these questions, use the Data Filter to select and include just the 2017 responses.
   a. Create a crosstabulation of employment status by sex, and report on what you find.
   b. Create a crosstabulation of full versus part-time employment status by gender, and report on what you find.
   c. Compare the distribution of time spent sleeping across the employment categories. Report on what you find.
   d. Now change the data filter to include all rows. Compare the distribution of time spent on personal email in 2003, 2007, and 2017. Comment on your findings.
6. *Scenario:* The data table **Sleeping Animals** contains information about the sizes, life spans, and sleeping habits of various mammal species. The last few columns are ordinal variables classifying the animals according to their comparative risks of being victimized by predators and the degree to which they sleep in the open rather than in enclosed spaces.
   a. Create a crosstabulation of predation index by exposure index, and report on what you find.
   b. Compare the distribution of hours of sleep across values of the danger index. Report on what you find.
   c. Create a scatterplot of total sleep time and life span for these animals. What does the graph tell you?
   d. Compute the correlation between total sleep time and life span for these animals. What does the correlation tell you?

7. *Scenario:* Let's return to the data table **FAA Bird Strikes CA**. The FAA includes categorical variables pertaining to the number of birds struck, the size of the birds struck, and the general weather conditions.
   a. Create a crosstabulation of number of birds struck versus sky conditions (**Sky**), and report on what you find.
   b. Create a crosstabulation of number of birds struck versus the precipitation conditions (**Precip**), and report on what you find.
   c. Investigate the relationship between the number of birds struck and the speed of the aircraft. Write a sentence to describe that relationship.
   d. Investigate the relationship between the number of birds struck and the height of the aircraft. Write a sentence to describe that relationship.
8. *Scenario:* Every ten years, the United States conducts a census of the population, gathering considerable data about the nation and its residents. The data table called **USA Counties** contains demographic, economic, commercial, educational, and other data about each of the 3,143 counties in the United States as of 2010.
   a. Create a scatterplot of median household income (Y) versus percent of the population with a bachelor's degree. Comment on what you see.
   b. Compute and report the line of best fit for these data. Use that line to estimate the median household income in counties with 25% of the population holding bachelor's degrees.
   c. Create and report on a scatterplot between the percentage of households where a foreign language is spoken in the home (**foreign_spoken_at_ home**) and the percentage of households with a foreign-born member (**foreign_born**). How do you explain the distinctive pattern in the graph?
   d. Compute and explain the correlation coefficient for the two variables in item c above.
   e. Estimate the line of best fit using the population as determined by the 2010 US Census as Y and the 2000 population count as X. Think about the slope of this line. What does it tell us about what happened to the average of US counties' populations between 2000 and 2010?
   f. The point representing Cook County, Illinois, is distinctive in that it lies below the red estimated line (2000 population was 5,194,675). According to this fitted line, what was unusual about Cook County in comparison to other counties of the United States?

# Chapter 5: Review of Descriptive Statistics

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## Overview

The prior four chapters introduced several foundational concepts of data analysis and have also led you through a series of illustrative analyses. In this chapter, we pause to pull together what we have learned about descriptive analysis. This is the first in a series of short review chapters, each of which shares the common goal of recapitulating recent material and calling upon you, the reader, to apply the principles, concepts, and techniques that you have recently studied.

In the review chapters, you will find fewer step-by-step instructions. Instead, there are guiding questions to remind you of the analytical process that you'll want to follow. Refer to earlier chapters if you have forgotten how to perform a task. In this and all future chapters, do your computer work within a **JMP Project**. The examples in this and later review chapters are based on the World Development Indicators, collected and published by the World Bank.

## The World Development Indicators[1]

The World Bank was established in 1944 to assist with the redevelopment of countries after World War II. It has evolved into a group of five institutions concerned with interconnected missions of economic development. One important goal of its work is the alleviation of poverty worldwide, and as part of that mandate, the World Bank annually publishes the World Development Indicators (WDI) gathered from 215 nations. Earlier, we looked at some of the WDI data about birth rates and life expectancy.

In the public sector as well as in business, policy makers rely on accurate, current data to gauge progress and to evaluate the impact of policy decisions. The WDI data informs policy-making by

many agencies globally, and the World Bank's annual data collection and reporting play an important role in the U.N.'s Millennium Development Goals.

### Sustainable Development Goals

At the start of the millennium, the United Nations sponsored a Millennium Summit, which led to the adoption of the Millennium Declaration by 189 member states. The declaration laid out an ambitious set of goals including commitments to "combat poverty, hunger, disease, illiteracy, environmental degradation, and discrimination against women. The MDGs are derived from this Declaration, and all have specific targets and indicators." [2]

The Millennium Development Goals include "8 goals, 21 targets, and 60 indicators for measuring progress between 1990 and 2015, when the goals are expected to be met."[3] More recently, the MDG's have evolved into Sustainable Development Goals (SDGs). Some of these indicators are identical to the World Bank's WDIs.

The eight goals, listed below, are:

- Eradicate extreme poverty and hunger
- Achieve universal primary education
- Promote gender equality and empower women
- Reduce child mortality
- Improve maternal health
- Combat HIV/AIDS, malaria, and other diseases
- Ensure environmental sustainability
- Develop a Global Partnership for Development

We return to our introductory example from Chapter 1 about variation in life expectancies worldwide, and further investigate variables that might give us insight into why people in some regions tend to live long than in others. We will initially analyze some of the WDI measures from the year 2016 to explore and to understand their variability, as well as plausible associations between variables. In doing so, we will engage in an extended exercise in statistical reasoning and review several of the descriptive techniques that were presented in the first four chapters.

## Questions for Analysis

In Chapter 1, we speculated about possible factors that might contribute to variation in life expectancies. At that time, we wondered about the impacts of education, health care, basic sanitation, nutrition, political stability, and wealth. Relying on several of the WDIs in our data, we will focus on four *constructs* as measured by the variables listed in Table 5.1. Notice that several of these variables are indirect measures of the constructs; statisticians sometimes refer to such variables as *proxies* to indicate that they "stand in" for a difficult-to-measure concept or attribute. Many of the WDIs are reported for all or nearly all the nations, while others are reported more

sparsely. For this chapter, we will use indicators that are reported for the large majority of the 215 countries. We will also examine life expectancies in different regions of the world.

Note that **WDI** contains data through 2018 for just a few of the 42 columns. In this analysis, we will focus on 2016 because we have data for the variables of interest for most nations.

**Table 5.1: Ten Variables[4] Used in This Chapter**

| Construct | Variable |
|---|---|
| Wealth | Gross Domestic Product per capita<br>Poverty headcount ratio at national poverty lines (% of population)<br>World Bank Income Group |
| Health Care | Health expenditures per capita<br>Percent of children ages 12–23 months receiving DPT immunization |
| Education | Percent of eligible children enrolled in primary education<br>Ratio of female to male students enrolled secondary education (if genders are treated equitably, the ratio = 1)<br>Percent of eligible children enrolled in tertiary (post-high school) education |
| Nutrition | Prevalence of undernourishment (% of population)<br>Food production index: covers food crops that are considered edible and that contain nutrients. |

*Our research questions in this chapter are:*

- *How did life expectancy vary around the world in 2016?*
- *How did each of the variables listed in the table above vary in 2016?*
- *How do we best describe the co-variation between each factor and life expectancy?*

## Applying an Analytic Framework

In any statistical analysis, it is essential to be clear-minded about the research questions and about the nature of the data that we will be analyzing. We have mentioned these before, and this is a good time to review them in the context of a larger research exercise.

## Data Source and Structure

We know these indicators are published by the World Bank for the purpose of monitoring economic development, identifying challenges, and assessing policy interventions. The figures are determined by analysts at the World Bank using estimates gathered and provided by governmental agencies from each country. Each of the annual indicators is best described as **observational** data, as opposed to experimental or survey.

In this data table, we have 42 variables, or data series, observed for 215 countries for each year from 1990 through 2018. Hence, we have 29 years of repeated observations for 215 countries, giving us 6,235 rows. Many cells of the table are empty, reflecting difference in national statistical infrastructure or, in some cases, simply reflecting periods before the World Bank began monitoring a development indicator. We have relatively few observations for 2018.

Recall the difference between cross-sectional data (many observations selected at one time from a population or process), and time series or longitudinal data (regularly repeated observations of a single observational unit). In this table, we have a combination: we have 29 sets of repeated cross-sectional samples.

## Observational Units

Each row in the data table represents a country in a particular year. We will be dealing with aggregated totals and rates rather than with individual people.

## Variable Definitions and Data Types

Before we dive into analysis, it is critical to understand what each variable represents, as well as the type of each variable. We need to understand the measurements in order to interpret them. We need to know the data types because that guides the choice of summary or descriptive techniques that are applicable.

**Table 5.2: Data Types and Definitions for All Variables**

| Col. name | Extended Description | Data Type[5] |
|---|---|---|
| life_exp | Mean life expectancy of a baby born in observation year. Estimated number of years of life assuming no change in patterns of mortality from the year of birth onward. | C |
| Region | Each country is assigned to one of 7 regions. | N |
| gdp_pc | Gross Domestic Product per capita, expressed in constant (inflation-adjusted) 2010 U.S. dollars. Equals annual GDP divided by midyear estimate of population size. | C |

| Col. name | Extended Description | Data Type[5] |
|---|---|---|
| pov_headcount | Poverty headcount ratio at national poverty lines (% of population). | C |
| Income Group | The World Bank defines five groups of countries according to per capita income in each nation. Among the highest-earning countries, they distinguish between countries that are members of the Organisation for Economic Co-operation and Development (OECD) and those that are not. OECD members tend to be "developed" nations. | O |
| hlth_exp_pcap | Health expenditures per capita. This is the sum of all public and private expenditures expressed in 2006 purchasing power parity (PPP) rates, divided by national population for the year. | C |
| imm_dpt | Percent of children ages 12–23 months receiving DPT immunization. DPT immunization is a combination of vaccines to combat three serious infectious diseases, identified by their initials: diphtheria, pertussis, and tetanus. DPT immunization has proven effective in eradicating or reducing incidence of these life-threatening conditions in children. | C |
| prim_enroll | Percent of eligible children enrolled in primary education.[6] | C |
| sec_ftom | School enrollment, secondary (gross), gender parity index (GPI). | C |
| ter_enroll | Percent of eligible children enrolled in tertiary (post-high school) education. | C |
| undernour | Prevalence of undernourishment (% of population). | C |
| food | Food production index: covers food crops that are considered edible and that contain nutrients. | C |

## Preparation for Analysis

This chapter relies entirely on data found in the JMP data table called **WDI**. Open the data table now. In the chapter, we are going to work with about one-fourth of the columns. Instructors might want to assign additional exercises incorporating other columns, and inquisitive readers may want to apply the approach of this chapter to further explore additional columns.

We can easily restrict our attention to the columns listed in Table 5.2 by just selecting the relevant columns as we work through menus. However, we also want to graph and compute statistics using only those rows containing 2016 data. For this, we will apply the **Data Filter** and subset the table so that we are only working with the 2016 data.[7]

1. In the **WDI** data table window, open the thumbnail histograms within the column headings. Then select **Row ► Data Filter**. Filter by **Year**, and **Select, Show,** and **Include** the 215 rows in which **Year** equals 2016. Before moving to the next step, select different years and notice how the histograms change, paying attention in later years to the absence of data in some columns. Minimize the Data Filter.
2. **Tables ► Subset.** The default setting is to use the **Selected Rows,** which is what we want. Change the name of the subset to **WDI2016** and click **OK.** Now we have a data table restricted to 2016.

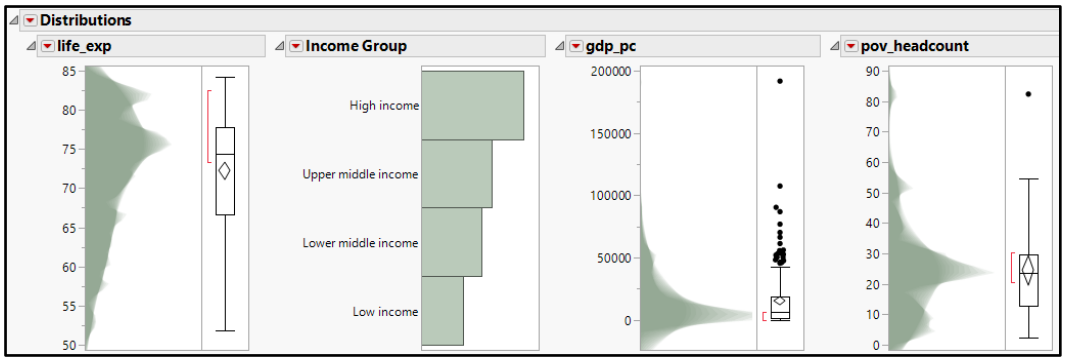
## Univariate Descriptions

Our purpose in this analysis is to more fully understand the variability in life expectancy around the world by looking at how it does or does not co-vary with other indicators. As a first step, we will look at the distributions of several variables while taking advantage of the way that JMP interconnects all open graphs. Most of the variables listed in Table 5.1 are continuous, so we can anticipate examining a series of histograms looking for center, shape, and dispersion of each variable. The first illustration focuses on the Wealth measures and also includes one ordinal variable, **Income Group**.

1. Select **Analyze ► Distribution**. Choose the following columns in this order: **life_exp**, **Income Group**, **gdp_pc**, and **pov_headcount**. Click **OK**. This will produce the four side-by-side analyses.
2. Hold down the Ctrl key and click the red triangle next to **life_exp**. Choose **Histogram Options ► Shadowgram** to more clearly suggest the shape of each distribution.
3. Click the red triangle next to the column for the one ordinal variable. Choose **Histogram Options ► Separate Bars** to indicate that each category is distinct from the others.

At this point, the graphs in the Distributions report should look like Figure 5.1.

**Figure 5.1: Life Expectancy and Wealth Variables**




Among the three continuous variables, we immediately notice three quite different shapes: life expectancy is strongly skewed negatively (that is, the long tail reaches to lower values) with multiple peaks. GDP per capita is also strongly skewed, but it is skewed positively, with numerous outliers. Finally, the percentage of the population living in poverty is right skewed but with multiple small peaks and one outlier.

The distribution of **Income Groups** shows increasing numbers of countries as we move from low to high. Unlike the other variables in the full data table, "income group" is not a naturally occurring attribute that changes each year, but rather a designation assigned by the World Bank based on gross national income per capita.

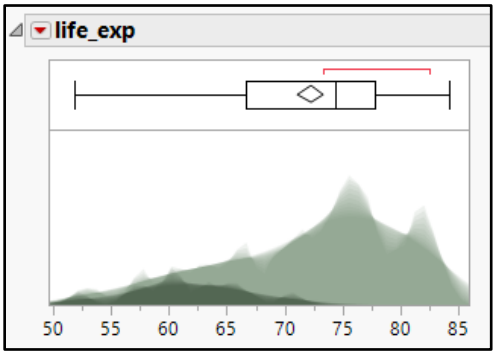
4. To facilitate further comparison, let's stack the four Distributions so that the graphs are oriented horizontally. Click the red triangle next to **Distributions** and choose **Stack**.

Look back at the continuous variables. You should be able to write a few brief sentences to characterize the center (mean, median) and dispersion (standard deviation, range, interquartile range) of each variable.

In the next section of this chapter, we will more systematically investigate how each of these indicators varies with life expectancy. Within this set of univariate distributions, we can get some inkling of the patterns of variation.

5. Before looking at the remaining variables, move your cursor into the **Income Group** bar chart, and click the bar corresponding to **Low income** countries. The shadowgram for life expectancy will look like Figure 5.2.

**Figure 5.2: Distribution of Life Expectancy with Low Income Nations Shaded**




We see that the poorest countries make up a large portion of the lower tail of the distribution. Whereas the entire world has many countries with life expectancies concentrated roughly between 75 and 82 years, the lower income countries have a relatively flat distribution with three modal peaks, and they nearly all have life expectancies between 50 and 72 years.

6. Click through the other bars of the **Income Group** graph and observe the changes in the remaining graphs.
7. Now move on to the other sets of variables, repeating this analysis by generating groups of descriptive analysis for **life_exp** plus the variables measuring each of the constructs listed in Tables 5.1 and 5.2.

Specifically, generate the distributions for these variable lists:

- **life_exp**, **hlth_exp_pcap**, **imm_dpt**
- **life_exp**, **prim_enroll**, **sec_ftom**, **ter_enroll**
- **life_exp, undernour, food**

Once again, examine the distributions for noteworthy features and for insights into possible relationships with life expectancy. Consider these questions:

- Which variables have symmetric distributions, and what might account for the symmetry?
- Which variables have skewed distributions, and what might account for the skewness?
- Which variable have distributions with multiple modes or peaks? What might account for the multiple peaks?
- Which variables have an unusual number of outlying observations?
- Which variables appear to be most closely associated with variation in life expectancy?

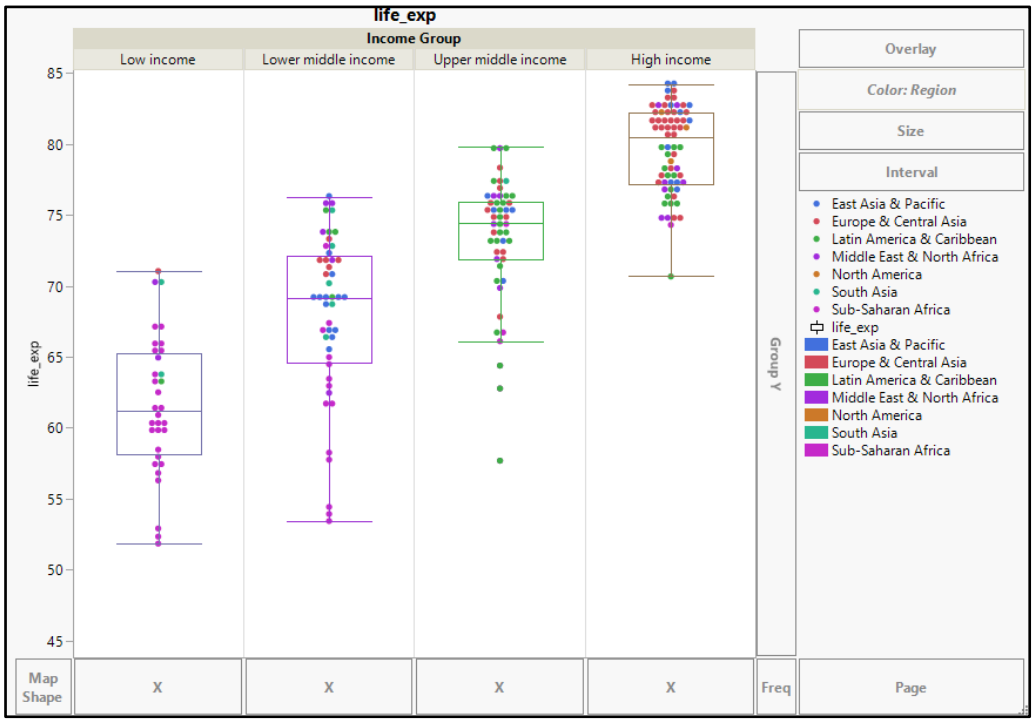
# Explore Relationships with Graph Builder

We are now ready to investigate the nature and strength of relationships between life expectancy and the ten other variables. In addition, because we have selected variables to represent four constructs, we will find it useful to examine how pairs of these variables co-vary.

In Chapter 1, we became aware that life expectancy varies differently in different regions of the world. In this section, we will use region to color-code countries while investigating how other variables relate to life expectancy.

1. Clear all row states in the **WDI2016** subset.
2. **Graph ► Graph Builder**. Drag **life_exp** to the **Y** drop zone.
3. Drag **Region** to the **Color** drop zone.
4. Drag **Income Group** to the **Group X** drop zone.
5. Finally, move your mouse over the graph, right-click, and **Add a Box Plot** to the graph. The result will look like Figure 5.3.

**Figure 5.3: Variability in Life Expectancy by Income Group**




What do you notice in this graph? The most important features include:

- Life expectancy, indicated by the medians and the location of the boxes, generally increases with income group.

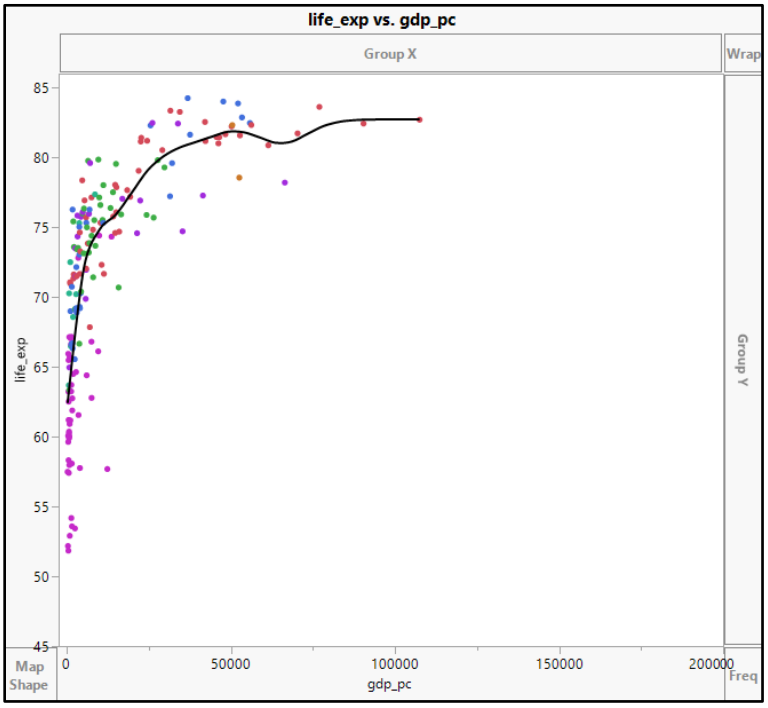
- The two low-income groups also show more dispersion; low income countries differ more than higher-income countries.
- The coloration of points seems to indicate an association between income level and region: for example, Sub-Saharan Africa dominates the lower-income groups and is mostly absent from the High Income groups.

The remaining nine variables are all continuous, and can all be analyzed in similar fashion. We will illustrate the approach using GDP per capita, and leave the remaining analyses to the reader.

6. To clear the **Graph Builder** report, click the **Start Over** button in the upper left. Alternatively, you can replace the current variables by dragging a new column directly over the current one in the drop zones or open a new **Graph Builder** from the menu.
7. Again, drag **life_exp** to the vertical **Y** drop zone.
8. Now drag **gdp_pc** to the **X** drop zone on the horizontal axis
9. Drag **Region** to the **Color** drop zone.

By default, JMP places a *smoother* on the scatterplot to trace a pattern formed by the points. You can leave the smoother in place or remove it by clicking on the smoother icon, which is second from the left on the icon bar. In this graph (Figure 5.4) we see that there is a distinct and strong curvilinear relationship. Countries with higher per capita GDP tend to have longer life expectancies, but the relationship is better described as a curve than as a line. In Chapter 19, we will learn to build models for curvilinear relationships.

**Figure 5.4: Relationship Between Life Expectancy and GDP per Capita**



10. Drag **pov_headcount** to the **X** drop zone and drop it directly over **gdp_pc** to replace the horizontal variable. How does life expectancy vary with the relative size of the population in poverty? Does this make sense to you?
11. Continue to swap out **X** variables proceeding through our list of candidates, as listed below. For each variable, make note of the form, direction, and strength of the relationship with life expectancy, and think about which graphs do or do not surprise you.
    - **pov_headcount**
    - **hlth_exp_prcap**
    - **imm_dpt**
    - **prim_enroll**
    - **sec_ftom**
    - **ter_enroll**
    - **undernour**
    - **food**
12. JMP's **Column Switcher** provides a quick way to swap out variables. Click the red triangle next to **Graph Builder** and choose **Redo ►Column Switcher**.
13. Click on the column name that is currently on the X axis, and then highlight other columns that you want to examine.
14. A new panel opens; press the right-pointing arrow to automatically cycle through the columns.

By now you have rather quickly looked at ten relationships, nine of which were between pairs of continuous variables. These nine visual inspections should lead to the conclusions summarized in Table 5.3. We might generate graphical and numerical summaries to describe the covariation of life expectancy with any of these variables, but based on the preliminary explorations, it might be most useful to focus for now on the clear association between life expectancy and the ordinal variable **Income Group** and on the strongest linear associations: access to improved sanitation, secondary school enrollments, and the percentage of people living in rural areas.

**Table 5.3: Summary of Graph Builder Explorations**

| Column | Form | Direction | Strength |
|---|---|---|---|
| gdp_pc | curvilinear | positive | strong |
| pov_headcount | linear/ curvilinear | negative | weak-moderate |
| hlth_exp_pcap | curvilinear | positive | strong |
| imm_dpt | positive | linear (?) | weak |
| prim_enroll | none (?) | vertical | weak |
| sec_ftom | curvilinear (?) | positive | weak |
| ter_enroll | curvilinear | positive | moderate-strong |
| undernour | curvilinear | negative | moderate |
| food | linear | negative | very weak |

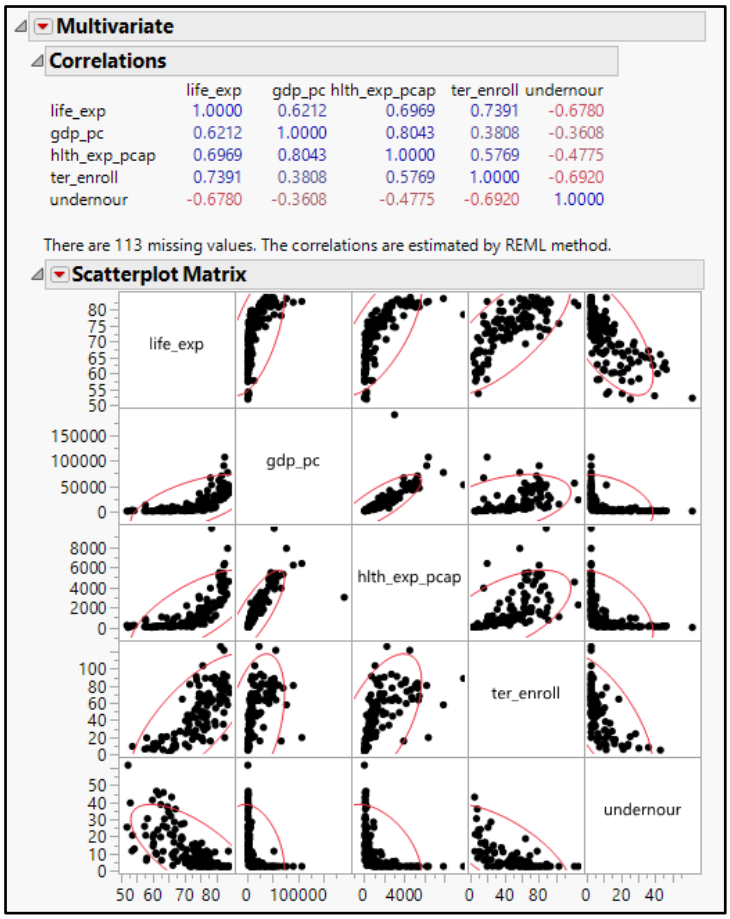
## Further Analysis with the Multivariate Platform

Recall from Chapter 4 that the Multivariate platform computes correlations and generates multiple small scatterplots for pairs of continuous columns. As we narrow our investigation, we can summarize the relationships using correlations.

1. **Analyze ► Multivariate Methods ► Multivariate.** In this order, cast **life_exp**, **gpdpc**, hlth_exp_pcap, **ter_enroll** and **undernour** into the **Y** role.
2. Click the red triangle next to **Scatterplot Matrix** and select **Density Ellipses**.

Your results should look like Figure 5.5.

**Figure 5.5: Multivariate Summary of Three Relationships**




Tertiary enrollments have the strongest correlation with life expectancy ($r$ = 0.7391), followed by health care expenditures per capita ($r$ = 0.6969), prevalence of undernutrition ($r$= –0.6780), and gdp per capita ($r$ = 0.6212). The red ellipses in the scatterplot matrix help emphasize the comparative strength of the relationships. The matrix arrangement also makes it clear that there is also a strong positive association between gdp and health care expenditures, and a strong negative association between tertiary enrollments and undernutrition. Thinking ahead, because of these associations, this set of variables will not enable us to logically distinguish between, say, the relationship of longer life expectancy to gdp and health care expenditures. If the wealthiest countries, which also spend the most on health care, have long lives, we can't say if the key factor is the health care or the wealth itself.

## Further Analysis with Fit Y by X

To generate both graphical and numerical descriptive summaries for bivariate relationships, the simplest approach is to use the **Fit Y by X** platform. It takes into account the data types of whichever variables we select for the X and Y roles and performs appropriate analysis.

1. Use **Analyze ▶ Fit Y by X** to cast life expectancy as **Y**, and income group, gdp per capita, health care expenditures per capita, tertiary enrollment, and undernourishment prevalence as **X**. Note that you can select one Y and five X columns to generate four analyses at once.

   Look at the five graphs in the report window. The left-most graph has an ordinal horizontal axis, with points (unjittered) vertically arrayed within each category. The other three graphs are scatterplots. For the first graph, we want to compute quantiles, means, and standard deviations by income group; for the others, a simple linear regression equation is the most suitable descriptive summary.
2. Click the red triangle next to **Oneway Analysis of life_exp by Income Group.** Select **Quantiles**. Click the hotspot again and select **Means and Std Dev**. This will produce the tables shown in Figure 5.6.

**Figure 5.6: Summary Statistics of Life Expectancy by National Income Group**

**Quantiles**

| Level | Minimum | 10% | 25% | Median | 75% | 90% | Maximum |
|---|---|---|---|---|---|---|---|
| Low income | 51.835 | 54.259 | 58.1385 | 61.163 | 65.2835 | 69.0102 | 71.051 |
| Lower middle income | 53.428 | 58.0092 | 64.625 | 69.125 | 72.133 | 75.308 | 76.253 |
| Upper middle income | 57.681 | 66.6647 | 71.8633 | 74.46994 | 75.95725 | 77.3169 | 79.831 |
| High income | 70.673 | 75.4788 | 77.16585 | 80.508 | 82.25854 | 82.92585 | 84.22683 |

**Means and Std Deviations**

| Level | Number | Mean | Std Dev | Std Err Mean | Lower 95% | Upper 95% |
|---|---|---|---|---|---|---|
| Low income | 33 | 61.638212 | 4.924115 | 0.8571784 | 59.892197 | 63.384227 |
| Lower middle income | 47 | 67.973062 | 5.8684746 | 0.856005 | 66.250013 | 69.69611 |
| Upper middle income | 50 | 73.362954 | 4.3773054 | 0.6190445 | 72.118937 | 74.60697 |
| High income | 67 | 79.582017 | 3.0121889 | 0.3679974 | 78.847286 | 80.316748 |

Missing Rows 18

Notice that the measures of location (mean and quantiles) consistently show an association between rising life expectancy and rising income levels.

Now we want to find regression lines for the continuous relationships. You could use the red triangles to add a linear fit to each graph separately, but it's more efficient to do the following:

3. Hold the Ctrl key on your keyboard (command key on a Mac) and click the red triangle next to **Bivariate Fit of life_exp By gdp_pc**. Select **Fit Line**. Pressing the Ctrl key while clicking a red triangle applies the choice to all similar analyses open in the report window.

4. Within the results, you should locate three estimated linear fits with their associated R-square values as follows (I have rounded the estimated slopes and intercepts):
   - life_exp = 68.42 + 0.00027 (gdp_pc) R-square = 0.42
   - life_exp = 67.44 + 0.0031 (hlt_exp_pcap) R-square = 0.47
   - life_exp = 66.24 + 0.17 (ter_enrolll) R-square = 0.51
   - life_exp = 77.47 – 0.48 (undernour) R-square = 0.48

## Summing Up: Interpretation and Conclusions

With observational data, we cannot draw definitive cause-and-effect conclusions. At this point in your study of statistics, we must limit ourselves to descriptive statements about patterns observed within the available data for 2016. There are many ways one might verbally summarize what we have seen in these analyses. Below are several statements that characterize the descriptive findings in this chapter:

- Across the World Bank income groups, there was a strong tendency for life expectancies in higher-income countries to exceed those in lower-income countries. In the well-off countries, life expectancies were consistently close to 80 years on average. Lower-income countries exhibited far more variability, but at the lowest end of the income scales, life expectancies were in the vicinity of 61 years.
- The strongest associations are curvilinear, and they are between life expectancy and GDP and health care spending per capita. The numerical summaries of linear fitted models do not reveal much useful information at this point, but the visualizations clearly indicate that small increases in these variables are associated with large increases in life expectancy among the countries with the lowest values of these two explanatory variables.
- The strongest linear association is between life expectancy and tertiary enrollments. A one percent increase in tertiary enrollment is associated with a difference of approximately 0.17 years (2 months) of life expectancy.
- Prevalence of undernourishment has a negative association with life expectancy. This makes sense, in that to the extent that residents of a nation face hunger and poor nutrition, the less likely it is that they will live long lives.

## Visualizing Multiple Relationships

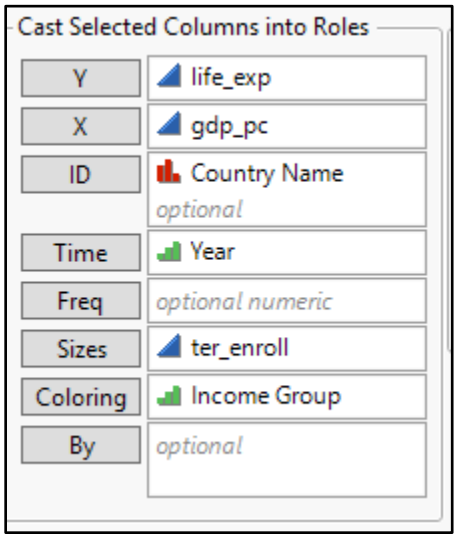
To conclude this chapter, we briefly adopt a more ambitious descriptive task: visualizing six WDI variables in *one* screen graphic. In the previous section, we noted that life expectancy varied predictably with three different continuous variables in the year 2010. This in turn raises other questions: Were these patterns unique to 2010, or did they appear in other years? To what degree are, for example, secondary enrollments related to the relative size of the rural population? How does access to sanitation compare across income groups?

1. Return to the original **WDI** table that we opened at the beginning of the chapter and clear the row states.

   For most of the chapter, we have been working only with rows of data from 2016, and this table contains observations from 1990 through 2018. In this next graph, we will use all available years.

2. We can construct a bubble plot (see Graph menu) with the following variable assignments (see Figure 5.7 below): **Y** is **life_exp**, **X** is **gdp_pc**, **ID** is **Country Name**, **Time** is **year**, bubble **Sizes** are determined by **ter_enroll**, and **Coloring** is **Income Group**. Click **OK**.

   **Figure 5.7: Role Assignments for a Six-Variable Bubble Plot**

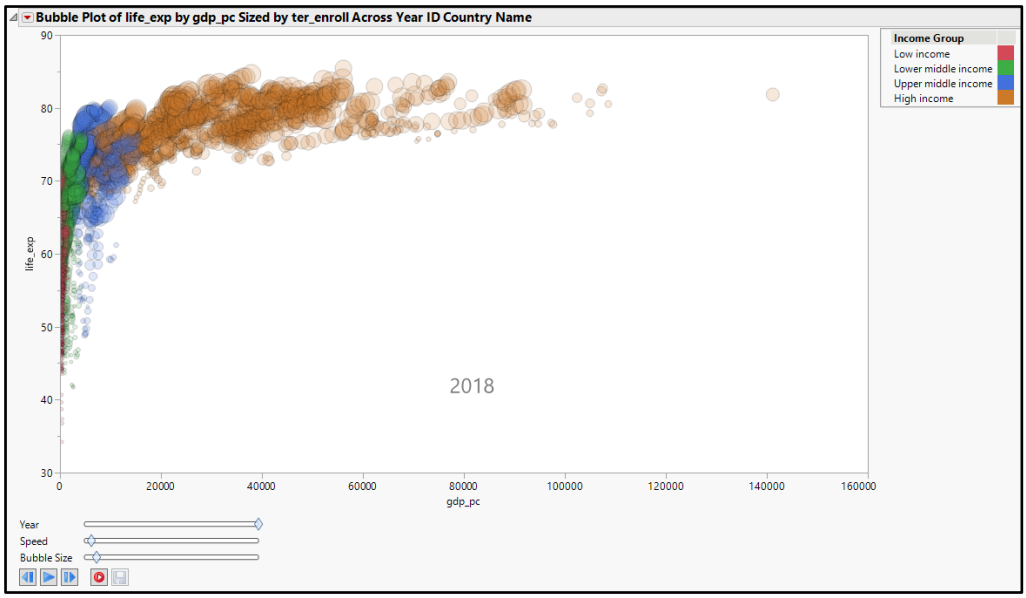



3. You will now see a bubble plot with **life_exp** on the vertical axis and **gdp_pc** on the horizontal. The default bubble sizes are a little too large for optimal viewing of what comes next, so make them slightly smaller by grabbing the slider control next to **Bubble Size** in the lower left portion of the window and slide it slightly leftward. The bubbles will shrink.




4. Click the red triangle next to **Bubble Plot**, and choose **Trail Bubbles ▶ All**.
5. Just below the Bubble Size slider control, you'll see three animation controls; press the "Play" button  in the middle, and watch the patterns unfold over the years from 1990 through 2018. Figure 5.8 shows the state of the world in 2018.

**Figure 5.8: Animated Bubble Plot for Multivariate Insight**




This graph contains most of the variables that we have investigated over the past several pages, but there are two major differences between this bubble plot and the previous visual displays. First, all variables are represented simultaneously by taking advantage of several visual *primitives*: position, size, color, and motion. Second, the entire picture is animated over time, powerfully representing changes as they occurred.

All countries have experienced increases in GDP per capita and in life expectancy over the three decades. It is also clear that the poorest nations have seen big increases in life expectancy without corresponding growth in GDP per capita. In contrast, the highest income nations have enjoyed rapid GDP growth and most gains in life expectancy.

The bubble plot opens the door for complex multivariate descriptive summaries of large volumes of data. It helps us think about what has happened, and about patterns warranting further investigation.

[1] World Bank History, https://www.worldbank.org/en/about/archives/history, accessed August 29, 2019.

[2] Millenium Development Goals, *World Health Organization*, https://www.who.int/topics/millennium_development_goals/about/en/, accessed August 29, 2019.

[3] Millenium Development Goals (MDG) Monitoring, *UNICEF*, https://www.unicef.org/statistics/index_24304.html, accessed August 29, 2019.

[4] See Table 5.2 for fuller definitions of each variable.

[5] C = Continuous, O = Ordinal, N = Nominal

[6] The three enrollment indicators are calculated by taking the total enrollment at each grade level divided by the number of children who are the appropriate age for that level. Accordingly, some countries have enrollment percentages in excess of 100% because of overage or underage children in school. Often, this occurs when students repeat grades.

[7] We could rely on the Data Filter alone in this chapter, but with so many tasks to follow, there is a chance that the reader might inadvertently clear the filter. With a subset of the table, all should go smoothly.

# Chapter 6: Elementary Probability and Discrete Distributions


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## Overview

The practical analysis of data occurs within the context of models and some of the most valuable models come from the world of probability. Some of the patterns that we observed in conducting descriptive analysis reflect or mimic patterns that are characteristic of specific probability models or theories. This chapter offers a brief introduction to a few probability models that will help build bridges to the statistical topics of later chapters.

## The Role of Probability in Data Analysis

In the prior chapters, we have discussed ways to summarize collections of data visually and numerically. At this point, our attention turns from the analysis of empirical information—data collected through observation—to the consideration of some useful theoretical constructs and models. *Probability theory* is a large and useful body of material and is the subject of entire courses in its own right. In an elementary applied statistics course and in this book, probability is especially germane in two respects. First, sometimes the goal of data analysis is to approximate the likelihoods of events or outcomes that are important to a decision-maker. For example, people who plan communications or transportation networks want to know the likelihood of traffic at different network locations during the day.

Second, probability theory provides the bridge between purely descriptive analysis and *statistical inference*, or the practice of drawing general conclusions from a single sample. Whenever we rely on one sample to represent a population, we need to be aware that our sample is one of many possible samples that we might have taken. Each sample imperfectly represents the parent population, and samples vary in their representativeness. Because it's impossible to know for certain just how representative any sample is, sampling inherently risks drawing erroneous conclusions about a population. To manage the risk, we will (eventually) use probability theory.

In this chapter, we will confine our attention to a few probability topics that pertain to these two uses. If you are taking an introductory statistics course, your principal text might include additional topics. The next section of this chapter selectively reviews some of the elementary vocabulary and rules of probability theory. It does not provide an exhaustive treatment but recapitulates some of the terms and ideas covered by your major text.

## Elements of Probability Theory

In the next section of this chapter, we will look at some data collected by Professor Marianne Rasmussen of the University of Southern Denmark who observed the behavior of dolphins off the coast of Iceland. Let's use this activity—observing dolphins in their typical comings and goings—as an illustrative framework for defining some terms and concepts.

Professor Rasmussen's basic activity was to watch the behavior of dolphins at different times of day and to record their activities. She identified three basic activities: traveling quickly, feeding, or socializing. Dolphins tend to engage in these activities in groups, so she also tallied how many dolphins engaged in each activity. She also noted the time of day: morning, noon, afternoon, or evening.

Consider the activity of observing the behavior of one dolphin. In the language of probability, an *event* is an observable result of an activity. So, we might define the event *A* as that of observing a dolphin who is traveling quickly, or an event *B* as observing a feeding dolphin. (By the same token, we could call these events *T* or *F.*) In Dr. Rasmussen's study, we have three possible events of interest.

Similarly, we can define events as the outcomes of noting the time of day, and we have four possible events. For each of the two research activities in this study, we have pre-defined lists of possible events (traveling, feeding, or socializing and morning, noon, afternoon, and evening). For the purposes of this study, anytime she spots a dolphin, it will be doing one of the three defined activities, and she does her watching four times a day.

## Probability of an Event

We will assign a numerical value to the *probability of an event* as being proportional to its relative frequency of occurrence in the long run. For any event *A* we can write *Pr(A)* to represent this value. There are just two rules in assigning probabilities to events:

- The probability of an event is a value between 0 and 1, that is, $0 \leq Pr(A) \leq 1$. An event that is impossible has a probability of 0, and an event that is certain to happen has a probability of 1.
- For any activity, if *S* is the complete set of all possible outcomes, $Pr(S) = 1$. For the dolphins, this means that if Dr. Rasmussen sees a dolphin, it will surely be traveling, feeding, or socializing.

Relying on these two rules, we can always define an event $\bar{A}$ as the *complement* of event A. The complement of an event refers to all possible outcomes excluding the event, and $Pr(\bar{A}) = 1 - Pr(A)$. To illustrate, if event *A* is seeing a feeding dolphin, then event $\bar{A}$ is seeing a dolphin who is not feeding, or rather, one who is traveling or socializing.

## Rules for Two Events

In many situations, we are interested in two events at a time, and ask questions about the interaction or co-occurrence of two events. This section briefly reviews several important definitions and possible relationships between two events.

- **Mutually exclusive events:** If two events cannot possibly occur together, we call them mutually exclusive or *disjoint*. In the dolphin study, socializing and traveling are mutually exclusive.
- **Joint probability:** Joint probability refers to the simultaneous occurrence of two events. Often it is expressed as the probability of events *A and B* occurring together, or the probability of the *intersection* of events *A* and *B.*
- **Union of two events:** The union of two events refers to the situation in which one or the other event occurs, including their joint occurrence. Often it is expressed as *A or B.*
- **Conditional Probability:** Sometimes the occurrence or non-occurrence of event *B* affects the likelihood of event *A*. Conditional probability is the probability of one event under the condition that another event has occurred. For example, in the dolphin study, let's designate event *A* as "observing a dolphin who is feeding." In general, there is some probability at any time of observing a feeding dolphin; call it *Pr(A).* We might ask whether that probability is the same at different times of day. For instance, we might want to know the probability of observing a feeding dolphin if we are looking at noon. Customarily, we would speak of the conditional probability of feeding *given that* it is noon, and we would write this as *Pr(A|B)* or *Pr(Feeding|Noon).*

  **NOTE:** Conditional probability is different from joint probability. When we ask about the probability that we observe a dolphin feeding *and* it is noon, the implication is that we don't know either the behavior or the time. When we ask about the probability of

observing a feeding dolphin *given that* it is noon our situation is different. We know that it's noon, but we don't yet know what the dolphin is doing.

- **Independence:** Two events are *independent* if the occurrence of one provides no information about the occurrence of the other. More formally, if $P(A|B) = P(A)$, we say that events *A* and *B* are independent. So, if dolphins are more likely to eat at one time of day than another, we would find that the event feeding is not independent of the event noon.

## Assigning Probability Values

We have learned that a probability value is a number between 0 and 1, and it's time to discuss how we determine which value should attach to which events. There are three basic methods for assigning probabilities—they are known as the Classical, Relative Frequency, and Subjective methods. The approach that we use depends on the circumstances of the particular activity we are conducting.

- The *Classical* method applies if we know and can list the possible outcomes of an activity, and we can assume that all basic outcomes share the same likelihood. If there are *n* equally likely outcomes, then the probability of each outcome is $1/n$. Among other things, the classical method is applicable to games of chance (like card games) and to activities like simple random sampling.
- Computations using classical probabilities often refer to events that can happen in multiple ways. So, for example, when rolling two dice, we can think of the elementary outcomes for each die as being the six distinct faces. In dice games, we sum the spots on the two dice. In the context of the game, the events of interest—rolling a seven, or doubles, or eleven—are not equally likely, but the fundamental outcomes for each die do share the same probability of 1/6.
- The *Relative Frequency* method applies to situations in which we know and can list the possible outcomes but cannot assume that the outcomes are equally likely. This method relies on empirical observation of numerous repetitions of the situation, and then computing relative frequencies. Observing dolphin behavior is one such situation. Other examples include rates of disease incidence, batting averages, and college graduation rates.
- The *Subjective* method applies to situations in which we lack information about possible outcomes or their relative frequency or both. In such instances, expert opinion forms the basis for probability assignments. Problems requiring expert opinion are beyond the scope of an introductory course and will receive no further attention here.

For elementary classical problems involving one or two events, a calculator is the tool of choice. For relative frequency problems, we are well served to use software like JMP to construct univariate frequency tables or bivariate contingency tables. We will use the dolphin example to illustrate.

# Contingency Tables and Probability

1. Select **File** ▶ **Open**. Select the **Dolphins** data table.

   This data table (Figure 6.1) is different from those we have seen before. The rows here actually represent the behavior of 189 dolphins but consolidate observations of the same time and behavior into a single row. The column *Groups* shows the *joint frequency* of each pair of time and behavior events. For instance, throughout the entire observation period, Dr. Rasmussen observed six dolphins traveling in the morning and 28 feeding in the morning.

**Figure 6.1: Dolphins Data Table**

Dolphins*

Dolphins
Notes Groups of dolp
Note This is one of the

Columns (3/0)
Activity
Period
Groups

Rows
All rows 12
Selected 0
Excluded 0
Hidden 0
Labelled 0

| | Activity | Period | Groups |
|---|---|---|---|
| 1 | Travel | Morning | 6 |
| 2 | Feed | Morning | 28 |
| 3 | Social | Morning | 38 |
| 4 | Travel | Noon | 6 |
| 5 | Feed | Noon | 4 |
| 6 | Social | Noon | 5 |
| 7 | Travel | Afternoon | 14 |
| 8 | Feed | Afternoon | 0 |
| 9 | Social | Afternoon | 9 |
| 10 | Travel | Evening | 13 |
| 11 | Feed | Evening | 56 |
| 12 | Social | Evening | 10 |

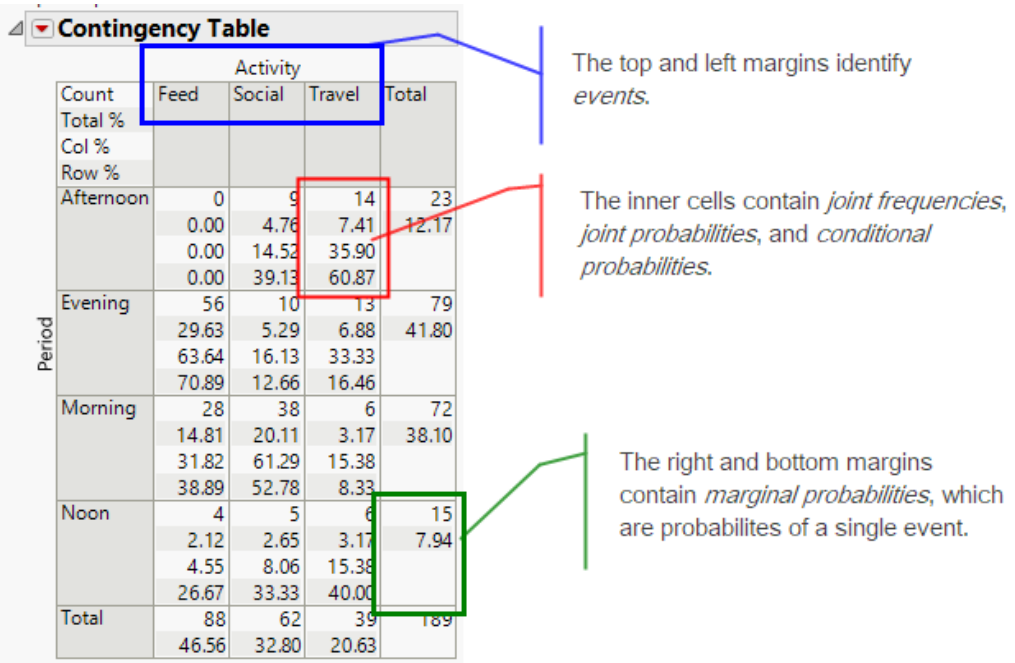
   When we analyze these columns, we will just need to specify that the rows represent frequencies rather than individual observations.

2. Select **Analyze** ▶ **Fit Y by X**. You have worked with this dialog box (not shown) many times by now. With this data table, we will use the **Freq** role for the first time.
3. Cast **Activity** as **Y** and **Period** as **X** and then cast **Groups** as **Freq**.

We could have reversed the X and Y roles, but here we want to think about the logic of the situation. It's reasonable to think that animals naturally incline toward different behaviors at different times of day. In terms of dependence and independence, it makes more sense to ask if a dolphin's behavior depends on the time of day than it does to ask if the time of day depends on the behavior.

Because we are interested in assigning probability values, we will focus first on the contingency table in the output report. Look closely at the table, which is reproduced with annotations in Figure 6.2.

**Figure 6.2: Contingency Table for Dolphins Data**




We will think of the table as having three main regions, as indicated in Figure 6.2. This study defined seven different events, which are all named in the left and upper margins of the table. To find the probability (relative frequency) of any one event, we look in the opposite margin. For instance, 88 of 189 dolphins were observed to be feeding. The relative frequency was 46.56%, so we find that *Pr(Feed)* = 0.4656.[1]

Within the interior cells, the upper number is the joint frequency showing the number of times that Dr. Rasmussen observed a behavior at a particular time. The next number is the *joint probability*. In her observations, she did not see any dolphins feeding in the afternoon, but she did see dolphins feeding in the evening. The joint probability *Pr(Feed and Evening)* = 0.2963.

The upper left cell of the table reports that the next two numbers within each cell are the column percentage and the row percentage. In terms of assigning probabilities, these are *conditional* values. A column percentage is the relative frequency of the row event within the column only. In other words, restricting ourselves to the condition represented by the column, it is the relative frequency of each row. Similarly, the row percentages give the probability of the column events conditioned by each row.

A word about notation. We denote "the conditional probability of event A given that event B is true" as $P(A|B)$. Notice that the character between A and B is a vertical line |, not a division slash /. Division is necessary to compute probabilities, but the | symbol means "given that" or "conditional upon," rather than "divided by."

So, *Pr(Noon|Feed)* = .0455, but *Pr(Feed|Noon)* = 0.2667.

With this contingency table, we can easily check for independence of any two events by comparing the conditional probability *Pr(A|B)* to the marginal probability *Pr(A)*. If they are equal, the events are independent; if they are unequal, the events are not independent. So, for instance, we have just noted that *Pr(Feed|Noon)* = 0.2667. Our earlier glance at the bottom margin of the table shows that *Pr(Feed)* = 0.4656. Since these two probabilities are unequal, we would conclude that feeding behavior is *not* independent of the noon condition.

Looking at the mosaic plot (not reproduced here) you might notice that feeding is much more common at evening than at any other time of day, and we might be tempted to conclude that dolphins tend to eat in the evening. Because we are looking at sample data rather than at definitive information about all dolphins always, this is an oversimplification, but it is a useful guide at this point in the course. In Chapter 12, we will see a more sophisticated way to interpret these results.

## Discrete Random Variables: From Events to Numbers

Thus far in this chapter, we have framed the research questions in terms of categorical variables. In some instances, the research variables are numerical rather than categorical: during a flu outbreak, how many children in the school system will become ill? In a random sample of customer accounts, how many billing errors will we find? In such cases, we are focusing on a *discrete* variable—one that has countable integer values. If we are talking about a flu outbreak, we may find 0, 1, 2, 3... sick children, and each of these values might occur with a different probability.

If the variable of interest takes on discrete numerical values rather than simple categories, we will refer to the *probability distribution* of the variable. In practice, we can often use one of several *theoretical discrete distributions* to build a model of such situations. It is well beyond the scope of this book to provide a full treatment of discrete random variables. Refer to your principal text for a full discussion of probability distributions. The following two sections assume that you are familiar with the concept of a random variable, you know which theoretical distribution you want to apply to a problem, and you want to use JMP either to calculate probability values for a random variable or to generate simulated random data conforming to a known discrete distribution. The goal here is to introduce how JMP handles discrete distributions, but not to provide an exhaustive treatment of the subject.

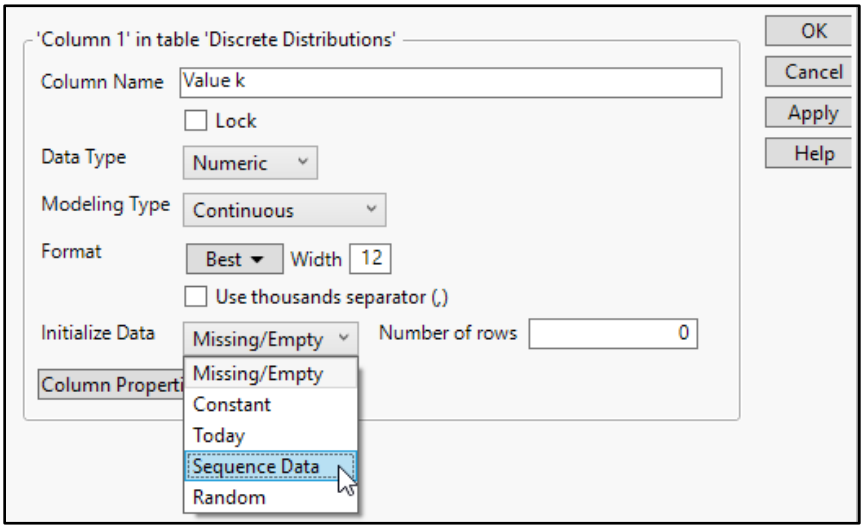
# Three Common Discrete Distributions

There are many theoretical discrete distributions. This section and the next demonstrate how JMP implements only three of them, but others work similarly. In this section, we will see how to build a data table that computes probability values for each distribution. In the next section, we will simulate random observations that conform to each distribution.

For these illustrations, we will consider three random variables, each of which can assume values starting at 0 up to a maximum value of 20. To begin, we will set up a new data table whose first column is a list of integers from 0 to 20.

1. Select **File ▶ New ▶ Data Table** (or Ctrl-N).
2. Double-click the word **Untitled** in the table panel and type in a new name for this data table: ***Discrete Distributions***.
3. In the **Columns** panel, highlight **Column 1**. Right-click and choose **Column Info**. Type the **Column name *Value k*** and click the **Missing/Empty ▼** button and choose **Sequence Data** as shown below in Figure 6.3.

**Figure 6.3: Creating a Column of Consecutive Integers**




4. This expands the dialog box adding some new options. Replace the default value of **From** with **0**, and replace the **To** value with **20** and click **OK**. The first column of the table will now contain the integers from 0 to 20.

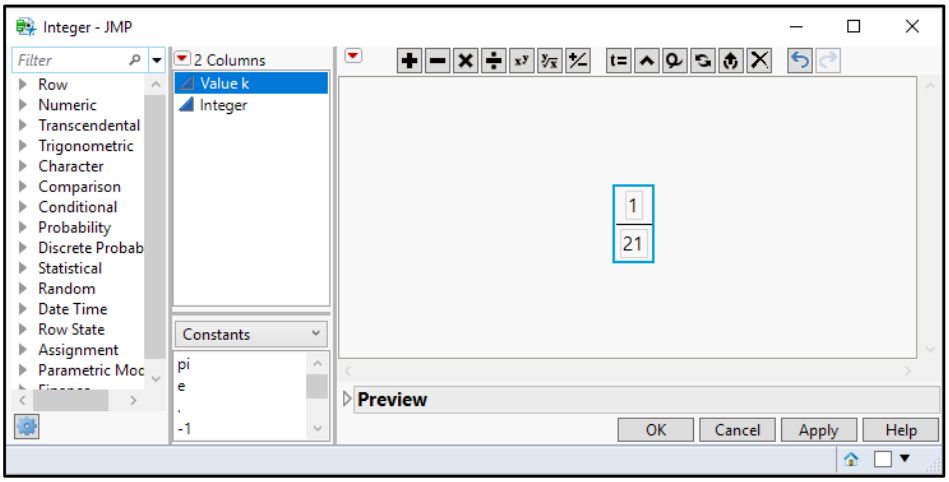
## Integer Distribution

The integer distribution (sometimes called the *uniform discrete distribution*) applies to situations in which there are *n* numerical outcomes of consecutive integers, each of which is equally likely. A roulette wheel is a good example of a process generating values following an integer distribution.

In our example, we will consider an integer random variable that assumes values of 0 to 20. With 21 possible values and the characteristic of equal probabilities, we don't need software to tell us that for each value the probability equals 1/21. Note that if we know the number of possible values (*n*), we can do the computation; we refer to *n* as the *parameter* of the integer distribution. We will start here though to demonstrate how JMP handles such calculations for simple and not-so-simple formulas.

1. Select **Cols ▶ New Columns**. Rename **Column 2** as **Integer**; click the **Column Properties**, select **Formula**. This will open the **Formula Editor** dialog box.
2. As shown below in Figure 6.4, within the red rectangle that initially contains the words **no formula**, type **1/21** and then click anywhere in the formula editor dialog box that is outside the blue outline where you just typed.

**Figure 6.4: Using the Formula Editor to Create a Simple Formula**




3. The formula editor changes appearance to show your typing as a numerator over a denominator. Click **OK** twice. The second column now shows the probability of each possible value of the random variable in column 1.

## Binomial

Binomial distributions describe random variables that represent the number of times we observe a binary outcome in repeated attempts. More specifically, a binomial variable is the number of observed outcomes ("successes") in *n* trials where the following is true:

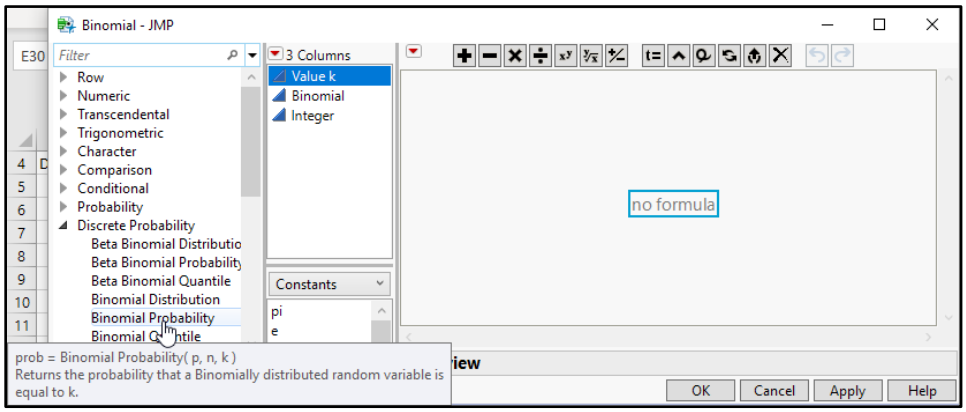
- Each trial is conducted in the same fashion.
- There are just two complementary outcomes possible on each trial, traditionally called *success* and *failure*. A success is the outcome of interest to the investigator.
- The probability, *p*, of the outcome of interest is the same each trial.
- Each trial is independent of the one before.

Imagine a survey researcher who intends to email 20 volunteers, asking each to complete a short online survey. She wants some idea of how many volunteers might proceed to take the survey. Imagine further that the probability that each volunteer takes the survey is 0.30 (researchers should be so lucky), and that all volunteers are strangers who do not consult with each other. The number of survey-takers is a binomial variable.

The computation of binomial probabilities is far more complex than for integer probabilities, but JMP does all of the work for us once we specify the two parameters of the binomial distribution, *n* and *p*. Following our hypothetical example, let's consider a binomial random variable with 20 trials and probability 0.3 of success.

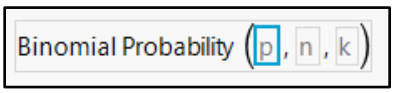
1. Set up a third column (**Cols ▶ New Columns**) named **Binomial**. Again, create a formula for this column just as you did in the previous step.
2. Instead of immediately typing in the formula, we will use a built-in function. In the **Functions** list of the formula editor, scroll down to find **Discrete Probability** and select **Binomial Probability.** (See Figure 6.5.)

**Figure 6.5: Locating the Discrete Probability Functions in the Formula Editor**




The binomial probability function (Figure 6.6) has three arguments: the probability of success on each trial, the number of trials, and the specific value of the variable to be evaluated.

**Figure 6.6: The Binomial Probability Function**




3. In the box labeled p, type the probability of success .3 and press Enter.
4. Select the box labeled n and type the number of trials 20 and press Enter.
5. Now select the box labeled k. Rather than typing a value, move the cursor to the Table Columns list in the upper left and click once on Value k. Click OK in the formula editor and OK in the Column Info dialog box.

The Binomial column now contains the probabilities corresponding to each possible value, *k*, of the random variable. Reading down the list of probabilities, it becomes clear that with these particular parameters of *n* and *p*, our survey researcher is more likely to obtain, say, 5 or 6

respondents than she is to succeed with 15 or 16. Because each value of *k* is mutually exclusive with other values, we can add probabilities to determine the likelihoods of any range of specific number of successes. For example, the probability of 2 or fewer = 0.0008 + 0.0068 + 0.0278 = 0.0354. There is less than a 0.04 probability that the researcher will succeed with 2 or fewer volunteers. If that were to happen, the researcher would be surprised and disappointed. Similarly, there is only about a 0.05 probability that more than 9 volunteers will complete the survey. If 10 or more complete the survey, the researcher should also be surprised and delighted.

> Had we wanted to JMP to add up the probabilities, we would compute *cumulative probabilities*—the probability of finding *k* or fewer successes in *n* trials—and we could have used the **Binomial Distribution** function instead of the **Binomial Probability** function.

## Poisson

Like the binomial distribution, the Poisson distribution refers to the number of times an event occurs. Unlike the binomial, the Poisson refers to the number of occurrences over a continuous span of time or distance, rather than over a discrete number of trials. Because of this difference, a Poisson-distributed variable has no inherent maximum. In other respects, its characteristics are the same as the binomial. In a Poisson distribution, the likelihood of a success is identical in all time periods, and one period is independent of the others.

A Poisson distribution has a single parameter, conventionally known as $\lambda$ (lambda), which is equal to the mean number of occurrences within one unit of time or distance. For the sake of this example, we will compute probabilities for a Poisson random variable with a mean rate of occurrence of $\lambda = 6$.

1. As you have done before, create another new (fourth) column and name it **Poisson**.
2. Click **Column Properties** and select **Formula**.
3. In the **Formula Editor** select **Poisson Probability** from among the **Discrete Probability** functions.
4. Within the function, replace **lambda** with **6**, and again select **Value k** as the argument **k**. Click **OK** in both open dialog boxes and look at your data table.

Notice that the values with the highest probabilities are $k = 5$ and $k = 6$. Though there is no maximum possible value, the probability of 20 successes is listed as $3.7250619e^{-6}$, or a tiny probability of 0.0000037. In other words, 20 successes would be nearly impossible.

The integer, binomial, and Poisson distributions are often included in introductory statistics courses. We have seen that the list of discrete distribution functions is quite long, but this brief introduction should give you some sense of how we can work with theoretical distributions. In the next section of this chapter, we see how JMP can be used to generate random values drawn from a specific underlying distribution. Simulation can clarify some characteristics of random variation and has applications in decision-modeling and design as we will see in upcoming chapters.

## Simulating Random Variation with JMP

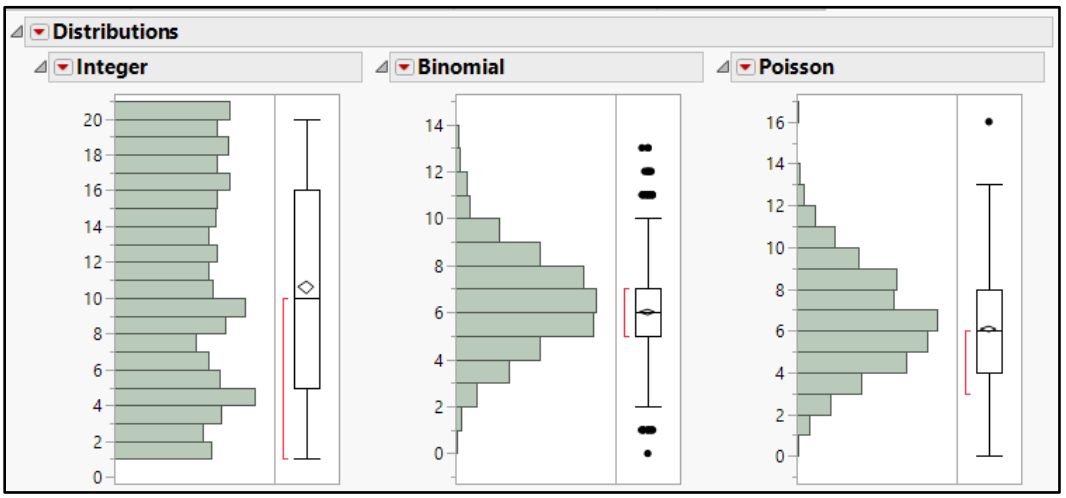
In this section, we will use the binomial and Poisson distributions to simulate random samples of 1,000 observations. We will open a new data table and populate it with artificial simulated observations.

1. Select **File** ▶ **New** ▶ **Data Table**. Create a new data table and name it **Simulated Data.**
2. Click the title Column 1, right-click and choose **Column Info**. Change the Column Name to **Observation**. This column will just identify the observations. Refer to Figure 6.3 to create a column of sequential integers from 1 to 1,000.
3. Create a new column named **Integer**.
4. Click the **Column Properties** button, choose **Formula**.
5. Now, from among the functions choose **Random** ▶ **Random Integer**.
6. Specify that **n1** = **20** and click **OK** in both dialog boxes.
7. Next create a new column and name it **Binomial**.
8. Click the **Column Properties** button, choose **Formula**.
9. Now from among the functions choose **Random** ▶ **Random Binomial**.
10. Specify that **n** = **20** and **p** = **0.3** and click **OK** in both dialog boxes.

Before examining the simulated data, we will simulate some Poisson observations. By this point, you should be able to create a new column called **Poisson** on your own.

11. When you are ready to enter the formula, select the **Random Poisson** function with a parameter of **lambda** equal to *6*.
12. Finally, we will examine the pattern of variation in the simulated data generated by these three probability models. **Analyze** ▶ **Distribution**. Cast the **Integer**, **Binomial** and **Poisson** columns as **Y**, and click **OK**. The histograms should look like Figure 6.7.

**Figure 6.7: Comparison of Simulated Sets of Binomial and Poisson Random Data**



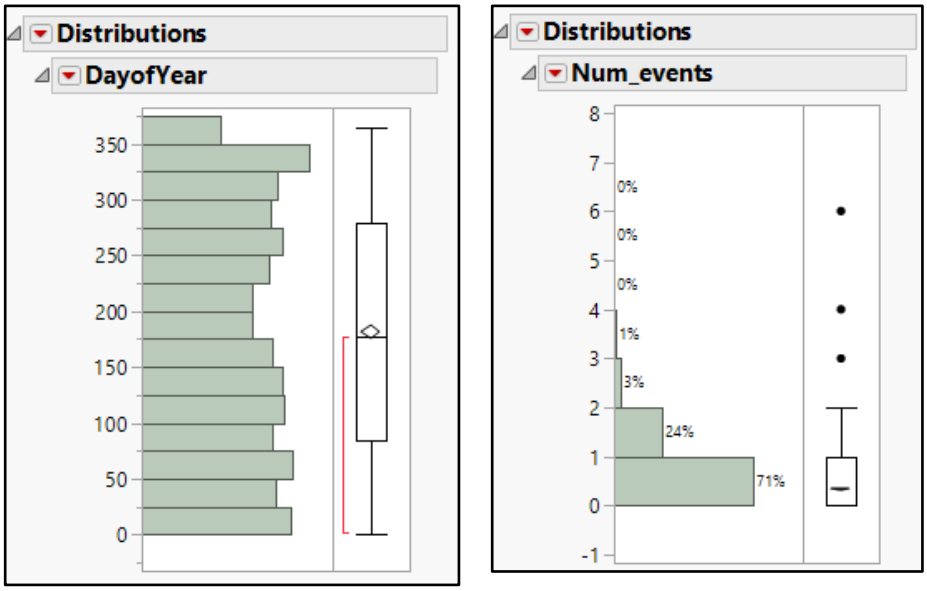
The Y axis represented the number of successes in each simulated observational unit. Because this is a simulation, your results could look different from the graph shown here, but the overall pattern in the distribution should be similar. Look at these histograms, especially noting the similarities and differences in their shapes. These graphs illustrate the long-run nature of random variation. Even though it is impossible to predict the value of any single observation, we can reliably characterize the long-run pattern for a random variable. In the Integer graph, all outcomes occur with very similar (though not identical) frequencies. In contrast, in both the binomial and Poisson distributions (for example) 4 to 8 successes occur frequently, and more than 12 successes are possible, but quite rare.

## Discrete Distributions as Models of Real Processes

The Pipeline and Hazardous Materials Program (PHMSA) is an office of the U.S. Department of Transportation. PHMSA monitors all disruptions in the natural gas pipelines supplying natural gas throughout the United States, maintaining detailed records about all major incidents. We have some PHMSA data in the **Pipeline_Safety** data table for the period from 2010 through May 2019. Each row of the table represents an incident in which a natural gas pipeline was somehow disrupted.

The ninth column of the table represents the day of the year (1 through 366) on which the disruption occurred. Figure 6.8 is a histogram of the frequency with which each day of the year appears within the table.

**Figure 6.8: Pipeline Disruptions per Day of the Year (2010 through May 2019)**




Comparing this distribution on the left to the Integer distribution shown in the left-most panel of Figure 6.7, we could provisionally suggest that pipeline disruptions occur with almost equal likelihoods on any day of the year.

Recall that our data table only has dates on which disruption did happen, but not on days without incident. The data spans the period from January 1, 2010 through May 31, 2019, which comprises 3,438 days. In all, 1,154 incidents were reported.

A tally of the data shows that there were incidents on 983 of those days—meaning that 2,455 days had no pipeline disruptions at all. Using this observed information, the sample mean of events per day is 1,154/3,438= 0.3357 disruptions per day. Might a Poisson model be a reasonable approximation of this process?

To answer that question, we could look at the frequencies from a different perspective. Using JMP, I tabulated the number of events occurring each day during the sample period, and then counted the number of days on which 0, 1, 2… events occurred.[2] The right-hand panel of Figure 6.8 displays that distribution. Within the sample, there were no disruptions at all on 71% of the days. If we use JMP to compute the probability of 0 successes in a Poisson distribution with $\lambda$ = 0.3357, we find a theoretical probability of 0.7148, or 71.5% compared to the observed relative frequency of 71%. To this limited extent, the theory matches closely to the empirical observations.

In the next few chapters, we will make further use of simulated data to help us visualize important statistical concepts and develop deeper understanding of the ideas, tools, and techniques of statistical investigations. Before that, though, we will extend our thinking about probability distributions to include *continuous* variables.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* In these questions, we return to the **NHANES 2016** data table containing survey data from almost 10,000 people in the U.S. in 2016. For this analysis, we will focus on just the following variables:

    - **RIDRETH1:** Respondent's racial or ethnic background
    - **DMDMARTL:** Respondent's reported marital status

    For each of the following questions, use JMP to create a suitable contingency table, and report on the probability of sampling one person that satisfies the specified events. What is the probability that we sample an individual who reported being the following:

    a. Mexican American?
    b. Mexican American or never married?
    c. Mexican American and never married?
    d. Never married, given that they are Mexican American?

    Finally, use your responses to consider the question of independence:

    e. Are the events "Never married" and "Mexican American" independent? Explain.

2. *Scenario:* The following questions were inspired by an article on alcohol consumption and risky behaviors (Hingson *et al.,* 2009) based on responses to a large national survey. The data table **BINGE** contains summary frequency data calculated from tables provided in the article. We'll focus on the self-reported frequency of binge drinking (defined as five or more drinks in a two-hour period for males and four or more drinks in a two-hour period for females) and whether the respondents reported having ever been involved in a car accident after drinking.

   For each of the following questions, use JMP to create a suitable contingency table, and report on the probability of sampling one person who satisfies the specified events. What is the probability that we sample an individual who reported:

   a. binge drinking at least once a week?
   b. never binge drinking?
   c. having had an accident after drinking?
   d. having had an accident after drinking or binge drinking at least once a week?
   e. having had an accident after drinking given binge drinking at least once a week?
   f. binge drinking at least once a week given having had an accident after drinking?

   Finally, use your responses to consider the question of independence:

   g. are binge drinking at least once a week and having had an accident after drinking independent events? Explain.

3. *Scenario:* In this scenario we return to the data from the American Time Use Survey (**TimeUse**). Our data table contains observations from nearly 16,000 respondents in three different years, giving us 43,191 rows of data. Use the Data Filter to select and include just the 2017 responses. Suppose we select one 2017 respondent at random. What is the probability that we have chosen an individual who:

   a. is not in the labor force?
   b. works part-time?
   c. works part-time or is female?
   d. works part-time, given that she is female?

   Finally, use your responses to consider the question of independence:

   e. Are the events "works part time" and "female" independent? Explain.

4. *Scenario:* This scenario continues the analysis of the **Pipeline Safety** data table. Each row of the table represents an incident in which a natural gas pipeline was somehow disrupted. For each of the following questions, use JMP to create a suitable contingency table, and report on the probability of sampling one disruption incident that satisfies the specified events. What is the probability that we sample an incident that:

   a. was caused by equipment failure? (HINT: column **CAUSE**)
   b. ignited?
   c. required evacuation of the public in the area?
   d. required evacuations, given that the pipeline ignited?
   e. involved ignition or an explosion?

Now we return to the issue of how well a Poisson model fits these data. Figure 6.8 reported that 1 incident occurred 24% of the time, 2 incidents 3% of the time, and 5 incidents did not occur within our sample.

f. Use JMP to compute probabilities of 1, 2, and 5 incidents assuming a Poisson distribution with $\lambda = 0.3357$. Compare the probabilities to the empirical percentages and comment on your findings.

5. *Scenario:* The next few questions refer to data from the **Hubway** data. This table contains observations of 55,230 individual bike trips using a bike rental service in Boston, Massachusetts. People can register with the program or simply rent a bike as needed. Individual trips within the data table are labeled as taken by *Registered* or *Casual* users. For each of the following questions, use JMP to create a suitable contingency table and report on the probability of sampling one trip that satisfies the specified events. What is the probability that we sample a trip:
   a. by a registered user?
   b. by a female rider?
   c. by a female who is a registered user?
   d. by a female, given that the rider is registered?
   e. that began at South Station – 700 Atlantic Avenue?
   f. that ended at Boston Public Library – 700 Boylston St?
   g. that started at South Station and ended at the Boston Public Library?
   h. that ended at the Boston Public Library given that it began at South Station?
6. *Scenario:* The data table called **NC Births** contains a random sample of 1,000 individuals drawn from a larger data set from the state of North Carolina.[3] In 2004, the state gathered and released an anonymized data from a large number of recent births. The data table includes several attributes of the mother, father, and baby.

   For each of the following questions, use JMP to create a suitable contingency table and report on the probability of sampling one person who satisfies the specified events. What is the probability that we sample an individual who is a:
   a. smoker with a premature baby
   b. smoker with a low birth weight baby
   c. mature mom and smoker

   Finally, use your responses to consider the question of independence:
   d. Are the events “smoker” and “low birth weight” independent? Explain.

7. *Scenario:* Recall the FAA bird-strike data that we examined in a prior chapter. In that instance, we used a subsample of data from three California airports. We have the full data set called **FAA Bird Strikes**. In this question, we focus on the extent to which two observed variables appear to follow an Integer Distribution.
    a. The column labeled **Month** indicates the month of the bird-strike event. Create a histogram of that column. Do strikes occur with equal frequency across the twelve months of the year? Explain your thinking.
    b. The column labeled **DayofWeek** indicates the day of the week (Sunday = 1, Saturday = 7) of the bird-strike event. Create a histogram of that column. Do strikes occur with equal frequency across the seven days of the year? Explain your thinking.

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[1] For the sake of clarity, this section uses labels from the data table to identify events rather than using symbols like A and B. **Feed** refers to the event "observed a dolphin feeding."

[2] The summary is stored in the data table called Pipeline_Safey By (DATE). To construct the histogram, it was necessary to adjust the summary to reflect the number of dates on which no incidents occurred.

[3] This data table comes from *OpenIntro Statistics, 2nd Ed*. (2012) by David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel, available online at http://www.openintro.org/stat/textbook.php.

# Chapter 7: The Normal Model

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## Overview

The prior chapter presented some probability models that can represent discrete data. The focus of this chapter is on one essential model widely applicable to continuous data: the normal model. In addition to usefully representing many real-world phenomena, the normal model underlies many traditional inferential techniques and can help us understand standard approaches to inference.

## Continuous Data and Probability

In Chapter 6, we developed a model of discrete random variables based on the concept of a probability value associated with individual possible values of a random variable. In this chapter, we will consider *continuous* random variables. Continuous data has the property that there are an infinite number of possible values between any two possible values. For example, the winning time in the next Olympic marathon is a continuous variable: no matter how thinly we slice the imaginable values, it is always possible that the winner's time will fall between two values that we imagined. As a practical matter, we tend to measure continuous variables discretely—the official scores might report race times to the nearest $100^{th}$ second—but that does not change the fact that time is continuous.

Because of the nature of continuous data, our discrete model of probability distributions does not apply to continuous data. We cannot list all possible values of a continuous random variable, and because of the infinitesimal nature of the variables, each possible value has a probability of zero. We need a different approach.

## Density Functions

For a continuous random variable X, we will think about the probability that X lies between two values instead of thinking about X assuming one specific value. In other words, rather than solving for Pr(X = a), we will solve for Pr(a ≤ X ≤ b). Fortunately, the software handles all of the computational issues, but it's helpful to have some basic grounding in the underlying theory.

With continuous random variables, probability values are derived from the *density function* that describes the variable. From the standpoint of data analysis, density functions *per se* have little direct application. You might think of a density function as representing the concentration of probability near different possible values of X.

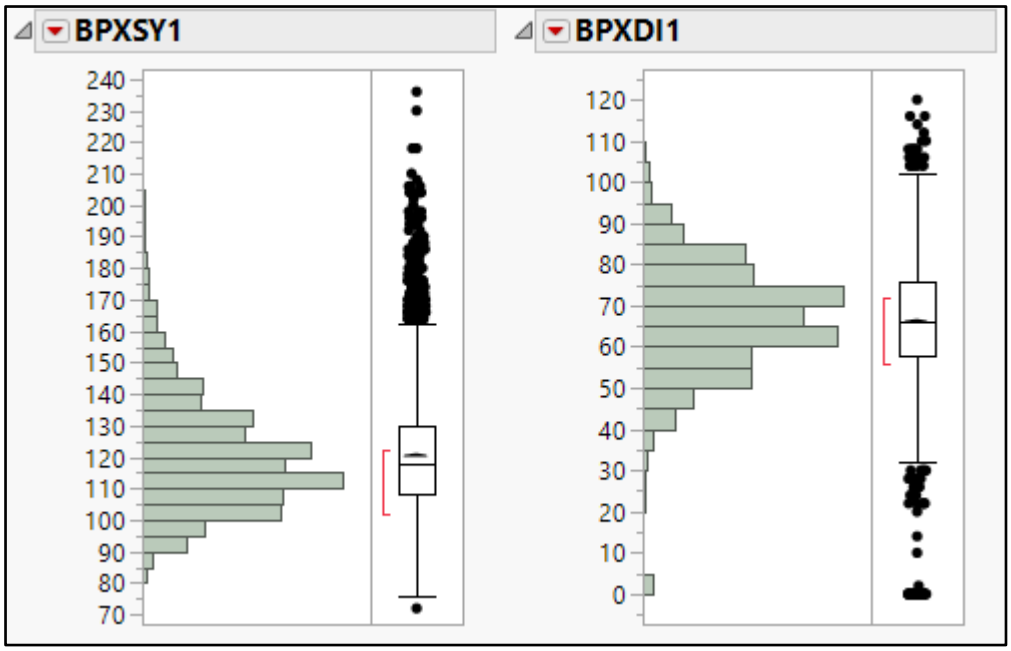
To help illustrate this concept, we will take advantage of one of the graphing features of JMP and look at some histograms in a new way.

1. Open the **NHANES 2016** data table once again.

We have worked with this table before; it contains health-related measurements and survey responses gathered in 2016 from 9,971 people in the United States. Two of the columns in the table contain the blood pressure readings of the adults in the sample. Blood pressure—the force exerted by flowing blood on the walls of arteries and veins—is a continuous variable. By standard practice, it is measured to the nearest whole number, but it is continuous nonetheless. When JMP constructs a histogram of these measurements, it groups similar values into bins, and we see a relatively small number of bars in the graph.

2. Select **Analyze ▶ Distribution**. Cast the last two columns **BPXSY1** and **BPXD1** as **Y** and click **OK**. Your results will look like Figure 7.1.

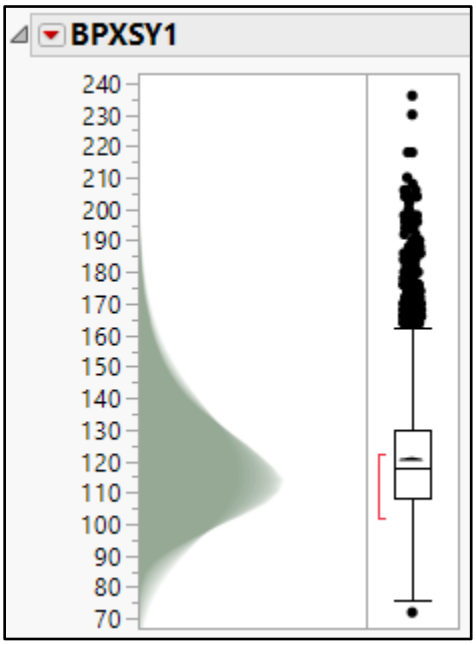
**Figure 7.1: Histograms of Blood Pressure Measurements**




The bars in these histograms are the default choices selected by JMP. In Chapter 3, we learned to adjust the number and therefore the width of the bars. The bars are an artificial convention used to represent the shape of the distribution. JMP provides another way to depict a continuous distribution that better conveys the concept of density.

3. Click the red triangle next to **BPXSY1**. Select **Histogram Options ▶ Shadowgram**. The graph should now look like Figure 7.2.

**Figure 7.2: Shadowgram of Systolic Blood Pressure Data**




We have seen shadowgrams several times now. Be aware that they are a unique feature in JMP. When one varies the bin width in the histogram, the apparent shape of the distribution changes. The software averages the effects of many possible bin widths, and the shadowgram is the visual result.

In Figure 7.2, the jagged steps of the histogram are replaced with a smooth, shadowy density graph. Both graphs clearly show that systolic blood pressures are most common near 110 to 120 mmHg[1] and fall off in frequency at lower and higher pressures.

4. Convert the histogram of diastolic pressure (**BPXD1**) to a shadowgram and compare the two ways of showing the data. Do you notice the difference in visual impressions created by the two graphs? The shadowgram should leave the impression of a variable with smoothly changing likelihoods.

With continuous values, probability is computed as the area under the density curve between two points *a* and *b*. Some density curves have infinite domains; others have minima and maxima. For all density functions, the area under the entire curve equals 1, analogous to the property that probabilities for a discrete variable sum to 1. As you advance through your study of statistics and encounter an increasing portfolio of applications, you will eventually work with several commonly used classes of density functions. In this chapter, we will focus on the *normal* (also known at the *Gaussian*) density function, which has wide application and is particularly relevant to the analysis of data from simple random samples.

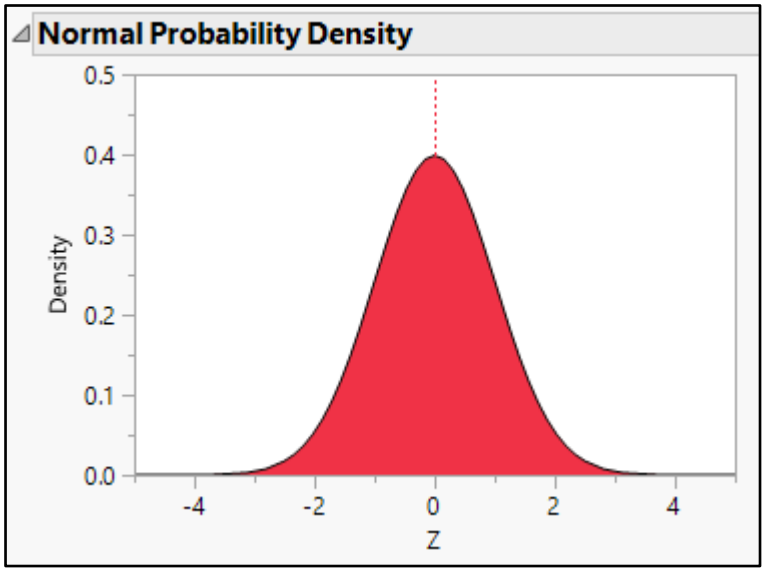
# The Normal Model

As in Chapter 6, we will rely on your principal textbook to provide an extensive introduction to normal distributions. In this section, we will review several attributes of the normal distributions that are essential for practical data analysis. Though we can use normal distributions in many ways, one frequent application is as a model for the population or process from which we are sampling data. Specifically, we often want to ask whether a normal model suitably describes observations of a random variable.

There are an infinite number of normal distributions, each uniquely identified by its mean $\mu$ and standard deviation $\sigma$. Figure 7.3 shows the graph of a normal distribution known as the *Standard Normal Distribution* whose mean equals 0 and standard deviation equals 1. The standard normal variable is commonly represented by the letter Z and is the basis for z-scores. This curve illustrates several of the major properties of all normally distributed random variables:

- If we know that a variable follows a normal distribution with mean = $\mu$ and standard deviation = $\sigma$, we know its shape, center, and spread; in short, we know everything that we need to know to find probabilities.
- The distribution is symmetric, with its mean at the center. Half of the area lies on either side of the mean.
- The mean and median of the distribution are equal.
- Nearly the entire area of the density curve lies within three standard deviations of the mean.
- The density function has an infinite domain. The tails of the curve (the ends nearest the X axis) approach the axis asymptotically.

**Figure 7.3: A Normal Probability Density Function**



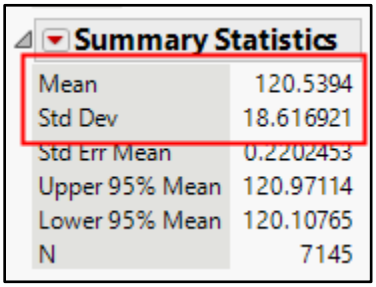
# Normal Calculations

There are two classes of problems that we typically solve using a normal distribution:

- For X values *a* and *b,* we want to solve for *Pr(a ≤ X ≤ b)*. In these problems, we are solving for *cumulative probability.*
- For a probability value, *p*, we want to solve to find *x** such that *Pr(X ≤ x*)* = *p*. In these problems, we are solving for *inverse cumulative probability.*

For the sake of illustration, consider a normal distribution that shares the same mean and standard deviation as the sample of systolic blood pressure measurements. Figure 7.4 displays the summary statistics for our sample of 7,145 readings. Although there were almost 10,000 people in the sample, researchers do not record blood pressures for children. The sample mean is 120.539 mgHg, and the sample standard deviation is 18.617mmHg. For the sake of this illustration, then, we will consider a normal distribution with $\mu$ = 120.539 and $\sigma$ = 18.617, which from now on we will write as X ~ N*(120.539, 18.617)*.

**Figure 7.4: Summary Statistics for the Systolic Blood Pressure Data**




## Solving Cumulative Probability Problems

Let us introduce the variable X ~ *N(120.539, 18.627)* as a model for adult systolic blood pressure and use this model to estimate the proportion of adults whose blood pressure falls within specific intervals. The American Heart Association advises that "normal" systolic blood pressure is below 120 mgHg.[2] Suppose we want to use this model to estimate the percentage of adults who have systolic pressure below 120 mgHg.

Recall that cumulative probability is defined at the probability that a variable is less than or equal to a given value. In JMP 15, there are two methods that we can use to find probabilities with normal distributions. The first method uses a specialized calculator with a visual interface. In the second approach, we set up a small data table and use cumulative probability density functions to compute whatever probability values we want to find. The first method is easier to learn, but the second provides additional capability and control. We will see both methods applied to the same problem.

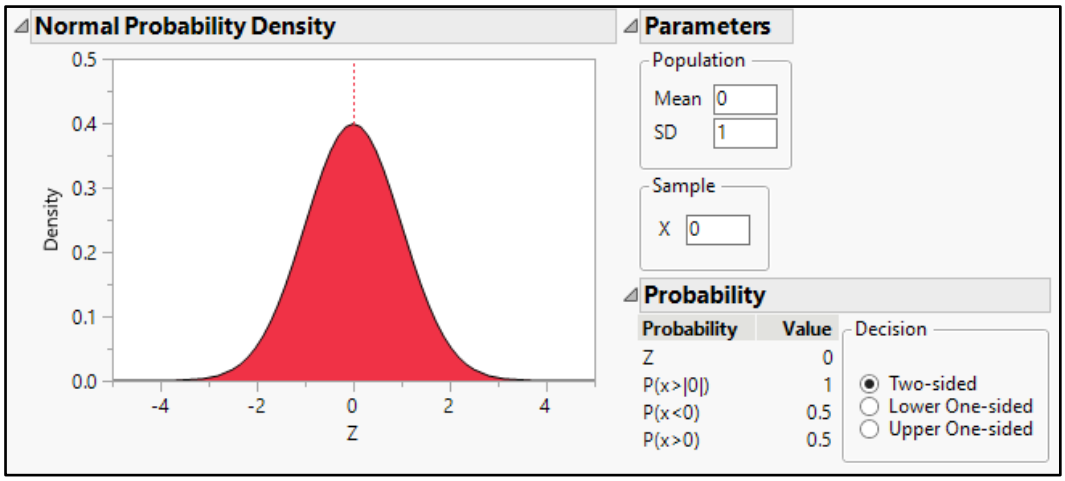
### The "Easy" Way

All normal distributions share the same shape but differ with respect to mean and standard deviation. JMP's calculator takes advantage of that fact to compute areas under the curve. We will specify the two *parameters* of the distribution as well as the X value of interest (120 mgHg), and the calculator will do the rest.

1. Choose Help ► Sample Data. This opens the Sample Data Index.
2. Near the lower right, choose Teaching Scripts and then Teaching Demonstrations.
3. Scroll to the bottom of the list and select **UtilNormalDistribution.**

You should now see the window shown in Figure 7.5. There are three numerical inputs at the upper right (Mean, SD, and X) and the results appear in the graph and in the Probability area of the window.

**Figure 7.5: The Initial State of the Normal Distribution Calculator**



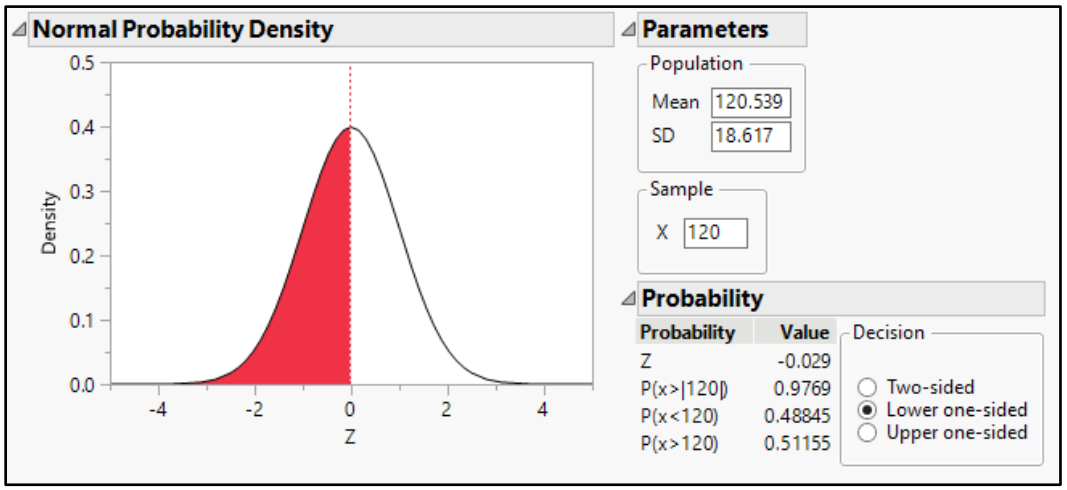

4. In this example, want to analyze a model normal population distribution with the parameters cited above, namely a mean of 120.539 mgHg and a standard deviation of 18.617 mgHg. Enter those values in the corresponding boxes.
5. We are seeking the area to the left of a blood pressure of 120. This will be our **Sample** value of X.
6. Finally, we want to graph the area to the left of X = 120, so click the radio button marked **Lower One-Sided**.

> When referring to continuous distributions, the portions of the curve that approach the horizontal axis are often called "tails;" you will see references to the lower and upper tails. "Lower" indicates the Z-values farthest to the left from 0, not to the height of the density curve.

When you are finished, your screen will look like Figure 7.6. From the graph, we see that slightly less than half of this theoretical model of adults in the U.S. have systolic pressures less than 120

mgHg. The list of probabilities shows that P(x < 120) = 0. 48845, or about 49% of adults, have normal systolic numbers.

**Figure 7.6: The Problem Solved**




## The Less-Easy, but More Powerful Way

We will use the Normal Distribution function in JMP to find *Pr(X ≤ 120)*. To do so, follow these steps:

1. Select **File ► New ► Data Table.**
2. Rename **Column 1** as **Systolic** and initially create the column with two empty rows.
3. Move your cursor into the first empty cell of the **Systolic** column and type **120**.
4. Add a new column called **CumProb (Cols ► New Columns)** and under **Column Properties** choose **Formula**.
5. From the list of **Functions** select **Probability ► Normal Distribution** (be careful not to choose Normal Density) and complete the formula as shown in Figures 7.7 and 7.8. Click **OK** twice.

> Take extra care in entering the arguments for the Normal Distribution function. For the first argument choose the column **Systolic**. Then type a comma ( **,** ) and enter the mean. Again, type a comma and enter the standard deviation.

In the new data table, you will now see the value 0.4884514233. This indicates that for this specific normal distribution, 48.8% of the distribution lies to the left of 120 mmHg, or that according to this model, approximately 49% of adults have systolic readings in the normal range.

**Figure 7.7: Creating a Column Using the Normal Probability Function**

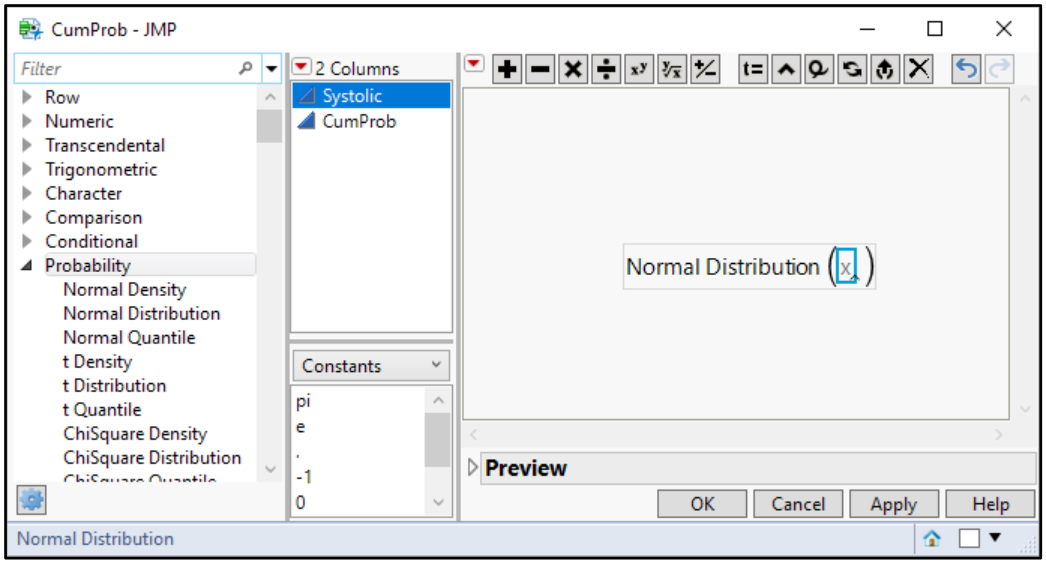
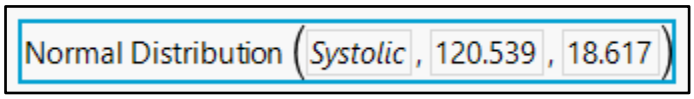



**Figure 7.8: Normal Distribution Function with Arguments**




Let's consider another example. Systolic pressures above 140 are generally considered high. According to this model, what proportion of the population has readings in the high range?

6. Within the data table, enter the value **140** in the second row of the **Systolic** column.
7. The corresponding value in the **CumProb** column is now **0.8520658956**.

This is the probability of observing a reading less than 140; the probability of the complementary event is approximately 1 − 0.852 = 0.148. In other words, in a population described by this model, approximately 14.8% of the people have high blood pressure.

As a final example, we will find the probability between two X values. In our data set, the first and third quartiles were 108 and 130, respectively. Let's calculate the probability between these two values using our model.

8. In the first row of the data table, replace **120** with **130**.
9. Replace **140** with **108** in the second row.

The **CumProb** column will now contain the (rounded) values 0.694 and 0.250. Because 69.4% of the distribution is less than 130, and 25.0% is less than 108, we conclude that 69.4% – 25.0% = 44.4% of the theoretical normal distribution lies between 106 and 128 mmHg.

At this point, we should note that this model is not a perfect match to our observed data with respect to the quartiles. In the empirical data table, 75% of the observations fell below 130, but

only 69.4% of the model distribution lies below 128. On the other hand, 25% of the actual observations are less than 108 mmHg, which is a nearly perfect match to our model. Fully one-half of the sample observations are between 106 and 128 mmHg, and only about 44% of the model is between the two numbers. Soon, we will consider the question of how closely a model corresponds to an observed table of data. First, let's see how inverse probability calculations work.

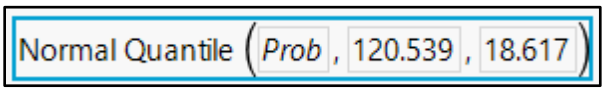
## Solving Inverse Cumulative Problems

Suppose that public health officials wanted to address the group of adults with the highest systolic blood pressure, and as a first step, want to identify the top 10% of such individuals. According to our model, we already know that that about 14.8% of the population has blood pressure above 140 mmHg; at what pressure reading have we reduced the group to 10%?

We want to find $x^*$ such that $Pr(X > x^*) = 0.10$. We do this with the *Normal Quantile* function in JMP. This function returns a value of $x^*$ that satisfies the cumulative probability condition $Pr(X \leq x^*) = p$ for any probability value ($p$) that we choose. So, our first step is to recognize that the problem that we want to solve is the equivalent of solving for $Pr(X \leq x^*) = 0.90$. In other words, we want to find the 90$^{th}$ percentile of the distribution.

1. Add a column to our data table. Name the column **Prob** and type the value **0.9** in the first row.
2. Add another new column named **Inverse** and choose **Column Properties ► Formula**.
3. As we did earlier, choose **Probability** from the list of functions. This time, select the **Normal Quantile** function, and supply the arguments as shown in Figure 7.9.

**Figure 7.9: Normal Quantile Function to Compute Inverse Probabilities**




In the new column of the data table, you will now see the value 144.3976. This value divides the distribution with 90% to the left of the value and 10% to the right. In short, in our model population, approximately 10% of the pressure readings are above 144.4 mmHg.

Sometimes we want to use inverse probability to identify the middle $p$% of a normal distribution. For example, earlier we referred to the first and third quartiles of our sample data. We know that the quartiles mark off the middle half of the data. We might similarly want to find the middle 50% of this normal distribution. One way to do this is to recognize that the middle 50% is defined as the region that is 25% below the center and 25% above the center, defined by the 50 – 25 = 25$^{th}$ percentile and the 50 + 25 = 75$^{th}$ percentile. Just as we did for the 90$^{th}$ percentile, we can use our formula to find the 75$^{th}$ and 25$^{th}$ percentiles.

4. In the data table, change the value **0.9** in the **Prob** column to **0.75**.
5. Enter the value **0.25** in the second row of the **Prob** column.

There will be two new values in the **Inverse** column that round to 133.1 and 108.0, corresponding to the quartiles of the theoretical normal distribution. In the observed data, the first quartile was 108 and the third was 130. Here (consistent with our earlier calculations), we see a match between the observed and theoretical first quartile and a more disparate contrast between the third quartiles. The degree of correspondence brings us to the next topic in this chapter.

# Checking Data for the Suitability of a Normal Model

We have already noted that there are times when we will want to ask how well a normal distribution can serve as a model of a real-world population or process. Because there are often advantages to using a normal model, the practical question often boils down to asking if our actual data deviate too grossly from a normal distribution. If not, a normal model can be quite useful.

In this chapter, we will introduce a basic approach to this question, based on the comparison of observed and theoretical quantiles (that is, the types of comparisons that we have just made in the prior section). In Chapter 9, we will return to this topic and refine the approach.

## Normal Quantile Plots

In the previous section, we computed three different quantiles (percentiles) of the normal variable X ~ *N(120.539, 18.617).* We also have a large data table containing well over 7,000 observations of a variable that shares the same mean and standard deviation as *X*. We also know the medians (50$^{th}$ percentile) for each. If we compare our computed quantiles to the observed quantiles, we see the following:

**Table 7.1: Comparison of Observed and Theoretical Quantiles**

| Percentile Value | Observed Value BPXSY1 | Computed Value of *X* |
|---|---|---|
| 25 | 108 | 107.98 |
| 50 | 118 | 120.54 |
| 75 | 130 | 133.10 |
| 90 | 146 | 144.40 |

The observed and computed values are similar, though not identical. If **BPXSY1** were normally distributed, the values in the right two columns of Table 7.1 would match perfectly.

We could continue to calculate theoretical quantiles for other percentiles and continue to compare the values one pair at a time. Fortunately, there is a more direct and simple way to carry out the comparison—a *Normal Quantile Plot* or *NQP*.[3]

In JMP, a normal quantile plot is a scatterplot with values of the observed data on the vertical axis and theoretical percentiles on the horizontal.[4] If the normal model were to match the observed data perfectly, the points in the graph would plot out along a 45° diagonal line. For this reason, the plot includes a red diagonal reference line. To the extent that the points deviate from the line, we see imperfections in the fit. Let's look at two examples.

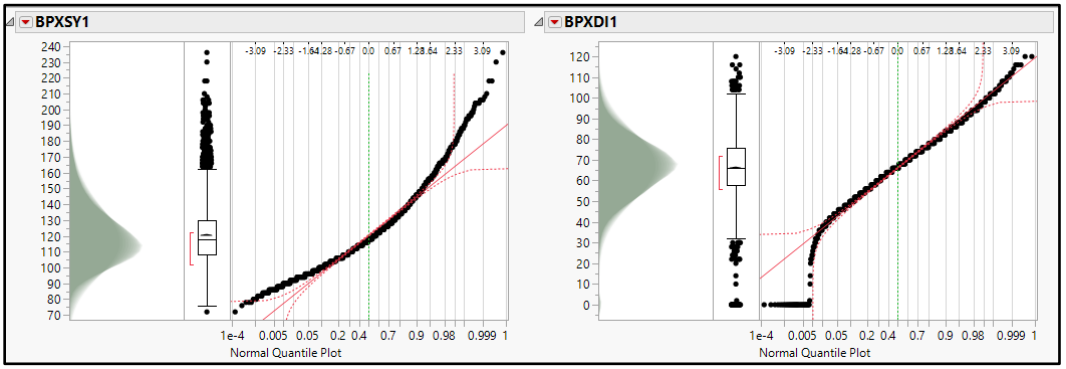
1. Return to the **NHANES 2016 Distribution** report window.
2. Hold down the Ctrl key and click the red triangle next to **BPXSY1**; select **Normal Quantile Plot**.

> Remember that pressing the Ctrl key while clicking a red triangle applies changes or selections to all open panels.

Figure 7.10 shows the plots for both blood pressure columns. Neither shows a perfectly straight diagonal pattern, but the plot of diastolic pressure on the right more closely runs along the diagonal for most of the distribution. The normal model fits poorly in the tails of the distribution but fits well elsewhere.

Recall that we have 7,145 observations here. The shadowgrams and box plots show that the diastolic distribution is much more symmetric than the systolic. Both samples have many outliers

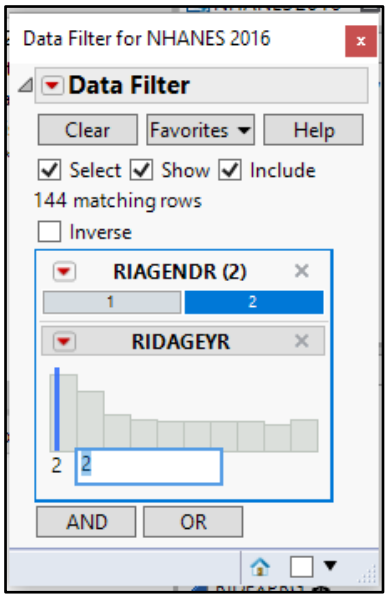
**Figure 7.10: Normal Quantile Plots for Blood Pressure Data**




As an example of a quite good fit for a normal model, let's look at some other columns from a subset of the **NHANES 2016** data table, selecting just the two-year-old girls in the sample.

3. Select **Rows ► Data Filter.** We need to specify two conditions to select the rows corresponding to two-year-old girls. Within the dialog box, use the Ctrl key to choose **RIAGENDR** and **RIDAGEYR** and click **Add**. Check the **Show** and **Include** boxes (shown in Figure 7.11).
4. For gender, click the **Female** button.
5. Click the 0 to the left of **RIDAGEYR** and change it to a 2; do the same with the 80 to the right side. This will select the two-year-old girls.

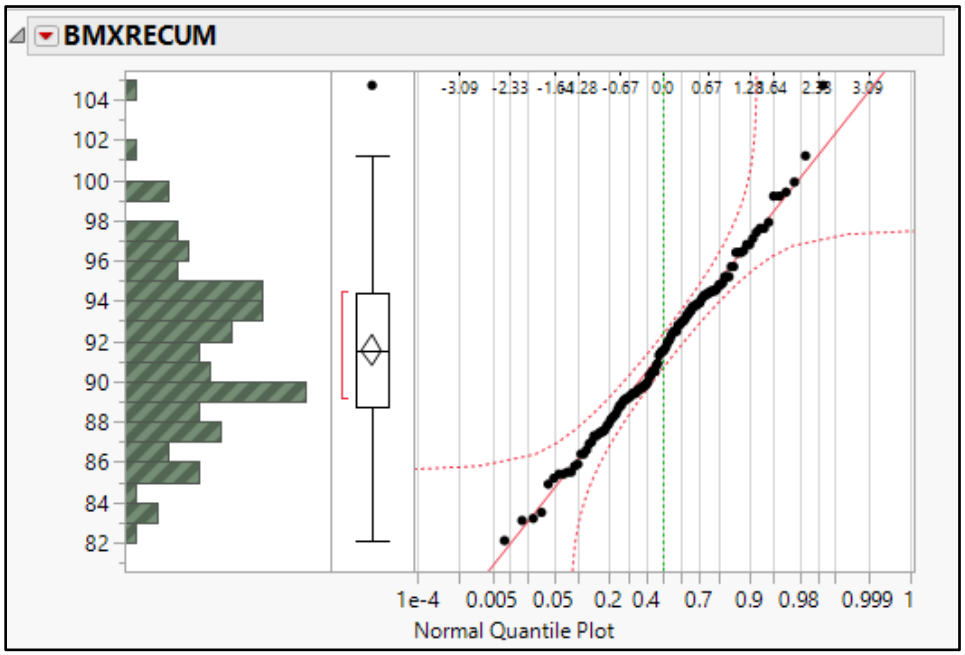
**Figure 7.11: Completed Data Filter Dialog Box**




Scroll down the rows of the data table and you will find a relatively small number of rows selected. In fact, there are just 144 two-year-old girls in this sample of almost 10,000 people.

6. Select **Analyze ► Distribution**. Cast **BMXRECUM** as **Y**. This column is the recumbent (reclining) height of these two-year-old girls. In our data table, we have measurements for 122 of the 144 children.
7. Create a normal quantile plot for the data; it will look like Figure 7.12. Here we find that the points are very close to the diagonal, suggesting that the normal model would be very suitable in this case.

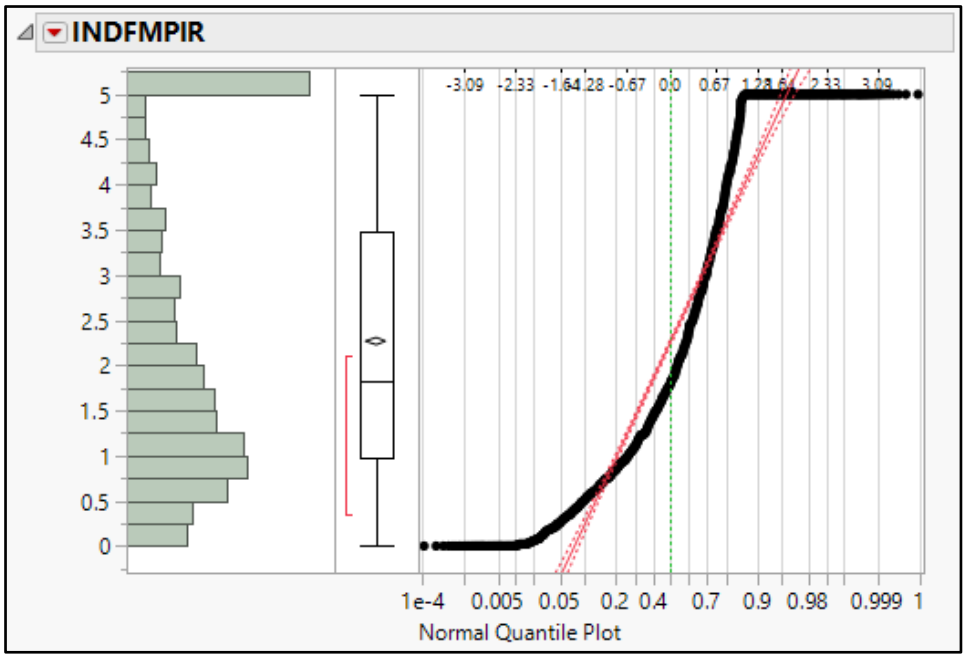
**Figure 7.12: Histogram and Normal Quantile Plot for Recumbent Length**




8. As a final example in this section, clear all row states and perform all the steps necessary to create a normal quantile plot for **INDFMPIR**. The correct result is shown in Figure 7.13.

This column is equal to the ratio of family income to the federally established definition of poverty. If the ratio is less than 1, this means that the family lives in poverty. If the ratio is more than 5.0, **NHANES** records the value as equal to 5; there are 1,120 such individuals in this subsample.

**Figure 7.13: Distribution of Family Income Poverty Ratio**




In contrast to the prior graphs, we should conclude that a normal distribution is a poor model indeed for this variable. One could develop a normal model, but the results calculated from that model would rarely come close to matching the reality represented by the observed data.

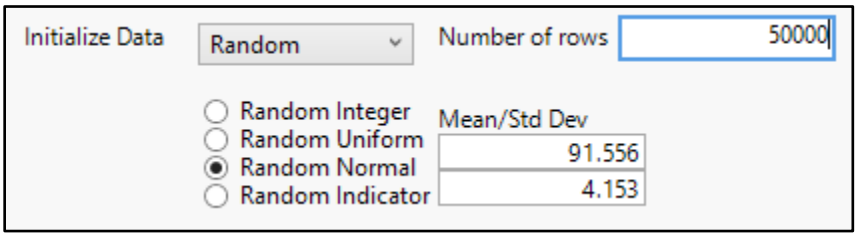
## Generating Pseudo-Random Normal Data

We conclude this chapter in much the same way that we concluded Chapter 5. We might sometimes want to generate a simulated sample of random data that conforms to an approximate normal distribution. We could use the **Random** function as we did in Chapter 6, but JMP provides an even easier method.

To illustrate, we will use the current data table and create some simulated data for a normal random variable with a mean and standard deviation matching the recumbent length measurements. Look back at the distribution of **BMXRECUM** and you will see that the mean length was 91.556 cm and the standard deviation was 4.153.

1. Select **File ► New ► New Data Table**.
2. Double-click in the title of **Column 1**. Name the new column **Normal_Sim**. In the dialog box, change the **Initialize Data** from **Missing/Empty** to **Random**.
3. As shown in Figure 7.14, select **Random Normal** and enter a mean value of **91.556** and a standard deviation value of **4.153**, and specify 50,000 rows to provide a large sample.

**Figure 7.14: Filling a New Column with Normal Random Data**




4. Analyze the distribution of this new column and generate a normal quantile plot.

Because each simulation is different, your results will be unique. However, you should find that the normal quantile plot generally, if imperfectly, follows the red diagonal line. You will also find that the sample mean is approximately 91.56, and the sample standard deviation is approximately 4.15. Why? Remember that probability theory is about the long-run (or large sample) patterns that emerge from many replications. This simulation shows that the shape, center, and dispersion of a random variable are predictable even when a single measurement is not.

The summary statistics of your simulated sample will almost always deviate slightly from the population parameters that you specified. The same is true of real random samples, and that simple and unavoidable fact sets the stage for the next chapter.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* We will continue to examine the World Development Indicators data in **BirthRate 2017**. This time, we will see if we can use a normal model to represent one or more variables.
   a. Complete the steps necessary to create a normal quantile plot for the **Fertil** column. **Fertil** is the mean number of children that women have during their child-bearing years in each country. Report on what you see in the graph and what you conclude about the suitability of a normal model.
   b. Consider a random variable $X \sim N(2.711, 1.321)$. Such a variable has the same mean and standard deviation as **Fertil**. For **Fertil**, 10% of observations fell below 1.438 children. What proportion of X lies to the left of 1.438?
   c. Using the same X, find $Pr(X > 5.5)$. Compare your result to the observed data for **Fertil**.
   d. Find the third quartile for X and compare the result to the third quartile for **Fertil**.

2. *Scenario:* In the late 19th century, the physicist Albert A. Michelson did ground-breaking research on the speed of light. Some of his early measurements are in the data table **Michelson 1879.**
   a. Complete the steps necessary to create a shadowgram of the **Velocity** data. Paste a copy of the graph in your report and comment on what you see.
   b. Construct a normal quantile plot for the **Velocity** column. Do you think that a normal model is suitable for this variable? Explain.
   c. "Measurement error" is sometimes assumed to be normally distributed. What does this set of data suggest about that assumption?
3. *Scenario:* In Chapter 2, we first looked at the **Concrete** data table containing experimental data. The point of the experiment was to develop a concrete mixture that has high compressive strength.
   a. Complete the steps necessary to create a normal quantile plot for the **Compressive Strength** column. Report on what you see in the graph and what you conclude about the suitability of a normal model.
   b. Consider a random variable $X \sim N(35.818, 16.706)$. Such a variable has nearly the same mean and standard deviation as **Compressive Strength**. Use this normal model and find its 10th, 25th, 50th, 75th, and 90th percentiles. Compare your computed values to the observed corresponding percentiles and report what you find.
4. *Scenario*: Normal models are frequently used to describe the fluctuations in stock market prices. Open the data table called **Stock Index Weekly**. This table contains the weekly values, volumes, and percentage changes in six major international stock market indexes for all of 2008. In the fall of 2008, world financial markets suffered precipitous declines in value.
   a. Construct NQPs for the last five columns in this table. Which column would you say is best described by a normal model? Explain your thinking.
   b. Which column would you say is least described by a normal model? Explain your thinking.
   c. Identify the mean and standard deviation of a normal distribution that might be suitable for modeling the weekly changes in the HangSeng index. Use your model to estimate the percentage of weeks that the index lost value (that is, had a weekly change that was less than or equal to zero).
   d. Compare your result in the prior question to the actual percentage of weeks that the HangSeng lost value in 2008. Comment on the comparison.

5. *Scenario:* In this scenario, we return to the data from the American Time Use Survey (**TimeUse**). Our data table contains observations from approximately 16,000 respondents in 2003, 2007, and 2017. Use the Data Filter to select and include just the 2003 responses.
    a. Complete the steps necessary to create a shadowgram of the **Sleeping** data (that is, time spent sleeping). Paste a copy of the graph in your report and comment on what you see.
    b. Construct a normal quantile plot for the **Sleeping** column. Do you think that a normal model is suitable for this variable? Explain.
    c. Repeat steps a and b for the column containing the ages of the respondents.
    d. Repeat steps a, b, and c for the 2017 data. Comments on changes (if any) that you observe.
6. *Scenario:* In this scenario, we return to the **NIKKEI225** data table with values from the Tokyo Stock Exchange index. We will focus on the monthly closing value of the index, the daily percentage change, and the daily volume (number of shares traded).
    a. Use the **Distribution** platform to summarize the monthly adjusted closing values and the monthly percentage change. Comment on interesting features of these two distributions.
    b. The percentage changes are calculated from the data in the **Adj Close** column, and the histogram appears to be more normal than the corresponding one for the monthly adjusted closing figures. In fact, computing percentage changes is one strategy that analysts use to transform time series into a nearly normal variable. Construct normal quantile plots for both **Close** and **Change%** and report on your findings.
    c. Would a normal model be an apt description of the daily volume variable? If so, which normal model fits this variable?
7. *Scenario:* In this scenario, we return to the World Development Indicators data (**WDI**). Use the **Data Filter** to **Show** and **Include** only data from the year **2017**.
    a. Two columns measure the prevalence of telecommunications technology around the world. They are the number of mobile cellular subscriptions per 100 people (**cell**) and the number of fixed broadband Internet subscribers per 100 people (**broadband**). Construct histograms and NQPs for the two columns. Which column is better summarized as an approximate normal distribution? Explain your reasoning.
    b. Using the mean and standard deviation of the more normal column, use JMP to estimate the percentage of countries with 70 or fewer subscribers per 100 people. Explain the approach that you took.
    c. As in part b, estimate the percentage of countries with 150 or more subscribers per 100 people.
    d. Approximately 90% of countries have *at least* X subscribers per 100 people. Find X and explain how you did so.

8. *Scenario:* The **States** data table contains measures and attributes for the 50 U.S. states plus the District of Columbia. Two of the columns are **poverty** (percentage of individuals living below the poverty level) and **med_income_17** (median income per capita).
   a. Complete the steps necessary to create NQPs for these two columns.
   b. Comment on what you see in the graphs and what you conclude about the suitability of the normal model for these columns.
9. *Scenario:* In this scenario, we return to the Federal Aviation Administration data on wildlife strikes (**FAA Birdstrikes**).
   a. Prepare normal quantile plots for height and speed. Which column is better described by a normal model? Explain your thinking.
   b. We know from the **Quantiles** for **Speed** that 25% of the strikes occur at speeds of 120 mph or less. Use the normal model with mean and standard deviation equal to the **Summary Statistics** for **Speed** to estimate the percentage of the distribution below 120. Comment on the comparison.
   c. From the **Quantiles**, we also know that about 10% of the bird strikes occur at speeds in excess of 210 mph. What is the area of the same normal curve above 210? How does this compare to the observed 90$^{th}$ percentile?
   d. Compare your findings in parts b and c. How do you explain the comparisons between the observed data and the normal approximations?

---

[1] mmHg is the abbreviation for millimeters of mercury (chemical symbol Hg), based on the now old-fashioned mercury gauges used to measure blood pressure.

[2] See "Highlights from the 2017 Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults" at https://www.heart.org/-/media/data-import/downloadables/hypertension-guideline-highlights-flyer-ucm_497841.pdf.

[3] Some texts present Normal Probability Plots (NPP) as an alternative. We interpret both in similar fashion.

[4] These are the default axis settings.

# Chapter 8: Sampling and Sampling Distributions

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## Overview

Although current technology makes it possible to analyze population-level data nearly in real time, a great deal of statistical analysis begins with carefully selected samples from a population. The use of sample data introduces a layer of uncertainty for analysts who need to make conclusions about the parent population. This chapter is devoted to two topics: the relationship between the *method* of sampling and the amount of uncertainty, and the nature of the uncertainty created by simple *random sampling.*

## Why Sample?

Early in Chapter 2, we noted that we often analyze data because we want to understand the variation within a population or a process, and that usually we work with just a subset of the target population or process. Such a subset is known as a *sample*, and in this chapter, we focus on the variability across possible samples from a single population.

It is usually impossible or impractical to gather data from every individual within a population. In Chapter 2, we presented the following sequence that describes the typical situation in statistical studies:

- We are interested in the variation of one or more attributes of a population.
- We cannot gather data from every single individual within the population.
- Hence, we choose some individuals from the population and gather data from them *in order to generalize about the entire population.*

We gather and analyze data from the sample of individuals instead of doing so for the whole population. The individuals within the sample are *not* the group that we are ultimately interested in knowing about. We really want to learn about the variability within the population, and to do so, we learn about the variability within the sample. Before drawing conclusions, though, we also need to know something about the variability among samples that we might have taken from this population. While many potential samples probably do a very good job of representing the population, some might do a poor job. Variation among the many possible samples is apt to distort our impression of what is happening in the population itself.

Ultimately, the trustworthiness of statistical inference—the subject of the chapters following this one—depends on the degree to which any one sample might *misrepresent* the parent population. When we sample, we run the risk of unluckily drawing a misrepresentative sample, and therefore it's important to understand as much as we can about the size and nature of the risk. Our ability to analyze the nature of the variability across potential samples depends on which method we use to select a sample.

## Methods of Sampling

Chapter 2 also introduced several standard approaches to sampling. In an introductory course, it is valuable to recognize that there are different reasonable methods. It is also valuable to understand that each method carries different risks. In this chapter, we will focus the discussion on simple random samples (SRS) and the ways in which sample proportions and sample means tend to vary in different samples from one population.

This chapter should help you understand the nature of simple random sampling. It might also help you become a more critical consumer of statistical studies by teaching you to ask questions about the implications of sampling methods other than simple random sampling. So-called complex sampling methods, such as stratification and clustering, can also lead to excellent representative samples, but they are generally considered to be beyond the scope of an introductory course. Non-random methods do not reliably yield representative samples, even though they are widely used. We will concentrate on the SRS in this chapter to develop a good foundation in the logic of statistical inference but do discuss complex methods briefly as well.

In the next part of this chapter, we will use JMP to select random samples from a *sampling frame*, a complete list of individuals within a population. Then, we will make some visual and numerical comparisons between the known population distribution and the distribution of a

random sample. We will then extend our examination of sampling by using the simulation capabilities of JMP to visualize the variability across many possible random samples. We will begin with an artificially small population from which we will select some simple random samples. Our goal is twofold: first, simply to learn how JMP can help us select a random sample, and second, to develop an understanding of the ways in which a sample might mislead a researcher. If we were truly working with such a small population, we would clearly include every observation in the analysis. Our second goal here is subtle but important—we are starting small so that we can actually examine all possible samples from this population.

## Using JMP to Select a Simple Random Sample

The concept of a random sample is straightforward. Choose a subset of a population, taking steps to ensure that all equally sized subsets share the same chance of selection. In practice, there are two common obstacles to random sampling: we often don't know the precise size or the identities of all members of the population. While common, these obstacles are easily overcome, but the resulting samples are not simple random samples. In this section, we will confine our attention to two ways in which we can use JMP to select a simple random sample.

To choose *n* observations as a simple random sample, we need a way to gather or list all elements in the population. This is known as a *sampling frame*—the set from which our random subset is drawn. We also need some type of randomizing algorithm or method.

By way of illustration, let's revisit the **Military**[1] data once again. This data table includes categorical data gathered from the 1,048,575 active-duty personnel from four branches of the U.S. military, excluding the Coast Guard, in April 2010. We will treat this group as the sampling frame. For the sake of this example, we will suppose that we want to select a 100-person random sample from the population for further follow-up study.

1. Open the **Military** data table.
2. Select **Analyze ► Distribution.** Examine the distributions of the first three columns in the table: **grade**, **branch**, and **gender**.

We see that approximately 78% of the individuals are enlisted personnel, that 6% are in the Air Force, and that 13% are female. We will keep these population proportions in mind as we examine some simple random samples. Remember that our goal here is to gain insight in the extent to which different methods of sampling produce samples that mirror the population.

> We have used random functions in several prior chapters. JMP generates pseudo-random numbers using an algorithm that we can control. In prior chapters, if you followed the instructions to generate a column of random values, you would obtain different values than the ones shown in the chapter.
>
> For this exercise, we introduce a method for you to control the starting point of the algorithm. In this way, your sample will match the one shown here, and you will be able to reproduce the same sample again in the future. To carry out the instructions below, you will

need to download and install an *Add-In* to augment JMP's built-in functions. This is a task that you only need to complete once.

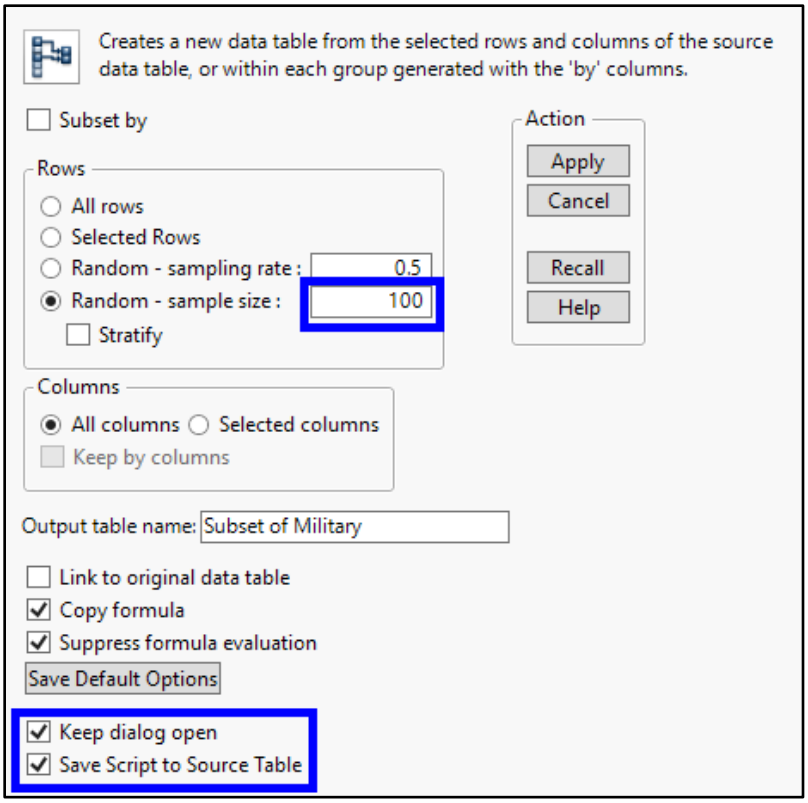
Before going further, visit the JMP User Community website at community.jmp.com. Search the term Random Seed Reset. You should find one result. Click on the boldface phrase Random Seed Reset to see details. Click the download button and save the script in a convenient location on your computer.

Once the Add-In archive file is downloaded, run it to install the add-in. You will see a security warning; click **Install**.

Finally, click the Add-Ins menu and you should see the item Random Seed Reset. If you do not see it, return to the location of the saved archive and try to install it once more.

3. Choose **Add-Ins ► Random Seed Reset**. This simple dialog box asks for an integer to "seed" the algorithm. We can enter any integer, so in honor of the year of the first release of JMP, we will use 1989.
4. Choose **Tables ► Subset**. Complete the dialog box as shown in Figure 8.1, requesting **100** randomly selected rows. Check **Keep dialog open** and **Save Script to Source Table**. Then click **Apply.**

**Figure 8.1: Selecting a Simple Random Sample**



If we did not already have the full data table available to us in electronic form (for **Military** or some other population, like all of the names listed in the local city telephone directory), or if we didn't intend to use the current data columns but wanted to contact the respondents with other questions, we might use JMP to generate a list of random integers, and then use those numbers to select our sample.

You should now see a new data table titled **Subset of Military**, with 100 rows randomly selected from the sampling frame. The new table contains all columns from the main table but includes only 100 rows randomly selected from the population.

Because we set a random seed, your list of 100 personnel will match mine. If you set a different seed, your sample will differ from the samples other readers obtain and from the one used as an illustration here. Now let's compare the descriptive statistics for **grade**, **branch**, and **gender** in the random sample to the actual population parameters:

5. Create a distribution for the columns **grade**, **branch**, and **gender.**
6. Compare the proportion of personnel who are enlisted, who are in the Air Force, and who are female to the population proportions.

It is very likely that you will notice first that the proportions won't match exactly, but that they are similar in magnitude. The author's subset contained 73 enlisted personnel, 7 Air Force, and 15 women. Perhaps you have 70% or 85% enlisted, but you probably don't have 50% or 100% enlisted. Let's try it again with a different subset.

7. Earlier we left the **Subset** dialog box open. Return to it now and click **Apply** to choose a different subset. Repeat steps 4 and 5.

Table 8.1 summarizes the findings in this exercise, rounded to one decimal place. Your results should be the same, unless you have experimented with different seeds. Even in that case, they should be similar to these:

**Table 8.1: Comparison of the Population and Random Samples for Selected Categories**

| Data Table | % Enlisted | % Air Force | % Female |
|---|---|---|---|
| Military (n=1,048,575) | 78.0 | 6.2 | 13.2 |
| Subset 1 (n=100) | 82.0 | 5.0 | 14.0 |
| Subset 2 (n=100) | 75.0 | 6.0 | 20.0 |

In practice, we typically draw one random sample, collect and summarize the sample data, and then attempt to draw reasonable conclusions about a population. Here we see that sample data

is apt to be "close" to an accurate match with a population, but not exact. We want to ask just how much difference it makes to look at one sample rather than a different one.

With more than one million rows in this **Military** table, there are a vast number of different 100-person samples that we might have randomly selected. In the next section, we will begin to think about how the variability across those many samples can blur our perception of the patterns of variation within the population. We will start by simulating repeated subsets of a fixed size from one population. Because the number of possible samples can truly be enormous with even modest-sized populations, we will automate the process until we have a large enough group of subsets to see a pattern.

## Variability Across Samples: Sampling Distributions

All samples vary, and the relevant variation is different depending on whether we are looking at quantitative continuous data or qualitative ordinal or nominal data. In this section, we will look more closely at how we can gauge and characterize the *sampling distribution* of the sample proportion for qualitative, categorical data and the sampling distribution of the sample mean for continuous data.

> Don't be thrown by the terminology here. We want to describe the variability of a sample statistic (like the proportion of Air Force personnel) across all possible samples. Statisticians call such a distribution a *sampling distribution* to distinguish it from the distribution within one sample or the distribution in an entire population.

### Sampling Distribution of the Sample Proportion

To illustrate how different samples will yield different results, we will focus just on the **gender** column in the data. We know that 13.221% of the more than one million personnel are women. That value—13.221%—is a population parameter. We want to think about what would happen if we did not have the entire population data set and needed to use a simple random sample in order to estimate the parameter. How much might our estimates deviate from the actual value?

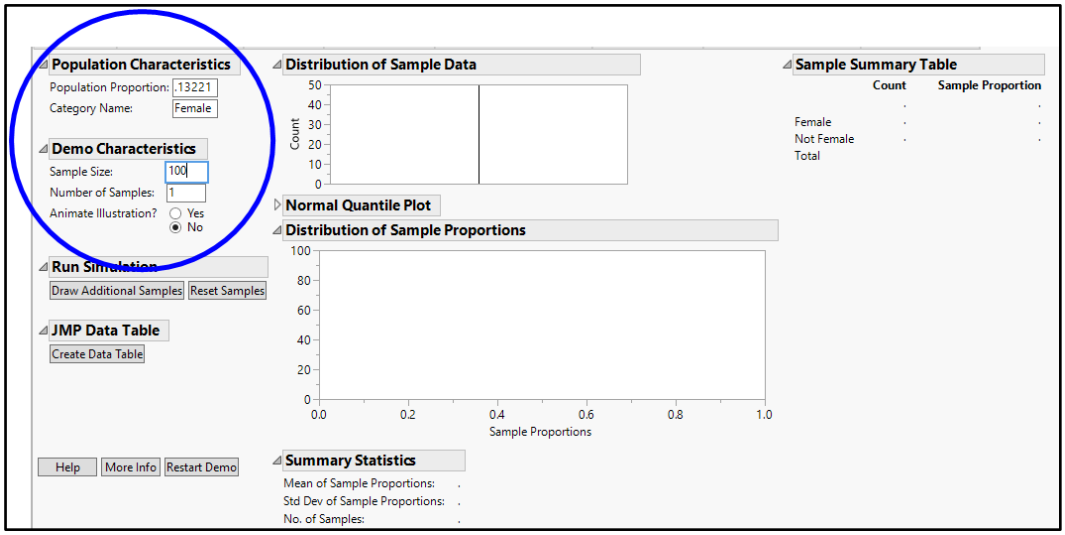
In this instance, there are far too many possible outcomes for us to list them all. Instead, we'll use a JMP *script* that has been written to simulate situations just like this.

A script is a file containing a sequence of JMP commands written in the JMP Scripting Language (JSL). Scripts are particularly useful when we want to run through a series of commands repeatedly, as in this case, where we will generate numerous random samples and tabulate the results. You can extend your JMP capabilities by learning to use and write scripts, but that is a subject for another book. In this chapter, we will just run two scripts to illustrate patterns of sampling variability. These scripts were developed by Amy Froelich of Iowa State University and Bill Duckworth of Creighton University. They were programmed by Brady Brady and Julian Parris of SAS Institute.

1. Select **Help ► Sample Data**. Click the disclosure arrow next to **Teaching Scripts** and then the arrow next to **Interactive Teaching Modules.** Select the module called **Sampling Distribution of Sample Proportion**.

Choosing this file opens the interactive simulator. (See Figure 8.2.) With this tool, we can simulate repeated sampling to see how samples vary. In this way, we can develop an approximate picture of the sampling distribution. For our first simulations, we will simulate the process of selecting 100-person SRSs.

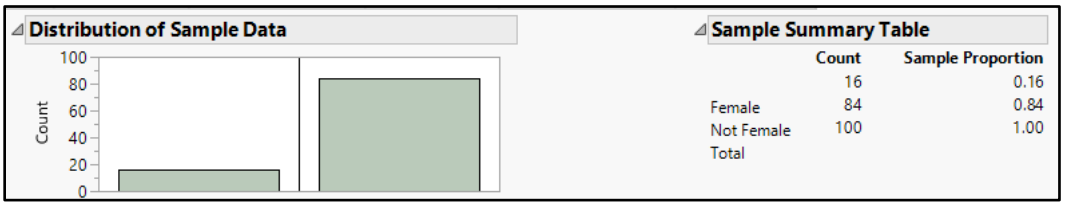
**Figure 8.2: The Simulator Window for Sample Proportions**




2. In the upper left of the window, find the **Population Characteristics** (circled in Figure 8.2) and change the **Population Proportion** from **0.2** to **0.13221**.
3. Change the **Category Name** to **Female**.
4. Change the **Sample Size** to **100**.
5. Click the **Draw Additional Samples** button to generate a single random sample of 100 military personnel from this population with a proportion of 13.221% females.

Now, look at the upper right of the window to see the summary for your first simulated sample. Each time we draw a sample, we will see a different result. Figure 8.3 displays one typical sample result. In this instance, our sample of 100 people contained 16 women for a sample proportion of 0.16 (16 %). This one sample yields a high estimate of the population proportion, but other samples should be closer to 13%.

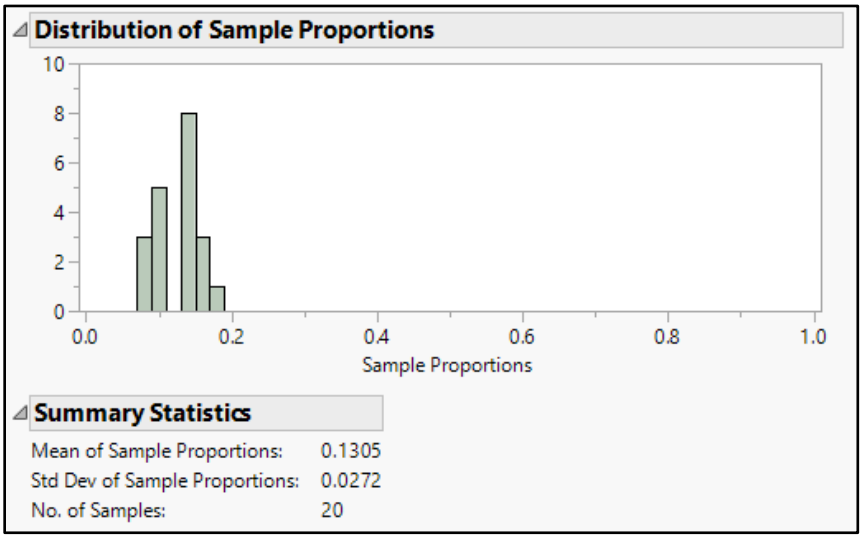
**Figure 8.3: Results for One Simulated Sample**




6. Click **Draw Additional Samples** 19 more times for a total of 20 random samples, keeping your eyes on the bar chart labeled **Distribution of Sample Data**. Notice how the bars oscillate up and down, indicating the variability across different samples.

By now, you might also have noticed that the large region titled **Distribution of Sample Proportions** is starting to take shape. Figure 8.4 shows the results after one set of 20 samples; your graph will not match this one exactly, but will very likely be similar in center, shape, and spread. At this point in this simulation, all of the sample proportions have fallen between approximately 0.08 and 0.19, with a concentration at around 0.12–0.14. We can see in the summary statistics below the graph that the mean of the sample proportions from all 20 samples is 0.131—just slightly lower than that population proportion of 0.13221.

**Figure 8.4: Distribution of Simulated Sample Proportions after 20 Repetitions**



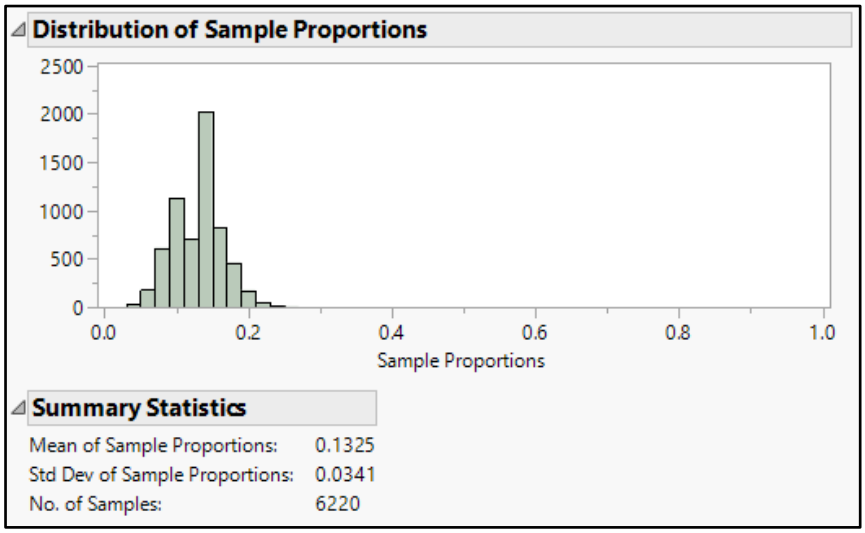

Let's keep adding more simulated samples but pick up the pace.

7. In the center left of the simulator window under **Demo Characteristics**, choose **Number of Samples = 100** and click **Yes** next to **Animate Illustration?**
8. Click **Draw Additional Samples** and watch the histogram of Figure 8.4 morph into a different picture.
9. Repeat this step once more.

10. Let's turn off the animation and generate several thousand samples. Select **Number of Samples ► 1000** and click the radio button **No** just below the number of samples. Click **Draw Additional Samples** until you have drawn more than 6,000 samples.

The sampling distribution graph should now look very much like Figure 8.5. There are several things to notice in these results. The distribution of observed sample proportions is symmetric with a strong central concentration near 0.132. The vast majority of sample proportions fell between 0.08 and 0.2. The summary statistics inform us that the mean proportion after 6,220 repetitions was 0.132—very nearly matching the population proportion. In a single sample of 100 military personnel, we will never have exactly 13.221% females because that is impossible, but out of all possible samples, the mean sample proportion is 13.221%.

**Figure 8.5: Results After More than 6,000 Replications**




## From Simulation to Generalization

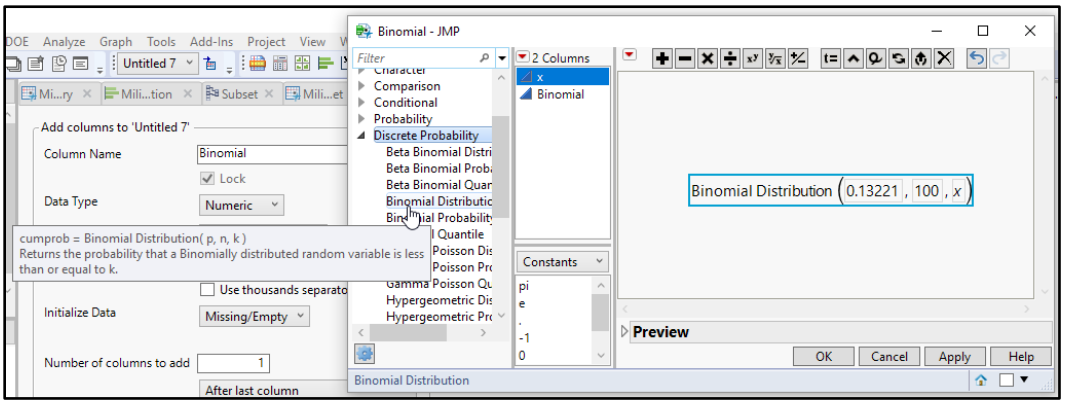
The distribution in Figure 8.5 might be reasonably approximated by a normal distribution. In fact, the normal approximation has long been the traditional approach in analyzing sampling distributions owing to its computational simplicity in an era before software. It is common for introductory textbooks to continue the tradition, but with software like JMP, it is practical to take a more direct and more accurate approach. If we take 100-person random samples from a very large population with a population proportion of 0.13221, then we can use the binomial distribution to describe the theoretical sampling distribution.[2] If we define $\hat{p}$ as the sample proportion of females in a 100-person random sample of military personnel, then the distribution of $\hat{p}$ is described by a binomial distribution with parameters $n = 100$ and $p = 0.13221$. In this way, we can compute the likelihood of drawing a random sample that lies a specific distance from the population proportion.

For example, the simulation showed that although the population proportion is 13.2%, it is possible to draw a simple random sample that could mislead us into estimating that the

population proportion is 10% or less. But how likely is that? In other words, how likely is it that an investigator would observe 10 or fewer women in a sample of 100? Alternatively, we might ask the question this way: "If the population actually contains 13.2% women, how surprised should we be if our sample of 100 contains fewer than 10% women?"

1. Create a new empty data table.
2. Rename the first column **x** and enter the value **10** into the top row.
3. Add a new column named **Binomial** containing the **Discrete Probability** formula **Binomial Distribution** (0.13221, 100, x). Figure 8.6 shows the formula editor dialog box. You might want to refer to Chapter 6 to refresh your memory about how to create a column like this.

**Figure 8.6: Computing a Cumulative Binomial Probability**




After you click **OK** to set up the column, you should find the value 0.2146 (rounded) in your data table. This is the probability of observing ten or fewer women in a sample of 100 according to the binomial model. So, in this example, more than 21%--or about one-fifth—of all possible samples from this population have 10 or fewer women.

## Sampling Distribution of the Sample Mean

Thus far in the chapter, the examples have focused on categorical data. What happens when the variable of interest is continuous? Let's consider some data from the Bureau of Transportation Statistics (BTS), an office of the U.S. Department of Transportation. BTS monitors all airline travel throughout the United States, maintaining detailed records about all scheduled commercial flights.

For this example, we will focus on one of the many variables that BTS tracks, namely the total number of minutes that a flight is late in arriving at its destination. Time is a continuous variable, measured here in minutes. If a flight is late, the value is a positive number. If it arrives early, the "delay" is negative.

1. Open the data table called **Airline delays 3**. This table lists all domestic flights during the month of March 2019.
2. Analyze the distribution of the column **Arr Delay**.

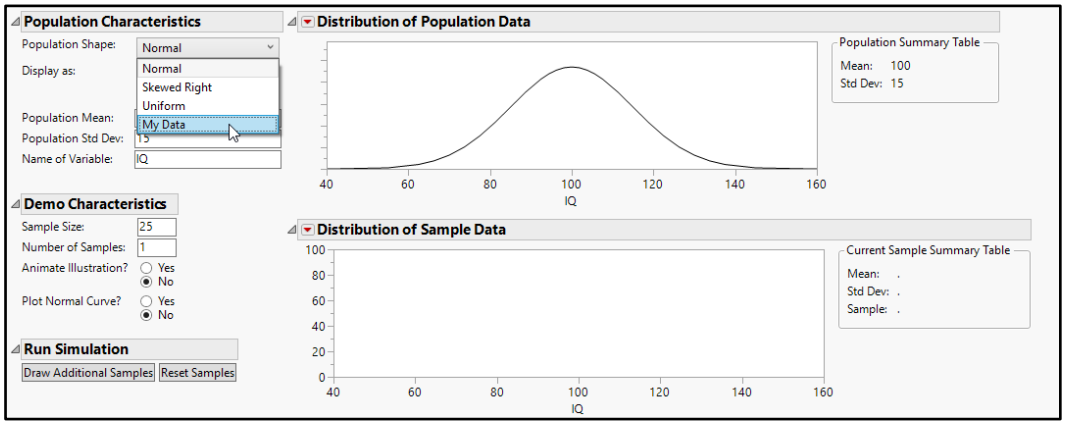
This column is strongly skewed, which makes sense. Most flights arrive early or within 15 minutes of their scheduled arrival time, but some delays can be quite lengthy.

As we did earlier, let's initially assume that this data table represents an entire population of more than 600,000 flights. Within the population, the mean arrival delay is 3.289 minutes with a standard deviation of 46.93 minutes. We will use simulation to examine the sampling variation from this population.

Specifically, we will look at variation in the *sample means* among many simulated samples from this population. From the distribution report, we have arrival delay data in 618,505 rows, and we can use those rows to inform another simulation script. For this illustration, we will consider samples of size $n = 200$.

3. For the sake of reproducibility, let's set a new random seed. Under **Add-Ins,** choose **Random Seed Reset**. This time, choose your own seed—any integer greater than 1. For the record, I have chosen 2067314.
4. As we did early, select **Help ► Sample Data**. Under **Teaching Scripts,** expand the list of **Interactive Teaching Modules** and this time select the module called **Sampling Distribution of Sample Means**. This simulator (shown on the next page in Figure 8.7) works very much like the one for proportions, but this one works with the continuous data and the sample mean. As before, the goal is to investigate the variability in sample means across different possible samples.
5. We first need to specify the **Population Characteristics**. Unlike the prior example where we typed in the parameters of the population distribution, here we will rely on the empirical data in the **Airline delays 3** data table to define the population attributes. From the drop-down list next to **Population Shape**, choose **My Data.**

**Figure 8.7: Sampling Distribution Simulator for a Sample Mean**




6. This opens a dialog box from which we can choose a numeric column from the active data table. Or select an unopened table. Choose **Arr Delay** and click **OK**.
7. After you do this, notice that the simulator has automatically filled in the three boxes for the mean, standard deviation, and variable name. The top panel in the simulator shows the population distribution.

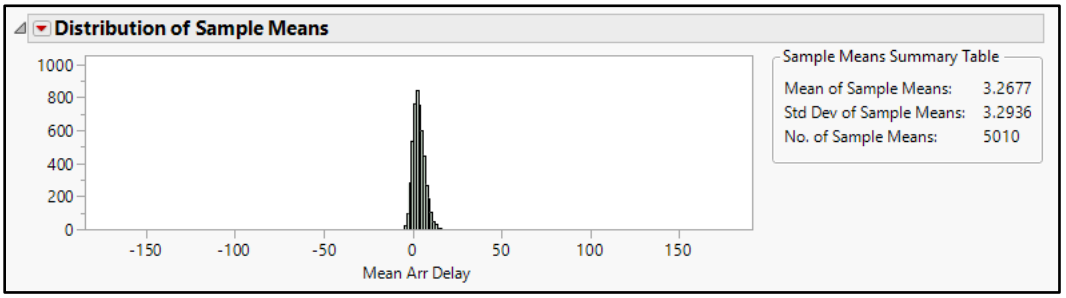
8. Now type **200** into the **Sample Size** box, indicating that we want to simulate drawing one sample of 200 flights.
9. This simulator operates like the one we used for proportions. Click the **Draw Additional Samples** button a few times to create 10 samples and look at the three histograms in the simulator.

The top distribution is the entire population looking more symmetric than in our initial distribution report, but still right skewed. The middle panel shows the current sample, and the lower panel shows the distribution of means from the samples drawn thus far. Within each sample, we see a right-skewed distribution of values, reflecting the shape in the parent population. When we look at the **Distribution of Sample Means**, we find that the sample means are starting to cluster tightly to the right of 0 minutes, near 3 minutes.

10. Increase the number of samples from **1** to **1,000,** being sure not to animate the illustration. Generate 5,000 additional samples. (Depending on your computer's speed, this might take a while.)

Each group of 5,000 simulated samples is different, but your results probably look very much like those shown below in Figure 8.8. If you chose your own random seed, your graph should be very similar in shape, with slightly different summary statistics. Despite the skewed population and samples, the means of the samples follow a much more symmetric shape.

**Figure 8.8: Typical Results of More than 5,000 Simulations**




Moreover, we notice in the **Means Summary Table** that the mean of the sample means is 3.27 minutes. In other words, samples from this population yield sample means whose values cluster rather closely near the actual population mean of 3.29 minutes. This is very important; not only do samples tend to have means close to the true population mean, they vary in a predictable way. So even though we cannot know whether one sample will perfectly match the population mean, we can describe the pattern and extent of variation across possible samples. Thanks to the *Central Limit Theorem*, we know even more about the variation in sample means.

## The Central Limit Theorem

The symmetric shape of the distribution and the clustering near the population mean happen to be quite predictable. The *Central Limit Theorem* (CLT) applies to the variation among sample means for continuous data. The theorem essentially says that regardless of the shape of the parent population, for sufficiently large samples, the distribution of all possible sample means is

approximately normal in shape, with a mean value of $\mu$ (the population mean) and a standard deviation of $\sigma/\sqrt{n}$. To distinguish the standard deviation of the sampling distribution from the standard deviation of the population in this book, we will restrict the term *standard deviation* to refer to the variation among individual cases in a sample or population, using the symbols *s* and $\sigma$ respectively. In reference to a sampling distribution, we will use the term *standard error*. In this section, we are asking about variability of the sample mean, so we will refer to the *standard error of the mean*, denoted as $\sigma_{\bar{x}}$.

To summarize:

$\sigma$ denotes the standard deviation of a population

s denotes the standard deviation on one sample

$\sigma_{\bar{x}}$ denotes the standard error of the sample mean

More precisely, the sampling distribution is described by a *t distribution*. The *t* distributions are another family of mound-shaped symmetrical distributions, similar in shape to normal curves but with proportionately more area in the tails. Like the normal distributions, there are a large number of *t* distributions, each uniquely and completely identified by one parameter known as *degrees of freedom* (or *df*). For a *t* distribution, we calculate degrees of freedom as $df = n - 1$.

According to the CLT, the means of all possible 200-observation random samples from this population will follow a *t* distribution with 199 degrees of freedom. Their distribution will have a mean of 3.2888 and a standard error of $46.9318/\sqrt{200}$ = 3.32.

As shown in Figure 8.8, after 5,000 simulated samples, the means have approximated a curve that is normal in shape. As the CLT predicts, the mean of these samples is quite close to 3.2888. Now let's check the standard error.

Look at your screen to the right of the histogram of the distribution of sample means. (Refer back to Figure 8.8.) Using the CLT, we just calculated the standard error of the mean to be 3.32 minutes. This is the spread of the theoretical distribution of all possible sample means. In the simulation illustrated here, the standard deviation of the collection of 5,010 simulated sample means was about 3.29 minutes. Although not perfect, over the span of just 5,010 samples, the CLT does an impressive job of informing us about the ways in which sampling introduces error into our estimates of the population mean. If we were to continue simulating samples, the CLT predicts that three things will happen:

- the mound shape of the histogram will approach the normal distribution shape
- the mean of the sample means will approach 3.29
- the standard deviation of the sample means will approach 3.32

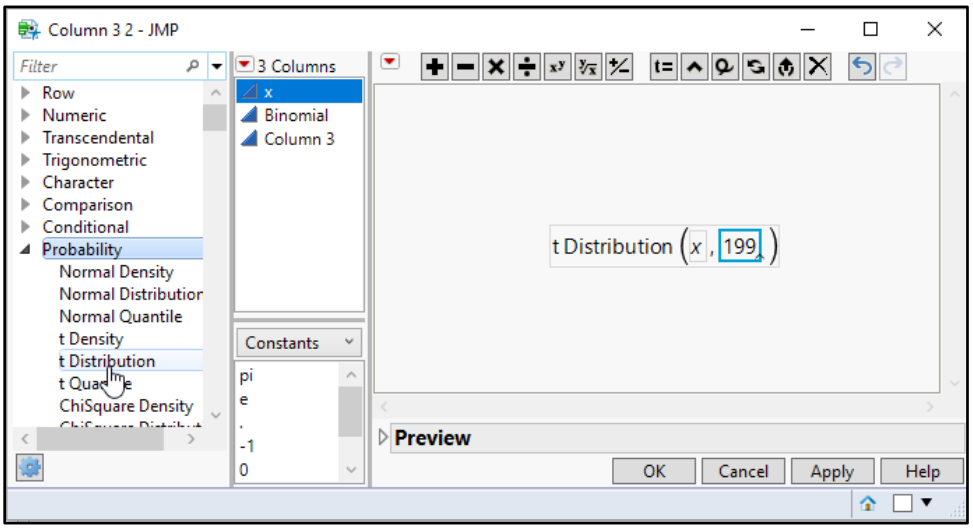
Armed with the CLT and an understanding of *t* distributions, we can estimate how often and how far we are likely to be misled by our samples. With a normal model, we could confidently say that 95% of all observations lie within almost two standard deviations of the mean. Similarly,

with a normal model, we could say that 95% of all possible samples have means within about two standard *errors* of the mean.

JMP has a *t* distribution function that we can use to calculate the proportion of sample means lying more than two standard errors from the mean, assuming a sample size of 200.

1. Return to the data table that we created to make the binomial computation, and type **–2** into the first row of the **x** column and **2** into the second row of the **x** column.
2. Add a new column called **t** and edit the formula as illustrated in Figure 8.9. Create a formula using the **t Distribution** function is listed under the **Probability** group. Complete the formula as shown.

**Figure 8.9: Creating a Formula using the *t* Distribution Function**




The results of this command appear in your new data table. (See Figure 8.10.)

**Figure 8.10: Results of *t* Distribution Calculations**

| | x | Binomial | t |
|---|---|---|---|
| 1 | -2 | 0 | 0.0234300002 |
| 2 | 2 | 0.000091027 | 0.9765699998 |

This shows that about 2.3% of the *t* distribution with 199 *df* lies to the left of -2, and the complementary 97.7% lies below +2, meaning that another 2.3% is to the right of +2. Therefore, we can conclude that approximately 4.6% of all possible samples of 199 incidents would have sample means more than two standard errors from the true population mean. In other words, any one sample might mislead us, but about 95% of all possible samples would mislead us by no more than 2 standard errors, or 6.6 minutes.

## Stratification, Clustering, and Complex Sampling (Optional)

As noted previously, simple random sampling requires that we can identify and access all N elements within a population. Sometimes this is not practical, and there are several alternative strategies available. It's well beyond the scope of this book to discuss these strategies at length, but many of the JMP analysis platforms anticipate the use of these methods, so a short introduction is in order.

Two common alternatives break a population up into groups, and then use random sampling methods to gather a probability sample based on the groups. Fundamentally, we group in one of two ways:

- *Stratification: Strata* (the singular form is stratum) are logical or natural subgroups within a population such that individuals within a stratum tend to share similar characteristics, but those characteristics are likely to vary across strata. In a *stratified sample,* we identify strata within the population and then randomly sample from each stratum. For example, imagine a public health study focusing on diet and blood pressure among adults in the U.S. We might reasonably stratify the population by gender and age group.
- *Clustering: Clusters* are often geographical or arbitrary groupings within a population such that there is no identifiable reason to expect individuals within a cluster to be more or less homogenous than individuals across clusters. In a *cluster sample*, we identify clusters within a population, randomly sample a small number of clusters, and then proceed to sample individuals from those clusters.

For example, many telephone surveys in the United States cluster the population by area code and randomly select a group of area codes, and subsequently use computers to randomly dial seven-digit telephone numbers. In such a process, where the sample is built up in stages, the area codes are referred to as the Primary Sampling Units (PSUs).

*Complex sampling* refers to methods that combine stratification and clustering. Whether a sample is complex, stratified, or clustered, analysis of the sample data generally will require *weighting* if we want to generalize from the sample to the entire population. In a simple random sample of *n* individuals from a population with N members, each of the individuals represents the same fraction of the total population (*n*/N). However, consider what happens when a pollster randomly calls households within two different area codes. The entire state of Wyoming (population of approximately 0.5 million) is covered by area code 307. In New York City, area code 212 covers part of the borough of Manhattan that is home to about 1.6 million people. If the pollster were to phone 10 people in each of these two area codes, the Wyoming subsample would be representing about 50,000 people apiece while the Manhattan subsample would be representing nearly three times that number.

When we use a data table assembled from a complex sample, we will typically find one or more weighting variables—sometimes identified as *post-stratification weights*—and a PSU column. Most of the analysis platforms in JMP provide the option of specifying a **weight** column, to be used if our data came from a complex sample.

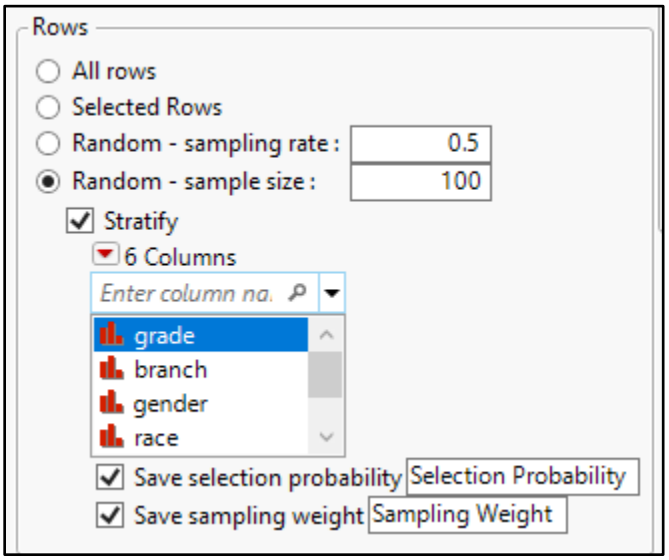
Several of the data tables used in this book come from survey organizations that use complex sampling methods. Among these examples are the NHANES data and the American Time Use Survey data.

Clustering and stratification both rely on random selection. Looking back at our military personnel table, you can understand how we might proceed if we wanted, say, to cluster by military branch. We would establish a data table listing the branches, randomly subset those tables, and proceed through the process.

The Subset platform enables us to stratify using columns already within the data table. Let's stratify by **grade**; there are three grades in this table. To illustrate the concept of sampling weights, we will choose 100 individuals from each grade, recognizing that with varying numbers of personnel in each grade, a sample with an equal number of rows from each grade will misrepresent the military.

1. Select the **Military** data table.
2. Choose **Tables ▶ Subset**. Specify a random sample size of **100** and select the **Stratify** check box, indicating we want 100 observations from each stratum.

**Figure 8.11: Starting to Create a Stratified Random Sample**




3. As shown above in Figure 8.11, select **grade** as the stratification column.
4. Select the two options to save the selection probabilities and sampling weights and click **OK**. You will see a new data grid containing the stratified sample.

Notice the two new columns to the far right of the table and notice that this table has 300 rows rather than the 100 we specified. In this illustration, the first 100 rows are officers; they all have the sample selection probability and sampling weights. If you scroll down the rows, you should notice that the values in the two new columns change when you come to the warrant officers, and then change again when you come to the enlisted personnel. In the full data table, there are

211.525 officers or approximately 20% of the population. In contrast, there are only 19,179 warrant officers, or just fewer than 2% of the population. Our sample contains 100 officers out of the 211,525 in the population, and the **Sampling Probability** equals exactly that: 100/21,525 = 0.000473. For warrant officers, the sampling probability is just about ten times higher. Inversely, in our sample, 100 officers will represent all 211,525 officers in the population, so each officer represents 211,525/100 = 2115.25 officers, which is what the **Sampling Weight** indicates. In round numbers, each sampled officer is standing in for more than 2000 peers, but each warrant officer is standing in for only about 200 peers. Hence, in later analysis, JMP can adjust its calculations to reflect that imbalance.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* You live in a community of 23,847 households. You want to select a simple random sample of 250 of these households from a printed list of addresses.
    a. Set the Random Seed to 1776. Use JMP to generate a random list of 250 integers between 1 and 23,847. Report the first 10 and last 10 integers on the list.
    b. Suppose that from prior research you estimate that 18% of these households (approximately 4,300) have no Internet service. Use what you have learned in this chapter to estimate the probability that a random sample of 250 households would contain only households with Internet service.
    c. Estimate the probability that an SRS of 250 households would include 25 or fewer homes without Internet service.
    d. Is there any realistic chance that an SRS of 250 households would be made up entirely of homes without Internet service? Explain your thinking.

2. *Scenario:* Recall the World Development Indicators data in **BirthRate 2017**.
    a. First, analyze the full table to determine the actual proportion of all countries that are in Sub-Saharan Africa. What is that proportion?
    b. Now find and report the mean and standard deviation of the infant mortality rate for all countries.
    c. Select a random sample of 30 countries. In your sample, what was the proportion of Sub-Saharan African countries? What was the mean infant mortality rate? How do your sample statistics compare to the values that you found in parts a and b? In your own words, explain the discrepancies.

3. *Scenario*: You want to develop a deeper understanding of sampling variability for categorical data by making further use of the sampling distribution of the sample proportion simulator script. We will imagine that you want to conduct a large-scale survey to learn what proportion of the population favors a proposed public policy. Complete the following activities, and report briefly on what you find.
    a. Set the population characteristics as follows: The **Population Proportion** is **0.4**, and the sample size is **1,000**. Set the Category Name to **Favor**. Generate 5,000 samples by repeatedly running the simulation and describe the distribution of sample proportions.
    b. Reset the simulator and keep the population proportion set at 0.4 but change the sample size to 250. Again, generate 5,000 simulated samples and report on what you find. If we reduce the sample size by a factor of ¼, what happens to the center, shape, and spread of the sampling distribution?
    c. Reset once again, and change the population proportion to 0.95, keeping the sample size at 250. Generate 5,000 samples, and compare your results to those obtained in part b.
    d. In major public opinion polls, pollsters often choose sample sizes around 250 or around 1,000. What difference does the size of the sample make?
    e. If you were conducting an important public opinion survey and were concerned about the potential adverse impacts of sampling variation, would you be better off sampling from a population in which there was considerable uniformity (say, in which 95% of the population agreed on an issue) or a population that was split on the issue? Explain how this exercise shapes your response.
4. *Scenario*: You want to develop a deeper understanding of sampling variability for continuous data by making further use of the sampling distribution of the sample means simulator script. In this scenario, you are investigating the number of seconds required for a website to respond after a user issues a request. Complete the following activities, and report briefly on what you find.
    a. Set the population characteristics as follows: the **Population Shape** is **Normal**, the **Population Mean** is **15** seconds, the **Standard Deviation** is **3.3** seconds, and the **Name of the Variable** is **Response Time**. The **Sample Size** is **1,000**. Generate 10,000 samples and describe the distribution of sample means.
    b. Reset the simulator and keep the other settings the same as in part a, but change the sample size to 250. Again, generate 10,000 simulated samples and report on what you find. If we reduce the sample size by a factor of ¼, what happens to the center, shape, and spread of the sampling distribution?
    c. Reset once again and change the population standard deviation to 6.6 seconds, keeping the sample size at 250. Generate 10,000 samples and compare your results to those obtained in part b.
    d. Now reset the simulator, change the **Population Shape** to **Right-Skewed,** and repeat the other settings from part a. How do your results compare to those obtained in part a?
    e. Once again, reset and change the shape to Uniform to model a population in which all values have the same probability.

f. In the prior questions, we have varied the shape of the parent population, the size of our samples, and the amount of variability within the population (standard deviation). Write a few sentences summarizing how each of these changes impacts the nature of sampling variability.

5. *Scenario:* Recall the **FAA BirdStrikes** data.
   a. First, analyze the full table to determine the actual proportion of all strikes that occurred for California strikes. What is that proportion?
   b. Now find and report the mean and standard deviation of the cost of repairs for all strikes.
   c. Select a random sample of 100 observations. In your sample, what was the proportion of California flights? What was the mean cost of repairs? How do your sample statistics compare to the values that you found in parts a and b? In your own words, explain the discrepancies.
6. *Scenario:* The city of Boston, Massachusetts, introduced a bike sharing program in 2007. Using technology installed in the bike racks and bicycles, the Hubway system can track usage patterns and habits. The data table titled **Hubway** contains data drawn from more than 55,000 actual bike trips between and among nearly 100 different rack locations in the city.
   a. First, analyze the full table to determine the actual mean **rider_age** (third to last column) for all trips. What is the population mean?
   b. How would you describe the shape of the distribution of ages?
   c. Using the Central Limit Theorem, what do we expect the sampling distribution of the mean to be (center, shape, spread)?
   d. Use the sampling distribution of the mean simulator to simulate samples of size $n$ = 50, using the **Hubway** data table to define the population characteristics. Run 10,000 simulations.
   e. Describe the pattern of variation among the 10,000 simulated sample means, noting the center, shape, and spread of the sampling distribution.
   f. Using the lower histogram in the simulator window, and without making any calculations, how likely do you think it would be to select a random sample in which the mean rider age was more than 35 years old? Less than 30 years? More than 45 years? Explain your thinking.

[1] This section is inspired by an example in *OpenIntro Statistics, 2nd Ed.* (2012) by David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel, available online at http://www.openintro.org/stat/textbook.php.

[2] Because the population is finite in size, one more properly should use the *hypergeometric* distribution to compute the exact probabilities. This is because the probability of success changes after each individual is selected, since the size of the remaining pool from the population is decreased by 1. With a population this large in comparison to the sample size, the impact is negligible. We studied the binomial in an earlier chapter, so we will use it.

# Chapter 9: Review of Probability and Probabilistic Sampling

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The Normal and *t* Distributions ..... 164
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When Samples Surprise Us: Ordinary and Extraordinary Sampling Variability ..... 167
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## Overview

The past three chapters have focused on probability—a subject that forms the crucial link between descriptive statistics and statistical inference. From the standpoint of statistical inference, probability theory provides a valuable framework for understanding the variability of probability-based samples. Before moving forward with inference, this short chapter reviews several of the major ideas presented in Chapters 6 through 8, emphasizing two central concepts: (a) probability models are often very useful summaries of empirical phenomena, and (b) sometimes a probability model can provide a framework for deciding if an empirical finding is unexpected. In addition, we will see another JMP script to help visualize probability models. As we did in Chapter 5, we will use the **WDI** data table to support the discussion.

## Probability Distributions and Density Functions

We began the coverage of probability in Chapter 6 with elementary rules concerning categorical events, then moved on to numerical outcomes of random processes. In this chapter, we will start the review by remembering that we can use discrete probability distributions to describe the variability of random outcomes of a discrete random variable. A discrete distribution essentially specifies all possible numerical values for a variable as well as the probability of each distinct value. For example, a rural physician might track the number of patients reporting flu-like symptoms to her each day during flu season. The number of patients will vary from day to day, but the number will be a comparatively small integer.

In some settings, we know that a random process will generate observed values that are described by a formula, such as in the case of a binomial or Poisson process. For example, if we intend to draw a random sample from a large population and count the number of times a single

categorical outcome is observed, we can reliably use a distribution to assess the likelihood of each possible outcome. The only requirement is that we know the size of the sample and we know the proportion of the population sharing the categorical outcome. As we will see later, we know that approximately one-third of the countries in the WDI table are classified as "high income" nations. The binomial distribution can predict the likelihood of selecting a random sample of, say, 12 nations and observing a dozen high-income countries.

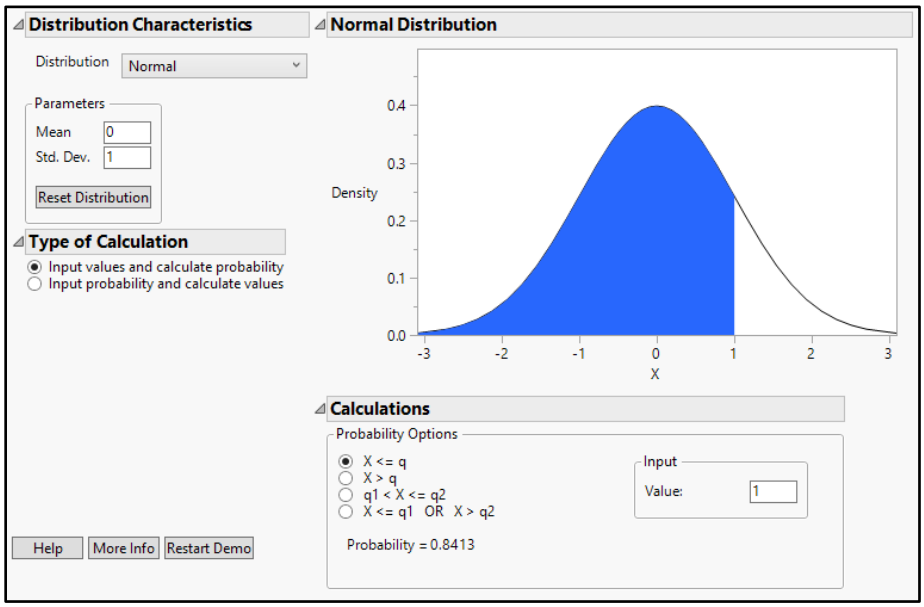
## The Normal and *t* Distributions

When the variable of interest is continuous, we use a *density function* to describe its variation. In conventional parlance, a continuous density function is said to specify the distribution of a variable or family of variables. Chapter 7 presented an in-depth treatment of the normal model and some of its uses. The normal model is one notable example of a theoretical continuous distribution, that is, a distribution that can be characterized mathematically. In Chapter 8 we encountered Gosset's *t* distribution, similar to the normal in that it is symmetric and mound-shaped. These are two continuous distributions that underlie many of the techniques of inference, and yet there are many more distributions used regularly by statisticians and data analysts.

For a quick visual refresher on the distributions do the following:

1. **Help ► Teaching Scripts ► Interactive Teaching Modules ► Distribution Calculator**. Figure 9.1 shows the initial view of this distribution calculator.

**Figure 9.1: Distribution Calculator**




Initially, the calculator displays the standard normal distribution. Remember that all normal distributions are specified by two parameters—a mean and a standard deviation. There are two

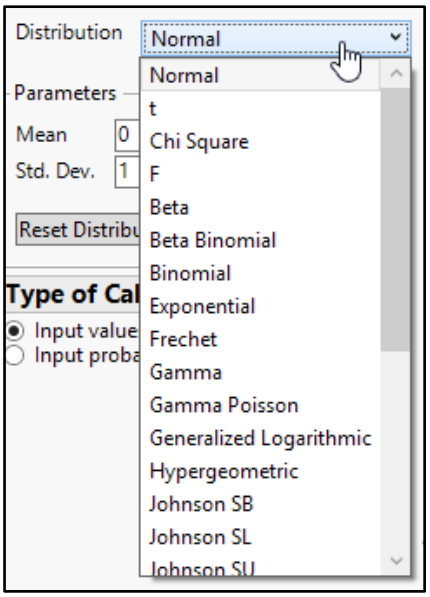
types of calculations we might want to perform. Either we want to know the probability corresponding to values of the random variable (X), or we want to know the values of X that correspond to specified probabilities. The left side of the dialog box allows the user to enter the parameters and type of calculation.

Below the graph is an input panel to select regions of interest (less than or equal to a value, between two values, and so on) and numerical values either for X or for probability, depending on the type of calculation. Spend a few moments exploring this calculator for normal distributions.

This calculator can perform similar functions for many discrete and continuous distributions, including the few that we have studied. To review and rediscover those distributions, try this:

2. In the upper left of the dialog box, click the button next to **Distribution**. This opens a drop-down list of all the distributions available for the calculator, as shown in Figure 9.2.

**Figure 9.2: Available Distributions**




As you scroll through the list, you will find the t, normal, binomial, Poisson, and uniform distributions as well as others that are unfamiliar. Select the familiar ones and notice the parameters of each as well as their characteristic shapes. Curious readers may wish to investigate other distributions in the list as well.

## The Usefulness of Theoretical Models

In statistical applications and in introductory statistics courses, we tend to put these theoretical models to work in two ways. In some instances, we might have a model that reliably predicts how random values are generated. For example, we know how to recognize a binomial or Poisson process. We also know when the Central Limit Theorem is relevant. In other instances, we might simply observe that sample data are well-approximated by a known family of distributions (for example, Normal), and we might have sound reasons to accept the risks from using a single formula rather than using a set of data to approximate the population. In such instances, using the model instead of a set of data can be a more efficient way to draw conclusions or make predictions. In the context of statistical inference, we will often use models to approximate sampling distributions. That is, the models will help us characterize the nature of sampling variability and therefore, sampling error.

Let's consider one illustration of using a model to approximate a population and refer back to Application Scenario 7 of Chapter 7, which in part worked with the cellphone use rates among countries in the **WDI** data table. We will elaborate on that example here.

1. Open the **WDI** data table.
2. Select **Analyze ► Distribution**. Select the column **cell** (number of mobile cellular subscriptions per 100 people).

Because cell phone usage has changed considerably over the 29-year period, we will focus on just one year, 2017. We can apply a data filter within the **Distribution** report to accomplish this.

3. Click the red triangle next to **Distributions** and choose **Script ► Local Data Filter** to **Show** and **Include** only data from the year **2017**.
4. Finally, add a normal quantile plot. The comparatively straight diagonal pattern of the points suggests that a normal distribution is an imperfect but reasonable model for this set of 200 observations.

The **Distribution** command reports selected quantiles for this column. We can see, for example, that 75% of the reporting countries had 129.859 subscriptions or fewer per 100 people. Suppose we wanted to know the approximate value of the 80$^{th}$ percentile. We might use a JMP command to report the 80$^{th}$ percentile from the sample, but we can quickly approximate that value using the normal distribution calculator. From the **Distribution** report, we know that the sample mean = 107.567 subscriptions, and the sample standard deviation = 40.904 subscriptions.

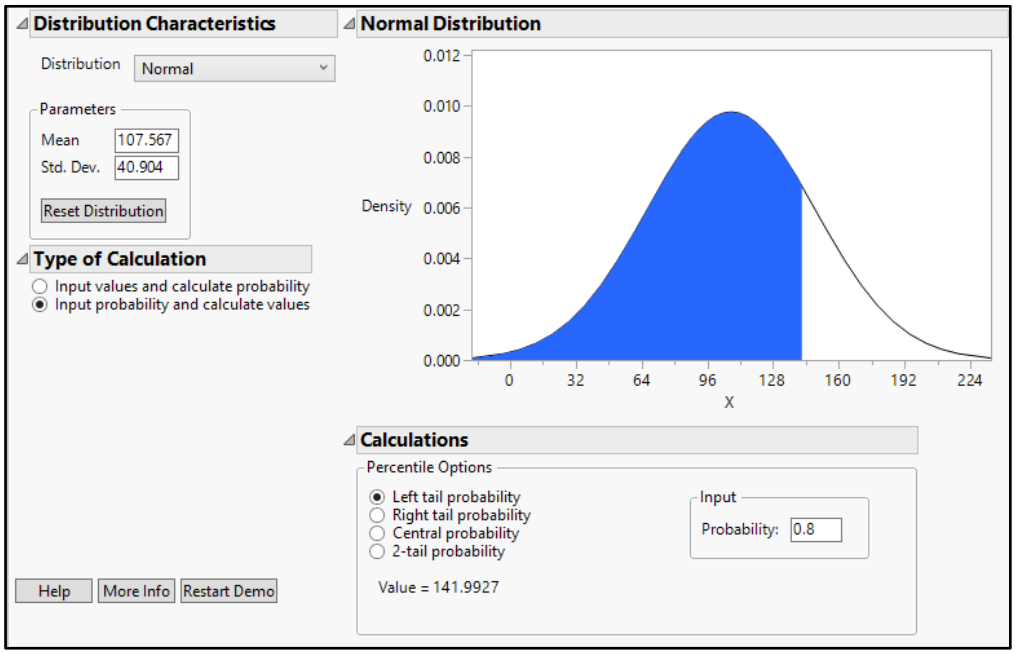
Follow these steps to calculate the 80$^{th}$ percentile of a theoretical normal population with the same mean and standard deviation.

5. Return to the **Distribution and Probability Calculator** and re-select **Normal** distribution in the upper left.
6. In the **Parameters** panel, enter **107.567** and **40.904** as the mean and standard deviation.

7. Under **Type of Calculation**, select **Input probability and calculate quantiles**.
8. Finally, under **Calculations**, select **Left tail probability**, and type **0.8** in the **Probability** box.

The completed dialog box is shown in Figure 9.3. In the lower right panel, we find that the theoretical 80$^{th}$ percentile is 141.99 subscriptions. If you repeat this exercise to locate some of the other percentiles that are reported in the distribution report, you will see that the theoretical normal model is a better approximation for quantiles near the center of the distribution than in the tails.

**Figure 9.3: Using a Normal Model to Approximate a Set of Data**




## When Samples Surprise Us: Ordinary and Extraordinary Sampling Variability

Chapter 8 was devoted to probabilistic sampling and to the idea that samples from a population will inevitably differ from one another. Regardless of the care taken by the investigator, there is an inherent risk that any one sample might poorly represent a parent population or process. Hence, it is important to consider the extent to which samples will tend to vary. The variability of a sample statistic is described by a *sampling distribution.*

In future chapters, sampling distributions will be crucial underpinnings of the techniques to be covered but will slip into the background. They form the foundation of the analyses, but like the foundations of buildings or bridges, they will be metaphorically underground. We will rely on them, but often we won't see them or spend much time with them.

As we move forward into statistical inference, we will shift our objective and thought processes away from summarization toward judgment. In inferential analysis, we frequently look at a sample as *evidence* from a somewhat mysterious population, and we will want to think about the type of population that would have yielded such a sample. As such, we will begin to think about samples that would be typical of a particular population and those that would be unusual, surprising, or implausible.

All samples vary, and the shape, center, and variability depend on whether we are looking at quantitative continuous data or qualitative ordinal or nominal data. Recall that we model the sampling variation differently when dealing with a sample proportion for qualitative, categorical data and the sample mean for continuous data. Let's consider each case separately.

## Case 1: Sample Observations of a Categorical Variable

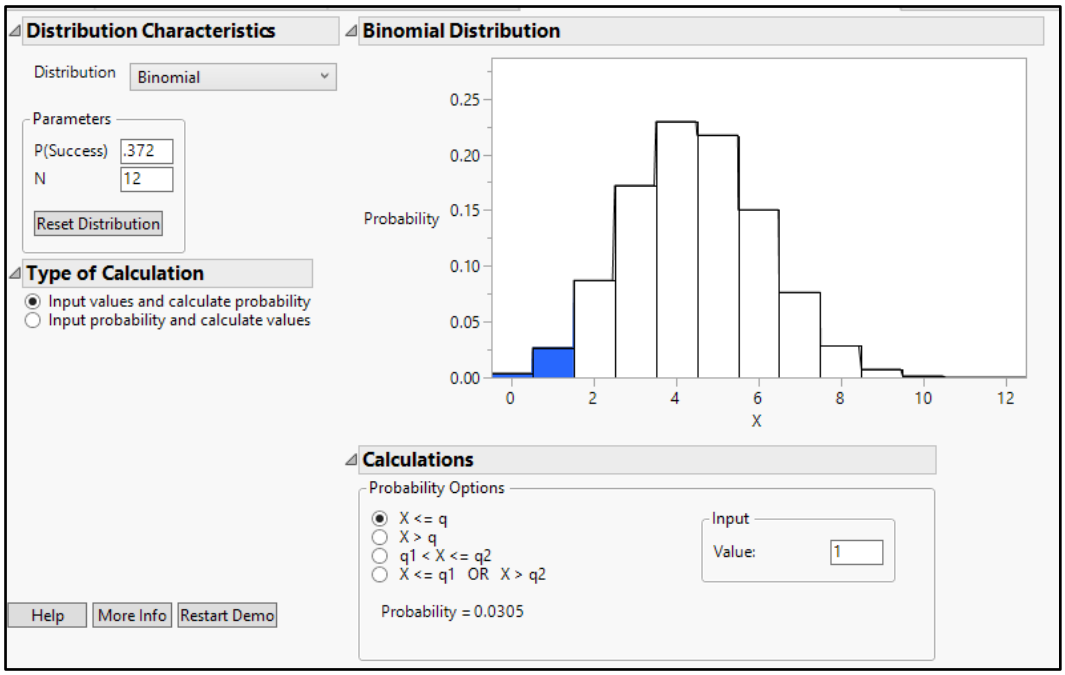
To illustrate, consider the WDI data table once again. The table contains 42 different columns of data gathered over 29 years. Of the 215 countries, more than one-third are classified as high income (80 of 215 nations = 0.372). If we were to draw a random sample of just twelve observations from the 6,235 rows, we would anticipate that approximately four of the twelve sample observations would be high-income nations. We should also anticipate some variation in that count, though. Some samples might have four high-income countries, but some might have three, or five, or perhaps even six. Is it likely that a single random sample of 12 would not have any high-income countries? Could another sample have 12? Both are certainly possible, but are they plausible outcomes of a truly random process?

The process of selecting 12 nations from a set of 6,235 observations and counting the number of high-income nations is approximately a binomial process. It is only approximately binomial because the probability of success changes very slightly after each successive country is chosen; for the purposes of this discussion, the approximation is sufficient.[1] As such, we can use the binomial distribution to describe the likely extent of sampling variation among samples of 12 countries.

1. In the probability calculator, select the **Binomial** distribution.
2. Specify that the probability of success is **0.372** and a sample size, **N**, of **12.**
3. Choose **Input values and calculate probability**.

Your screen will look like Figure 9.4. The binomial histogram (reflecting the discrete number of successes in twelve trials) is initially set to show the probability of obtaining one or fewer high-income countries. This probability is represented by the blue shading, and at the bottom of the dialog box, we find that it is 0.0305. The other Probability Options in the Calculations panel allow us to find probabilities greater than a value, between two values, or in the tails of the distribution.

**Figure 9.4: A Binomial Distribution**




4. Below the histogram, click the radio button marked **X > q**, and enter a **6** into the **Input** panel box **Qa**.
5. The probability that a 12-country sample will contain more than six high-income countries is 0.1134. Repeat step 3 for seven, eight, and nine countries; notice that the probability becomes vanishingly small.

Such random samples might be logically possible but are so unlikely as to be considered nearly impossible. It is quite likely that this population would generate samples with between, for example, three and six high-income countries, but unlikely to yield a sample with zero or 10 high-income nations.

## Case 2: Sample Observations of a Continuous Variable

For this example, we will move away from the **WDI** data and return to the **Airline delays 3** data that we recently investigated in Chapter 8. For this illustration, we will work with the column called **DISTANCE**, which is the number of miles between the origin and destination airports. As before, we will investigate the variability of sample means to get a better understanding of potential sampling error. Equipped with the knowledge of the true population mean, we want to see how often random samples might have a surprisingly small or large mean.

1. Open the **Airline delays 3** data table.
2. Create a distribution report for the column **DISTANCE**.

This column is right-skewed because no flight can have a negative distance. Many are less than 1,000 miles, but a few are considerably longer.

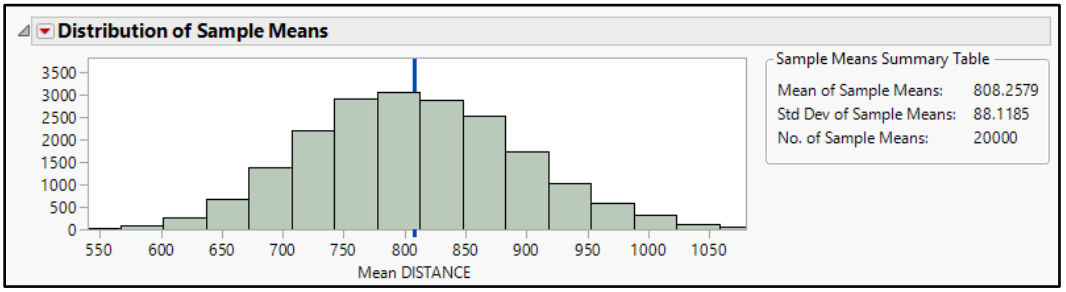
Usually, we think of samples as coming from a very large population or from an ongoing process. This data table represents 632,075 flights during one month in North America. We can think of commercial air travel as a continuous process. For the sake of this discussion, imagine that this sample is representative of the ongoing process.

If all flights were the same length, then any sample we might draw would provide the same evidence about the process. In fact, a sample of n=1 is sufficient to represent the process. Our challenge arises from the very fact that individual flights vary; because of those individual differences, each sample will be different.

In Chapter 8, we used the **Sampling Distribution** simulator to simulate thousands of random samples of size n = 200 from the data table. This process is sometimes known as *bootstrap sampling* or *resampling*, in which we repeatedly sample and replace values from a single sample until we have generated an empirical sampling distribution.

For this illustration, let's consider random samples of size 45 flights. Readers who want to repeat the simulation using the **DISTANCE** column should go back to the section of Chapter 8 entitled **"**Sampling Distribution of the Sample Mean," and regenerate perhaps 10,000 iterations until the histogram is relatively smooth and mound-shaped. Be forewarned that this can take a long time. Bootstrapping from such a large data table can consume computing resources. The results of one such simulation appear in Figure 9.5. The histogram has been revised to more clearly show the horizontal axis. The blue vertical line at 807.3 miles marks the mean of all flights.

**Figure 9.5: A Bootstrap Sampling Distribution**




So what message are we supposed to take away from this graph and the resampling exercise? Consider these messages for a start:

- Different 45-observation samples from a given population have different sample means.
- Though different, sample means tend to be "near" the mean of the population.

- We can model the shape, center, and variability of possible sample means.
- Even though **DISTANCE** in the original sample is very strongly skewed to the right, repeated samples consistently generate sample means that vary symmetrically—with an approximately normal distribution predicted by the Central Limit Theorem. Specifically, the CLT says that the sampling distribution will approach *~N(807.3, 87.8)*.

Very importantly, the histogram establishes a framework for judging the rarity of specific sample mean values. Approximately two-thirds of all possible samples of 45 flights commonly have mean distances between approximately 720 and 900 miles. Sample means will very rarely be smaller than 630 miles or longer than 980 miles. It would be stunning to take a sample of 45 incidents and find a sample mean less than 600 miles.

We cannot conclude that a result is surprising or rare or implausible without first having a clear sense of what is typical. Sampling distributions tell us what is surprising and what is not.

## Conclusion

Probability is an interesting and important subject in its own right. For the practice of data analysis, it provides the foundation for the interpretation of data gathered by random sampling methods. This chapter has reviewed the content of three earlier chapters that introduced readers to a few essential elements of probability theory. Taken together, these chapters form the bridge between the two statistical realms of description and inference.

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[1] The correct distribution to apply here is the *Hypergeometric* distribution, another discrete distribution that applies when you are sampling without replacement. The probability of success changes following each observation. In this particular case where the population size is so much larger than the sample size, the impact of the change in probabilities is minuscule. Inasmuch as this is a review chapter, I decided to resist the introduction of a new distribution.

# Chapter 10: Inference for a Single Categorical Variable

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## Overview

Beginning with this chapter, we enter the realm of inferential statistics, where we generalize from a single sample to draw conclusions about an entire population or process. When statisticians use the term *inference*, it refers to two basic activities. Sometimes we want to ask *whether the parameter is or is not equal to a specific value*, and sometimes we want to *estimate the value of a population parameter.* In this chapter, we will restrict our attention to categorical dichotomous data and focus on the proportion of a population that falls in one category.

## Two Inferential Tasks

We will discuss the ideas of *significance testing* (often called *hypothesis testing*) and of *confidence interval* estimation. Along the way, we will illustrate these techniques and ideas using data tables with two formats common to categorical data. We often find categorical data summarized in a frequency table, but sometimes we have detailed case-by-case data.

The main message of the previous chapter is that sampling always carries the risk of inaccurately representing the larger group. We cannot eliminate the possibility that one sample will lead us to an incorrect conclusion about a population, but we can at least calibrate the risk that it will. To do so, it's important to consider the form and origin of our data before rushing ahead to generalize from a sample. There are also other considerations to take into account and these are the topic of the next section.

## Statistical Inference Is Always Conditional

Under a wide range of common conditions, good data generally leads to good conclusions. Learning to reason statistically involves knowing what those conditions are, knowing what constitutes good data, and knowing how to move logically from data to conclusions. If we are working with sample data, we need to recognize that we simply cannot infer one characteristic (parameter) of a population without making some assumptions about the nature of the sampling process and at least some of the other attributes of the population.

As we introduce the techniques of inference for a population proportion, we start with some basic conditions. If these conditions accurately describe the study and the data, the conclusions that we draw from the techniques are most likely trustworthy. If the sample departs substantially from the conditions, we should avoid generalizing from the sample to the entire population.

At a minimum, we should be able to assume the following about our sample data:

- Each observation is independent of the others. Simple random sampling assures this, and it is possible that in a non-SRS we preserve independence.
- Our sample is no more than about 10% of the population. When we sample too large a segment of the population, essentially, we no longer preserve the condition of independent observations.

We will take note of other conditions along the way, but these form a starting point. In our first example, we will look again at the pipeline safety incident data. This is not a simple random sample, even though it might be reasonable to think that one major pipeline interruption is independent of another. Because we are sampling from an ongoing nationwide process of having pipelines carry natural gas, it is also reasonable to assume that our data table is less than 10% of all disruptions that will ever occur. We will go ahead and use this data table to represent pipeline disruptions in general.

## Using JMP to Conduct a Significance Test

The goal of significance testing is to make a judgment based on the evidence provided by sample data. Often the judgment implies action: is this new medication or dosage effective? Does this strategy lead to success with clients or on the field of play?

In the first chapter dealing with probability, we used data from the Pipeline and Hazardous Materials Program (PHMSA) of the U.S. Department of Transportation. The PHMSA monitors disruptions to the network of pipelines that transport natural gas. With respect to pipeline disruptions, public safety authorities might ask “Do more than 10% of disruptions involve ignition?”

The process of significance testing is bound up with the scientific method. Someone has a hunch, idea, suspicion, or concern and decides to design an empirical investigation to put the hunch to the test. As a matter of rigor and in consideration of the risk of erroneous conclusions, the test

approximates the logical structure of a proof by contradiction. The investigator initially assumes the hunch to be incorrect and asks if the empirical evidence makes sense or is credible given the contrary assumption.

The hunch or idea that motivates the study becomes known as the *Alternative Hypothesis* usually denoted as $H_a$ or $H_1$. With categorical data, the alternative hypothesis is an assertion about possible values of the population proportion, *p*. There are two formats for an alternative hypothesis, summarized in Table 10.1.

**Table 10.1: Forms of the Alternative Hypothesis**

| Name | Example | Usage |
|---|---|---|
| One-sided | Ha: p > .10 | Investigator suspects the population proportion is more than a specific value. |
| | Ha: p < .10 | Investigator suspects the population proportion is less than a specific value. |
| Two-sided | Ha: p ≠ .10 | Investigator suspects the population proportion is not equal to a specific value. |

The alternative hypothesis represents the reason for the investigation, and it is tested against the *null hypothesis ($H_0$)*. The null hypothesis is the contrarian proposition about *p* asserting that the alternative hypothesis is untrue.

To summarize: we suspect, for example, that more than 10% of gas pipeline disruptions ignite. We plan to collect data about actual major pipeline incidents and count the number of fires. To test our suspicion, we initially assume that *no more than 10%* of the disruptions burn. Why would we begin the test in this way?

We cannot look at all incidents, but rather can observe only a sample. Samples vary, and our sample will imperfectly represent the entire population. So, we reason in this fashion:

- Assume initially that the null hypothesis is true.
- Gather our sample data and compute the sample proportion. Take note of the discrepancy between our sample proportion and the null hypothesis statement about the population.
- Knowing what we know about sampling variation, estimate the probability that a parent population like the one described in the null hypothesis would yield up a sample similar to the one that we have observed.

- Decision time:
  - If a sample like ours would be extraordinary under the conditions described, then conclude that the null hypothesis is not credible and instead believe the alternative.
  - If a sample like ours would not be extraordinary, then conclude that the discrepancy that we have witnessed is plausibly consistent with ordinary sampling variation.

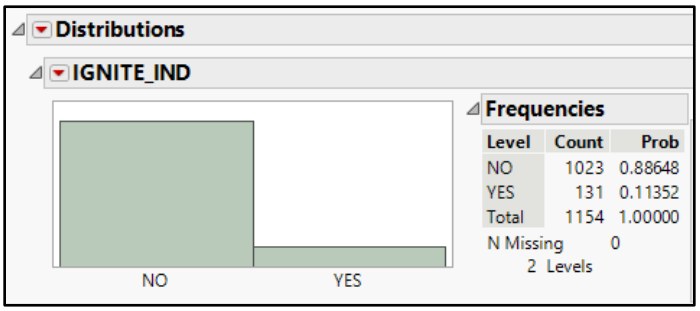
**NOTE:** When we decide to reject the null hypothesis, it is fair to say that the data discredit the null assumption. However, if we decide against rejecting the null hypothesis, the sample data do *not* "prove" that the null hypothesis is true. The sample data simply are insufficient to conclude that the null hypothesis is false. At first, this difference might appear to be one of semantics, but it is essential to the logic of statistical inference.

In this example, suppose that our investigator wonders if the proportion of disruptions that ignite exceeds 0.10. Formally, then, we are testing these competing propositions:

| | | |
|---|---|---|
| Null hypothesis: | $H_0$: $p \leq .10$ | (no more than 10% of disruptions ignite) |
| Alternative hypothesis: | $H_a$: $p > .10$ | (more than 10% of disruptions ignite) |

1. Open the data table **Pipeline Safety.**
2. Select **Analyze ► Distribution** to summarize the **IGNITE_IND** (for "ignite indicator") column. Your results look like Figure 10.1, which was prepared in **Stacked** orientation to save space.

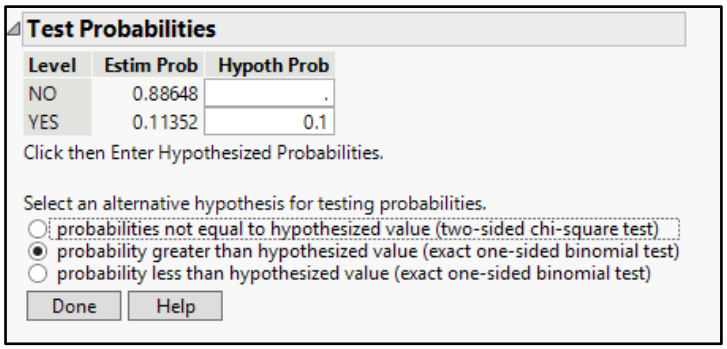
**Figure 10.1: Distribution of IGNITE_IND Column**




In our sample of 1,154 incidents, slightly more than 11% involved ignition. Given that disruptions had an ignition rate above 10%, should we be prepared to conclude that more than 10% of all past and future disruptions did or will ignite?

3. Click the red triangle beside **IGNITE_IND** and choose **Test Probabilities**. This opens a new panel within the report, as shown in Figure 10.2.

**Figure 10.2: Specifying a Significance Test**




4. The upper portion of this panel displays the observed proportions of **Yes** and **No** within the sample, and leaves space for us to enter the proportions as stated in the null hypothesis. Just type **.1** in the box in the second row.
5. The lower portion of the panel lists the three possible formulations of the alternative hypothesis. Click the middle option because our alternative hypothesis asserts that the population proportion is greater than 0.10, and then click **Done**.

There are several common ways to estimate the size of sampling error in significance tests. Most introductory texts teach a "by-hand" method based on the normal distribution because of its computational ease. Because software can do computation more easily than people can, JMP does not use the normal method, but rather selects one of two methods, depending on the alternative hypothesis. For a two-sided hypothesis, it uses a *chi-square* distribution, which we will study in Chapter 12. For one-sided hypotheses, it uses the discrete binomial distribution rather than the continuous normal. With categorical data, we always count the number of times an event is observed (like ignition of gas leaks). The binomial distribution is the appropriate model, and the normal distribution is a good approximation that is computationally convenient when there is no software available.

Your results should look like those in Figure 10.3. Look first to the bottom row of the results. There you will see the alternative hypothesis (**p1** refers to the number that you typed in the

white box above), the proportion based on the **Yes** values, and the **p1** value that you entered. The right-most number—the *P*-value—is the one that is informative for the test.

**Figure 10.3: Results of a Significance Test**

**Test Probabilities**

| Level | Estim Prob | Hypoth Prob |
|---|---|---|
| NO | 0.88648 | 0.9 |
| YES | 0.11352 | 0.1 |

| Binomial Test | Level Tested | Hypoth Prob (p1) | p-Value |
|---|---|---|---|
| Ha: Prob(p > p1) | YES | 0.10000 | 0.0712 |

A *P*-value is a conditional probability. The assumed condition is that the null hypothesis is true, and the probability refers to the chance that an SRS would produce a result at least this far away from the hypothesized value. In this example, we interpret the *P*-value by following this logic: Our sample proportion was 0.11352, which is more than the hypothesized value of 0.10 by nearly 0.014. Assuming that the null hypothesis is true (that is, at most, 10% of all gas-line breaks ignite) and assuming we take an SRS of 1,154 incidents, the probability that we would find a discrepancy of 0.014 or more is 0.0712. This probability equates to an event that occurs 7 times out of 100.

Our decision comes down to the question, "which do we find more credible?"

- The null hypothesis is true, and our relatively high sample proportion is the result of chance.
- The reason our sample proportion is 11% is simply that the null is incorrect, and the true proportion is higher than 10%.

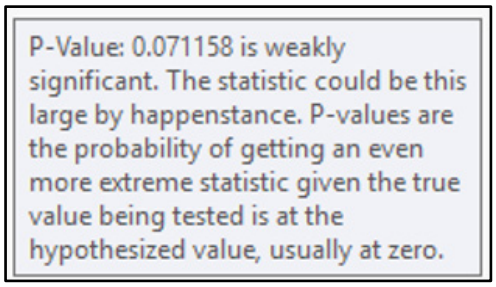
How shall we incorporate the *P*-value of 0.0712 into our decision-making? Though it has been customary to reject the null hypothesis when the *P*-value is smaller than 0.05, this practice has come under scrutiny.[1] The reject/do not reject decision should account for (a) the consequences of mistakenly rejecting a true null hypothesis and (b) the decision maker's tolerance for risking such an error.

The customary practice simplifies the weighing of risks and relies on a conventional standard. The standard is represented by the symbol alpha ($\alpha$) and $\alpha$ represents our working definition of credibility. Conventionally, the default value of $\alpha$ is 0.05, and there is a simple decision rule. If the *P*-value is greater than or equal to $\alpha$, we conclude that the null hypothesis is credible. If the *P*-value is less than $\alpha$, we reject the null hypothesis as being not credible. When we reject the null hypothesis because of a very small *P*-value, we say that the result is *statistically significant*.

JMP places an asterisk (*) beside *P*-values that are less than 0.05 and colors them for emphasis. This convention applies to *P*-values in all types of significance tests. The rule is simple, but it can be hard to remember or interpret at first. The designers of JMP know this and have incorporated a reminder, as shown in this next step.

6. Move your cursor in a very small circle over the value 0.0712. Closely read the message that pops up on the screen. (See Figure 10.4.)

**Figure 10.4: Pop-up Explanation of P-value**




## Confidence Intervals

A significance test provides a yes/no response to an inferential question. For some applications, that is sufficient to the purpose. In other instances, though, the decision maker might need to estimate a proportion rather than simply determine whether it exceeds a specified limit.

We already know how to calculate the proportion of ignition incidents within our sample (about 0.11 or 11%), but that only describes the sample rather than the entire population. We shouldn't say that 11% of all disruptions involve ignition, because that is overstating what we know. What we want to say is something like this: "Because our sample proportion was 0.81 and because we know a bit about sampling variability, we can confidently conclude that the population proportion is between ___ and ___."

We will use a *confidence interval* to express an inferential estimate. A confidence interval estimates the parameter, while acknowledging the uncertainty that unavoidably comes with sampling. The technique is rooted in the sampling distributions that we studied in the prior chapter. Simply put, rather than estimating the parameter as a single value between 0 and 1, we estimate lower and upper bounds for the parameter and tag the bounded region with a specific degree of confidence. In most fields of endeavor, we customarily use confidence levels of 90%, 95%, or 99%. There is nothing sacred about these values (although you might think so after reading statistics texts); they are conventional for sensible reasons. Like most statistical software, JMP assumes we will want to use one of these confidence levels, but it also offers the ability to choose any level.

## Using JMP to Estimate a Population Proportion

We already have a data table of individual incident reports. In this section, we will analyze the data in two ways. First, we will see how JMP creates a confidence interval from raw data in a data table. Next, we will see how we might conduct the same analysis if all we have is a summary of the full data table.

## Working with Casewise Data

The Pipeline Safety data table contains raw casewise data—that is, each row is a unique observation. We will use this data table to illustrate the first method.

1. In the **Distribution** report, click the red triangle beside **IGNITE_IND** and choose **Confidence Intervals ► 0.95.**

A new panel will open in the results report. The confidence interval estimate for the entire population appears in the table row beginning with the word **Yes**, as highlighted in Figure 10.5. We find the sample proportion in the **Prob**(ability) column and the endpoints of the confidence interval in the **Lower CI** and **Upper CI**. Finally, the confidence level[2] of 0.95 appears in the column labeled **1-Alpha**. Assuming that our sample meets the conditions noted earlier, and given that 11.4% of the observed disruptions involved ignition, we can be 95% confident that the population-wide proportion of disruptions with ignition is between about 0.096 and 0.133.

**Figure 10.5: A Confidence Interval for a Proportion**

**Confidence Intervals**

| Level | Count | Prob | Lower CI | Upper CI | 1-Alpha |
|---|---|---|---|---|---|
| NO | 1023 | 0.88648 | 0.866882 | 0.903517 | 0.950 |
| YES | 131 | 0.11352 | 0.096483 | 0.133118 | 0.950 |
| Total | 1154 | | | | |

Note: Computed using score confidence intervals.

This interval estimate acknowledges our uncertainty in two respects: the width of the interval says, "this interval probably captures the actual parameter," and the confidence level says, "there is about a 5% chance that the sample we drew unfortunately produced an interval estimate that excludes the actual parameter."

Notice that the sample proportion of .11352 is just about midway (but not exactly) between the upper and lower CI boundaries. That is not an accident; confidence intervals are computed by adding and subtracting a small *margin of error* to the sample proportion. There are several common methods for estimating both the center of the interval and the margin of error. One common elementary method takes the number of observed occurrences divided by the sample size ($\hat{p}$) as the center and computes the margin of error based on percentiles of the Normal distribution. JMP uses an alternative method, as noted beneath the intervals. JMP uses "score confidence intervals," which are sometimes called the *Wilson Estimator*. The Wilson estimator does not center the interval at $\hat{p}$ but at a slightly adjusted value.

For most practical purposes, the computational details behind the interval are less important than the assumptions required in connection with interpreting the interval. If our sample meets the two earlier conditions, we can rely on this interval estimate with 95% confidence if our sample size is large enough. Specifically, this estimation method captures the population proportion within the interval about 95% of the time if:

> The sample size, *n*, is sufficiently large such that $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$.

## Bootstrap Confidence Intervals (Optional; Requires JMP Pro)

What if our sample size is too small to meet this requirement? Imagine that our sample contained just 20 pipeline disruptions. With a rough estimate of 11% of incidents involving ignition, $n\hat{p} = 2.2$, which falls short of 5. In such a circumstance, we might well turn to a bootstrapping to estimate the sampling distribution of the proportion.

Earlier we mentioned bootstrapping in connection with the simulators to model sampling distributions. In recent years, especially in the field of machine learning, the bootstrap approach has grown in usage. We can apply the bootstrap to create a confidence interval.

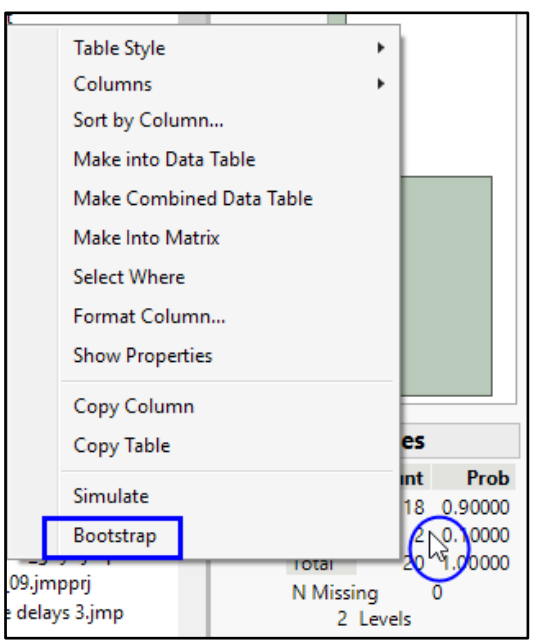
To illustrate, let's first draw a simple random sample of 20 disruptions from our data table. We will then use that sample to demonstrate the bootstrap method.

1. For reproducibility, set a random seed of **34097**. We learned to do this in Chapter 8. Recall that you choose **Add-Ins ► Random Seed Reset**. Enter the new seed value in the dialog box.
2. Make the **Pipeline_Safety** data table the active tab or window.
3. **Tables ► Subset**. Select a random sample size of 20 and click **OK**.

You will now have a data table with 20 rows.

4. **Analyze ► Distribution**. Cast **IGNITE_IND** as **Y**.
5. Move your cursor over the count or probability column in the Frequencies table in the distribution report and right-click, as shown in Figure 10.6. Select **Bootstrap**.

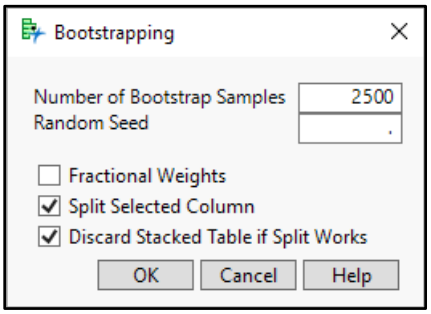
**Figure 10.6: Launching the Bootstrap**




6. Bootstrapping is the process of repeatedly choosing rows from a data table with

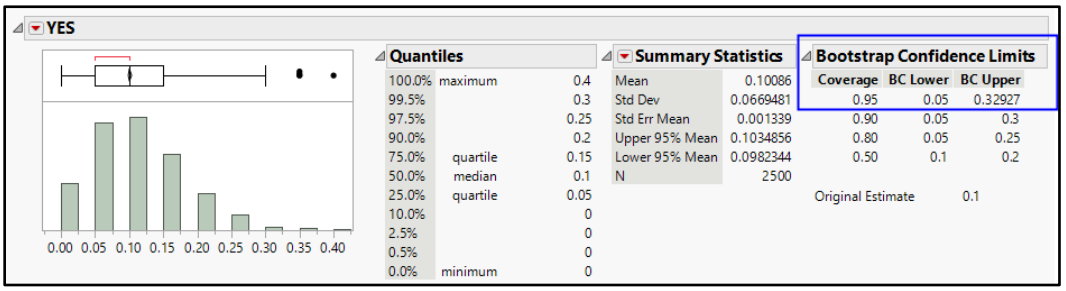
replacement. The default is to choose 2,500 samples (See Figure 10.7) of $n = 20$ observations. Click **OK**. We can then examine the distribution of the proportion of YES results to create confidence intervals.

**Figure 10.7: The Bootstrapping Dialog Box**




7. A new data table will open containing the relative frequencies from our first subset as well as 2,500 more bootstrap samples. Scroll through the table and notice that the proportion of YES results varies between 0.05 and 0.20. Then click the green arrow next to **Distribution** in the table pane of the data window.
8. Figure 10.8 shows the distribution of YES for the bootstrapped results as well as four confidence intervals.

**Figure 10.8: Bootstrap Confidence Limits**




In this example, the bootstrap confidence interval is [0.05, 0.329]. By comparison to the first interval we estimated, [0.096, 0.133], the bootstrap method generates a much wider interval to cover 95% confidence. Think of the extra width as an "uncertainty premium" that we pay to work under less-than-ideal conditions. The bootstrap provides less precision, but allows us to create a useful interval from a small sample.

## Working with Summary Data

Now suppose that we did not have the complete table of data but had read a report or article that summarized this aspect of pipeline disruptions. Such a report might have stated that in the period under study, 131 pipeline disruptions ignited out of 1,154 reported incidents. So, we do not know which incidents involved fires, but we do know the total number of ignitions and the total number of incidents.

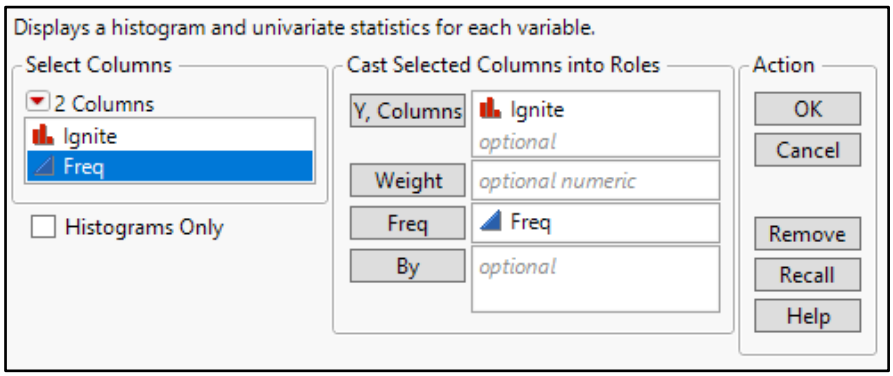
1. Create a new data table that matches Figure 10.9.

**Figure 10.9: A Table for Summary Data**

| | Ignite | Freq |
|---|---|---|
| 1 | NO | 1023 |
| 2 | YES | 131 |

2. Select **Analyze ► Distribution**. As shown in Figure 10.10, cast **Ignite** into the **Y** role and identify **Freq** as containing the frequency counts.

**Figure 10.7: Analyzing a Distribution with Summary Data**




The rest of the analysis is exactly the same as described earlier. Clicking the red triangle next to **Ignite** opens a menu to generate the confidence interval or conduct a significance test.

## A Few Words about Error

Whatever we conclude, there is some chance that we have come to the wrong conclusion about the population proportion. This has nothing to do with computational mistakes or data entry errors (although it's always wise to double check) and everything to do with sampling variability. When we construct a 95% confidence interval, we are gambling that our one sample is from among the 95% whose sample proportions lie a modest distance from the actual parameter. By bad luck, we might draw a sample from among the other 5% of possible samples.

With hypothesis tests, the situation is slightly different because we can go astray whether we reject the null hypothesis or not. If our *P*-value is very small, there is still a chance that we are rejecting the null hypothesis when we should not. This is called a *Type I* Error, and hypothesis testing is really designed to minimize the risk of encountering this situation.

Alternatively, a larger *P*-value (which leads us to persist in believing the null hypothesis) also carries the risk of a different (*Type II*) error whereby we should reject the null hypothesis, but do not because our evidence does not convince us to do so. If you can come to understand what these errors imply and why they are important to decision-making, you will more quickly come to incorporate the logic of significance testing in your own thinking.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* We will continue our analysis of the pipeline disruption data.
   a. Construct and interpret a 95% confidence interval for the proportion of pipeline disruptions that involved explosions (**EXPLODE_IND**).
   b. Now construct and interpret a 90% confidence interval for the same parameter (that is, proportion of explosions).
   c. In a sentence or two, explain what happens when we lower the confidence level.
   d. Based on your confidence intervals, would it be credible to conclude that only 5% of all disruptions involved explosions? Explain your reasoning.
2. *Scenario:* You live in a community of 23,847 households. You have selected a simple random sample of 250 of these households from a printed list of addresses. Suppose that, from prior figures, you previously believed that 18% of these households have no Internet service. You suspect that more homes have recently acquired service, so that the number is now less than 18%. In your sample of 250 households, you find 35 with no service.
   a. Does this situation satisfy the conditions required for statistical inference? Explain.
   b. Construct a 95% confidence interval for the proportion of the community without Internet service.
   c. Perform a significance test using the alternative hypothesis that is appropriate to this situation. Report on your findings, and explain your conclusion.
   d. Now suppose that your sample had been four times larger, but you found the same sample proportion. In other words, you surveyed 1,000 homes and found 140 with no Internet service. Repeat parts a and b, and report on your new findings. In general, what difference does a larger sample seem to make?

3. *Scenario:* Recall the data in **NHANES 2016**. Researchers asked school-aged respondents (635 of 9,336 participants) about physical education in their schools. Recall that NHANES is a national project in the United State that randomly samples approximately 10,000 individuals, gathering comprehensive data about health, nutrition, and lifestyles. There is a new sample every two years.
   a. Does the NHANES data set satisfy the conditions for inference? Explain.
   b. Construct and interpret a 95% confidence interval for the population proportion of all school children who have Physical Education (PE) during the school day. (Column **PAQ744**). **NOTE**: Of the 635 school children, one did not know about PE. Use the Local Data Filter to hide and exclude that person so that we have binary responses for our sample.
   c. Construct and interpret a 99% confidence interval for the same proportion using the same column. Explain fully how this interval compares to the one in part a.
   d. Repeat part b using a 90% confidence level.
   e. What can you say in general about the effect of the confidence level?

4. *Scenario:* Recall the study of traffic accidents and binge drinking introduced in Chapter 5. The data from the study were in the data table called **Binge**. Respondents were people ages 18 to 39 who drink alcohol.
   a. Assume this is an SRS. Does this set of data otherwise satisfy our conditions for inference? Explain.
   b. In that study, 213 of 3,801 respondents reported having ever been involved in a car accident after drinking. If we can assume that this is a random sample, construct and interpret a 95% confidence interval for the population proportion of all individuals who have ever been in a car accident after drinking.
   c. In the same study, 485 respondents reported that they binge drink at least once a week. Based on this sample, is it credible to think that approximately 10% of the general population binge drinks at least once a week? Explain what analysis you chose to do and describe the reasoning that leads to your conclusion.

5. *Scenario:* Recall the study of dolphin behavior introduced in Chapter 5; this summary data is in the data table called **Dolphins**.
   a. Assume this is an SRS. Does this set of data otherwise satisfy our conditions for inference? Explain.
   b. Assuming that this is a random sample, construct and interpret a 95% confidence interval for the population proportion of dolphins who feed at a given time in the evening.
   c. Based on the data in this sample, would it be reasonable to conclude that if a dolphin is observed in the morning, there is greater than a 0.5 probability that the dolphin is socializing?

6. *Scenario:* The data table **MCD** contains daily stock values for the McDonald's Corporation for the first six months of 2019. One column in the table is called **Increase**, and it is a categorical variable indicating whether the closing price of the stock increased from the day before.
   a. Using the data in the table, develop a 90% confidence interval for the proportion of days that McDonald's stock price increases. Report on what you find and explain what this tells you about daily changes in McDonald's stock.
   b. Now construct a 95% confidence interval for the proportion of days that the price increases. How, specifically, does this interval compare to the one you constructed in part a?

7. *Scenario:* The data table **TimeUse** contains individual responses to a survey conducted by the American Time Use Survey in 2003, 2007, and 2017. The survey used complex sampling methods as described in Chapter 8. In all, there were 43,191 respondents, including 10,223 interviewed in 2017. For these questions, use Data Filter to restrict your analysis to the 2017 respondents. Respondents' ages ranged from 15 to 85 years.
   a. Does this data table satisfy the conditions for inference? Explain.
   b. Within the chapter, all of the categorical variables that we have analyzed have been dichotomous. We can also create confidence intervals for categorical variables that have more than two levels. Construct and interpret a 95% confidence interval for the population proportion whose employment status (**empstat**) is "Not in labor force".
   c. Construct and interpret a 99% confidence interval for the same proportion using the same column. Explain fully how this interval compares to the one in part a. In general, what can you say about the effect of the confidence level?
   d. Using the 95% confidence interval, can we conclude that less than 40% of Americans in 2017 between the ages of 15 and 85 were not in the labor force? Explain your reasoning.
8. *Scenario:* In this exercise, we return to the **FAA Bird Strikes** data table. Recall that this set of data is a record of events in which commercial aircraft collided with birds in the United States.
   a. Does this data set satisfy the conditions for inference? Explain.
   b. Construct and interpret a 95% confidence interval for the population proportion of flights where just one bird was struck.
   c. Construct and interpret a 99% confidence interval for the same proportion using the same column. Explain fully how this interval compares to the one in part a. What can you say in general about the effect of the confidence level?
   d. Based on this sample, is it credible to think that approximately 90% of reported bird strikes involve just one bird? Explain what analysis you chose to do and describe the reasoning that leads to your conclusion.

9. *Scenario:* The data table **Hubway** contains observations of 55,230 individual bike trips using the short-term rental bike-sharing system in Boston, Massachusetts. People can register with the program or simply rent a bike as needed. Within the data table, individual trips are either by **Registered** or **Casual** users.
    a. Does this data table satisfy the conditions for inference? Explain.
    b. Estimate (90% CI) the proportion of all Hubway trips taken by Casual users.
    c. Of the 55,230 trips in the sample, 2,136 of them started at the Boston Public Library. Construct and interpret a 95% confidence interval for the proportion of all Hubway trips originating at the Library.
    d. Suppose that the data table had only 5,523 rows (one-tenth the sample size) and 214 trips starting at the Library. Using this summary data, construct and interpret a 95% confidence interval for the proportion of all Hubway trips originating at the Library.
    e. Compare the intervals in parts c and d. How does the size of the sample affect the confidence interval?

---

1 See Wasserstein, R.L. and Lazar, N.A. 2016, "The ASA Statement on p-Values: Context, Process, and Purpose." The American Statistician, 70:2, 129-133, DOI: 10.1080/00031305.2016.1154108.

[2] "1-Alpha" is admittedly an obscure synonym for "confidence level." It is the complement of the significance level discussed earlier ($\alpha$).

# Chapter 11: Inference for a Single Continuous Variable

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## Overview

Chapter 10 introduced inference for categorical data, and this chapter does the same for continuous quantitative data. As in Chapter 10, we will learn to conduct hypothesis tests and to estimate a population parameter using a confidence interval. We already know that every continuous variable can be summarized in terms of its center, shape, and spread. There are formal methods of inference for a population's center, shape, and spread based on a single sample. In this chapter, we will just deal with the mean.

## Conditions for Inference

Just as with categorical data, we rely on the reasonableness of specific assumptions about the conditions of inference. Because the conditions are so fundamental to the process of generalizing from sample data, we begin each of the next several chapters by asking "what are we assuming?"

Our first two assumptions are the same as before: observations in the sample are independent of one another, and the sample is small in comparison to the entire population (usually 10% or less). We should also be satisfied that the sample is probabilistic and therefore can reasonably represent the population. If we know that our sample data violates one or more of these conditions, then we should also understand that we will have little ability to make reliable inferences about the population mean.

The first set of procedures that we will encounter in this chapter relies on the sampling distribution of the mean, which we introduced in Chapter 8. We can apply the *t*-distribution

reliably if the variable of interest is normally distributed in the parent population, or if we can invoke the Central Limit Theorem because we have ample data. If the parent population is highly skewed, we might need quite a large *n* to approach the symmetry of a *t*-distribution; the more severe the skewness of the population, the larger the sample required.

## Using JMP to Conduct a Significance Test

We will return to the pipeline safety data table and focus on the ages of the pipes that were disrupted in the period. According to a report from the U.S. Department of Energy in 2017,[1] public utilities around the United States have placed a priority on replacing and updating pipelines. The report (p. 5) states that "The age and type of pipeline infrastructure in service throughout LDC systems varies widely across the U.S. In some areas, portions of the natural gas distribution system can be more than 100 years old." As of 2015, the vast majority of the local distribution company main gas lines were plastic, but there was still a sizeable number of miles of cast iron and bare steel pipes that are subject to corrosion. These tend to be older pipes that were placed into service many years ago.

Our sample does not contain data about the entire infrastructure, but only provides data about disruptions that have occurred. For the sake of this illustration, suppose that there is a national goal of reducing the mean age of pipelines known to leak, and that the target is less than 40 years, and we want to ask if this goal has been met. We could use this set of data to test the following hypotheses:

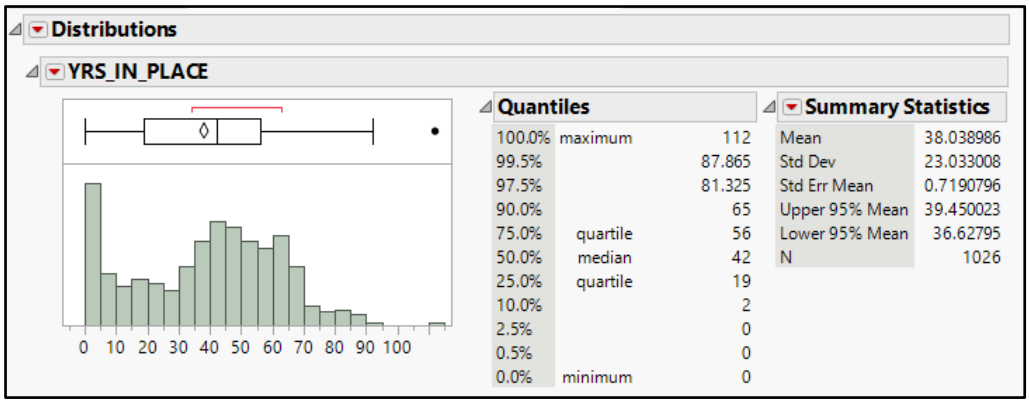
$H_0$: $\mu \geq 40$ [null hypothesis: the goal has not been met]

$H_a$: $\mu < 40$ [alternative hypothesis: the goal has been met]

As before, we will assume that the observations in the table are a representative sample of all disruptions, that the observations are independent of one another, and that we are sampling from an infinite process.

1. Open the **Pipeline Safety** data table.
2. Select **Analyze ► Distribution**. Select **YRS_IN_PLACE** as **Y** and click **OK**.
3. Click the red triangle next to **Distributions** and **Stack** the results (shown in Figure 11.1).

**Figure 11.1: Distribution of Pipeline Years in Place**



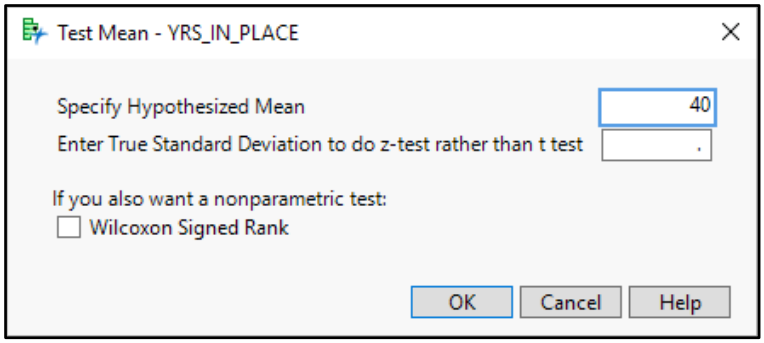

Before going further, it is important to consider the remaining conditions for inference. The histogram reveals a multi-peaked, right-skewed distribution, most likely revealing a sample that comes from a similarly shaped population. Because our sample has 1,026 observations, we will rely on the Central Limit Theorem to justify the use of the *t*-distribution.

In the context of our hypotheses, we note that the mean of the sample is a bit longer than 38 years, but the median is 42 years.

If we had a smaller sample and wanted to investigate the normality condition, we would construct a normal quantile plot by clicking on the red triangle next to **YRS_IN_PLACE** and selecting that option.

4. Click the red triangle next to **YRS_IN_PLACE** and select **Test Mean**. This opens the dialog box shown in Figure 11.2. Type **40** as shown and click **OK**.

**Figure 11.2: Testing a Mean**

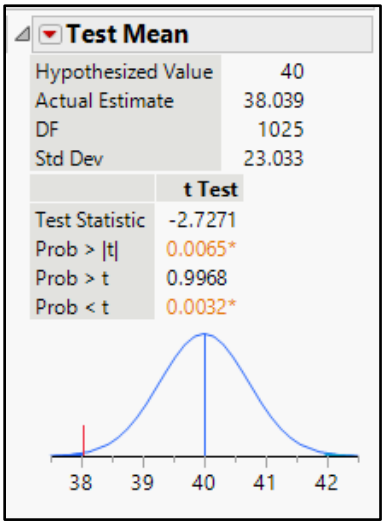



Notice that the dialog box has two optional inputs. First, there is a box into which you can **Enter True Standard Deviation to do z-test rather than t test.** In an introductory statistics course, one might encounter a textbook problem that provides the population standard deviation, and therefore one could base a test on the Normal distribution rather than the *t*-distribution (that is, conduct a z-test). Such problems may have instructional use, but realistically, one never knows

the population standard deviation. The option to use the z-test has virtually no practical applications except possibly when one uses simulated data.

The second option in the dialog box is for the **Wilcoxon Signed Rank** test; we will discuss that test briefly in the next section. First, let's look at the results of the t test in Figure 11.3.

**Figure 11.3: T Test Results**




In the test report, we find that this test compares a hypothesized value of 40 minutes to an observed sample mean of 38.039, using a *t*-distribution with 1,025 degrees of freedom. The sample standard deviation is 23.033 minutes.

We then find the **Test Statistic**, which JMP computes using the following formula:

$$t = \frac{(\overline{x} - \mu_0)}{s_{\overline{x}}} = \frac{(38.039 - 40)}{\left(\frac{23.033}{\sqrt{1025}}\right)} = -2.721$$

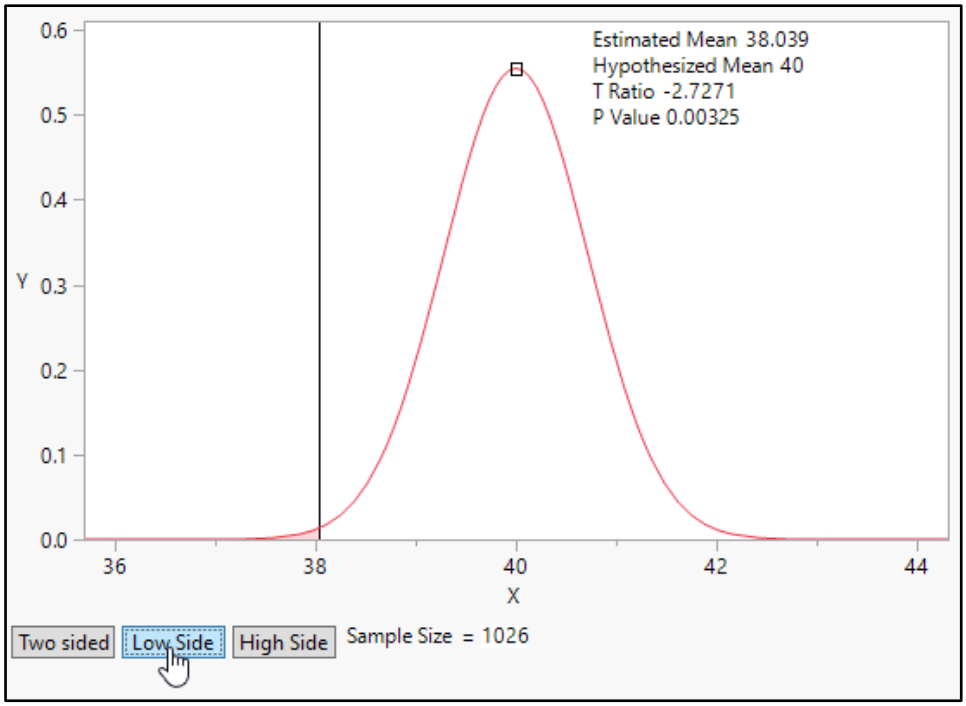
Below the test statistic are three *P*-values, corresponding to the three possible formulations of the alternative hypothesis. We are doing a less-than test, so we are interested in the last value shown in red: 0.0032. This probability value is so small that we should reject the null hypothesis. In view of the observed sample mean, it is not plausible to believe that the population mean is 40 or higher. The graph at the bottom of the panel depicts the situation. Under the null hypothesis that the population mean is truly 40 minutes, the blue curve shows the likely values of sample means. Our sample mean is marked by the red line, which sits -2.7 standard errors to the left of 180. If the null hypothesis were true, we are very unlikely to see a sample mean this low.

## More About *P*-Values

*P*-values are fundamental to the significance tests that we have seen thus far, and they are central to all statistical testing. You will work with them in nearly every remaining chapter of this book, so it is worth a few moments to explore another JMP feature at this point.

1. Click the red triangle next to **Test Mean** and select **PValue animation**. This opens an interactive graph (Figure 11.4) intended to deepen your understanding of *P*-values.

**Figure 11.4: Interactive P-Value Visualization Tool**




The red curve is the theoretical sampling distribution assuming that the population mean equals 40, and the vertical black line marks our sample mean. The shaded pink area below the curve to the left of the black line represents the *P*-value. In the current example, our alternative hypothesis was $\mu < 40$, and because our sample mean was so much smaller than 40, we concluded that the null hypothesis was far less credible than this alternative. We can use this interactive graph to visualize the connection between the alternative hypothesis and the *P*-value.

2. Click the **Low Side** button, corresponding to the less-than alternative.

This illustrates the reported *P*-value of 0.00325. Clicking the other two buttons in turn shows where the probability would lie in either of the other two forms of the alternative.

Suppose that our initial hypothesized value had not been 40 years, but a lower value. How would that have changed things?

3. Notice that there is a small square at the peak of the curve. Place your cursor over that peak, click and drag the cursor slowly to the left, essentially decreasing the hypothesized value from 40 years to a smaller amount.

As you do so, the shaded area grows. Using the initial value of 40, the results of this test cast considerable doubt on the null hypothesis. Had the null hypothesis referred to a smaller number, then our results would have been less closely associated with the low side alternative.

4. Continue to move the center of the curve until the *P*-value decreases to just about 0.05—the conventional boundary of statistical significance.

From this interaction we can see that our sample mean of 38.039 would no longer be statistically significant in a low side test (that is, with a less-than alternative) if the hypothesized mean were about 39.2 or less.

What if our sample had been a different size?

5. Click the number **1026** next to **Sample Size** in the lower center of the graph. Change it to **513** (one-half the original sample size) and observe the impact on the graph and the *P*-value.
6. Change the sample size to **2,052** and observe the impact.

The magnitude of the *P*-value depends on the shape of the sampling distribution (assumed to be a *t*-distribution here), the size of the sample, the value of the hypothesized mean, and the results of the sample. The size of the *P*-value is important to us because it is our measure of the risk of Type I errors in our tests.

Ideally, we want to design statistical tests that lead to correct decisions. We want to reject the null hypothesis when the alternative is true, and we want to retain the null hypothesis when the alternative is false. Hypothesis testing is driven by the desire to minimize the risk of a Type I error (rejecting a true null hypothesis). We also do well to think about the capacity of a particular test to discern a true alternative hypothesis. That capacity is called the *power of the test*.

## The Power of a Test

Alpha ($\alpha$) represents the investigator's tolerance for a Type I error. There is another probability known as beta ($\beta$), which represents the probability of a Type II error in a hypothesis test. The *power of a test* equals $1 - \beta$. In planning a test, we can specify the acceptable $\alpha$ level in advance, and accept or reject the null hypothesis when we compare the *P*-value to $\alpha$. This is partly because the *P*-value depends on a single hypothesized value of the mean, as expressed in the null hypothesis.

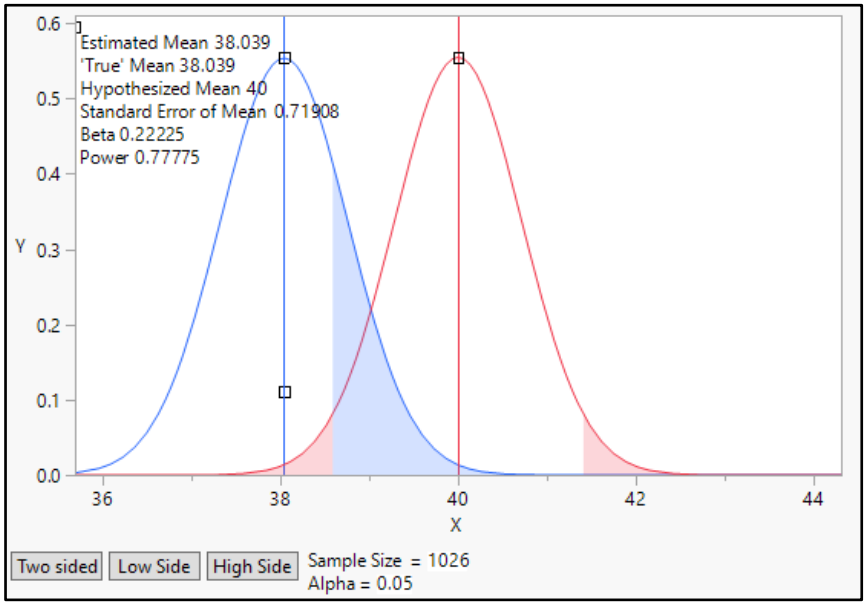
The situation is not as simple with $\beta$. A Type II error occurs when we do not reject the null hypothesis, but we should have done so. That is to say, the alternative hypothesis—which is always an inequality—is true, but our test missed it. In our example, where the alternative

hypothesis was $\mu < 40$, there are an infinite number of possible means that would satisfy the alternative hypothesis. To estimate $\beta$, we need to make an educated guess at an actual value of $\mu$. JMP provides another animation to visualize the effects of various possible $\mu$ values.

For this discussion of power, we will alter our example and consider the two-sided version of the significance test: is the mean different from 40 years?

1. Return to the distribution report. Click the red triangle next to **Test Mean** and select **Power animation**. This opens an interactive graph (Figure 11.5 below) that illustrates this concept of power. The two-sided alternative is the default setting.

**Figure 11.5: The Power Animation**




2. Initially, the annotations on the graph are far in the upper left corner. Grab the small half-square just above the words "Estimated Mean" and slide all of text to the right as shown in Figure 11.5 so that you can see the blue curve more clearly.

This graph is more complex than the earlier one. We initially see two distributions. The red one (on the right), centered at 40, shows the sampling distribution based on the assumption expressed in the null hypothesis. The blue curve on the left shows an alternative sampling distribution (a parallel universe, if you like) based on the possibility that $\mu$ actually is equal to 38.039, the value of the sample mean.

This graph initially lays out one pre-sampling scenario of two competing visions. The red vision depicts the position of the null hypothesis: the population mean is truly 40, and samples vary around it. The blue vision says, "Here's another possible reality: the population mean just so happens to be 38.039." We will be able to relocate the blue curve, and thereby see how the power of our test would change if the population mean were to be truly any value we choose.

The graph initially displays a two-sided test. In the red curve, the pink-shaded areas now represent not the *P*-value but rather $\alpha = .05$. That is, each tail is shaded to have 2.5% with 95% of the curve between the shaded areas. If we take a sample and find the sample mean lying in either pink region, we will reject the null hypothesis. In advance of sampling, we decided to make the size of those regions 5%. This is the rejection rule for this particular test.

The blue-shaded area shows $\beta$. As it is depicted, with the population mean really at 38.039, the null hypothesis *is* false and should be rejected. Among the possible samples from the blue universe (which is where the samples would really come from) about 22% of them are outside of the rejection region of the test, and in those 22% of possible samples, we would erroneously fail to reject the null hypothesis. The power of this test is 0.78, which is to say that if the true mean were really 38.039 minutes, this test would correctly reject the null hypothesis in 78% of possible samples. Not bad. Now let's interact with the animator.

3. First, let's change **Alpha** from **0.05** to **0.01**. Click **0.05** and edit it.

Notice what happens in the graph. Because alpha is smaller, the pink-shaded area is smaller, and the light blue shading gets larger. There is a trade-off between $\alpha$ and $\beta$: if we tighten our tolerance for Type I error, we expose ourselves to a greater chance of Type II error and diminish the power of a test.

4. Restore **Alpha** to **0.05** and now double the **Sample Size** to **2052**. You will need to rescale the vertical axis (grab the upper portion with the hand tool) to see the full curves after doing this.

Doubling the size of the sample radically diminishes beta and substantially increases power from 0.78 to 0.97. So, there is a lesson here: another good reason to take large samples is that they make tests more powerful in detecting a true alternative hypothesis.

5. Finally, grab the small square at the top of the blue curve, and slide it to the right until the true mean is at approximately 39.5 years.

This action reveals yet another vertical line. It is green and it represents our sample mean. More to the point, however, see what has happened to the power and the chance of a Type II error. If the mean truly was 39.5 minutes, the alternative is again true and the null hypothesis is false: the population mean is not 40 as hypothesized, but half a year less.

When the discrepancy between the hypothesized mean and the reality shrinks, the power of the test falls, and the risk of Type II error increases rapidly. This illustrates a thorny property of power, and one that presents a real challenge. Hypothesis testing is better at detecting large discrepancies than small ones. If our null hypothesis is literally false but happens to be close to the truth, there will be a relatively good chance that a hypothesis test will lead us to the incorrect conclusion. Enlarging the sample size would help, if that is feasible and cost-effective.

## What If Conditions Are Not Satisfied?

The pipeline data table contains a large number of observations so that we could use the t test despite the indications of a highly skewed population. What are our options if we do not meet the conditions for the t test?

If the population is reasonably symmetric, but normality is questionable, we could use the Wilcoxon Signed Rank test. This is an option presented in the **Test Means** command, as illustrated in Figure 11.2. Wilcoxon's test is identified as a *nonparametric test*, which just indicates that its reliability does not depend on the parent population following any specific probability distribution. Rather than testing the mean, the test statistic computes the difference between each observation and the median, and ranks the differences. This is why symmetry is important: in a skewed data set, the gap between mean and median is comparatively large.

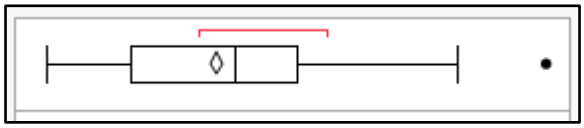
We interpret the *P*-values just as we do in any hypothesis test, and further information is available within the JMP Help system. At this point, it is useful to know that the t test is not the only viable approach. If the sample is small and strongly skewed, then inference is risky. If possible, one should gather more data. If costs or time constraints prohibit that avenue, and if the costs of indecision exceed those of an inferential error, then the techniques illustrated above might need to inform the eventual decision (however imperfectly) with the understanding that the estimates of error risk are inaccurate. Another alternative is bootstrapping, which we will discuss shortly in the context of confidence intervals.

## Using JMP to Estimate a Population Mean

Just as with categorical data, we might want to estimate the value of the population parameter—in this case, the mean of the population. We will continue to work with the pipeline data and the mean age of pipes that leaked. We will start by looking back at the initial **Distributions** report that we created earlier.

You might not have noticed previously, but JMP always reports a confidence interval for the population mean, and it does so in two ways. Box plots include a *Mean Confidence Diamond* (see Figure 11.6) and the **Summary Statistics** panel reports the **Upper 95% Mean** and **Lower 95% Mean**. The mean diamond is a visual representation of the confidence interval, with the left and right (or lower and upper) points of the diamond corresponding to the lower and upper bounds of the confidence interval.

**Figure 11.6: Box Plot Detail Showing Mean Confidence Diamond**



Now look at the **Summary Statistics** panel, shown earlier in Figure 11.1. If we can assume that the conditions for inference have been satisfied, we can estimate with 95% confidence that the nationwide mean age of pipes that were disrupted in the U.S. is between 36.63 and 39.45 years.

Notice that JMP calculates the bounds of the interval anytime we analyze the distribution of a continuous variable; it is up to us whether the conditions warrant interpreting the interval.

What if we wanted a 90% confidence interval, or any confidence level other than 95%?

1. Click the red triangle next to **YRS_IN_PLACE** and select **Confidence Interval ► 0.90**.

**Figure 11.7: 90% Confidence Intervals**

**Confidence Intervals**

| Parameter | Estimate | Lower CI | Upper CI | 1-Alpha |
|---|---|---|---|---|
| Mean | 38.03899 | 36.85514 | 39.22284 | 0.900 |
| Std Dev | 23.03301 | 22.22794 | 23.90381 | 0.900 |

A new panel opens in the results report. (See Figure 11.7.) The report shows the boundaries of the 90% confidence interval for the population mean. It also shows a confidence interval for the population standard deviation—not only can we estimate the population's mean ($\mu$), but we can similarly estimate the standard deviation, $\sigma$.

In this case, it's not entirely clear that we meet the conditions for inference: the sample is highly skewed, and one might wonder if observations are genuinely independent. If, for example, there is earthquake activity in a region, perhaps many pipelines share a common root cause for leaks. If so, then the disruptions did not occur independently. One alternative is bootstrapping, illustrated in the next optional section.

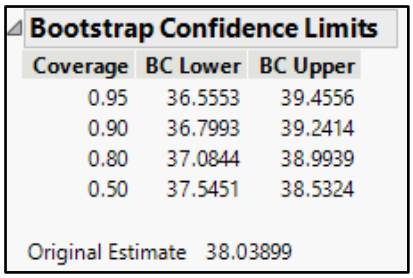
## Bootstrapping (Optional; Requires JMP Pro)

As noted in Chapter 10, when our sample data do not satisfy the traditional conditions for inference, we can use bootstrapping to produce an empirical sampling distribution for the population mean and permit a judgment about the likelihood of the one observed sample mean. Of particular value is that the bootstrap generates a reasonable estimate of the standard error.

We proceed much as we did in Chapter 10.

1. Hover the cursor over the table of values just under **Summary Statistics** within the distribution report, right-click and choose **Bootstrap**.
2. In the Bootstrapping dialog box, accept the default number of samples. For reproducibility, set a random seed of **84,346** and click **OK**.
3. Once the new data table of bootstrapped sample results has opened, click the green arrowhead next to **Distribution**. Figure 11.8 shows the resulting bootstrap confidence intervals for the **Mean**.

**Figure 11.8: Bootstrap Confidence Intervals**



**Bootstrap Confidence Limits**

| Coverage | BC Lower | BC Upper |
|---|---|---|
| 0.95 | 36.5553 | 39.4556 |
| 0.90 | 36.7993 | 39.2414 |
| 0.80 | 37.0844 | 38.9939 |
| 0.50 | 37.5451 | 38.5324 |

Original Estimate 38.03899

Once again, we obtain several intervals at different confidence levels. With a sample as large as this one, we notice that the bootstrapped intervals are quite similar to the ones that we obtained earlier.

## Matched Pairs: One Variable, Two Measurements

The *t*-distributions underlying most of this chapter were the theoretical contribution of William Gosset, so it is appropriate to conclude this chapter with some data originally collected by Gosset, whose expertise was in agriculture as it related to the production of beer and ale, the industry in which he worked as an employee of the Guinness Brewery.

This particular set of data contains measurements of crop yields from eleven different types of barley (referred to as "corn" in Gosset's day) in a *split plot* experiment. Seeds of each type of barley were divided into two groups. Half of the seeds were dried in a kiln, and the other half were not. All of the seeds were planted in eleven pairs of farm plots, so that the kiln-dried and regular seeds of one variety were in adjacent plots.

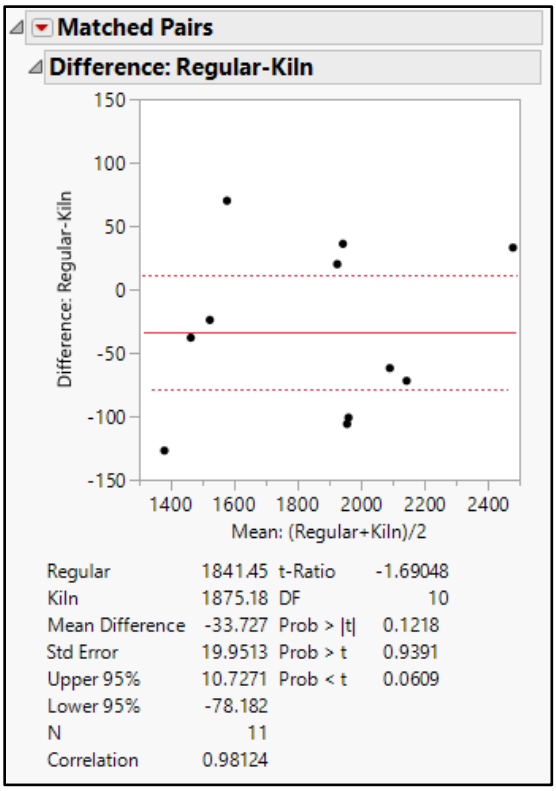
As such, the data table contains 22 measurements of the same variable: crop yield (pounds of barley per acre). However, the measurements are not independent of one another, because adjacent plots used the same variety of seed and were presumably subject to similar conditions. The point of the experiment was to learn if kiln-dried seeds increased the yields in comparison to conventional air-dried seeds. It is important that the analysis compares the results variety by variety, isolating the effect of drying method.

The appropriate technique is equivalent to the one-sample t test, and rather than treating the data as 22 distinct observations, we treat it as 11 *matched pairs* of data. Rather than analyze the individual values, we analyze the *differences* for each pair. Because we have a small sample, the key condition here is that the population differences be approximately normal, just as in the one-sample t test.

Open the data table **Gosset's Corn**. The first column shows the yields for the 11 plots with kiln-dried seeds, and the second column shows the corresponding yields for the regular corn. Glancing down the columns, we see that the kiln-dried values are sometimes larger and sometimes smaller than the others.

1. Select **Analyze ► Specialized Modeling ► Matched Pairs**. Choose both columns and select them as **Y, Paired Response** and click **OK**. The results are shown in Figure 11.9.

**Figure 11.9: The Results of the Matched Pairs Command**




This report provides a confidence interval and *P*-values for the mean of the pairwise differences. First, we see a graph of the 11 differences in comparison to the mean difference of -33.7 pounds per acre. There is variation, but on average, the kiln-dried seeds yielded nearly 34 pounds more per acre within the sample (the differences are calculated as regular – kiln). Now we want to ask what we can infer about the entire population based on these results.

Next note the confidence interval estimate. With 95% confidence, we can conclude that the yield difference between regular and kiln-dried seeds is somewhere in the interval from -78.182 to +10.7271. Because this interval includes 0, we cannot confidently say that kiln-drying increases or decreases yield in the population, even though we did find an average increase in this one study.

On the lower right side of the report, we find the results of the significance test. We are interested in the bottom *P*-value, corresponding to the less-than test (that is, our alternative hypothesis was that kiln-dried yields would be greater than regular, so that differences would be negative). Based on this *P*-value, the results fall slightly short of conventional significance; we do not reject the null hypothesis.

# Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* We will continue our analysis of the pipeline disruption data. The columns **EST_COST_PROP_DAMAGE** and **EST_COST_GAS_RELEASED** contain the dollar value of property and natural gas lost as a consequence of the disruption.
   a. Do each of these columns satisfy the conditions for statistical inference? Explain.
   b. Report and interpret a 95% confidence interval for the mean property value costs of a pipeline disruption.
   c. Construct and interpret a 90% confidence interval for the same parameter.
   d. In a sentence or two, explain what happens when we lower the confidence level.
   e. Construct and interpret a 99% confidence for the dollar cost of natural gas loss in connection with a pipeline disruption.
   f. Conduct a test using an alternative hypothesis that, on average, a pipeline disruption loses less than $50,000 worth of gas. Report on your findings.
   g. Use the *P*-value animation tool to alter the null hypothesis just enough to change your conclusion. In other words, at what point would the data lead us to the opposite conclusion?
   h. Use the power animator and report on the power of this test.

2. *Scenario:* Recall the data table called **Michelson 1879**. These are Michelson's recorded measurements of the speed of light. Because the measurements were taken in different groups, we will focus only on his last set of measurements. Before doing any further analysis, take the following steps.
   - Open the data table.
   - Select **Rows ► Data Filter**. Highlight **Trial#** and click **Add**.
   - Select the **Show** and **Include** check boxes and select **fifth** from the list. Now your analysis will use only the 20 measurements from his fifth trial.
     a. Assume this is an SRS. Does the **Velocity** column satisfy our conditions for inference? Explain.
     b. Construct and interpret a 90% confidence interval for the speed of light, based on Michelson's measurements.

c. Michelson was a pioneer in measuring the speed of light, and the available instruments were not as accurate as modern equipment. Today, we generally use the figure 300,000 kilometers per second (kps) as the known constant speed. Put yourself in Michelson's shoes and conduct a suitable hypothesis test to decide whether 300,000 kps would have been a credible value based on the measurements that he had just taken.

d. Use the *P*-value animation tool to alter the null hypothesis so that Michelson might have believed the null hypothesis. In other words, at what point would his data no longer be statistically distinguishable from the null hypothesis value?

3. *Scenario:* Recall the data in the **NHANES 2016** data table. Specifically, remember that we have two measurements for the height of very young respondents: **BMXRECUM** and **BMXHT**, the recumbent (reclining) and standing heights.

   a. In a sentence or two, explain why you might expect these two measurements of the same individuals to be equal or different.

   b. Perform a matched pairs test to decide if the two ways of measuring height are equivalent. Interpret the reported 95% confidence interval and explain the test results. What do you conclude?

4. *Scenario:* The data table **Airline Delays** contains a sample of flights for four airlines (carriers) destined for three busy airports. Using the data filter as described in Scenario 2 above, select just the sample of Delta Airlines flights flying into Hartsfield-Jackson Atlanta International Airport for analysis. Our goal is to analyze the delay times, expressed in minutes. Recall that an early arrival is expressed as a negative delay.

   a. Assume this is an SRS of many flights by the selected carrier over a long period of time. Have we satisfied the conditions for inference about the population mean delay? Explain.

   b. Report and interpret the 95% confidence interval for the mean flight delay for Delta Airline flights bound for Atlanta.

   c. Let L stand for the lower bound of the interval and U stand for the upper bound. Does this interval tell you that there is a 95% chance that your flight to Chicago will be delayed between L and U minutes? What *does* the interval tell you?

   d. Using a significance level of 0.05, test the null hypothesis that the population mean delay is zero minutes against a one-sided alternative that the mean delay is less than zero (that is, on average flights are early).

   e. Now suppose that the true mean of the delay variable is really -2 minutes. Use the power animation to estimate the power of this test under that condition. Explain what you have found.

5. *Scenario:* We have extensive flight data in the data table **Airline delays 3**, which contains data for every North American flight in the month of March 2019. Our goal is to analyze the departure delay times, expressed in minutes.
   a. Assume this is an SRS. Have we satisfied the conditions for inference about the population of flights?
   b. Before performing further analysis, use the data filter to hide and exclude canceled flights (**cancelled** = 1). "Departure Delay" refers to the time difference between the scheduled departure and the time when the aircraft's door is closed and the plane pulls away from the gate. Develop a 95% confidence interval for the mean departure delay **DepDelay** and interpret the interval.
   c. Sometimes there can be an additional delay between departure from the gate and the start of the flight. "Wheels off time" refers to the moment when the aircraft leaves the ground. Develop a 95% confidence interval for the mean time between scheduled departure and wheels-off (**WheelsOff**). Report and interpret this interval. (HINT: use Match Pairs.)
   d. Now use a matched-pairs approach to estimate (with 95% confidence) the mean difference between departure delay and wheels off delay. Explain what your interval tells you.

6. *Scenario:* In this exercise, we return to the **FAA Bird Strikes** data table. Recall that this set of data is a record of events in which commercial aircraft collided with birds in the United States.
   a. Do the columns **SPEED** and **COST_REPAIRS** satisfy the conditions for statistical inference? Explain.
   b. Report and interpret a 95% confidence interval for the mean speed of flights when birds were struck.
   c. Construct and interpret a 99% confidence interval for the same parameter.
   d. In a sentence, explain what happens when we increase the confidence level.
   e. The full data table has more than 35,000 rows, but we only have repair cost data for 1,885 rows. Nevertheless, conduct a test using an alternative hypothesis that, on average, flight repairs caused by bird strikes cost more than $100,000. Report on your findings.
   f. Use the *P*-value animation tool to alter the null hypothesis just enough to change your conclusion. In other words, at what point would the data lead us to the opposite conclusion?
   g. Use the power animator and report on the power of this test.

7. *Scenario:* In 2004, the state of North Carolina released to the public a large data set containing information about births recorded in this state. Our data table **NC Births** is a random sample of 1,000 cases from this data set. In this scenario, we will focus on the birth weights of the infants (**weight**) and the amount of weight gained by mothers during their pregnancies (**gained**). All weights are measured in pounds.
    a. Do the data columns labeled weight and gained satisfy the conditions for inference? Explain.
    b. Test the hypothesis that the mean amount of weight gained by all NC mothers was more than 30 pounds in 2004. Explain what you did and what you decided. Use a significance level of 0.05.
    c. Construct and interpret a 95% confidence interval for the mean birth weight of NC infants in 2004.
8. Scenario: The data table **TimeUse** contains individual responses to a survey conducted by the American Time Use Survey in 2003, 2007, and 2017. The survey used complex sampling methods as described in Chapter 8. In all, there were 43,191 respondents, including 10,223 interviewed in 2017. Respondents' ages ranged from 15 to 85 years. For these questions, use Data Filter to restrict your analysis to the 2017 respondents.
    a. The column labeled **sleeping** contains the number of minutes of sleep per night reported by the respondents. Conventional wisdom holds that adults average approximately 8 hours (480 minutes) of sleep. Perform a significance test of the hypothesis that the population mean of **sleeping** equals 480 versus a two-sided alternative. Explain your conclusion.
    b. Construct and interpret a 95% confidence interval for the population mean number of minutes people spent reading and responding to personal email (column **email**) in 2017.
    c. Construct and interpret a 99% confidence interval for the mean using the same column. Explain fully how this interval compares to the one in part b. In general, what can you say about the effect of the confidence level?
    d. Using the 95% confidence interval, can we conclude 95% of adults spent between 1.65 and 2.13 minutes daily on personal email? Explain your reasoning.
    e. (Advanced) As noted above, this sample was not a simple random sample, but rather a complex sample. As such, we should properly apply the sampling weight located in the second column of the table. Reopen the **Distribution** dialog box, cast **email** into the Y role, and select **wt06** as the **Weight**. Re-construct the 95% CI for **email**. Comment on how these results compare to your earlier results.

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[1] Office of Energy Policy and Systems Analysis, "Natural Gas Infrastructure Modernization Programs at Local Distribution Companies: Key Issues and Considerations." Washington DC: U.S. Department of Energy. Available at https://www.energy.gov/sites/prod/files/2017/01/f34/Natural%20Gas%20Infrastructure%20Modernization%20Programs%20at%20Local%20Distribution%20Companies--Key%20Issues%20and%20Considerations.pdf . Accessed July 1, 2019.

# Chapter 12: Chi-Square Tests

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## Overview

This chapter takes us further into inferential analysis of categorical data. Two prior chapters have concentrated on categorical variables. In Chapter 6, we learned to compute relative frequencies, which we then used to illustrate concepts and rules of probability. In Chapter 10, we studied dichotomous categorical variables to learn about confidence intervals and hypothesis tests for a single population proportion.

In this chapter, we will consider two basic issues. First, we will learn to draw inferences about a categorical variable that can take on more than two possible values. Second, we will use sample data to infer whether two categorical variables are independent or related. The thread that unifies most of the discussion in this chapter is another family of continuous density functions called *chi-square* distributions. The test statistics that we will encounter in this chapter follow the shape and proportions of a chi-square ($\chi^2$) distribution, enabling us to calibrate the likelihood of sampling error.

## Chi-Square Goodness-of-Fit Test

In Chapter 10, we worked with the **Pipeline Safety** data table to learn about estimating a population proportion. In that chapter, we looked at the **IGNITE_IND** column—a dichotomous variable indicating whether natural gas ignited in connection with a reported disruption. At this point, let's consider another variable in the data table that indicates the nature of the leak. The **LEAK_TYPE** column shows five different values: **SEAL OR PACKING**, **PINHOLE**, **OTHER, CRACK,** and **CONNECTION FAILURE**. Suppose we naively wondered whether all leak types are equally likely to occur.

Here is how we will approach the question. First, we will specify a hypothesis reflecting our question and express the hypothesis as a model describing the causes. There are four categories,

and we are provisionally asserting that all four categories happen with equal probability in the United States. In short, our hypothesis becomes this:

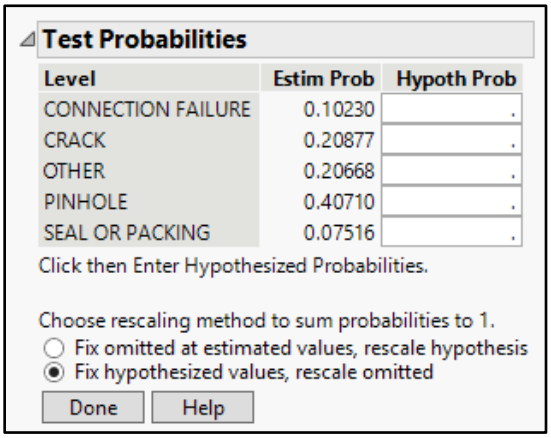
$H_0$: *Pr*(**SEAL OR PACKING**) = *Pr*(**PINHOLE**) = *Pr*(**OTHER**) = *Pr*(**CRACK**) = *Pr*(**CONNECTION FAILURE**) = *0.20*

Next, we will examine the sample data in comparison to the model. As usual, the software does the heavy work, but the underlying logic is simple. For this variable, our sample size happens to be *n* = 479. Before we look at the distribution of the observed data, we can note that if our null hypothesis is true, all five leak types share the same likelihood, then we anticipate observing that approximately 1/5 = 0.20 x 479 = 95.8 incidents are classified as each of the four types.

Obviously, we won't have observed 95.8 instances of any type—that's not possible. If the model is credible, though, we should discover somewhere in the vicinity of 96 seal or packing leaks, 96 pinhole leaks, and so on. As usual in statistical testing, we will need to decide precisely what qualifies as "in the vicinity" of 95.8. If some of the observed frequencies deviate too far from 95.8, we will conclude that there is a mismatch between the model and the empirical data. If the observed frequencies are all close to 95.8, we will conclude that the model is credible. Appropriately, the name of the test that we will use is the *goodness-of-fit* test. The goal of this test is to decide how well the model fits with the data.

1. Open the **Pipeline Safety** data table.
2. Select **Analyze ► Distribution**. Cast **LEAK_TYPE** as **Y** and click **OK**.
3. Click the red triangle next to **LEAK_TYPE** and select **Test Probabilities**. You should see a new panel like the one shown in Figure 12.1.

**Figure 12.1: Test Probabilities Panel**




4. Our hypothesized probabilities are all 0.20, so go ahead and enter **.20** into each of the five cells in the **Hypoth Prob** column and click **Done**. The new results are in Figure 12.2.

**JMP** anticipates that we might have hypothesized different probabilities for each category, so we separately enter each probability value. In fact, we could just enter a **1** into each cell, and JMP will rescale all values to equal 0.2; in this example, we will take the trouble to enter the hypothesized probabilities to reinforce the fact that we are testing the observed relative frequencies against these specific values. If we had a model with different null values, the procedure would be the same, and we would enter the appropriate values. The software needs to ensure that the sum of the probabilities is 1, and it needs our instructions about how to correct things if it isn't. By selecting the **Fix hypothesized values, rescale omitted** option, we are telling JMP to lock in the numbers that we enter, and change any omitted cell to balance the sum to 1.

JMP reports two test statistics and their associated *P*-values. Under the null hypothesis with sufficiently large samples, both test statistics asymptotically follow an approximate chi-square distribution with $k - 1$ degrees of freedom (DF), where $k$ refers to the number of categories for the variable in question. In this discussion, we will consider Pearson's chi-square;[1] both test results are interpreted in the same way.

**Figure 12.2: The Results of the Goodness-of-Fit Test**

**Test Probabilities**

| Level | Estim Prob | Hypoth Prob |
|---|---|---|
| CONNECTION FAILURE | 0.10230 | 0.2 |
| CRACK | 0.20877 | 0.2 |
| OTHER | 0.20668 | 0.2 |
| PINHOLE | 0.40710 | 0.2 |
| SEAL OR PACKING | 0.07516 | 0.2 |

| Test | ChiSquare | DF | Prob>Chisq |
|---|---|---|---|
| Likelihood Ratio | 156.1017 | 4 | <.0001* |
| Pearson | 163.2025 | 4 | <.0001* |

Method: Fix hypothesized values, rescale omitted

The reported test statistic was computed using this formula:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

where $O_i$ is the observed frequency of category $i$ and $E_i$ is the expected frequency of category $i$ assuming that the null hypothesis is true. In this example, the expected frequency for each of the four categories is 95.8, as explained earlier.

Pearson's chi-square statistic aggregates the discrepancies between the observed and expected counts for each category. If the data perfectly corresponded to the model, the numerator of each term in the above sum would be zero, and the test statistic would equal zero. To the extent that the data and the model do not correspond, the statistic grows.

When the null is true, there will be some small aggregated discrepancy due to sampling error and to the fact that observed frequencies are integers. That aggregated discrepancy follows a chi-square distribution, enabling JMP to compute a *P*-value for the test statistic. When the discrepancy is so large as to cast doubt on the null hypothesis (as reflected in a very tiny *P*-value), we reject the null. That is the case in this instance. The *P*-value is reported as being smaller than 0.0001, so we reject the null hypothesis and conclude that our initial model of equal likelihoods does *not* fit with the data.

### What Are We Assuming?

This chi-square test has comparatively few assumptions. As with other forms of inference, we can generalize to the population if we can assume that the sample is representative of the population, and hence we need to believe that the observations in the sample are independent of one another. If the sample is randomly selected, we can safely assume independence.

The test can be unreliable if the *expected frequency* of any one category is less than 5. This condition involves both the number of categories as well as the sample size. Note that JMP will compute and report test results whether this condition is satisfied. It is up to the analyst to make the judgment about whether the conditions warrant interpretation of the test.

## Inference for Two Categorical Variables

Let's carry the analysis a step further. In Chapter 10, we looked at the likelihood of ignition in connection with a major disruption, and we estimated that approximately 11% of incidents involved ignition. Is the risk of ignition somehow related to the type of leak? Are some leak types more likely to ignite than others? In other words, are **IGNITE** and **LEAK_TYPE** *independent*? You might remember that we discussed independence back in Chapter 5 and used the conditional probabilities from a contingency table to make decisions about whether two events were independent. In Chapter, 5 the subject was the behavior of dolphins, and we considered two events at a time. With the current example, we will ask a broader question: are the variables independent of one another?

One word of caution: with *bivariate analysis*—analyzing two variables together—we can only use data table rows where neither value is missing. We need "complete cases." As you will note in this example, although the data table has 1,154 rows, we are missing much of the leak type information. As such, we have complete data from only 479 rows. Among other consequences, the overall rate of ignition will drop considerably.

## Contingency Tables Revisited

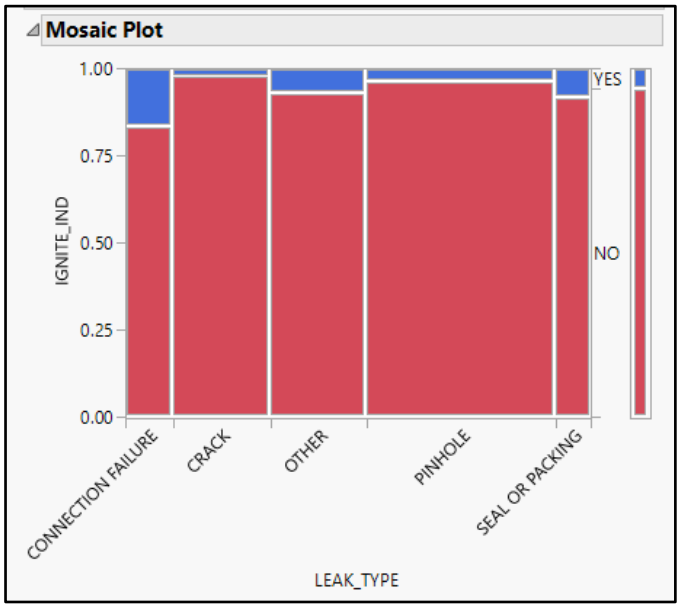
Before considering the inferential question, let's start with a review of contingency tables by making one for this example. We created several contingency tables in Chapter 5.

1. Select **Analyze ► Fit Y by X**. Choose **IGNITE_IND** as **Y** and **LEAK_TYPE** as **X.**

The results window shows a mosaic plot, a contingency table, and test results. Let's look initially at the mosaic plot and contingency table, shown in Figures 12.3 and 12.4. As you look at these two figures, remember that they both display the same information.

In the mosaic plot, the vertical dimension represents whether there was ignition and the horizontal represents the various incident types. The visual prevalence of red[2] shows that ignition occurs in a small minority of incidents. At the far right, we see the marginal breakdown with about 6% of the incidents did not ignite. The plot shows that the prevalence of ignition within the sample observations seems to vary a bit. Notably, cracks were relatively unlikely to have ignited, but connection failures were more likely to ignite among the incidents reported in this sample.

**Figure 12.3: Mosaic Plot for Pipeline Data**




Now let's consider the contingency table and track down the corresponding numerical values. The table provides the specific values that underlie the visual impression conveyed by the mosaic plot.

**Figure 12.4: Basic Contingency Table for Ignite Versus Leak Type**

**Contingency Table**

| LEAK_TYPE | Count / Total % / Col % / Row % | IGNITE_IND NO | YES | Total |
|---|---|---|---|---|
| CONNECTION FAILURE | | 41 | 8 | 49 |
| | | 8.56 | 1.67 | 10.23 |
| | | 9.07 | 29.63 | |
| | | 83.67 | 16.33 | |
| CRACK | | 98 | 2 | 100 |
| | | 20.46 | 0.42 | 20.88 |
| | | 21.68 | 7.41 | |
| | | 98.00 | 2.00 | |
| OTHER | | 92 | 7 | 99 |
| | | 19.21 | 1.46 | 20.67 |
| | | 20.35 | 25.93 | |
| | | 92.93 | 7.07 | |
| PINHOLE | | 188 | 7 | 195 |
| | | 39.25 | 1.46 | 40.71 |
| | | 41.59 | 25.93 | |
| | | 96.41 | 3.59 | |
| SEAL OR PACKING | | 33 | 3 | 36 |
| | | 6.89 | 0.63 | 7.52 |
| | | 7.30 | 11.11 | |
| | | 91.67 | 8.33 | |
| Total | | 452 | 27 | 479 |
| | | 94.36 | 5.64 | |

As in Chapter 6, we will look in the margins of the table for the categories and their marginal frequencies, and at the inner cells for joint and conditional frequencies. To find the probability (relative frequency) of any one category, we look in the opposite margin. For example, 452 of 479 incidents did not ignite. The relative frequency was 94.36%, so we find that *Pr(No)* = 0.9436.

Within the interior cells, the upper number is the joint frequency showing the number of incidents with or without ignition for each disruption type. The next number is the *joint relative frequency*. Thus, in the upper left cell, we see that 41 of the 479 incidents were connection failures without ignition for a joint relative frequency of 8.56% of all incidents.

The upper left cell of the table reports that the next two numbers within each cell are the column percentage and the row percentage. These are *conditional* values. A column percentage is the relative frequency of the row event within the column only. In other words, restricting ourselves to the condition represented by the column, it is the relative frequency of each row. Similarly, the row percentages give the probability of the column events conditioned by each row.

The contingency table provides an informative summary of the sample data. Within the sample, the relative frequency of ignition plainly varies across the different incident types. Based on this evidence, can we conclude a similar pattern of variation occurs in the entire population of incidents?

## Chi-Square Test of Independence

The chi-square test of independence uses sample data to test the null hypothesis that two categorical variables are independent of one another. In this example, if ignition is independent of leak type, then gas is just as likely to ignite from one type of leak as another. Or, to put it another way, knowing the type of leak would give us no information about the chance of ignition.

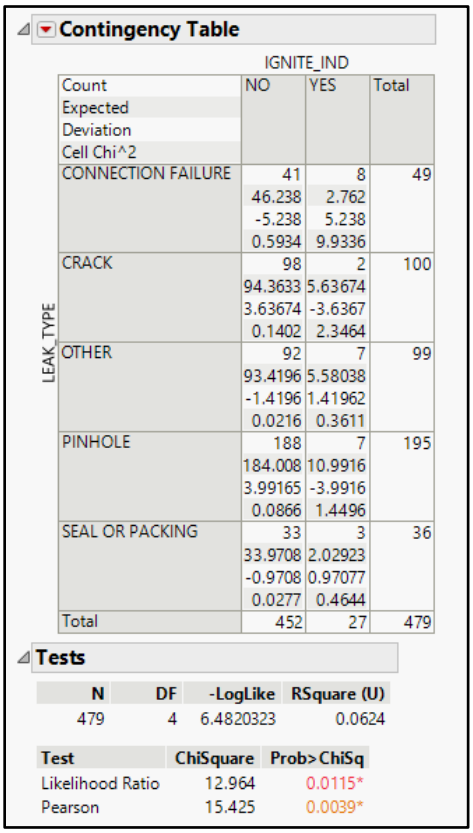
In this test, the null hypothesis is always the same:

> $H_0$: The two variables are independent.

Naturally, the alternative hypothesis is that they are not independent, but rather related in some way. This test works much the same as the goodness-of-fit test. Our null hypothesis establishes expectations about the joint frequencies that we would observe if the null were true. We can then compare the actual observed values to the expected values, aggregate the discrepancies, and gauge the size of the accumulated differences using a chi-square distribution. To understand the test more fully, do this:

1. Press the Alt key and click the red triangle next to **Contingency Table**. This opens a dialog box titled **Select Options and click OK** (not shown here). Clear the options **Total %**, **Col %**, and **Row %**.
2. Select **Expected**, **Deviation**, and **Cell Chi Square** and then click **OK**.

**Figure 12.5: Modified Contingency Table with Chi-Square Results**



**Contingency Table**

| LEAK_TYPE | Count / Expected / Deviation / Cell Chi^2 | IGNITE_IND NO | IGNITE_IND YES | Total |
|---|---|---|---|---|
| CONNECTION FAILURE | | 41<br>46.238<br>-5.238<br>0.5934 | 8<br>2.762<br>5.238<br>9.9336 | 49 |
| CRACK | | 98<br>94.3633<br>3.63674<br>0.1402 | 2<br>5.63674<br>-3.6367<br>2.3464 | 100 |
| OTHER | | 92<br>93.4196<br>-1.4196<br>0.0216 | 7<br>5.58038<br>1.41962<br>0.3611 | 99 |
| PINHOLE | | 188<br>184.008<br>3.99165<br>0.0866 | 7<br>10.9916<br>-3.9916<br>1.4496 | 195 |
| SEAL OR PACKING | | 33<br>33.9708<br>-0.9708<br>0.0277 | 3<br>2.02923<br>0.97077<br>0.4644 | 36 |
| Total | | 452 | 27 | 479 |

**Tests**

| N | DF | -LogLike | RSquare (U) |
|---|---|---|---|
| 479 | 4 | 6.4820323 | 0.0624 |

| Test | ChiSquare | Prob>ChiSq |
|---|---|---|
| Likelihood Ratio | 12.964 | 0.0115* |
| Pearson | 15.425 | 0.0039* |

After modifying the contingency table in this way, your results should match those in Figure 12.5.

Let's look closely at the numbers within the cell in the upper left of the table that correspond to connection failure leaks that did not ignite. There are four numbers in the cell, as there are in each cell of the table. The legend again appears in the upper left corner of the table.

Each cell contains the observed count, the expected count, the deviation ($O - E$), and the cell chi-square value. We will consider these one at a time. As in Figure 12.4, we see that 41 of the 479 incidents were, in fact, connection failure leaks where gas did not ignite. Next is the expected count of 46.238, computed in the following way.

Among all observed incidents 452 of 479, or 94.36325%, did not ignite. If the null hypothesis is true—that in the on-going process of pipeline disruptions ignition is independent of leak type—then we would expect 94.36325% of all incidents to be free from ignition, regardless of leak type. If so, then 94.36325% of the 49 connection failures would have no ignition. 94.36325% of 49 equals 46.238.

The deviation value of –5.238 is just the difference between the observed and expected frequencies: *Oi – Ei.* Finally, the cell Chi^2 is $(O_i - E_i)^2 / E_i$, or $-5.238^2/46.238$ = 0.5934.

Each cell's deviation contributes to the aggregated total chi-square value, as reported at the bottom of the panel. The test statistic, Pearson's chi-square, for the two-dimensional contingency table is 15.425. In a contingency table, the number of degrees of freedom equals (number of rows – 1) x (number of columns – 1). Here, we have five rows and two columns, so DF = 4. If the null hypothesis is true, the chance of obtaining a chi-square statistic of 15.425 or larger is about 0.0039. Assuming a conventional significance level of $\alpha = 0.05$, we would conclude that this evidence is sufficient to reject the null. The discrepancies between the observed and hypothesized frequencies are compelling evidence to reject the null hypothesis and to conclude that leak type and ignition are related.

### What Are We Assuming?

The assumptions for this test are the same as for the goodness-of-fit test. We can rely on the results if our observations are independent and if the expected count in each cell exceeds 5.

Notice that, unlike with the t tests of earlier chapters, there is no assumption here about the shape or nature of the population distribution. For this reason, chi-square tests are also sometimes classified as "distribution-free" or *nonparametric* methods. The reliability of the test results is not dependent on how the categorical variables are distributed in the entire population.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions. For each question, explain which test you have chosen, report the relevant test results, and interpret the results in the context of the scenario.

1. *Scenario:* Before we leave the pipeline data, let's ask three more questions:
   a. Are disruptions equally likely to occur in all regions of the United States?
   b. Are the variables **EXPLODE_IND** and **LEAK_TYPE** independent?
   c. Are the variables **CAUSE** and **IGNITE_IND** independent?

2. *Scenario:* The main example in Chapter 5 was based on the data table called **Dolphins**. Please note that this data table contains frequencies rather than raw data; refer to Figure 5.2 for instructions about creating the contingency table.
   a. Are the variables **Activity** and **Period** independent?
   b. Are dolphins equally likely to be observed feeding, socializing, and traveling? (Again, recall that the data table shows frequencies rather than raw data.)

3. *Scenario:* We will continue to examine the World Development Indicators data in **BirthRate 2017**. We will focus on three categorical variables in the data table.
    - **Region**: Global region of the country in question
    - **Provider**: Source of maternity leave benefits (public, private, or a combination of both)
    - **MatLeave90+**: Mandated length of maternity leave—fewer than 90 days, or 90 days and more

    a. Are the variables **Provider** and **Region** independent?
    b. Are the variables **Provider** and **MatLeave90+** independent?
    c. Are the variables **MatLeave90+** and **Region** independent?

4. *Scenario:* In this scenario, we return to the **NHANES 2016** data table containing survey data from a large number of people in the U.S. in 2016. For this analysis, we will focus on just the following variables:
    - **RIDRETH1**: Respondent's racial or ethnic background
    - **DMDMARTL**: Respondent's reported marital status

    a. Are these two categorical variables independent?

5. *Scenario:* The following questions were inspired by a 2009 article on alcohol consumption and risky behaviors (Hingson *et al.* 2009) based on responses to a large national survey. The data table **BINGE** contains summary frequency data calculated from tables provided in the article. We'll focus on the self-reported frequency of binge drinking (defined as five or more drinks in a two-hour period for males and four or more drinks in a two-hour period for females) and whether the respondents reported having ever been involved in a car accident after drinking.
    a. Are the two variables independent? Explain.

6. *Scenario:* Let us once again look at the **Sleeping Animals** data table.
    a. Are mammalian species equally likely to be classified across the five categories of the exposure index?
    b. Are all mammalian species equally likely to be classified across the five categories of the predation index?
    c. Are the variables **Predation** and **Exposure** independent? Explain.

7. *Scenario:* For these questions, we return to the **TimeUse** data table. Use the data filter to confine the analysis to the responses from 2017.
    a. Are employment status and sex independent? Explain.
    b. Are full/part-time working status and marital status independent? Explain.
    c. Are employment status and marital status independent? Explain.
    d. Are marital status and sex independent? Explain.

8. *Scenario:* Open the **NC Births** data table. Recall that this data table contains a simple random sample of 1,000 births in North Carolina in 2004.
   a. Use a chi-square test of goodness of fit to test the hypothesis that a woman is as likely to give birth to a girl as to a boy. Explain what you did and what you concluded.
   b. Are baby gender and whether the mother smoked independent? Explain.
   c. Are low birth weight and whether the mother smoked independent? Explain.
   d. Are low birth weight and mature mother independent? Explain.

9. *Scenario:* Open the **FAA Bird Strikes CA** data table, which contains data from three airports in California: Los Angeles, Sacramento, and San Francisco. For each question below, we want to know if the attributes in question varied by location (that is, is the response variable independent of airport?). Conduct a test of independence and report on your conclusions, explaining the reasoning for your conclusion. For all tests, use a significance level of 0.05.
   a. Wildlife strikes occur during different phases of flights. Does the frequency of strikes in each phase vary by airport? (NOTE: You should recode the **PHASE_OF_FLT** column as we did in Chapter 4.)
   b. Does the airline operator involved in the incident vary by airport? Be sure to check for sample size conditions before drawing a conclusion.
   c. **BIRDS_STRUCK** is an ordinal variable describing the number of birds struck during the flight. Is this variable independent of the airport?
   d. Does the size of the bird(s) struck (**SIZE**) vary by airport?

---

[1] There is more than one method of computing a chi-square statistic. Pearson's is the statistic commonly presented in introductory statistics texts.

[2] On your screen, JMP is using the default mosaic plot color scheme of blue for Yes and red for No; readers of the book will see a grayscale version of this image in which it will be hard to differentiate blue and red. In each column, the larger shaded area is red, corresponding to "NO".

# Chapter 13: Two-Sample Inference for a Continuous Variable

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## Overview

At the end of the prior chapter, we studied the matched pairs design to make inferences about the mean of differences within a population. In this chapter, we take up the comparison of samples drawn independently from two different populations and learn to estimate the differences between population means and variances. The procedures discussed here assume that samples are randomized (and therefore representative) and are comparatively small subsets of their respective populations.

## Conditions for Inference

We will first focus on the centers of the two populations and assume that measurements are independent both within and across samples. As we did in the last chapter, we begin with cases in which the populations are either approximately normal or in which the Central Limit Theorem applies. Then, we will see a nonparametric (also known as *distribution-free*) approach for non-normal data. Finally, we will learn to compare the spread of two distributions as measured by the variance.

## Using JMP to Compare Two Means

We have used the **NHANES 2016** data to illustrate many statistical concepts, and we will return to it again now. Recall that the National Health and Nutrition Examination Survey is a biennial cross-sectional sample conducted by the U.S. Centers for Disease Control. Within this large data table there are several measurements typically taken during a routine physical exam. For some

of these measurements, we expect to find differences between male and female subjects. Men tend to be taller and heavier than women, for example. But are their pulse rates different? In our first illustration, we will estimate the difference (if any) between the resting pulse rates of adult women and men.

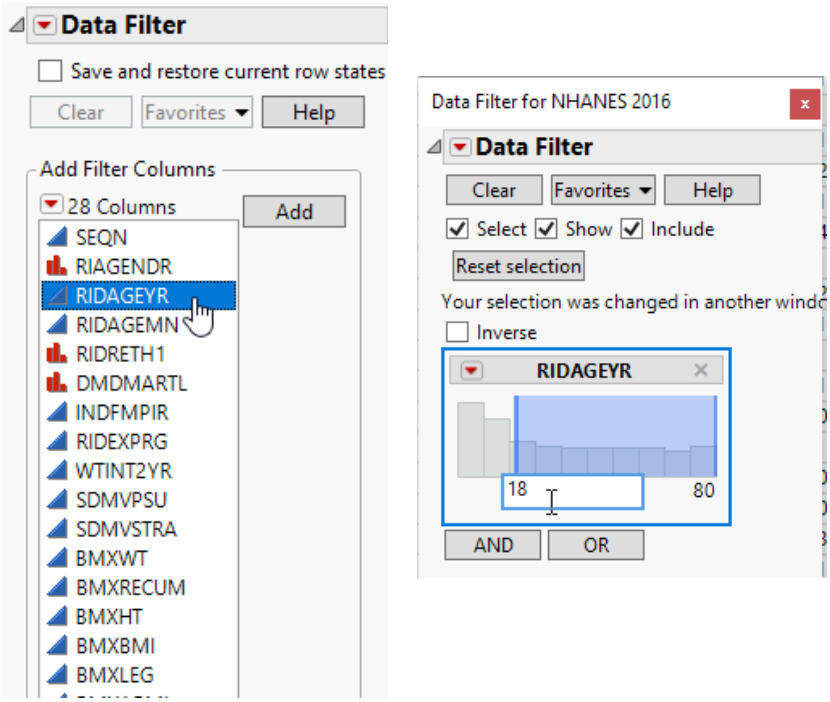
## Assuming Normal Distributions or CLT

1. Open the **NHANES 2016** data table.

The subjects in the **NHANES 2016** data table range in age from infants up. Our first step is to exclude subjects under the age of 18 years. We will use the Data Filter to select just the adults.

2. Select **Rows ► Data Filter.** As shown in the left side of Figure 13.1, choose the **RIDAGEYR** column and click **Add**.

**Figure 13.1: Using the Data Filter to Restrict Analysis to Adults**

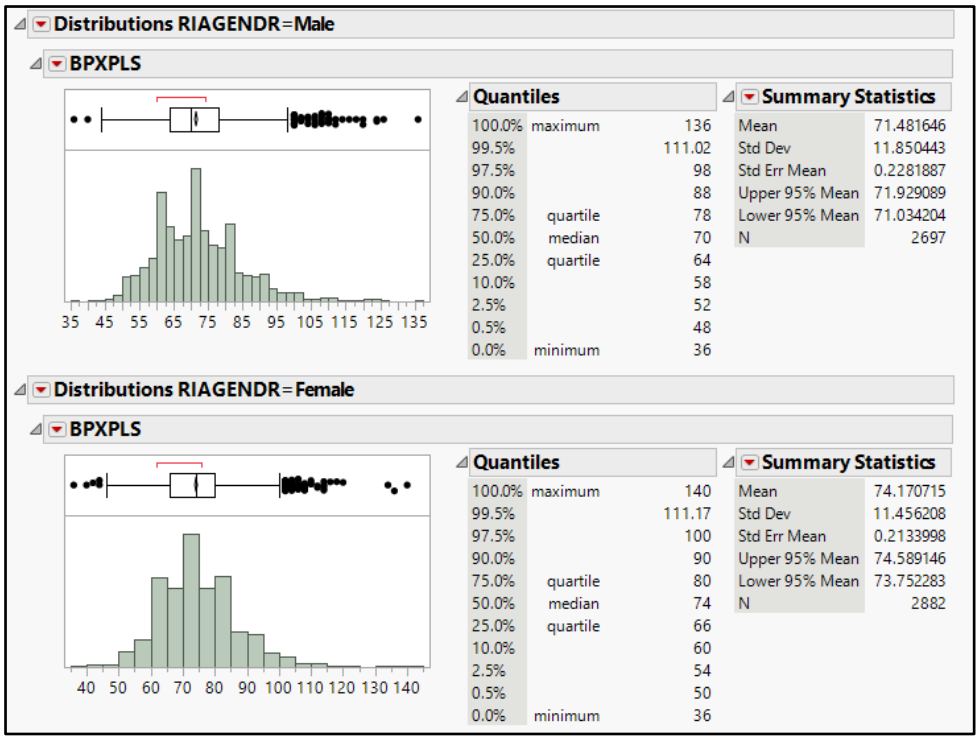



3. In the filter, check the boxes as shown and edit the minimum age to equal 18.

Before performing any inferential analysis, let's look at the pulse rate data, separating the observations for females and males.

4. Select **Analyze ► Distribution**. Cast **BPXPLS** (pulse rate) into **Y** and **RIAGENDR** into **By**.
5. In the **Distribution** report, click the red triangle next to **Distributions**, and select **Stack** to create the report shown in Figure 13.2.

**Figure 13.2: Distribution of Pulse Rate by Gender**



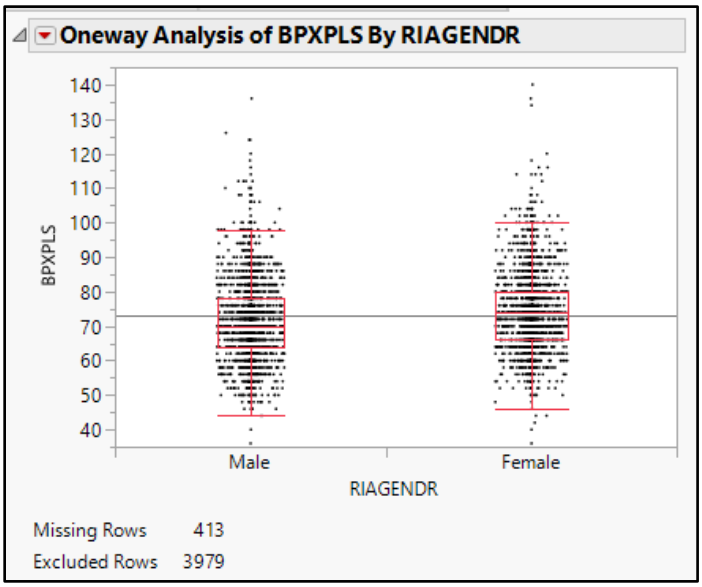

A quick glance suggests that each distribution is reasonably symmetric with a central peak and similar measures of dispersion. Given that fact as well as the large size of the samples, we can conclude that we are on solid ground using a t test. As in the one-sample case, the t test is reliable when the underlying population is approximately normal or when the samples are large enough for the Central Limit Theorem to apply. We have very large samples, so we will rely on the CLT to proceed with the analysis.

Conceptually, we are about to investigate a statistical model. The model says "gender may account for some of the variability in individual pulse rates." The conventional language that we use is to say that we are "fitting" a model for Y as a function of X. As you will see in the Fit Y by X dialog box, we will cast pulse rate as a response to a varying factor, gender.

6. Select **Analyze ▶ Fit Y by X**. Select **BPSXPLS** as **Y, Response** and **RIAGENDR** as **X, Factor,** and click **OK**.
7. Move your cursor over the graph and right-click to open a menu. Choose **Marker Size ▶ 0, Dot**. With more than 2,000 observations for each gender, the default dot size makes it difficult to see the individual points.
8. Click the red triangle next to **Oneway Analysis**, and select **Display Options ▶ Points Jittered**. This randomly spaces out the points horizontally for each group, adding further resolution to the image.

9. Click the same red triangle again and select **Display Options ▶ Box Plots**. When you have done this, your report looks like Figure 13.3.

**Figure 13.3: Bivariate Comparison of Pulse Rates**



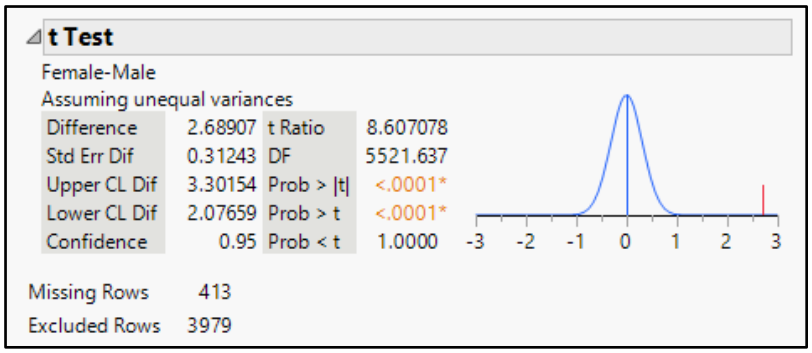

In this graph, we see side-by-side box plots of the observations. The box plots illustrate the symmetry of the two distributions and suggest that they have similar centers and dispersion. Though this graph re-displays some of the information from Figure 13.2, here we get a clearer sense of just how the points are distributed.

The **Fit Y by X** platform allows us to test whether there is a statistically significant difference between the mean pulse rates of men and women based on these two large subsamples, and to estimate the magnitude of the difference (if any).

10. Click the red triangle once more, and select **t Test**. This produces the results shown in Figure 13.4.

You might have noticed that, unlike the one-sample t test, this command gives us no option to specify a hypothesized difference of means. Usually, the two-sample t test has as its default the null hypothesis of no difference. In other words, the default setting is that the difference in means equals 0. This version of the two-sample t test assumes that the variances of the two populations are unequal; we will discuss that assumption shortly.

**Figure 13.4: Results of a Bivariate t Test**




In the results panel, we see that there was a difference of 2.69 beats per minute between the females and males in the sample. We also find the 95% confidence interval for the difference in population means. Based on these two samples of women and men, we estimate that in the population at large, women's pulse rates are 2.08 to 3.30 beats per minute higher than men's.

We also see a *t*-distribution centered at 0 and a red line marking the observed difference within the sample. The observed difference of nearly 2.7 beats is more than 8.61 standard errors from the center—something that is virtually impossible by chance alone. Whether or not you expected to find a difference, these samples lead us to conclude that there is one.

## Using Sampling Weights (Optional Section)

The **NHANES 2016** data comes from a complex sample, and the data table includes sampling weights. To conduct inferential analysis properly with surveys like **NHANES**, we should apply the post-stratification weights. The weights do alter the estimated means and standard errors and therefore lead to slightly different conclusions. We interpret weighted results exactly as described above, and the software makes it very easy to apply the weights.

1. Click the red triangle next to **Oneway Analysis** and choose **Redo ▶ Relaunch**. This re-opens the launch dialog box with **BPXPLS** cast as **Y** and **RIAGENDR** as **X**. Add **WTINT2YR** as **Weight**.

All other instructions and interpretations are as described above. Notice that the t test results have changed. While still quite significant, the estimated gender difference in pulse rate is not quite as large as we previously thought. Applying the sampling weights provides more accuracy, but is a topic that is typically excluded from an introductory course.

## Equal Versus Unequal Variances

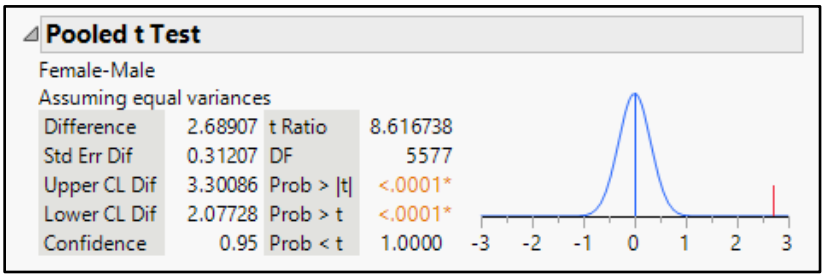
The t test procedure we just performed presumed that the variance in pulse rates for women and men are unequal and estimated the variance and standard deviation separately based on the two samples. That is a conservative and reasonable thing to do.

If we can assume that the populations not only share a common normal shape (or that the Central Limit Theorem applies because both samples are large) but also share a common degree

of spread, we can combine the data from the two samples to estimate that shared variance. With smaller samples, there is some gain in efficiency if we can combine, or *pool*, the data to estimate the unknown population variances. Later in this chapter, we will see a way to test whether the variances are equal or not. At this point, having noted that the samples seem to have similar spread, let's see how the pooled-variance version of this test operates.

2. Return to the first **Oneway Analysis** report. Click the red triangle, and select **Means/Anova/Pooled t**. This produces the results shown in Figure 13.5.

**Figure 13.5: The Pooled Variance t Test**




This command produces several new items in our report, all within a panel labeled **Oneway Anova**. We will devote all of Chapter 14 to ANOVA (Analysis of Variance) and will not say more about it here. Locate the heading **Pooled t Test** and confirm that the report includes the phrase **Assuming equal variances**. We read and interpret these results exactly as we did in the unequal variance test.

Because the two sample variances are very similar and the samples are so large, these results are extremely close to the earlier illustration. The estimated pooled standard error (**Std Err Dif**) is just slightly smaller, so the resulting confidence interval is a bit narrower in this procedure.

## Dealing with Non-Normal Distributions

Let's look again at the pipeline safety data, confining our attention to just two states of the United States—Texas and Oklahoma, the states with the largest number of incidents in this data table. Moreover, to illustrate the challenge of non-normal data from samples too small to rely on the Central Limit theorem, we will also analyze only those disruptions that occurred in 2018. We will compare the cost of gas released, assuming that disruptions and their repairs are independent in the two regions over time.

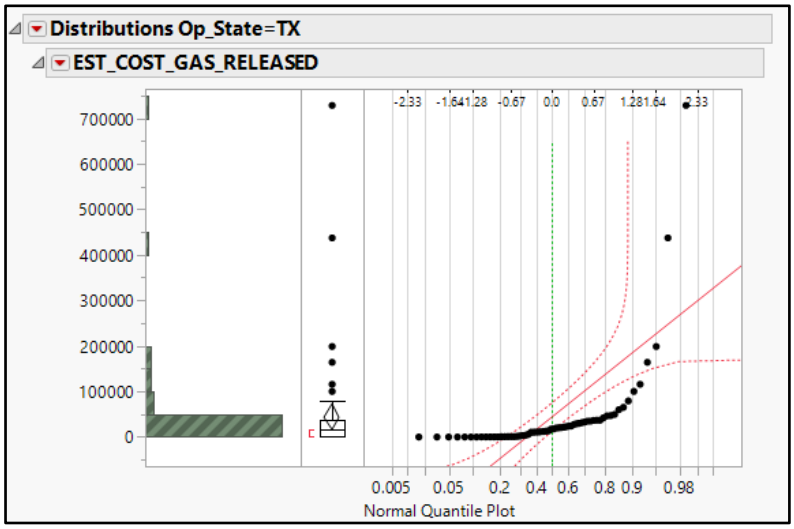
1. Open the **Pipeline Safety** data table.
2. Use the Data Filter to show and include only those leaks occurring in Oklahoma or Texas (**Op_State** column; state abbreviations are **OK** and **TX**) **AND** select, show, and include rows where **IYEAR = 2018**.
3. Select **Analyze ► Distribution**. Select **EST_COST_GAS_RELEASED** as **Y, Op_State** as **By,** and click **OK**. This creates two histograms, one for each region.

According to the reported quantiles and summary statistics, the observed incidents in Oklahoma had a median cost of $11,600 and a mean of $12,495. In contrast, the median was $17,485.50

and the mean was $44,625.57 in Texas. This is apparently a large difference, but is it statistically significant? We have only 19 observations from Oklahoma and 62 from Texas. Before running a t test as before, we should check for normality in these two small samples. The histograms are strongly skewed to the right, but let's also look at a normal quantile plot.

4. Hold the Ctrl key, click the red triangle next to **Distributions**, and check **Normal Quantile Plot**; the Texas results are shown in Figure 13.6.

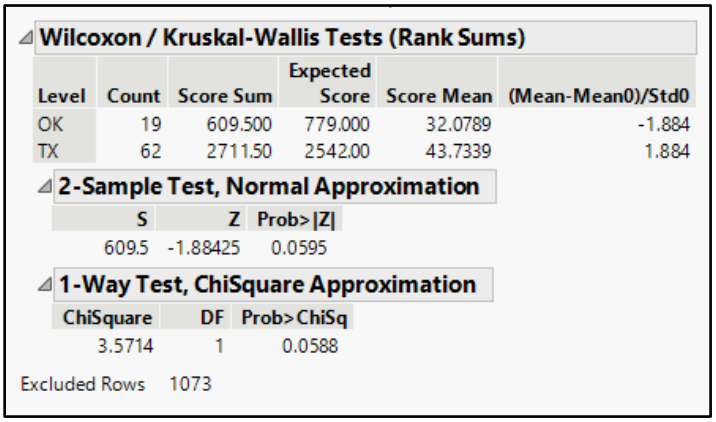
**Figure 13.6: Normal Quantile Plot for a Strongly Skewed Distribution**




We have two skewed distributions with relatively small samples. We cannot invoke the Central Limit Theorem as needed to trust the t test, but we do have nonparametric options available. It is beyond the scope of most introductory courses to treat nonparametric alternatives in depth, but we will illustrate one of them here because it is commonly used. The null hypothesis of the test is that the two distributions of **EST_COST_GAS_RELEASED** are centered at the same dollar value.

5. Select **Analyze ► Fit Y by X**. The **Y, Response** column is **EST_COST_GAS_RELEASED** and the **X, Factor** column is **Op_State**. Click **OK**.
6. Click the red triangle next to **Oneway Analysis** and select **Display Options ► Box Plots.** This provides an easy visual comparison of the two distributions.
7. Click the red triangle again, and select **Nonparametric ► Wilcoxon Test**. This generates the report shown in Figure 13.7.

**Figure 13.7: Results of a Two-Sample Wilcoxon Test**



**Wilcoxon / Kruskal-Wallis Tests (Rank Sums)**

| Level | Count | Score Sum | Expected Score | Score Mean | (Mean-Mean0)/Std0 |
|---|---|---|---|---|---|
| OK | 19 | 609.500 | 779.000 | 32.0789 | -1.884 |
| TX | 62 | 2711.50 | 2542.00 | 43.7339 | 1.884 |

**2-Sample Test, Normal Approximation**

| S | Z | Prob>\|Z\| |
|---|---|---|
| 609.5 | -1.88425 | 0.0595 |

**1-Way Test, ChiSquare Approximation**

| ChiSquare | DF | Prob>ChiSq |
|---|---|---|
| 3.5714 | 1 | 0.0588 |

Excluded Rows 1073

When we have two samples, as we do here, JMP conducts a Wilcoxon test, which is the equivalent of another test known as the Mann-Whitney U Test. With more than two samples, JMP performs and reports a Kruskal-Wallis Test (as we will see in Chapter 14). The common approach in these tests is that they combine all observations and rank them. They then sum up the rankings for each group (region, in this case) and find the mean ranking. If the two groups are centered at the same value, then the mean ranks should be very close. If one group were centered at a very different point, the mean ranks would differ by quite a lot.

Naturally, the key question is, "how much of a difference is considered significant?" Suffice it to say that JMP uses two conventional methods to estimate a *P*-value for the observed difference in mean ranks. Both methods use approximations to a standard distribution; the first test relies on a Normal model, and the second on a Chi-Square model. The two *P*-values appear in the lower portion of Figure 13.7 and are quite similar. The alpha risk—the risk of a Type I error—is very near the conventional 5% level. Depending on the context of the decision, a decision-maker may or may not find the observed difference statistically significant. In a published article, an author might say that the difference is significant at the 0.10 level but not at the 0.05 level.

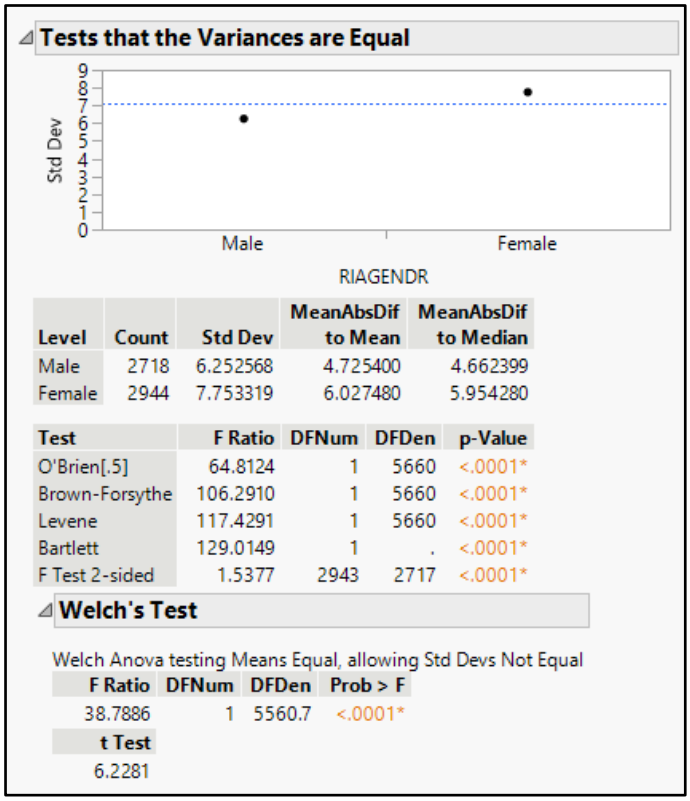
## Using JMP to Compare Two Variances

In our earlier discussion of the t test, we noted that there are two versions of the test, corresponding to the conditions in which the samples share the same variance or have different variances. Both in the t tests and in techniques we will see later, we might need to decide whether two or more groups have equal or unequal variances before we choose a statistical procedure. The **Fit Y by X** platform in JMP includes a command that facilitates the judgment.

Let's shift back to the **NHANES 2016** data and consider the Body Mass Index for the adult men and women (ages 18 and older) in the samples. BMI is a measure of body fat as a percentage of total weight. A lean person has lower BMI than an overweight person. We will test the hypothesis that adult women and men in the US have the same mean BMI, and before choosing a test, we will compare the variances of the two populations.

1. Return to the **NHANES 2016** data table. Make sure that you are still showing and including respondents 18 years of age and older.
2. Select **Analyze ► Fit Y by X.** Our **Y** column is **BMXBMI** and our **X** column is **RIAGENDR.**[1] Click **OK**.
3. Click the red triangle next to **Oneway Analysis** and select **UnEqual Variances.** This opens the panel shown in Figure 13.8.

**Figure 13.8: Testing for Equal Variances**




| Level | Count | Std Dev | MeanAbsDif to Mean | MeanAbsDif to Median |
|---|---|---|---|---|
| Male | 2718 | 6.252568 | 4.725400 | 4.662399 |
| Female | 2944 | 7.753319 | 6.027480 | 5.954280 |

| Test | F Ratio | DFNum | DFDen | p-Value |
|---|---|---|---|---|
| O'Brien[.5] | 64.8124 | 1 | 5660 | <.0001* |
| Brown-Forsythe | 106.2910 | 1 | 5660 | <.0001* |
| Levene | 117.4291 | 1 | 5660 | <.0001* |
| Bartlett | 129.0149 | 1 | . | <.0001* |
| F Test 2-sided | 1.5377 | 2943 | 2717 | <.0001* |

**Welch's Test**

Welch Anova testing Means Equal, allowing Std Devs Not Equal

| F Ratio | DFNum | DFDen | Prob > F |
|---|---|---|---|
| 38.7886 | 1 | 5560.7 | <.0001* |

| t Test |
|---|
| 6.2281 |

We find a simple graph comparing the sample standard deviations (*s*) of the two groups, with the numerical summaries below it. For the women, $s = 7.75$, and for the men, $s = 6.25$. Because the variance is the square of the standard deviation, we can determine that the sample variances are approximately 60.1 and 39.1, respectively. Based on these sample values, should we conclude that the population variances are the same or different?

JMP performs five different tests for the equality, or *homogeneity,* of variances. Although they all lead to the same conclusion in this example, that is not always the case. For two-sample comparisons, the most commonly used tests are probably Levene's test and the F test two-sided. The null hypothesis is that the variance of all groups is equal, and the alternative hypothesis is that variances are unequal.

In this example, the *P*-values fall well below the conventional significance level of 0.05. That is, there is enough evidence to reject the null assumption of equal variances, so we can conclude that this set of data should be analyzed with an unequal variance t test. You can perform the test as one of the Application exercises at the end of the chapter.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions. Be sure to examine the data to check the conditions for inference, and explain which technique you used, why you chose it, and what you conclude.

1. *Scenario:* Let's continue analyzing the data for adults in the **NHANES 2016** data table. For each of the following columns, estimate the difference between the male and female respondents using a 95% confidence level.
    a. What is the estimated difference in Body Mass Index (BMI)?
    b. What is the estimated difference in systolic blood pressure?
    c. What is the estimated difference in diastolic blood pressure?
    d. What is the estimated difference in waist circumference?

2. *Scenario:* We will continue our analysis of the pipeline disruption data in Oklahoma and Texas. The column **EST_COST_PROP_DAMAGE** contains the dollar value of property damage as a consequence of the disruption.
    a. Report and interpret a 95% confidence interval for the difference in mean property value costs of a pipeline disruption.
    b. Test the hypothesis that there is no difference in the mean property damage costs between the two states.

3. *Scenario:* Go back to the full **Pipeline Safety** data table. Clear row states to show and include all states and years. We will compare the age of pipelines in locations controlled by the operator and those that are public rights of way.
    a. Use the Distribution platform to inspect the distributions of YRS_IN_PLACE by LOCATION_TYPE. Write a sentence comparing the two distributions. Does your inspection of the distributions give the impression that they are similar or different?
    b. Estimate the difference in years in place for lines on Operator-Controlled Property versus Pipeline Right-of-Way. Assume for the moment that utility companies can more easily perform maintenance on property that they control. Interpret your findings in light of this assumption.
    c. Test the hypothesis that the variances are equal in comparing pipeline for leaks versus ruptures. Explain your conclusion.

4. *Scenario:* Parkinson's disease (PD) is a neurological disorder affecting millions of older adults around the world. Among the common symptoms are tremors, loss of balance, and difficulty in initiating movement. Other symptoms include vocal impairment. We have a data table, **Parkinsons** (Little, McSharry, Hunter, & Ramig, 2008), containing a number of vocal measurements from a sample of 32 patients, 24 of whom have PD.

   Vocal measurement usually involves having the patient read a standard paragraph, or sustain a vowel sound, or do both. Two common measurements of a patient's voice refer to the wave form of sound: technicians note variation in the amplitude (volume) and frequency (pitch) of the voice. In this scenario, we will focus on just three measurements.

   - **MDVP:F0(Hz)**: Vocal fundamental frequency (baseline measure of pitch)
   - **MDVP:Jitter(Abs)**: Jitter refers to variation in pitch
   - **MDVP:Shimmer(db)**: Shimmer refers to variation in volume

     a. Examine the distributions of these three columns, distinguishing between patients with and without PD. Do these data meet the conditions for running and interpreting a two-sample t test? Explain your thinking.

     b. Use an appropriate technique to decide whether PD and non-PD patients have significantly different baseline pitch measurements.

     c. Use an appropriate technique to decide whether PD and non-PD patients have significantly different jitter measurements.

     d. Use an appropriate technique to decide whether PD and non-PD patients have significantly different shimmer measurements.

5. *Scenario:* The data table **Airline Delays** contains a sample of flights for four airlines destined for three busy airports. Our goal is to analyze the delay times, expressed in minutes, and compare two of the airports. First, use the Data Filter to show and include only those flights destined for Atlanta and Chicago.

   a. Assume this is an SRS of many flights by the selected carriers and airports over a long period of time. Is the variance of delay the same for both airports?

   b. What can you conclude about the difference in mean delays for these two airlines?

6. *Scenario:* Let's look again at the **TimeUse** data. For each of the following questions, evaluate the extent to which conditions for inference have been met, and then report on what you find when you use an appropriate technique.

   a. Estimate the mean difference in the amount of time men and women spent sleeping in 2017.

   b. Estimate the mean difference in time that individuals devoted to email in 2007 versus 2017.

   c. What can you conclude about the difference in the mean amount of time spent by women and men socializing on a given day across all years?

7. *Scenario:* This question looks at the **NC Births** data. For each of the following questions, evaluate the extent to which conditions for inference have been met, and then report on what you find when you use an appropriate technique.
   a. Do babies born to mothers who smoke weigh less than those born to non-smokers?
   b. Estimate the mean weight gain among pregnant smokers and non-smokers.
   c. Do smokers and non-smokers deliver their babies after the same number of weeks, on average?

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[1] If you want to use sampling weights in the analysis, use the weight column **WTINT2YR**.

# Chapter 14: Analysis of Variance

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## Overview

Previous chapters introduced the concept of bivariate inference and the idea of a response and factor variables. In Chapter 12, we worked with two categorical variables, and in Chapter 13, we worked with a continuous response and a dichotomous factor (to represent two independent samples). This chapter introduces several techniques that we can use when we have a *continuous response variable* and one or more categorical *factors* that may have several categorical values.

## What Are We Assuming?

As a first example, consider an experiment in which subjects are asked to complete a simple crossword puzzle, and our response variable is the amount of time required to complete the puzzle. We randomly divide the subjects into three groups. One group completes the puzzle while rock music plays loudly in the room. The second group completes the puzzle with the recorded sound of a large truck engine idling outside the window. The third group completes the puzzle in a relatively quiet room. We refer to the different sound conditions as *treatments*.

Our question might be this: does the mean completion time depend on the treatment for the group? In other words, are the means are the same for the three groups or not?

We might model the completion time for each individual ($Y_{ik}$) as consisting of three separate elements: an overall population mean, plus or minus a *treatment effect*, plus or minus *individual variation*. We might anticipate that the control group's times might enable us to estimate a baseline value for completion times. Further, we might hypothesize that rock music helps some people work more quickly and efficiently, and therefore that individuals in the rock music condition might complete the puzzle faster, on average, than those in the control group. Finally,

regardless of conditions, we also anticipate that individuals are simply variable for numerous unaccountable (or unmeasured) reasons.

Symbolically, we express this as follows:

$$Y_{ik} = \mu + \tau_k + \varepsilon_{ik}$$

where:

$\mu$ represents the underlying but unknown population mean,

$\tau_k$ is the unknown deviation from the population mean (that is, the treatment effect) for treatment level k, and

$\varepsilon_{ik}$ is unique individual variation, or error.

Our null hypothesis is that all groups share a single mean, which is to say that the treatment effect for all factor levels ($\tau_k$) is zero. We can express this null hypothesis in this way:

$H_0$: $\tau_1 = \tau_2 = \cdots = \tau_k = 0$ versus the complementary alterative hypothesis that $H_0$ is not true.

The first technique we will study is called *one-way analysis of variance* (ANOVA), which is a reliable method of inference when our study design satisfies these three assumptions:

- All observations are independent of one another.
- The individual error terms are normally distributed.[1]
- The variance of the individual errors is the same across treatment groups.

These three conditions form the foundation for drawing inferences from ANOVA. In practice, it is helpful, though not necessary to have samples of approximately equal size. With varying sample sizes, the technique is particularly sensitive to the equal variance assumption, which is to say that the statistical results can be misleading if our sample data deviate too far from approximately equal variances. Therefore, an important step in conducting an analysis of variance is the evaluation of the extent to which our data set challenges the validity of these assumptions.

## One-Way ANOVA

Our first example comes from the United Nation's birth rate data, a source that we have used before. In this instance, we will look at the variation in the birth rate around the world. Specifically, birth rate is defined as the number of births annually per 1,000 population. We will initially ask whether differences in birth rates are associated with one particular matter of public policy: whether maternity leave is provided by the state, by the private sector, or by a combination of public and private sectors.

Note that this is observational data, so we have not established the different factor levels experimentally or by design. Open the **BirthRate 2017** data table.

We will start simply by examining the distribution of the response and factor variables, in part to decide whether we have satisfied the conditions needed for ANOVA, although we will evaluate the assumptions more rigorously below. Even before looking at the data, we can conclude that the observations are independent. It is difficult to think of reasons that either the birth rates or type of provider in one country would be affected by those in another.

1. Select **Analyze ► Distribution**. Cast the columns **BirthRate** and **Provider** into the **Y, Columns** role, and click **OK**.

When we look at the **BirthRate** histogram, we find that the variable is positively skewed in shape; this could signal an issue with the normality assumption, but we might have large enough subsamples that we can rely on the Central Limit Theorem to proceed with the analysis.

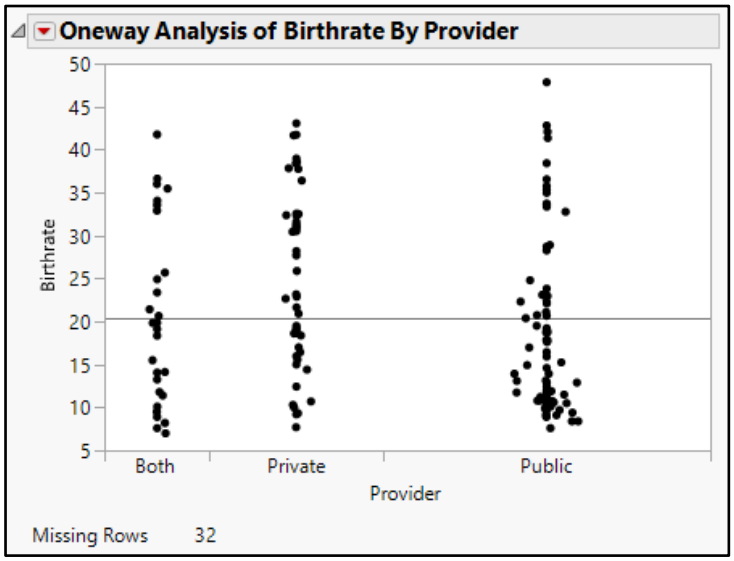
2. Click on any of the bars in the **Provider** graph to select the rows that share one provider type. When you do so, note the corresponding values in the **BirthRate** distribution. Does there seem to be any consistent pattern in birth rates that corresponds to how maternity leave benefits are provided? ANOVA will help clarify the situation.

To begin the ANOVA, we return to the **Fit Y by X** analysis platform.

3. Select **Analyze ► Fit Y by X**. Cast **BirthRate** into the **Y** role and **Provider** into the **X** role and click **OK**.

The default output is a simple scatterplot showing the response values across the different treatment groups. Figure 14.1 has been modified to jitter the points for visual clarity.

**Figure 14.1: One-Way ANOVA Initial Graph (with Points Jittered)**



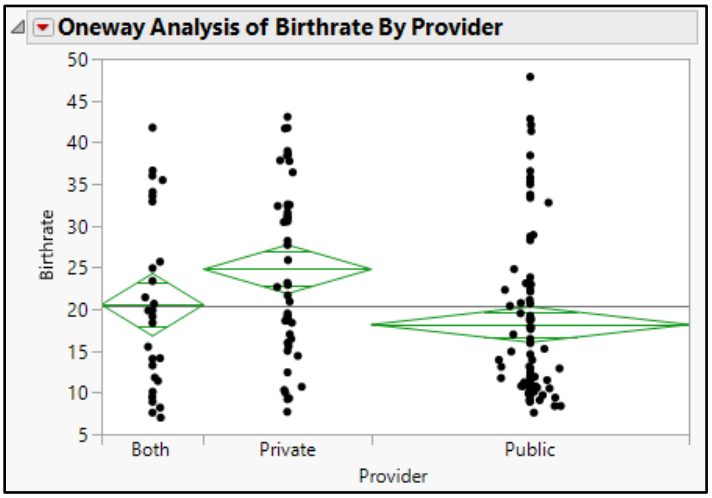

This graph reveals at least two things of interest. The dispersion of points within each group appears comparable, which suggests constant variance and some indication of a treatment effect—that countries with publicly provided maternity-leave benefits might have somewhat

lower birth rates on average than those with privately provided benefits. In the graph, notice the concentration of points above the overall mean line in the **Private** group and the large number of points below the line for the **Public** group. Let's look more closely.

4. Click the red triangle next to **Oneway Analysis of BirthRate by Provider** and select **Means/Anova**.

As shown in Figure 14.2, you will see a green *means diamond* superimposed on each set of points. In this case, the diamonds have different widths that correspond to differences in sample size.

**Figure 14.2: Graph Including Means Diamonds**




The center line of the diamond is the sample mean, and the upper and lower points of the diamond are the bounds of a 95% confidence interval. The horizontal lines near the top and bottom of the diamonds are *overlap marks*; these are visual boundaries that we can use to compare the means for the groups. Diamonds with overlapping areas between the overlap marks indicate insignificant differences in means. In this picture, the means diamond for the **Public** group does not overlap the **Private** group, suggesting that we do have significant differences here. Before we go too much further with the analysis of variance, we need to check our assumptions.

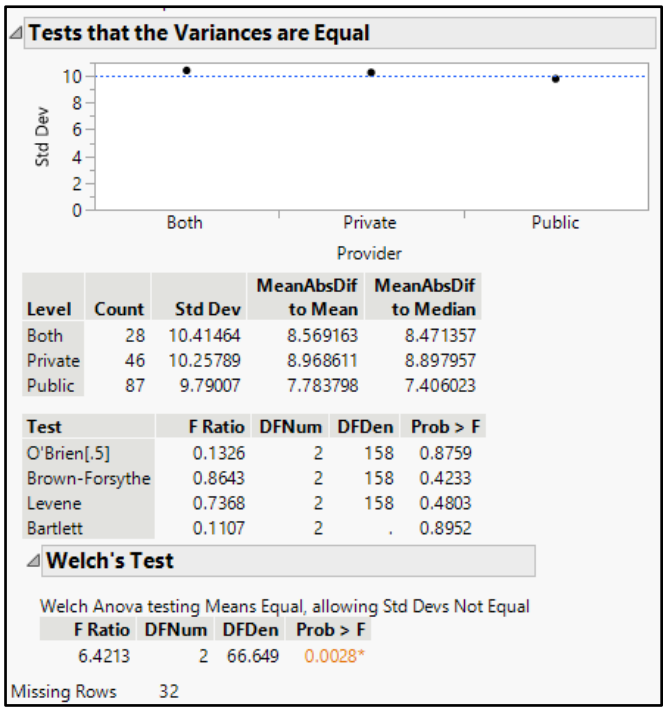
## Does the Sample Satisfy the Assumptions?

The independence assumption is a logical matter; no software can tell us whether the observations were collected independently of each other. Independence is a function of the study and sampling design, and, as noted above, we can conclude that we have satisfied the independence condition.

1. We can test for the equality of variances by again clicking the red triangle and choosing **UnEqual Variances**, which generates the result in Figure 14.3.

Direct your attention to the lower table titled **Test**. There are several standard tests used to decide whether we have satisfied the equal variance condition. In all cases, the null hypothesis is that the variance of all groups is equal, and the alternative hypothesis is that variances are unequal. Levene's test is probably the most commonly used, and we will use it here.

**Figure 14.3: Tests for Equality of Variances**




| Level | Count | Std Dev | MeanAbsDif to Mean | MeanAbsDif to Median |
|---|---|---|---|---|
| Both | 28 | 10.41464 | 8.569163 | 8.471357 |
| Private | 46 | 10.25789 | 8.968611 | 8.897957 |
| Public | 87 | 9.79007 | 7.783798 | 7.406023 |

| Test | F Ratio | DFNum | DFDen | Prob > F |
|---|---|---|---|---|
| O'Brien[.5] | 0.1326 | 2 | 158 | 0.8759 |
| Brown-Forsythe | 0.8643 | 2 | 158 | 0.4233 |
| Levene | 0.7368 | 2 | 158 | 0.4803 |
| Bartlett | 0.1107 | 2 | . | 0.8952 |

**Welch's Test**

Welch Anova testing Means Equal, allowing Std Devs Not Equal

| F Ratio | DFNum | DFDen | Prob > F |
|---|---|---|---|
| 6.4213 | 2 | 66.649 | 0.0028* |

Missing Rows 32

In this example, the *P*-values for all four tests are quite large by conventional standards. That is, there is insufficient evidence to reject the null assumption of equal variances, so we can feel safe in continuing to assume that the sample satisfies the equal variance assumption.

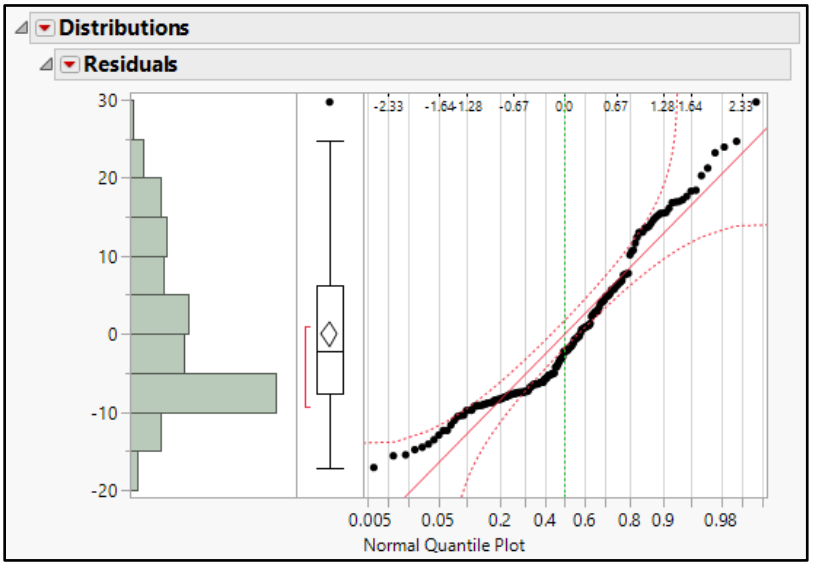
When we test for equality of variances, JMP also reports the results of Welch's test. This test adjusts for the inequality of variance to provide an alternative to the standard one-way analysis of variance. In other words, Welch's test is a fallback option in case the variances are unequal; we will say more about this in a few pages.

As for the normality assumption, we should not reflexively rely on the Central Limit Theorem applying in this case. None of the subgroups is especially large. The **Both** group has a sample size of 28, and the distribution of birth rates is moderately skewed. It is prudent to check whether the individual error terms might be normally distributed. We have no way to measure the individual errors directly, but we can come close by examining the deviations between each individual measurement and the mean of its respective group. We call these deviations *residuals* and can examine their distribution as an approximation of the individual random error terms.

To analyze the residuals, we proceed as follows:

2. Click the red triangle next to **Oneway**, and select **Save ► Save Residuals**.
   The new column is called Birthrate centered by Provider. The terminology "centered by" indicates that the column values are computed by subtracting the mean of the provider subsamples from the original birth rates, which is precisely what a residual is.
3. In the data table, select **Birthrate centered by Provider**. Right-click, select **Column Info**, and change the column name to **Residuals**.
4. With the data table highlighted, select **Analyze ► Distribution.** Choose the new residuals column as **Y, Columns**, and click **OK**.
5. Click the red triangle next to **Residuals** and select **Normal Quantile Plot**. We can use the normal quantile plot (Figure 14.4) to judge whether these residuals are decidedly skewed or depart excessively from normality.

**Figure 14.4: Normality Check for Residuals**




The reliability of ANOVA is not dramatically affected by data that is non-normal. We say that the technique is *robust* with respect to normality. If the residuals are generally unimodal and symmetric, we can proceed.

Figure 14.4 shows the distribution of residuals in this case, and based on what we see here, we can conclude that the normality assumption is in question. Given the robustness of the technique and the single-peaked distribution, we will proceed with the one-way analysis of variance to illustrate the method.

## Factorial Analysis for Main Effects

In the scatterplot with means diamonds, we found visual indications that the mean of the **Public** provision group might be lower than the other two groups. We find more definitive information

below the graph. Having validated the assumptions, it is time to look back at the **Oneway Anova** report. We find three sets of statistics, shown below in Figure 14.5. At this point in our studies, we will focus on just a few of the key results.

**Figure 14.5: One-Way ANOVA Results**

**Oneway Anova**

**Summary of Fit**

| | |
|---|---|
| Rsquare | 0.076727 |
| Adj Rsquare | 0.06504 |
| Root Mean Square Error | 10.0336 |
| Mean of Response | 20.4665 |
| Observations (or Sum Wgts) | 161 |

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Ratio | Prob > F |
|---|---|---|---|---|---|
| Provider | 2 | 1321.862 | 660.931 | 6.5651 | 0.0018* |
| Error | 158 | 15906.349 | 100.673 | | |
| C. Total | 160 | 17228.211 | | | |

**Means for Oneway Anova**

| Level | Number | Mean | Std Error | Lower 95% | Upper 95% |
|---|---|---|---|---|---|
| Both | 28 | 20.5366 | 1.8962 | 16.792 | 24.282 |
| Private | 46 | 24.7870 | 1.4794 | 21.865 | 27.709 |
| Public | 87 | 18.1595 | 1.0757 | 16.035 | 20.284 |

Std Error uses a pooled estimate of error variance

The **Summary of Fit** table provides an indication of the extent to which our ANOVA model accounts for the total variation in all 155 national birth rates. The key value here is **Rsquare**,[2] which is always between 0 and 1, and describes the proportion of total variation in the response variable that is associated with the factor. Here, we find that only 7.7% of all variation in birth rates is associated with the variation in provision of maternity benefits; that's not very impressive. Other factors must account for the other 92.3% of the variation.

In the **Analysis of Variance** table (like the tables described in your main text), we find our test statistic (6.5651) and its *P*-value (0.0018). Here, the *P*-value is very small, indicating statistical significance. We reject the null hypothesis of equal treatment effects and conclude that at least one treatment has a significant effect.

Looking at the last table, **Means for Oneway Anova**, we see that the confidence interval for the mean of the private group is higher than the mean for the public group, and that the interval for the "both" group overlaps the other two. For a more rigorous review of the differences in group means, we can also select a statistical test to compare the means of the groups.

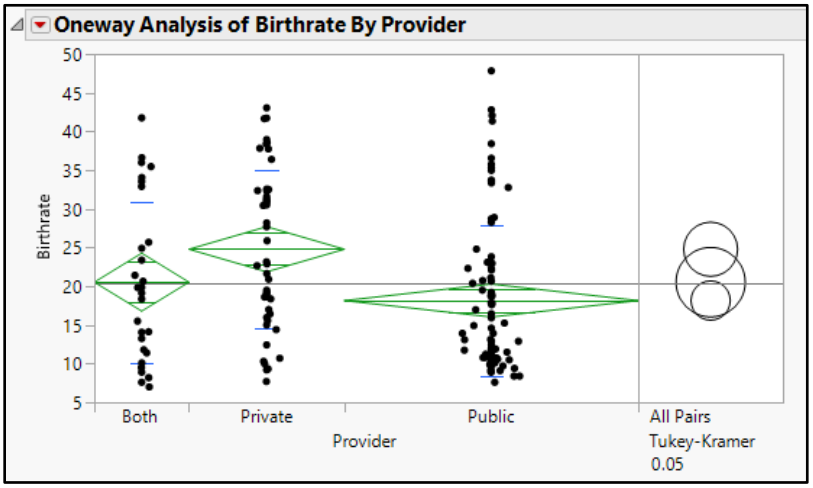
We have several common methods available to us for such a comparison. With two factor levels, it makes sense to use the simple two-sample t test comparison, but with more than two levels, the t tests will overstate differences, and therefore we should use a different approach here.

In this study, we do not have a control group as we might in a designed experiment. Because there isn't a control, we will use *Tukey's HSD* (Honestly Significant Difference), the most

conservative of the approaches listed. If one of the factor levels had referred to a control group, we would use *Dunnett's method* and compare each of the other group means to the mean of the control group.

1. Click the **Oneway** red triangle once again, and choose **Compare Means ► All Pairs, Tukey's HSD.**

**Figure 14.6: Graphical Representation of Tukey's HSD**

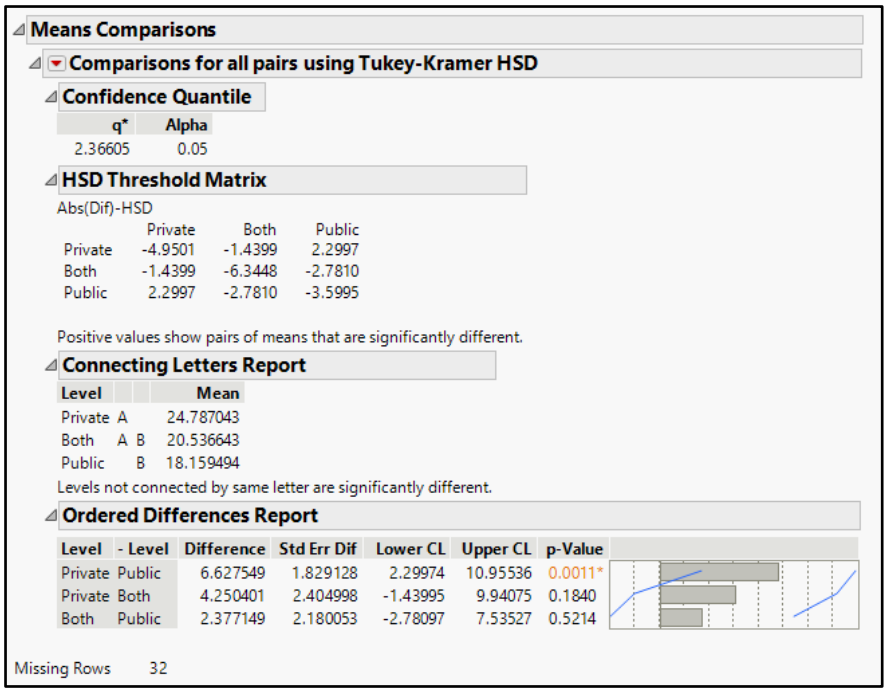



In the upper panel (shown above in Figure 14.6), we now see a set of three overlapping rings representing the three groups. Note that the upper ring (**Private** group) overlaps the middle **Both** ring, that the middle and lower rings overlap, but that the **Private** and **Public** rings do not overlap.

Click the uppermost (**Private**) ring. Notice that it turns red, as does this middle (**Both**) ring, but the lower **Public** ring becomes blue. This indicates a significant difference between **Private** and **Public**, but no significant difference between **Private** and **Both**. In other words, based on this sample, we cannot discern a difference in birth rates between the **Private** and **Both** groups, but we do discern a difference between **Private** and **Public**.

We will scroll down to find the numerical results of Tukey's test, shown here in Figure 14.7.

**Figure 14.7: Tukey HSD Results**



**Means Comparisons**

**Comparisons for all pairs using Tukey-Kramer HSD**

**Confidence Quantile**

| q* | Alpha |
|---|---|
| 2.36605 | 0.05 |

**HSD Threshold Matrix**

Abs(Dif)-HSD

| | Private | Both | Public |
|---|---|---|---|
| Private | -4.9501 | -1.4399 | 2.2997 |
| Both | -1.4399 | -6.3448 | -2.7810 |
| Public | 2.2997 | -2.7810 | -3.5995 |

Positive values show pairs of means that are significantly different.

**Connecting Letters Report**

| Level | | | Mean |
|---|---|---|---|
| Private | A | | 24.787043 |
| Both | A | B | 20.536643 |
| Public | | B | 18.159494 |

Levels not connected by same letter are significantly different.

**Ordered Differences Report**

| Level | - Level | Difference | Std Err Dif | Lower CL | Upper CL | p-Value |
|---|---|---|---|---|---|---|
| Private | Public | 6.627549 | 1.829128 | 2.29974 | 10.95536 | 0.0011* |
| Private | Both | 4.250401 | 2.404998 | -1.43995 | 9.94075 | 0.1840 |
| Both | Public | 2.377149 | 2.180053 | -2.78097 | 7.53527 | 0.5214 |

Missing Rows 32

How do we read these results? The bottom portion of the output displays the significance levels for estimated differences between each group. We find one significant *P*-value, which is associated with the difference between the **Private** and the **Public** provision of benefits. We can confidently conclude that mean birth rates are different (apparently higher) in countries with private provision of benefits than in those with public provision of benefits but cannot be confident about any other differences based on this sample.

Look at the **Connecting Letters Report** just above the bottom portion. Here the idea is simple: each level is tagged with one or two letters. If a mean has a unique letter tag, we conclude that it is significantly different from the others. If a level has two tags, it could be the same as either group. In this analysis, **Private** is A and **Public** is B, but **Both** is tagged as A B. So, we conclude that the means for public and private are different from one another but cannot make a definitive conclusion about the **Both** category.

## What If Conditions Are Not Satisfied?

In this example, our data were consistent with the assumptions of normality and equal variances. What happens when the sample data set does not meet the necessary conditions? In most introductory courses, this question probably does not arise or if it does, it receives very brief treatment. As such, we will deal with it in a very limited way. Thankfully, JMP makes it quite easy to accommodate “uncooperative” data.

Earlier we noted that the assumption of equal variances is a critical condition for reliable results in an ANOVA, and we saw how to test for the equality of variances. If we find that there are significant differences in variance across the groups, we should not rely on the ANOVA table to draw statistical conclusions. Instead, JMP provides us with the results of **Welch's ANOVA**, a procedure that takes the differences in subgroup variation into account.

1. Look back in your output for the **Test that the Variances are Equal** (which appears in Figure 14.3 above).

At the bottom of that output, you will find the results of Welch's test. In our example, that test finds F = 6.4213 with a *P*-value of 0.0028. We would decide to reject the null hypothesis and conclude that the group means are not all equal. With Welch's test, there are no procedures comparable to Tukey's HSD to determine where the differences lie.

If the residuals are not even close to looking normal—multimodal, very strongly skewed, or the like—we can use a *nonparametric* equivalent test. Nonparametric tests make no assumptions about the underlying distributions of our data (some statisticians refer to them as "distribution-free" methods).

2. Return to the red triangle, and select **Nonparametric ► Wilcoxon Test**.

Now look at the results (not shown here), labeled **Wilcoxon/Kruskal-Wallis Tests**. First a note on terminology: strictly speaking, the Wilcoxon test is the nonparametric equivalent of the two-sample t test, while the Kruskal-Wallis test is the equivalent of the one-way ANOVA.

These tests rely on chi-squared, rather than the F, distribution to evaluate the test statistic. Without laying out the theory of these tests, suffice it to say that the null hypothesis is this: the distributions of the groups are centered[3] in the same place. A *P*-value less than 0.05 conventionally means that we will decide to reject the null, as we do in this example.

> For users of JMP Pro, bootstrapping is another alternative when conditions are not satisfied. Use of the bootstrap for Analysis of Variance is well beyond the scope of an introductory statistics course. Interested readers should consult the Bootstrapping section within the JMP Documentation Library.

## Including a Second Factor with Two-Way ANOVA

Earlier we noted that our model accounted for relatively little of the variation in national birth rates, and that other factors might also play a role. To investigate the importance of a second categorical variable, we can extend our Analysis of Variance to include a second factor. The appropriate technique is called *Two-Way ANOVA.*

In some countries, new mothers typically have lengthy maternity leaves from their jobs, while in others, the usual practice is to offer much briefer time off from work. Let's use the variation in this aspect of prevailing national policies as an indication of a country's commitment to the well-being of women who give birth. We will focus on the factor called **MatLeave90+,** a dichotomous

variable indicating whether a country provides 90 days or more of maternity leave for new mothers.

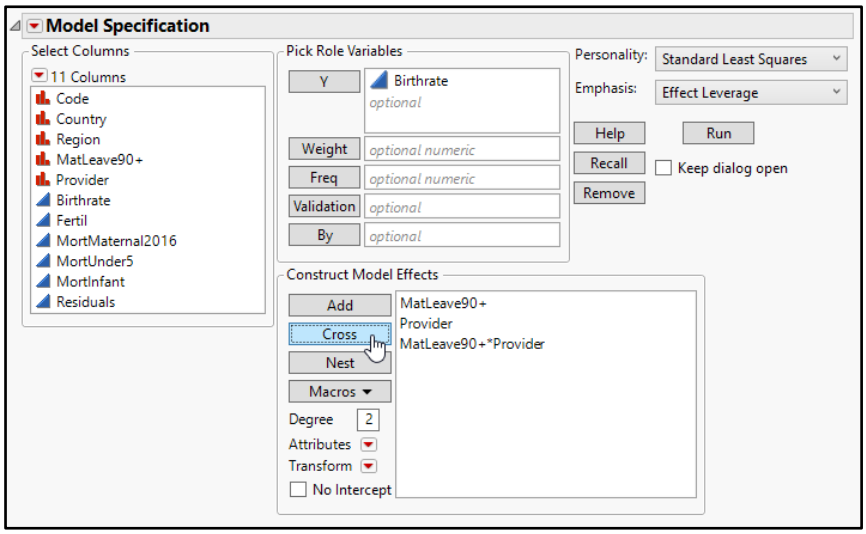
1. As we did before, use **Analyze ▶ Distribution**. This time, select **BirthRate**, **Provider**, and **MatLeave90+** as the **Y, Columns.**
2. Click on different bars in the three graphs, looking for any obvious visual differences in birth rates in countries with different insurance and maternity leave practices. Think about what you see in this part of the analysis.

Before performing the two-way analysis, we need to consider one potential complication. It is possible that the duration of the maternity leave and the provider of benefits each have separate and independent associations with birth rate, and it is also possible that the impact of one variable is different in the presence of the other. For example, perhaps the duration of the leave is relevant where it is privately provided, but not where it's publicly provided.

Because of this possibility, in a two-way ANOVA, we investigate both the *main effects* of each factor separately as well as any possible *interaction effect* of the two categorical variables taken together. When we specify the two-way model, we will identify the main factors and their possible interactions. Because we are no longer modeling Y as a function of a single X, we depart from the familiar **Fit Y by X** platform in favor of a platform with more flexibility.

3. Now let's perform the two-way ANOVA. Select **Analyze ▶ Fit Model**. As shown in Figure 14.8, cast **Birthrate** as the **Y** variable, and then highlight both **Provider** and **MatLeave90+** and click **Add**.
4. Now highlight the same two categorical columns again in **Select Columns** and click **Cross**. Crossing the two categorical variables establishes an *interaction term* with six groupings (three provider groups times two leave duration groups).
5. Now click **Run** to perform the analysis.

**Figure 14.8: Fit Model Dialog Box for Two-Way Model**



This command generates extensive results, and we will now focus on a few critical elements that will be illustrated in the next several pages. In a two-way analysis, it is good practice to review the results in a sequence like the one we follow here. We see in the Analysis of Variance table (about halfway through the report) that we have a significant test (F = 3.8133, *P*-value = 0.0028), so we might have found some statistically interesting results. First, we evaluate assumptions (which are the same as for one-way ANOVA), look at the significance of the entire model, then check to determine whether there are interaction effects, and finally, evaluate main effects.

The **Fit Model** results are quite dense and contain elements that go beyond the content of most introductory courses in statistics. We will look very selectively at portions of the output that correspond to topics that you are likely to see in an introductory course. If your interests lead you to want or need to interpret portions that we bypass, you should use the context-sensitive **Help** tool ( ? ) and just point at the results that you want to explore.

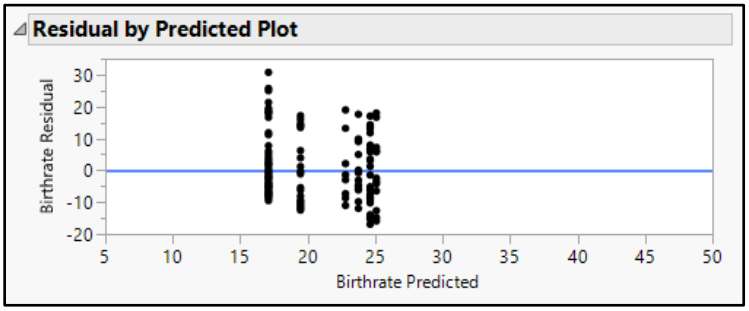
## Evaluating Assumptions

As before, we can reason that the observations are independent of one another and should examine the residuals for indications of non-normality or unequal variances.

1. Scroll down in the report until you find the **Residual by Predicted Plot** (Figure 14.9).

This graph displays the computed residuals in six vertical groupings, corresponding to the six possible combinations of the levels of both factors. We use this graph to make a judgment as to whether each of the groups appears to share a common variance; in this case, the spread of points within each group appears similar, so we can conclude that we have met the constant variance assumption. If the spreads were grossly different, we should be reluctant to go on to evaluate the results any further.

**Figure 14.9: Plot of Residuals Versus Predicted Means**




Let's also check the residuals to see how reasonable it is to assume that the individual error terms are normally distributed.

2. We can again save the residuals in the data table by clicking on the red triangle at the top of the analysis and selecting **Save Columns ▶ Residuals**.
3. As you did earlier, use **Analyze ▶ Distribution** to create a histogram and normal probability plot of the **Residual Birthrate** column (not shown).

You should see that the residuals are single-peaked and somewhat skewed. While not perfectly normal, there is nothing to worry about here.

## Interaction and Main Effects

Next, we should evaluate the results of the test for the interaction term. We do so because a significant interaction can distort the interpretation of the main effects, so before thinking about the main effects, we want to examine the interaction. An interaction exists when the relationship between one factor and the response is *different* across the groups of the second factor.

1. Return to the Fit Model report. At the very bottom, locate the **Parameter Estimates and Effect Tests** table, as shown in Figure 14.10.

This lower table lists the results of three separate F-tests, namely the test for equality of group means corresponding to different provider arrangements, different lengths of maternity leave, and the interaction of the two factors. The null hypothesis in each test is that the group means are identical.

**Figure 14.10: Tests of Main and Interaction Effects**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 22.137814 | 0.961458 | 23.03 | <.0001* |
| MatLeave90+[No] | 1.5906374 | 0.961458 | 1.65 | 0.1001 |
| Provider[Both] | -1.0052 | 1.50784 | -0.67 | 0.5060 |
| Provider[Private] | 2.7188839 | 1.309211 | 2.08 | 0.0395* |
| MatLeave90+[No]*Provider[Both] | 0.0780819 | 1.50784 | 0.05 | 0.9588 |
| MatLeave90+[No]*Provider[Private] | -1.819502 | 1.309211 | -1.39 | 0.1666 |

**Effect Tests**

| Source | Nparm | DF | Sum of Squares | F Ratio | Prob > F |
|---|---|---|---|---|---|
| MatLeave90+ | 1 | 1 | 270.62696 | 2.7370 | 0.1001 |
| Provider | 2 | 2 | 485.80349 | 2.4566 | 0.0891 |
| MatLeave90+*Provider | 2 | 2 | 293.59287 | 1.4847 | 0.2298 |

Each row of the table reports the number of parameters (**Nparm**), corresponding to one less than the number of factor levels for each main factor, and the product of the nparms for the interaction. We also find degrees of freedom, the sum of squares, and the F-ratio and *P*-value for each of the three tests.

We first look at the bottom row (the interaction) and see that the F statistic is small with a moderately high *P*-value. Therefore, we conclude that we cannot reject the null for this test, which is to say that there is no significant interaction effect. Whatever difference exists (if any) in birth rates associated with the length of maternity leave, it is the same across the different provider conditions. Ordinarily, having found no evidence of an interaction, an investigator would probably re-run the model without the interaction term. However, to illustrate additional visualizations of main and interaction effects, we will leave the term in the model for now.

Finally, we consider the main effects where we find that neither *P*-value falls below the customer 5% milestone. However, depending on the context, they may be of interest to a researcher or policy maker. Both factors might have main effects, so the natural next step is to understand what those effects are.

The parameter estimates table shows a significant intercept term and a significant difference when the provider is private. The effect of having shorter maternity leaves approaches significance at the 10% level.

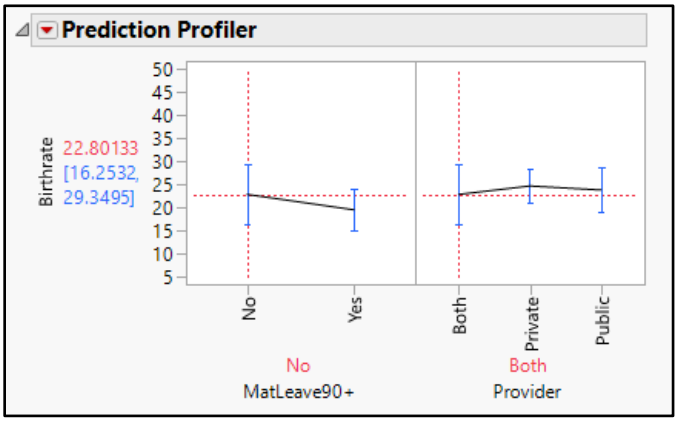
For a visual display of the differences in group means we can look at *factor profiles*.

2. Click the red triangle at the very top of the report (next to **Response Birthrate)** and select **Factor Profiling ▶ Profiler.**

JMP generates two graph panels that display confidence intervals for the mean birth rates for each main factor level grouping.

3. Use the Grabber (Hand) tool to stretch the vertical axes to magnify the scale of the two graphs, or enlarge the graphs by dragging the lower right corner diagonally. The result will look like something like Figure 14.11.

**Figure 14.11: Prediction Profiler for Main Effects**




The profiler graphs are easy to interpret. In the left panel, we see confidence intervals for mean birth rate in the countries with maternity leaves less than 90 days and those with 90 days or more. Similarly, the right panel compares the intervals for the different provider arrangements. As with other JMP graphs, the panels here are linked and interactive.

The red dashed lines indicate the level settings for the profiler. Initially, we are estimating the birth rate for countries with shorter maternity leaves that are provided jointly by the public and private sectors. On the far left, we see the confidence interval of approximately 16.25 to 29.35 births per 1,000 persons. Notice also that, once we control for shorter maternity leave nations, the confidence intervals for all providers overlap.

4. To see how the mean birth rate prediction changes, for example, if we look at countries with shorter maternity leaves that are publicly provided, just use the arrow tool and click on the blue interval corresponding to **Public**.

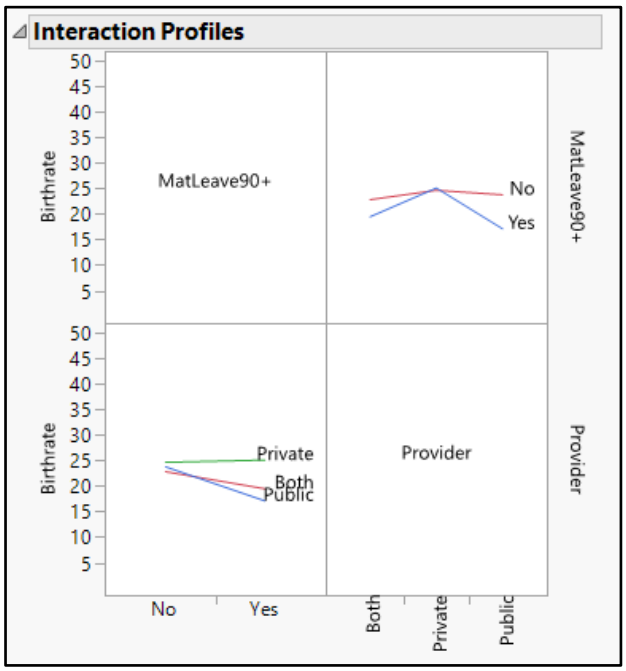
The prediction changes to approximately 18.85 to 28.67.

5. Experiment by clicking on the different interval bars until you begin to see how the **BirthRate** estimates change with different selections in the Profiler. Notice especially what happens when you focus on countries that do have longer maternity leaves.

We can also construct a profiler for interaction effects. In this example, the interaction is insignificant, but we can still learn something from the profiler graph.

6. Click the red triangle again and select **Factor Profiling ▶ Interaction Plots**. Enlarge the plot area or use the Grabber to adjust the vertical axes so that you can see the plotted lines clearly as in Figure 14.12.

**Figure 14.12: Interaction Plots for the Two-Way Analysis of Variance**



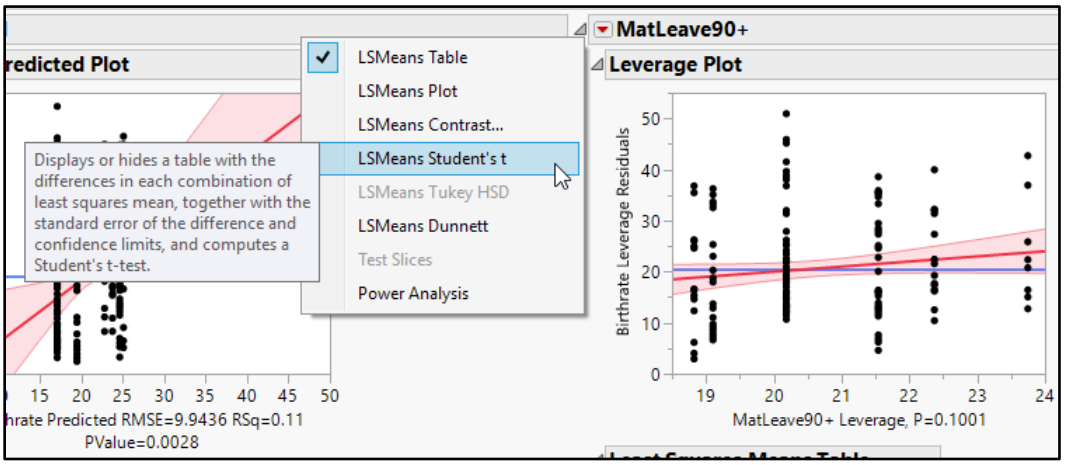

The interaction profiles show the mean birth rates corresponding to both categorical variables. In the lower left graph, the different provider arrangements are represented by three color-coded lines, and we see that regardless of provider, the mean **BirthRate** is lower for those countries with longer maternity leaves (**MatLeave90+ = Yes**). The plot in the upper right swaps the arrangement. In this graph, we have two color-coded lines, corresponding to the two different levels of maternity leave durations. The lines display the different birth rate mean values for the three different provider systems.

When there is no interaction, we will find lines that are approximately parallel. If there is a significant interaction, we will see lines that cross. The intersection shows that the pattern of different means for one categorical factor varies depending on the value of the other factor.

If we want more precision than a small graph offers, we can once again look at multiple comparisons just as we did for the one-way ANOVA. Because this command performed three tests (two main effects and one interaction), we will need to ask for three sets of multiple comparisons.

7. Return to the top of the output, scroll to the right slightly to locate the **Leverage Plots,**[4] and click the red triangle next to **MatLeave90+**. Because this categorical variable is dichotomous, we can compare the means using a two-sample t test. Choose **LSMeans Student's t** as shown in Figure 14.13.

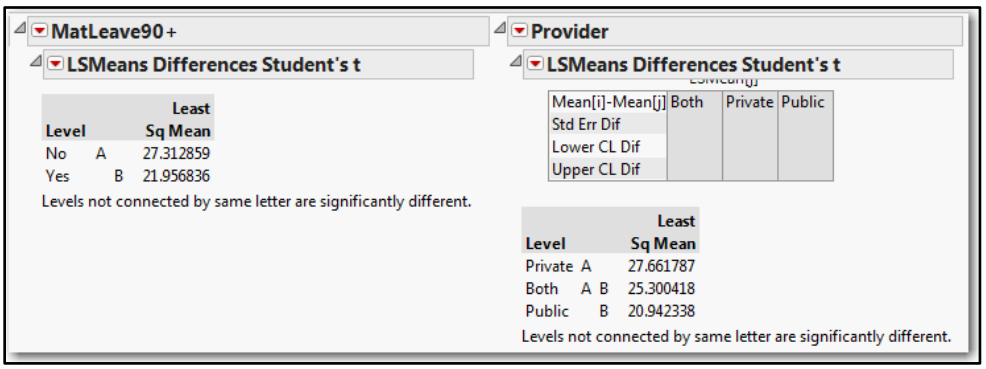
**Figure 14.13: Requesting Multiple Comparisons for Two-Way ANOVA**




8. Click the red triangle next to **Provider** and select **LSMeans Tukey HSD.**

Now look below each of the Leverage Plots and locate the connecting letters reports (shown here in Figure 14.14; scroll down and right a bit to adjust the report in this fashion).

**Figure 14.14: Connecting Letters Report for the Two Factors**



Here we see the significant contrasts: there is no significant difference in means between countries with longer and shorter maternity leaves. There is a conclusive difference between countries that provide the benefits privately and those that do so publicly.

## Application

Now that you have completed all of the activities in this chapter, use the techniques that you have learned to respond to these questions. In all problems, be sure to evaluate the reasonableness of the ANOVA assumptions and comment on your conclusions.

1. *Scenario:* We will continue to examine the World Development Indicators data in the data table **BirthRate 2017**. We will broaden our analysis to work with other columns in that table:
    - **Region**: Country's geographic region
    - **Fertil**: Number of births per woman
    - **MortMaternal2016**: Maternal mortality, estimated deaths per 100,000 births in 2016
    - **MortUnder5**: Deaths, children under 5 years per 1,000
    - **MortInfant**: Deaths, infants per 1,000 live births

    a. Perform a one-way ANOVA to decide whether birth rates are consistent across regions of the world. If you conclude that the rates differ, determine where the differences are.
    b. Perform a one-way ANOVA to decide whether fertility rates are consistent across regions of the world. If you conclude that the rates differ, determine where the differences are.
    c. Perform a two-way ANOVA to see how maternal mortality varies by length of maternity leaves and provider. If you conclude that the rates differ, determine where the differences are, testing for both main and interaction effects.
    d. Perform a two-way ANOVA to see how mortality under five years of age varies by length of maternity leaves and provider. If you conclude that the rates differ, determine where the differences are, testing for both main and interaction effects.
2. *Scenario:* A company that manufactures insulating materials wants to evaluate the effectiveness of three different additives to one material in their product line. The company performs an industrial experiment to evaluate the additives. For the sake of this example, these additives are called **Regular, Extra**, and **Super**. The company now uses the regular additive, so that serves as the control group.

    The experimenter has a temperature-controlled apparatus with two adjoining chambers. He can insert the experimental insulating material in the wall that separates the two chambers. Initially, he sets the temperature of chamber A at 70°F and the temperature of the other chamber (B) at 30°F. After a pre-determined amount of time, he measures the temperature in chamber A, recording the change in temperature as

the response variable. A good insulating material prevents the temperature in chamber A from falling far.

Following good experimental design protocols (see Chapter 18) he repeats the process in randomized sequence thirty times, testing each insulating material a total of 10 times.

We have the data from this experiment in a file called **Insulation**.

a. Perform a one-way ANOVA to evaluate the comparative effectiveness of the three additives.
b. Which multiple comparison method is appropriate in this experiment and why?
c. Should the company continue to use the regular additive or change to one of the others? If they should change, which additive should they use?

3. *Scenario:* How do prices of used cars differ, if at all, in different areas of the United States? Our data table **Used Cars** contains observational data about the listed prices of three popular compact car models in three different metropolitan areas in the U.S. The cities are Phoenix, AZ; Portland, OR; and Raleigh-Durham-Chapel Hill, NC. The car models are the Chrysler PT Cruiser Touring Edition, the Honda Civic EX, and the Toyota Corolla LE. All of the cars are two years old.
   a. Perform a two-way ANOVA to see how prices vary by city and model. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   b. Perform a two-way ANOVA to see how mileage of these used cars varies by city and model. If you find differences, perform an appropriate multiple comparison and explain where the differences are.

4. *Scenario:* Does a person's employment situation influence the amount that he or she sleeps? Our **TimeUse** data table can provide a foundation for analysis of this question. Initially, show and include only observations from 2017 for parts a and b.
   a. Perform a one-way ANOVA to see how hours of sleep vary by employment status. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   b. Perform a two-way ANOVA to see how hours of sleep vary by employment status and sex. Be sure to model potential interaction effects. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   c. Clear row states to include all years. Perform a two-way ANOVA to see how hours of socializing vary by year and sex. Be sure to model potential interaction effects. If you find differences, perform an appropriate multiple comparison and explain where the differences are.

5. *Scenario:* In this scenario, we return to the **Pipeline Safety** data table to examine the consequences of pipeline disruptions. Clear row states in case you have saved row states in a previous analysis.
   a. Perform a one-way ANOVA to see how the dollar amount of property damage caused by a disruption varies by **CAUSE**. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   b. Perform a two-way ANOVA to see how the dollar amount of property damage caused by a disruption varies by **LEAK_TYPE** and **IGNITE_IND**. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   c. Perform a two-way ANOVA to see how dollar amount of property damage varies by type of leak type and cause. If you find differences, perform an appropriate multiple comparison and explain where the differences are.

6. *Scenario:* A company manufactures catheters, blood transfusion tubes, and other tubing for medical applications. In this particular instance, we have diameter measurements from a newly designed process to produce tubing that is designed to be 4.4 millimeters in diameter. These measurements are stored in the **Diameter** data table.

   Each row of the table is a single measurement of tubing diameter. There are six measurements per day, and each day, one of four different operators was responsible for production using one of three different machines. The first 20 days of production were considered Phase 1 of this study. Following Phase 1, some adjustments were made to the machinery, which then ran an additional 20 days in Phase 2. Use the data filter to analyze just the Phase 1 data.
   a. Perform a one-way ANOVA to see how the tubing diameters vary by Operator. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   b. Perform a one-way ANOVA to see how the tubing diameters vary by Machine. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   c. Perform a two-way ANOVA to see how the tubing diameters vary by Operator and Machine. If you find differences, perform an appropriate multiple comparison and explain where the differences are.
   d. Explain in your own words what the interaction plots tell you in this case. Couch your explanation in terms of the specific machines and operators.

7. *Scenario:* JMP ships illustrative data tables with the software. One of these is the **Popcorn** data table. It is one of the few realistic but artificial examples that we will use in this book, but because it is based on the work of three highly influential statisticians,[5] it appears here.
    a. Perform a series of three one-way ANOVAs to decide if **Yield** varies by **popcorn type,** by the **amount of oil used (oil amt),** or **size of batch**. Verify assumptions; if you find differences, perform an appropriate multiple comparison and explain where the differences are.
    b. Perform a series of two-way ANOVAs to see if there are any interactions among factors, and report on your findings.

---

[1] Alternatively, we might express this assumption as saying that the individual responses are normally distributed within each group.

[2] We have seen Rsquare before in Chapter 4, and will work with it again in Chapters 15 and 16 when we study regression analysis.

[3] Different nonparametric procedures take different approaches to the concept of a distribution's "center." None use the mean *per se* as the One-way ANOVA does.

[4] Leverage plots are discussed more fully in Chapter 15.

[5] The Popcorn data table is inspired by experimental data from Box, Hunter, Hunter (1978).

# Chapter 15: Simple Linear Regression Inference

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## Overview

In Chapter 4, we learned to summarize two continuous variables at a time using scatterplot, correlations, and line fitting. In this chapter, we will return to that subject, this time with the object of generalizing from the patterns in sample data in order to draw conclusions about an entire population. The main statistical tool that we will use is known as *linear regression analysis.* We will devote this chapter and three later chapters to the subject of regression.

Because Chapter 4 is now many pages back, we will begin by reviewing some basic concepts of bivariate data and line fitting. Then, we will discuss the fundamental model used in simple linear regression. After that, we will discuss the crucial conditions necessary for inference, and finally, we will see how to interpret the results of a regression analysis.

## Fitting a Line to Bivariate Continuous Data

We introduced regression in Chapter 4 using the data table **Birthrate 2017**. This data table contains several columns related to the variation in the birth rate and the risks related to childbirth around the world as of 2017. In this data table, the United Nations reports figures for 193 countries. Let's briefly revisit that data now to review some basic concepts, focusing on two measures of the frequency of births in different nations.

In the next several chapters, we will expand our investigation of regression analysis and will return to the illustrative data tables several times. For this reason, you should work through the examples in a JMP Project. By creating and saving a project in this chapter, your workflow in the coming chapters will be more efficient. At this point, create a new project called Chap_15.

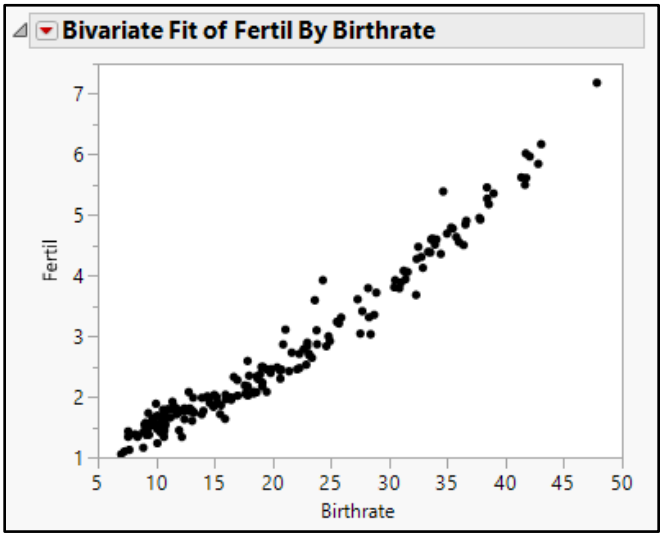
1. Open the **Birthrate 2017** data table now. If any row states were previously set and saved, clear them now.

As we did in Chapter 4, let's look at the columns labeled **BirthRate** and **Fertil**. A country's annual birth rate is defined as the number of live births per 1,000 people in the country. The fertility rate is the mean number of children that would be born to a woman during her lifetime. We plotted these two variables in Chapter 4; let us do that again now.

2. Select **Analyze ► Fit Y by X**. Cast **Fertil** as **Y** and **BirthRate** as **X** and click **OK**.

Your results will look like those shown in Figure 15.1.

**Figure 15.1: Relationship Between Birth Rate and Fertility Rate**




This is the same graph that we saw in Figure 4.13. Again, we note that general pattern is upward from left to right: fertility rates increase as the birth rate increases, although there are some countries that depart from the pattern. The pattern can be described as linear, although there is a mild curvature at the lower left. We also see that a large number of countries are concentrated in the lower left, with low birth rates and relatively low maternal mortality.

In Chapter 4, we illustrated the technique of line-fitting using these two columns. Because these two columns really represent two ways of thinking about a single construct ("how many

babies?"), let us turn to a different example to expand our study of simple linear regression analysis.

We will return to the NHANES data and look at two body measurement variables. Because adult body proportions are different from children and because males and females differ, we will restrict the first illustrative analysis to male respondents ages 18 and up.

3. Open the data table called **NHANES 2016**. If row states were saved from an earlier session, clear them now. In the coming chapters, we will work through several illustrations using NHANES data.

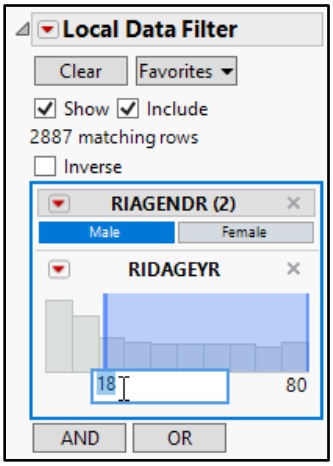
This table contains respondents of all ages, male and female. To isolate adult males, we might be inclined to apply a data filter to the entire table, but because we will want to use other respondents in later examples, let's wait and apply a local data filter within the context of the first regression analysis. The global data filter temporarily affects the whole data table; if you resave the data table, the row states are saved with it. A local data filter affects the rows used within a single analysis report but does not alter the underlying data table.

Now we can begin the regression analysis. We will examine the relationship between waist circumference and body mass index, or BMI, which is the ratio of a person's weight to the square of height. In the data table, waist measurements are in centimeters, and BMI is kilograms per square meter. In this analysis, we will see if there is a predictable relationship between men's waist measurements and their BMIs.

We begin the analysis as we have done so often, using the **Fit Y by X** platform.

4. Select **Fit Y by X**. Cast **BMXBMI** as **Y** and **BMXWAIST** as **X** and click **OK.** The scatterplot contains points for the entire NHANES sample. Let's filter the rows to restrict the analysis to adult males.
5. Click the red triangle next to **Bivariate Fit** and select **Local Data Filter.**
6. While pressing the Ctrl key, highlight **RIAGENDR** and **RIDAGEYR**, and click the dark plus sign.
7. In the **Data Filter** (see Figure 15.2), make sure that the **Show** and **Include** options are checked.
8. Then click **Male** under **RIAGENDR** to include just the male subjects.
9. Finally, click the number 0 on the left of **RIDAGEYR** and replace it with 18. This sets the lower bound for **RIDAGEYR** to be just 18 years. We want to select any respondent who is a male age 18 or older.

**Figure 15.2: Selection Criteria for Males Age 18 and Older**




We have restricted the analysis to the 2,887 rows of male respondents who are 18 years of age and older. You might have noticed that the resulting graph has far fewer points and very clearly shows a positive linear trend, but the plot is difficult to decipher because there are so many points crowded together, with many of them plotted on top of each other. As very large data tables have become more common, we need strategies for making interpretable graphs with densely packed points. Do the following to make the graph more readable.

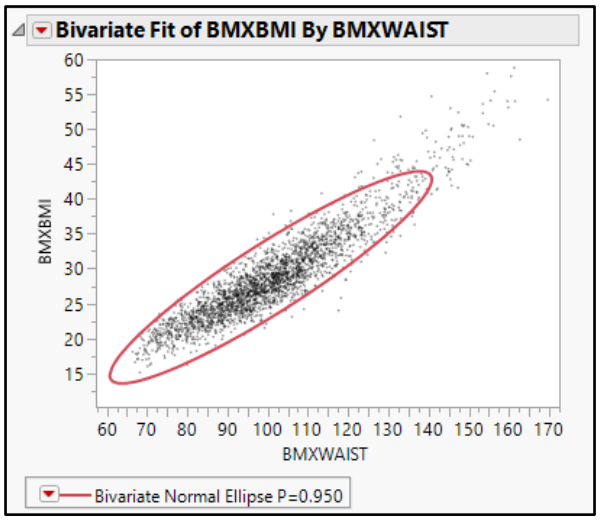
10. Right-click anywhere in the plot region, and select **Marker Size ► 0, Dot**. This makes each point much smaller, reducing the overlap.
11. Right-click again and select **Transparency**. Within the dialog box, set the transparency parameter to 0.4. This makes dot partially transparent, so that overplotted points appear darker than single points.

This graph (see Figure 15.3, which reflects the next two numbered steps as well) illustrates the first thing that we want to look for when planning to conduct a linear regression analysis—we see a general linear trend in the data. Think of stretching an elliptical elastic band around the cloud of points; that would result in a long and narrow ellipse lying at a slant, which would contain most, if not all, of the points. In fact, we can use JMP to overlay such an ellipse on the graph.

12. Click the red triangle next to **Bivariate Fit** and select **Density Ellipse ► 0.95**.

The resulting ellipse appears incomplete because of the default axis settings on our graph. We can customize the axes to show the entire ellipse using the grabber to shift the axes.

**Figure 15.3: A Linear Pattern of BMI Versus Waist**




13. Move the grabber tool near the origin on the vertical axis and slide upward until you see a hash mark below 15 appear on the Y axis. Do the same on the horizontal axis until the waist value of 60 cm appears on the X axis.

This graph is a typical candidate for linear regression analysis. Nearly all of the points lie all along the same sloped axis in the same pattern, with consistent scatter. Before running the regression, let's step back for a moment and consider the fundamental regression model.

## The Simple Regression Model

When we fit a line to a set of points, we do so with a model in mind and with a provisional idea about how we came to observe the points in our sample. The reasoning goes like this. We speculate or hypothesize that there is a linear relationship between Y and X such that whenever X increases by one unit (centimeters of waist circumference, in this case), then Y changes, on average, by a constant amount. For any specific individual, the observed value of Y could deviate from the general pattern.

Algebraically, the model looks like this:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

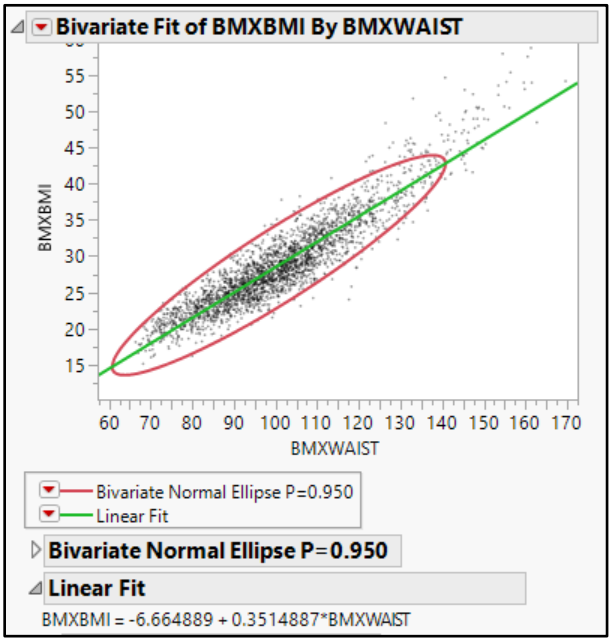
where $Y_i$ and $X_i$ are the observed values for one respondent, $\beta_0$ and $\beta_1$ are the intercept and slope of the underlying (but unknown) relationship, and $\varepsilon_i$ is the amount by which an individual's BMI departs from the usual pattern. We envision $\varepsilon_i$ as purely random noise. In short, we can express each observed value of $Y_i$ as partially reflecting the underlying linear pattern, and partially reflecting a random deviation from the pattern. Look again at Figure 15.3. Can you

visualize each point as lying in the vicinity of a line? Let's use JMP to estimate the location of such a line.

1. Click the red triangle next to **Bivariate Fit** and select **Fit Line**.

Now your results will look like Figure 15.4 below. We see a green *fitted line* that approximates the upward pattern of the points.

**Figure 15.4: Estimated Line of Best Fit**




Below the graph, we find the equation of that line:

$$\text{BMXBMI} = -6.664889 + 0.3514887 * \text{BMXWAIST}$$

The slope of this line describes how these two variables co-vary. If we imagine two groups of men whose waist circumferences differ by 1 centimeter, the group with the larger waists would average BMIs that are approximately 0.35 $kg/m^2$ higher. As we learned in Chapter 4, this equation summarizes the relationship among the points in this sample. Before learning about the inferences that we might draw from this, let's refine our understanding of the two chunks of the model: the linear relationship and the random deviations.

## Thinking About Linearity

If two variables have a linear relationship, their scatterplot forms a line or at least suggests a linear pattern. In this example, our variables have a *positive* relationship: as X increases, Y increases. In another case, the relationship might be negative, with Y decreasing as X increases.

But what does it mean to say that two variables have a *linear* relationship? What type of underlying dynamic generates a linear pattern of dots?

As noted earlier, linearity involves a constant change in Y each time X changes by one unit. Y might rise or fall, but the key feature of a linear relationship is that the shifts in Y do not accelerate or diminish at different levels of X. If we plan to generalize from our sample, it is important to ask if it is reasonable to expect Y to vary in this particular way as we move through the domain of realistically possible X values.

### Random Error

The regression model also posits that empirical observations tend to deviate from the linear pattern, and that the deviations are themselves a random variable. We will have considerably more to say about the random deviations in Chapter 16, but it is very useful at the outset to understand this aspect of the regression model.

Linear regression analysis does not demand that all points line up perfectly, or that the two continuous variables have a very close (or "strong") association. On the other hand, if groups of observations systematically depart from the general linear pattern, we should ask if the deviations are truly random, or if there is some other factor to consider as we untangle the relationship between Y and X.

## What Are We Assuming?

The preceding discussion outlines the conditions under which we can generalize using regression analysis. First, we need a logical or theoretical reason to anticipate that Y and X have a linear relationship. Second, the default method[1] that we use to estimate the line of best fit works reliably. We know that the method works reliably when the random errors, $\varepsilon_i$, satisfy four conditions:

- They are normally distributed.
- They have a mean value of 0.
- They have a constant variance, $\sigma^2$, regardless of the value of X.
- They are independent across observations.

At this early stage in the presentation of this technique, it might be difficult to grasp all the implications of these conditions. Start by understanding that the following might be red flags in a scatter plot with a fitted line:

- The points seem to bend or oscillate predictably around the line.
- There are a small number of outliers that stand well apart from the mass of the points.

- The points seem snugly concentrated near one end of the line but fan out toward the other end.
- There seem to be greater concentrations of points distant from the line, but not so many points concentrated near the line.

In this example, none of these trouble signs is present. In the next chapter, we will learn more about looking for problems with the important conditions for inference. For now, let's proceed assuming that the sample satisfies all of the conditions.

## Interpreting Regression Results

There are four major sections in the results panel for the linear fit (see Figure 15.5), three of which are fully disclosed by default. We have already seen the equation of the line of best fit and discussed its meaning. In this part of the chapter, we will discuss the three other sections in order.

**Figure 15.5: Regression Results**

**Linear Fit**

BMXBMI = -6.664889 + 0.3514887*BMXWAIST

**Summary of Fit**

| | |
|---|---|
| RSquare | 0.856221 |
| RSquare Adj | 0.856165 |
| Root Mean Square Error | 2.356141 |
| Mean of Response | 28.75412 |
| Observations (or Sum Wgts) | 2600 |

**Lack Of Fit**

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Ratio |
|---|---|---|---|---|
| Model | 1 | 85887.64 | 85887.6 | 15471.35 |
| Error | 2598 | 14422.54 | 5.6 | Prob > F |
| C. Total | 2599 | 100310.18 | | <.0001* |

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | -6.664889 | 0.28848 | -23.10 | <.0001* |
| BMXWAIST | 0.3514887 | 0.002826 | 124.38 | <.0001* |

### Summary of Fit

Under the heading **Summary of Fit**, we find five statistics that describe the fit between the data and the model.

- **RSquare** and **RSquare Adj** both summarize the strength of the linear relationship between the two continuous variables. The R-square statistics range between 0.0 and 1.0, where 1.0 is a perfect linear fit. Just as in Chapter 4, think of R square as the proportion of variation in Y that is associated with X. Here, both statistics are

approximately 0.86, suggesting that a male's waist measurement could be a very good predictor of his BMI.

- **Root Mean Square Error** (RMSE) is a measure of the dispersion of the points from the estimated line. Think of it as the sample standard deviation of the random noise term, $\varepsilon$. When points are tightly clustered near the line, this statistic is relatively small. When points are widely scattered from the line, the statistic is relatively large. Comparing the RMSE to the mean of the response variable (next statistic) is one way to assess its relative magnitude.
- **Mean of Response** is just the sample mean value of Y.
- **Observations** is the sample size. In this table, we have complete waist and BMI data for 2600 men. Although we selected 2,887 rows with the data filter, 287 of the rows were missing a value for one or both of the columns in the regression model.

## Lack of Fit

The next heading is **Lack of Fit**, but this panel is initially minimized in this case. Lack of fit tests typically are considered topics for more advanced statistics courses, so we only mention them here without further comment.

## Analysis of Variance

These ANOVA results should look familiar if you have just completed Chapter 14. In the context of regression, ANOVA gives us an overall test of significance for the regression model. In a one-way ANOVA, we hypothesized that the mean of a response variable was the same across several categories. In regression, we hypothesize that the mean of the response variable is the same regardless of X—that Y does not vary in tandem with X.

We read the table just as we did in the previous chapter, focusing on the F-ratio and the corresponding *P*-value. Here F is over 15.471 and the *P*-value is smaller than 0.0001. This probability is so small that it is highly unlikely that the computed F-ratio came about through the error associated with random sampling. We reject the null hypothesis that waist circumference and BMI are unrelated, and conclude that we have found a statistically significant relationship.

Not only can we say that the pattern describes the sample, we can say with confidence that the relationship generalizes to the entire population of males over age 17 in the United States.

## Parameter Estimates and t Tests

The final panel in the results provides the estimated intercept and slope of the regression line and the individual t tests for each. The slope and intercept are sometimes called the *coefficients* in the regression equation, and we treat them as the *parameters* of the linear regression model.

In Figure 15.6, we reproduce the parameter estimate panel, which contains five columns. The first two columns—**Term** and **Estimate**—are the estimated intercept and slope that we saw earlier in the equation of the regression line.

**Figure 15.6: Parameter Estimates**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | -6.664889 | 0.28848 | -23.10 | <.0001* |
| BMXWAIST | 0.3514887 | 0.002826 | 124.38 | <.0001* |

Because we are using a sample of the full population, our estimates are subject to sampling error. The **Std Error** column estimates the variability attributable to sampling. The **t Ratio** and **Prob>|t|** columns show the results of a two-sided test of the null hypothesis that a parameter is truly equal to 0.

> Why do we test the hypotheses that the intercept and slope equal zero? Think about what a zero slope represents. If X and Y are genuinely independent and unrelated, then changes in the value of X have no influence or bearing on the values of Y. In other words, the slope of a line of best fit for two such variables should be zero. For this reason, we always want to look closely at the significance test for the slope. Depending on the study and the meaning of the data, the test for the intercept may or may not have practical importance to us.

In a simple linear regression, the ANOVA and t test results for the slope will always lead to the same conclusion[2] about the hypothesized independence of the response and factor variables. Here, we find that our estimated slope of 0.351 kg/m$^2$ change in BMI per 1 cm increase in waist circumference is very convincingly different from 0: in fact, it's more than 124 standard errors away from 0. It's inconceivable that such an observed difference is the coincidental result of random sampling.

## Testing for a Slope Other Than Zero

In some investigations, we might begin with a theoretical model that specifies a value for the slope or the intercept. In that case, we come to the analysis with hypothesized values of either $\beta_0$ or $\beta_1$ or both, and we want to test those values. The **Fit Y by X** platform does not accommodate such significance tests, but the **Fit Model** platform does. We used **Fit Model** in the prior chapter to perform a two-way ANOVA. In this example, we will use it to test for a specific slope value other than 0.

We will illustrate with an example from the field of classical music, drawn from an article by Prof. Jesper Rydén of Uppsala University in Sweden (Rydén 2007). The article focuses on piano sonatas by Franz Joseph Haydn (1732–1809) and Wolfgang Amadeus Mozart (1756–1791) and investigates the idea that these two composers incorporated the *golden mean* within their compositions. A sonata is a form of instrumental music that follows a standard structure. In the first part, known formally as the exposition, the composer introduces a melodic theme in one key and then a second longer theme in a related but different key. After the exposition comes a second portion called development, which elaborates upon the basic melodies, developing them more fully, offering some variations, Finally, both themes recur in the recapitulation section, but this time, both tunes are in the same key as the opening of the piece.

Some music scholars believe that Haydn and Mozart strove for an aesthetically pleasing but asymmetric balance in the lengths of the melodic themes. More specifically, these scholars hypothesize that the composers might have structured their sonatas (deliberately or not) so that the relative lengths of the two themes approximated the golden mean. We will refer to the first, shorter theme as a and the second theme as b.

The golden mean (sometimes called the golden ratio), characterized and studied in the West at least since the ancient Greeks, refers to the division of a line into a shorter segment *a*, and a longer segment *b*, such that the ratio of *a:b* equals the ratio of *b:(a+b).* Equivalently,

$$\frac{a}{b} = \frac{b}{(a+b)} = \phi \approx 0.61803.$$

We have a data table called **Mozart** containing the lengths, in musical measures, of the shorter and longer themes of 29 Mozart sonatas. If, in fact, Mozart was aiming for the golden ratio in these compositions, then we should find a linear trend in the data. Moreover, it should be characterized by this line:

$$a = 0 + 0.61803(b)$$

So, we will want to test the hypothesis that $\beta_1 = 0.61803$ rather than 0.

1. Open the data table called **Mozart**.
2. Select **Analyze ► Fit Model**. Select **Parta** as **Y**, then add **Partb** as the only model effect, and run the model.

Both the graph and the **Summary of Fit** indicate a strong linear relationship between the two parts of these sonatas. Figure 15.7 shows the parameter estimates panel from the results.

**Figure 15.7: Estimates for Mozart Data**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 1.3596328 | 2.882715 | 0.47 | 0.6410 |
| Partb | 0.6259842 | 0.030851 | 20.29 | <.0001* |

Rounding the estimates slightly we can write an estimated line as *Parta = 1.3596 + 0.626(Partb)*. On its face, this does not seem to match the proposed equation above. However, let's look at the t tests. The estimated intercept is not significantly different from 0, so we cannot conclude that the intercept is other than 0. The hypothesized intercept of 0 is still credible.

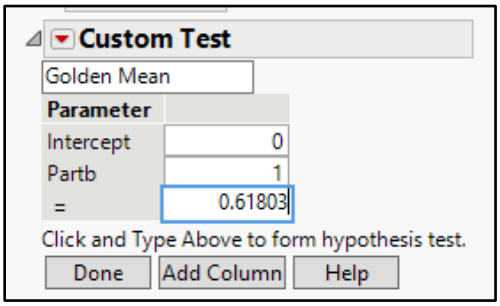
Now look at the results for the slope. The estimated slope is 0.6259842, and its standard error is 0.030851. The reported *t*-ratio of about 20 standard errors implicitly compares the estimated slope to a hypothesized value of 0. To compare it to a different hypothesized value, we will want to compute the following ratio:

$$\frac{estimate - hypothesized}{std.error} = \frac{0.6259842 - 0.61803}{0.030851} = 0.2578$$

We can have JMP compute this ratio and its corresponding *P*-value as follows:

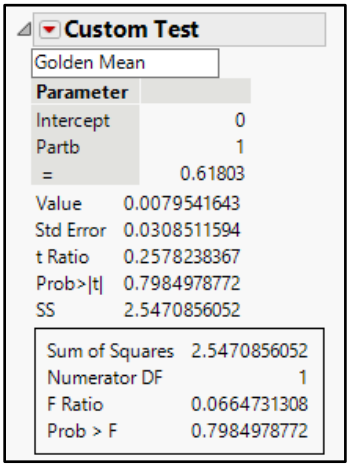
3. Click the red triangle next to **Response Parta** and select **Estimates ► Custom Test**. Scroll to the bottom of the results report where you will see a panel like the one shown in Figure 15.8.
4. The upper white rectangle is an editable field for adding a title; type **Golden Mean** in the box.
5. In the box next to **Partb**, change the **0** to a **1** to indicate that we want to test the coefficient of **Partb**.
6. Finally, enter the hypothesized value of the golden mean, **.61803** in the box next to **=**, and click the **Done** button.

**Figure 15.8: Specifying the Column and Hypothesized Value**




The **Custom Test** panel now becomes a results panel, presenting both a t test and an F test, as shown in Figure 15.9. As our earlier calculation showed, the estimated slope is less than 0.26 standard errors from the hypothesized value, which is very close. Based on the large *P*-value of 0.798, we fail to reject the null hypothesis that the slope equals the golden mean.

**Figure 15.9: Custom Test Results**




In other words, the golden mean theory is credible. As always, we cannot prove a null hypothesis, so this analysis does not definitively establish that Mozart's sonatas conform to the

golden mean. This is an important distinction in the logic of statistical testing—our tests can discredit a null hypothesis with a high degree of confidence, but we cannot confirm a null hypothesis. What we can say is that we have put a hypothesis to the test, and it is still plausible.

Before ending this session, be sure to save your project. If prompted to save unsaved documents within the project, save them all.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* Return to the **NHANES 2016** data table.
    a. Using the Data Filter, show and include female respondents ages 18 and older. Perform a regression analysis for BMI and waist circumference for adult women and report your findings and conclusions.
    b. Is waist measurement a better predictor (in other words, a better fit) of BMI for men or for women?
    c. Perform one additional regression analysis, this time looking only at respondents under the age of 17. Summarize your findings.

2. *Scenario:* High blood pressure continues to be a leading health problem in the United States. In this problem, continue to use the **NHANES 2016** data table. For this analysis, we will focus on just the following variables:
    - **RIAGENDR:** Respondent's gender
    - **RIDAGEYR**: Respondent's age in years
    - **BMXWT**: Respondent's weight in kilograms
    - **BPXPLS**: Respondent's resting pulse rate
    - **BPXSY1**: Respondent's systolic blood pressure ("top" number in BP reading)
    - **BPXD1**: Respondent's diastolic blood pressure ("bottom" number in BP reading)

    a. Investigate a possible linear relationship of systolic blood pressure versus age. What, specifically, tends to happen to blood pressure as people age? Would you say there is a strong linear relationship?
    b. Perform a regression analysis of systolic and diastolic blood pressure. Explain fully what you have found.
    c. Create a scatterplot of systolic blood pressure and pulse rate. One might suspect that higher pulse rate is associated with higher blood pressure. Does the analysis bear out this suspicion?

3. *Scenario:* We will continue to examine the World Development Indicators data in **BirthRate 2017**. We will broaden our analysis to work with other variables in that file:
   - **MortUnder5**: Deaths, children under 5 years per 1,000 live births
   - **MortInfant**: Deaths, infants per 1,000 live births

   a. Create a scatterplot for **MortUnder5** and **MortInfant**. Report the equation of the fitted line and the R-square value and explain what you have found.
4. *Scenario:* How do the prices of used cars vary according to the mileage of the cars? Our data table **Used Cars** contains observational data about the listed prices of three popular compact car models in three different metropolitan areas in the U.S. All of the cars are two years old.
   a. Create a scatterplot of price versus mileage. Report the equation of the fitted line and the R-square value and explain what you have found.
5. *Scenario*: Stock market analysts are always on the lookout for profitable opportunities and for signs of weakness in publicly traded stocks. Market analysts make extensive use of regression models in their work, and one of the simplest ones is known as the *random* (or *drunkard's*) *walk* model. Simply put, the model hypothesizes that over a relatively short period of time, the price of a share of stock is a random deviation from its price on the prior day. If $Y_t$ represents the price at time $t$, then $Y_t = Y_{t-1} + \varepsilon$. In this problem, you will fit a random walk model to daily closing prices for McDonald's Corporation for the first six months of 2019 and decide how well the random walk model fits. The data table is called **MCD**.
   a. Create a scatterplot with the daily closing price on the vertical axis and the prior day's closing price on the horizontal. Comment on what you see in this graph.
   b. Fit a line to the scatterplot and test the credibility of the random walk model. Report on your findings.
6. *Scenario*: Franz Joseph Haydn was a successful and well-established composer when the young Mozart burst upon the cultural scene. Haydn wrote more than twice as many piano sonatas as Mozart. Use the data table **Haydn** to perform a parallel analysis to the one we did for Mozart.
   a. Report fully on your findings from a regression analysis of **Parta** versus **Partb**.
   b. How does the fit of this model compare to the fit using the data from Mozart?

7. *Scenario*: Throughout the animal kingdom, animals require sleep, and there is extensive variation in the number of hours in a day that different animals sleep. The data table called **Sleeping Animals** contains information for more than 60 mammalian species, including the average number of hours per day of total sleep. This will be the response column in this problem.
    a. Estimate a linear regression model using gestation as the factor. Gestation is the mean number of days that females of these species carry their young before giving birth. Report on your results and comment on the extent to which gestational period is a good predictor of sleep hours.
    b. Now perform a similar analysis using brain weight as the factor. Report fully on your results and comment on the potential usefulness of this model.

8. *Scenario*: For many years, it has been understood that tobacco use leads to health problems related to the heart and lungs. The **Tobacco Use** data table contains recent data about the prevalence of tobacco use and of certain diseases around the world.
    a. Using cancer mortality (**CancerMort**) as the response variable and the prevalence of tobacco use in both sexes (**TobaccoUse**), run a regression analysis to decide whether total tobacco use in a country is a predictor of the number of deaths from cancer annually in that country.
    b. Using cardiovascular mortality (**CVMort**) as the response variable and the prevalence of tobacco use in both sexes (**TobaccoUse**), run a regression analysis to decide whether total tobacco use in a country is a predictor of the number of deaths from cardiovascular disease annually in that country.
    c. Review your findings in the earlier two parts. In this example, we are using aggregated data from entire nations rather than individual data about individual patients. Can you think of any ways in which this fact could explain the somewhat surprising results?
9. *Scenario*: In Chapter 2, our first illustration of experimental data involved a study of the compressive strength of concrete. In this scenario, we look at a set of observations all taken at 28 days (4 weeks) after the concrete was initially formulated. The data table is **Concrete28**. The response variable is the **Compressive Strength** column, and we will examine the relationship between that variable and two candidate factor variables.
    a. Use **Cement** as the factor and run a regression. Report on your findings in detail. Explain what this slope tells you about the impact of adding more cement to a concrete mixture.
    b. Use **Water** as the factor and run a regression. Report on your findings in detail. Explain what this slope tells you about the impact of adding more water to a concrete mixture.

10. *Scenario*: Prof. Frank Anscombe of Yale University created an artificial data set to illustrate the hazards of applying linear regression analysis without looking at a scatterplot (Anscombe 1973). His work has been very influential, and JMP includes his illustration among the sample data tables packaged with the software. You'll find **Anscombe** both in this book's data tables and in the JMP sample data tables. Open it now.
    a. In the upper left panel of the data table, you will see a red triangle next to the words **The Quartet**. Click the triangle, and select **Run Script**. This produces four regression analyses corresponding to four pairs of response and predictor variables. Examine the results closely, and write a brief response comparing the regressions. What do you conclude about this quartet of models?
    b. Now return to the results and click the red triangle next to **Bivariate Fit of Y1 By X1**; select **Show Points** and re-interpret this regression in the context of the revised scatterplot.
    c. Now reveal the points in the other three graphs. Is the linear model equally appropriate in all four cases?

11. *Scenario*: Many cities in the U.S. have active used car markets. Typically, the asking price for a used car varies by model, age, mileage, and features. The data table called **Used Cars** contains asking prices (**Price**) and mileage (**Miles**) for three popular budget models; all cars were two years old at the time the data were gathered, and we have data from three U.S. metropolitan areas. All prices are in dollars. In this analysis, **Price** is the response and **Miles** is the factor.
    a. Because the car model is an important consideration, we will begin by analyzing the data for one model: the Civic EX. Use a **Local Data Filter** to isolate the Civic EX data for analysis. Run a regression. How much does the asking price decline, on average, per mile driven? What would be a mean asking price for a two-year-old Civic EX that had never been driven? Comment on the statistical significance and goodness-of-fit of this model.
    b. Repeat the previous step using the Corolla LE data.
    c. Repeat one more time using the PT Cruiser data.
    d. Finally, compare the three models. For which set of data does the model fit best? Explain your thinking. For which car model are you most confident about the estimated slope?

12. *Scenario:* We will return to the World Development Indicators data in **WDI**. In this scenario, we will investigate the relationship between access to improved sanitation (the percent of the population with access to sewers and the like) and life expectancy. The response column is **life_exp** and the factor is **sani_acc**.
    a. Use the **Local Data Filter** to **Show** and **Include** only the observations for the **Year 2015**, and the **Latin America & Caribbean Region** nations. Describe the relationship that you observe between access to improved sanitation and life expectancy.
    b. Repeat the analysis for **East Asia & Pacific** countries in 2010.
    c. Repeat the same analysis for the countries located in **Sub-Saharan Africa**.
    d. How do the three regression models compare? What might explain the differences in the models?

13. *Scenario:* The data table called **USA Counties** contains a wide variety of measures for every county in the United States based on the 2010 U.S. Census.
    a. Run a regression casting **sales_per_capita** (retail sales dollars per person, 2007) as **Y** and **per_capita_income** as **X**. Write a short paragraph explaining why county-wide retail sales might vary with per capita income, and report on the strengths and weaknesses of this regression model.

---

[1] Like all statistical software, JMP uses a default method to line-fitting that is known as *ordinary least squares estimation*, or *OLS.* A full discussion of OLS is well beyond the scope of this book, but it's worth noting that these assumptions refer to OLS in particular, not to regression in general.

[2] They will have identical *P*-values and the F-ratio will be the square of the *t* ratio.

# Chapter 16: Residuals Analysis and Estimation


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## Overview

Regression analysis is a powerful and flexible technique that we can reliably apply under certain conditions. Given a data table with continuous X-Y pairs, we can always use the ordinary least squares (OLS) method to fit a line and generate results like those we saw in Chapter 15. However, if we want to draw general conclusions or use an estimated regression equation to estimate, predict, or project Y values, it is critical that we decide whether our sample data satisfy the conditions underlying OLS estimation. This chapter explains how to verify whether we are on solid ground for inference with a particular regression analysis, then covers three types of inference using a simple linear regression. The techniques of this chapter are also relevant in later chapters.

## Conditions for Least Squares Estimation

Recall that the linear regression model is

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where $Y_i$ and $X_i$, are the observed values for one case, $\beta_0$ and $\beta_1$ are the intercept and slope of the underlying relationship, and $\varepsilon_i$ is the amount by which an individual response deviates from the line.

OLS will generate the equation of a line approximating $\beta_0 + \beta_1 X_i$ so that the first condition is that the relationship between X and Y really is linear. Additionally, the model treats $\varepsilon_i$ as a normal random variable whose mean is zero, whose variance is the same all along the line, and whose observations are independent of each other. Thus, we have four other conditions to verify before using our OLS results. Also, the logic of the simple linear model is that X is the only variable that accounts for a substantial amount of the observed variation in Y.

In this chapter, we will first learn techniques to evaluate the degree to which a sample satisfies the OLS conditions, and then learn to put a regression line to work when the conditions permit. We will consider all of the conditions in sequence and briefly explain criteria for judging whether they are satisfied in a data table, as well as pointing out how to deal with samples that do not satisfy the conditions.

## Residuals Analysis

Several of the conditions just cited refer directly to the random error term in the model, $\varepsilon_i$. Because we cannot know the precise location of the theoretical regression line, we cannot know the precise values of the $\varepsilon_i$s. The random error term values are the difference between the observed values of Y and the position of the theoretical line. Although we don't know where the theoretical line is, we do have a fitted line, and we will use the differences between each observed $Y_i$ and the fitted line to approximate the $\varepsilon_i$ values.

These differences are called *residuals*. Again, a residual is the difference between an observed value of Y and the fitted value of Y corresponding to the observed value of X. In a scatterplot, it is the vertical distance between an observed point and the fitted line. If the point lies above the line, the residual is positive, and if the point lies below the line, the residual is negative. By examining the residuals after fitting a line, we can judge how well the sample satisfies the conditions. To illustrate the steps of a residual analysis, we will first look at the **NHANES** data again and look at the residuals from the regression that we ran in Chapter 15.

Because we saved the project in Chapter 15, we can now re-open that work and continue the analysis without having to repeat the steps to run the regression.

1. Open the project called **Chap_15**. **File ► Save Project As**. Before moving on, save this project as **Chap_16**. This will preserve the entire project from the prior chapter, and new work will be preserved in this project.
2. Locate the tab titled NHANES 2016 - Fit Y by X of BMXBMI by BMXWAIST. This contains the local data filter and regression analysis from Chapter 15. This should look familiar, but you might want to review Figures 15.4 and 15.5.
3. Click the red triangle below the graph, next to **Linear Fit,** and select **Plot Residuals.**
4. Click the same red triangle again and click **Save Residuals**.

The **Plot Residuals** option creates five graphs at the bottom of the report. These graphs help us judge the validity of the OLS conditions for inference. We will start with the linearity condition.

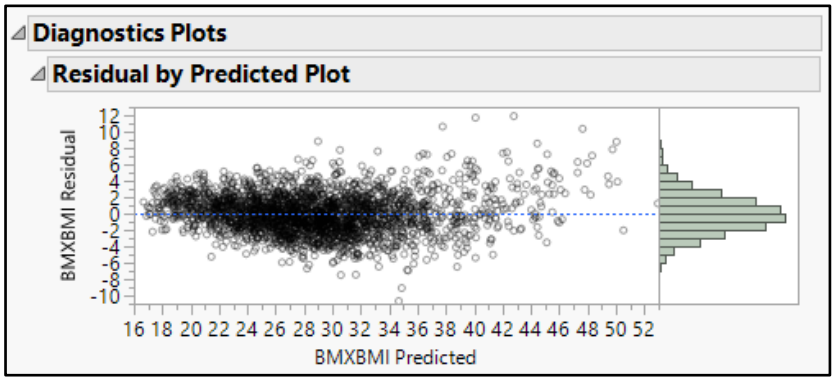
> Like the original scatterplot that we saw in Chapter 15, these plots contain 2,600 points densely packed into a small space. For the purposes of residual analysis, we can leave them as is, or modify them slightly for clarity. As we did earlier, you can adjust the point size and transparency to make the plot easier to read.
>
> Alternatively, right-click on any graph region and select **Customize**. Choose **Marker** and change the marker shape to an open circle and set **Transparency** to **0.4**. The figures below reflect this approach with small marker sizes.

## Linearity

In the prior chapter, we discussed the concept of linearity, noting that if a relationship is linear, then Y changes by a constant each time X changes by one unit. Look at the **Bivariate Fit** report with the fitted line. The scatterplot and fitted line in this example do suggest a generally linear pattern.

Sometimes mild curvature in the points is difficult to detect in the initial bivariate fit scatterplot, and we need a way to magnify the image to see potential non-linear patterns. One simple way to do this is to look at a plot of residuals versus fitted values; scroll down in the report window until you find the **Diagnostics Plots** heading. The first graph (shown in Figure 16.1) is the plot that we want.

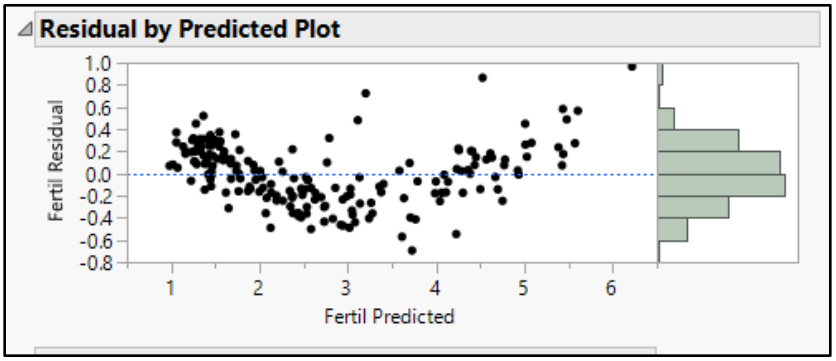
**Figure 16.1: Residuals Versus Predicted Values Plot**




In this graph the fitted (predicted) values of BMXBMI are on the horizontal and the residuals are on the vertical axis. For the most part in this case, we see that the points are scattered evenly above and below the horizontal line at 0, without any sign of curvature. If the relationship were non-linear, we would see the points bending upward or downward. For example, think back to the first scatterplot we made in Chapter 15. (Refer back to Figure 15.1.) The plot of fertility rate versus birth rate from the **Birthrate 2017** data showed a very slight upward curve. Figure 16.2 is the residual plot from that regression, and it shows such a curvilinear pattern. Notice that on the

left side of the graph, corresponding to predicted fertility values from 0 to 2, the large majority of residuals are positive. Between 2 and 5, most of the residuals are negative, and then for predicted fertility rates greater than 5, the residuals are positive. Taken together, the overall trend is a U-shaped curve. As it happens, this plot also shows non-constant variance, and we will discuss that issue in a later section.

**Figure 16.2: Residual by Predicted Plot Showing Non-linearity**




## Curvature

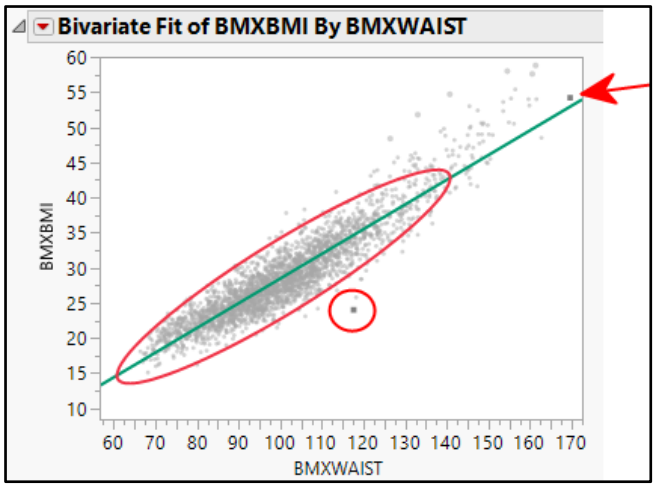
What do we do when we find a curved, rather than a straight line, pattern? We have several options, which are discussed at length in Chapter 19. For the moment, the most important thing to recognize is that we should not interpret or apply the results of a simple linear regression if the residual plot indicates curvature. Without a linear pattern, the results will be of very limited use and could lead to very poor decisions.

## Influential Observations

Another issue related to linearity is that of influential observations. These are points that are outliers in either the X or the Y directions. Such points tend to bias the estimate of the intercept, the slope, or both.

For example, look back at the scatter plot with the fitted line on your screen (with slightly enlarged points here as Figure 16.3). Notice the highlighted points in the upper right corner of the graph and look closely at the point marked with an arrow. Point your cursor at this dot, and the hover label indicates that it represents a man whose waist circumference was 169.6 cm, or 66.8 inches, which was the largest in the sample. His BMI of 54.2 kg/m$^2$—the eighth highest in the sample—lies very close to the line. The OLS method is sensitive to such extreme X values, and consequently the line nearly passes through the point.

**Figure 16.3: Some Influential Observations**




As another example, find the point that is circled in Figure 16.3. This represents a man whose waist measurement is in the fourth quartile and BMI is in the first quartile—not an outlier in either dimension. However, this point is a bivariate outlier because it departs from the overall linear pattern of the points. His BMI is low considering his waistline.

With outliers such as these, we can take one or more of the following steps, even in the context of an introductory statistics course.

- If possible, verify the accuracy of the data. Are the measurements correct? Is there a typographical error or obvious mistake in a respondent's answer?
- Verify that the observation really comes from the target population. For example, in a survey of a firm's regular customers, is this respondent a regular customer?
- Temporarily exclude the observation from the analysis and see how much the slope, intercept, or goodness-of-fit measures change. We can do this as follows:

1. Click the red triangle next to **Bivariate Fit of BMXBMI by BMXWAIST** and choose **Redo ► Automatic Recalc**. If there is no checkmark next to **Automatic Recalc**, click it.
2. Now click the scatterplot and click once on the point that is shown circled in Figure 16.3 above. You will see that it is the observation in row 3,140, and it appears darkened. Clicking on the point selects it.
3. Now right-click and choose **Row Exclude.**[1] This excludes the point from the regression computations, but leaves it appearing in the graph.

When we omit this point from consideration, the estimated regression line is

$$\text{BMXBMI} = -6.686835 + 0.3517473*\text{BMXWAIST}$$

as compared to this line when we use all 2600 observations:

BMXBMI = -6.664889 + 0.3514887*BMXWAIST

In a large sample, omitting one outlier will not have an enormous impact, but when we eliminate this outlier, the intercept decreases by about 0.02 $kg/m^2$, but the slope barely changes. A point near the mass of the data with a large residual has little effect on the slope (or on conclusions about whether the two variables are related) but can bias the estimate of the intercept. Omitting that observation very slightly improves the goodness of fit measure R square from 0.856 to 0.857. This makes sense because we are dropping a point that does not fit the general pattern.

What about that point in the far upper-right corner, highlighted with an arrow (Row 1,943)? When we exclude it, the estimated equation becomes

BMXBMI = -6.652788 + 0.3513638*BMXWAIST

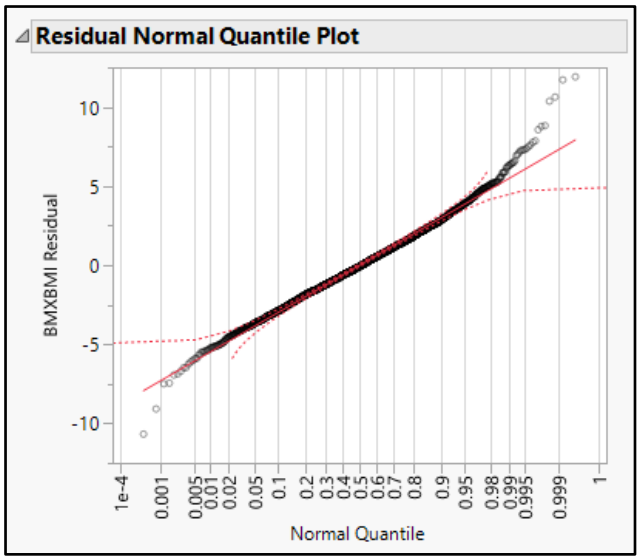
This changes the equation very little, although R square falls a bit farther to 0.855. These points are not especially influential, but in smaller samples, we need to watch for influential points.

## Normality

The set of standard residual plots includes two ways to ask whether the residuals follow an approximate normal distribution. The first is a histogram appearing in the margin of the first graph of residuals by predicted values. As shown in Figure 16.1, these residuals form a mound-shaped, symmetric pattern.

The last graph in the **Bivariate Fit** report is a Normal Quantile Plot. (See Figure 16.4.) Though the residuals do not track perfectly along the red diagonal line, this graph provides no warning signs of non-normality. Here we are looking for any red flag that would suggest that the residuals are grossly non-normal. We want to be able to continue to interpret the results and would stop for corrective measures if the residuals were heavily skewed or multimodal.

**Figure 16.4: Normal Quantile Plot for the BMI Residuals**




What happens when residuals do *not* follow a normal distribution? What types of corrective measures can we take? Generally speaking, non-normal residuals might indicate that there are other factors in addition to X that should be considered. In other words, there could be a lurking factor that accounts for the non-random residuals. Sometimes it is helpful to transform the response variable or the factor using logarithms. In Chapter 18, we will see how we can account for additional factors and use transformations.

## Constant Variance

Look back at Figures 16.1 and 16.2. In an ideal illustration, we would see the residuals scattered randomly above and below the horizontal zero line. Figure 16.1, even though it's not a perfect illustration, comes closer to the ideal than Figure 16.2. In Figure 16.2, not only do the residuals curve, but they increase in variability as you view the graph from left to right. On the left side, they are tightly packed together, and as we slide to the right, they drift farther and farther from the center line. The variability of the residuals is not constant, but rather it appears to grow predictably from left to right.

The constant variance assumption goes by a few other names as well, such as *homogeneity of variance* or *homoscedasticity.* Valid inference from a regression analysis depends quite strongly on this condition. When we have non-constant variance (or, you guessed it, *heterogeneity of variance* or *heteroscedasticity*) we should refrain from drawing inferences or using the regression model to estimate values of *Y.* Essentially, this is because *heteroscedasticity* makes the reported standard errors unreliable, which in turn biases the *t*- and *F*-ratios.

What to do? Sometimes a variable transformation helps. In other cases, multiple regression (Chapter 18) can ameliorate the situation. In other instances, we turn to more robust regression techniques that do not use the standard OLS technique. These techniques go beyond the topics typically covered in an introductory course.

## Independence

The final condition for OLS is that the random errors are independent of each other. This is quite a common problem with time series data, but much less so with a cross-sectional sample like NHANES. Much like the other conditions, we check for warning signs of severe non-independence with an inclination to continue the analysis unless there are compelling reasons to stop.

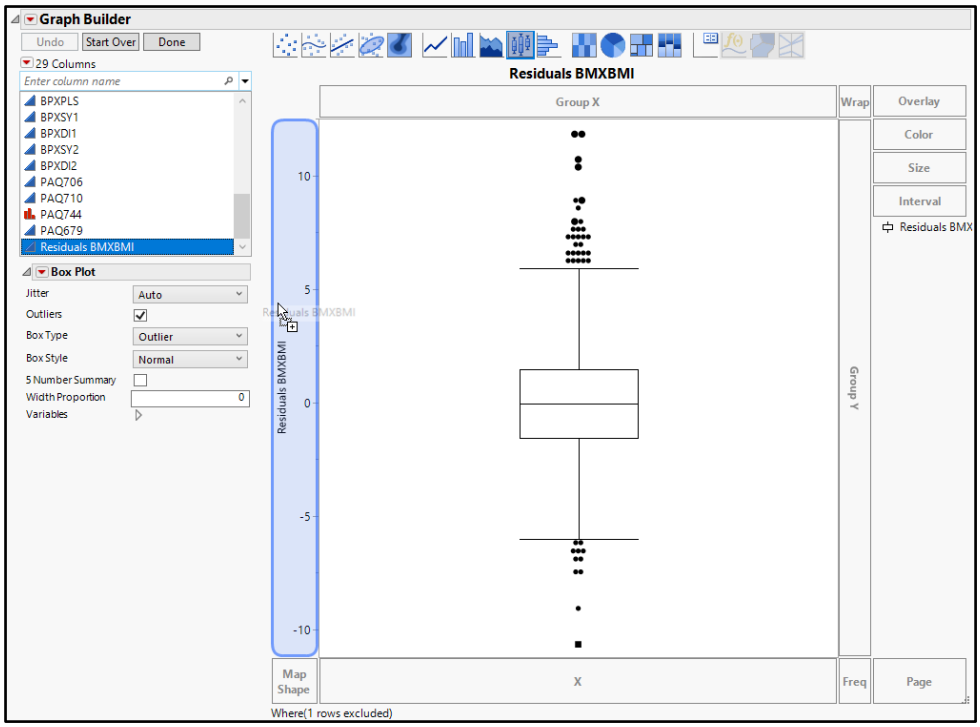
In regression, the assumption of independent errors means that the sign and magnitude of one error is independent of its predecessor. Using the residuals as proxies for the random errors, we will reason that if the residuals are independent of one another, then when we graph them in sequence, we should find no regular patterns of consecutive positive or negative residuals.

Within the standard set of **Diagnostics Plots**, there is a **Residual by Row Plot**. This is the graph that we want. Though if the sample is large, it will be easier to detect consecutive clusters of positive and negative residuals if the points in the graphs are connected in sequence. We can make our own version of the graph to better serve the purpose. This is the reason for saving the residuals earlier. The result of that request is a new data table column called **Residuals BMXBMI**.

To illustrate this method, we will use both our current example as well as some time series data. In both cases, we want to graph the residuals in sequence of observation, looking for repeated patterns of positive and negative residuals. There are multiple ways to accomplish this; we will use the **Overlay Plot** here because it enables us easily to connect residuals with lines to clearly visualize their sequence.

1. Select **Graph ► Graph Builder**. As shown in Figure 16.5, drag **Residuals BMXBMI** to the **Y** drop zone and **SEQN** to the **X** drop zone.

**Figure 16.5: Starting to Plot Residuals in Sequence using Graph Builder**

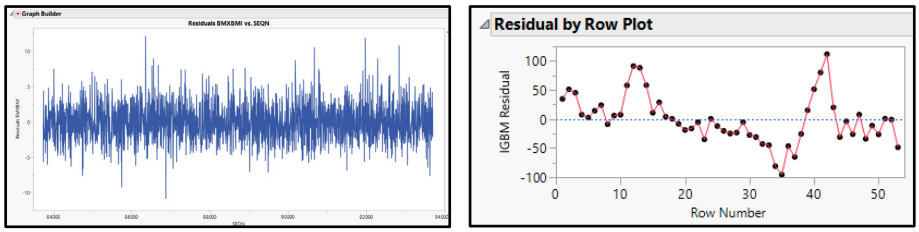



2. In the menu bar, click the Line button  to create the graph shown in the left portion of Figure 16.6.

The graph of the BMI residuals has a jagged, irregular pattern of ups and downs, indicating independence. The sign and magnitude of one residual does little to predict the next residual in order.

In contrast, look at the overlay plot on the right in Figure 16.6, based on data from the **Stock Index Weekly** data table. The residuals come from a regression using the weekly closing index value for the Madrid Stock Exchange as the response variable, and the index from the London Stock Exchange as the factor. Here we see a group of positive values, followed by a group of negatives, then more positives. The line oscillates in a wave, suggesting that there is a non-random pattern to the weekly residual values.

**Figure 16.6: Two Sets of Residuals in Sequence**




# Estimation

Once we have confirmed that our sample satisfies the conditions for inference, we can use the results in three estimation contexts. This section explains the three common ways that regression results can be used and introduces ways to use JMP in all three contexts.

- In some studies, the goal is to estimate the magnitude and direction of the linear relationship between X and Y, or the magnitude of the intercept, or both.
- In some situations, the goal is to estimate the *mean value of Y* corresponding to (or conditioned on) a given value of X.
- In some situations, the goal is to estimate an *individual value of Y* corresponding to a given value of X.

Earlier, we estimated this regression line based on the 2016 probabilistic sample:

BMXBMI = -6.664889 + 0.3514887*BMXWAIST

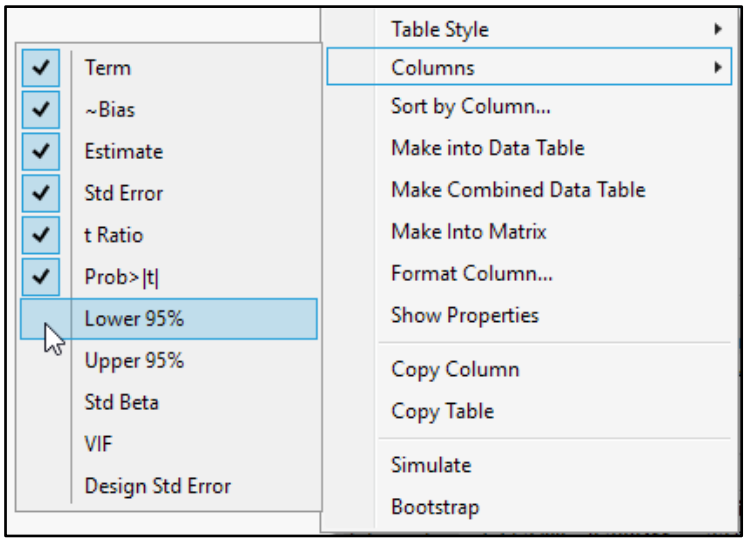
Note that our estimates of the intercept and slope depend on the measurements of the individuals in the 2016 sample. Another group of people would yield their own measurements, and hence our estimates would change. To put it another way, the parameter estimates in the equation are estimated with some degree of sampling error. The sampling error also cascades through any calculations that we make using the estimated line. For example, if we want to estimate BMI for men whose waistlines are 77 cm (about 30 inches), that calculation is necessarily affected by the slope and intercept estimates, which in turn depend on the sample that we work with.

## Confidence Intervals for Parameters

Rather than settle for point estimates of $\beta_0$ and $\beta_1$, we can look at confidence intervals for both. By default, JMP reports 95% confidence intervals, and we interpret these as we would any confidence interval. Confidence intervals enable us to more fully express an estimate and the size of potential sampling error.

1. Return to the **Fit Y by X** report, hover the cursor over the table of parameter estimates, and right-click. This brings up a menu, as shown in Figure 16.7.

**Figure 16.7: Requesting Confidence Intervals for Parameter Estimates**




Remember: In JMP, right-click menus are context sensitive. If you right-click over the title **Parameter Estimates**, you will see a different menu than if you right-click over the table below that title. Make sure that the cursor is over the table before right-clicking.

2. Select **Columns ► Lower 95%.**
3. Right-click once again and select **Columns ► Upper 95%.**

The revised table of parameter estimates should now look like Figure 16.8 below. On the far right, there are two additional columns representing the upper and lower bounds of 95% confidence intervals for the intercept and slope.

**Figure 16.8: Confidence Intervals for Parameters**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| | Lower 95% | Upper 95% |
|---|---|---|---|---|---|---|
| Intercept | -6.652788 | 0.289411 | -22.99 | <.0001* | -7.220287 | -6.085289 |
| BMXWAIST | 0.3513638 | 0.002836 | 123.90 | <.0001* | 0.3458029 | 0.3569247 |

How do we interpret the intervals for the slope and intercept? Consider the slope. Based on this sample of individuals, this regression provides an estimate of the amount by which BMI varies for a 1-cm increase in waist circumference among all adult males. The point estimate is, most likely, inaccurate. That is, we don't really know that the marginal increase in BMI is 0.3514 per centimeter of waist circumference. It is more accurate and genuine to say that we are 95% confident that the slope is between approximately 0.346 and 0.357.

## Confidence Intervals for Y|X

As noted above, we can use the estimated equation to compute one or more *predicted* (or *fitted*) values of Y corresponding to specific values of X. Such an estimate of Y is sometimes called *Y-hat,* denoted $\hat{Y}$. Computing a *Y-hat* value is a simple matter of substituting a specific value for X and then solving the regression equation. If we want to build an interval around a fitted value of Y, we first need to look to the context of the estimate.

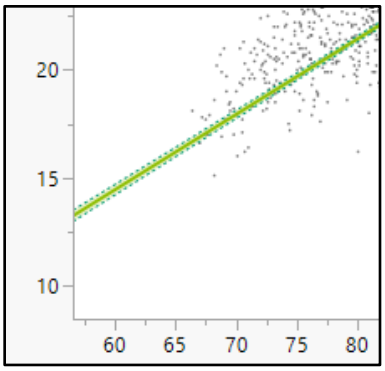
A *confidence interval* for Y|X estimates the mean value of Y given the value of X. In our current model of BMI, for example, we might want to estimate the mean BMI among men with 77 cm waists. The point estimate would be

$$\text{BMXBMI} = -6.664889 + 0.3514887 * 77 \approx 20.4 \text{ kg/m}^2$$

This value, which is the location of the estimated line at X = 77, becomes the center of the confidence interval, and then JMP computes an appropriate margin of error to form the interval. In the next chapter, we will discuss this technique in greater depth. As an introductory illustration, we will see how JMP constructs a *confidence band* along the entire length of the estimated line.

1. In the **Fit Y by X** report, click the red triangle next to **Bivariate Normal Ellipse** and click **Remove Fit.**
2. Click the red triangle next to **Linear Fit** and select **Confid Curves Fit**. Repeat to select **Confid Shaded Fit.**
3. Enlarge the graph by clicking and dragging the lower right corner downward and rightward. This will help you see the confidence bands. Figure 16.9 shows an enlarged detail of the graph, showing the fitted line and confidence region near BMXWAIST = 77.

**Figure 16.9: Fitted Line Plot with Confidence Bands**




In addition to visualizing the fitted values and confidence limits, we can also save the computed values for each observed value of X.

4. Click the red triangle next to **Linear Fit** and select **Save Predicteds**. Repeat to select **Mean Confidence Limit Formula**.

Now look at the **NHANES 2016** data table. There you will find three new columns containing predicted (fitted) values for BMI as well as 95% upper and lower confidence limits.

NOTE: In this final step, we illustrate the procedure for saving estimated values. You should notice in the data table that most rows contain values, even though we estimated the equation using only male respondents. The local data filter was instrumental in estimating the slope and intercept for the linear model, but the calculation of fitted values and confidence intervals is unaffected by the local filter.

## Prediction Intervals for Y|X

Finally, we consider the situation in which one might want to estimate or predict a single observation of Y given a value of X. Rather than asking about the mean BMI for all men with a 77-centimeter waist, we are asking about one man in particular. Conceptually, the procedure is the same as the prior example, but it differs in one critical detail. The magnitude of the sampling error is greater when predicting an individual value than when estimating the mean value of Y.

1. Return to the **Fit Y by X** report. Just as before, click the red triangle next to **Linear Fit**.
2. Select **Confid Curves Indiv** and then repeat to select **Confid Shaded Indiv**.
3. Finally, also select **Indiv Confidence Limit Formula**. You do not need to resave the fitted values because they will be the same as for the confidence interval estimate.

Now look at your scatterplot (not shown here). You should see a new shaded region bounded by *prediction bands,* which are much wider than the confidence bands from Figure 16.9. Both shaded regions are centered along the line of best fit, but the prediction band reflects the substantially greater uncertainty attending the prediction of an individual value rather than the mean of values.

# Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions. Notice that these problems are continuations of the corresponding scenarios and questions posed at the end of Chapter 15. Each problem begins with a residual analysis to assess the conditions for inference.

1. *Scenario:* Return to the Bivariate Fit of BMXBMI By BMXWAIST report.
    a. Using the local data filter again, show and include adult females 18 and older. Perform a regression analysis for BMI and waist circumference for adult women, evaluate the OLS conditions for inference, and report your findings and conclusions.
    b. If you can safely use this model, provide the 95% confidence interval for the slope and explain what it tells us about BMI for adult women.
    c. If you can safely use this model, save the predicted values and mean confidence limits. Then look at Row 21 in the data table; this respondent was a 30-year-old woman with a waist circumference of 90.7 cm. Use the new columns of the data table to report an approximate 95% confidence interval for the mean BMI for women whose waist measurements are 90.7 cm.

2. *Scenario:* High blood pressure continues to be a leading health problem in the United States. In this problem, continue to use the **NHANES 2016** table. For this analysis, we will focus on just the following variables:
    - **RIAGENDR:** Respondent's gender
    - **RIDAGEYR**: Respondent's age in years
    - **BMXWT: Respondent's weight in kilograms**
    - **BPXPLS**: Respondent's resting pulse rate
    - **BPXSY1**: Respondent's systolic blood pressure ("top" number in BP reading)
    - **BPXD1**: Respondent's diastolic blood pressure ("bottom" number in BP reading)
        a. Perform a regression analysis with systolic BP as the response and age as the factor. Analyze the residuals and report on your findings.
        b. Perform a regression analysis of systolic and diastolic blood pressure and then evaluate the residuals. Explain fully what you have found.
        c. Create a scatterplot of systolic blood pressure and pulse rate. One might suspect that higher pulse rate is associated with higher blood pressure. Does the analysis bear out this suspicion?

3. *Scenario:* We will continue to examine the World Development Indicators data in **BirthRate 2017**. We will broaden our analysis to work with other variables in that file:
    - **MortUnder5: Deaths, children under 5 years per 1,000 live births**
    - **MortInfant: Deaths, infants per 1,000 live births**
        a. Create a scatterplot for **MortUnder5** and **MortInfant**. Run the regression and explain what a residual analysis tells you about this sample.

4. *Scenario:* How do the prices of used cars vary according to the mileage of the cars? Our data table **Used Cars** contains observational data about the listed prices of three popular compact car models in three different metropolitan areas in the U.S. All of the cars are two years old.
    a. Create a scatterplot of price versus mileage. Run the regression, analyze the residuals, and report your conclusions.
    b. If it is safe to draw inferences, provide a 95% confidence interval estimate of the amount by which the price falls for each additional mile driven.
    c. You see an advertisement for a used car that has been driven 35,000 miles. Use the model and the scatterplot to provide an approximate 95% interval estimate for an asking price for this individual car.
5. *Scenario*: Stock market analysts are always on the lookout for profitable opportunities and for signs of weakness in publicly traded stocks. Market analysts make extensive use of regression models in their work, and one of the simplest ones is known as the *Random,* or *Drunkard's*, *Walk* model. Simply put, the model hypothesizes that over a relatively short period of time, the price of a particular share of stock will be a random deviation from the prior day. If $Y_t$ represents the price at time $t$, then $Y_t = Y_{t-1} + \varepsilon$. In this problem, you will fit a random walk model to daily closing prices for McDonald's Corporation for the first six months of 2019 and decide how well the random walk model fits. The data table is called **MCD**.
    a. Create a scatterplot with the daily closing price on the vertical axis and the prior day's closing price on the horizontal. Comment on what you see in this graph.
    b. Fit a line to the scatterplot and evaluate the residuals. Report on your findings.
    c. If it is safe to draw inferences, is it plausible that the slope and intercept are those predicted in the random walk model?
6. *Scenario*: Franz Joseph Haydn was a successful and well-established composer when the young Mozart burst upon the cultural scene. Haydn wrote more than twice as many piano sonatas as Mozart. Use the data table **Haydn** to perform an analysis parallel to the one we did for Mozart.
    a. Evaluate the residuals from the regression fit using **Parta** as the response variable.
    b. Compare the Haydn data residuals to the corresponding residual graphs using the Mozart data; explain your findings.

7. *Scenario*: Throughout the animal kingdom, animals require sleep, and there is extensive variation in the number of hours in the day that different animals sleep. The data table called **Sleeping Animals** contains information for more than 60 mammalian species, including the average number of hours per day of total sleep. This will be the response column in this problem.
    a. Estimate a linear regression model using gestation as the factor. Gestation is the mean number of days that females of these species carry their young before giving birth. Assess the conditions using residual graphs and report on your conclusions.

8. *Scenario*: For many years, it has been understood that tobacco use leads to health problems related to the heart and lungs. The **Tobacco Use** data table contains data about the prevalence of tobacco use and of certain diseases around the world.
    a. Using Cancer Mortality (**CancerMort**) as the response variable and the prevalence of tobacco use in both sexes (**TobaccoUse**), run a regression analysis and examine the residuals. Should we use this model to draw inferences? Explain.
    b. Using Cardiovascular Mortality (**CVMort**) as the response variable and the prevalence of tobacco use in both sexes (**TobaccoUse**), run a regression analysis and examine the residuals. Should we use this model to draw inferences? Explain.

9. *Scenario*: In Chapter 2, our first illustration of experimental data involved a study of the compressive strength of concrete. In this scenario, we look at a set of observations all taken at 28 days (4 weeks) after the concrete was initially formulated; the data table is called **Concrete 28**. The response variable is the **Compressive Strength** column, and we will examine the relationship between that variable and two candidate factor variables.
    a. Use **Cement** as the factor, run a regression analysis, and evaluate the residuals. Report on your findings in detail.
    b. Use **Water** as the factor, run a regression analysis, and evaluate the residuals. Report on your findings in detail.

10. *Scenario*: Prof. Frank Anscombe of Yale University created an artificial data set to illustrate the hazards of applying linear regression analysis without looking at a scatterplot (Anscombe 1973). His work has been very influential, and JMP includes his illustration among the sample data tables packaged with the software. You'll find **Anscombe** both in this book's data tables and in the JMP sample data tables. Open it now.
    a. In the upper left panel of the data table, you will see a green triangle next to the words **The Quartet**. Click the triangle, and select **Run Script**. This produces four regression analyses corresponding to four pairs of response and predictor variables. Plot the residuals for all four regressions and report on what you find.
    b. For which of the four sets of data (if any) would you recommend drawing inferences based on the OLS regressions? Explain your thinking. Suggestion: in each of the four Bivariate Fit reports, click the red triangle next to **Bivariate Fit** and choose **Show Points**.

11. *Scenario*: Many cities in the U.S. have active used-car markets. Typically, the asking price for a used car varies by model, age, mileage, and features. The data table called **Used Cars** contains asking prices (**Price**) and mileage (**Miles**) for three popular budget models; all cars were two years old at the time the data were gathered, and we have data from three U.S. metropolitan areas. All prices are in dollars. In this analysis, **Price** is the response and **Miles** is the factor.
    a. As in the prior chapter, run three separate regressions (one for each model). This time, analyze the residuals and report on your findings.
    b. Finally, compare the analysis of residuals for the three models. Which one seems to satisfy the conditions best?

12. *Scenario:* We will return to the World Development Indicators data in **WDI**. In this scenario, we will investigate the relationship between access to improved sanitation (the percent of the population with access to sewers and the like) and life expectancy. The response column is **life_exp**, and the factor is **sani_acc**.
    a. Use the **Local Data Filter** to **Show** and **Include** only the observations for the **Year 2015**, and the **Latin America & Caribbean Region** nations. Evaluate the residuals and report on how well the data satisfy the OLS conditions.
    b. Repeat the residuals analysis for **East Asia & Pacific** countries in 2015.
    c. Now do the same one additional time for the countries located in **Sub-Saharan Africa**.
    d. How do the three residuals analyses compare? What might explain the differences?

13. *Scenario:* The data table called **USA Counties** contains a wide variety of measures for every county in the United States based on the 2010 Census
    a. Run a regression casting **sales_per_capita** (retail sales dollars per person, 2007) as **Y** and **per_capita_income** as **X**. Analyze the residuals and report on your findings.

---

[1] You can accomplish the same effect by pressing Ctrl + e.

# Chapter 17: Review of Univariate and Bivariate Inference

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## Overview

The unifying thread connecting Chapters 10 through 16 has been *statistical inference*, or, how do we assess and manage the risks of drawing conclusions based on limited probabilistic samples? Before moving on to the final portion of the book where we continue with further topics in the practice of inference, we pause again to review the fundamental concepts, assumptions, and techniques of conventional inference methods. As in the earlier review chapters, we will use the World Development Indicators (WDI) data for our investigations. We will revisit some of the descriptive analysis that we performed in Chapter 5 to extend the investigation into the realm of inference.

As in the earlier review chapters, the objective is to recapitulate and draw connections between important concepts. The emphasis in this chapter is on the choice of techniques, assessment of assumptions, and interpretation of results. We will not review each previous technique but focus on a few needed to pursue some specific research questions. We will also see some "next steps" that anticipate later topics.

## Research Context

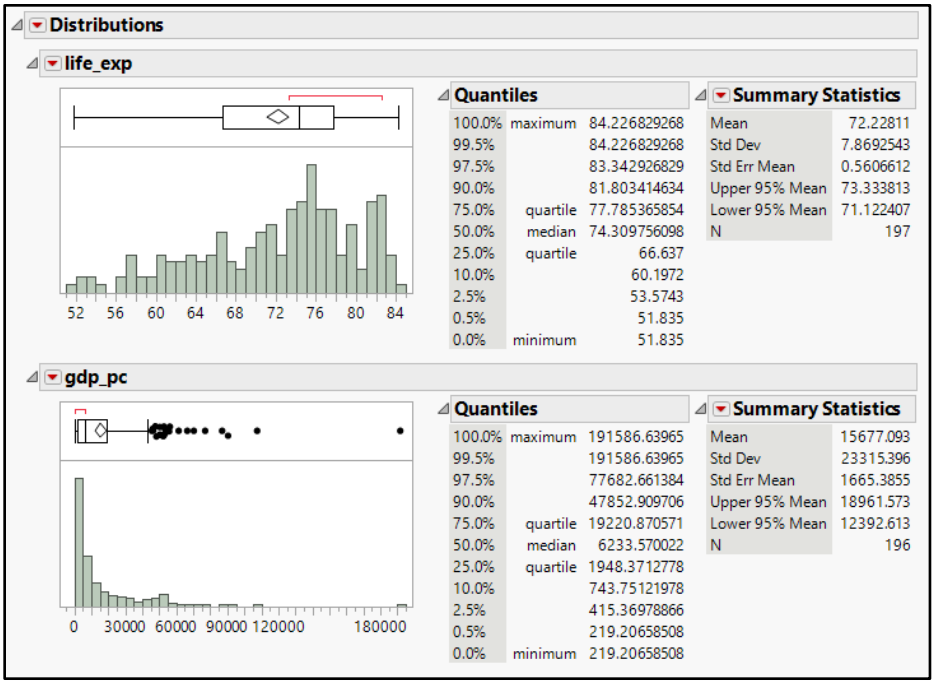
In Chapter 5, some of our investigations centered on the variability of life expectancy around the world in 2010 and some factors that might account for or be associated with that variation. In this chapter, we will apply the techniques of inference to the same data and take a closer look at the ways in which differences in national income are associated with changes in life expectancy at birth. To focus the discussion, we will restrict the investigation to just three variables: life expectancy, income group, and GDP per capita. We will treat life expectancy as the response variable and both income group and GDP per capita as potential factors.

# One Variable at a Time

Our eventual goal is to investigate two bivariate models (a) to determine whether there are statistically significant relationships in either or both models and (b) if so, to decide whether we can use the models to understand how differences in wealth can impact life expectancy. With a quantitative (continuous) response variable and an ordinal variable like income group, we can anticipate that we will eventually consider applying one-way ANOVA to test their hypothesized relationship. Because GDP per capita is also continuous, simple linear regression is the tool of choice for the second pair. Each of these techniques depends on particular characteristics of the sample data and how it was gathered. Hence, we will start with some basic explorations and estimation for each variable in our study.

1. Open the **WDI** data table.
2. To use only the observations from 2016, use the global **Data Filter** to **Show and Include** cases where **Year = 2016.**
3. Use the **Distribution** platform to create summary graphs and statistics for each of the three variables**.**

**Figure 17.1: Distributions of Life_Exp and gdp_pc**




For the two quantitative variables, in anticipation of inference, we should almost instinctively look to the histograms and box plots to get a sense of symmetry, outliers, and peaks. The summary statistics provide further detail about center and dispersion.

As we see in Figure 17.1, both distributions are skewed in opposite directions, with GDP per capita being more strongly skewed with numerous outliers. The life expectancy distribution has multiple peaks, which suggests key subgroup differences.

Furthermore, we can say with 95% confidence that in 2010, the mean life expectancy was between 68.3 and 71.1 years. With the same degree of confidence, the mean income per capita (in constant 2000 US dollars) was between $6,024.91 and $9,343.06.

International agencies define the income groups, so the variation is by design. All five groups are similar in size, with between 31 and 54 nations assigned to each category.

# Life Expectancy by Income Group

## Checking Assumptions

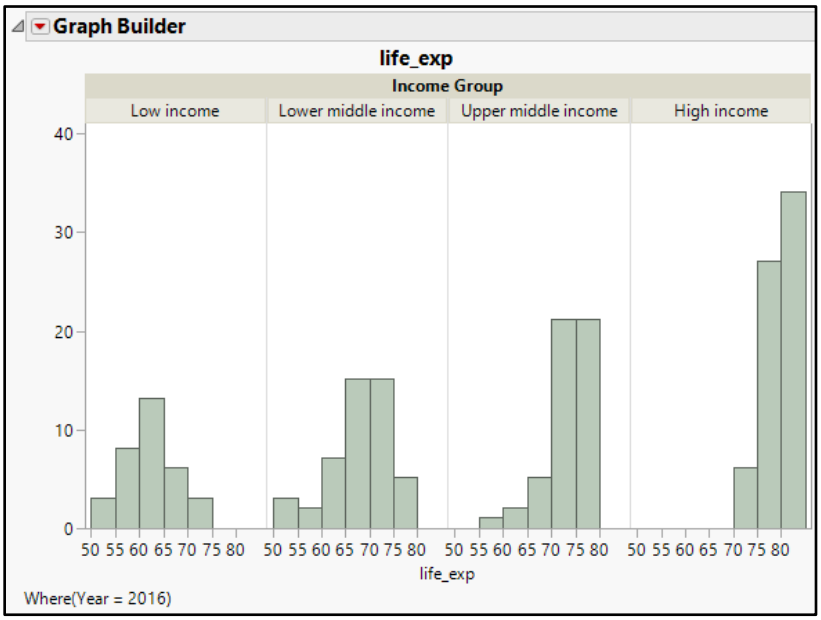
Both ANOVA and ordinary least squares regression provide reliable inferences under specific conditions. In the case of ANOVA, we want the response variable to be approximately normal within each subgroup of the sample and for variances to be similar for each group. For linear regression, we look for a linear relationship between the two variables, normality in the residuals after first estimating the regression line,[1] and constant variance throughout predicted values of Y. We will consider the linear regression model in the next section. In this part of the review, we will examine the relationship (if any) between life expectancy and income group, and first consider whether life expectancy can be described as approximately normal.

Earlier, we noted the presence of multiple peaks in the histogram of life expectancy. Might the peaks reflect differences across income groups? Furthermore, might the strongly left-skewed distribution in the aggregate be composed of four separate, nearly-normal distributions in the four income groups?

There are several ways that we might investigate this. Here is one easy exploratory approach:

1. Open **Graph Builder** and drag **life_exp** into the **X** drop zone.
2. From the center of the menu bar across the top of the graph, choose the histogram tool.
3. Now drag **Income Group** in the **Group X** drop zone, producing a graph similar to Figure 17.2.

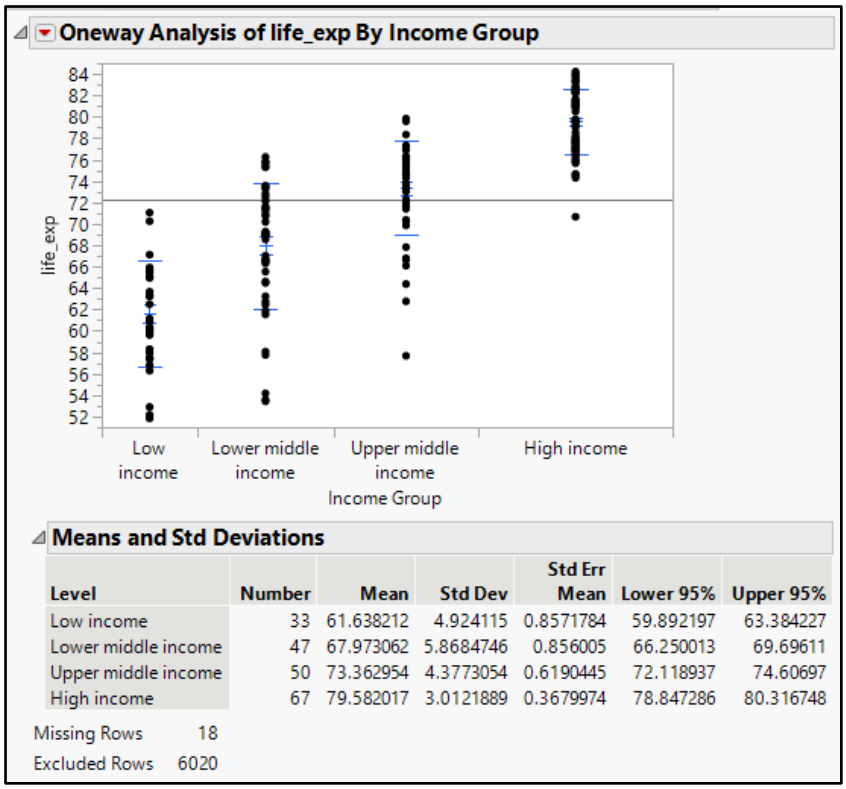
**Figure 17.2: Checking Normality for Life Expectancy by Income Group**




All four histograms are single-peaked, and in the higher income groups, they are left-skewed. We can also see considerable variability in the dispersion of life expectancy values, though it is difficult to judge equality of variances just by looking at these graphs. Perhaps an easier way to evaluate the assumptions is by looking at the means and standard deviations in the **Oneway** report from the **Fit Y by X** platform.

4. In the **Fit Y by X** platform, cast **life_exp** as **Y** and **Income Group** as **X.**
5. In the **Oneway** report, click the red arrow and choose **Means and Std Dev** to produce the revised report shown in Figure 17.3.

**Figure 17.3: Inspecting Standard Deviations to Check Equal Variance Assumption**




**Means and Std Deviations**

| Level | Number | Mean | Std Dev | Std Err Mean | Lower 95% | Upper 95% |
|---|---|---|---|---|---|---|
| Low income | 33 | 61.638212 | 4.924115 | 0.8571784 | 59.892197 | 63.384227 |
| Lower middle income | 47 | 67.973062 | 5.8684746 | 0.856005 | 66.250013 | 69.69611 |
| Upper middle income | 50 | 73.362954 | 4.3773054 | 0.6190445 | 72.118937 | 74.60697 |
| High income | 67 | 79.582017 | 3.0121889 | 0.3679974 | 78.847286 | 80.316748 |

Missing Rows 18

Excluded Rows 6020

The graph shows the individual data values of life expectancy for each income group. In pale blue you should see a short line at the mean value with error bars above and below the mean. Additionally, you will see longer lines one standard deviation above and below the means. Visually, it is evident that the dispersion of the high-income group is smaller than the other groups, whose standard deviations are similar. From the table below the graph, we find that the first three groups have standard deviations between approximately 4.4 and 5.9 years; the last group's standard deviation is only about 3.0. This should signal to us that we might anticipate problems with the assumption of equal variances as well as the normality assumption.

We can formally test for the equality of variances (refer back to Chapter 14 for more details) as follows:

6. Click the red triangle next to **Oneway** and select **Unequal Variances**.

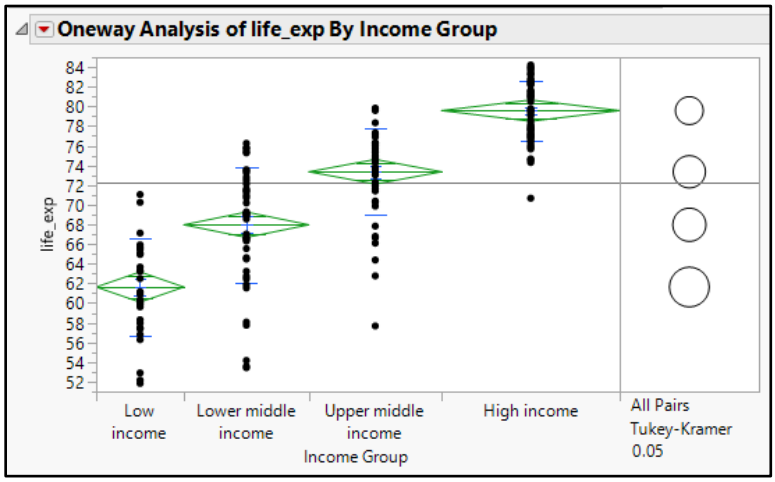
As we did in Chapter 14, consult the results of Levene's test, and you will see that there is statistically significant evidence at the 0.0012 level that the variances should not be considered equal. Hence, ANOVA results might be affected by the discrepancies in variance. With unequal variances, Welch's test is a common alternative to ANOVA, and we note that Welch's ANOVA does find a significant difference among the group means.

## Conducting an ANOVA

In this case, it is probably better to rely on the Welch test than ANOVA, but by way of reviewing the method, let's produce an ANOVA report as well and explore where the differences in group means might occur.

1. Click the red triangle again and select **Means/Anova**.
2. Again, this time choose **Compare Means ▶ All Pairs, Tukey HSD**.

**Figure 17.4: Visual Summary of the ANOVA and Tukey's HSD**




Figures 17.4 and 17.5 are selected from the One-way report and will be discussed separately. In Figure 17.4, looking at the means diamonds from left to right as well as the clicking on the comparison circles on the right side, we see that the means of all four income groups are significantly different from each other.

**Figure 17.5: The Summary of Model Fit and ANOVA Table**

**Oneway Anova**

**Summary of Fit**

| | |
|---|---|
| Rsquare | 0.678858 |
| Adj Rsquare | 0.673866 |
| Root Mean Square Error | 4.493986 |
| Mean of Response | 72.22811 |
| Observations (or Sum Wgts) | 197 |

**Analysis of Variance**

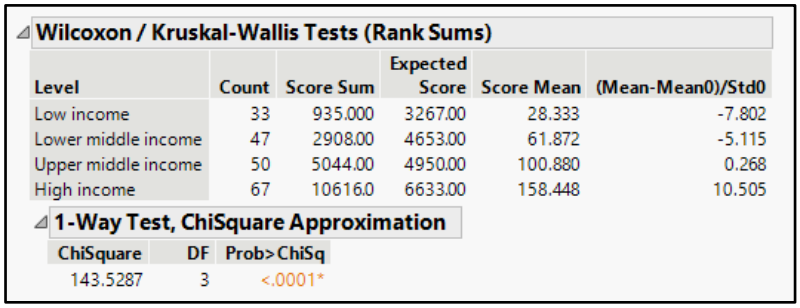
| Source | DF | Sum of Squares | Mean Square | F Ratio | Prob > F |
|---|---|---|---|---|---|
| Income Group | 3 | 8239.521 | 2746.51 | 135.9932 | <.0001* |
| Error | 193 | 3897.811 | 20.20 | | |
| C. Total | 196 | 12137.332 | | | |

Because of the departures from constant variance and normality, we should be reluctant to draw conclusions from this report, but if conditions had been met, we would take note of the significant F-Ratio found in Figure 17.5 (indicating very little likelihood of a Type I error) and the

Root Mean Square Error as a goodness-of-fit measure. The *P*-value of this test is comparable to that in Welch's Test.

Recall from Chapter 14 that when normality cannot be assumed, we should adopt a nonparametric alternative, specifically the Wilcoxon test. Figure 17.6 displays the report from that test. In this instance, the test statistic follows a chi-square distribution (discussed in Chapter 12) and indicates that we should reject a hypothesis that mean life expectancy is the same in all four income groups.

**Figure 17.6: Report of the Wilcoxon/Kruskal Wallis Test**



**Wilcoxon / Kruskal-Wallis Tests (Rank Sums)**

| Level | Count | Score Sum | Expected Score | Score Mean | (Mean-Mean0)/Std0 |
|---|---|---|---|---|---|
| Low income | 33 | 935.000 | 3267.00 | 28.333 | -7.802 |
| Lower middle income | 47 | 2908.00 | 4653.00 | 61.872 | -5.115 |
| Upper middle income | 50 | 5044.00 | 4950.00 | 100.880 | 0.268 |
| High income | 67 | 10616.0 | 6633.00 | 158.448 | 10.505 |

**1-Way Test, ChiSquare Approximation**

| ChiSquare | DF | Prob>ChiSq |
|---|---|---|
| 143.5287 | 3 | <.0001* |

## Life Expectancy by GDP per Capita

Comparing mean life expectancy across income groups provides a rather crude approach to the relationship between long life and financial resources. We do have a quantitative measure of income, namely, GDP per capita. The variable is measured in inflation-adjusted US dollars, using the year 2000 as the base year. With two quantitative variables and a logical expectation that higher income can provide people with at least some of the preconditions for longer lives, the obvious technique to apply is linear regression.

We will initially use **Graph Builder** to check for a possible linear relationship.

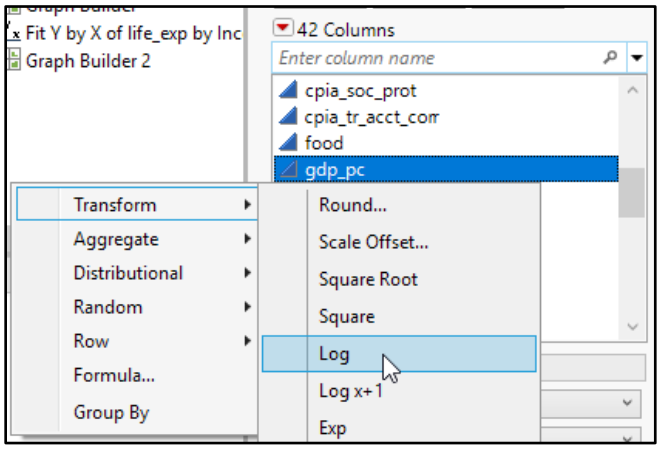
1. Open **Graph Builder**, and place **life_exp** on the **Y** axis and **gdp_pc** on the **X** axis.

The resulting scatterplot is quite obviously *non*-linear. The default smoother rises nearly vertically on the left side of the graph, curves, and then levels off as GDP increases. Such a pattern is typical of *logarithmic growth*, and we can often model such a pattern by taking the logarithm of the X variable; doing so is known as *transforming* the variable.

Within the **Graph Builder**, we can temporarily transform a variable to see how such an adjustment makes a useful difference. Try this:

2. Move the cursor into the **Variables** panel that lists the column names, select **gdp_pc**, and right-click.
3. Choose **Transform ► Log**, as shown in Figure 17.7.

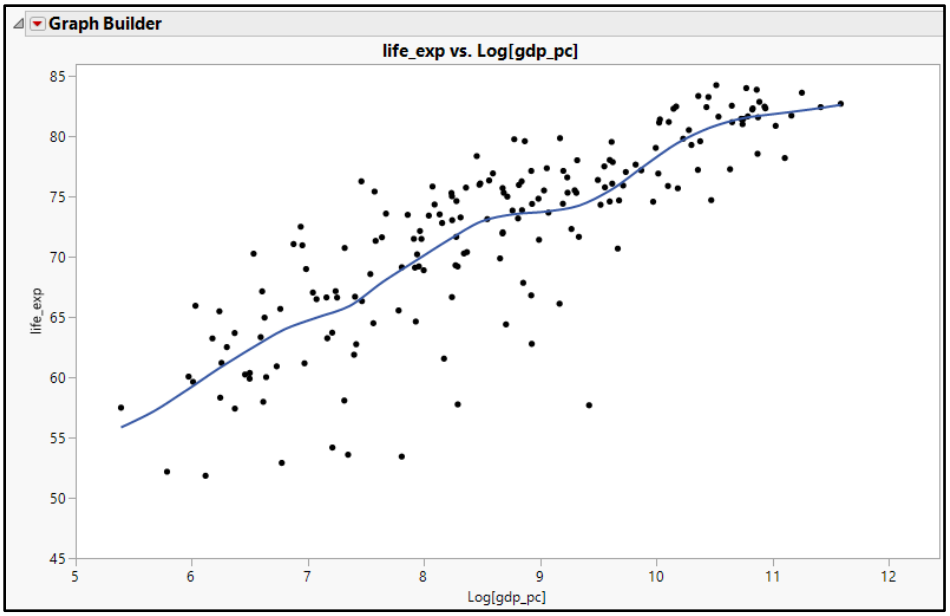
**Figure 17.7: Making a Temporary Transformed Variable**




Doing so creates a new temporary variable just after **gdp_pc** in the variable list, shown in italics as **Log[gdp_pc]**.

4. Now drag the temporary variable into the **X** drop zone, placing it just over the axis number values so that it replaces **gdp_pc** as the variable on the horizontal axis.

**Figure 17.8: Scatterplot Using life_exp Versus Log (gdp_pc)**




The resulting new graph (Figure 17.8) is much more linear than the initial graph. We have surely seen stronger and more distinctly linear patterns, but compared to the initial graph, we can describe this as linear. To continue with our review, let's return to the Fit Y by X platform for more extended analysis.

5. Open the **Fit Y by X** dialog box and cast **life_exp** into the **Y, Response** role.
6. Hover over the variable name **gdp_pc**, and again apply the log transformation. Be sure to hover over the name, and not the blue triangular continuous variable icon.
7. Cast the new transformed variable as **X, Factor** and click **OK**.
8. In the **Bivariate Fit** report, click the red triangle and choose **Fit Line**.
9. Finally, click the red triangle below the graph, next to **Linear Fit**, and choose **Plot Residuals**.

In Chapter 15, we learned to interpret the series of residual graphs and the regression table, looking for potential violations of the OLS conditions. In these graphs, we find problems with both normality and equal variances (homoscedasticity); consequently, the estimates of mean square error, confidence intervals, and the t test will be in doubt, but we are still on safe ground to interpret the estimated slope. Consulting the report, we find that the estimated equation is **life_exp = 33.893863 + 4.4299595*Log[gdp_pc]**. To estimate life expectancy for a given country, we would take the natural log of GDP per capita, substitute it into the equation, and find our estimate.

How do we interpret these estimated parameters? In this analysis, the intercept has little real-world meaning. If we take its simplest literal meaning, the estimated intercept says that in a nation where the natural log of GDP = 0 (that is, GDP per capita is $1), life expectancy on average would be about 29.9 years, but there is no country that has such a low national income. In our data, the country with the lowest GDP per capita in 2016 was Burundi, with a GDP value of only $219.21.

The slope tells us that for each 1 unit increase in the natural log of per capita income, life expectancy increases by approximately 4.4 years. But, as we noted at the outset, this is not a linear relationship—the 4.4-year increase does not correspond to fixed increases in GDP per capita, but rather to increases in the log of GDP per capita.

Our regression line allows us to estimate that the mean life expectancy among countries with log[gdp_pc] = 6 is approximately 60.47 years. For the log of GDP to be 6, GDP per capita would be $403.43. The step from log values of 6 to 7 corresponds to per capita income of $1,096.63, an increase of more than $693. But the next step, from 7 to 8, requires an increase of $1,884.33 in income—hardly a constant rate of increase. Hence, interpretation of this regression line involves a more nuanced approach.

## Conclusion

Before drawing inferences from sample data, it is important to select a technique appropriate to the type of data at hand, to review the conditions (if any) required by the technique, and to assess whether the data is suitable for drawing inferences. If a hypothesis test is to be done, it is crucial to know what hypothesis is to be tested. If the conditions are met and the data are suitable, we follow well-defined steps to conduct the inferential procedures and interpret the estimates and/or evaluate test results. For all the statistical tests that we have studied (and will study in coming chapters), we will want to consult the *P*-values and decide if the risk of a Type I error is tolerable.

[1] Normality in the residuals can also be thought of as normality in the conditional distribution of Y at each observed value of X. Hence, the normality of Y is of some interest.

# Chapter 18: Multiple Regression


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## Overview

The past two chapters have introduced most of the major concepts important to the practice of regression analysis. At this point, we will extend the discussion to consider situations in which a response variable Y has simultaneous linear relationships with more than one independent factor. We have been interpreting R square as the fraction of the variation in Y associated with variation in X. We have seen examples where a single factor accounts for most of the variation in Y, but we have also seen examples where one X accounts for a small portion of Y's variation. In such cases, we might want to use *multiple regression analysis* to investigate additional factors. Conceptually and in terms of software skills, multiple regression is just an extension of simple regression with one major new concept, known as collinearity. We will begin by introducing the basic model.

## The Multiple Regression Model

Multiple regression substantially expands our ability to build models for variation in Y by enabling us to incorporate the separate effects of more than one factor, and also by enabling us to examine ways in which factors might interact with one another. The multiple regression model accommodates more than one predictive factor:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_k X_{ki} + \varepsilon_i$$

where $Y_i$ is an observed value of the response variable, each $X_{ji}$ is an observed value for a distinct factor variable, and each of the $\beta$s is a parameter of the model. As in the simple model, $\varepsilon_i$ is the amount by which an individual response deviates from the model.

As our first illustration, we will return to a subset of the **Concrete** data. Recall that Prof. Yeh (1998) conducted an experiment to investigate the compressive strength of concrete, measured in a unit call megaPascals (MPa). Compressive strength increases as concrete cures, or ages, after mixture. To simplify our example, we will restrict our attention to a sample of concrete batches that were all measured after curing 28 days.

Concrete is a mixture of water, aggregates (stones, pebbles, sand), and cementitious material. Cementitious materials serve as the binders that hold the mixture together as the water evaporates in curing and are the key source of compressive strength. The most common such binder is called Portland cement, and manufacturers use various cementitious additives to achieve desired properties depending on the structural application.

In our central example, we will focus on three ingredients: water, Portland cement, and slag—a waste product of iron smelting, which is another cementitious material. We will build a multiple regression model to predict the compressive strength of concrete after 28 days. Along the way, we will point out similarities and differences with simple regression analysis. All of the prior conditions and caveats still apply—independent observations, linear relationships, and assumptions about random errors. In addition, we will add one new concern called *collinearity*, and we will use a new JMP analysis platform.

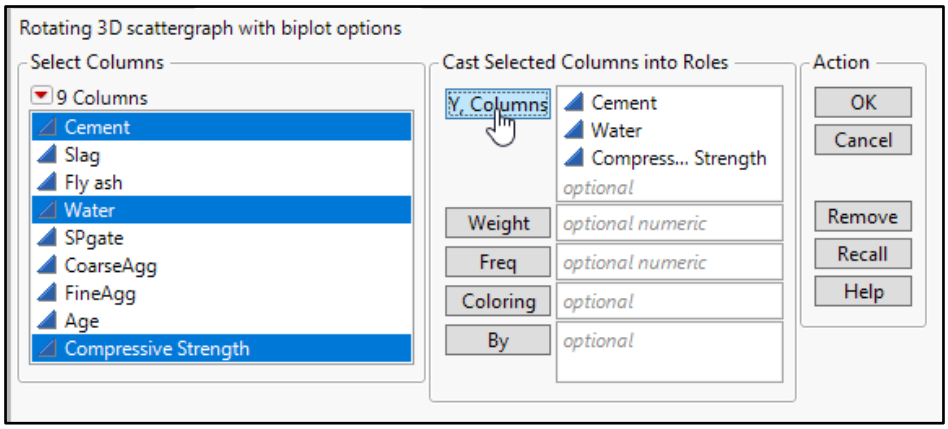
## Visualizing Multiple Regression

To begin, let's consider the influence of water and Portland cement. In general, the compressive strength of concrete comes from the cementitious ingredients; additional water tends to diminish the compressive strength during curing.

In a simple bivariate regression model, we visualized the relationship between X and Y using a scatterplot. With two independent variables, we could think of the relationship between the response variable and the independent variables in three-dimensional space. JMP makes this easy.

1. Open the data table called **Concrete 28.**
2. Select **Graph ► Scatterplot 3D.** Select the columns **Cement, Water**, and **Compressive Strength**, and cast them into the **Y** role (as shown in Figure 18.1). Then click **OK**.

**Figure 18.1: Creating a 3D Scatterplot**




This creates the graph shown below in Figure 18.2. In the graph, you will notice three axes corresponding to the three columns. Initially, the perspective in this graph is analogous to standing on a balcony looking down into a cube-shaped room from one corner. The points are hanging in space; their heights above the floor are set at the observed compressive strength, and their locations reflect the amount of cement and water in the mixture.

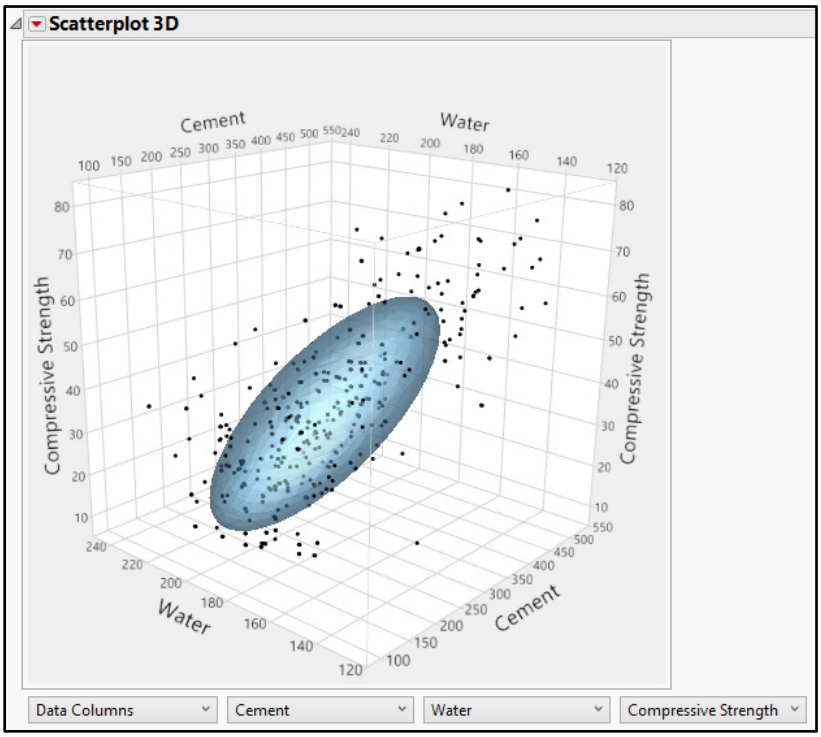
JMP enables us to change our perspective interactively.

3. Move the cursor over the grid (into the "room"). You will note that the cursor shape now becomes indicating that we can now rotate the graph.
4. Click and slowly drag the cursor from left to right. The cloud of points will appear to revolve around the vertical axis. Release the mouse button.
5. Now hold the Shift key on your keyboard and click and slowly drag the cursor a short distance from left to right. Release the mouse button and watch as the graph now spins on its own.

As you watch the spinning graph, notice that the points generally form an *ellipsoid*, which is the three-dimensional version of an ellipse. You might want to imagine a soft rubber ball that has been squashed.

6. To visualize the ellipse, click the red triangle, and select **Normal Contour Ellipsoids**. In the next dialog box, just click **OK**.

**Figure 18.2: Initial Rendering of 3D Scatterplot with Density Ellipsoid**




Notice also that from some perspectives, the points may appear quite randomly scattered, but from other perspectives, the ellipsoidal shape of the point cloud is very apparent. This image conveys one very important concept in multiple regression: if Y is related simultaneously to two independent Xs, and if we do not look at both X dimensions appropriately, we may fail to detect the nature of the relationship.

In the next section, we will fit a model with three independent (X) variables. Once we move beyond three dimensions, our graphical choices need to change. We will shortly see how JMP provides graphs that help us see what is going on in the analysis. At this point, though, we are ready to perform a multiple regression analysis, and before we move to three independent variables, let's estimate a model corresponding to Figure 18.2 and see how the visual display corresponds to the multiple regression model.

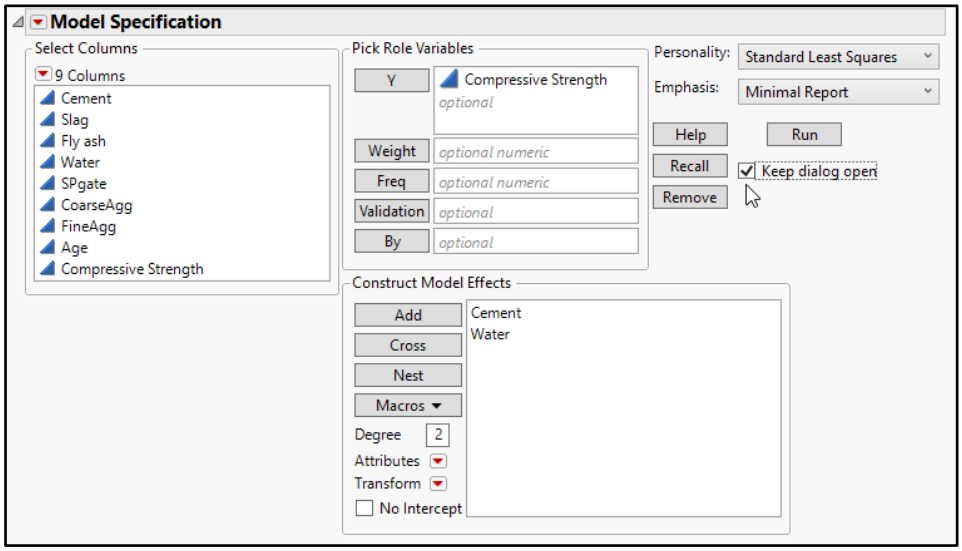
## Fitting a Model

Now let's estimate the model proposed in the previous section.

1. Select **Analyze ► Fit Model.**
2. Now select **Compressive Strength** as **Y**.

In our prior regressions, we used both the **Fit Y by X** and the **Fit Model** platforms. The **Fit Y by X** platform is designed for bivariate analysis and cannot be used for multiple regression analyses. We have seen this dialog box (Figure 18.3) before, but let's pause now to note how it is organized. As always, available columns appear at the left. In the upper center, we can choose the Y variable,[1] and in the lower center, we select the factors for the model. The upper right offers some choice of estimation approaches.

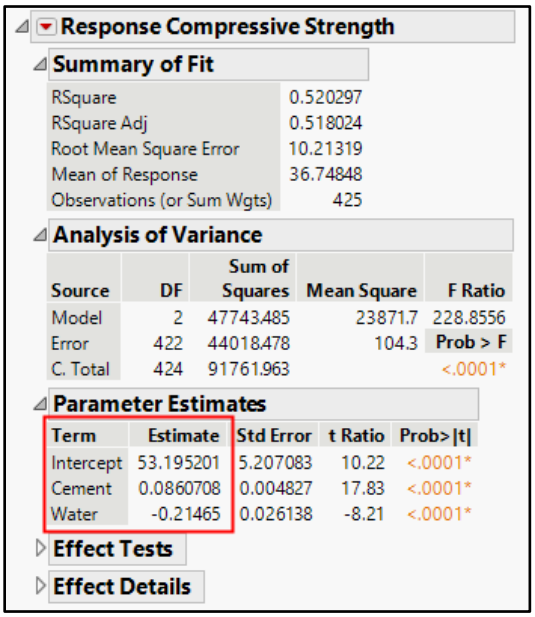
**Figure 18.3: The Fit Model Dialog Box for This Regression**




3. In the upper right corner, select **Minimal Report** as the **Emphasis**.
4. Then, highlight **Cement** and **Water** and click **Add**.
5. Check the box marked **Keep dialog open** and click **Run**.

Figure 18.4 displays a portion of the results.[2] Although the results window initially looks different from the regressions that we have seen thus far, it is very similar. Let's first look at the familiar panels in the output and then learn to interpret the newer parts.

**Figure 18.4: Detail from Multiple Regression Report**



**Response Compressive Strength**

**Summary of Fit**

| | |
|---|---|
| RSquare | 0.520297 |
| RSquare Adj | 0.518024 |
| Root Mean Square Error | 10.21319 |
| Mean of Response | 36.74848 |
| Observations (or Sum Wgts) | 425 |

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Ratio |
|---|---|---|---|---|
| Model | 2 | 47743.485 | 23871.7 | 228.8556 |
| Error | 422 | 44018.478 | 104.3 | Prob > F |
| C. Total | 424 | 91761.963 | | <.0001* |

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 53.195201 | 5.207083 | 10.22 | <.0001* |
| Cement | 0.0860708 | 0.004827 | 17.83 | <.0001* |
| Water | -0.21465 | 0.026138 | -8.21 | <.0001* |

**Effect Tests**

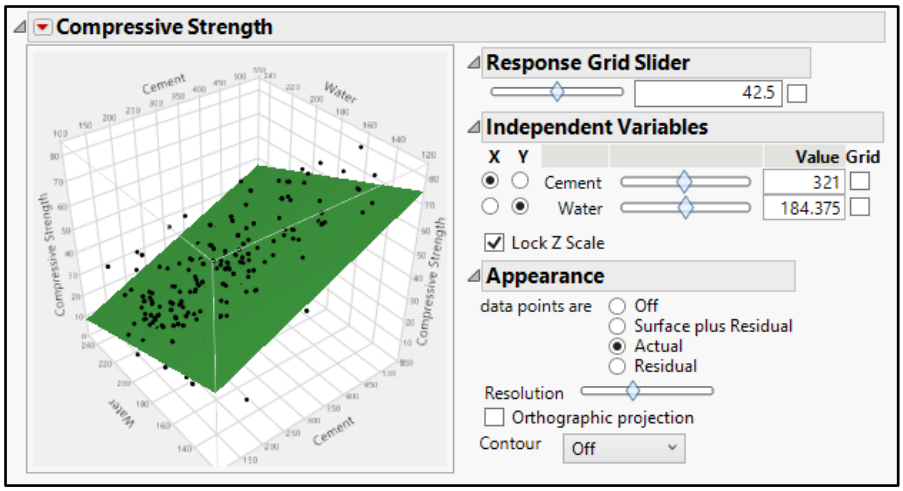
**Effect Details**

The next section discusses the result reports in detail. Before looking closely at all of the reported statistics, let's consider the coefficients in this model and visualize what their signs tell us about how cement and water together influence compressive strength. Look at the lower portion of Figure 18.4 and locate the estimated slopes. The estimated coefficient of **Cement** is approximately 0.086. The positive coefficient tells us that for each additional kilogram of cement in the mixture, compressive strength increases by 0.086 MPa—if the amount of water remains unchanged. Similarly, the negative coefficient of **Water** indicates that, for a given quantity of cement, additional water weakens compressive strength.

Unlike a simple bivariate regression, this regression model represents a plane in three-dimensional space. The plane slopes upward along the **Cement** dimension and downward along the **Water** dimension. To see this, do the following in the **Fit Model** report:

6. Click the red triangle at the top of the results report. Select **Factor Profiling ► Surface Profiler**. This opens the panel shown in Figure 18.5 below.
7. Select the **Lock Z Scale** option.
8. Click the disclosure arrow next to **Appearance** and select **Actual**.

**Figure 18.5: An Estimated Regression Plane**




Now you can move the cursor over the 3D grid and rotate the graph, just as you did earlier. You should be able to observe how the actual observed points scatter above and below the estimated regression plane. As you rotate the graph, notice the slope and orientation of the green surface—it rises as the values along the cement axis increase, and it declines along the water axis.

## Visualizing Model Predictions

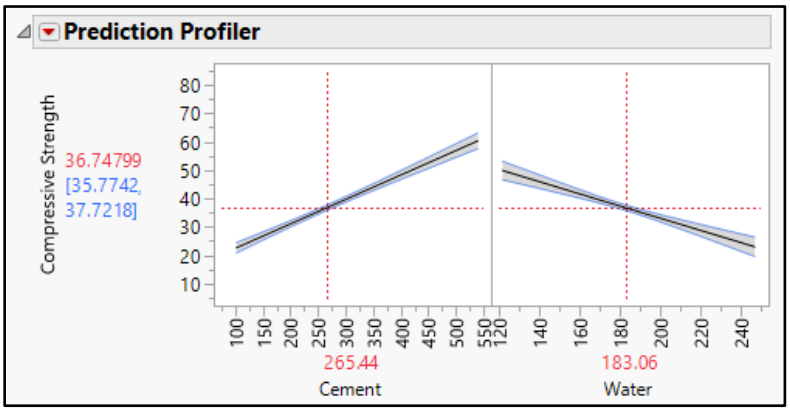
For another perspective on the estimated formula—the Prediction Formula—we can open a **Prediction Profiler** much as we did in Chapter 14 with our two-way ANOVA model. The model that we have just estimated hypothesizes that the factors make separate, independent contributions to compressive strength, and we have just viewed a rendering of the estimated regression plane.

Sometimes it is also helpful to view the model from the perspective of each predictor while holding the other constant. JMP's Prediction Profiler does just that.

1. At the top of the report, click the red triangle next to **Response Compressive Strength**. Select **Factor Profiling ► Profiler**.

This opens a new panel at the end of the report like the one shown in Figure 18.6. Initially we are looking at two cross-sections of the plane from Figure 18.5. The left graph represents the effect of **Cement** and the right graph **Water**. Notice that the cement line slopes upward and the water line downward, consistent with the signs of the regression coefficients. There is a dotted red vertical line at the mean value of each X variable.

**Figure 18.6: Prediction Profiler for the Cement-Water Model**




**Compressive Strength** is on the vertical axis, and the plot reports a predicted value of 36.74799 MPa, with a confidence interval from 35.77 to 37.72 with that mixture of cement and water. The horizontal red dashed line corresponds to predicted value.

If you slide either red vertical line left or right, you can see the change in compressive strength predicted by the formula. As you move away from the mean value of the predictors, the estimated value of Y changes, and the confidence interval becomes wider.

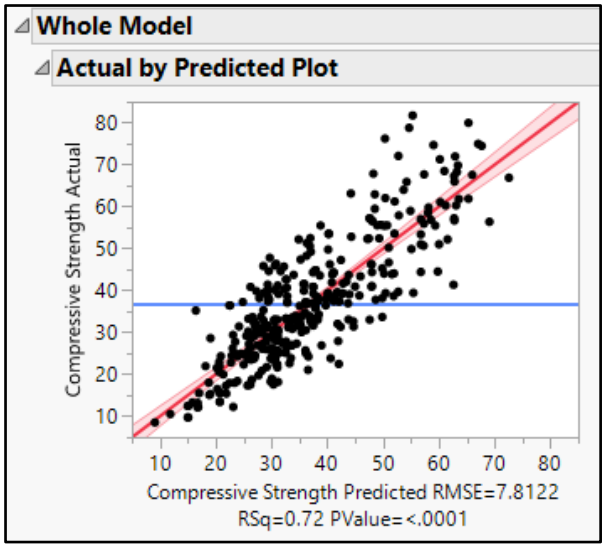
## A More Complex Model

We also note that this first model has a modest R square of 0.52, and the points scatter well away from the estimated plane. We can improve the model by adding the third predictor variable: slag. Slag is another "cementitious" component of concrete mixes that serves to bind the mixture together.

1. Return to the **Fit Model** dialog box (still open) and **Add Slag** to the list of **Model Effects**.
2. Change the **Emphasis** from **Minimal Report** to **Effect Leverage**; we do this to request additional output that helps us decide which factors are influential.
3. **Run** the model.

The first portion of the results is a group of four graphs. Let's first consider the graph on the far left, bypassing the others until a later section. The first graph (Figure 18.7) is essentially a visualization of the goodness of fit of the model. On the horizontal axis, we find the predicted (fitted) values computed by our model; on the vertical axis, we find the actual observed values of compressive strength.

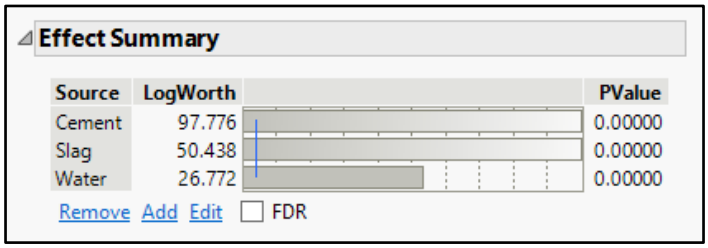
**Figure 18.7: Actual by Predicted Plot for the Concrete Strength Model**




If our model perfectly accounted for all variation in compressive strength, the points would lie exactly on a 45-degree line. To the extent that the model is imperfect and therefore over- or underestimates compressive strength, the points will scatter in the vicinity of the red line. In this graph, we can see that the model performs reasonably well, with predicted values regularly falling close to the actual observed values.

Just below the Actual by Predicted plot in Figure 18.7 you will find the Effect Summary Panel. (See Figure 18.8.) This lists the three explanatory factors in the model in order of *LogWorth*. LogWorth is equal to –log(P-value), so the smaller the *P*-value, the greater the LogWorth. In this case, we see that we are most confident of the effect of cement in accounting for compressive strength. Slag and Water also influence compressive strength.

**Figure 18.8: Effect Summary Plot**




Scroll further and find a plot of residuals versus predicted values; skip that for the moment, and we will discuss it in the next section. Now scroll to the bottom of the report and look at the three

panels—all of which we have seen before. We interpret these statistics as in the simple case with a few notable exceptions:

- In the **Summary of Fit** panel, we note that now the **RSquare** and **RSquare Adj** statistics are unequal. Because each additional X variable inflates R square, we prefer the adjusted R square because it is a more useful measure when comparing models with different numbers of factors.
- Although the **ANOVA** table is identical in structure to its counterpart in the bivariate context, the unstated null hypothesis has changed. In multiple regression, our null hypothesis is that there is no association between Y and any of the X variables.
- The **Parameter Estimates** panel has expanded, with a row for the intercept and additional rows for each of the X variables in the model.

There are two important changes in interpretation here. First, the coefficients for the slope estimates now represent the estimated change in Y corresponding to a one-unit change in $X_j$ *assuming that all other X values remain unchanged.* So, for example, the estimated coefficient of **Cement** in this model is approximately 0.112. We will interpret this by saying that the compressive strength of a mixture increases by 0.112 megaPascals for each additional 1 kg. of cement *if* there is no change in the amount of slag or water in the mixture.

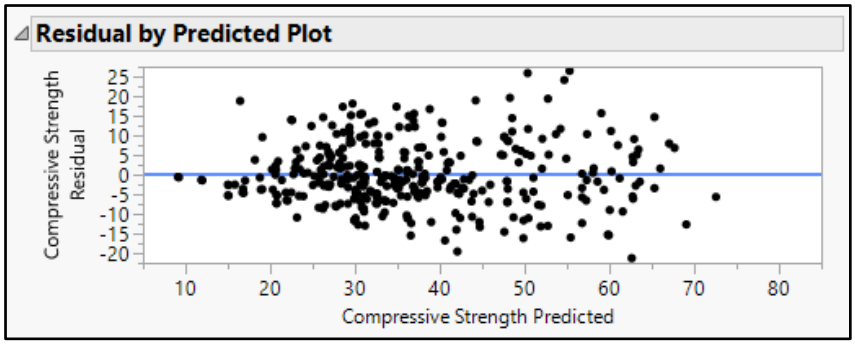
Similarly, the negative coefficient of **Water** means that, holding the amount of cement and slag constant, adding water will weaken the mixture.

The second change is the way we interpret the *t*-statistic and corresponding *P*-value. As before, a very small *P* indicates statistical significance. We now note whether a variable makes a significant contribution to the model once all the other variables have been accounted for. In this example, all the individual t tests are significant, meaning that each of our three X variables is significant in the presence of the other two. As noted above, the bar chart in the Effect Summary panel displays the variables in order of significance.

## Residuals Analysis in the Fit Model Platform

The conditions for OLS regression with a multiple regression model are the same as for simple regression. The **Effect Leverage** emphasis automatically generates a plot of residuals versus predicted (Fitted) value, as shown in Figure 18.9. We interpret this graph just as we did in Chapter 16. We see some indication of heteroscedasticity: The points are concentrated closer to the zero line on the left side of the graph, and drift away from zero as we move to the right.

**Figure 18.9: Residual Versus Predicted Plot for Three-Variable Compressive Strength Model**



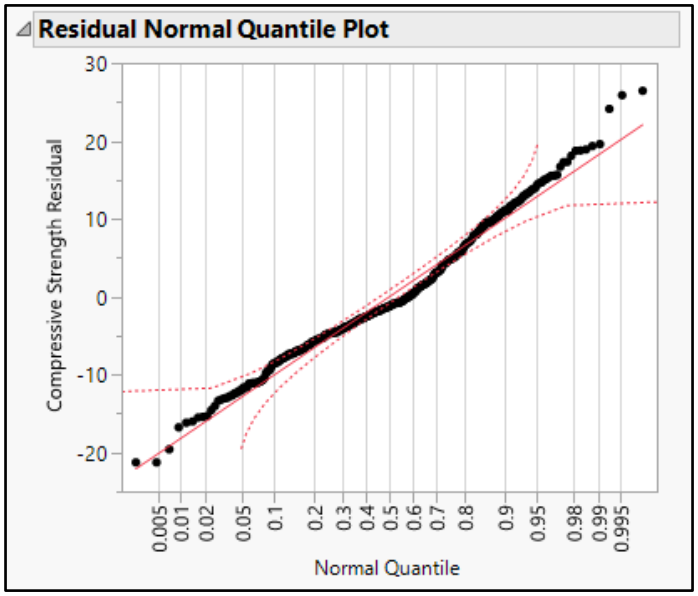

We might be able to come closer to a homoscedastic pattern by including more variables in the model, or by using some approaches illustrated later in Chapter 19. Stay tuned.

JMP does provide more sophisticated ways to deal with heteroscedasticity, but these go beyond the usual topics covered in an introductory course. Where heteroscedasticity is severe, we should be very reluctant to draw inferences from a least squares regression.

The ordinary least squares method is robust with respect to the normality condition. The quickest way to look for substantial non-normality is with a Normal Quantile Plot.

1. At the top of the results, click the red triangle next to **Response Compressive Strength** and select **Row Diagnostics ► Plot Residual by Normal Quantiles**.

**Figure 18.10: Residual Normal Quantile Plot for Three-Variable Compressive Strength Model**




The distribution of the residuals from this model (Figure 18.10) is mildly asymmetric, with departures from normality near the center. As such, we should be cautious about drawing

conclusions based on this sample and our least squares regression estimates. Heteroscedasticity is a considerable problem, and normality is in doubt as well. Because this is a multiple regression model, there is one new issue to consider: collinearity.

## Using a Regression Tree Approach: The Partition Platform

We have just estimated a model using three of the eight available predictor variables in the data table. We were guided by logic and some rudimentary understanding of the composition of concrete. An alternative approach is to supplement our limited knowledge with the signals that the data can provide. This is the essence of machine, or statistical, learning, whereby we seek patterns within a set of data to guide us.

This chapter cannot provide a full introduction to machine learning,[3] but this is an opportune moment to introduce one commonly applied method known as a Regression Tree, implemented in JMP in the **Partition** platform. This platform has several data mining capabilities, and we will use it here to select informative variables from our list of eight candidate variables. In Chapter 19, we will revisit it for model-building when the response variable is dichotomous.

The basic strategy in regression trees is to partition or divide data rows iteratively in such a way as to account for variation in a response variable. At each iteration, the algorithm searches through a list of factors, essentially recoding each factor as a binary categorical variable in order to separate observations into two groups such that the within-group variation is minimal, and the between-group variation is maximal. The algorithm then selects the factor that yields the two most distinct groups. At the next stage, it seeks the next most informative split, and so on. In this way, we can identify influential variables. Depending on the research question, we might use the results of the partition tree to make estimates, or we might estimate a multiple regression model using the variables identified with the Partition platform.

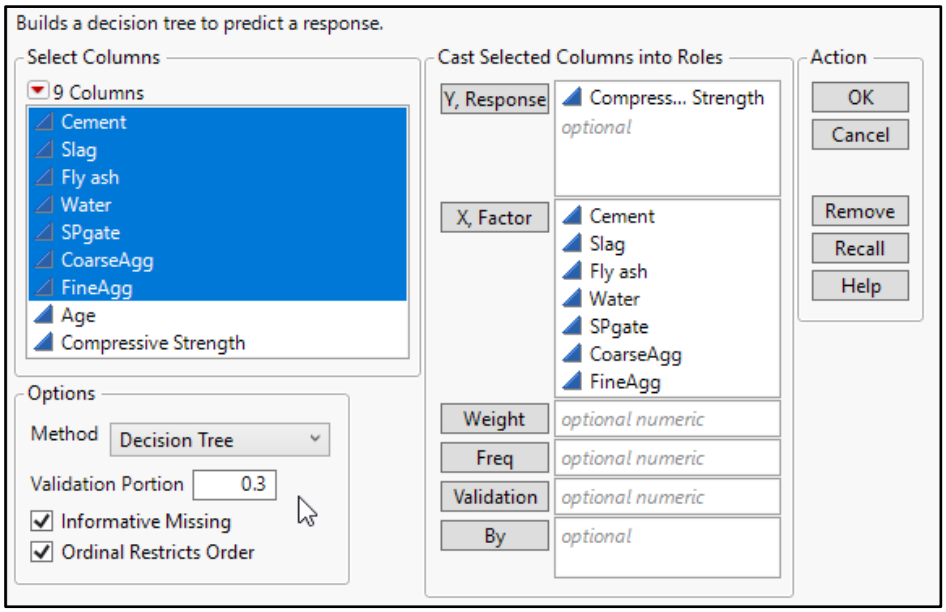
The variables identified in this way depend on the contents of the current sample. In statistical model-building, we don't want to use one sample to both develop and deploy a model. As such, current best practices in data mining call for us to divide our sample data into at least two subsamples: one to "train" or develop a model, and another to validate the results. Simply put, due to sampling variation, we may find a variable to be informative in one sample but not another. For this introductory illustration, we will rely on the Partition platform to reserve 30% of our sample for validation purposes. The method will use most of the sample data to identify influential factors, and then see if the same factors are comparably informative in the validation portion of the sample.

The validation portion is selected randomly. For the sake of reproducibility, first we set a random seed.

1. **Add-Ins ► Random Seed Reset**. Set the seed to 27513.[4]
2. **Analyze ► Predictive Modeling ► Partition**. This opens the dialog box shown in Figure 18.10.

3. Cast **Compressive Strength** as **Y, Response** and the columns from **Cement** through **FineAgg** as **X, Factor**. We omit **Age** because all the observations in this sample are 28 days old. **Age** does not vary, so it cannot be informative.
4. In the **Options** panel, use the default method of **Decision Tree**, and type **0.3** in the box labeled **Validation Portion**. Leave the other options as they are and click **OK**.

**Figure 18.11: Launching the Partition Platform**



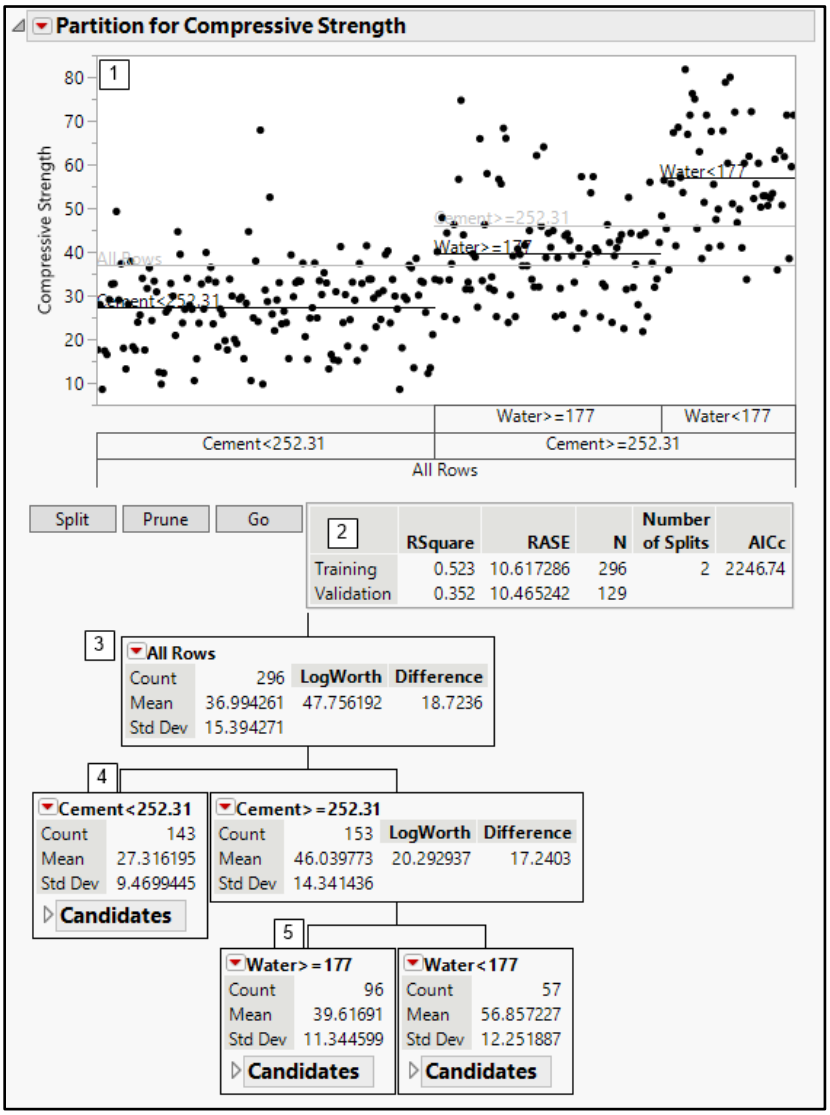

The Partition report now opens, initially showing a jittered plot with the response variable on the Y axis. Before describing the report further, click the **Split** button twice. You should then see the report shown in Figure 18.12. This report is complex, so let's examine it in sections.

The plot at the top (labeled 1 in the figure) shows the individual data values of the training subsample, divided horizontally into four groups corresponding to the two splits. Notice the gray horizontal line labeled "All Rows" at a value of approximately 37. This is the grand mean of the training sample.

As shown at the bottom of the graph, the observations in the left half are those rows for which the quantity of cement was less than 252.31. Though it is a bit hard to read, there is a black line at approximately 27.3 marked **Cement < 252.31**. The right half of the plot, rows with more cement, are further divided according to the amount of water.

Below the graph, in the rectangle labeled 2, is a summary of the tree model at this stage of the partitioning process, comparing the training and validation results. The RSquare column shows that the tree accounts for 52.3% of the variation within the 296 rows that make up the training sample, and just 35.2% of the variation in the 129-row validation sample. We have not seen RASE previously; it is analogous to the Root Mean Square Error in regression, and relatively small values are preferred. AICc[5] is also new to us, and for now it is sufficient to know that smaller values are preferred to larger values.

**Figure 18.12: Partition Report After Two Splits**




The table labeled 3 summarizes the full training set: 296 rows, with a mean compressive strength of 36.99 MPa and a standard deviation of 15.39 MPa.

The first branch of two *nodes* is labeled 4, showing the first split and its result. 143 rows of training data had cement quantities less than 252.31 kg, with a mean compressive strength of 27.316 MPa. The remaining 153 rows have a mean strength of 46.04 MPa. This difference in means is the largest possible of any split based only on the quantity of cement in the mixture.

The second split yields two more nodes (label 5 in the figure). Here, we split the 153 higher-cement mixtures according to the water content of the mix. There are 96 rows with greater than

or equal to 177 kg of water, and their mean strength is 39.62 MPa. There are 57 rows with less than 177 kg of water, with a mean strength of 56.86 MPa.

You can continue to create more splits, and each subsequent node will contain an ever-decreasing number of observations. In the early splits, the R square value will tend to increase in both the Training and Validation sets, but at some point, the increases for the Training set will slow, and the Validation R square may tend to decline.

Our goal in using the tree here is to identify the most informative variable for a multiple regression model. After ten splits or so, the gap between the two R squares stabilizes, and we can identify the variables that figure in the largest number of splits. These variables are top candidates for a multiple regression model.

5. Click the red triangle next to **Partition for Compressive Strength** and select **Column Contributions**.

The new panel that opens lists **Cement**, **Water**, and **Slag** as explaining a substantial amount of the variation in **Compressive Strength**. As such, these three variables are worth further investigation. Before we go too much further with the investigation, though, there is a new issue to contemplate, and that is the topic of the next section.

# Collinearity

When building a model with multiple X variables, it is important that the X variables vary independently of each other within our sample. If any of the X variables is highly correlated with any subset of Xs—a condition called *multicollinearity* or sometimes just *collinearity*—then the estimates of the regression coefficients will likely be distorted and sometimes lead to nonsensical results. Explanatory factors that are highly correlated are essentially redundant. If one is in the model, the next one will not contribute very much additional information. To put it another way: if the experimenters increased the slag content of the mixture every time they increased the cement component, they would never be able to determine how much the strength is affected by each ingredient.

Note two important aspects of collinearity. First, it is a property of a sample rather than the parent population. Whether or not the factors are correlated in the population, we will have a problem if they are correlated in the sample. Second, it can occur when any subset of Xs is correlated with any other subset of Xs. This can mean that two factors are correlated or that several Xs jointly are correlated with another X. In this section, we will consider two regression models—one without collinearity and one with.

## An Example Free of Collinearity Problems

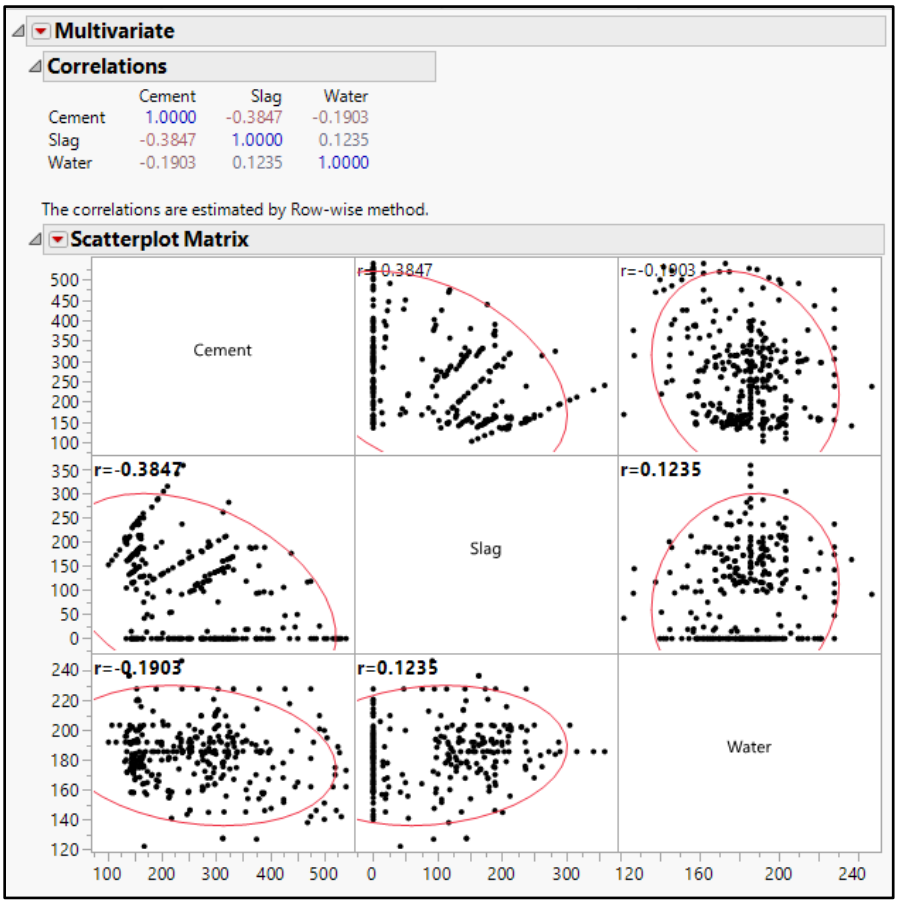
In a well-designed experiment like this one, the experimenter will have taken care to vary the X values independently. After all, in an experiment, the researcher selects and controls the X values. In observational or survey studies, collinearity is frequently a problem, and we will look at such an example shortly. First, let's see how we can check for its presence or absence.

JMP provides several ways of detecting collinearity, some of which are quite complex. In this chapter, we will look at some that are quite easy to use and interpret. First, prior to running a multiple regression, it is advisable to look at possible bivariate correlations between all pairs of X columns. We do this as follows:

1. Select **Analyze** ▶ **Multivariate Methods** ▶ **Multivariate**. Select the three *X* columns (**Cement**, **Slag** and **Water**) as **Y** variables in this platform and click **OK**.

This creates the results shown in Figure 18.13, which displays both scatterplots and a correlation matrix for each pair of the *X* variables. As we can see, none of the relationships are strong[6]—and that is good news. It indicates that no pair of Xs is strongly correlated; the strongest correlation is -0.3847, between **Cement** and **Slag**. The scatter plots confirm the absence of strong linear relationships. The ellipses superimposed on the plots are all quite round.

**Figure 18.13: Correlations and Scatterplot Matrix for Concrete Data**

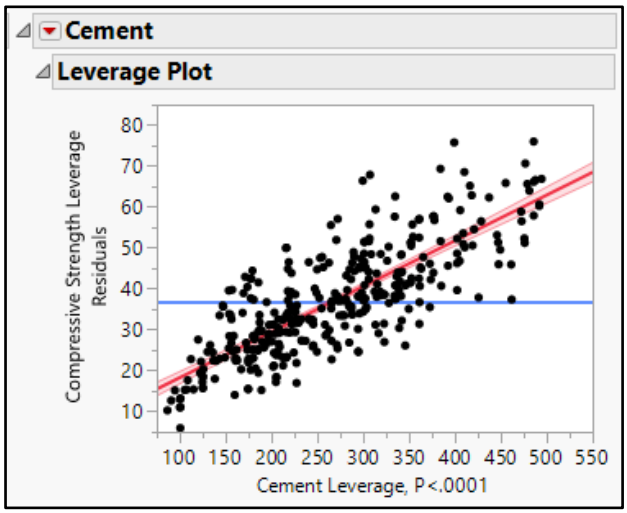



Second, a common indicator of collinearity is that the *F* test has a small *P*-value indicating a significant model, but the *P*-values for the individual t tests are all larger. Find these in the Fit

Model report. In this example, the individual tests are all quite significant and consistent with the *F* test.

A third diagnostic tool is the **Leverage Plot** that JMP creates for each factor variable at the top of the regression results. Figure 18.14 shows the leverage plot for the **Cement** factor. The major purpose of a leverage plot is to visually distinguish significant factors from non-significant ones, but these plots can also help show us whether a variable is collinear with other X variables.

**Figure 18.14: A Leverage Plot Showing No Collinearity**




The horizontal blue line represents a reduced regression model excluding the cement variable. The sloped red line with confidence bands represents the full model. Consider a non-outlying single point in the graph. The vertical distance between the point and the fitted red line is the residual for that observed value of *Y*. The vertical distance between the point and the horizontal blue line represents the residual in a model that excludes **Cement** from the group of factors. Because the cement variable is significant in this regression, the red line (that is, the full model) "carries" the points better than the blue line. In other words, the model is better when we include cement as a predictor variable.

Figure 18.14 typifies the type of pattern that we are looking for. The red confidence curves exclude the blue horizontal line, except for the small region in the center of the graph where they overlap. If collinearity were a problem in this regression, we would note that the red dashed curves would largely enclose the blue horizontal lin,e and the points would clump together near the center of the graph.

Another useful approach relies on a diagnostic statistic called the Variance Inflation Factor (VIF). When we have collinearity among X variables, the least squares technique tends to overestimate, or inflate, the standard errors of the estimated regression slopes. As shown in

Figure 18.15, JMP computes a VIF for each factor in a model to measure how much the standard errors are inflated by each individual factor.

2. Return to the **Parameter Estimates** panel of the results.
3. Move the cursor over the table of estimates and right-click.
4. Select **Columns ▶ VIF**. The parameter estimates table will include VIFs.

**Figure 18.15: Variance Inflation Factors for the Three-Variable Model**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| | VIF |
|---|---|---|---|---|---|
| Intercept | 42.930012 | 4.026775 | 10.66 | <.0001* | . |
| Cement | 0.1116149 | 0.003976 | 28.08 | <.0001* | 1.2029719 |
| Water | -0.233898 | 0.020024 | -11.68 | <.0001* | 1.0407662 |
| Slag | 0.0812208 | 0.004687 | 17.33 | <.0001* | 1.1773698 |

There is no absolute standard for interpreting VIFs. A VIF can never be less than 1, but there is no upper limit. Values close to 1 indicate very little inflation that might be attributed to collinearity. In general practice, values over 10 may signal a problem with collinearity, as we will see in the next example.

## An Example of Collinearity

In this example, we will use our NHANES data to build a model to predict systolic blood pressure for adult men. As we have done before, we will start by selecting only the observations of adult males:

1. Open the **NHANES 2016** data table.
2. Select **Rows ▶ Data Filter**. As we have done several times, filter the rows to select, show, and include males ages 18 years and older.

Let's initially examine a model that uses a man's age (in years), height, weight, and body mass index (BMI) to predict systolic blood pressure. Note that the BMI is computed directly from height and weight, so it is surely collinear with these two variables. We are deliberately specifying a model with substantial collinearity so that we can learn to detect the problem in the output results. Before running the regression, let's look at the correlations among these predictors.

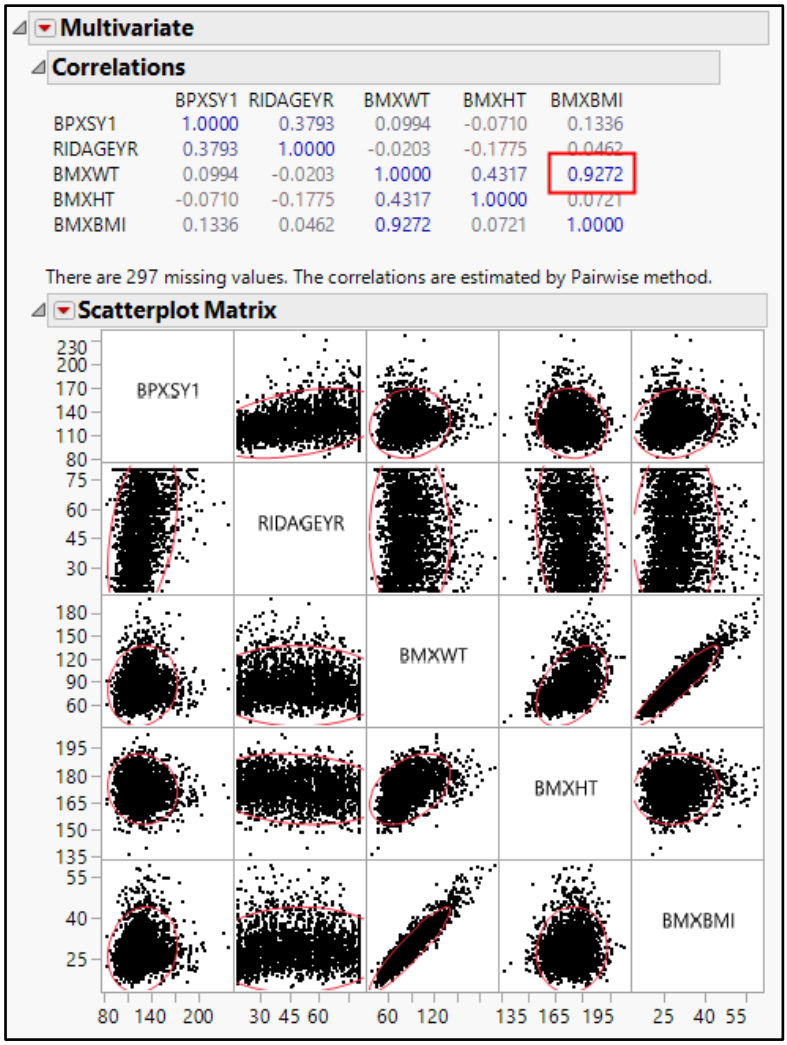
3. Select **Analyze ▶ Multivariate Methods ▶ Multivariate**.
4. Select the following five columns as **Y** variables: **BPXSY1, RIDAGEYR, BMXWT, BMXHT, BMXBMI**. Click **OK**.

The results now look like Figure 18.16. Notice the very strong positive linear association between weight and BMI. This makes sense, since the BMI formula has weight in the numerator:

$$BMI = \frac{Weight\ (kg)}{[Height(m)]^2}$$

In other words, BMI is a linear function of weight. This will cause problems estimating the parameters of a multiple regression model.

**Figure 18.16: Correlations in NHANES Data**




5. Select **Analyze ► Fit Model**. The **Y** variable is **BPXSY1**.
6. Now **Add** the columns **RIDAGEYR, BMXWT, BMXHT, BMXBMI**, to the model.
7. Choose the **Effect Leverage Emphasis** and click **Run**.

With four variables in the model, we get four leverage plots. In your own results, look first at the leverage plot for age. It looks like the plots that we saw earlier; the red confidence curves mostly exclude the blue horizontal line and the points scatter along the length of the red line of fit.

**Figure 18.17: Three Leverage Plots Showing Collinearity**

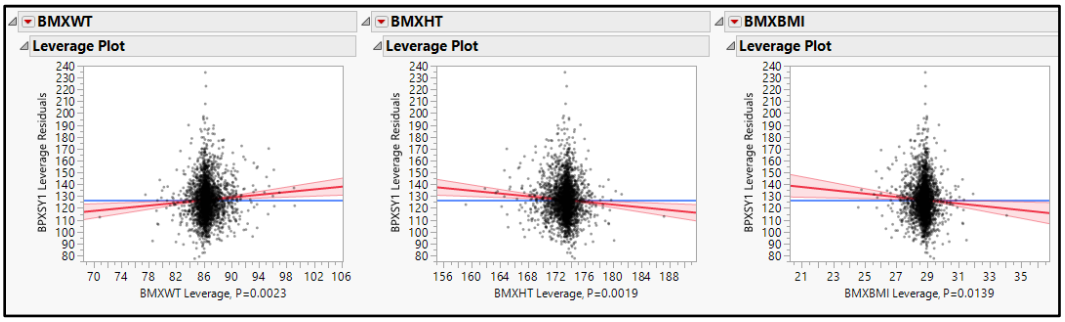



Figure 18.17 shows modified versions of the plots for weight, height, and BMI. Notice two key features. In these graphs, the markers are set at the smallest size to reduce overplotting, and the X axes have been altered to spread the points out. First, the red confidence curves nearly *include* the blue horizontal lines in the three graphs. This indicates that we cannot confidently conclude that the model including these four variables performs any better than the model without one or more of them. Additionally, the points have migrated to the middle of the graph, so that they are all stacked up nearly vertically. In the presence of the other explanatory variables, weight and BMI are redundant and do not contribute additional explanatory information.

A quick look at the Effect Summary indicates that age is quite definitively informative, but as a group, height, weight, and BMI have comparatively little effect.

Once we display the Variance Inflation Factors in the Parameter Estimates table, we see considerable inflation in the estimated standard errors, as shown in Figure 18.18. The three variables which we knew to be related all have very large VIFs. Note also that the estimated slope of the BMI variable is negative, but in the univariate correlation, BMI was positively related to systolic blood pressure.

**Figure 18.18: Variance Inflation Factors in a Model with Collinearity**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| | VIF |
|---|---|---|---|---|---|
| Intercept | 202.5862 | 32.94531 | 6.15 | <.0001* | . |
| RIDAGEYR | 0.3506951 | 0.017315 | 20.25 | <.0001* | 1.040594 |
| BMXWT | 0.5678229 | 0.186032 | 3.05 | 0.0023* | 152.82195 |
| BMXHT | -0.586536 | 0.188806 | -3.11 | 0.0019* | 22.017866 |
| BMXBMI | -1.397211 | 0.567914 | -2.46 | 0.0139* | 124.38715 |

## Dealing with Collinearity

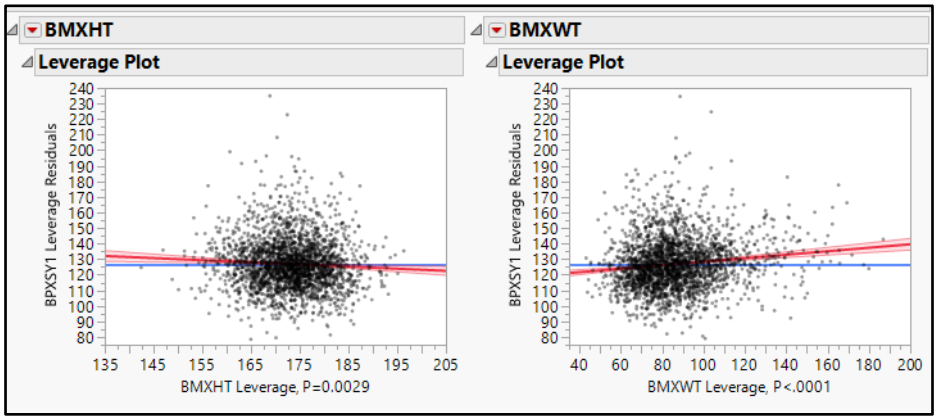
Once we find it, what do we do about it? There are two basic strategies: we can omit all but one of the collinear variables, or we can *transform* some variables to eliminate the collinearity. We will discuss transformations in Chapter 19. For now, we would fit other models using combinations of the remaining variables. In this section, we consider a model without the BMI variables because they had the most inflated variances.

1. At this point, return to the **Fit Model** platform and specify a model for systolic blood pressure using only age, height, and weight as explanatory factors.

Figure 18.19 shows part of the new leverage plot for this simpler, three-variable model. This plot shows that the collinearity problem has been ameliorated: the points are now spread farther along the fitted line. The VIFs (not shown here) are all less than 1.3, and all three variables are statistically significant. The Effect Summary shows that age and weight are informative, and that height also contributes some explanatory value once BMI is dropped from the model.

However, we still have heteroscedasticity and normality is now in question. Additionally, the model only accounts for approximately 16% of the variation in systolic blood pressure.

**Figure 18.19: An Improved Leverage Plot**




In short, we have dealt with the collinearity issue but still have a deficient model. In general, we do not want to include a variable with little predictive value. The previous model that included age, height, weight, and BMI performs no better than the one excluding BMI. Additionally, the model with substantial collinearity had an estimated coefficient with a questionable sign.

## Evaluating Alternative Models

With multiple regression analysis, we often want to compare two or more competing models. In our concrete example, for instance, we earlier estimated a model with three significant factors, reasonably well-behaved residuals, and which accounted for about 72% of the observed variation in compressive strength. Perhaps a four-variable model would do better?

The process of comparing models can easily devolve into a chaotic flurry of trial-and-error attempts. When estimating several models for a single Y variable, take these points into consideration:

- Does each independent factor make logical sense in the model? Are there solid theoretical reasons to include each factor? If not, why include nonsense factors?
- Does the model under consideration have acceptable residual plots? Does an alternative model more closely meet the conditions for inference? Other things being equal, a model with "good" residuals is preferable.
- Are there signs of collinearity? If so, deal with it. A model with collinearity has limited utility in understanding relationships between factors and a response.
- Do the signs of the estimated coefficients all make sense? Remember to apply the logic of "assuming that the other variable(s) are held constant, what effect do I expect when this independent variable increases by one unit?"
- Are the individual t tests significant for all factors? A variable with a non-significant t test should be removed from the model.
- Among the acceptable models, which one has the smallest root mean square error? In other words, which has the smallest total unexplained variation?
- Which model has the highest adjusted R square?

To illustrate, let's return to the concrete example and add a variable.

1. Locate the **Fit Least Squares** report for the **Concrete 28** data. Click the red triangle next to **Response Compressive Strength** and select **Model dialog**. We just want to add another variable to the list of independent X variables.

Fly ash is yet another cementitious by-product of coal burning. Those with expertise in concrete formulation would expect that the addition of fly ash will increase compressive strength. Let's estimate the amount by which fly ash increases the strength of concrete. Then we will work through the bullet points listed above to compare this model to the earlier three-variable model.

2. Highlight **Fly Ash** and click **Add**. Then click **Run**.

Take a moment to review the results of this regression, looking back at the bulleted list of questions above. Your comparison should be similar to that summarized in Table 18.1. Sometimes we will find that one model is better in one respect, but not in another. In this case, the second model equals or outperforms the first model.

**Table 18.1: A Comparison of Two Models**

| Criterion | First Model (Cement, Water, Slag) | Second Model (Cement, Water, Slag, Fly Ash) | Comments |
|---|---|---|---|
| All factors make sense? | Yes | Yes | |
| Acceptable Residual Plots? | Yes; mild heteroscedasticity | Yes; mild heteroscedasticity | Both sets of residuals look normal in shape |
| Evidence of Collinearity? | No | No | Leverage plots show no problem; VIFs small |
| Estimated signs make sense? | Yes | Yes | Water has negative coefficient; others positive |
| All t tests significant? | Yes | Yes | |
| Smallest Root MSE? | RMSE = 7.812 | RMSE = 7.236 | Second model is better |
| Largest adjusted RSquare? | Adj$R^2$ = .718 | Adj$R^2$ = .758 | Second model is better |

It is also worth comparing the estimated coefficients in the two analyses to note another important concept in multiple regression. As you compare the coefficients of cement, slag, and water—the variables included in both models—you will notice that they are very similar in both regressions, differing only by about ± 0.02. The intercepts of these two models are, however, quite different. In the second model, the intercept has shrunk from 42.93 to about 29.34.

If we omit a variable that should be included in the model—a type of common *specification error*—then the effect of that variable sometimes gets "lumped into" the estimated intercept or the coefficient of another independent variable or both. Once we include the relevant variable, its effect is appropriately assigned to the variable.

## Using the Formula Depot to Compare Models (Optional; Requires JMP Pro)

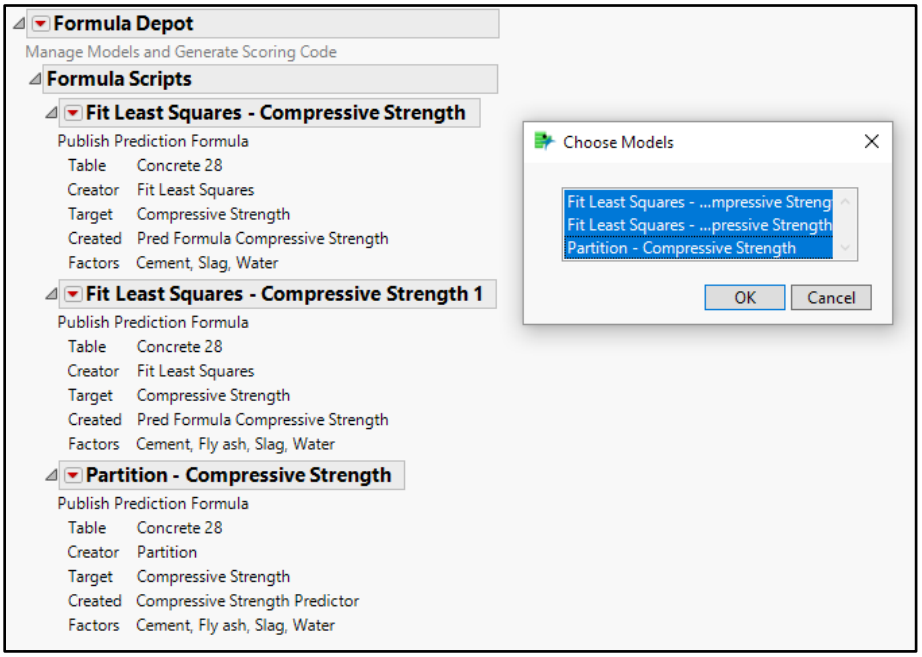
JMP Pro offers the Formula Depot platform that, among other capabilities, facilitates comparisons of different models.[7] To use the Formula Depot, we first publish the estimated models to the depot. Because some readers may not be using JMP Pro, we have made no mention of this step until now. However, as part of a normal workflow for a JMP Pro user, for each estimated model one should take this step:

1. Click the model report red triangle and choose **Save Columns ► Publish Prediction Formula**. This is an option for both the Fit Model and Partition platforms as well as several others that we have not yet seen. Go back in your project and locate the two regression models as well as the Partition Decision Tree reports and publish the prediction formulas.

For this example, I have selected all three models even though the focus of this chapter is multiple regression. Regression and decision trees are rarely covered in an introductory course, but since we created one, we might as well see how it compares with multiple regression modeling.

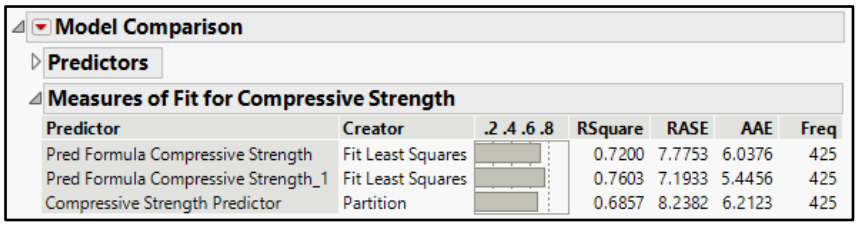
2. A new tab will open, as shown in Figure 18.20. Click the Formula Depot red triangle and choose **Model Comparison**.
3. This opens a small window listing key features of the three estimated models. Highlight all three and click **OK**.

**Figure 18.20: The Formula Depot**




We find a new brief report that summarizes several key performance measures for our three models (Figure 18.21).

**Figure 18.21: Model Comparison Report**



**Model Comparison**

**Predictors**

**Measures of Fit for Compressive Strength**

| Predictor | Creator | .2 .4 .6 .8 | RSquare | RASE | AAE | Freq |
|---|---|---|---|---|---|---|
| Pred Formula Compressive Strength | Fit Least Squares | | 0.7200 | 7.7753 | 6.0376 | 425 |
| Pred Formula Compressive Strength_1 | Fit Least Squares | | 0.7603 | 7.1933 | 5.4456 | 425 |
| Compressive Strength Predictor | Partition | | 0.6857 | 8.2382 | 6.2123 | 425 |

The Model Comparison report lists the three prediction formulas, noting the platform used to build the models, their R square values (shown graphically and numerically), and three additional statistics:

- RASE: Square root of the mean squared error. Smaller values are better.
- AAE: Average absolute error. Smaller values are better.
- Freq: Frequency counts (sample size) used to build each model.

The models are listed in the order we created and saved the prediction formulas. The best-performing of the three is the second regression model, using cement, water, and slag to predict compressive strength. Of the three models, it has the highest R square and the lowest RASE and AAE.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions. Notice that several of these problems are continuations of the corresponding scenarios and questions posed in the prior two chapters. In each multiple regression analysis, you should examine the residuals for possible violations of regression assumptions and also look for indications of collinearity. When you report your findings, be sure to discuss your conclusions about assumptions and about collinearity.

1. *Scenario:* Return to the **NHANES 2016** data table. Select, include, and show only female respondents ages 18 and older.
    a. Perform a regression analysis for adult women with BMI as Y and use waist circumference and height as the independent variables. Discuss the ways in which this model is or is not an improvement over a model that just uses waist circumference as a predictor.
    b. If you can safely use this model, provide the 95% confidence interval for the slope and explain what it tells us about BMI for adult women.

c. Perform another regression analysis to predict BMI using waist circumference and wrist circumference as independent variables. Evaluate this model in comparison to the prior model. Report your findings.

d. Choose another variable that makes sense to you and perform a third regression analysis. Evaluate this model in comparison to the prior models, and report what you find.

2. *Scenario:* High blood pressure continues to be a leading health problem in the United States. In this problem, continue to use the **NHANES 2016** table. For this analysis, we will focus on females between the ages of 12 and 19.

   a. Create a 3D scatterplot of systolic blood pressure (the "top" number in a BP reading), age in years, and weight. Describe what you observe after rotating the scatterplot.

   b. Perform a regression analysis with systolic BP as the response and age (years) and weight as the factors. Report on your findings.

   c. Perform a regression analysis with systolic BP as Y and age (years), weight, and diastolic blood pressure as the independent variables. Explain fully what you have found.

   d. Use the Profiler and write a short, qualitative sentence to describe the difference in Systolic blood pressure between 12-year-old females and 19-year-old females, holding weight and diastolic pressure constant. Does your finding make sense in light of your answer to parts b and c?

   e. Use the Partition platform to identify good candidate variables for use in the next step. Report the variables that you have selected.

   f. Review the full list of columns within the data table and develop another model to predict systolic BP using whichever columns make sense to you. Report on your findings.

3. *Scenario:* We will continue to examine the World Development Indicators data in **BirthRate 2017**. Throughout the exercise, we will use **BirthRate** as the **Y** variable. We will broaden our analysis to work with other variables in that file:

   **MortUnder5**: Deaths, children under 5 years per 1,000 live births
   **MortInfant**: Deaths, infants per 1,000 live births
   **MortMaternal2016**: Number of maternal deaths in childbirth per 100,000 live births

   a. Initially estimate a model using all three of the above-listed variables. Evaluate your model and report on your findings.

   b. Drop the weakest of the three variables from the model and re-estimate using a reduced model with two independent variables. Report on your findings, comparing these results to those for the prior model.

   c. Use the Profiler to explore the independent effects of the variables in your model. Briefly describe what you see.

4. *Scenario*: Throughout the animal kingdom, animals require sleep, and there is extensive variation in the number of hours in the days that different animals sleep. The data table called **Sleeping Animals** contains information for more than 60 mammalian species, including the average number of hours per day of total sleep. Use this as the response column in this problem.
    a. As in the prior chapters, first estimate a linear regression model using Gestation as the factor. Gestation is the mean number of days that females of these species carry their young before giving birth. Evaluate this regression.
    b. Now add a second independent variable: brain weight. Re-run the regression and evaluate this model in comparison to the first.
    c. Finally, estimate a model that also includes body weight as a third explanatory variable. Is this model an improvement over the earlier models? Explain your reasoning.

5. *Scenario*: Industrial production refers to the manufacture of goods within a country. In this scenario we look at industrial production in India, as measured by India's Index of Industrial Production (IIP). The IIP is a figure that compares a weighted average of industrial output from different sectors of the economy (for example, mining, agriculture, and manufacturing) to a reference year in which the index is anchored at a value of 100. As a result, we can interpret a value of the IIP as a percentage of what industrial output was in the base year. So, for example, if the IIP equals 150, we know that production was 50% higher at that point than it was in the base year. The data table **India Industrial Production** contains IIPs for six different segments of the Indian economy. All figures are reported monthly from April 2012 through May 2019.
    a. First, develop a simple linear regression model to estimate monthly General IIP using one of the sector-specific component indexes. Select the X column by first determining which of the available columns has the highest correlation with the general index. Report on your model.
    b. Now add a second X variable, using one of the remaining columns that makes sense to you. Report on your findings.
    c. What problem have you encountered in finding an appropriate column to add to your original model? Why should we anticipate encountering this problem?

6. *Scenario*: In the United States, the management of solid waste is a substantial challenge. Landfills have closed in many parts of the nation, and efforts have been made to reduce waste, increase recycling, and otherwise control the burgeoning stream of waste materials. Our data table **Maine SW** contains quarterly measurements of the tonnage of solid waste delivered to an energy recovery facility in Maine. The data come from five municipalities in the state.
    a. Bucksport is a lovely coastal community with a population of approximately 5,000 people. Use the available data from other Maine cities and towns to develop a multiple regression model for the quarterly tonnage of solid waste in Bucksport. Report on your choice of a model.
    b. It seems unlikely that the volume of solid waste in one town can "cause" the generation of solid waste in another community. How can you explain the relationships that your model represents?

---

[1] We actually can choose more than one Y for separate models, but we will restrict our coverage of the topic to one response variable.

2 For pedagogical reasons, the **Effect Summary** and **Lack of Fit** panels have been hidden here. As noted in a prior chapter, the lack of fit statistics are beyond the scope of most introductory courses.

3 For a comprehensive source on the subject, see Hastie, Tibshirani, and Friedman, 2009.

[4] Why this value? It is the ZIP code for Cary, NC, home of JMP.

5 The abbreviation stands for the corrected Akaike's Information Criterion. For an extensive explanation, see "Likelihood, AICc, and BIC" in the JMP Documentation Library book *Fitting Linear Models*.

[6] In the correlation matrix, very strong positive values appear in dark blue and very strong negative correlations are dark red. The color gradient goes from dark red through gray to dark blue. Weak or insignificant correlations are pale gray.

[7] For full coverage of the Formula Depot platform and its capabilities, see Chapter 12 of *Predictive and Specialized Modeling* in the JMP Documentation Library on your **Help** menu.

# Chapter 19: Categorical, Curvilinear, and Non-Linear Regression Models


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## Overview

In the past several chapters, we have worked extensively with regression analysis. Two common threads have been the use of continuous data and of linear models. In this chapter, we introduce techniques to accommodate categorical data and to fit several common curvilinear patterns. Throughout this chapter, all the earlier concepts of inference, residual analysis, and model fitting still hold true. We will concentrate here on issues of *model specification*, which is to say selecting variables and functional forms (other than simple straight lines) that reasonably and realistically are suitable for the data at hand.

## Dichotomous Independent Variables

Regression analysis operates upon numerical data, but for some response variables, a key factor is categorical. We can easily incorporate categorical data into regression models with a *dummy*, or *indicator, variable*—a numeric variable created to represent the different states of a categorical variable. In this chapter, we will illustrate the use of dummy variables for *dichotomous* variables, which are variables that can take on only two different values.

For our first example, we will return to the sonatas of Haydn and Mozart and some data that we first analyzed in Chapter 15 (Ryden, 2007). In the sonata form, the composer introduces a melody in the exposition, and then elaborates and recapitulates that melody in a second portion of the piece. The theory that we tested in Chapter 15 was that Haydn and Mozart divided their

sonatas so that the relative lengths of the two portions approximated the golden ratio. We hypothesized that our fitted line would come close to matching this line:

$$\text{Parta} = 0 + 0.61803(\text{Partb})$$

In this chapter, we will ask whether the two composers did so with the same degree of consistency. We will augment our simple regression model of Chapter 13 in two ways. First, we will just add a dummy variable to indicate which composer wrote which sonata.

1. Open the data table called **Sonatas**.

This data table combines the observations from the two tables **Mozart** and **Haydn**. The dummy variable is column 4. Notice that the variable equals 0 for pieces by Haydn and 1 for pieces by Mozart.

2. Select **Analyze ► Fit Model**. Cast **Parta** as **Y**, add **Partb** and **Mozart** to the model, then run the model. Check **Keep dialog open**. The parameter estimates appear in Figure 19.1.

**Figure 19.1: Estimated Coefficients for the Model with a Dummy Variable**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | -0.339905 | 2.112592 | -0.16 | 0.8727 |
| Partb | 0.585374 | 0.027251 | 21.48 | <.0001* |
| Mozart | 5.1794059 | 1.797016 | 2.88 | 0.0054* |

The fitted equation is:

$$\widehat{Parta} = -0.34 + 0.585(Partb) + 5.179(Mozart)$$

How do we interpret this result? Remember that **Mozart** is a variable that can equal only 0 or 1. Therefore, this equation actually expresses two parallel lines:

$$\text{Haydn Sonatas:} \quad \widehat{Parta} = -0.34 + 0.585(Partb)$$
$$\text{Mozart Sonatas:} \quad \widehat{Parta} = (5.179 - 0.34) + 0.585(Partb)$$

Both lines have the same slope but different intercepts. If we want a model that allows for two intersecting lines, we can introduce an *interaction* between the independent variable and the dummy variable, as follows:

3. Return to the **Fit Model** dialog box, which is still open in its own tab.
4. Under **Select Columns** in the upper left, highlight both **Partb** and **Mozart** in the **Select Columns** box and click **Cross** under **Construct Model Effects**. You will now see the term **Partb*Mozart** listed in the **Construct Model Effects** panel.
5. Now click **Run**.

Figure 19.2 shows the parameter estimates for this model, which fits the data slightly better than the previous model. With an interaction, JMP reports the coefficient in terms of adjusted

values of the variables, subtracting the sample mean for each variable. We note that all estimates except for the intercept are statistically significant.

**Figure 19.2: Results with a Dummy Interaction Term**

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 0.6247019 | 2.109818 | 0.30 | 0.7682 |
| Partb | 0.5630538 | 0.028647 | 19.65 | <.0001* |
| Mozart | 5.4173436 | 1.754171 | 3.09 | 0.0030* |
| (Partb-74.4063)*(Mozart-0.45313) | 0.1150727 | 0.055546 | 2.07 | 0.0426* |

Just as the first model implicitly represented two intercept values, this model has two intercepts and two slopes: there is a Mozart line and a Haydn line. The default report is a bit tricky to interpret because of the subtraction of means in the interaction line, so let's look graphically at the fitted lines. We will take advantage of yet another useful visual feature of JMP that permits us to re-express this regression model using the original dichotomous categorical column, **Composer**.[1]

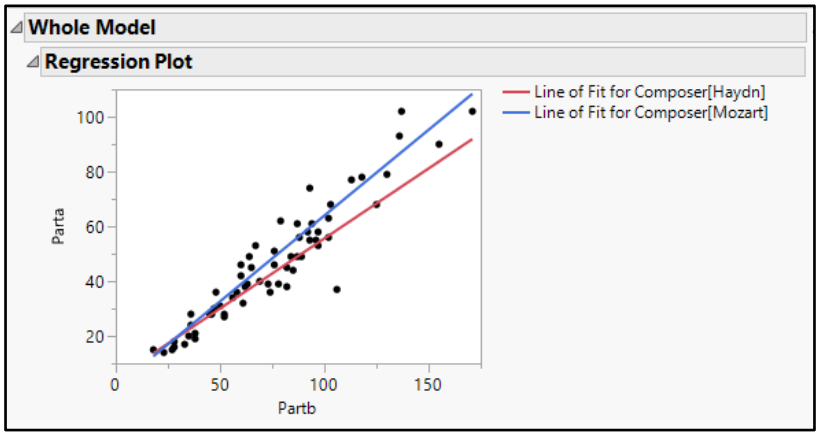
6. Return again to the **Fit Model** dialog box and highlight **Mozart** and **Partb*Mozart** within the list of model effects. Click the **Remove** button.

7. Select the **Composer** column and add it to the model effects.

8. Highlight **Partb** and **Composer** under **Select Columns** and click **Cross**. The model now lists **Partb**, **Composer**, and **Partb*Composer**. Click **Run**.

Because this regression model has just a single continuous independent variable, JMP automatically plots the regression fit above the scatterplot of fits versus observed values. Your output should now include the graph shown here in Figure 19.3. The red line represents the Haydn sonatas, and the blue represents those by Mozart.

**Figure 19.3: Estimated (Fitted) Lines for the Two Composers**



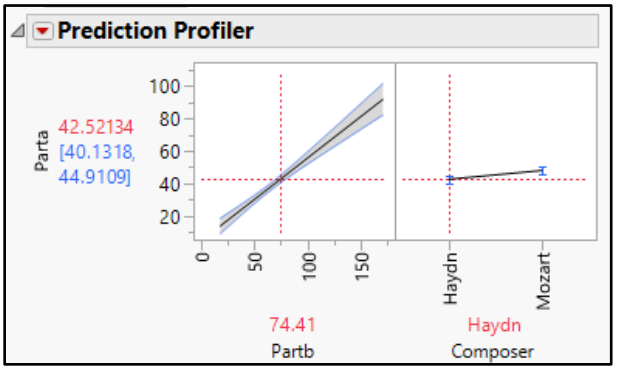
In the table of **Parameter Estimates**, we do find discrepancies between this model and the estimates shown in Figure 19.2. Now, the estimated coefficient of Partb is the same as before, but the others are different. The *t*-ratios and *P*-values are all the same, but the parameter estimates have changed. This is because JMP automatically codes a dichotomous categorical variable with values of -1 and +1 rather than 0 and 1. Hence, the coefficient of the composer variable is now -2.708672, which is exactly one-half of the earlier estimate. This makes sense, because now the difference between values of the dummy variable is twice what it was in the 0-1 coding scheme.

## Seeing Interaction Effects with the Prediction Profiler

In the prior chapter we used the Profiler to visualize the separate effects of factors in a multiple regression model. This model, containing both a categorical dichotomous factor plus an interaction term, presents a good opportunity to understand more about regression modeling.

1. Click the **Response Parta** red triangle, and choose **Factor Profiling ► Profiler**.

**Figure 19.4: Prediction Profiler for Model with Interaction Term**




This opens the Profiler, shown with its initial settings in Figure 19.4. The vertical red line in the Partb graph is set at its mean value, and the Composer graph set at Haydn. Among Haydn's sonatas, those with theme b averaging 74.41 measures would, on average, have a theme a approximately 42.5 measures long.

2. Now grab the red dashed line in the Composer graph and switch to Mozart.

Notice two changes: the estimated interval for Parta increases, and the slope of the line in the Partb graph becomes steeper.

This vividly illustrates the meaning of interaction: the relationship between Parta and Partb differs depending on the composer. It is not a simple matter that Mozart's themes tended to be longer overall.

What happens when we change the settings for Partb length?

3. Leaving the right-hand red dashed line at Mozart, grab the vertical line in the Partb graph and slide it all the way to the right to a value of 171. Notice that the right-hand graph shows an even larger difference between the two composers.

4. Now slide the Partb vertical line all the way to its minimum. The difference between the two composers nearly vanishes.

## Dichotomous Dependent Variable

Next, we consider studies in which the dependent variable is dichotomous. In such cases there are at least two important shortcomings of ordinary least squares estimation and a linear model. First, because Y can take on only two values, it is illogical to expect that a line provides a good model. A linear model will produce estimates that reach beyond the 0-1 limits. Second, because all observed values of Y will always be either 0 or 1, it is not possible that the random error terms in the model will have a normal distribution or a constant variance.

With a dichotomous dependent variable, one better approach is to use *logistic regression.*[2] Estimating a logistic model is straightforward in JMP. To illustrate, we will turn to an example from the medical literature (Little, McSharry et al. 2008) on Parkinson's disease (PD). PD is a neurological disorder affecting millions of adults around the world. The disorder tends to be most prevalent in people over age 60, and as populations age, specialists expect the number of people living with PD to rise. Among the common symptoms are tremors, loss of balance, muscle stiffness or rigidity, and difficulty in initiating movement. There is no cure at this time, but medications can alleviate symptoms, particularly in the early stages of the disease.

People with PD require regular clinical monitoring of the progress of the disease to assess the need for changes in medication or therapy. Given the nature of PD symptoms, though, it is often difficult for many people with PD to see their clinicians as often as might be desirable. A team of researchers collaborating from University of Oxford (UK), the University of Colorado at Boulder, and the National Center for Voice and Speech in Colorado was interested in finding a way that people with PD could obtain a good level of care without the necessity of in-person visits to their physicians. The team noted that approximately 90% of people with PD have "some form of vocal impairment. Vocal impairment might also be one of the earliest indicators for the onset of the illness, and the measurement of voice is noninvasive and simple to administer" (ibid., p. 2). The practical question, then, is whether a clinician could record a patient's voice—even remotely—and by measuring changes in the type and degree of vocal impairment, draw valid conclusions about changes in progress of PD in the patient.

In our example, we will use the data from the Little, McSharry *et al.* study in a simplified fashion to differentiate between subjects with PD and those without. The researchers developed a new composite measure of vocal impairment, but we will restrict our attention to some of the standard measures of voice quality. In our sample, we have 32 patients; 24 of them have PD and eight do not.

Vocal measurement usually involves having the patient read a standard paragraph or sustain a vowel sound or both. Two common measurements of a patient's voice refer to the wave form of sound: technicians note variation in the amplitude (volume) and frequency (pitch) of the voice. We will look at three measurements:

- MDVP:Fo(Hz): vocal fundamental frequency (baseline measure of pitch)[3]
- MDVP:Jitter(Abs): *jitter* refers to variation in pitch
- MDVP:Shimmer(db): *shimmer* refers to variation in volume

In this example, we will simply estimate a logistic regression model that uses all three of these measures and learn to interpret the output. Interpretation is much like multiple regression, except that we are no longer estimating the absolute size of the change in Y given one-unit changes in each of the independent variables. In a logistic model, we conceive of each X as changing the probability, or more specifically the odds-ratio, of Y. Fundamentally, in logistic regression, the goal of the model is to estimate the likelihood that Y is either "yes" or "no" depending on the values of the independent variables.

More specifically, the model estimates $\ln[p/(1-p)]$, where $p$ is the probability that Y = 1. Each estimated coefficient represents the change in the estimated log-odds ratio for a unit change in X, holding other variables constant—hardly an intuitive quantity! Without becoming too handcuffed by the specifics, consider at this point that a positive coefficient only indicates a factor that increases the chance that Y equals 1, which in this case means that a patient has PD.

JMP produces logistic parameter estimates using a *maximum likelihood estimation* (*MLE*)[4] approach rather than the usual OLS (ordinary least squares method). For practical purposes, this has two implications for us. First, rather than relying on F or t tests to evaluate significance, JMP provides chi-square test results, which are more appropriate to MLE methods. Second, the residual analysis that we typically perform in a regression analysis is motivated by the assumptions of OLS estimation. With MLE, the key assumption is the independence of observations, but we no longer look for linearity, normality, or equal variance.

1. Open the **Parkinson's** data table.

   Each row in the table represents one subject (person) in the study. The **Status** column is a dummy variable where 1 indicates a subject with PD. Hence, this is our dependent variable in this analysis. In the data table, **Status** is numeric but has a nominal modeling type.
2. Select **Analyze ► Fit Model**. Select **Status** as **Y** and add **MDVP:Fo(Hz)**, **MDVP:Jitter(Abs)**, and **MDVP:Shimmer(dB)** as model effects. Select with care, as there are several jitter and shimmer columns.
3. In the upper right of the dialog box, change the Target Level to 1 to specify that we want our model to predict the presence of Parkinson's disease (that is, **Status = 1**). Click **Run**.

Notice that as soon as you specify **Status** as the dependent variable, the personality of the **Fit Model** platform switches from **Standard Least Squares** to **Nominal Logistic**.

Part of the logistic regression report appears in Figure 19.5. Next, we review selected portions of several of the panels within the results. For complete details about each portion of the output, consult the JMP Help menu. You can click the Help menu, or select the Help tool from the Tools menu and then hover and click over any portion you want to learn about.

**Figure 19.5: Logistic Regression Results for the Parkinson's Model**

**Nominal Logistic Fit for Status**

**Effect Summary**

| Source | LogWorth | PValue |
|---|---|---|
| MDVP:Shimmer(dB) | 1.119 | 0.07603 |
| MDVP:Fo(Hz) | 0.580 | 0.26318 |
| MDVP:Jitter(Abs) | 0.008 | 0.98263 |

Remove Add Edit ☐ FDR

Converged in Gradient, 7 iterations

**Iterations**

**Whole Model Test**

| Model | -LogLikelihood | DF | ChiSquare | Prob>ChiSq |
|---|---|---|---|---|
| Difference | 6.123813 | 3 | 12.24763 | 0.0066* |
| Full | 11.870912 | | | |
| Reduced | 17.994725 | | | |

| | |
|---|---|
| RSquare (U) | 0.3403 |
| AICc | 33.2233 |
| BIC | 37.6048 |
| Observations (or Sum Wgts) | 32 |

**Fit Details**

**Lack Of Fit**

**Parameter Estimates**

| Term | Estimate | Std Error | ChiSquare | Prob>ChiSq |
|---|---|---|---|---|
| Intercept | 1.27397634 | 3.7917666 | 0.11 | 0.7369 |
| MDVP:Fo(Hz) | -0.020608 | 0.0192815 | 1.14 | 0.2852 |
| MDVP:Jitter(Abs) | -1570.7363 | 72190.346 | 0.00 | 0.9826 |
| MDVP:Shimmer(dB) | 15.150141 | 12.698194 | 1.42 | 0.2328 |

For log odds of 1/0

**Covariance of Estimates**

**Effect Likelihood Ratio Tests**

| Source | Nparm | DF | L-R ChiSquare | Prob>ChiSq |
|---|---|---|---|---|
| MDVP:Fo(Hz) | 1 | 1 | 1.2519399 | 0.2632 |
| MDVP:Jitter(Abs) | 1 | 1 | 0.00047388 | 0.9826 |
| MDVP:Shimmer(dB) | 1 | 1 | 3.14769625 | 0.0760 |

## Effect Summary

We have seen this panel in least squares regression analyses. It lists the three factors in order of statistical significance, and we see that none are significant at the 0.05 level (represented by the blue vertical line), but shimmer is significant at the 10% level.

### Whole Model Test

This portion of the output compares the performance of our specified model with a naïve default model with no predictor variables. The naïve model essentially posits that, based on the sample ratio of 8/32 or 25% of the patients having no disease, there is a probability of 0.75 that a tested patient has PD. In this panel, the chi-square test is analogous to the customary F test and in this analysis, we find a significant result. In other words, this model is an improvement over simply estimating a 0.75 probability of PD.

We also find **RSquare (U)**, which is analogous to R square as a goodness-of-fit statistic. This model provides an unimpressive value of 0.34, indicating a lot of unexplained variation. Such a result is common in logistic modeling due in part to the fact that we are often using continuous data to predict a dichotomous response variable.

### Parameter Estimates

As noted earlier, we do not interpret the parameter estimates directly in a logistic model. Because each estimate is the marginal change in the log of the odds-ratio, sometimes we do a bit of algebra and complete the following computation for each estimated $\beta_i$:

$$e^{\hat{\beta}} - 1$$

The result is the estimated percentage change in the odds corresponding to a one-unit change in the independent variable. Of course, we want to interpret the coefficients only if they are statistically significant. To make judgments about significance, we look at the next panel of output.

### Effect Likelihood Ratio Tests

The bottom panel contains the analogs to the t tests we usually consult in a regression analysis, except that these are based on chi-square rather than t-distributions. We interpret the *P*-values as we do in other tests, although in practice we might tend to use higher alpha levels, given the discontinuous nature of the data. In this example, it appears that there is no significant effect associated with jitter, but shimmer (variation in vocal loudness) does look as if it makes a significant contribution to the model in distinguishing PD patients from non-PD patients. We should hold off drawing any conclusion about the base frequency (**MDVP:Fo(Hz)**) until running an alternative model without the jitter variable. For now, we will suspend the pursuit of a better model for Parkinson's disease prediction, and move on to our next topic.

## Curvilinear and Non-Linear Relationships

Earlier in this book, we have seen examples in which there is a clear relationship between X and Y, but the relationship either is a linear combination of powers or roots or are non-linear (for example, when the X appears in an exponent). In some disciplines, there are well-established theories to describe a relationship, such as Newton's laws of motion. In other disciplines, non-linear relationships might be less clear. In fact, researchers often use regression analysis to

explore such relationships. This section of the chapter introduces two common functions that frequently do a good job of fitting curvilinear data.

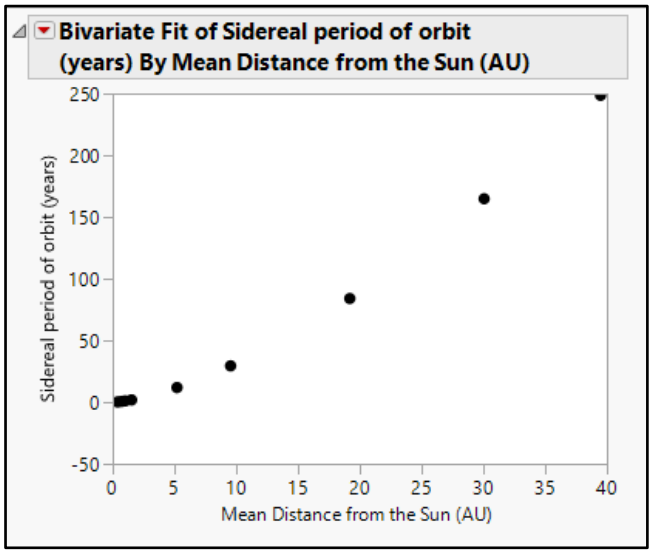
## Quadratic Models

We begin with a simple descriptive example of a well-known quadratic relationship. A *quadratic* function is one that includes an $X^2$ term. Quadratic functions belong to the class of *polynomial functions*, which are defined as any functions that contain several terms with X raised to a positive integer power. Let's see how this plays out in our solar system.

1. Open the **Planets** data table. This table contains some basic measurements and facts about the planets. For clarity in our later graphs, the table initially selects all rows and assigns markers that are larger than usual.
2. Select **Analyze ► Fit Y by X**. Cast **Sidereal period of orbit (years)** as **Y** and **Mean Distance from the Sun (AU)** as **X** and click **OK**.

The *sidereal period of orbit* is the time, in Earth years, that it takes for a planet to complete its orbit around the sun. *AU* refers to an astronomical unit, which is defined as the distance from the Earth to the sun. So, our response variable is the length of a planet's orbit relative to Earth, and the regressor is the mean distance (as a multiple of the Earth's distance) from the sun. Figure 19.6 shows the scatterplot.

**Figure 19.6: Orbital Period Versus Distance from the Sun**

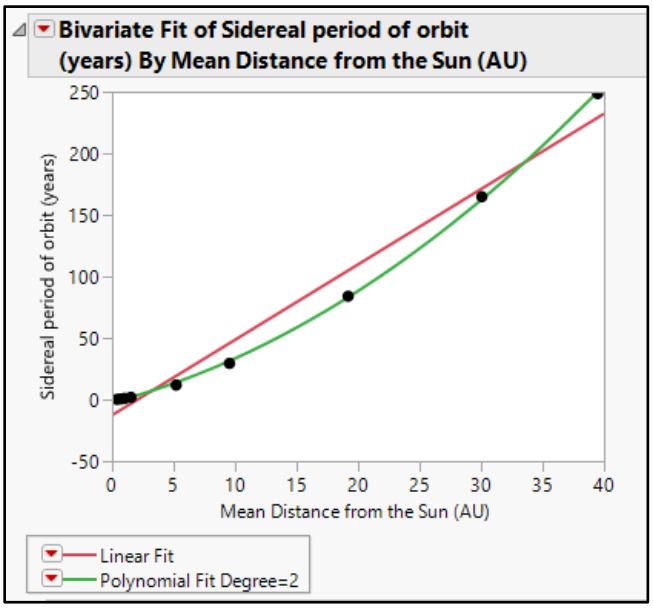



The relationship between these two columns is quite evident, but if we fit a straight line, we will notice that the points curve. If we fit a quadratic equation, we get a near-perfect fit.

3. Click the red triangle and select **Fit Line**.
4. Now click the red triangle again and select **Fit Polynomial ► 2, quadratic.**

Your analysis report should now look like Figure 19.7. Although the linear model is a good fit, the quadratic model more accurately describes this relationship.

**Figure 19.7: Quadratic and Linear Models for Planet Data**



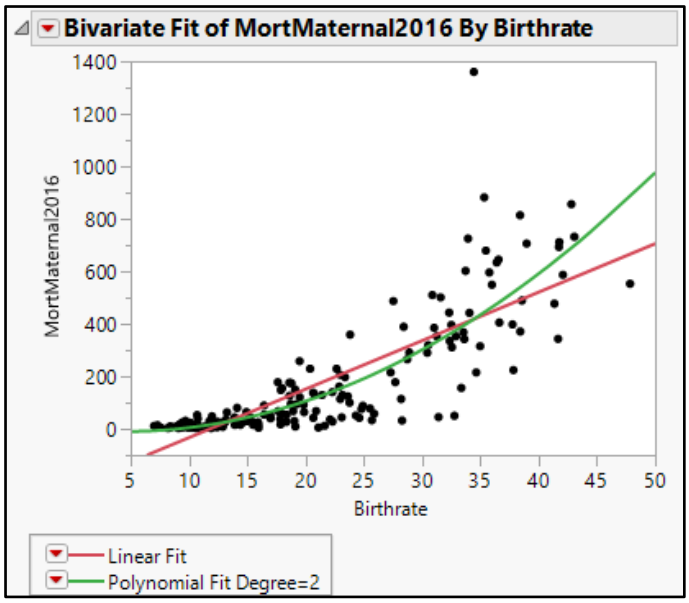

We can use quadratic models in many settings. We introduced regression in Chapter 4 using some data in the data table **Birthrate 2017**. This data table contains several columns related to the variation in the birth rate and the risks related to childbirth around the world as of 2017. In this data table, the United Nations reports figures for 193 countries. You might remember that we saw a distinctly non-linear relationship in that data table. Let's look once more at the relationship between a nation's birth rate and its rate of maternal mortality.

5. Open the **Birthrate 2017** data table.

6. Select **Analyze ► Fit Y by X**. Cast **MortMaternal2016** as **Y** and **BirthRate** as **X** and click **OK**.

We have seen this graph at least twice before. Your results will look like those shown in Figure 19.8, which shows the relationship with both linear and quadratic fits. When we last encountered this graph, we concluded that it would be unwise to fit a linear model, and that we should return to it again in this chapter. Here we are.

**Figure 19.8: Relationship Between Maternal Mortality and Birth Rate**




If we were to fit a line, we would find that a straight-line model is a poor characterization of this pattern. A line will underestimate maternal mortality in countries with very low or very high birth rates, and substantially overstate it in countries with middling birth rates. Given the amount of scatter in the graph, we are not going to find a perfect fit, but let's try the quadratic model as an alternative to a line.

7. Click the red triangle to fit a line.

8. Click the red triangle once more and select **Fit Polynomial ► 2, quadratic**.

The goodness-of-fit statistic, R square, is similar for the two models, but the quadratic model captures more of the upward-curving relationship between these two variables. Interpretation of coefficients is far less intuitive than in the linear case, because we now have coefficients for the X and the $X^2$ terms, and because whenever X increases by one unit, $X^2$ also increases.

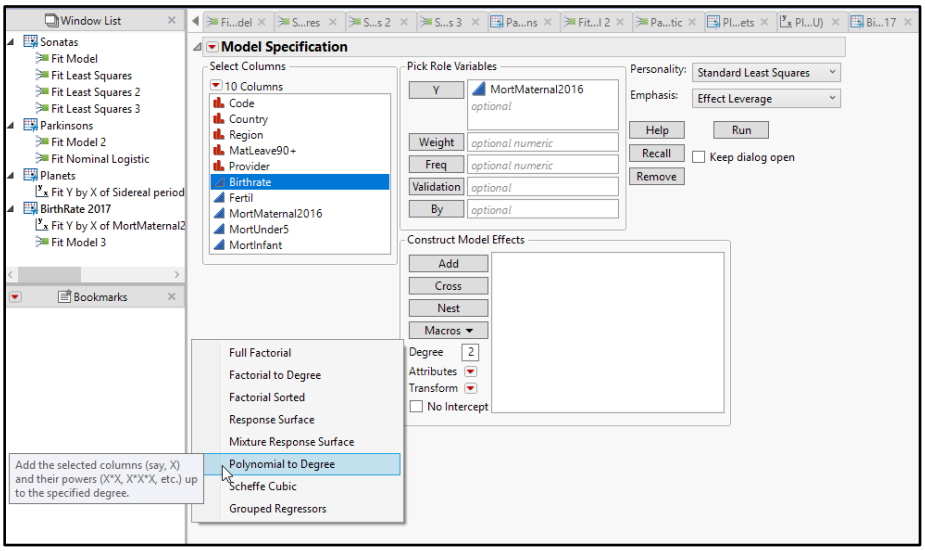
In the parameter estimates for the squared term, notice that JMP subtracts the mean (20.8442) from Birthrate before squaring. This step is referred to as centering. If you imagine a parabola formed by $X^2$, centering moves the bottom of the parabola to zero.

Thus far we have used the **Fit Y by X** analysis platform, which restricts us to a single independent variable. If we want to investigate a multivariate model with a combination of quadratic and linear terms and additional variables, we can use the **Fit Model** platform.

9. Select **Analyze ► Fit Model.** Again, cast **MortMaternal2016** as **Y.**

10. Highlight **BirthRate**, but don't click anything yet. Notice the **Macros** button, and the **Degree** box below it. By default, degree is set at 2. Click **Macros**, as shown in Figure 19.9.

**Figure 19.9: Building a Polynomial Model**




Among the macros, you will see **Polynomial to Degree**. This choice specifies that you want to add the highlighted variable and its powers up to and including the value in the **Degree** box. In other words, we want to specify a model with X and $X^2$.

We can use this command to build a polynomial model with any number of factors. If you set **Degree** = 4, the model would include terms for **BirthRate**, **BirthRate2**, **BirthRate3**, and **BirthRate4**.

11. Select **Polynomial to Degree** from the drop-down menu. You will see the model now includes **BirthRate** and **BirthRate*BirthRate**.

12. Click **Model**.

The model results (not shown here) are fundamentally identical to the ones that we generated earlier. The **Fit Model** platform provides more diagnostic output than the **Fit Y by X** platform, and we can enlarge our model to include additional independent variables.

## Logarithmic Models

Of course, there are many curvilinear functions that might be suitable for a particular study. For example, when a quantity grows by a constant *percentage* rate rather than a constant absolute amount, over time, its growth will fit a logarithmic[5] model. To illustrate, we will look at the number of cell phone users in Sierra Leone, a Sub-Saharan African nation that is among the poorest in the world. The population was approximately 7.5 million people in 2017, with a rate of 87.66 cell subscriptions per 100 people.

1. Open the data table called **Cell Subscribers**.

This is a table of stacked data, representing 215 countries from 1990 through 2018. We will start by extracting a subset table containing the Sierra Leone (**Country Code = SLE**) data.

2. Select **Rows ► Row Selection ► Select Where**. Select rows where **CountryCode** equals **SLE** and click **OK**.

We want to analyze only the 29 selected rows. We could exclude and hide all others, but let's just subset this data table and create a new 29-row table with only the data from Sierra Leone.

3. Select **Tables ► Subset**. Click the radio buttons corresponding to **Selected Rows** and specify that you want **All Columns**.

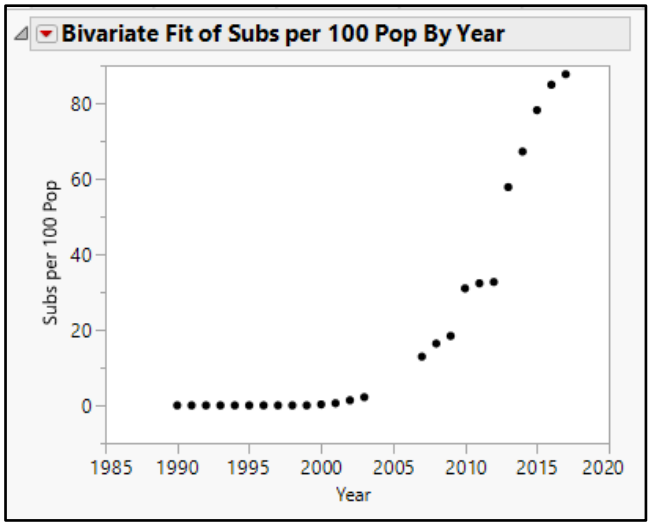
4. In the **Output table name** box, type **SLE Cell Subscribers**.

5. Click **OK**.

As we did earlier, we will learn to create a logarithmic model using both curve-fitting platforms.

6. Select **Analyze ► Fit Y by X**. Cast **Subs per 100 pop** as **Y** and **Year** as **X**. The result looks like Figure 19.10.

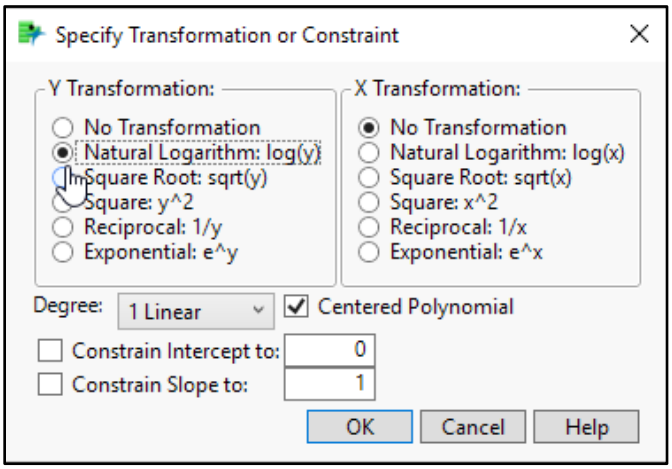
**Figure 19.10: Growth of Cellular Subscriptions in Sierra Leone**




Notice the pattern of these points. For the first several years there is a very flat slope at or near zero subscribers, followed by rapid growth. This distinctive pattern is a common indication of steady *percentage* growth in a series of values.

7. Click the red triangle and select **Fit Special**. As shown in Figure 19.11, we will transform the response variable by taking its natural log and leave X alone.

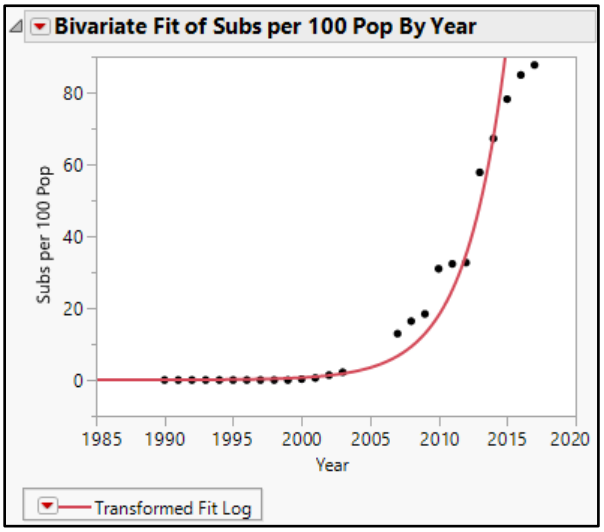
**Figure 19.11: Log Transformation of Y**




You will receive an alert message saying, "Transformation has 10 invalid arguments." There are 10 years with zero cell subscribers, and the natural log of 0 is undefined.

The scatterplot now looks like Figure 19.12. Here we see a smooth fitted curve that maps closely to the points and follows that same characteristic pattern mentioned above. The fitted equation is **Log(Subs per 100 Pop) = -662.7366 + 0.3311572*Year**. The fit is not perfect but should provide a sense of how we can fit a different curved equation to a set of points.

**Figure 19.12: A Log-Linear Model—Log(Y) Versus X**




The coefficient of **Year** tells us how the logarithm of subscribers increases each year. How can we use this coefficient to estimate the annual percentage growth rate of cellular phone subscriptions in Sierra Leone from 1990 through 2017? We need to do some algebra starting

with a model of steady annual percentage growth. If we let $Y_0$ represent the initial value of Y and let *r* equal the unknown annual growth rate, then the number of subscribers per 100 population at any future time, *t*, equals

$$Y_t = Y_0(1+r)^t$$

If we take the log of both sides of this equation, we get the following:

$$\ln(Y_t) = \ln(Y_0) + t(\ln(1+r))$$

Notice that because $Y_0$ is a fixed value, this function is linear in form. The intercept of the line is $\ln(Y_0)$, the slope of the line is $\ln(1+r)$, and the independent variable is *t.* Moreover, we solve for the annual rate of increase, *r*, as follows:

$$\hat{\beta}_1 = \ln(1+r)$$
$$e^{\hat{\beta}_1} = 1+r$$
$$r = e^{\hat{\beta}_1} - 1$$

So, in this model, the estimated slope is .3311572. If we raise *e* to that power and subtract 1, we find the rate: $e^{.3311572} - 1 = 1.3926 - 1 = .3926$, or 39.26% per year.

We can also estimate a log-linear model using the **Fit Model** platform. As we did earlier, we will transform the response variable to equal the log of the number of subscriptions. This time we will use a different strategy to transform Y because the Transform feature under Construct Model Effects will stall out if we try to take logs of zero values. Either we need to hide and exclude the zeros, or do this.

8. Select **Analyze ► Fit Model**.
9. Under **Select Columns**, right-click **Subs per 100 Pop** then **Transform ► Log.**
10. Cast the new temporary column as Y and add Year to the model.
11. **Run** the model.

By this point, you should be familiar enough with the results from the **Fit Model** platform that you can recognize that they match those reported from **Fit Y by X**. This command also gives us leverage and residual plots and permits the addition of other regressors to build more complex multiple regression models.

## More Non-Linear Functions

This chapter has introduced two common non-linear transformations that are often covered in a first course in statistics. Beyond quadratic and logarithmic models, there are many more curvilinear functions. Curious readers should investigate the **Fit Curve** platform in JMP. You will find the platform via the **Analyze** menu as the first option under **Specialized Modeling**.

**Fit Curve** expands upon the familiar **Fit Y by X** platform by offering numerous alternative curvilinear models. For each model chosen, JMP will estimate model parameters and maintain a running comparison of the model results. Although some of these functions will likely be unfamiliar to introductory readers, it is very much worth knowing that there are other ways forward if linear, quadratic, and logarithmic models do not fulfill the needs of an investigation.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* Return to the **NHANES2016** data table. Exclude and hide respondents under age 18, leaving both female and male adults.
    a. Perform a regression analysis for adult respondents with BMI as Y and use waist circumference and gender as the independent variables. The relevant columns are **BMXBMI, BMXWAIST**, and **RIAGENDR**. Discuss the ways in which this model is or is not an improvement over a model that uses only waist circumference as a predictor.
    b. Perform a regression analysis extending the previous model, and this time, add an interaction term between waistline and gender. Report and interpret your findings.

2. *Scenario:* High blood pressure continues to be a leading health problem in the United States. In this problem, continue to use the **NHANES2016** table. For this analysis, we will focus on just the following variables and focus on adolescent respondents between the ages of 12 and 19.
    a. Perform a regression analysis with systolic BP (**BPXSY1**) as the response and gender, age, and weight (**RIAGENDR**, **RIAGEYR**, **BMXWT**) as the factors. Report on your findings.
    b. Perform a regression analysis with systolic BP as Y and gender, age, weight, and diastolic blood pressure as the independent variables. Explain fully what you have found.
    c. Extend this second model to investigate whether there is an interaction between weight and gender. Use the Profiler in your investigation and report on what you find.

3. *Scenario:* We will continue to examine the World Development Indicators data in **BirthRate 2017**. Throughout the exercise, we will use **BirthRate** as the **Y** variable. We will broaden our analysis to work with other variables in that file:

    a. Earlier, we estimated a quadratic model with maternal mortality as the independent variable. To that model, add the categorical variable **MatLeave90+**, which indicates whether there is a national policy to provide 90 days or more maternity leave. Evaluate your model and report on your findings.
    b. Investigate whether there is any interaction between the maternity leave dummy and the rate of maternal mortality. Report on your findings, comparing these results to those we obtained by using the model without the categorical variable.

4. *Scenario*: The United Nations and other international organizations monitor many forms of technological development around the world. Earlier in the chapter, we examined the growth of mobile phone subscriptions in Sierra Leone. Let's repeat the same analysis for two other countries using the **Cell Subscribers** data table.

   HINT: Rather than starting with a subset of the full table, launch your analysis with **Fit Y by X,** and then use the **Local Data Filter** to switch countries.

   a. Develop a log-linear model for the growth of cell subscriptions in Denmark. Compute the annual growth rate.
   b. Develop a log-linear model for the growth of cell subscriptions in Malaysia. Compute the annual growth rate.
   c. Develop a log-linear model for the growth of cell subscriptions in the United States. Compute the annual growth rate.
   d. We have now estimated four growth rates in four countries (including Sierra Leone). Compare them and comment on what you have found.

5. *Scenario*: Earlier in the chapter, we estimated a logistic model using the Parkinson's disease (PD) data. The researchers reported on the development of a new composite measure of phonation, which they called **PPE**.

   a. Run a logistic regression using **PPE** as the only regressor, and **Status** as the dependent variable. Report the results.
   b. Compare the results of this regression to the one illustrated earlier in the chapter. Which model seems to fit the data better? Did the researchers succeed in their search for an improved remote indicator for PD?

6. *Scenario*: Occasionally, a historian discovers an unsigned manuscript, musical composition, or work of art. Scholars in these fields have various methods to infer the creator of such an unattributed work, and sometimes turn to statistical methods for assistance. Let's consider a hypothetical example based on the **Sonatas** data.

   a. Run a logistic regression using **Parta** and **Partb** as independent variables and **Composer** as the dependent variable. Report the results.
   b. (Challenge) Suppose that we find a composition never before discovered. Parta is 72 measures long and Partb is 112 measures long. According to our model, which composer is more likely to have written it? Why?

7. *Scenario*: The United Nations and other international organizations monitor many forms of technological development around the world. Earlier in the chapter, we examined the growth of mobile phone subscriptions in Sierra Leone. Let's look at one year and examine the relationships between adoption levels for several key communication technologies. Open the data table called **SDG_Technology** and use the global data filter to show and include data for the year 2017.

    a. Develop linear and quadratic models for the number of cellular subscribers per 100 population, using telephone lines per 100 population as the independent variable. Which model fits the data better?
    b. Develop linear and quadratic models for the number of cellular subscribers per 100 population, using internet users per 100 population as the independent variable. Which model fits the data better?
    c. Develop linear, quadratic, and cubic (polynomial, 3rd degree) models for the number of cellular subscribers per 100 population, using ATMs per 100,000 adult population as the independent variable. Which model fits the data best?

8. *Scenario*: The data table called **Global Temperature and CO2** contains estimates and readings for two important environmental measures: the concentration of carbon dioxide (CO2 parts per million) in the atmosphere and mean global temperature (degrees Fahrenheit).

    a. Using all available years' data, develop linear, quadratic, and log-linear models for the CO2 concentration as Y, and Year as X. Which model fits the data best?
    b. Use the Data Filter to include and show just the years since 1880, and repeat the previous question using the global temperature variable as Y. Which model fits the data best?

---

[1] You might wonder why we bother to create the dummy variable if JMP will conduct a regression using nominal character data. Fundamentally, the software is making the 0-1 substitution and saving us the trouble. This is one case where the author believes that there is value in working through the underlying numerical operations. Furthermore, the 0-1 approach initially makes for slightly clearer interpretation.

[2] Logistic regression is one of several techniques that could be applied to the general problem of building a model to classify, or differentiate between, observations into two categories. Logistic models are sometimes called *logit* models.

[3] MDVP stands for the Kay-Pentax Multi-Dimensional Voice Program, which computes these measures based on voice recordings.

[4] For further background, see Freund, R., R. Littell, et al. (2003). *Regression Using JMP*. Cary, NC: SAS Institute Inc. Or see Mendenhall, W. and T. Sincich (2003). *A Second Course in Statistics: Regression Analysis*. Upper Saddle River, NJ: Pearson Education.

[5] If one transforms Y by taking the log and leaves X untouched, this is called a log-linear model.

# Chapter 20: Basic Forecasting Techniques

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## Overview

For much of this book, we have examined variation across individuals measured simultaneously. This chapter is devoted to variation over time, working with data tables in which columns represent a single variable measured repeatedly and rows represent regularly spaced time intervals. In many instances, *time series* data exhibit common, predictable patterns that we can use to make forecasts about future time periods. As we move through the chapter, we will learn several common techniques for summarizing variation over time and for projecting future values.

> It is important to note that a time series data set, by definition, contains measurements at pre-determined intervals, like daily or hourly. This is different from event data that also occurs over time, but is only recorded when an event occurs (like a keystroke, an earthquake, or a patient arriving in an emergency room).

## Detecting Patterns Over Time

Typically, we classify patterns in a time series as a combination of the following:

- **Trend**: Long-term movement upward or downward.
- **Cycle**: Multi-year oscillations with regular frequency
- **Season**: Repeated, predictable oscillations within a year
- **Irregular**: Random movement up or down, akin to random disturbances in regression modeling

To illustrate these patterns, we will first consider industrial production in India. India is one of the largest and fastest-growing economies[1] in the world. Industrial production refers to the manufacture of goods within the country and is measured in this example by India's *Index of Industrial Production* (IIP). The IIP is a figure that compares a weighted average of industrial output from different sectors of the economy (for example, infrastructure, basic goods, consumer durables, and so on) to a reference year in which the index is anchored at a value of 100. As a result, we can interpret a value of the IIP as a percentage of what industrial output was in the base year. So, for example, if the IIP equals 150, we know that production was 50% higher at that point than it was in the base year.

1. Open the table called **India Industrial Production**.

This table contains IIPs for different segments of the Indian economy. All figures are reported monthly from April 2012 through May 2019. In this section, we will develop some models to forecast one quarter (three months) beyond the available data.

Our first step will be to hide and exclude the last three rows of the table. Standard practice in time series analysis is to hold some observations in reserve while building predictive models, and then test the performance of the models comparing forecasts to later observations.

2. Highlight rows 84 through 86, then select **Rows ► Hide and Exclude** to reserve these rows to test the models.

Let's look at the variability in production of primary goods (processed goods such as paper, lumber, and textiles valued for their use in the manufacture of other products). We will look at the variation using a *Time Series Chart*, which simply plots each observed data value in order. We will also look at the patterns (if any) of autocorrelation in the series, which is the extent to which monthly values of the index of basic goods are correlated with prior values.

3. Select **Analyze ► Specialized Modeling ► Time Series**. As shown in Figure 20.1, select **Primary** as the **Y, Time Series**, and cast **Month#** in the role of **X, Time ID**.
4. In the lower left of the dialog box, change the number of **Forecast Periods** to 3.
5. Click **OK.**

**Figure 20.1: Launching the Time Series Platform**

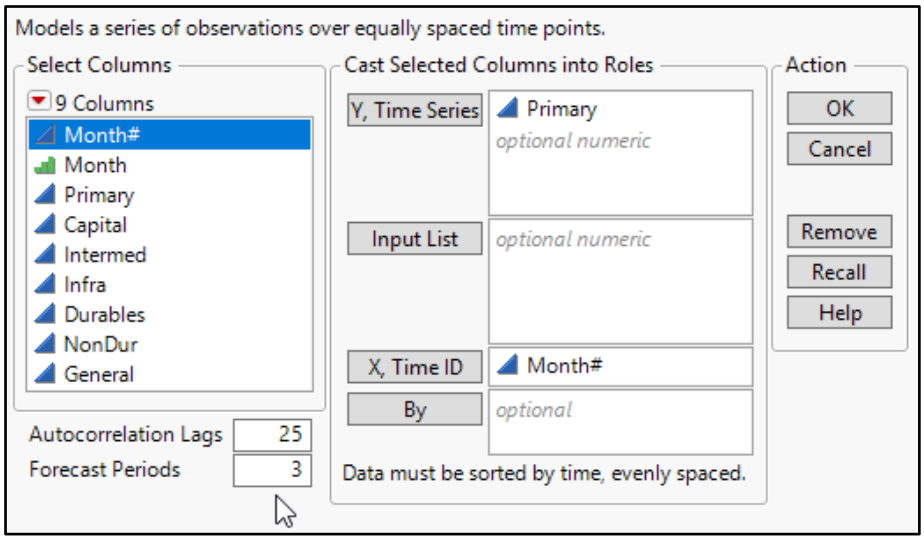
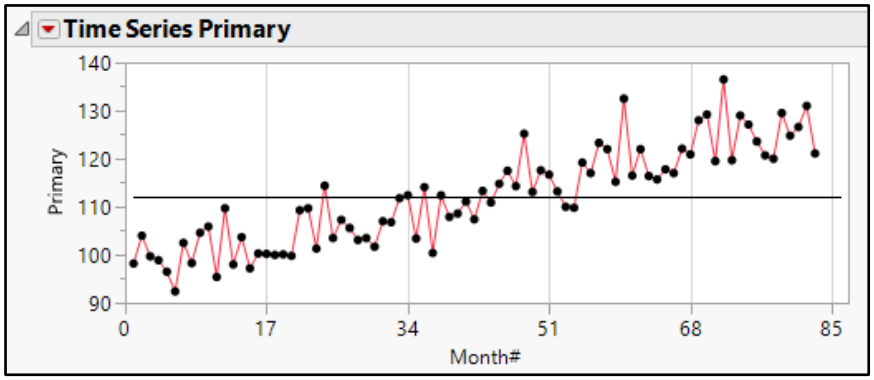



**Figure 20.2: Time Series Graph of Production of Primary Goods**




Look at the first simple time series chart shown in Figure 20.2. The horizontal axis shows the months beginning in April 2012, and the vertical axis shows the values of the IIP for primary (basic) goods. Because we have hidden and excluded months 84 through 86, the plot ends at month 83. The upward trend is visible, but it is not a steady linear trend; it has some undulation that suggests seasonal and cyclical components. Also visible is a sawtooth pattern of ups and downs from month to month, indicating some irregular movement. In short, this first example exhibits all four of the elements often found in time series data.

Below the graph are two other graphs and a table of diagnostic statistics. Although they go beyond the typical content of an introduction to time series techniques, these are a basic feature of the JMP Time Series platform. The two graphs illustrate the extent to which each observation of the time series is correlated with prior observations in recent prior periods. If a time series is stationary *(*without trend), the autocorrelation plot on the left side will have bars of random length and sign. A nonstationary series will have bars that gradually diminish in length. The

section about autoregressive models later in the chapter has more to say about using previous values of Y to forecast future values.

In the next two sections of the chapter, we will introduce two common methods for smoothing out the irregular sawtooth movement in a time series, both to better reveal the other components and to develop short-term forecasts. We will also introduce some statistics to summarize the performance of the models.

It is valuable to understand at the outset that the techniques of time series analysis share a characteristic logic. In each of the techniques presented in this chapter, we will apply the method to a sample of observations to generate retrospective "forecasts" for all or most of the observed data values. We then compare the forecasts to the values that were actually observed, essentially asking "if we had been using this forecasting method all along, how accurate would the forecasts have been?" We will then select a forecasting method for continued use, based in part on its past performance with a given set of data.

# Smoothing Methods

As a first step in time series analysis, it is often helpful to neutralize the effects of the random irregular component and thereby more clearly visualize the behavior of the other three components. Perhaps the simplest method of summarizing a time series is known as a *moving average*. As the name suggests, the method relies on computing means. We begin by computing the average of the first few observations from, say, $y_1$ through $y_m$. That average then corresponds either to the chronological center of the first $m$ observations or serves as the "prediction" for $y_{m+1}$. We then move along in the series, computing the average of observations $y_2$ through $y_{m+1}$. This continues until we have passed through the entire sample, continually taking averages of $m$ observations at a time, but always a different group of observations.

## Simple Moving Average

Moving averages are commonly used, perhaps chiefly for their simplicity and computability if not for their usefulness as a forecasting method. Simple smoothing methods are most appropriate when the time series is stationary, showing no particularly strong trend or seasonal components. As such, it is not well-suited to this particular time series, and we will see why. Nevertheless, we will apply it to this series both to demonstrate the technique and to understand its advantages and its shortcomings.

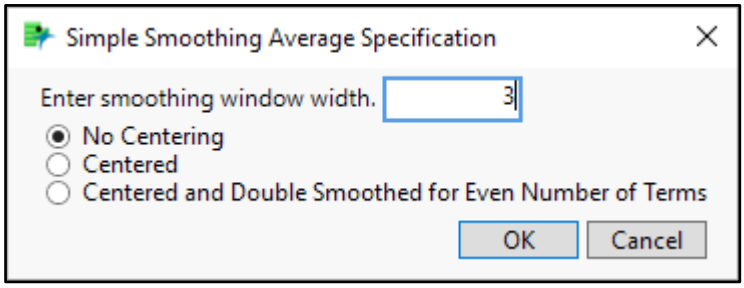
To introduce the concept of a smoothing method, we first look at the Simple Moving average in which each estimate is the mean of a relatively small number of recent observed values. The goal is to smooth out irregularities to make the trend, if any, more visible. To illustrate, we will compute an average for 3 months at a time. Our first estimate will be the mean of months 1, 2,

and 3. The second estimate will be based on months 2, 3, and 4. This is the reason for the name "moving averages": the averaging window moves as we progress through time.

1. Click the red triangle at the top of the time series report window and choose **Smoothing Model ► Simple Moving Average**.
2. Where noted, enter a **3** (for 3 months) as the smoothing window width. Leave the **No Centering** radio button alone.

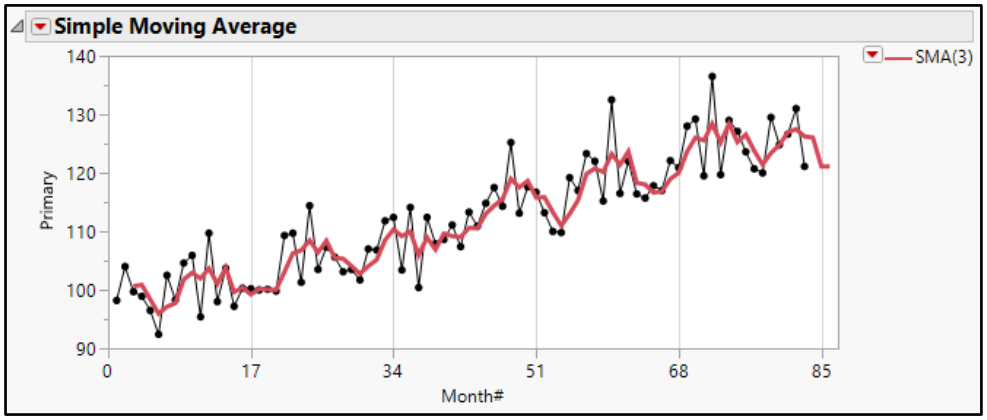
Conventionally, moving averages are either "centered" or "uncentered." A centered moving average locates the estimated value at the center of the averaging window. In this case, the first estimated value of a centered moving average would correspond to month 2. Otherwise, the estimate would correspond to the last month in the averaging window.

**Figure 20.3: Specifying a Moving Average Model**




The result appears at the bottom of the report and is shown in Figure 20.4, which has been modified for clarity.

**Figure 20.4: A Three-Month Moving Average**




The red **SMA(3)** line (for Simple Moving Average based on 3 observations) begins at Month 3 and is far less erratic than the initial time series plot. Notice also that the moving average line extends three months beyond the final observed month of the time series. These are the three forecast values. Importantly, because there is a steady upward trend in the data, a backward-looking moving average will predictably underestimate future values.

## Simple Exponential Smoothing

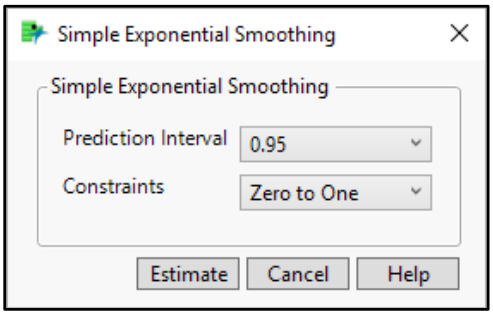
Another aspect of simple moving averages is that they rely on a relatively small subset of recent data and ignore earlier observations. In this case, although we have 111 observations in the data table, each estimate is based on only three values. *Exponential smoothing* methods address this shortcoming. JMP offers six variations of exponential smoothing, and we will consider three of them.

Exponential smoothing uses a weighted combination of all prior values to compute each successive estimate. The computed values incorporate a weight, or smoothing constant, traditionally selected through an iterative trial-and-error process. In exponential smoothing, each forecast is a weighted average of the prior period's observed and forecast values. Your principal text probably presents the formulas used in various exponential smoothing methods, so we will move directly into JMP.

The software finds the optimal smoothing constant for us. Let's see how we can use the method with the same set of Indian production data.

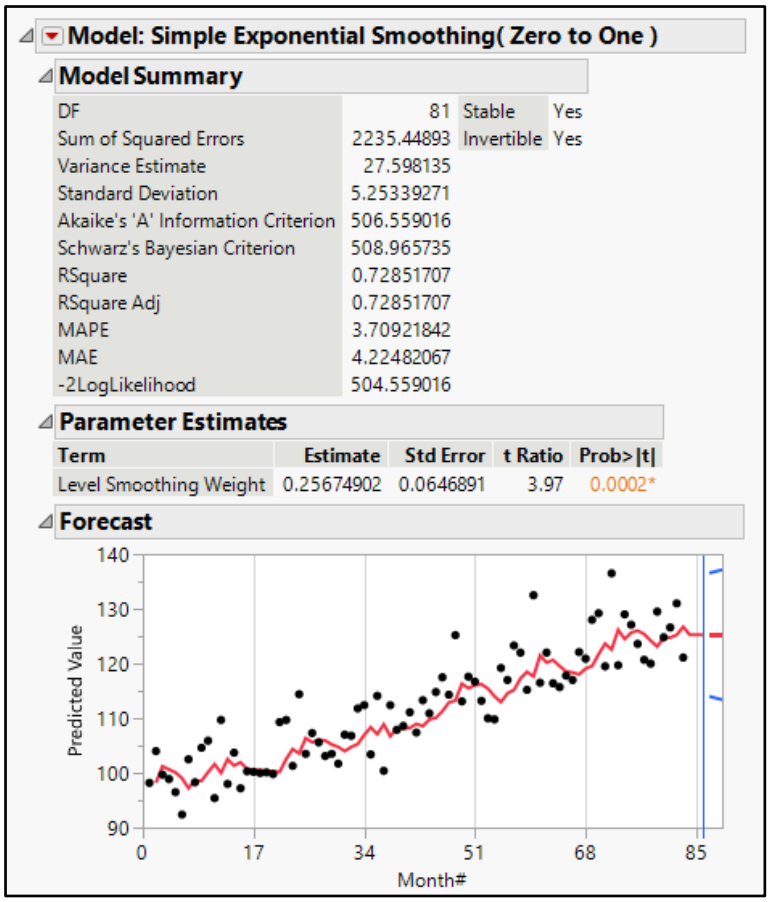
1. Click the red triangle next to **Time Series Primary** and select **Smoothing Model ► Simple Exponential Smoothing.**
2. This opens the dialog box shown in Figure 20.5. By default, JMP reports 95% confidence intervals for the forecast values, and constrains the smoothing constant to be a value between 0 and 1. Just click **Estimate**.

**Figure 20.5: Specifying a Simple Exponential Smoothing Model**




Several panels will open in the report. Look just below the **Simple Moving Average** panel and you will find the results shown in Figure 20.6. For this book, the smoothed and forecast lines have been thickened for clarity.

**Figure 20.6: Simple Exponential Smoothing Report**




The statistics in the model summary enable us to evaluate the fit and performance of the model and are directly comparable to subsequent models that we will build. In this introductory discussion of time series analysis, we will focus on just four of the following summary measures.

| | |
|---|---|
| Variance Estimate | Sometimes called MSE, a measure of the variation in the irregular component. Models with small variance are preferred. |
| RSquare | Goodness of fit measure, just as in regression. Values close to 1 are preferred. |
| MAPE | Mean Absolute Percentage Error. Expresses the average forecast error as a percentage of the observed values. Models with small MAPE are preferred. |
| MAE | Mean Absolute Error (sometimes called MAD—mean absolute deviation). Average of the absolute value of forecast errors. Models with small MAE are preferred. |

Under **Parameter Estimates** we also see the optimal smoothing constant value of .3845. This value minimizes the model variance and other measures of forecast error.

In the Forecast graph, we see that the model does smooth the series, and to the far right we see the forecasts for the next three periods represented by the short red line bracketed by the blue confidence interval. We can save the forecast values in a new data table as follows:

3. Click the red triangle next to **Model: Simple Exponential Smoothing (Zero to One)** and select **Save Columns.**

This creates an entirely new data table containing the observed and forecast values, as well as residuals, standard errors, and confidence limits. We had requested forecasts for three periods into the future. Because the original data table contained three observations that we hid and excluded from calculations, we get forecasts for those three months and then three additional months beyond.

4. Save this new data table as **Ch19_SES**.

Notice that each of the forecasts is the same value, represented in the graph by the horizontal red line—reflecting the problem with applying this simple approach to forecasting a series with a strong trend. For that, we need a method that captures the trend component. Holt's method does just that.

## Linear Exponential Smoothing (Holt's Method)

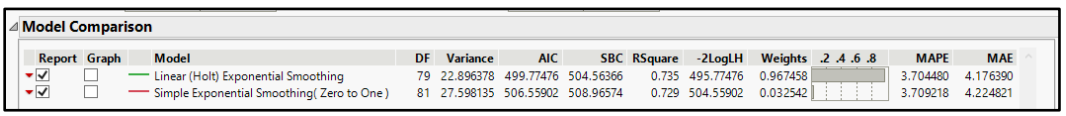
*Linear exponential smoothing,* developed by Charles Holt in 1957, expands the model of exponential smoothing by incorporating a linear trend element. Operationally, we proceed very much as we did in generating a simple exponential smoothing model.

1. Return to the top of the Time Series report window and again, click the uppermost red triangle and select **Smoothing Models ► Linear Exponential Smoothing.**
2. In the specification panel, accept the default settings and click **Estimate**.

Scroll to the very bottom. In your graph of the linear exponential smoothing model, notice that the forecast for the next three periods does rise slightly, in contrast to the simple model in which the forecasts are flat. This method estimates two optimal smoothing weights, one for the level of the index and another for the trend. We also find the same standard set of summary statistics reported, and near the top of the results window, we see a **Model Comparison** panel, as shown in Figure 20.7. The panel on your screen has a horizontal scroll bar at the bottom, and you cannot see the full width unless you enlarge the panel as has been done in Figure 20.7.

Each time we fit a different model, JMP adds a row to this table, enabling easy comparison of the different methods.

**Figure 20.7: Comparing the Two Exponential Smoothing Models' Performance**



Model Comparison

| Report | Graph | Model | DF | Variance | AIC | SBC | RSquare | -2LogLH | Weights | .2 .4 .6 .8 | MAPE | MAE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ☑ | ☐ | Linear (Holt) Exponential Smoothing | 79 | 22.896378 | 499.77476 | 504.56366 | 0.735 | 495.77476 | 0.967458 | | 3.704480 | 4.176390 |
| ☑ | ☐ | Simple Exponential Smoothing( Zero to One ) | 81 | 27.598135 | 506.55902 | 508.96574 | 0.729 | 504.55902 | 0.032542 | | 3.709218 | 4.224821 |

In this case we find that Holt's method has a smaller variance than simple exponential smoothing. On the other hand, the two methods have nearly the same R square and MAPE, and Holt's model has slightly better MAE.

In both smoothed graphs, we do see oscillations that might indicate a seasonal component. Neither simple nor linear exponential smoothing incorporates a seasonal component, but Winters' method does offer an explicit consideration of seasons.

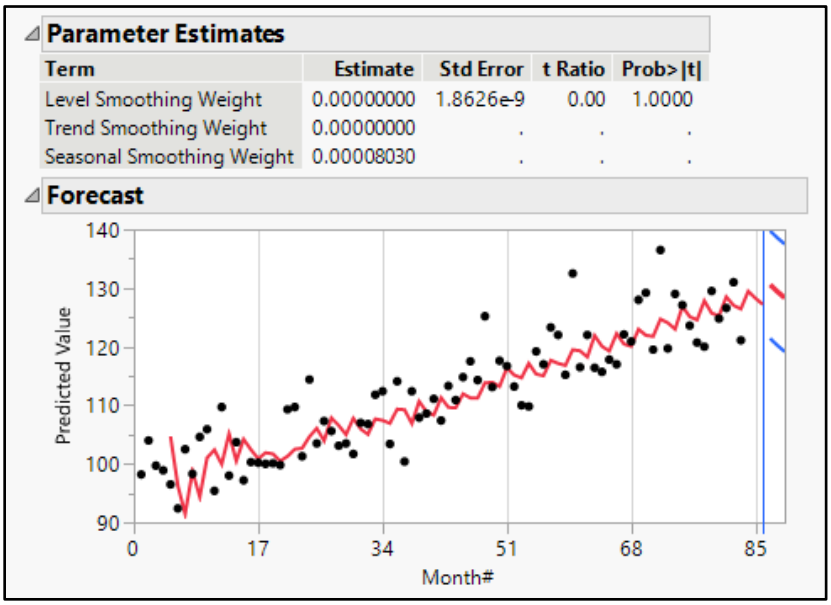
## Winters' Method

In time series estimation, there are two fundamental approaches to seasonality. One set of models is *additive,* treating each season as routinely deviating by a fixed amount above or below the general trend. The other family of models is *multiplicative*, positing that seasonal variation is a fixed percentage of the overall trend. In JMP, the implementation of Winters' method, named for Peter Winters who proposed it in 1960, is an additive model. Let's see how it works, and as a starting point, we will assume there are four seasons per year, each consisting of three months.

1. Click the red triangle next to **Time Series Primary** again and select **Smoothing Models ► Winters Method.**
2. In the specification panel, change the **Observations per Period** from 12 to 3, accept the other default settings and click **Estimate**.

The resulting parameter estimates and graph are shown in Figure 20.8. The graph is enlarged to more clearly show the forecast for the next three months. Notice in the parameter estimates that there are now three weights, corresponding to the three time series components. In the graph, we see a more refined set of future estimates that continue the upward trend but show a dip in production further into the future.

**Figure 20.8: Results of Winters' Method**




Now look at the comparison of the three models on your screen. Of the three models, the Winters Model has the smallest variance and lowest (preferred) AIC. On the other hand, Winters model has the worst R square, MAPE, and MAE. It is not uncommon to find such tradeoffs in the comparative statistics—which is one reason we consider multiple performance measures.

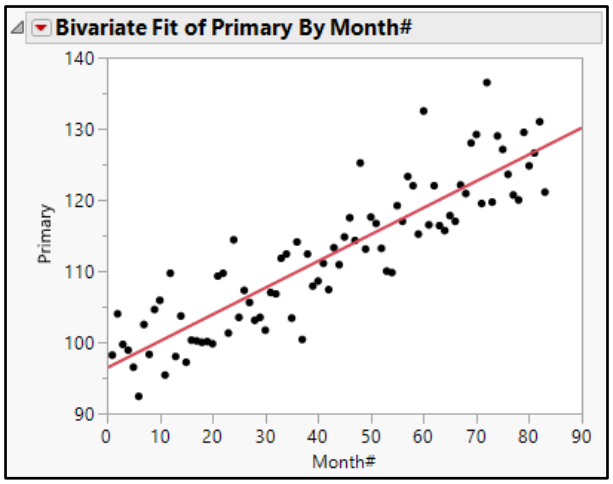
## Trend Analysis

For some time series, we can use the techniques of simple regression to fit a model that uses time as the X variable to describe or predict the general trend of Y. In the current example, we have seen that although there is clearly some irregular variation and sign of season and cycle, the production of basic goods in India has followed a steady upward trend in the years represented in our data table.

For simplicity of interpretation, we will use the column called **Month#**, which starts at 1 in the first month and runs for 114 months. We can fit the model using the observed 114 months and generate forecasts simply by extrapolating a straight line.

1. Select **Analyze ► Fit Y by X**. Cast **Primary** as **Y** and **Month#** as **X**.
2. In the report window, click the red triangle and select **Fit Line**. Your results should look like Figure 20.9.

**Figure 20.9: A Linear Trend Analysis**



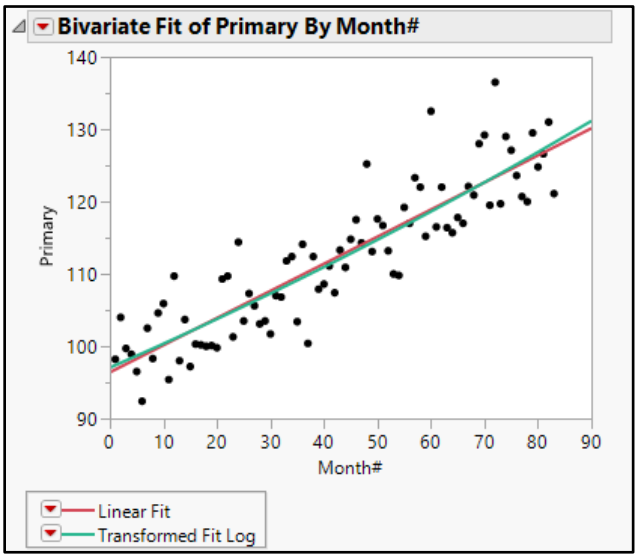

The slope coefficient in this model is approximately 0.37, meaning that with each passing month, the index has increased by 0.37 on average. The goodness-of-fit statistic, R square, is moderately high (0.79). We can compare these results to the performance of our exponential smoothing models. The Bivariate platform does not report MAPE and MAE, but it does produce a variance estimate, which we find in the ANOVA table as the Mean Square Error value of 22.11. Compared to the R square and variances of the exponential smoothing models, the simple linear trend is a reasonable model with a superior goodness of fit and low, but not minimum, variance.

If production of primary goods increases at a near-constant percentage rate, we would expect that a log-linear model will fit better than a simple linear model. For this reason, it is worth a moment to estimate a log-linear fit to see whether we can improve the model.

3. Click the red triangle next to **Bivariate Fit of Primary by Month#** and select **Fit Special** and choose **Natural Logarithm: log(y)** as the **Y Transformation**. Your results should look like Figure 20.10.

Visually, this model is barely different from the linear trend, and the model summary measures bear out that impression. R square improves very slightly from 0.789 to 0.795. Because of the log transformation, we cannot directly compare the variances. If you scroll to the very bottom of the report, you will find a panel labeled Fit Measured on Original Scale. This panel lists the Root Mean Square Error (RMSE) as 4.69, which is marginally superior to the RMSE of the linear model (4.70).

**Figure 20.10: A Quadratic Trend Analysis**




## Autoregressive Models

When we use regression analysis to model the trend of a time series, the dependent variable is the observed data $Y_t$ and the independent variable is just the time period, $t$. An alternative approach is to model $Y_t$ as a function of one or more prior observations of Y. As with exponential smoothing, there are many variations on this basic idea of an autoregressive model. In this section, we will examine one such model from the general class of AutoRegressive Integrated Moving Average, or ARIMA, models. These are also sometimes called Box-Jenkins models, after the two statisticians, George Box and Gwilym Jenkins, who developed the techniques in the 1960s and early 1970s.

The ARIMA analysis properly is performed on a stationary (without trend) time series. If there is a trend, the Box-Jenkins approach first transforms Y into a stationary series, usually by means of taking differences. That is, if Y were a steadily increasing series, we might find that the series based on first differences (that is, $Y^* = Y_t - Y_{t-1}$) is a stationary series.

An ARIMA model has three parameters as follows:

| Parameter | Role Within the ARIMA Model | Description |
|---|---|---|
| p | Autoregressive order | The number of lagged (prior) values of Y to include in the model. A model that only uses one prior period has an AR order=1. If the model includes the two most recent periods, it has AR order=2, and so on. |

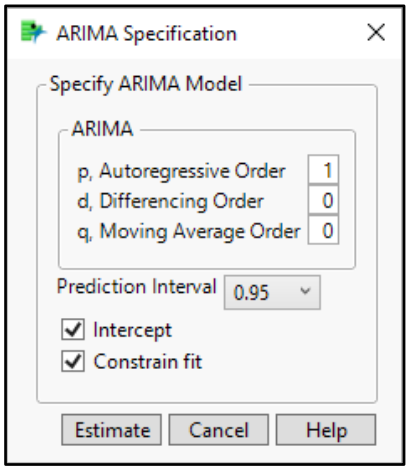
| Parameter | Role Within the ARIMA Model | Description |
|---|---|---|
| d | Differencing order | The number of times Y is differenced to generate the stationary series Y*. |
| q | Moving average order | The number of lagged values of error terms (residuals) to include in the model. |

The typical notation for an ARIMA model references all three parameters, or at least the nonzero parameters. Thus, an AR(1) model uses one lagged value of the observed values of Y. An AR(1,1) model uses one lagged value of the first differences of Y, and so forth.

Our time series plainly is *not* stationary, so we will want to use differencing to "flatten" the curve. However, to illustrate the techniques, we will first estimate an AR(1) model, and then an AR(1,1) model. We will compare the results of the two models to each other and then to our exponential smoothing estimates.

1. Return to the **Time Series** results window.
2. Once again, click the red triangle at the top of the results window and now select **ARIMA**.
3. This opens a new window; complete it as shown in Figure 20.11 and click **Estimate**.

**Figure 20.11: Specifying an AR(1) Model**




Autoregressive models are regression models. The AR(1) model results appear at the bottom of the **Time Series** report window (see Figure 20.12). The model can be expressed as the following equation:

$$Y_t = \beta_0 + \beta_1 Y_{t-1}$$

**Figure 20.12: Parameter Estimates for This AR(1) Model**

**Parameter Estimates**

| Term | Lag | Estimate | Std Error | t Ratio | Prob>\|t\| | Constant Estimate | Mu |
|---|---|---|---|---|---|---|---|
| AR1 | 1 | 0.75477 | 0.071967 | 10.49 | <.0001* | 27.4599557 | 111.97683 |
| Intercept | 0 | 111.97683 | 2.880492 | 38.87 | <.0001* | | |

In this case, both the estimated intercept and slope are significant, and we see that, at the margin, the estimated index in the next period is 75.5% of the prior period's index.

4. As with the smoothing methods in the Time Series platform, we can also save the computations from this model by clicking on the red triangle at the top of the **Model: AR(1)** results panel and choosing **Save Columns**. Save the table as **CH19_AR1**.

Before discussing the summary statistics for this model, let's estimate the AR(1,1) model and then compare the results. Recall that this next model uses the monthly change in the index (the "first difference") rather than the index itself as the basic element in the model.

5. Click the red triangle at the very top of the **Time Series** report and select **ARIMA** again.
6. This time, set both the **Autoregressive Order** and the **Differencing Order** (see Figure 20.11) to 1. The parameter estimates for this model should look like Figure 20.13.

**Figure 20.13: Parameter Estimates for AR(1,1) Model**

**Parameter Estimates**

| Term | Lag | Estimate | Std Error | t Ratio | Prob>\|t\| | Constant Estimate | Mu |
|---|---|---|---|---|---|---|---|
| AR1 | 1 | -0.7114708 | 0.0777302 | -9.15 | <.0001* | 0.51879323 | 0.30312713 |
| Intercept | 0 | 0.3031271 | 0.3251260 | 0.93 | 0.3540 | | |

We cannot directly compare these estimates to those for the AR(1) model because prior to estimating the regression, JMP calculated the first differences. At the margin, the incremental change in the monthly *change* in the index is about 71% less than the prior month. The negative coefficient leads to oscillations because if one month has a positive change, the forecast will be for a reduction in the following period and if a month has a negative change, the forecast will be positive. Although the computations are made for the relatively stationary series of first differences, JMP reconstructs the time series for the sake of the forecast graph, showing the general upward trend.

Now, let's compare our autoregressive models to each other and to the smoothing models. Everything we need is in the Model Comparison panel, as shown in Figure 20.14. Because the panel is so wide, we have eliminated some of the reported statistics,[2] focusing on the ones that we have discussed earlier.

**Figure 20.14: Comparing Five Time Series Methods**

**Model Comparison**

| Model | DF | Variance | RSquare | AIC Rank | SBC Rank | MAPE | MAE |
|---|---|---|---|---|---|---|---|
| Winters Method (Additive) | 76 | 21.931709 | 0.723 | 1 | 1 | 3.821894 | 4.300888 |
| Linear (Holt) Exponential Smoo... | 79 | 22.896378 | 0.735 | 2 | 2 | 3.704480 | 4.176390 |
| ARI(1, 1) | 80 | 25.760955 | 0.750 | 3 | 3 | 3.819508 | 4.313244 |
| Simple Exponential Smoothing... | 81 | 27.598135 | 0.729 | 4 | 4 | 3.709218 | 4.224821 |
| AR(1) | 81 | 45.563225 | 0.552 | 5 | 5 | 4.709015 | 5.270115 |

Because every time series is different, we cannot say categorically that one method will fit better than another. In this example, Winters Method performs best by three criteria (Variance, AIC and SBC), Holt's method by two criteria (MAPE and MAE), and the AR(1,1) model outperforms the others in terms of R square. We can eliminate the AR(1) model from consideration, but the others all seem to lead to comparable results.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* Let's continue our analysis of the Indian Index of Industrial Production.
   a. We used Winters' method with an assumed season of three months. Re-estimate using this approach with a six-month "season" (that is, two different seasons per year). Compare the performance of this model to the other model.
   b. Fit a log-linear model using the **Fit Y by X** platform. Does this perform any better than the quadratic model?
   c. We used simple AR(1) and AR(1,1) models. Try fitting an AR(2,1) model to this data series and report on what you find.

2. *Scenario:* Our **India Industrial Production** data table also contains several other indexes, including one for consumer durable goods. Durables are long-term purchases made by households, including items such as appliances and automobiles. Before building any models, make sure that the last three rows of the table are hidden and excluded.
   a. Use several of the techniques presented in this chapter to develop a six-month forecast for the index for consumer durables. Explain which technique seems most suitable for this time series and justify your choice.
   b. Save the forecasts using your chosen approach. Then, compare the forecast values from rows 84 through 86 to the observed (but hidden) values in our original table. How well does the model actually forecast the consumer durables index for those three months?

3. *Scenario*: The United Nations and other international organizations monitor many forms of technological development around the world. In Chapter 16, we estimated the annual growth of mobile phone subscriptions in Sierra Leone. Let's apply some of our newer techniques to the data in the **Cell Subscribers** data table.
   a. Of all the methods we saw in this chapter, we should not use Winters' method for this set of data. Why not?
   b. Use several of the techniques presented in this chapter to develop a three-year forecast for the number of cell subscriptions in Denmark. Explain which technique seems most suitable for this time series and justify your choice.
   c. Repeat part b for the annual series in Malaysia and report on your findings.
   d. Repeat part b for the annual series in the United States and report on your findings.
   e. Did you decide to model all four countries using the same time series technique? Discuss the insights that you have drawn about time series analysis from this exercise.

4. *Scenario:* In prior chapters, we examined cross-sectional data about fertility and birth rates. In this exercise, we will look at fertility time series for several countries. The time series are all in the data table **Fertility**; unlike the other examples in the chapter, these figures are reported every five years rather than annually or more frequently. Within the table, the data through 2014 are actual observed figures, but the 2015–2019 data are U.N. estimates. Accordingly, the rows corresponding to 2015 have been hidden and excluded in the saved data table.

   Suggestion: start with Graph Builder, plotting **Fertil** versus **Year** for all countries. Use the Local Data Filter to choose one country at a time to see the general growth pattern. Also remember that there will be no seasonal component to these series because they aggregate five years at a time.
   a. Develop a time series model for the fertility rate in Brazil. Report on your choice of a model.
   b. Develop a time series model for the fertility rate in the Russian Federation and report on your model.
   c. Develop a time series model for the fertility rate in India and report on your model.
   d. Develop a time series model for the fertility rate in China and report on your model.
   e. Develop a time series model for the fertility rate in Saudi Arabia and report on your model.
   f. The U.N. estimate for Brazil in 2015 is a fertility rate of 1.74 children per woman. Based on your analysis, which of our time series techniques might the U.N. have used to develop this estimate?

5. *Scenario*: In the United States, the management of solid waste is a substantial challenge. Landfills have closed in many parts of the nation, and efforts have been made to reduce waste, increase recycling, and otherwise control the burgeoning stream of waste materials. Our data table **Maine SW** contains quarterly measurements of the tonnage of solid waste delivered to an energy recovery facility in Maine. The data come from five municipalities in the state.
   a. Develop a time series model for quarterly solid waste delivered from Bangor. Report on your choice of a model.
   b. Develop a time series model for quarterly solid waste delivered from Bucksport. Report on your choice of a model.
   c. Develop a time series model for quarterly solid waste delivered from Enfield. Report on your choice of a model.
   d. Develop a time series model for quarterly solid waste delivered from Orono. Report on your choice of a model.
   e. Develop a time series model for quarterly solid waste delivered from Winslow. Report on your choice of a model.
   f. Did the same method work best for each of the five communities? Speculate about the reasons for your finding.

6. *Scenario*: The United Nations, World Bank and other international organizations monitor many forms of technological development around the world. Elsewhere we have looked at other Sustainable Development Goals. In this exercise, we will study carbon dioxide emissions between 2000 and 2016. The precise measure is kilograms of CO2 per constant 2010 United States dollars of GDP. The relevant data table is called **SDGCO2**. The time series column is called **CO2**.

   Suggestion: start with Graph Builder, plotting **CO2** versus **Year** for all countries. Use the Local Data Filter to choose one country at a time to see the general growth pattern. In this exercise, we will consider the so-called BRIC nations (Brazil, Russia, India, China) and the U.S.A.
   a. Develop a time series model for the **CO2** in Brazil. Report on your choice of a model.
   b. Develop a time series model for the **CO2** in the Russian Federation. Report on your choice of a model.
   c. Develop a time series model for the **CO2** in India. Report on your choice of a model.
   d. Develop a time series model for the **CO2** in the China. Report on your choice of a model.
   e. Develop a time series model for the **CO2** in the United States. Report on your choice of a model.
   f. Did the same method work best for each of the five nations? Speculate about the reasons for your finding.

7. *Scenario*: Our data table called **MCD** contains daily share prices for McDonald's common stock for the first six months of 2019. The **Daily Close** column is the price of one share of stock on any given day.
   a. Create a time series plot of the **Daily Close** column. Looking at this series, which time series modeling method or methods seem particularly inappropriate? Explain.
   b. Develop a time series model for the daily closing price and forecast the next five days beyond the series. Report on your choice of a model.

8. *Scenario*: Our data table called **Stock Index Weekly** contains weekly broad market indexes and trading volumes for six major stock markets around the world. Specifically, we have observations for every week of 2008, a year in which financial markets across the globe shuddered as a consequence of lending practices in the United States. The six market indexes are for the markets in Tokyo (NIK225), London (FTSE), New York (SP500), Hong Kong (HangSeng), Madrid (IGBM), and Tel Aviv (TA100).
   a. Compute correlations for the index values of all six stock markets. Which market is most closely correlated with the S&P 500 index?
   b. Develop separate time series models for the S&P 500 and the market that you identified in part a. Report on your findings.
   c. Does the same method work best for the New York market and the market that you identified in part a? What do you think explains this result?
   d. Develop a time series forecast for weekly trading volume in the Hong Kong market for the first three weeks of 2009. How confident are you in these forecasts?

9. *Scenario*: The data table **Global Temperature and CO2** contains annual measures of two variables: the concentration of carbon dioxide in the earth's atmosphere and the mean global temperature in degrees Fahrenheit.
   a. Create a time series plot of the **Concentration** column. Looking at this series, which time series modeling method or methods seem well suited to the data? Explain.
   b. Using one of the ARIMA or smoothing methods in the Time Series platform, develop a 5-year forecast for CO2 concentrations.
   c. Repeat parts a and b using the temperature data.

---

[1] In 2018, the International Monetary Fund projected India to rank 3rd in the world in terms of Gross Domestic Product Purchasing Power Parity (GDP-PPP) and first in terms of annual growth rate.

[2] As with other JMP reports, users can select table columns by hovering over the table, right-clicking once, and choosing Columns. This opens a list of column names, and the user can specify the contents of the reported table.

# Chapter 21: Elements of Experimental Design

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## Overview

In Chapter 2, we discussed the difference between *experimental* and *observational* studies. In an experiment, the investigator creates a controlled setting in which she or he can deliberately manipulate the values of some variables (*factors* or *independent* variables) and measure the values of one or more *response* (or *dependent*) variables. We can confidently draw inferences about causation based on well-designed experiments.

In contrast, researchers exercise little or no control in observational studies. They gather data, sometimes after the fact, about events or subjects and then use methods that we have studied to analyze the evidence of interdependence among variables. Even the best observational study does not permit us to conclude that one variable causes another.

The data tables that accompany this book typify observational and experimental research as well as survey research. Often, we have no choice but to use observational data because a properly designed experiment would be prohibitively expensive, would raise serious ethical issues, or would be impractical or impossible for one reason or another. For example, engineers concerned with buildings that can withstand earthquakes have no ability to randomly assign some cities to an earthquake. Even if it were possible to do so, it is unethical for public health officials to randomly expose some citizens to pollution or disease. This chapter is devoted exclusively to the design of experimental studies.

## Why Experiment?

When we can feasibly conduct an experiment, we should. If we need to establish the efficacy of a new disease treatment, we intentionally need to control relevant factors so that we can assess and understand their impact on the course of a disease or on patients' symptoms. If we need to optimize a manufacturing process or understand the relative importance of several factors in producing a measurable outcome, experimentation is the most reliable course of action. In this chapter, we will begin with a simplified version of the concrete experiment first presented in Chapter 2. Concrete is a mixture of ingredients that always include water, cement, a coarse aggregate like gravel, and a fine aggregate like sand. The investigator, Dr. Yeh, was interested in predicting the *compressive strength* of concrete, particularly when substituting other ingredients for the material called *fly ash,* a by-product of coal combustion. Compressive strength refers the ability to withstand pressures exerted by a structure and is a key property in engineered foundations of buildings and bridges. A concrete mixture with insufficient compressive strength could crumble under its load. Yeh theorized that the compressive strength would vary predictably as a result of the mixture of cement, water, and other component materials, and the goal of the experiment was to develop a reliable model of compressive strength. With such a model, it is possible to specify the right mixture to achieve a required strength.

In this chapter, after a brief discussion of the purposes of experimental design and some definitions, we will learn several standard approaches to DOE—the Design of Experiments. As we do so, our emphasis is on the structure of the experiments and the generation of data. We will not analyze the data that a design would generate, but instead devote this chapter to some of the fundamental concepts and issues in the design process. The concrete data that we have analyzed thus far was gathered through several carefully designed experiments. We have already done some analysis. The goal now is to understand how and why the data were gathered.

It is also important to note that DOE is a large and complex topic. This single chapter cannot deal with all aspects. The purpose of this chapter is to introduce the reader to some foundational concepts. Readers who seek more extensive coverage of DOE should consult the *Design of Experiments Guide* within the JMP Documentation Library.

## Goals of Experimental Design

DOE refers to the systematic process of planning the variation among factors so that suitable data can be collected for analysis in such a way that an investigator is able to generalize from their experiment. In other words, one major objective in the statistical design of experiments is to ensure that our experiments yield useful data that can be analyzed to draw valid conclusions and make better decisions.

Box, Hunter, and Hunter (1978) laid out a process of progressive experimentation by moving between generating ideas and then gathering facts, which would in turn generate new insight and ideas. In so doing, they described different goals of experiments. Hoerl and Snee (2002) summarize the process by identifying three experimental contexts, each of which is

characterized by its own fundamental question, although any experiment might be a hybrid of the three categories.

- *Screening*: Which variables have the largest effect on the response variable?
- *Characterization*: How large are the effects of the few most important variables? Are there any important interactions among these variables? Are relationships linear or curvilinear?
- *Optimization*: At what values of the independent and control variables does the response variable attain a desired minimum, maximum, or target value?

In the current example, the ultimate goal is characterization: to build a model that predicts the compressive strength of a particular mixture of concrete. How do we do this? There are many components that might influence strength: types of coarse and fine aggregates, water, cement, and other additives. A screening experiment would seek to identify a few key factors affecting the strength and indicate their relative importance. A characterization experiment would focus on factors identified by screening and seek to understand specifically how they collectively affect the strength. Moreover, it might also assess possible interactions whereby the effect of one factor changes depending on the values of other factors. With the relevant variables and relationships identified, we would finally use an optimization approach to fine-tune the variables to design a batch of cement with the desired compressive strength for a particular structure. We will begin with some standard approaches to screening experiments and define some important terms.

## Factors, Blocks, and Randomization

We will refer to the components that might affect compressive strength as *factors*. In general, a factor is a variable that can affect the response variable of interest. Factors might be categorical or numerical. The strategy in a screening experiment is to test several factors to discover which ones have the greatest impact on the response variable. When we design an experiment, we ideally consider all potential sources of variation in the response variable. We generally do this in several ways within an experiment.

- We deliberately vary—that is, set the values of some factors (sometimes called *treatments*)—that we believe affect the response variable. To use the language that we have used earlier, we want to identify candidate Xs that cause variation in Y.
- We *control* other factors by holding them constant. For instance, because we understand that the strength of concrete depends on the temperature at which it cures, we might test all combinations at the same temperature.
- We often repeat, or *replicate*, measurements of the response variable for a given treatment condition through a series of *experimental runs*. We do this to gauge and control for measurement error.

- We might *block* some runs to account for other "nuisance" factors that cause similarities among runs. We block when we expect that some response measurements will be similar due to the shared factor being blocked.
- We *randomize* experimental runs to account for those factors that we cannot control, block, or manipulate. Thus, if we expect that changes in barometric pressure could affect the outcome, we randomly assign the sequence of testing different formulations.

## Multi-Factor Experiments and Factorial Designs

It might be natural to think that one should experiment by holding all factors constant except for one, and systematically alter the values or levels of that single factor. So, for example, we might want to test varying amounts of sand in concrete mixtures with all other components being held at given quantities. However, such an approach is very inefficient and more importantly, it can conceal any combined influence of two or more factors. Perhaps the sand quantity is not nearly as important as the ratio of sand to coarser aggregate. When two variables combine to lead to a result, we say that there is an *interaction* between the two, and if there is an important interaction to be found, we will never find it by examining only one factor or the other. We need to vary multiple factors at a time to truly understand what is going on.

Another way to conceive of the issue is in terms of a three-dimensional *response surface*. Imagine that we have a response variable and two factors. If both factors were to contribute to the response in a simple additive and linear fashion (more of component A increases strength, more of component B weakens strength), then it would not matter if we vary A and B individually or jointly. The response variable would vary by strictly increasing one direction and decreasing in the other. The three variables together would form a rigid flat plane, and by varying A and then B, we would essentially be sliding up or down the plane in one direction or the other.

In reality, the situation is rarely so simple. Non-linear relationships and interactions occur frequently in many contexts. Consider, for example, how often a nutrient is beneficial in moderate quantities but hazardous at very low or very high amounts. The same nutrient might be beneficial for people with one health condition, but detrimental for those with a different condition. If we fail to include interaction effects in our model, we will not understand what is going on.

In complex situations like these, where the response values vary on a surface that is curved in one or both dimensions, we would need to explore a great deal of the surface to develop a more complete model. If the surface had multiple peaks and we vary only one factor at a time, it would be analogous to traversing the surface only from north to south or only from east to west. We would be characterizing one narrow region of the surface and could potentially miss the peaks completely.

Although single-factor experiments can be inadequate, multi-factor experiments can become very expensive and time-consuming. Let's suppose that we have identified ten possible factors affecting compressive strength. One conventional experimental strategy for screening is to test each factor at two different levels initially—high and low, large and small, or with and without. A

*full factorial* experimental design enables us to measure the response variable at all possible combinations of all factor levels. In this case, a full factorial design would require us to create enough concrete blends to cover all possible combinations of the ten different factors. That would require $2^{10}$ or 1,024 different designs. If we were to replicate each formulation ten times to estimate mean strength, we would need to have 10,240 experimental runs—at whatever cost of materials and time that would involve.

There are several DOE strategies to accommodate many factors far more efficiently than described above. In practice, researchers can usually rely on their experiences with similar situations to select factors that are most likely to affect the outcomes of interest. Additionally, we typically design experiments to seek answers to specific questions; therefore, a designer allows the research questions to guide the selection of factors and interactions to investigate.

To illustrate the fundamental logic and approach of factorial designs, we will begin with a modest number of factors. The concrete example is a good introductory context because there are just a few factors that are well-known to affect compressive strength. In this example, there is no need for screening because experts know which factors to vary. In general, a given factor might have several possible levels or values. For our first example, let's consider just three factors with two levels each. Specifically, we will design a full-factorial experiment as follows:

- **Coarse Aggregate**: the amount (kg) of crushed stone per one cubic meter of concrete. The levels tested are 850 kg and 1100 kg.
- **Water**: the amount of water (kg) per one cubic meter of concrete. The two levels will be 150 kg and 200 kg.
- **Slag**: Slag is a by-product in the production of iron from iron ore. Pulverized slag is a binding agent often used in combination with or as a substitute for Portland cement, which is the most common cementitious ingredient in concrete. Slag is also measured in kg/cubic meter, and the amounts to test are 0 and 150 kg.

We will develop an experimental design that ensures that we test all the combinations of factors and levels. Since we have three factors with two levels each, there are $2^3 = 8$ combinations in all. We can tabulate the eight possible experimental conditions as shown in Table 18.1.

We could mix up eight batches of concrete (holding other ingredients at fixed levels) and measure the compressive strength of each batch. Typically, though, it is wiser to test several batches of each formulation and to observe the compressive strength of each. We refer to such repeated trials as *replication*. Replication is a way to accommodate measurement error, to control its effects as well as to estimate its magnitude. So, suppose we decide to replicate each batch four times and compute the average for a total of five batches of each mixture. In all, we would measure 5 x 8 = 40 experimental runs.

**Table 20.1: Eight Experimental Conditions**

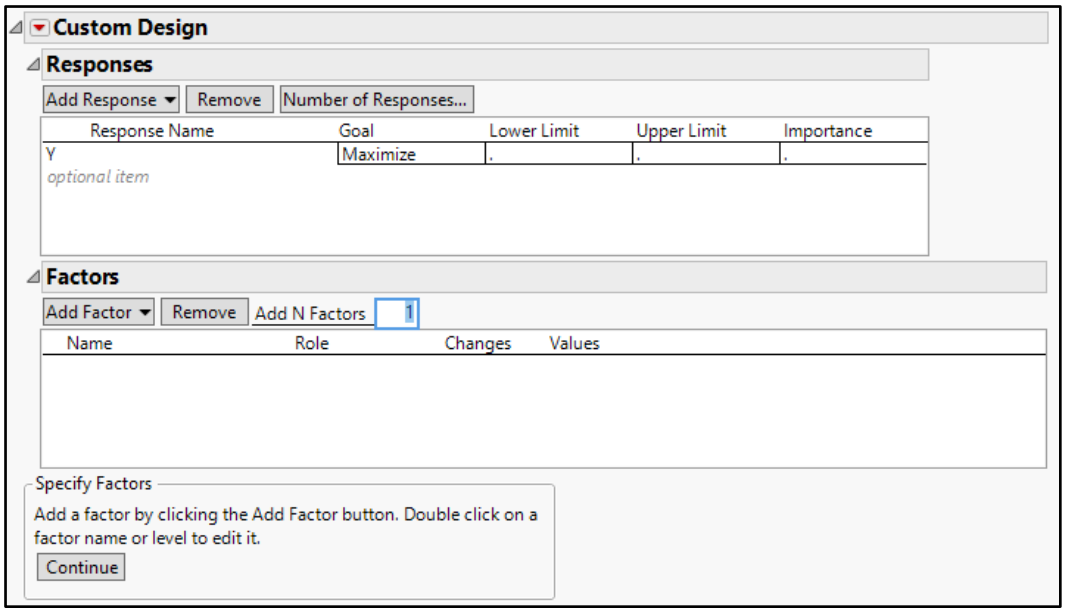
| Formulation | CoarseAgg | Water | Slag |
|---|---|---|---|
| **A** | 850 | 150 | 0 |
| **B** | 850 | 150 | 150 |
| **C** | 850 | 200 | 0 |
| **D** | 850 | 200 | 150 |
| **E** | 1100 | 150 | 0 |
| **F** | 1100 | 150 | 150 |
| **G** | 1100 | 200 | 0 |
| **H** | 1100 | 200 | 150 |

We might mix five batches of formulation A, then five batches of formulation B, and so on. However, this might compromise the validity of any conclusions we draw about the formulations because of the possibilities of such things as experimenter fatigue, changing air pressure, unexpected circumstances, or the like. A better approach would be to *randomize* the sequence of runs.

We will use JMP to create our sequence and build a suitable design, called a *full factorial design*. In a full factorial design, we test all factors at the specified levels and collect enough data to examine all possible interactions among the factors. The **DOE** menu in JMP includes numerous designs and methods, and it is well beyond the scope of this chapter to illustrate them all. For clarity, simplicity, and flexibility, we will do most of our examples using the **Custom Design** platform. The goal in this chapter is to understand how DOE works, rather than extensively exploring JMP's capabilities.

1. Select **DOE ► Custom Design**. In the dialog box (see Figure 21.1), we need to identify a response variable and all the factors we will be testing. In addition, we will indicate that we want to replicate the experimental runs four times.

**Figure 21.1: Full Factorial Design Dialog Box**




Because JMP eventually randomizes the sequence of trials, each reader of this book could generate a different sequence. For the sake of instructional clarity in this first example, we will standardize the randomization process by setting the random seed value as we have done before. I have chosen 222 as the seed.

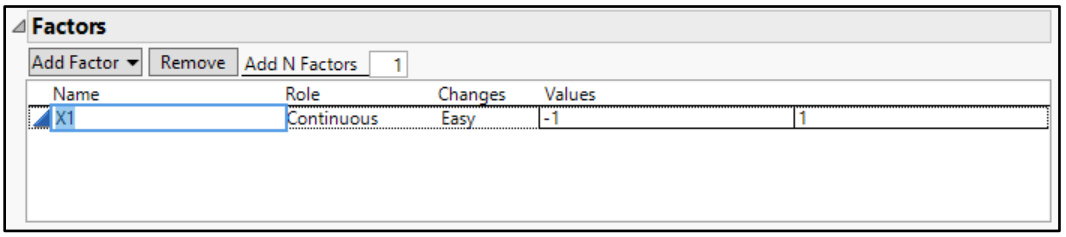
2. Click the red triangle next to **Custom Design** and select **Set Random Seed**. Enter the value **222** when prompted.

We will start by naming the response variable and setting the goal in this phase of the experiment.

3. Double-click the **Y** under the heading **Response Name** and type **Compressive Strength**.
4. Our eventual goal is to characterize the relationship between strength and factors so that we can achieve a target strength. Click **Maximize** under **Goal** and select **Match Target**. For now, leave the **Lower Limit**, **Upper Limit,** and **Importance** cells blank.

   Now let's move on to entering the factors for this experiment. In our initial list of factors, we have identified three continuous variables. In the empty **Factors** panel, we need to enter our three factors.
5. Click the Add Factor button and select **Continuous**. As soon as you do this, JMP creates an entry for a factor that it provisionally labels as **X1**, with experimental values of -1 and 1. (See Figure 21.2.)

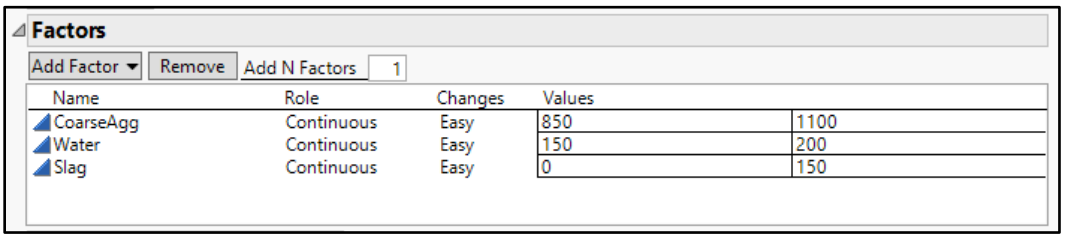
**Figure 21.2: Specifying a Continuous Factor**




6. Double-click **X1** and change the factor name to **CoarseAgg**.
7. Click the **-1** under **Values** and change the low value to **850**. Similarly, change the high value to **1100**.
8. Click **Add Factor** and select **Continuous** again to set up our second factor, **Water**, with two levels of 150 and 200.
9. Click **Add Factor** and select **Continuous** to set up the final factor **Slag** with two levels of 0 and 150.

When you are finished, the **Factors** panel should look like Figure 21.3.

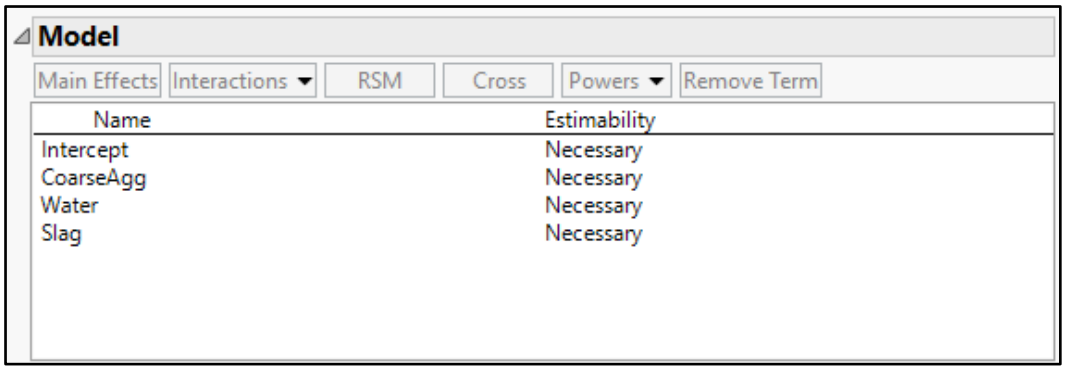
**Figure 21.3: The Completed Factors Panel for the Concrete Experiment**




In this chapter, we will use this list of three factors a few more times. Rather than re-enter the information repeatedly, we can save the three factors and their attributes as follows.

10. Click the red triangle next to **Custom Design** and choose **Save Factors**. This opens a new data table (not shown) containing the factor names and levels. Save it to your desktop or to the folder containing your projects as **Chap 21 Factors**.
11. Return to the **DOE – Custom Design** tab in your project and click the **Continue** button in the lower left of the dialog box. This adds four new panels, the second of which is titled Model Panel (Figure 21.4). Notice that the design will allow us to estimate an intercept and terms for the three continuous factors, that is, the main effects of the factors.
12. For a full factorial analysis, we will also want to plan for the interaction terms between that factors. Hence, click the button marked **Interactions** and select **2nd**. Now our model will capture main and interaction effects.

**Figure 21.4: Model Panel**




13. Scroll down further and find the Design Generation panel. Near the bottom, type **40** in the box next to **Number of Runs** to replace the 8, reflecting our intention to test each formulation five times for a total of 40 runs.
14. Click **Make Design**. Notice that the resulting panel shows the planned settings for the three variables over the 40 runs.
15. Now click the **Make Table** button at the bottom of the report.
16. A new data table opens containing our randomized sequence of 40 experimental runs. Figure 21.5 displays the first 19 rows of the table.

**Figure 21.5: The Custom Design**

Custom Design 3

Design Custom Design

Criterion D Optimal

- Model
- Evaluate Design
- Generalized Regression
- DOE Dialog

Columns (4/0)

- CoarseAgg *
- Water *
- Slag *
- Compressive Strength *

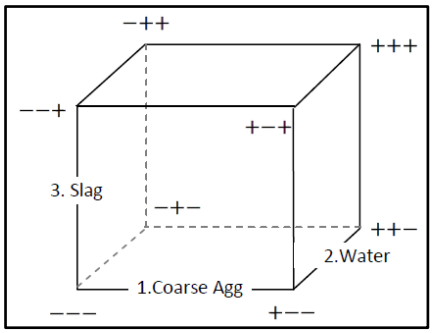
| | CoarseAgg | Water | Slag | Compressive Strength |
|---|---|---|---|---|
| 1 | 850 | 200 | 0 | • |
| 2 | 1100 | 150 | 0 | • |
| 3 | 850 | 150 | 150 | • |
| 4 | 850 | 200 | 0 | • |
| 5 | 1100 | 200 | 150 | • |
| 6 | 1100 | 200 | 150 | • |
| 7 | 1100 | 200 | 150 | • |
| 8 | 1100 | 200 | 150 | • |
| 9 | 1100 | 150 | 0 | • |
| 10 | 1100 | 200 | 150 | • |
| 11 | 1100 | 150 | 0 | • |
| 12 | 1100 | 150 | 0 | • |
| 13 | 1100 | 150 | 0 | • |
| 14 | 850 | 200 | 0 | • |
| 15 | 1100 | 200 | 150 | • |
| 16 | 850 | 150 | 150 | • |
| 17 | 850 | 200 | 0 | • |
| 18 | 850 | 200 | 0 | • |
| 19 | 850 | 150 | 150 | • |

The table contains 40 rows and four columns. The rows represent the factor settings for each of the runs. So, for example, the first run has the “low” setting for coarse aggregate and slag and the “high” setting for water. In our Table 20.1, this corresponds to Formulation C.

The first three columns of the data table show the settings of the factors, and the fourth column is ready for our measurements. Working from the sequence specified in this data table, we would then combine ingredients in each formulation, allow the concrete to cure, and measure compressive strength. We would then return to this data table, type in the compressive strength measurements, and estimate the model.

We can visualize this design as a cube, as shown in Figure 21.6. Each factor has two levels, denoted + or - for high and low values. Each corner of the cube corresponds to a different combination of the levels of the three factors. In our 40 runs, every corner will be measured five times.

**Figure 21.6: Visualizing the Full Factorial Design**




## Blocking

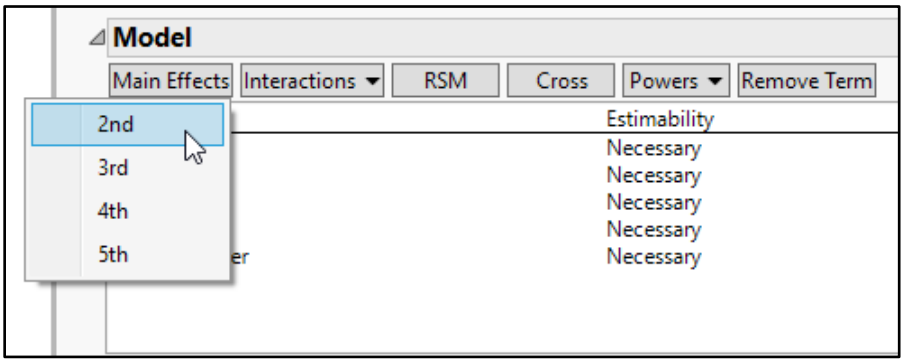
In this example, we assumed that a single experimenter performed all of the experimental runs. It is more likely that this work would be done in a team of people, each of whom might execute a share of the work. Suppose this experiment were being conducted by a four-member team, and we want to control for the possibility that different individuals could introduce an unintended factor into the mix.

We do this by *blocking* the experimental runs. A *block* is a group of runs that we expect to be relatively homogenous, but possibly different from other blocks. When we block, we eventually compare the results of each experimental condition within each block. In this case, it makes sense to block by person, enabling each experimenter to run several different formulations. We will randomly assign ten runs to each experimenter. Each team member will run different formulations, and again the experiment will have 40 runs.

1. Select **DOE ► Custom Design**. As before, we will specify that our response variable is **Compressive Strength** with a goal of **Match Target**.
2. Instead of re-typing all the factors, click the red triangle next to **Custom Design** and choose load **Factors**. When the **Open Data File** dialog box appears, locate the **Chap 21 Factors** data table file and open it. This reads in the prior factor specifications.
3. In the **Factors** panel, click **Add Factor** again and select **Blocking ► Other…** and specify **10** runs per each of the four blocks.
4. This adds a new factor, **X4**, to the model. Change the name of **X4** to **Team Member**. Later we will enter names of the four team members; for now, direct your attention to the **Model** panel.
5. Again, this is a full factorial experiment, so we want to be able to examine all interactions between pairs of the factors. Therefore, in the **Model** panel click the **Interactions** button and select **2nd** as shown in Figure 21.7.

You will see a warning that JMP does not support interactions involving blocks. Click **Continue**, and the model will include interactions for the three continuous factors.

**Figure 21.7: Specifying Second-Order Interactions**




6. Finally, in the **Design Generation** panel, under **Number of runs**, select the **User Specified** option and enter **40**.
7. Scroll back up to the list of factors. The **Team Member** blocking factor now shows four cells. Enter these names: **Jiao, Pranav, Daniel,** and **Li**.
8. Click **Make Design**.

This time, we still have a full factorial design with four replicates and a total of 40 runs, but now see that the runs are grouped into four blocks by team member. To compare your work to the example here, first set the random number seed once again.

9. Click the red triangle next to **Custom Design** and select **Set Random Seed**. Set the seed value to **5911**.
10. Scroll to the bottom of the **Custom Design** window and select **Make Table**. In the new data table (Figure 21.8) the random assignment of runs to team members is clear.

**Figure 21.8: Portion of a Data Table with Blocked Design**

Custom Design 2
Design: Custom Design
Criterion: D Optimal
Model
Evaluate Design
Generalized Regression
DOE Dialog

Columns (5/0)
CoarseAgg *
Water *
Slag *
Team Member *
Compressive Strength *

| | CoarseAgg | Water | Slag | Team Member | Compressive Strength |
|---|---|---|---|---|---|
| 1 | 850 | 150 | 0 | Daniel | • |
| 2 | 850 | 200 | 0 | Daniel | • |
| 3 | 850 | 200 | 150 | Daniel | • |
| 4 | 1100 | 200 | 0 | Daniel | • |
| 5 | 1100 | 150 | 0 | Daniel | • |
| 6 | 1100 | 150 | 150 | Daniel | • |
| 7 | 850 | 150 | 0 | Daniel | • |
| 8 | 1100 | 200 | 150 | Daniel | • |
| 9 | 850 | 150 | 150 | Daniel | • |
| 10 | 1100 | 150 | 150 | Daniel | • |
| 11 | 850 | 150 | 150 | Jiao | • |
| 12 | 1100 | 150 | 0 | Jiao | • |
| 13 | 850 | 200 | 150 | Jiao | • |
| 14 | 850 | 200 | 0 | Jiao | • |
| 15 | 1100 | 150 | 0 | Jiao | • |
| 16 | 1100 | 200 | 150 | Jiao | • |
| 17 | 850 | 200 | 0 | Jiao | • |
| 18 | 850 | 150 | 150 | Jiao | • |
| 19 | 850 | 150 | 0 | Jiao | • |

In this blocked design, we still have 40 runs in all, with four replicates of the eight possible formulations, and each block (team member) has an assignment of 10 formulations to test in randomized sequence. The analysis of the strength data would proceed as before, with the blocks accounted for.

## A Design for Main Effects Only

In this example, we have not considered any limits on the number of experimental runs to be performed. This is not the case for all experiments, particularly those with more factors or levels, or those with time constraints or with scarce or very costly materials.

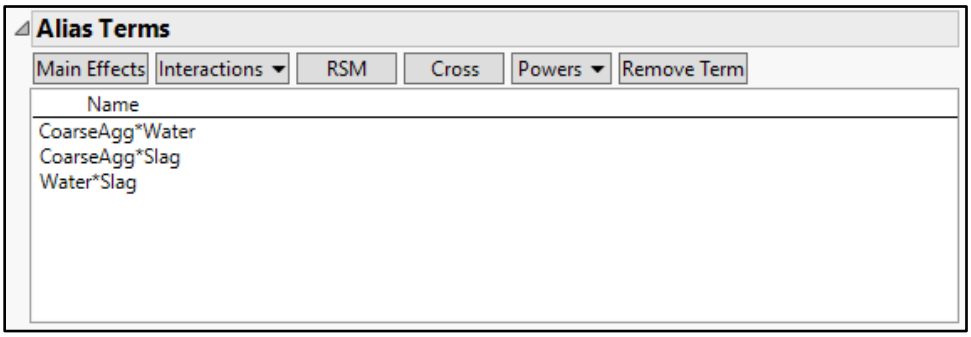
We might run fewer replicates to save time or materials, but then we would lose the advantages of replication. Instead, if we can confidently forego the testing of all possible interactions, we can reduce the total number of runs by using a *fractional* design. In a fractional design, we focus on the main effects of each factor and deliberately omit interaction combinations in exchange for the savings in experimental runs. We plainly collect less information than with a full factorial design, but we can still obtain useful data that we can analyze fruitfully.

We will illustrate with an example in which we reduce the number of experimental runs by half, generating only 20 runs. For simplicity in this example, we will not block by team member.

1. Select **DOE ► Custom Design**. By now, you are familiar with the dialog box. Name the response variable as Compressive Strength and reload the three continuous factors as you did in the previous example.

2. Click the red triangle next to **Custom Design** and **Set Random Seed** to 13326.
3. Before making the design, click the disclosure triangle next to Alias Terms. this opens the panel shown in Figure 21.9.

**Figure 21.9: Alias Terms**




Our model, as specified, will not detect the presence of interactions. Such effects may be confounded when our model is estimated, which is to say that our estimates of a main effect might be biased due to the omission of an interaction. We use the word "alias" to identify possible confounders. In this example, we read the first row of the table to mean that we will not be able to distinguish between main effects of slag quantify and the interaction (if any) of compressive strength and water. If, say, a high quantity of slag is associated with greater compressive strength, we won't be able to decide whether the slag causes greater strength or whether the interaction of coarse aggregate and water are the cause.

4. Change the User-Specified number of runs to 20 and click **Make Design**.
5. Click **Make Table**.

Look closely at the data table that you have generated. Notice that, for each factor we observe 10 runs with the high setting and 10 runs with the low setting. A close look reveals that some combinations are tested more often than others.

As with any compromise, we gain something and lose something in this design. We save time and resources by using only half of the possible runs. This comes at the loss of valuable information; in this design, we will not test all possible interactions, and thus some effects will be confounded with others. We might not be able to determine whether a particular response value is due to the main effects of the factors or to their interactions. As noted before, we refer to the confounded main effects and interactions as *aliases*.

Traditionally, there are other design options available when an investigator needs to produce useful results with comparatively few runs. Some approaches (*fractional designs*) test only some of the corner points shown in Figure 21.6. This chapter does not consider those designs, but it is important to understand that some experimental designs provide more information to an investigator than others. Readers who work in settings that use fractional designs are encouraged to learn more about such designs on their own.

## Definitive Screening Designs

Let us now shift to a different experimental context and a decision-maker at an early stage in her explorations. To set the scene, consider the common marketing practice of customer loyalty programs often used by retail chains, airlines, credit card companies, and others. Typically, customers provide basic data about themselves in exchange for discounts or other benefits. Some promotions are communicated on the company website and others pop up on customers' social media sites.

Consider an online clothing retailer that wants to maximize the response rate for a special offer on a jacket that has a list price of $150. At the time of this fictional experiment, the team has a screening question: which aspects of a promotion generate the highest net sales? After searching the literature and discussing the idea with the marketing team, she identifies the following factors to investigate in an initial screening experiment:

- Form of discount: dollars or percentage? (dichotomous)
- Cash value of discount: continuous from $7.50 to $60
- Shipping charge: continuous from $0 to $15.95
- Days until delivery: continuous from 1 to 5
- Promotional channel: website or social media? (dichotomous)
- Jacket color shown in image: blue or red? (dichotomous)
- Jacket alone or shown with a model? (dichotomous)

Some of these effects could be non-linear: for example, with a steep enough discount, the firm could sell many jackets but realize lower net revenue (sales plus shipping charge minus discount) than compared to a smaller discount. Additionally, there could also be interactions or second-order effects that could confound main effects.

With online technology, it is a straightforward matter to show varying combinations of ads to thousands of customers in the loyalty program. By randomly displaying ads to large numbers of customers, the retailer should be able to discern the most influential factors in the list. The findings might, in turn, suggest follow-on experiments with a different set of factors of values, but these seven factors are adequate to illustrate the idea of a *definitive screening design* (DSD).[1]

> "Definitive Screening Designs are a family of extremely efficient (small) screening designs for continuous or two-level categorical factors. The designs are particularly useful if you suspect active two-factor interactions or suspect that a continuous factor's effect on the response might exhibit strong curvature."[2]

We could use one of the experimental designs already covered in this chapter, but with seven factors, a conventional design would require an enormous number of runs just to identify main effects, let alone interactions. In the current example, a full factorial design with two levels for each factor would require $2^7 = 128$ runs. As we will soon see, the DSD can provide very useful information with far fewer runs.

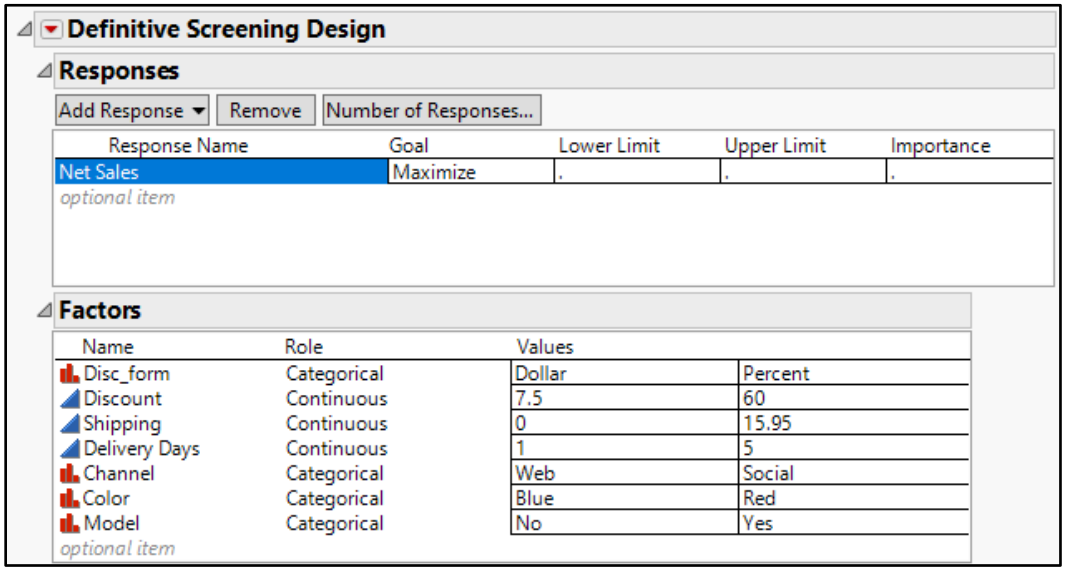
Additionally, the methods that we have seen thus far typically simplify continuous variables to two endpoints (low and high values). If a factor has a non-linear effect on the response, it is essential to investigate more than two values. We will create a DSD that includes a central value between the high and low so that we can look for curvature in a main effect.

Definitive screening designs have several key advantages over standard screening designs. They can discern non-linear effects of individual factors and will identify both main and two-factor interactions. For situations in which screening is the goal, DSDs offer an efficient approach for researchers.

We will illustrate the core concepts with a simplified example based on this marketing scenario. In a real application, we would want quite a large number of customers responding to each of the experimental treatments, so we would want many replicates. Here, I will ask for 10 simply to demonstrate the approach.

1. Select **DOE ► Definitive Screening ► Definitive Screening Design**. Name the response variable as **Net Sales** and leave the **Goal** as **Maximize**.
2. Enter the seven factors from the bulleted list above. Note that you need to distinguish between continuous and categorical factors. When you are finished, the dialog box should look like Figure 21.10.

**Figure 21.10: Launch Definitive Screening Design**




3. Click **Continue**.
4. To screen for non-linear effects, under **Design Options**, choose **Add Blocks with Center Runs to Estimate Quadratic Effects**. Leave the number of blocks at 2 and change the number of extra runs to 0. Click **Make Design**.

**Figure 21.11: A Definitive Screening Design for the Marketing Problem**

**Design**

| Run | Disc_form | Discount | Shipping | Delivery Days | Channel | Color | Model |
|---|---|---|---|---|---|---|---|
| 1 | Percent | 33.75 | 15.95 | 5 | Social | Red | Yes |
| 2 | Dollar | 33.75 | 0 | 1 | Web | Blue | No |
| 3 | Percent | 60 | 7.975 | 5 | Web | Red | No |
| 4 | Dollar | 7.5 | 7.975 | 1 | Social | Blue | Yes |
| 5 | Percent | 60 | 0 | 3 | Social | Blue | Yes |
| 6 | Dollar | 7.5 | 15.95 | 3 | Web | Red | No |
| 7 | Percent | 60 | 0 | 1 | Social | Red | No |
| 8 | Dollar | 7.5 | 15.95 | 5 | Web | Blue | Yes |
| 9 | Dollar | 60 | 15.95 | 1 | Social | Red | Yes |
| 10 | Percent | 7.5 | 0 | 5 | Web | Blue | No |
| 11 | Dollar | 60 | 0 | 5 | Web | Red | Yes |
| 12 | Percent | 7.5 | 15.95 | 1 | Social | Blue | No |
| 13 | Percent | 60 | 15.95 | 1 | Web | Blue | Yes |
| 14 | Dollar | 7.5 | 0 | 5 | Social | Red | No |
| 15 | Dollar | 60 | 15.95 | 5 | Social | Blue | No |
| 16 | Percent | 7.5 | 0 | 1 | Web | Red | Yes |
| 17 | Dollar | 33.75 | 7.975 | 3 | Web | Blue | No |
| 18 | Percent | 33.75 | 7.975 | 3 | Social | Red | Yes |

Notice that this design has 18 runs, in contrast to a full-factorial design of 128 runs. Moreover, look at the values tests for the three continuous factors. In a conventional design, the discount amounts would be set at 7.5 and 60 only. By testing some runs at a setting of 33.75, we can learn if there is a non-linear relationship between net sales and the amount of the discount. Notice also that the 18 runs break down to include 9 runs each for the dichotomous categorical factors and six runs each for the three levels of the continuous variables.

There is considerably more that can be said about DSDs, but at the introductory level of this book, we have probably gone far enough. Inasmuch as one key advantage of DSDs is the examination of non-linear effects, let's now turn to another class of experimental designs intended to map out such complex causal situations.

## Non-Linear Response Surface Designs

Recall the earlier discussion of response surfaces that might be non-linear. With multivariate linear models, we might imagine a smooth plane (or hyperplane) that describes the relationship between a response variable and multiple continuous factors. As noted in the section on definitive screening designs, we typically tested high and low values of factors with the implicit assumption that the response variable would change in linear fashion. For example, if the response increases when a factor level shifts from low to high, we might infer that at an intermediate factor level, the response variable would also take on an intermediate value. But what if the response function curves, so that between the two tested factor levels it declines?

Rather than thinking of the response variable as changing linearly along a hyperplane, we consider non-linear models with a response surface that might have hills or valleys. We can

design an experiment explicitly for the purpose of exploring possible curvature in the response surface by generating sufficient data to estimate models with quadratic or higher power polynomial terms.

We will return to the Custom Design platform to illustrate this concept using another marketing example. Consider a simpler two-factor experiment to evaluate a customer loyalty program. When a known customer makes a purchase, the retailer's system captures the details of each transaction to their data warehouse, tracking consumer purchasing patterns.

Among the common rewards programs are special offers generated at the register. A retailer might present the member-customers with a coupon for a discount on their next purchase of a specific product if they return to the store within so many days.

For this example, suppose that a retailer wants to conduct a two-factor experiment with its customer loyalty program. The response variable is *sales lift*, a common term for the increase in the amount customers spend in a given period above and beyond their normal purchases. For example, a grocery store could have a great deal of data about prior buying behavior. If customers receiving special offers end up spending an average of $20 per month more than customers not receiving the offers, the sales lift would be $20.

Suppose further that the two factors are the dollar value of the discount offered and the length of time allowed before the offer expires. With each of these factors, it might be reasonable to anticipate curvature in the response. A very small discount might not provide enough incentive, but too large a discount might offset any sales lift. Similarly, if the coupon expires too soon, customers might not schedule a return visit soon enough. If the coupon expires too far into the future, customers might forget about it and not use it. The retailer might want to dial in just the right combination of dollars and days to maximize the sales lift and will randomly distribute coupons with different offers over the next month. As a first approximation, we will assume that the response curves are quadratic (in other words, parabolic in shape).

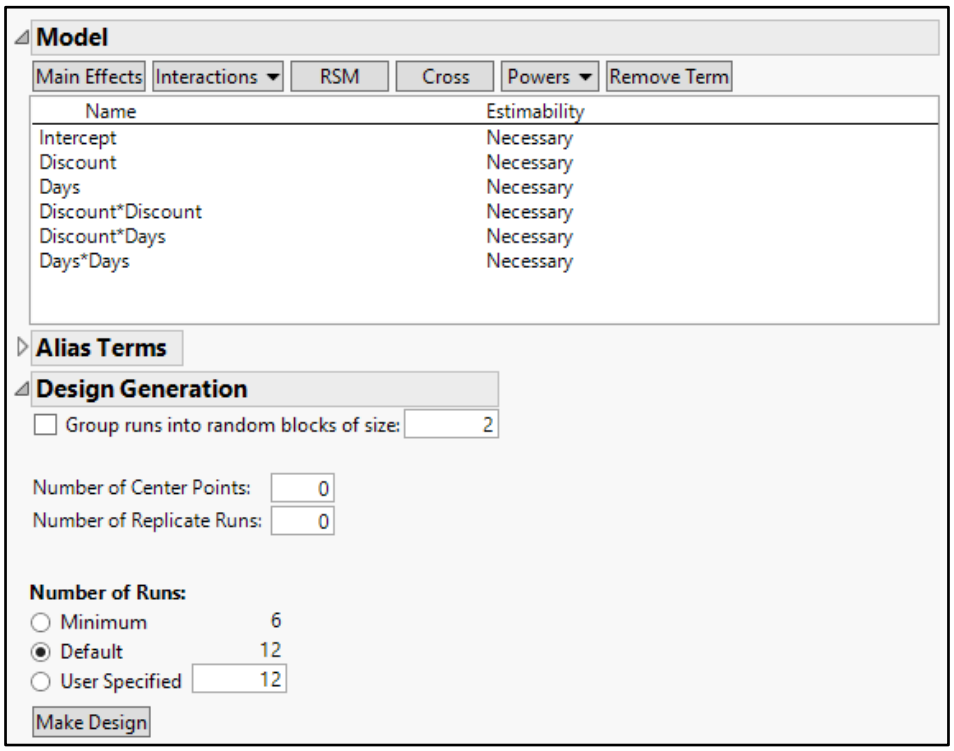
Because the factors are continuous and could conceivably be set at any number of levels, we need a strategy to select starting levels and to evaluate the quality of those choices. One measure of the design quality is the variance or standard errors of the coefficients that our design will ultimately enable us to estimate. JMP features an interactive graph called a *Prediction Variance Profile Plot* that enables us to visualize the effects of different choices and to compare alternative designs.

1. Select **DOE ► Custom Design**. Name the response variable **Sales Lift** and note that we want to maximize it.
2. Create two continuous factors. Name the first factor **Discount**, which ranges in value from -10 to -1 dollars, and the second factor is **Days** ranging from 2 to 30 days before the coupon expires. Click **Continue**.

In the designs that we have seen thus far, we would test a high and low value of each continuous factor with a linear model. The Model panel (not shown) should look familiar to you, indicating that we will estimate an intercept and coefficients for Discount and Days. Looking at the bottom of the report, this will require a minimum of 3 runs, with a recommendation of 9 runs.

3. Now click **RSM** (response surface methodology) in the Model panel (see Figure 21.12) and look to the bottom of the report. The minimum number of runs now doubles to 6, and the default number increases from 9 to 12 runs.

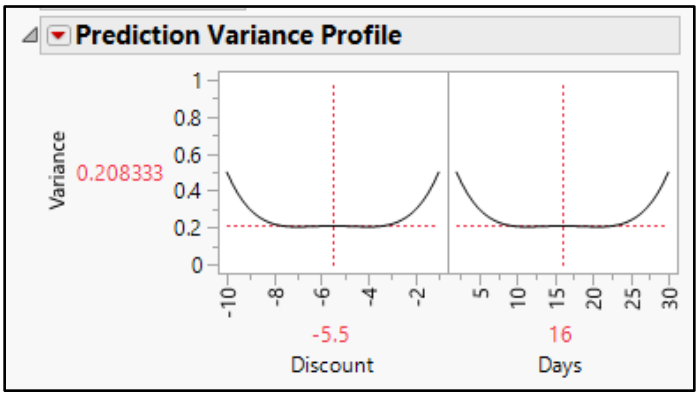
**Figure 21.12: Effect of Specifying a Model Using RSM**




4. Click **Make Design.**
5. As we have seen before, we now have a **Design** panel followed by a **Design Evaluation** panel listing several hidden items. We use design evaluation to characterize the likely variability of our estimates of the response variable.
6. Click the disclosure icon next to **Prediction Variance Profile**.

Prior to running the experiment, we cannot estimate the magnitude of the sampling error. However, the profiler can show us the relative size of the variance associated with specific values of the factors. Initially, we see that if both factors are set at the central points of the ranges we specified (indicated by the vertical red dashed crosshairs in Figure 21.13), then the resulting relative variance of prediction is 0.208333.

**Figure 21.13: Prediction Variance Profiler for the Coupon Experiment**



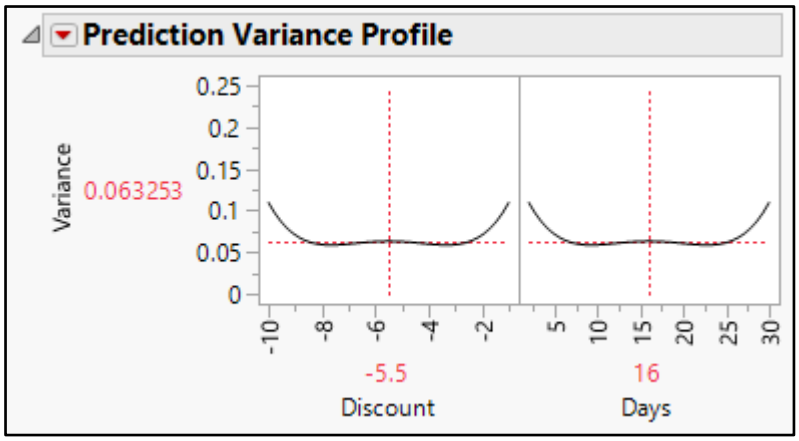

Notice in Figure 21.13 that the prediction variance reaches this value when the factors are set at their midpoint values. Typically, we would like the prediction variance to be relatively small over the entire range of factor values, and we can examine the profiler to discover the worst case for the prediction variance.

7. Move your cursor over the vertical red dashed line in the left-hand profiler plot and click and drag the line left and right. As you do so, notice that two things happen: the horizontal dashed line moves up and down, as does the entire profiler graph to the right.

   At most, with both factors set at their respective extremes, the relative variance rises to about 0.79. We could evaluate alternative designs against this standard by trying, for example, different initial factor levels, a cubic rather than quadratic model, or a different number of runs.
8. To see how the number of runs influences variability, click the **Back** button at the bottom of the report and change the number of runs from 12 to 48 and then click **Continue**.
9. As you did above, disclose the **Prediction Variance Profiler.**

   In this profile (see Figure 21.14), the variance profile is flatter than in the simpler design. The initial relative variance value is approximately 0.063, considerably smaller than before.
10. Once again, slide the red vertical line in the **Discount** graph within the **Prediction Variance** profiler. Notice that as you move to the right, the level and the shape of the other curve changes. Using 48 runs rather than 12, we reduce the magnitude and range of prediction variance. At extreme values of the factors, the variance is only 0.19.

**Figure 21.14: A Prediction Variance Profile with 48 Runs**




Once the investigators are satisfied with the level of risk as represented by the prediction variance, they could proceed to make the table, carry out the experiment, and collect response data.

## Application

In this session, we have not worked with pre-saved data tables, but instead have created data tables based on different DOE principles. Use what you have learned in this chapter to design some experiments suitable for the scenarios presented here. Each exercise specifies a random seed so that your design can be compared to a solution prepared by the author.

1. *Scenario:* We will continue our work on concrete compressive strength.
    a. Return to our initial full factorial design and add a fourth categorical factor representing the type of stone used as a coarse aggregate. Set the initial values as **Granite** or **Marble**. Leave all other settings as they were initially but set the random number seed to **27513**. Make the data table and report on the **Patterns** in the first five rows of the table.
    b. How did the addition of a fourth factor change the design of this experiment?
    c. Now modify the design to accommodate five blocks (for a five-person team). Report on your design.
    d. Modify the design to be a fractional factorial design with only half as many runs. What information do we give up with this design?

2. *Scenario:* We will continue our work on sales lift in a customer loyalty program experiment but screen for effects when we want to test a total of five factors. In this problem, we will assume we are still working at the screening stage of the experiment. Use the **DOE ► Definitive Design** command and set the random number seed initially at **1918**.
    a. In addition to the two factors of discount amount and days as used earlier, we will add three more two-level factors to be screened in a full-factorial design:
        - A categorical factor indicating whether the discount applies to a specific product or to the entire purchase.
        - A continuous factor indicating the minimum purchase dollar value required to qualify for the discount (for example, "$5 off your next purchase over $20"). This factor should range from $2 to $20.
        - A categorical factor indicating whether the discount is valid only at the same store or at any store in the chain.
    b. How many runs are required?
    c. Look back at the chapter example and briefly explain what happens when we move from a two-factor design to a five-factor DSD.
    d. Compare the variance prediction of this design with all factors set at their initial levels to the variance prediction within our first chapter example. Comment on what you find.
3. *Scenario:* There is a growing body of evidence to suggest that sleep plays an important role in the learning process. Some neuroscientists think that our brains consolidate newly acquired learning during the sleep process so that interrupted or insufficient sleep can prevent us from completing the full cycle of learning a new skill or set of concepts. This exercise is inspired by two articles reporting on studies of the effects of sleep on learning. (Stickgold and Walker, 2005; Walker, Brakefield, Morgan, Hobson, and Stickgold, 2002). The original experimental conditions have been modified slightly.

    This experiment is conducted with 72 right-handed adults (36 men and 36 women). Researchers teach subjects to perform a complex motor skills task consisting of repeating a five-number sequence at a computer keyboard. Subjects learn the sequence and then repeat it as many times as they can in a 30-second interval. Training consists of 12 repeated trials with 30-second breaks in between. Subjects are then retested later. Retesting consists of two trials within two minutes with the performance score being the average of the two trials; the dependent variable is the improvement in performance scores. Ultimately, we have 72 runs with an improvement score for each of the 72 subjects.

Half of the subjects received training at 10 a.m. with retesting at 10 p.m. the same day and again at 10 a.m. after a night's sleep (the Awake First condition). The other half of the subjects were trained at 10 p.m., allowed a night's sleep, and retested at 10 a.m. and 10 p.m. the next day (Sleep First). The groups were further differentiated in that 50% of the individuals in each group were awakened briefly during the night, interrupting their sleep ("Full Night" versus "Interrupted"). The goal was to identify what role sleep might play in learning.

Set the random seed to **4495**.

a. Create a custom design for 72 subjects (runs) using the three categorical factors listed below with a plan to estimate all 2nd order interactions. Make the data table, and then report the three factor values for the first five rows of the data table.
    - Gender
    - "Awake First" versus "Sleep First"
    - "Full Night" versus "Interrupted"

b. Now, let's consider altering the third factor by allowing the possibility that they are interrupted a varying number of times during the night. Specifically, now the third factor refers to the number of interruptions, which can be set at 0, 1, or 2. The simplest way to do this is to delete the factor and add a new, three-level discrete numeric factor. Make the data table and report on the ***factor values*** in the first three rows of the table.

c. Use the variance prediction profiler to compare the performance of this design to one in which researchers work with 144 subjects rather than 72. Comment on the advantages and disadvantages of doubling the sample size.

4. *Scenario:* Social scientists, opinion pollsters, and others engaging in survey research are often concerned with low response rates in household surveys. Some survey researchers offer incentives to encourage responses, and some have investigated the effectiveness of different types of incentives. A survey article (Singer, 2002) suggests that potential factors that affect response rates include the following:
    - Type of incentive (monetary versus non-monetary gifts)
    - Burden of the survey (length, complexity)
    - Whether incentive is offered at the time of contact or contingent upon the completion of the survey
    - Number of contacts made (that is, how many times the respondent is contacted about the survey)
    - Survey mode (mail, telephone, face-to-face, email[3])
    - Amount of money offered (for monetary incentives)
    - Guaranteed gift versus entry in lottery

For this exercise, set the random seed to **10501**.

a. Which of the factors are continuous and which are categorical?
b. For each of the categorical factors, identify (or define) plausible factor levels that might be suitable in an experimental design.
c. Consider an experiment in which the response variable is dichotomous[4]: respondents either complete a survey or they do not. If we were to create a full factorial design using all the factors and 2nd order interactions listed above, how many runs would be required?
d. Create a design that measures just the main effects of each factor. Specify 200 runs and report descriptively on the design.
e. Finally, create a definitive screening design for this study. To do so, you will need to modify the Survey Mode factor to make it dichotomous. Define the two levels as face-to-face or remote. Compare the minimum and maximum variance predictions of this method to the design in step d.

5. *Scenario:* Manufacturers frequently test materials and products to determine how long they will stand up to repeated use or to field conditions. This exercise is inspired by an article (Fernandes-Hachic and Agopyan, 2008) describing the use of a material called unplasticized polyvinyl chloride (uPVC), a plastic that is sometimes used in the manufacture of windows and home siding. Specifically, building products manufacturers in Brazil want to assess the durability of the product when exposed to natural weather in Sao Paulo over a period of five years. We will assume the role of designers for this experiment.

   The response variable is a standard measurement known as the Charpy Impact Resistance Index. The index is a continuous measurement of the amount of energy lost when a pendulum strikes the material from a fixed distance.

   The experimenters will formulate batches of uPVC while varying three components of the material (described below), measure the sample specimens using the Charpy Index at the time of manufacture, and then measure annually for five years. All sample specimens are white in color. During the five-year experiment, all specimens will be left outdoors at the natural weathering station in Piracicaba, Brazil and fixed at a 45° angle facing north.

   The experimental factors are these:

   - **Impact modifier.** These are chemicals added to the compound formulation. Four different modifiers will be used: acrylic (ACR), chlorinated polyethylene (CPE), acrylonitrile-butadiene-styrene (ABS), and metacrylate-butadiene-styrene (MBS).
   - **Thermal stabilizer.** These compounds influence the extent to which the uPVC expands and contracts at varying temperatures. The four formulations tested were: (1) a complex of lead, barium, and cadmium (Pd/Ba/Cd); (2) lead (Pb); (3) a complex of barium and cadmium (Ba/Cd); and (4) tin (Sn).

- **Anti-UV additive**. Titanium dioxide protects against degradation caused by ultraviolet radiation. Because this additive is quite costly, three small quantities will be tested: 3, 5, and 10 parts per hundred resin (phr). Treat this as a three-level discrete numeric factor.

  For this exercise, set the random seed value to **225196**.

a. Create a custom design to test main and 2nd order interaction effects. Make the data table, and report on Variance Prediction Profile. What challenges do you envision in carrying out this design?

b. In the actual situation, practical considerations limited the scale of this experiment, so that it was not feasible to use the default number of runs. Design and experiment with half the number of runs as in the design. Use the prediction profiler to compare the performance of the half- and full factorial designs. Comment on the effects of using the reduced design.

6. *Scenario:* A *Harvard Business Review* article described an email survey conducted by Crayola.com that illustrates DOE in marketing. (Almquist and Wyner, 2001). The response variable was simply whether the email recipient completed the survey questionnaire.[5] During a two-week period, surveys were emailed to customer addresses within the company database according to a design that considered the following factors:

   - **Subject Line**: The subject line of the email was either "Crayola.com Survey" or "Help Us Help You"
   - **Salutation:** The first line of the message was either "Hi [user name] :)", "Greetings! [user name]", or just "[user name]"
   - **Call to Action:** "As Crayola grows…" or "Because you are…"
   - **Promotion**: Three different conditions—(1) no offer; (2) $100 product drawing; (3) $25 Amazon.com gift certificate drawing
   - **Closing**: The email was signed either "Crayola.com" or "EducationEditor@Crayola.com"

   For this exercise, set the random seed to **688098**.

   a. First, create a full factorial design (main effects and interactions) with 10,000 runs (it's cheap to send email). Create the design and report the minimum variance profile prediction value.

   b. When Crayola conducted the experiment, they opted for fewer runs. Use the **Custom Design** again for main effects only and omit the "Closing" factor. Cut the number of runs to 5,000 and report the minimum variance profile prediction value.

   c. JMP has a platform to aid in design comparisons. Select **DOE ► Design Diagnostics ► Compare Designs**. Comment on the comparison of these two designs.

[1] A good theoretical article on this class of designs is Jones, B., and Nachtsheim, C. J. (2011). "A Class of Three-Level Designs for Definitive Screening in the Presence of Second-Order Effects." *Journal of Quality Technology 43:1–14*. Also, refer to Chapter 7 of the *Design of Experiments Guide* within the JMP Documentation Library.

[2] Sall, John, Ann Lehman, Mia Stephens, and Sheila Loring.2017. JMP Start Statistics: A Guide to Statistics and Data Analysis Using JMP 6th Ed. Cary NC: SAS Press.. p. 473.

[3] Email was not a major modality of surveying in 2002, but it is included in this exercise to illustrate the concepts of experimental design.

[4] Please note that JMP's DOE platform is designed for continuous response variables; this exercise is intended to illustrate how the number and complexity of factors influences the scale of a design.

[5] Please note that JMP's DOE platform is designed for continuous response variables; this exercise is intended to illustrate how the number and complexity of factors influences the scale of a design.

# Chapter 22: Quality Improvement

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## Overview

We expect to observe variation in the continuous functioning of natural and artificial processes. In this chapter, we will study several techniques commonly used in the world of business, often in the context of manufacturing and service operations. Each of these techniques plays a role in efforts to monitor, assure, and improve the quality of processes that provide services or manufacture goods. As you encounter these techniques, you will recognize concepts and approaches that we have seen earlier, but now we will combine them in different ways. Collectively, these techniques help managers control their operations in order to maintain consistent levels of quality and to discover opportunities for improvement.

## Processes and Variation

One of the core ideas in the realm of quality improvement is the distinction between **common cause** variation and **special** (or **assignable**) cause variation. Take, for example, the case of a company that manufactures plastic tubing for hospitals and medical uses. Tubing is made by an extrusion process where liquefied material is forced through an opening to form a continuous tube with pre-specified inside and outside diameters. By controlling the temperature, composition of the material, the rate of flow, and the cleanliness of the equipment, the manufacturer can assure a high degree of consistency in the diameters. Nevertheless, variation occurs after many hours of continuous high-speed operation.

One challenge in managing any process is to distinguish between variation that indicates a problem and variation that is simply part of the fluctuations of any human system. We will conceive of the routine chance variation that characterizes an imperfect process as being attributable to common causes. Special cause variation is typically attributable either to a problem that requires attention or a temporary, passing event. If special causes are rare and

most of the variation is attributable to common causes, we say that the process is "in control." If special cause variation is unacceptably frequent or pronounced, the process is "out of control," and the manager needs to intervene and restore the process to its desired state. Sometimes management will want to improve a process either by shifting its overall level or by understanding and reducing common cause variation.

## Control Charts

A control chart[1] is a time series graph of measurements taken from an ongoing process. Control charts are visual tools that were created in the 1930s so that supervisors on a factory floor could make real-time judgments about whether to intervene in a production process to adjust equipment or otherwise respond to a process that was slipping out of control. There are several types of control charts suitable for different data types and different analytical goals. The design and computations required for many control charts are quite simple, reflecting their historical origins as "quick and dirty" methods to track and improve production. In this section, we will use a single example to introduce several techniques.

Depending on the product or process being observed, we will first distinguish between charts for measurements and charts for counted events. If we are observing a process and measuring a key outcome (for example, the size or weight of a product), we will be charting continuous data, principally by charting means. On the other hand, if we are observing the frequency with which attributes occur (for example, defects of a certain type), then we will chart nominal data, typically by charting frequencies or relative frequencies.

A company manufactures catheters, blood transfusion tubes, and other tubing for medical applications. In this instance, we have diameter measurements from a newly designed process to produce tubing that is designed to be 4.4 millimeters in diameter. Open the data table **Diameter** and look at the data.

Each row of the table is a single measurement of tubing diameter. There are six measurements per day, and each day, one of four different operators was responsible for production using one of three different machines. During the study period, there were two prototypes, and the first 20 days' measurements were for the first prototype. The remaining readings were for the second prototype.

The first 20 days of production were considered Phase 1 of this study; following Phase 1, some adjustments were made to the machinery, which then ran an additional 20 days in Phase 2. In total, there are 240 individual observations (40 days x 6 measurements daily). In the language of quality and process control, a "phase" is a collection of consecutive measurements that correspond to a succession of changes to a process.

In analyzing the data, we will want to place ourselves in the mindset of the operations manager who wants to evaluate the effectiveness of the adjustments. Initially, we will do so by examining a *Run Chart* of individual diameter measurements during the two phases.

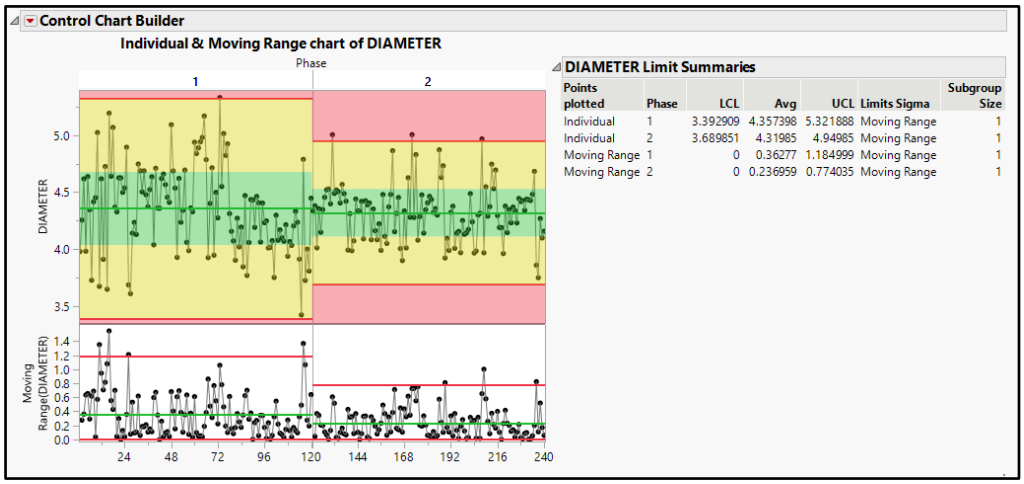
## Charts for Individual Observations

The simplest control chart plots each measurement over time. It is a time series graph of individual measurements. For this first example, we will use JMP's **Control Chart Builder**, which operates much like **Graph Builder**, which we have used throughout the book.

1. Unlike **Graph Builder**, **Control Chart Builder** has just three main graph zones corresponding to the phase of a study, the response variable, and one or more subgrouping variables. Select **Analyze ► Quality and Process ► Control Chart Builder**. Drag and drop **DIAMETER** into the **Y** graph zone, and **Phase** into the **Phase** zone.
2. When the control chart report opens, look at the left side of the report. In the Limits[1] section, check **Shade Zones**.
3. Click **Done**.

The result will look like Figure 22.1. The report window contains two graphs and a table of statistics. On the left side of the report, the upper graph is a run chart of individual measurements, with the Phase 1 results on the left and the Phase 2 results on the right. Below the graph of individual diameters is a Moving Range chart, which shows the differences between successive observations starting with the second row of the data table. Because each panel of the report provides different perspectives, we will discuss each of them in turn.

**Figure 22.1: Control Chart Builder for the Tubing Diameters**




A run chart is a time series graph of individual values with a line connecting each observation. Additionally, the charts in Figure 22.1 display three colored zones, each approximately one standard deviation in width. You will also find a green horizontal line at the mean diameter value and two red lines representing *upper* and *lower control limits.* Control limits are calculated at approximately two standard deviations above and below the mean. These limits are not exactly equivalent to two standard deviations but are based on easily computed approximations, reflecting the origins of control charts.

We refer to the colored zones by letter. Zone A is red, defining the region three standard deviations above or below the mean. Zone B (yellow) is between one and two standard

deviations and Zone C (green) is within one standard deviation of the mean. When a process is in statistical control, it varies randomly around the mean line and approximately 95% of the measurements fall within zones B and C.

In a control chart, we also look for other warning signs that variability is not purely random. In other words, we watch for indicators of assignable cause variation that signal a manager to consider an intervention to bring a process back under control.

Many of these warning signs come down to us from early work at Western Electric, and there is a set of rules that commonly suggest assignable causes. When we scrutinize a control chart, we are looking for specific indications of non-random variation[2] such as:

- Do an excessive number of points fall outside of control limits?
- Is there systematic clustering of consecutive points either above or below the center line?
- Does the series line consistently drift either upward or downward? An increase in sample ranges indicates greater inconsistency in the process.
- Is there a sudden shift in the level of the curve, either upward or downward? Such a shift might suggest a one-time shock to the process.

4. Click the red triangle next to **Control Chart Builder** and choose **Show Control Panel**. On the left side near the bottom of the window, notice the disclosure triangle next to **Warnings[1]**.
5. Click the disclosure triangle to reveal another disclosure triangle next to **Tests**. Click that triangle as well.
6. This opens a list of eight numbered tests with check boxes. Each test refers to a different Western Electric Rule. (For a full explanation of all of the Western Electric Rules, refer to the Tests section of the Control Chart Builder chapter in the JMP Documentation library).
7. As an illustration, check Test 2. This displays a red circle and the number 2 on the ninth point in a sequence when there are nine points in a row above or below the mean in Zone C or beyond.

8. Click **Done**.

Stepping back to look at the chart as a whole, we can see in the Phase 1 Individual run chart for the first prototype that the average diameter is slightly less than 4.4 mm (4.357 mm according to the **Diameter Limit Summaries** in the right panel), but that there is quite a lot of variation and instability in the process. We have one point in Zone A, and 12 violations of Test 2. For the second prototype, after adjustments to the process were made, the mean remained very nearly unchanged, but the control limits were drawn in closer to the mean. In other words, the common cause variation was reduced in magnitude. We now find three Zone A violations, but only six Test 2 instances.

The moving range chart also indicates that the Phase 2 modifications reduced process variability as measured by the range between observations. At a glance, we can see that measurement differences between consecutive observations tended to be considerably smaller for the second prototype than for the first.

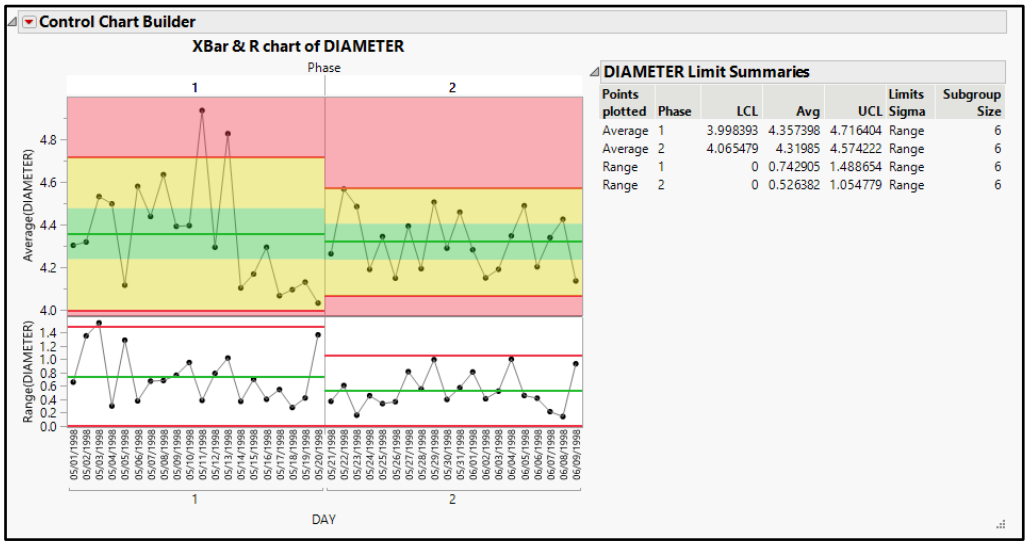
Before searching for possible special causes, let's take a closer look at the question of *stationarity*, or stability of the process. We will examine the degree of stationarity by looking at averages of several measurements at a time. Specifically, we will use an *X-bar Chart*, which graphs the means of successive samples through time.

## Charts for Means

Because this tubing example involves a continuous product, the individual diameter measurements could be subject to considerable sampling variation. In cases like this, it often makes sense to compute the mean of a sample of measurements and use the sample mean as a gauge of process stability. In our data table, we have six daily measurements all recorded by a different operator each day, so it is convenient to define subsamples by day (in other words, our sample size will be $n = 6$).

1. We want to modify the current Control Chart Builder graph and need to re-open the control panel. Click the red triangle next to **Control Chart Builder** and choose **Show Control Panel**.
2. Drag **Day** to the horizontal axis drop zone at the bottom; just drop it directly over the middle of the horizontal axis. The title, the graph, and the **Limit Summaries** change, as shown in Figure 22.2.
3. Again, check **Shade Zones**.
4. Click **Done**.

**Figure 22.2: XBar-R Chart**



An XBar chart, as the name implies, is a graph of sample means. In this example, we have six daily measurements of diameter over a period of 40 days. We are specifying that we want to chart the daily sample means for the two prototypes, so the chart will display 20 points within each phase.

Like the chart that we made in Figure 22.1, we again see zones and upper and lower control limits. There are different methods for establishing the control limits, and the default method in a JMP Individual and XBar charts is to base the limits on the observed range of values. On the left side of **Control Chart Builder**, note the **Limits** in the upper and lower graphs (that is, graphs [1] and [2]) refer to Range as the estimate of sigma. Because we specified phases in this study, the center lines and control limits are computed separately for each phase of the investigation.

A process cannot be in control if the variability is not constant throughout the observation period. In other words, there is a requirement of homoscedasticity in order to draw conclusions about whether a process is in control. Traditionally, there are two methods to judge the stability of the process variation: by charting the *sample ranges (R chart)* or by charting the sample *standard deviations (S chart)*.

The chief virtue of an R chart over an S chart in a factory setting is its computational simplicity: if an operator makes six sample measurements, then plotting the range of the sample requires only the subtraction of two measurements. Computing a sample standard deviation might have been an intrusive burden on a production worker when control charting was first introduced into factory work. With software, the S chart is readily available and probably the better choice.

The default setting, in a nod to history, is the R chart. Now, look again at your screen or Figure 22.2. You should develop the habit of interpreting these charts by looking first at the lower R chart to decide whether this manufacturing process exhibits stable variability. In the Range Chart, we see a center line that represents the mean of the ranges of the twenty samples. Above and below the center line are control limits and the colored zones described above.

As we look at this range chart in Phase 1, we note that the range of just one sample of the 20 lies above the UCL—exactly 5% of the observed samples. Of course, the LCL in the R-chart falls at 0, and it is impossible that any sample range would fall below the LCL. On the first several days, the variability appears more extreme than on later days, although the final day shows a range comparable to those earlier ones. Toward the right, there are six consecutive days with below-average ranges, which might suggest a special cause. Overall, though, there are no large red flags in the range chart.

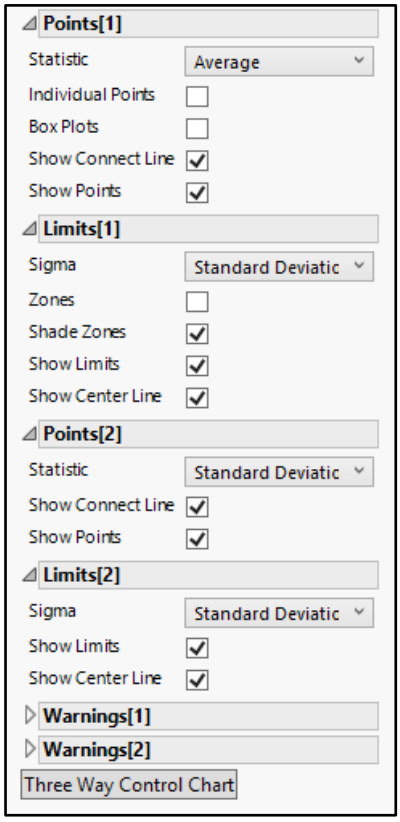
With a stable range, we can meaningfully interpret the XBar chart. Here again, there is a center line at the grand mean of the first 20 samples. The upper and lower control limits lie approximately three sigmas from that mean, where sigma is estimated based on the sample ranges.

In the first ten samples, the means fluctuate without any noteworthy pattern. Then we find two out of three in a row with means above the UCL, followed by seven consecutive means below the center line, and a gradual descent toward the LCL. A non-random pattern such as this one suggests that the process needs some adjustment.

Let's take a second look at the same data, this time using an S chart to evaluate the variability within the process. This is easily accomplished just by changing some settings in **Control Panel** on the left side of **Control Chart Builder**, shown above in Figure 22.3. The upper and lower graphs each display points and limits. These are set by default, but we can alter them. The limits in both graphs are based on observed ranges in each phase and the plotted points in the lower graph [2] are also based on ranges. We want to change the basis of computations from ranges to standard deviations.

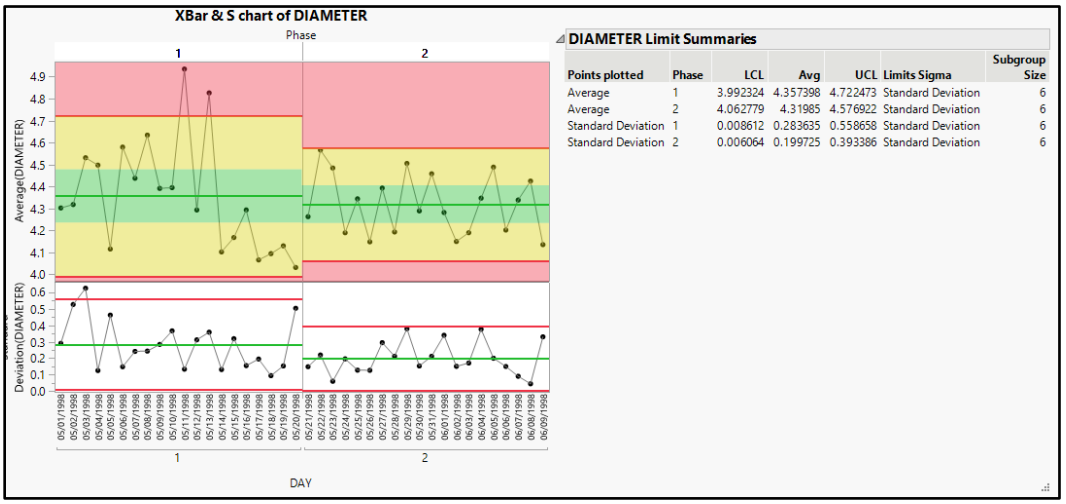
5. Again, click the red triangle and open the **Control Panel**.
6. Click the three instances where **Statistic** is set to **Range** and change them to **Standard Deviation.** (See Figure 22.3.)

**Figure 22.3: The Control Panel in Control Chart Builder**




The new report is shown below in Figure 22.4. The general visual impression is very much like the Xbar and range chart. Upon closer inspection, you should notice three key differences. First, the vertical scale in the S chart is different from that shown in the R chart because we are plotting sample standard deviations rather than ranges. Second, because the standard deviation is less sensitive to extreme values than the range, we no longer observe a run of consecutive six points below the center line in the lower graph. Finally, the plotted means in the XBar chart are identical to those in Figure 22.2, but the UCL and LCL are located slightly farther from the center line, reflecting a different (and more accurate) estimate of the process standard deviation.

**Figure 22.4: Xbar & S Chart for the Diameter Data**




The conclusions that we draw from these control charts are the same as those noted above. The first prototype testing indicates a process that is not in statistical control. At this point, management intervened to revise the process.

The S chart shows a smaller mean and standard deviation in Phase 2, indicating less process variability overall. Moreover, the standard deviation is more stable, consistently remaining within the newer and tighter control limits. This is a process with stable variability.

The XBar chart also shows improvement with respect to variability in the mean diameters. The charted line remains within the narrower control limits and the oscillations are random. However, recall that the nominal diameter of this tubing is 4.4 mm and the mean of these samples is smaller than for the first prototype, when the grand mean was 4.36 mm. Whether this is a problem for the firm depends on the specifications that they have established with and for their customers. At any rate, the revised process is more consistent than before but appears to be producing a product that does not conform to the intended specification of diameter.

## Charts for Proportions

For some processes, we might be concerned with a categorical attribute of a product or service rather than a measurable variable. For example, manufactured products might be unacceptable for distribution to customers because of one or more defects. Some situations might focus on defective items or units while others might focus on the number of defects. The two major types of attribute charts are known as *NP charts*, which illustrate simple counts of events, and *P charts*, which represent proportions or relative frequency of occurrence. To illustrate how control charting can help monitor attribute data, let us turn to some additional data about airline delays.

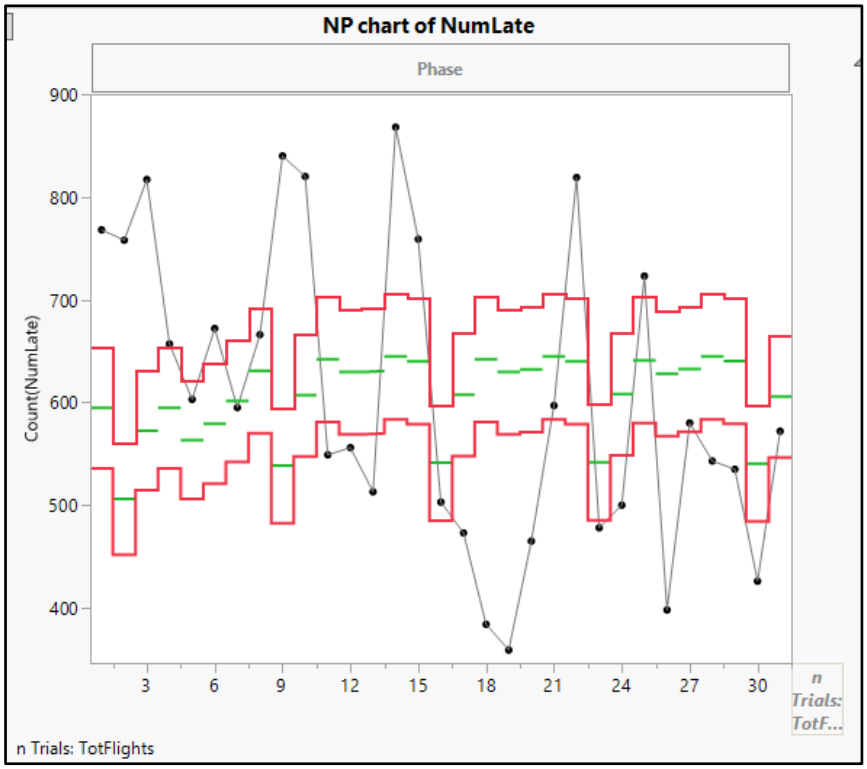
Air travel inevitably involves some chance that a flight will be delayed or canceled. Our data table **Airline delays 2** has daily summary information about flights operated by United Airlines that were scheduled to arrive at Chicago's O'Hare International Airport for one month. United is the largest carrier using O'Hare and accounted for just over 23% of all arrivals at the airport that

month. Conversely, O'Hare is United's largest hub, with nearly 12% of all United flights touching town there. In other words, we are looking at flights for the largest carrier at one of the busiest airports in the U.S. United was chosen to illustrate these control charts rather than to suggest that their performance is either superior or problematic.

The data table contains information about delayed, canceled, and diverted flights. Each row represents a single day, and we have both counts and relative frequencies for the occurrence of these flight disruptions. Let's first look at the number of delayed arrivals among all of the flights, taking into account that the total number of flights varied from day to day. The two columns of interest in this analysis will be the number of late flights on a given day (**NumLate**) and the total number of flights that day (**TotFlight**).

1. Open the data table **Airline delays 2.**
2. Select **Analyze ► Quality and Process ► Control Chart Builder**.
3. Click the **Shewhart Variables** button and select **Shewhart Attribute** to indicate we are dealing with a categorical (attribute) variable.
4. Drag **NumLate** to the **Y** drop zone and drag **TotFlight** to the **n Trials** drop zone in the lower right of the graph.
5. Under **Points**, change the **Statistics** to **Count**, and under **Limits**, change **Sigma** to **Binomial (P, NP)**. Your graph will now look like Figure 22.5.

**Figure 22.5: NP Chart for Flight Delays**



With attribute charts, there is no equivalent for the S or R charts. The NP chart is structured identically to other control charts shown earlier. In this chart, though, the center line and control limits are jagged due to the different sample sizes (number of flights) each day. We see a green center line indicating a varying average. We also see varying upper and lower control limits.

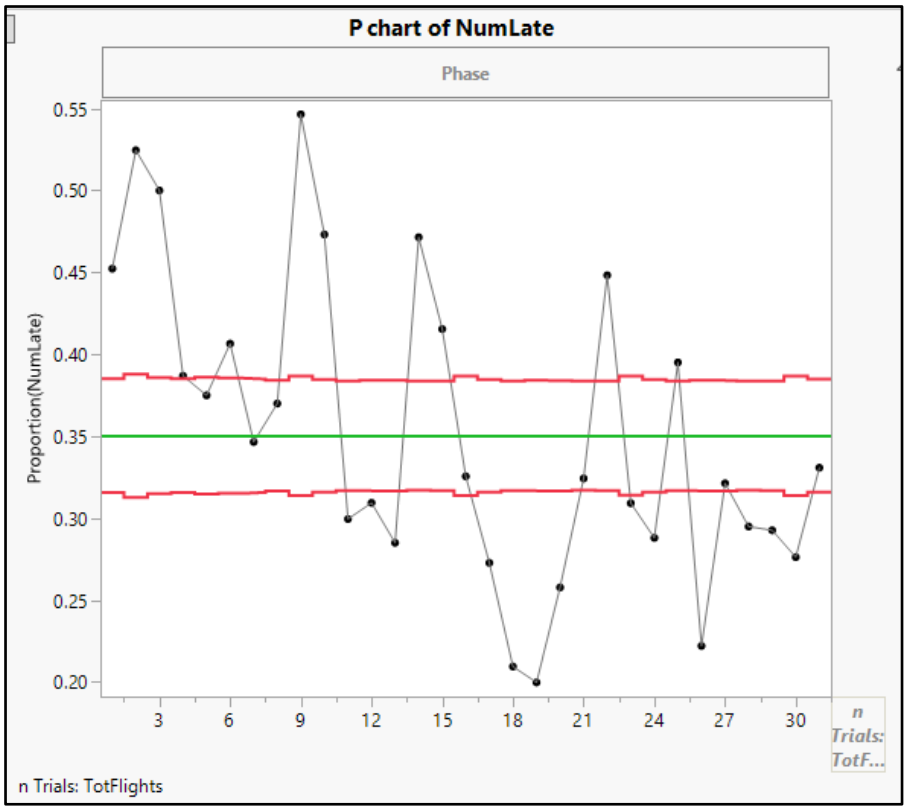
This chart shows very little consistency, and hence illustrates a process that is not in control. There are precious few days on which the number of delays falls within the control limits. With flight delays, values above or near the UCL are the troubling ones, and this process is an example of one that is out of statistical control. The control chart cannot reveal the reasons for the delays but does highlight that the process was less problematic late in the month.

An alternative to the NP chart is the P chart, which analyzes the *proportion* rather than absolute *counts* of late flights. To convert the NP chart into a P chart, one just needs to make one small change.

6. In the **Points Panel** in the lower left of the **Control Panel**, change the **Statistic** back to **Proportion**. The revised graph appears in Figure 22.6.

In this chart, we see that both the center line is continuous, but the control limit lines are jagged. The overall proportion of flights is based on all the data, and we can see that 35% of flights experienced a delay. Because the daily sample sizes are different, the limits are recomputed from left to right, reflecting changing estimates of the underlying sampling distribution. As before, flight delays were more common and more variable in the earlier part of the month.

**Figure 22.6: P Chart of Flight Delays**




## Variability Charts

When a process is out of control, it is natural to seek the assignable causes that account for the variability. There are several quality tools available for this purpose, and we introduce one of them—the *Variability Chart*—here. A variability chart displays a comparison of means and variability of a continuous variability across several groups. An example is the best way to illustrate.

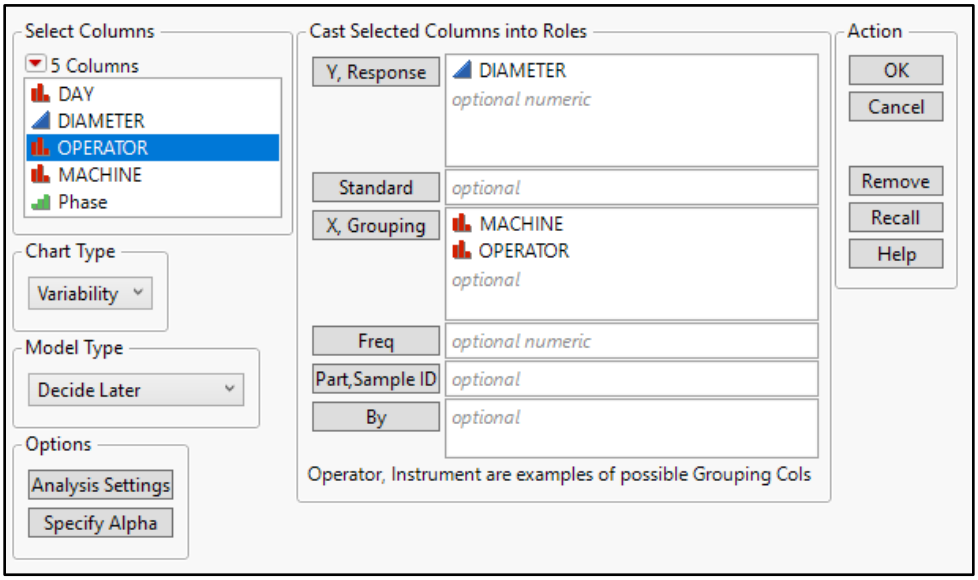
Let's return to the prior case of the tubing diameters and focus on the Phase 1 measurements. You may want to glance back at Figure 22.1 to see the situation. Recall that the process was out of control for prototype 1, and managers took corrective action to bring the process within control for the second prototype. Let's imagine that we are back in the first twenty days working on the first prototype and are trying to understand why the diameters are so erratic. Our data table includes two columns representing the machine and operator for each measurement. Can we associate the variability with one machine or one operator? We will look at both possible factors together, with a primary emphasis on the machine.

1. Return to the **Diameter** data table.
2. Select **Rows ► Data Filter** and Show and Include rows where **Phase** equals 1.

3. **Analyze ► Quality and Process ► Variability/Attribute Gauge Chart**.
4. Cast **DIAMETER** as **Y, Response**.
5. Cast **MACHINE** as **X, Grouping** and then add **OPERATOR** as a second X factor.

Your selections should look like Figure 22.7.

**Figure 22.7: Launching a Variability Chart for the Tubing Diameter Data**

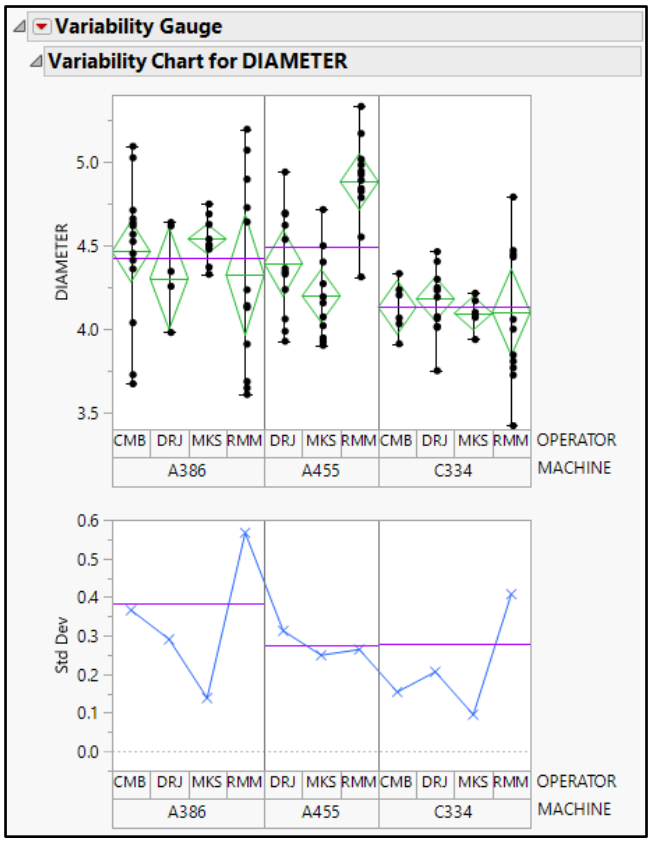



This will produce a two-chart report. In the upper chart, the vertical axis shows the diameter measurements. The chart is subdivided into nested panels. The primary division is by machine, with points plotted by operator for each machine.

To facilitate interpretation of the graphs, do this:

6. Alt-click the red triangle next to **Variability Gauge**.
7. Check **Show Group Means**, **Mean Diamonds**, and **Group Means of Std Dev**.
8. Click **OK** to produce the report shown in Figure 22.8.

**Figure 22.8: Variability Chart for the Tubing Diameter Data**




In the upper chart, we see that the mean diameters are similar for the first two machines, but that they are smaller for the third machine. Operator MKS appears to be the most consistent operator across all machines, and operator RMM is the least consistent.

The lower chart shows the patterns of variability by plotting the standard deviation for each operator within machine. Machine A386 is the most variable of the three, and there is an interaction between the operator and machine. On machine A455, the three operators who use it were similarly variable in their results, but variability was much less consistent across operators on the other two machines.

A knowledgeable manager could use the information in these charts to decide on a course of action by some combination of adjusting the machines and intervening with the operators. Presumably, this is what happened between Phases 1 and 2 to bring the process under control.

## Capability Analysis

In most settings, it is not enough that a process be in statistical control. It is also important that the results of the process meet the requirements of clients, customers, or beneficiaries of the process. To return to our earlier example, if the extrusion process consistently produces tubing that is either the wrong diameter or highly inconsistent, the tubing might not be suitable for the intended medical purposes.

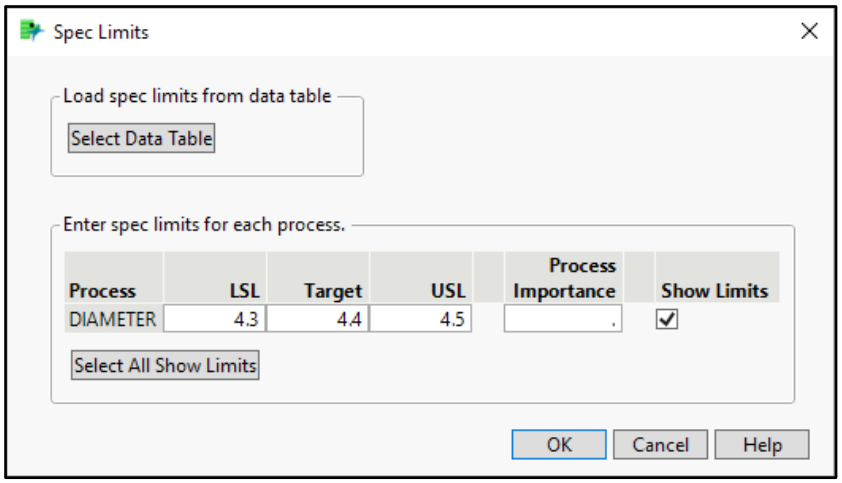
For a process that is in statistical control, *process capability* refers to the likelihood that the process as designed can meet the engineering or design criteria. We analyze capability only for numerical measurement data, not for attributes. Thus, for example, if our tubing manufacturer needs the tubing to have an average diameter of 4.4 mm with a tolerance of ± 0.1 mm, then once the process is in control (that is, Phase 2), they want the individual observations to remain in the range between 4.3 and 4.5 mm.

In a capability analysis, the fundamental question is whether the process is capable of delivering tubing that meets the standard. There are several methods for generating a capability analysis within JMP, and we will illustrate one that is an extension of the Phased XBar-S control chart that we made earlier.

Because we want to use only the Phase 2 data, we will hide and exclude the Phase 1 data.

1. Return to the **Data Filter** for the diameter data and select the **Phase 2** measurements to **Show** and **Include**. Now only the phase 2 data will be used in the calculations and graphs.
2. Select **Analyze ► Quality and Process ► Process Capability.**
3. In the Capability dialog box (not shown), select **DIAMETER** as **Y** and click **OK**.
4. This opens a second dialog box (Figure 22.9). Recall that we want to investigate whether this system can produce tubing that is consistently between 4.3 and 4.5 mm in diameter, with a mean of 4.4 mm. In this dialog box, enter the limits as shown in Figure 22.9 and click **OK**.

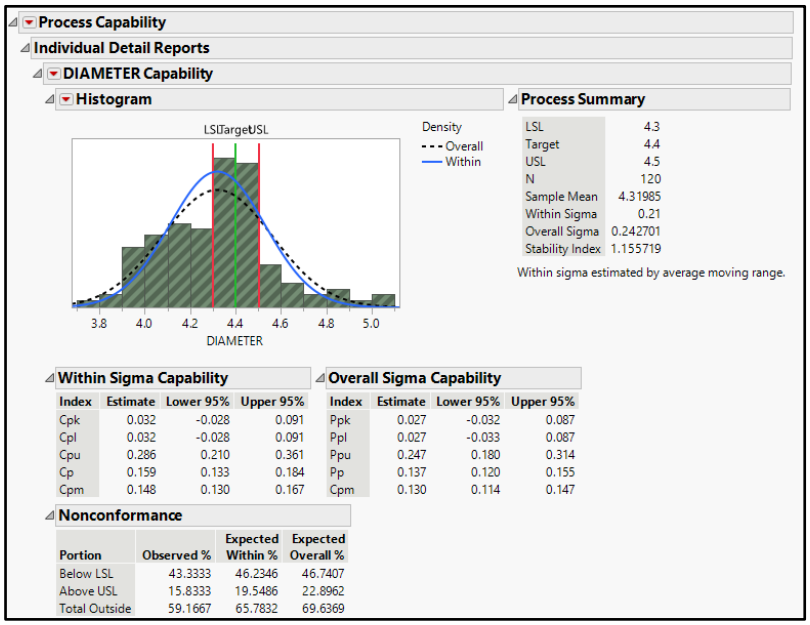
**Figure 22.9: Setting Capability Specifications**




5. This opens a **Process Capability** report. Click the top-most red triangle and request **Individual Detail Reports**.

The **Nonconformance** summary in the **Process Capability** report window (Figure 22.10) shows that slightly more than 43% of the Phase 2 diameters were below the lower specification of 4.3 and almost 16% were above the upper specification limit, for a total of 59% outside of the specifications. So, despite the fact that the process was under control in Phase 2, it does not appear to be meeting the producer's needs.

**Figure 22.10: Capability Analysis Results**



Direct your attention to the histogram at the top of the report. It shows the target marked with a green vertical line at 4.4 mm with two red lines at the upper and lower specification limits. We can see the comparatively small proportion of values falling within the desired specification limits, and this casts doubt on whether the system in phase 2 is capable of regularly producing the desired tubing diameters.

In addition to considering the proportion of observations within the specified limits, there are several common indexes of process capability, and we will look at just three of them. We should interpret these statistics only if we can assume normally distributed data and have a moderately large sample of independent observations. We might fall short on the issue of independent observations, because we might have operator effects here, but the normal quantile plot indicates that normality is a reasonable assumption.

The three indexes of process capability are Cp, Cpk, and Cpm. For this discussion it is less important to know the specific formulas[3] for these indexes than it is to understand what they measure. In a capable process, the large majority of observed measurements fall within the specification limits. In a capable process, all three indexes would be at least 1.33, indicating that all observations lie within the specification limits. Cpk and Cpm adjust for the possibility that the process is not centered on the target.

In this case, all three measures are considerably less than 1. This echoes the visual impression that we form from the graph.

What do we conclude about this process? It is still a work in progress: the good news about Phase 2 is that the firm has now brought the process into statistical control, in the sense that it is stable and predictable. On the other hand, there is further work to do if the firm needs tubing that is uniformly between 4.3 and 4.5 mm in diameter.

## Pareto Charts

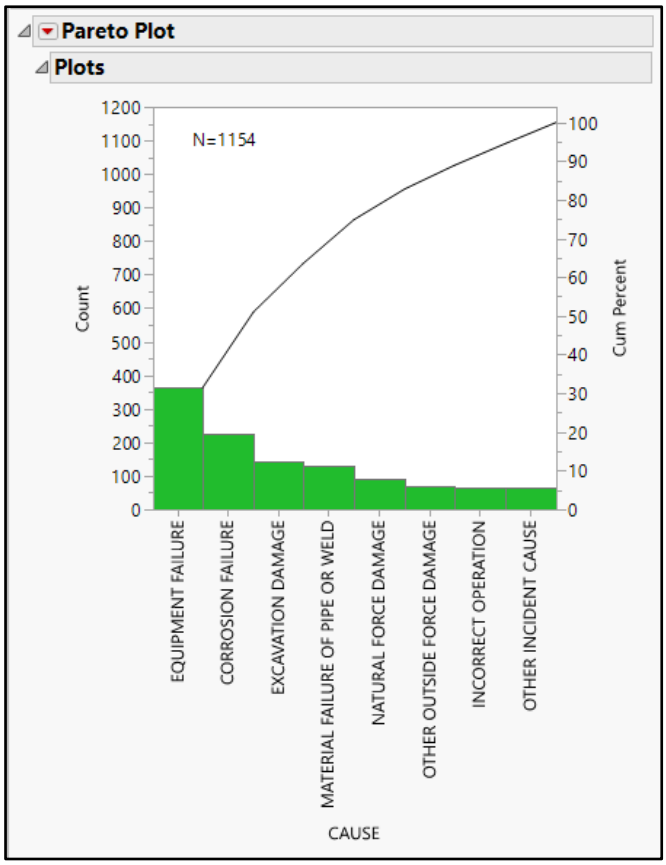
Yet another simple charting tool for attribute data is known as a *Pareto chart*. A Pareto chart is a bar chart showing the frequency or relative frequency of specific attributes, often those representing a problem or the cause of a problem in a process. The bars in the chart are always sorted from most to least frequent to draw attention to the most commonly occurring problems. Additionally, Pareto charts display the cumulative relative frequency of the various problem categories to enable managers to identify the critical few issues that account for the greatest proportion of defects.

To illustrate, let's look back at the pipeline disruption data. We have analyzed the data previously, focusing on several different variables. This time, we will examine the reported causes of pipeline disruptions.

1. Open the **Pipeline Safety** data table.
2. Select **Analyze ► Quality and Process ►Pareto Plot.** Select **CAUSE** as the **Y, Cause** column and click **OK**
3. Click the red triangle and request **N Legend**.

You will now see the graph that is shown in Figure 22.11. The data table contains eight categories of causes, listed on the horizontal axis in descending order of frequency. Looking at the left vertical axis, we can see that more than 350 of the 1154 incidents were attributed to equipment failure and approximately 225 due to corrosion failure.

**Figure 22.11: Pareto Plot of Pipeline Disruption Types**




The right vertical axis displays cumulative percent, or cumulative relative frequency. Thus, for example, you can see that the first four causes account for about 75% of the interruptions.

How is a Pareto plot useful? It can help prioritize preventive or corrective action. For example, incorrect operation accounts for 6% of all incidents, so that improved operator training might not be as beneficial as, say, addressing corrosion or excavation procedures.

Second, it might inform emergency preparedness. Recall that some disruptions lead authorities to evacuate the public from affected areas. We might ask whether some causes lead to more evacuations than others. We can modify the basic plot to group incidents by the column called EVAC, which is a NO/YES dichotomous nominal variable.

4. Click the red triangle and select **Script ► Redo ► Relaunch Analysis**. Select **EVAC** as the **X, Grouping** column and click **OK**.

In the new results, we see two separate panels corresponding to the incidents that did not or did require evacuations. Evacuations are comparatively uncommon but occur at a relatively high rate during excavations. This might lead to a deeper analysis of available data, or prompt decision-makers to reexamine excavations near natural gas lines. You should begin to see the potential uses of this type of chart in suggesting areas for inquiry about possible operational problems.

## Application

Now that you have completed all of the activities in this chapter, use the concepts and techniques that you have learned to respond to these questions.

1. *Scenario:* Let's continue our analysis of the tubing diameter data (**Diameter** data table). As before, we will use sample sizes of 6, corresponding to the six daily measurements.
    a. In Phase 1, we concluded that the process was not in control. Restrict your attention to Phase 1 data once more and create XBar-S charts **BY MACHINE**. Is there any indication that the problems are caused by one or more specific machines? Describe what you find and what you conclude.
    b. Now, create a set of similar control charts breaking out the data by **OPERATOR**. Do any of the operators seem to need attention? Explain.
    c. Let's refocus on just the Phase 2 data. Earlier, we concluded that the process was not capable of producing 4.4 mm tubing within tolerances of 0.1 mm. Perform a capability analysis to determine whether the process is capable within tolerances of 0.6 mm. Report your findings.
2. *Scenario:* Let's return to the airline data in **Airline delays 2**. In this exercise, we will focus on flight cancellations.
    a. Create a P chart for the proportion of canceled flights and report on what you see.
    b. Create an NP chart using the columns **NumCancelled** and **TotFlights** and report on what you see.

3. *Scenario:* Let's return to our analysis of the Indian Index of Industrial Production and look more closely at some of the sub-indexes for specific types of products. For this exercise, we will treat each three-month quarter as a sample and therefore specify a sample size of 3 observations.
    a. Create an XBar-R chart for **Primary** goods. Before going further, click the red triangle in the upper left and choose **Set Subgroup Size**; specify n = 3. This chart looks very different from the ones that we have seen earlier. Does this indicate a problem? (Hint: Think about what this variable represents.) Why is this chart so different from the ones in the chapter?
    b. Look at the R chart more closely. What pattern or patterns do you see, and what might account for them?

c. Create an XBar-R chart for **NonDur** (non-durable goods) also using a subgroup size of 3. How does this chart compare to the one for primary goods? What might explain the similarities and differences?

d. Look at the R chart more closely. What pattern or patterns do you see, and what might account for them?

4. *Scenario:* Let's continue our analysis of the **Pipeline Safety** data.
   a. Create a Pareto chart for **LEAK_TYPE**. Report on your findings.
   b. Create one or more control charts to examine variation in the **TOTAL_COST** by **Year**. Then click the red triangle in the upper left, choose **Local Data Filter**, and filter by **Op_State**. Click the red triangle next to **Op_State** and choose **Order by Count**. You will find that Texas, Oklahoma, California, and Nebraska are the four states with most incidents. Click through the different states and report on how the control charts compare.
   c. Create a variability chart for **TOTAL_COST** by **CAUSE** and **LEAK_TYPE**. Comment on what you find.
5. *Scenario*: In the United States, the management of solid waste is a substantial challenge. Landfills have closed in many parts of the nation, and efforts have been made to reduce waste, increase recycling, and otherwise control the burgeoning stream of waste materials. Our data table **Maine SW** contains quarterly measurements of the tonnage of solid waste delivered to an energy recovery facility in Maine. The data come from five municipalities in the state. Each part of this scenario asks you to use a sample size of 4. To do that, choose your Y variable, click the red triangle in the upper left, and choose **Set Sample Size**; specify n = 4.
   a. Create an appropriate control chart for annual solid waste delivered from Bangor (that is, use a sample size of 4 quarterly observations). Comment on important or noteworthy features of the process.
   b. Create an appropriate control chart for annual solid waste delivered from Bucksport (that is, use a sample size of 4 quarterly observations). Comment on important or noteworthy features of the process.
   c. Create an appropriate control chart for annual solid waste delivered from Enfield (that is, use a sample size of 4 quarterly observations). Comment on important or noteworthy features of the process.
   d. Create an appropriate control chart for annual solid waste delivered from Orono (that is, use a sample size of 4 quarterly observations). Comment on important or noteworthy features of the process.
   e. Create an appropriate control chart for annual solid waste delivered from Winslow (that is, use a sample size of 4 quarterly observations). Comment on important or noteworthy features of the process.

6. *Scenario*: The World Bank and other international organizations monitor many forms of technological development around the world. Elsewhere, we have looked at other Sustainable Development Indicators. In this exercise, we will study carbon dioxide emissions between 2000 and 2016. The relevant data table is called **SDGCO2**. The time series column is called **CO2**, which measures the number of metric tons of CO2 emitted per capita annually in the country. Let's examine worldwide CO2 emissions.
   a. Create an XBar-R chart for **CO2**, using **Year** as the **Subgroup**. Then click the red triangle in the upper left, choose **Script ► Local Data Filter**, and filter by **Region**. Click through the different regions and report on what you see in this group of control charts.
   b. Repeat this analysis with XBar-S charts. How does this analysis differ from the previous one?
   c. Make a variability chart of **CO2** by Region. Report on what you find.

7. *Scenario*: Let's look again at some airline delays data from Chicago. This time, we will consider all flights from March 2019 and focus on *departure* delays for United Airlines flights. We will need to filter the data found in **Airline delays 3**.

   Start by using the global Data Filter to include only flights where the **Carrier ID** is **UA**, the **Origin** is **ORD**, and **Cancelled** = **0**.
   a. The departure time of a flight refers to the difference, if any, between the scheduled departure and the time at which the plane rolls away from gate. Create an XBar-S chart for **DepDelay** grouped by **Date** and report on the extent to which this process seems to be under control.
   b. Create a new temporary variable as follows: highlight **WheelsOff** and right-click. Choose formula, and edit the formula to equal **WheelsOff – Sched Dep**. This will calculate the elapsed time between scheduled departure and the moment when the wheels leave the runway. Create an XBar-S chart for this variable by Date and report on the extent to which this process seems to be under control. How does it compare to the previous chart?
   c. Create a Pareto Plot for the column **Main Cause**. Describe what the plot tells you. Which causes combined account for more than half of all delays?
   d. Create a variability plot of **DepDelay** by **Main Cause**. Discuss your findings.
   e. (Challenge) Imagine that the airline would like to control departure delays with a goal of having 90% of all non-canceled *weekday* flights leave the gate 20 minutes or less from the scheduled time. Run a capability analysis and report on your findings. (Hint: the column called **Weekend** is a dummy variable equal to 1 for Saturday and Sunday flights.)

8. *Scenario*: The data table called **Earthquakes** contains earthquake measurements across the entire globe for one month of 2009. Although this refers to a natural process of geologic activity, we can use control charts to look for changes in earthquake activity. The key variable is **Magnitude**, which measures the size of the earthquake on the logarithmic scale developed by Charles Richter.
    a. Use a means control chart and sample size of 10 to monitor the magnitude of earthquakes during this period. To set the sample size, choose your Y variable and then click the red triangle in the upper left and choose **Set Sample Size**; specify n = 10. Comment on noteworthy features of the chart.
    b. Repeat the analysis with a sample size of 20. How does this change alter the analysis, if at all?

---

[1] Control charts are sometimes referred to as Shewhart charts after Walter Shewhart, an engineer at Bell Laboratories who introduced them in the 1930s.

[2] There are some standard criteria against which JMP can report important deviations. This list is intended to introduce the concept of specific tests by which to judge process stability.

[3] Here are the formulas in which USL and LSL are the upper and lower specification limits.

$$CP = \frac{USL - LSL}{6s}$$

$$CPK = \min\left(\frac{USL - \bar{X}}{3s}, \frac{\bar{X} - LSL}{3s}\right)$$

$CPM = \frac{\min(T-LSL, USL-T)}{3\sqrt{s^2+(\bar{X}-T)^2}}$, where T is the target mean.

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